The construction of four-dimensional quantum Yang–Mills theory

Draft of November 13, 2026

Abstract

For every closed subgroup $G \subset U(n_G)$ with simple nonabelian Lie algebra $\mathfrak{g}$ and every positive block penalty, the coefficient of the soft constraint that ties each coarse link to a predictor computed from the fine transports, we construct gauge-invariant local fields from four-dimensional Wilson measures. Exact scale integration and tuning to a common endpoint coupling give the full ultraviolet limit at fixed volume. Reflection positivity and chessboard bounds give volume-uniform control; compactness gives a limit along a refinement of any prescribed strictly increasing volume sequence. The limit decomposes into phases, each the Euclidean continuation of a faithful Wightman theory with unique vacuum. The fields carry a conserved stress tensor, an operator product expansion with spectator-independent coefficients and phase-uniform remainders, and distributional extensions through collisions. In a selected phase, the centered stress trace has a two-point function G_Θ with $r^8 \log^2(1/r)\, G_\Theta(r) \to 3 \dim_{\mathbb{R}} \mathfrak{g}/\pi^4$ as $r \downarrow 0$, and on no interval $(0, \delta)$ does G_Θ agree with a finite Wick profile, a finite sum of products of free mass-shell Laplace kernels. No mass gap, uniqueness of the volume limit or phase, or independence from the block penalty is proved. This paper was written entirely by AI agents from OpenAI's Lean code for the construction of four-dimensional quantum Yang–Mills theory (commit `3456e5e`). The Lean proof is kernel-checked, with only the standard axioms. The agents compared every numbered statement here with the Lean declarations used by the main theorem; the correspondence table is published separately. The prose arguments were written and reviewed by AI agents. They are not machine-checked and have not been refereed by human mathematicians (Section 1.1).

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Part I
The theorem and the architecture of its proof

1 The result

How can a lattice model give a quantum field theory in continuous space-time? We answer this question for four-dimensional Yang–Mills theory with a compact gauge group whose Lie algebra is simple and nonabelian, starting from Wilson's lattice gauge theory [36]. On a finite periodic lattice the variables are group elements attached to the links, and their law is an ordinary probability measure. Let $G \subset U(n_G)$ be a closed matrix group, $s \geq 1$ an integer and $\Lambda_s = (\mathbb{Z}/s\mathbb{Z})^4$. A configuration assigns $U(y,a) \in G$ to the positive link from the site y in the coordinate direction a, and $U(y,a)^{-1}$ to the reversed link. For $\beta \in \mathbb{R}$,

$$\begin{aligned} U_{y;ab} &= U(y,a)U(y+e_a,b)U(y+e_b,a)^{-1}U(y,b)^{-1}, \\ E_s^{\mathrm{W}}(U) &= \sum_{y\in\Lambda_s}\sum_{a<b}\bigl(1-n_G^{-1}\operatorname{Re}\operatorname{tr}U_{y;ab}\bigr), \qquad d\mu_{s,\beta}(U) = Z_{s,\beta}^{-1}e^{-\beta E_s^{\mathrm{W}}(U)}\prod_{y,a}dh(U(y,a))\,, \end{aligned} \tag{1.1}$$

where dh is probability Haar measure and $Z_{s,\beta}$ normalizes the measure. The plaquette holonomy $U_{y;ab}$ is the transport around an elementary square, and β penalizes its distance from the identity. The problem is to choose the inverse coupling and the local observables as functions of the lattice spacing so that their correlations converge as the spacing tends to zero and the volume grows, and to identify the limit. No measure on continuum connections is constructed here. What converges are the correlations of gauge-invariant local observables at distinct points; the continuum theory is the resulting family of correlation functions, from which Hilbert spaces and relativistic quantum fields are then reconstructed.

Main result. Let the real Lie algebra $\mathfrak{g}$ of G be simple and nonabelian; G need not be connected. Fix a block penalty $\mathsf{b} > 0$: the coefficient of the soft constraint that ties each coarse link to a predictor computed from the fine transports in the block-spin kernel (Definition 5.1). We construct one family of Wilson measures on tori of physical side $L_p \uparrow \infty$ with spacings $a_N = D_N^{-1} \downarrow 0$, together with continuous gauge-invariant local insertions on fixed finite stencils, all chosen before any sequence of volumes is prescribed. Its bare inverse couplings satisfy the one-loop running law

$$\frac{\beta_{N,p}}{\log(1/a_N)} \longrightarrow \kappa_G = \frac{11C_{\mathfrak{g}}}{24\pi^2} \qquad (N\to\infty), \quad \text{uniformly in } p\,, \tag{1.2}$$

where $C_{\mathfrak{g}}$ is the adjoint Casimir in the trace normalization of Definition 2.1 (for $SU(n)$, $\kappa_G = 11n^2/(12\pi^2)$; Remark 2.2). The family has the following properties; Theorem B in Section 3 states them precisely.

(i) At each fixed volume, correlations of the insertions, smeared against smooth tests supported away from coincident points and inside one period cell, converge along the full sequence of spacings.

(ii) Every prescribed strictly increasing sequence p of volume indices has a strictly increasing refinement q such that, with $P = \mathsf{p} \circ \mathsf{q}$ and the lattice correlations $C^{\mathrm{lat}}_{N,p}$ of Definition 2.4,

$$\lim_{k\to\infty}\Big(\lim_{N\to\infty} C^{\mathrm{lat}}_{N,P(k)}(v;f)\Big) = \int f(x) S_n(v;x)\,dx \tag{1.3}$$

for one table S of Euclidean correlation functions, symmetric under permutations, translations and signed permutations of the axes, reflection positive, smooth away from collisions and polynomially bounded near them (properties E1–E7 of Section 2.4). One refinement serves all insertions and all tests.

(iii) $S = \int_{\mathsf{X}} S^{\xi}\,d\nu(\xi)$ is an average of phases over a compact metric space with a fully supported probability measure, and each S^{ξ} is the Euclidean continuation of a faithful Wightman theory with unique vacuum.

(iv) The fields carry a conserved symmetric stress tensor, normalized by its action on the other fields. In a selected phase, the two-point function G_Θ of the centered trace of this tensor satisfies

$$\lim_{r\downarrow 0} r^8 \log^2(1/r)\, G_\Theta(r) = \frac{3\dim_{\mathbb{R}}\mathfrak{g}}{\pi^4} > 0\,, \tag{1.4}$$

and on no interval $(0,\delta)$ does G_Θ agree with a finite Wick profile, a finite sum of finite products of free mass-shell Laplace kernels (Definition 2.18).

(v) Nearby fields have an operator product expansion whose coefficients depend neither on the other fields nor on the phase, with remainders bounded uniformly in the phase. The correlations of each phase extend across coincident points by an explicit subtraction of Taylor jets.

Theorem B holds for every $\mathsf{b} > 0$. Section 8 states the existence theorem in the exact form of the formal statement (Theorem A), which asks only for some positive logarithmic slope, and deduces it from Theorem B.

What is not claimed. The Clay problem [18, §4] asks for existence and a positive mass gap. The construction gives Wightman theories with unique vacuum, one in each phase; no mass gap is proved or used. Uniqueness of the vacuum means that zero is a simple eigenvalue of the energy operator. It gives no lower bound on the rest of the spectrum: a positive operator with one-dimensional kernel may have spectrum accumulating at zero, as the energy of a free massless field does, and nothing stated in this paper excludes such a spectrum for the theories constructed here. The infinite-volume limit, the phase space and the selected phase are not shown to be unique, and independence from b is not proved. Section 3 lists the limitations next to Theorem B.

1.1 Source of the result and of this text

The mathematical result is due to OpenAI's Lean construction at commit `3456e5e` of the repository `yang-mills-group/OpenAI-YM-20261028` [24]. Theorem B records that construction; Section 8 states its top-level existence theorem and deduces it. Three kinds of checking are involved, and they should be kept apart.

- The Lean kernel checks the formal proof. Its kernel-check record reports no unproved placeholders in the proved theorem and only propositional extensionality, classical choice and quotient soundness as axioms.
- AI agents compared every numbered statement of this text with the Lean declarations used by the main theorem: hypotheses, definitions, quantifier order, normalizations and conclusions,

with usage determined from the compiled dependencies. The correspondence table is published separately; the source repository supplies a declaration-usage table.

- The prose arguments were written and reviewed by AI agents. They are not machine-checked and have not been refereed by human mathematicians.

A correct formal declaration does not by itself show that an English statement has kept its hypotheses and quantifiers, or that a prose proof proves that statement. This text was written entirely by AI agents under human direction, with separate writing and checking of statements and arguments, by the method described in *How the manuscript is written*. The work is carried out in collaboration with Martin Hairer's *humanise-yang-mills* project.

1.2 Earlier work and the method

Table 1 lists, for the main mechanisms of the proof, the closest earlier rigorous result and its use here. The comparison concerns how these operations are combined; it assigns none of them a new origin, and we make no priority claim for renormalization-group principles or physical predictions.

Table 1: Mechanisms, earlier results and their use here.

Mechanism	Earlier result	Use here
Nonlinear block averages, gauge fixing, stationary backgrounds, marginal curvature-square extraction	Bałaban: analytic stationary backgrounds [2, Theorem 1, Proposition 6]; symmetric block kernels and the marginal analysis [3, pp. 252–254, §§4–5]; large fields for small effective couplings [4, Theorem 1, p. 355], physical observables left open there (p. 356)	Running couplings and local composite fields controlled together on one tuned family
Four-dimensional ultraviolet construction	Magnen–Rivasseau–Sénéor [21]: pure $SU(2)$, trivial sector, regularized axial gauge, fixed infrared cutoff; detailed convergence deferred and infrared cutoff retained (p. 327), composite observables set aside (p. 329)	Infinite-volume gauge-invariant local fields, phases, stress tensor and products
Disjoint polymers covering the large-field region; selective exponentiation	Gawędzki–Kupiainen, critical φ^4_4 [16, p. 212, (4.8), pp. 218–221]	Forced activities covering all large-field sites
Polymer extraction, reblocking, analytic norms, Gaussian integration	Brydges–Yau [8, §§1–2, pp. 378–384]	Activities and their weighted norms
Clipped kinetic term, local quadratic terms, regular errors, covering sum, exact coupling reset	OpenAI's sigma-model papers: the exact map of [23, §5, pp. 38–55] and its adaptation [22, §3.3, pp. 23–28]	The coordinates of the effective densities
Reflection positivity, transfer matrix, chessboard estimates	Osterwalder–Seiler [27], [31, §2a, Theorem 2.1]; Fröhlich–Israel–Lieb–Simon [14, §4, Theorems 4.1–4.2], lattice version for local functions [31, Theorem 2.2]	Volume-uniform bounds; reflection Hilbert spaces
Analytic continuation of Euclidean correlations	Osterwalder–Schrader [25], with the growth and boundary-value issues addressed in [26]	Reconstruction in every phase, from slab and distribution estimates; reflection positivity alone does not give it
One-loop running	Gross–Wilczek, Politzer [17, 28]; background-field coefficient [1, §3.3]	The slope κ_G, on the exact block map

Mechanism	Earlier result	Use here
Operator products, trace anomaly, Taylor subtraction	Wilson [35, §II]; all-order perturbative OPE in Euclidean Yang–Mills theory with gauge-invariant closure and associativity [13]; Collins–Duncan–Joglekar [9, (3.19)–(3.21)]; Brunetti–Fredenhagen [7, §5.2]	A nonperturbative OPE at each finite truncation, uniform in the phase; the trace law; contact extensions

The renormalization group is organized as in the method of Bauerschmidt, Brydges and Slade [6]: an explicitly defined map, a finite-dimensional flow of couplings, norms in which the remaining coordinate contracts, and a global trajectory with a tuned initial parameter. What requires proof here is the compatibility of the one-loop slope with the exact gauge block map, the coupling reset and the tuning to a common endpoint, followed by continuum identifications on one family: control of composite fields, restoration of rotations through stress-tensor fluxes, the trace law through all radii and product bounds uniform over phases.

1.3 Reading guide

Section 2 defines the objects needed to state Theorem B, which is Section 3. Section 4 explains the route of the proof and the mechanisms that make it work, with a dependency map (Figure 2). Sections 5 and 6 state the principal results, each with the idea of its proof, and Section 7 deduces Theorem B. Part II gives the proofs. Table 2 lists the principal results; their tags appear in statement titles. Section 9 records the order in which constants are chosen. A first reading can skip the coordinates of the effective densities (Section 5.2), the marked decomposition (Section 5.5) and Section 9. The index of notation follows the appendices.

Table 2: Principal results and their proofs.

Tag	Statement	Part I	Proof in Part II
RG	Exact integration of one scale and coupling drift	Thm 5.5	Sections 11–13
MT	Common endpoint coupling and running slope	Thm 5.6	Section 14.1
UB	Volume-uniform bounds on lattice differences	Prop 5.8	Section 14.2
UV	Full fixed-volume ultraviolet limit	Thm 5.9	Section 14.3
IV	Volume refinement and Euclidean properties	Thm 6.1	Section 15
FC	Inclusions, derivatives and Maxwell dimensions	Prop 6.4	Lemma 15.5
MC	Maxwell comparison at short distances	Prop 6.7	Lemma 16.8
ST	Conserved stress tensor and its shell fluxes	Thm 6.10	Lemmas 16.17–16.18
SC	Differentiated comparison with stress insertions	Prop 6.11	Lemma 16.19
PH	Phase decomposition on the same fields	Thm 6.13	Section 17.1
RC	Wightman reconstruction in every phase	Thm 6.16	Section 17.2
TR	All-radii law for the centered stress trace	Thm 6.18	Section 18.1
NF	Exclusion of finite Wick profiles	Prop 6.19	Lemma 18.10
OP	Operator product expansion with error bounds	Thm 6.20	Section 19.1
CT	Extensions through collision diagonals	Thm 6.21	Section 19.2

1.4 Notation and conventions

Throughout, $\mathbb{N} = \{0, 1, \dots\}$, coordinate indices run from 1 to 4, e_a is the ath coordinate vector, and time is the fourth coordinate. For the relativistic statements write

$$\mathsf{m}(x) = (x^4)^2 - |\vec{x}|^2, \qquad \overline{V}_+ = \{\mathbf{p} : |\vec{\mathbf{p}}| \le \mathbf{p}^4\}\,.$$

Inner products are antilinear in the first variable, and spacetime integrals use Lebesgue measure. Correlations of arity zero are included; empty sums are zero and empty products are one. The group H_4 consists of the signed permutation matrices, and $\mathsf{s}_a^{\mathrm{refl}}$ denotes reflection in the ath coordinate.

2 Definitions

This section defines the objects needed to state the main theorem: the gauge group and the Wilson measure, families of lattice theories and the convergence of their separated correlations, the local fields, the Euclidean properties of the limit, its field spaces, Wightman theories, and the quantities of the trace and product statements. The renormalization-group objects that construct the family are defined in Section 5, where they are used.

2.1 The gauge group and the Wilson measure

The matrix normalization fixes both the Wilson energy and the coefficient of the logarithmic coupling flow.

Definition 2.1 (Gauge normalization)**.** Fix $n_G \ge 1$ and a closed subgroup $G \subset U(n_G)$ with simple nonabelian real Lie algebra

$$\mathfrak{g} = \{X \in M_{n_G}(\mathbb{C}) : \exp(tX) \in G \text{ for every } t \in \mathbb{R}\}, \qquad \langle X, Y\rangle_{\mathfrak{g}} = -n_G^{-1}\operatorname{Re}\operatorname{tr}(XY)\,.$$

For an orthonormal basis $(e_\alpha^{\mathfrak{g}})$, put

$$d_{\mathfrak{g}} = \dim_{\mathbb{R}} \mathfrak{g}, \qquad C_{\mathfrak{g}} = -d_{\mathfrak{g}}^{-1}\sum_{\alpha} \operatorname{tr}_{\mathfrak{g}}\big(\operatorname{ad}(e_\alpha^{\mathfrak{g}})^2\big),$$

$$b_0 = \frac{11 C_{\mathfrak{g}}}{48\pi^2}, \qquad \kappa_G = 2b_0\,.$$

Here $\operatorname{ad}(X)Y = [X, Y]$ and $\operatorname{tr}_{\mathfrak{g}}$ is the real operator trace on $\mathfrak{g}$. Simplicity means $[\mathfrak{g}, \mathfrak{g}] \ne 0$ and no proper nonzero real ideal; connectedness of G is not assumed. The adjoint maps are skew-adjoint for this metric, so nonabelianity gives $C_{\mathfrak{g}} > 0$ and $\kappa_G > 0$. For $u \in G$ the plaquette energy and the normalized chord are

$$\operatorname{pr}(u) = 1 - n_G^{-1}\operatorname{Re}\operatorname{tr} u, \qquad \operatorname{ch}(u) = n_G^{-1/2}\|u - 1\|_{\mathrm{F}}, \qquad \operatorname{ch}(u)^2 = 2\operatorname{pr}(u)\,; \tag{2.1}$$

both are invariant under conjugation and inversion, and the chord measures the distance of a plaquette holonomy from the identity.

Remark 2.2 (The normalization for $SU(n)$). For $G = SU(n)$ in its defining representation, $d_{\mathfrak{g}} = n^2 - 1$, $C_{\mathfrak{g}} = 2n^2$ and $\kappa_G = 11n^2/(12\pi^2)$. With the conventional Wilson coupling $\beta = 2n/g^2$, the running law (1.2) reads $g^{-2}(a) \sim (11n/24\pi^2)\log(1/a)$, the leading behaviour of the one-loop law $\mu\, dg/d\mu = -(11n/48\pi^2)g^3$ with $\mu = 1/a$.

Definition 2.3 (Wilson measure)**.** For an integer $s \geq 1$ and $\beta \in \mathbb{R}$, $\mu_{s,\beta}$ is the probability measure (1.1) on the configurations of $\Lambda_s = (\mathbb{Z}/s\mathbb{Z})^4$, with plaquette holonomies $U_{y;ab}$, Wilson energy $E_s^{\mathrm{W}}(U) = \sum_y \sum_{a<b} \mathrm{pr}(U_{y;ab})$ and normalizing constant

$$Z_{s,\beta} = \int e^{-\beta E_s^{\mathrm{W}}(U)} \prod_{y,a} dh(U(y,a)) \,.$$

The integrand of $Z_{s,\beta}$ is continuous and strictly positive on the compact configuration space, so $0 < Z_{s,\beta} < \infty$. The link variable $U(y,a)$ carries a row vector $\psi \in \mathbb{C}^{n_G}$ at y to $\psi U(y,a)$ at $y + e_a$, and the ordered product along a nearest-neighbour path $y_0, \dots, y_k$, with inverses on reversed links, carries row vectors from y_0 to y_k. A site gauge transformation $\gamma_{\mathrm{g}} : \Lambda_s \to G$ changes the frame at each site, $\psi \mapsto \psi\gamma_{\mathrm{g}}(y)^{-1}$, and acts on links by $U^{\gamma_{\mathrm{g}}}(y,a) = \gamma_{\mathrm{g}}(y)U(y,a)\gamma_{\mathrm{g}}(y+e_a)^{-1}$. In the product along a path the intermediate factors $\gamma_{\mathrm{g}}(y_i)^{-1}\gamma_{\mathrm{g}}(y_i)$ cancel, so the product is multiplied by $\gamma_{\mathrm{g}}(y_0)$ on the left and $\gamma_{\mathrm{g}}(y_k)^{-1}$ on the right: only the endpoints remain. Around a plaquette the path closes:

$$\begin{aligned} U^{\gamma_{\mathrm{g}}}_{y;ab} &= \gamma_{\mathrm{g}}(y)U(y,a)\gamma_{\mathrm{g}}(y{+}e_a)^{-1} \cdot \gamma_{\mathrm{g}}(y{+}e_a)U(y{+}e_a,b)\gamma_{\mathrm{g}}(y{+}e_a{+}e_b)^{-1} \\ &\quad \cdot \gamma_{\mathrm{g}}(y{+}e_a{+}e_b)U(y{+}e_b,a)^{-1}\gamma_{\mathrm{g}}(y{+}e_b)^{-1} \cdot \gamma_{\mathrm{g}}(y{+}e_b)U(y,b)^{-1}\gamma_{\mathrm{g}}(y)^{-1} = \gamma_{\mathrm{g}}(y)U_{y;ab}\gamma_{\mathrm{g}}(y)^{-1}. \end{aligned}$$

Hence the energy is gauge invariant, and invariance of Haar measure under left and right multiplication makes the measure gauge invariant. The size of a single link variable therefore carries no gauge-invariant information: the configuration $U(y,a) = \gamma_{\mathrm{g}}(y)\gamma_{\mathrm{g}}(y+e_a)^{-1}$, the transform of the identity configuration, has links far from 1 when γ_{g} varies rapidly, yet every plaquette holonomy equals 1. Fields are measured by gauge-invariant quantities such as plaquette chords. For a smooth $\mathfrak{g}$-valued field $\mathcal{A}$ and links $U(y,a) = \exp(h_{\mathrm{lat}}\mathcal{A}_a(h_{\mathrm{lat}}y))$ at a small spacing h_{lat}, expansion to second order gives

$$U_{y;ab} = 1 + h_{\mathrm{lat}}^2 F_{ab}(h_{\mathrm{lat}}y) + O(h_{\mathrm{lat}}^3), \qquad F_{ab} = \partial_a \mathcal{A}_b - \partial_b \mathcal{A}_a + [\mathcal{A}_a, \mathcal{A}_b] \,:$$

the deviation of the plaquette holonomy from 1, divided by the enclosed area h_{lat}^2, tends to the curvature, and $\mathrm{ch}(U_{y;ab}) = h_{\mathrm{lat}}^2 |F_{ab}(h_{\mathrm{lat}}y)|_{\mathfrak{g}} + O(h_{\mathrm{lat}}^3)$ in the norm of Definition 2.1. The measure is also invariant under the lattice isometries $x \mapsto hx + b$, $h \in H_4$, which transport links and invert reversed ones (Section 10.4).

2.2 Wilson families and their correlations

A Wilson family, indexed by $N, p \in \mathbb{N}$, consists of strictly increasing positive integer sequences (D_N) and (L_p), real couplings $\beta_{N,p}$, a set $\mathcal{T}$ of insertion labels, and complex insertions $I_{N,p}(v, y; U)$, functions of the configurations on $\Lambda_{L_p D_N}$ for $v \in \mathcal{T}$ and sites y. The index N controls the spacing and p the physical volume:

$$a_N = D_N^{-1}, \qquad \mu_{N,p} = \mu_{L_p D_N, \beta_{N,p}}, \qquad \hat{y} = a_N \big(r_{L_p D_N}(y^a)\big)_{a=1}^4 \,, \tag{2.2}$$

where $r_s(j)$ is the integer representative of $j \in \mathbb{Z}/s\mathbb{Z}$ in $(-s/2, s/2]$. The lattice has $L_p D_N$ sites per side at physical distance a_N, so its physical side is L_p; $\hat{y}$ places a site in the centered period cell.

Definition 2.4 (Lattice moments)**.** For $v \in \mathcal{T}^n$, sites $y \in \Lambda^n_{L_p D_N}$ and a test $f : (\mathbb{R}^4)^n \to \mathbb{C}$, define

$$\begin{aligned} M^{\mathrm{lat}}_{N,p}(v; y) &= \int \prod_{i=1}^n I_{N,p}(v_i, y_i; U)\, d\mu_{N,p}(U), \\ C^{\mathrm{lat}}_{N,p}(v; f) &= a_N^{4n} \sum_{y \in \Lambda^n_{L_p D_N}} f(\hat{y}) M^{\mathrm{lat}}_{N,p}(v; y) \,. \end{aligned}$$

The moment is the expectation of a product of insertions; the smeared correlation is its Riemann sum against a test, the factor a_N^{4n} being the lattice volume element. The main theorem concerns a family with the following regularity.

Definition 2.5 (Q1. Regular Wilson family)**.** A Wilson family is regular if (a) each insertion $U \mapsto I_{N,p}(v, y; U)$ is continuous and gauge invariant; (b) for each $v \in \mathcal{T}$ there is a finite $\Sigma_v \subset \mathbb{Z}^4$, independent of N, p, y, such that $I_{N,p}(v, y; U)$ depends only on the links with both endpoints in $y + \Sigma_v$ modulo $L_p D_N$; (c) for some $b_* > 0$, $\beta_{N,p}/\log(1/a_N) \to b_*$ as $N \to \infty$, uniformly in p.

The physical diameter of the stencil $y + \Sigma_v$ tends to zero with a_N, so the insertions are local. Condition (c) is a logarithmic growth law for the bare inverse coupling, uniform in the volume.

Correlations are tested on separated configurations. For positions in $(\mathbb{R}^4)^n$ put

$$\mathrm{Conf}_n = \{x : x_i \neq x_j \text{ for } i \neq j\}, \qquad \mathrm{sep}_\infty(x) = \min_{i \neq j} |x_i - x_j|_\infty ,$$

with $\mathrm{sep}_\infty(x) = +\infty$ for $n \leq 1$. A *separated test* is a smooth function on $(\mathbb{R}^4)^n$ with compact closed support in Conf_n; it need not be a product of one-point tests. It is *p-admissible*, or supported inside volume p, if its support lies in $((-L_p/2, L_p/2)^4)^n$. The following definition keeps the order of the two limits explicit.

Definition 2.6 (Ordered convergence)**.** A Wilson family converges to kernels K along $P : \mathbb{N} \to \mathbb{N}$ if, for every n, $v \in \mathcal{T}^n$ and separated test f, the test is eventually $P(k)$-admissible and there are numbers $C_p^{\mathrm{UV}}(v; f)$ with

$$C_{N,p}^{\mathrm{lat}}(v; f) \longrightarrow C_p^{\mathrm{UV}}(v; f) \quad (N \to \infty) \qquad \text{at every admissible } p,$$
$$C_{P(k)}^{\mathrm{UV}}(v; f) \longrightarrow \int f(x) K_n(v; x)\, dx \quad (k \to \infty) .$$

The inner limit uses the full sequence of spacings at every volume containing the support of the test; only the outer limit runs along P. At volumes not containing the support, C_p^{UV} is unconstrained. In Theorem B the same P serves every arity, list of labels and test (Figure 1).

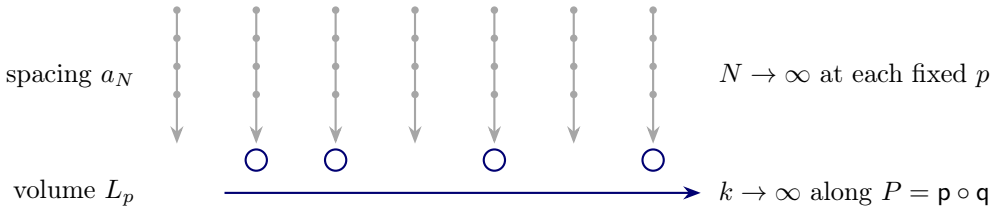


Figure 1: The two limits of Definition 2.6. At every fixed volume the full sequence of spacings is used (vertical arrows); the volume limit then runs along the retained indices $P(k)$ (circled), chosen once for all fields and tests.

2.3 Local fields

Two examples orient this subsection. The unit label has the constant insertion one at every sufficiently fine cutoff. The curvature square $c_{F^2} = \frac{1}{4}\sum_{a,b}\langle F_{ab}, F_{ab}\rangle_{\mathfrak{g}}$, the Yang–Mills density at a point, is read to leading order by the clover average $P^{\mathrm{cl}} = \frac{1}{4}\sum_{\square} \mathrm{pr}(U_{\square})$ over the 24 plaquettes with a corner at the site: since $\mathrm{pr} = \frac{1}{2}\mathrm{ch}^2$, the expansion of Section 2.1 gives each of them

$\frac{1}{2}h_{\mathrm{lat}}^4|F_{ab}|_{\mathfrak{g}}^2 + O(h_{\mathrm{lat}}^5)$ for the links of a smooth field, so $P^{\mathrm{cl}} = h_{\mathrm{lat}}^4 c_{F^2} + O(h_{\mathrm{lat}}^5)$ with F taken at the site. The construction itself uses the local sections below, normalized by a linear map (Definition 2.8).

In general, local fields are labelled by invariant polynomials in the Taylor coefficients of a connection. The labels use radial, or Fock–Schwinger, gauge, in which the radial component of the connection vanishes. In this gauge a smooth connection satisfies $\mathcal{A}_a(x) = \int_0^1 t\, x^b F_{ba}(tx)\, dt$, so its homogeneous part of degree $k+1$ is determined by the kth Taylor coefficients of the curvature at the origin.

Definition 2.7 (Polynomial labels)**.** Let $\mathcal{J}$ consist of formal $\mathfrak{g}$-valued arrays $\mathcal{A}_a(x) = \sum_{k\geq 0} \mathcal{A}_{k,a}(x)$, with $\mathcal{A}_{k,a}$ homogeneous of degree $k+1$ and $\sum_a x^a \mathcal{A}_{k,a}(x) = 0$. A coordinate of $\mathcal{A}_k$ has bidegree $(k+2, 1)$: engineering weight $k+2$ and factor count one. Put

$$V_d = \mathbb{C}[\mathcal{J}]^{\mathrm{Ad}(G)}_{\mathrm{weight}\leq d}\,,$$

with fixed bihomogeneous bases $b_{d,i}$ of bidegree (w_i, k_i), the supremum coefficient norm, conjugation and the induced H_4 action.

For instance, at $d = 4$ the retained part is $\mathcal{A}_0 + \mathcal{A}_1 + \mathcal{A}_2$: for a smooth connection in radial gauge, with the curvature and its derivatives taken at the origin and repeated indices summed,

$$\mathcal{A}_a(x) = \tfrac{1}{2}x^b F_{ba} + \tfrac{1}{3}x^b x^c \partial_c F_{ba} + \tfrac{1}{8}x^b x^c x^e \partial_c \partial_e F_{ba} + O(|x|^4)\,,$$

so the labels in V_4 are invariant polynomials in the curvature and its first two derivatives at the origin. Curvature has weight two and each derivative adds one; for instance c_{F^2} has bidegree $(4, 2)$. Three gradings play different roles below:

- the engineering weight w fixes the power of the spacing in the Taylor normalization (2.4);
- the factor count k fixes the scaling by the small-field threshold, through the operator $\mathsf{N} b_{d,i} = k_i b_{d,i}$ and $t^{\mathsf{N}} b_{d,i} = t^{k_i} b_{d,i}$;
- the degree d of the space V_d fixes the factor ℓ_m^d in the transfer of insertions under blocking and the physical prefactor a_N^{-d}.

A polynomial of weight at most $d \leq e$ lies in V_d and in V_e; its two insertions differ, since the normalization below depends on the degree.

Lattice holonomies are compared with smooth connection jets. For $\mathsf{j} = (\mathcal{A}_k)_k \in \mathcal{J}$, put $\mathcal{A}^{\leq d} = \sum_{k+2\leq d} \mathcal{A}_k$. The test holonomies at spacing h_{lat} are, on a positive link (y, a) with $y \in \mathbb{Z}^4$,

$$U(h_{\mathrm{lat}}, \mathsf{j})(y, a) = u(1), \qquad u'(t) = u(t) h_{\mathrm{lat}} \mathcal{A}_a^{\leq d}(h_{\mathrm{lat}}(y + te_a)), \qquad u(0) = 1_G\,, \tag{2.3}$$

and the jet dilation is $\mathrm{sc}_{h_{\mathrm{lat}}} \mathsf{j} = (h_{\mathrm{lat}}^{k+2} \mathcal{A}_k)_k$, so that $U(h_{\mathrm{lat}}, \mathsf{j}) = U(1, \mathrm{sc}_{h_{\mathrm{lat}}} \mathsf{j})$. For a function O of the links near the origin, its *Taylor map* $\mathrm{Tay}_d O$ is the part of weight at most d of the Taylor series at $\mathsf{j} = 0$ of $\mathsf{j} \mapsto O(U(1, \mathsf{j}))$. Lemma 10.2 constructs a *local section* $P_d^{\mathrm{loc}}[\mathbf{c}](y; U)$, linear in $\mathbf{c} \in V_d$ and polynomial in the real and imaginary parts of the link entries on a fixed finite stencil $y + \Sigma_d$; it is gauge invariant, commutes with complex conjugation, is H_4 covariant, sends $\mathbf{1}$ to one, and satisfies, for $0 \leq q \leq d$,

$$\frac{d^q}{dh_{\mathrm{lat}}^q} P_d^{\mathrm{loc}}[\mathbf{c}](0; U(h_{\mathrm{lat}}, \mathsf{j}))\bigg|_{h_{\mathrm{lat}}=0} = \frac{d^q}{dh_{\mathrm{lat}}^q} \mathbf{c}(\mathrm{sc}_{h_{\mathrm{lat}}} \mathsf{j})\bigg|_{h_{\mathrm{lat}}=0}\,. \tag{2.4}$$

Equivalently, $\mathrm{Tay}_d P_d^{\mathrm{loc}}[\mathbf{c}] = \mathbf{c}$: the local section is a right inverse of the Taylor map, which explains its name. A homogeneous label of weight w specifies the coefficient of h_{lat}^w; for example

$\mathrm{Tay}_4 P^{\mathrm{cl}} = c_{F^2}$. Smooth connections enter only as test configurations that define and normalize lattice observables; the labels assert nothing about differentiability of a continuum field.

An *insertion label* is a pair $v = (d, \mathbf{c})$ with $d \in \mathbb{N}$ and $\mathbf{c} \in V_d$; the set of insertion labels is $\mathcal{T} = \bigsqcup_d V_d$. The insertions of the construction are normalized at a fixed physical scale. Section 5 fixes, for each d, a reference level $m_{\mathrm{ref}}(d)$, the small-field threshold t_N at the cutoff (5.2), and, along the trajectories of Theorem 5.6, an invertible linear map $\mathsf{Z}_{N,p,d}$ of V_d, the terminal map: it sends a coefficient vector given at the cutoff to the coefficient vector of the blocked insertion at level $m_{\mathrm{ref}}(d)$ (Section 5.5). For $N < m_{\mathrm{ref}}(d)$ it is a fixed invertible map.

Definition 2.8 (Renormalized insertions)**.** For every N, p, $(d, \mathbf{c}) \in \mathcal{T}$ and site y, put

$$I_{N,p}((d,\mathbf{c}), y; U) = O^{(d)}_{N,p}(\mathbf{c}, y; U) = a_N^{-d} P_d^{\mathrm{loc}} \big[t_N^{-\mathsf{N}} \mathsf{Z}^{-1}_{N,p,d} \mathbf{c} \big](y; U)\,.$$

The unit coefficient is $\mathbf{u}_d = a^d_{m_{\mathrm{ref}}(d)} \mathbf{1}$.

The inverse terminal map makes the transferred coefficient equal to the prescribed $\mathbf{c}$ at the reference level, so $\mathbf{c}$ is specified at the fixed physical scale $a_{m_{\mathrm{ref}}(d)}$, in that level's lattice units; $t_N^{-\mathsf{N}}$ measures each factor in units of the small-field threshold, and a_N^{-d} converts to physical units. Because blocking can mix labels of different weight within V_d, the normalization is an invertible linear map of V_d rather than a factor for each label. As a first check, the insertion of $\mathbf{u}_d$ is the constant one for $N \geq m_{\mathrm{ref}}(d)$: the terminal map sends $a_N^d \mathbf{1}$ to $a^d_{m_{\mathrm{ref}}(d)} \mathbf{1}$ (Lemma 14.8), $t_N^{-\mathsf{N}} \mathbf{1} = \mathbf{1}$, and the section sends $\mathbf{1}$ to one. Part II follows the curvature square through these maps (Section 10.5); the following table summarizes the data of a smeared correlation.

Object	Role	Example
label $v = (d, \mathbf{c}) \in \mathcal{T}$	the local measurement	$(4, c_{F^2})$
insertion $I_{N,p}(v, y; U)$	its cutoff recipe on the stencil $y + \Sigma_d$	$a_N^{-4}\big(\lambda_{N,p} t_N^{-2} P_4^{\mathrm{loc}}[c_{F^2}](y; U) + \mu_{N,p}\big)$ for $N \geq m_{\mathrm{ref}}(4)$
separated test f	weights the positions in $C^{\mathrm{lat}}_{N,p}(v; f)$	$f_1(x_1) f_2(x_2)$, with f_1, f_2 of disjoint supports

Here $\lambda_{N,p} \neq 0$ and $\mu_{N,p}$ are real; the multiple of the unit accounts for mixing of the curvature square with the constant under blocking. The trace of the holonomy around a fixed lattice loop is also continuous, gauge invariant and local on a finite stencil, but it is not thereby an insertion of the family constructed here, and no limit is asserted for loops of fixed physical size, whose stencils grow as $a_N \to 0$.

2.4 Euclidean properties

A *table*, or family of Schwinger functions, assigns to each degree list $(d_i)_{i=1}^n$ and $x \in \mathrm{Conf}_n$ a complex multilinear function $S_n(c; x)$ of $c_i \in V_{d_i}$. For $v_i = (d_i, c_i)$ write $S_n(v; x)$, and for the word $w = (v_i(x_i))_{i=1}^n$ write $S(w)$; the product of words concatenates their lists. Conjugation and H_4 act on labels as in Definition 2.7, and $h \in H_4$ acts on positions linearly. Thus two insertion labels v_1, v_2 and two distinct points $x_1 \neq x_2$ select one number $S_2(v; x_1, x_2)$, and a separated test f on $(\mathbb{R}^4)^2$ selects the number $\int f(x) S_2(v; x)\, dx$. In the main theorem this number is the ordered limit (1.3): the full cutoff limit at each fixed volume, then the volume limit along a refinement that may depend on the prescribed sequence p and serves every label list and test. A table has no values on the collision diagonals. Values assigned there are invisible to separated tests (Step 7 of Section 7); a contact extension (Section 2.7) is instead a distribution on all of $(\mathbb{R}^4)^n$ that agrees with the table on separated tests, and Theorem 6.21 constructs one.

Definition 2.9 (E1–E3. Euclidean symmetries)**.** For every n, degree list, c and $x \in \mathrm{Conf}_n$:

(E1) $S_0 = 1$, $S_n(c_\pi; x_\pi) = S_n(c; x)$ for every permutation π of the slots, and $S_{n+1}(\mathbf{u}_d, c; x_0, x) = S_n(c; x)$;

(E2) $S_n(hc; hx) = S_n(c; x)$ for $h \in H_4$, and $S_n(\bar{c}; x) = \overline{S_n(c; x)}$;

(E3) $S_n(c; x + z) = S_n(c; x)$ for $z \in \mathbb{R}^4$, with $x + z = (x_i + z)_i$.

A word is positive-side for the axis a when all its positions have positive ath coordinate. For a word w, let $\theta_a w = \mathsf{s}_a^{\mathrm{refl}}(w^*)$, where w^* reverses the word and applies conjugation to its labels, and $\mathsf{s}_a^{\mathrm{refl}}$ reflects positions and acts on labels through H_4.

Definition 2.10 (E4–E5. Reflection positivity)**.** (E4) For every axis a, positive-side separated words w_i and $z_i \in \mathbb{C}$, $\sum_{i,j} \bar{z}_i z_j S(\theta_a(w_i)\, w_j) \geq 0$.

(E5) The same positivity holds for finite sums of words smeared against joint smooth compactly supported tests on positive-side separated configurations, and every mixed pairing is absolutely integrable.

For a fixed label list, D_x^k is the kth Fréchet derivative in the positions, with the product Euclidean norm. The last condition controls the approach to collisions uniformly over translations and over configurations of large diameter.

Definition 2.11 (E6–E7. Euclidean regularity)**.** (E6) Each $S_n(c; \cdot)$ is smooth on Conf_n.

(E7) For all n, d, c, k there are $C(n, d, c, k) \geq 0$ and an integer $A(n, d, c, k) \geq 0$ such that $\|D_x^k S_n(c; x)\| \leq C r^{-A}$ whenever $0 < r \leq 1$ and $\mathrm{sep}_\infty(x) \geq r$.

2.5 Field spaces and phases

A field is described in three successive ways. An insertion label (d, c) specifies a lattice insertion. Its class $[c]$ identifies labels with the same separated correlations: two labels represent the same field when every spectator sees the same answer. The response $\iota_d[c]$ records those correlations against every spectator list and places fields of all degrees in one space. Write $v(x) = (v_1(x_1), \ldots, v_n(x_n))$ for a spectator word.

Definition 2.12 (Observable fields)**.** For a table S put

$$\begin{aligned} \mathcal{N}_d(S) &= \big\{c \in V_d : S_{n+1}(c(x_0), v(x)) = 0 \text{ for every } n, v \text{ and } (x_0, x) \in \mathrm{Conf}_{n+1}\big\}, \\ F_d &= V_d/\mathcal{N}_d(S), \qquad \overline{[c]} = [\bar{c}], \qquad h[c] = [hc] \quad (h \in H_4), \\ \iota_d[c](v; x_0, x) &= S_{n+1}(c(x_0), v(x)), \qquad \mathcal{L} = \mathrm{span} \bigcup_d \iota_d F_d\,, \end{aligned}$$

with conjugation-invariant quotient norms on the finite-dimensional spaces F_d.

By E2, $\mathcal{N}_d(S)$ is invariant under conjugation and H_4, so both actions on F_d are well defined. By E1, correlations do not depend on representatives in any slot: for $\boldsymbol{A} = ([c_1], \ldots, [c_n])$ put $S_n(\boldsymbol{A}; x) = S_n(c; x)$. The unit field is $\mathbf{1}_d = [\mathbf{u}_d]$. Write $\mathbb{F} = \bigsqcup_d F_d$, $\deg(d, A) = d$ and $\iota(d, A) = \iota_d A$; the elements of $\mathcal{L}$ are the field labels.

Phases. The main theorem writes the limiting table as an average $S = \int_{\mathsf{X}} S^\xi \, d\nu(\xi)$ of tables S^ξ on the same field spaces, over a compact metric space X with a fully supported probability measure ν; the points ξ are the phases. A phase is not a set of configurations: Section 6.3 obtains the phases as the characters of a commutative algebra of operators.

2.6 Wightman theories

Write $\mathcal{S}_n = \mathcal{S}(\mathbb{R}^{4n};\mathbb{C})$ for the Schwartz space and $\mathcal{S} = \mathcal{S}_1$. Let $\mathcal{P} = SO^+(1,3) \ltimes \mathbb{R}^4$ be the proper orthochronous Poincaré group, where $SO^+(1,3)$ consists of the linear maps of determinant one that preserve m and $\overline{V}_+$; write $\gamma = (\Lambda_\gamma, z_\gamma)$ and $(\gamma \cdot f)(x) = f(\Lambda_\gamma^{-1}(x - z_\gamma))$.

Definition 2.13 (Relativistic datum)**.** A relativistic datum consists of a complex Hilbert space $\mathcal{H}$, a complex vector space $\mathcal{L}$ of field labels, a linear subspace $\mathcal{D} \subset \mathcal{H}$, a vacuum $\Omega \in \mathcal{D}$, and fields $\Phi(\ell, f) : \mathcal{D} \to \mathcal{D}$, linear operators that are complex bilinear in $(\ell, f) \in \mathcal{L} \times \mathcal{S}$; an antilinear map $\ell \mapsto \ell^*$; strongly continuous unitary translations $\mathsf{T}(z)$, $z \in \mathbb{R}^4$; a Borel projection-valued measure E on $\mathbb{R}^4$ with finite diagonal measures $\mu_\psi(B) = \langle \psi, \mathsf{E}(B)\psi\rangle$; a strongly continuous unitary representation U of $\mathcal{P}$ with $\mathsf{U}(1, z) = \mathsf{T}(z)$; and linear bijections $\rho_{\mathrm{lab}}(\gamma)$ of $\mathcal{L}$.

Definition 2.14 (Wightman theory)**.** A relativistic datum is a Wightman theory if:

(W0) $\langle \psi, \mathsf{T}(z)\psi\rangle = \int e^{i\mathbf{p}\cdot z}\, d\mu_\psi(\mathbf{p})$ for all ψ, z, with the Euclidean pairing; $\mathsf{E}(\mathbb{R}^4 \setminus \overline{V}_+) = 0$; and every vector fixed by all translations lies in $\mathbb{C}\Omega$.

(W1) $\|\Omega\| = 1$; $\mathcal{D}$ is dense and spanned by the vectors $\Phi(\ell_1, f_1)\cdots\Phi(\ell_k, f_k)\Omega$, $k \geq 0$; each $f \mapsto \Phi(\ell, f)\psi$ is continuous from $\mathcal{S}$ to $\mathcal{H}$; and $\langle \Phi(\ell, f)\psi, \chi\rangle = \langle \psi, \Phi(\ell^*, \bar{f})\chi\rangle$ for $\psi, \chi \in \mathcal{D}$.

(W2) ρ_{lab} is a representation; $\mathsf{U}(\gamma)\mathcal{D} \subset \mathcal{D}$, $\mathsf{U}(\gamma)\Omega = \Omega$ and $\mathsf{U}(\gamma)\Phi(\ell, f)\psi = \Phi(\rho_{\mathrm{lab}}(\gamma)\ell, \gamma \cdot f)\mathsf{U}(\gamma)\psi$ for $\psi \in \mathcal{D}$.

(W3) $[\Phi(\ell, f), \Phi(\ell', h)]\psi = 0$ for $\psi \in \mathcal{D}$ whenever $\mathsf{m}(x - y) < 0$ for all $x \in \operatorname{supp} f$, $y \in \operatorname{supp} h$.

Definition 2.15 (Faithful labels)**.** The labels are faithful if, for every nonzero $\ell \in \mathcal{L}$, some $f \in \mathcal{S}$ has $\Phi(\ell, f) \neq 0$.

Reconstruction uses one function in two limits: at imaginary times it equals the Euclidean correlation, and approaching real times from the ordered tube gives the Wightman distributions. We first specify the continuation, then its boundary value.

Definition 2.16 (Ordered-time continuation)**.** For a phase ξ, put $\mathcal{Z}_n = \{z \in \mathbb{C}^n : \operatorname{Im} z_1 < \cdots < \operatorname{Im} z_n\}$. A continuation $\mathsf{F}_n^\xi(\boldsymbol{A}; z, \vec{x})$, for $\boldsymbol{A} \in \mathbb{F}^n$, is holomorphic in $z \in \mathcal{Z}_n$ at each fixed real spatial configuration and satisfies

$$\mathsf{F}_n^\xi(\boldsymbol{A}; (it_i)_i, \vec{x}) = S_n^\xi(\boldsymbol{A}; (\vec{x}_i, t_i)_i) \qquad (t_1 < \cdots < t_n)\,.$$

Definition 2.17 (Boundary condition)**.** For a phase ξ, a relativistic datum $(\mathcal{H}_\xi, \mathcal{L}, \mathcal{D}_\xi, \Omega_\xi, \Phi_\xi, \ldots)$, the maps ι_d and a continuation F_n^ξ, require, for every n, $\boldsymbol{A} \in \mathbb{F}^n$, real $h_1 < \cdots < h_n$ and $f_i \in \mathcal{S}$, that the integrand below be integrable for every $r > 0$ and

$$\lim_{r\downarrow 0} \int \mathsf{F}_n^\xi(\boldsymbol{A}; (x_i^4 + irh_i)_i, (\vec{x}_i)_i) \prod_i f_i(x_i)\, dx \\ = \langle \Omega_\xi, \Phi_\xi(\iota(A_1), f_1)\cdots\Phi_\xi(\iota(A_n), f_n)\Omega_\xi\rangle\,.$$

A phase S^ξ is the *Euclidean continuation* of a Wightman theory when continuations satisfying Definitions 2.16 and 2.17 exist for that theory and the response maps of Definition 2.12.

2.7 The trace and product statements

For a phase ξ and a field $\Theta \in F_4$, the time-axis two-point function is

$$G_\Theta(r) = S_2^\xi(\Theta, \Theta; -re_4/2, re_4/2) \qquad (r > 0)\,.$$

The trace law is compared with free fields through the following class. Fix $m_{\mathrm{sh}} \geq 0$, integers $j_h, k_E \geq 0$ and a continuous real h_{mom} on $\mathbb{R}^3$ with $h_{\mathrm{mom}}(t\mathbf{k}) = t^{j_h} h_{\mathrm{mom}}(\mathbf{k})$ for $t > 0$.

Definition 2.18 (Finite Wick profiles)**.** Put $\omega_{m_{\mathrm{sh}}}(\mathbf{k}) = \sqrt{m_{\mathrm{sh}}^2 + |\mathbf{k}|^2}$ and, for $r > 0$,

$$K_{m_{\mathrm{sh}},k_E,h_{\mathrm{mom}}}(r) = \int_{\mathbb{R}^3} \frac{\omega_{m_{\mathrm{sh}}}(\mathbf{k})^{k_E} h_{\mathrm{mom}}(\mathbf{k})}{2\omega_{m_{\mathrm{sh}}}(\mathbf{k})} e^{-r\omega_{m_{\mathrm{sh}}}(\mathbf{k})}\, d^3\mathbf{k}\,.$$

A finite Wick profile is a function $W_{\mathrm{Wick}}(r) = \sum_{i=1}^{N} c_i \prod_{j=1}^{n_i} K_{m_{ij},k_{ij},h_{ij}}(r)$, $r > 0$, with $c_i \in \mathbb{C}$ and finitely many terms and factors of this kind; empty products are one.

The momentum integrals converge absolutely for $r > 0$. For a free scalar field of mass m_{sh} the time-axis two-point function is $(2\pi)^{-3} K_{m_{\mathrm{sh}},0,1}(r)$; derivatives give momentum factors, and Wick contractions give finite sums of products at the same radius.

The product expansion replaces a cluster of nearby fields by fields at one point, inside correlations with distant spectators. For finite index sets $I_{\mathrm{cl}}, I_{\mathrm{sp}}$, write $S^{\zeta}(\boldsymbol{A}, x; \boldsymbol{B}, y)$ for the correlation of the concatenated lists, in a fixed enumeration of $I_{\mathrm{cl}} \sqcup I_{\mathrm{sp}}$, where $\zeta \in \mathsf{X} \sqcup \{\mathrm{mix}\}$ indexes a phase or the mixture, $S^{\mathrm{mix}} = S$. A pair (x, o), with $x \in (\mathbb{R}^4)^{I_{\mathrm{cl}}}$ and $o \in \mathbb{R}^4$, is an $(r, M_{\mathrm{geo}}, \eta)$-cluster if $|x_i - o| \leq M_{\mathrm{geo}} r$ and $|x_i - x_{i'}| \geq \eta r$ for distinct slots; spectators $y \in (\mathbb{R}^4)^{I_{\mathrm{sp}}}$ are δ_{sp}-clear if $|o - y_j| \geq \delta_{\mathrm{sp}}$ and $|y_j - y_{j'}| \geq \delta_{\mathrm{sp}}$ for distinct slots. The expansion uses finite channel sets Ch_j, real fields $O_{j,\lambda} \in F_j$ forming bases of conjugation-invariant complements of F_{j-1} in F_j ($F_{-1} = 0$), and coefficient functions $c_{j,\lambda}(\boldsymbol{A}; x, o)$, all fixed once before spectators, states and truncations; Section 6.4 describes how they are read off. Put $|\boldsymbol{A}|_{\mathrm{deg}} = \sum_i \deg A_i$.

Definition 2.19 (Product remainder)**.** For $D \in \mathbb{N}$, $\boldsymbol{A} \in \mathbb{F}^{I_{\mathrm{cl}}}$ and $\boldsymbol{B} \in \mathbb{F}^{I_{\mathrm{sp}}}$, put

$$\mathcal{R}_D^{\zeta} = S^{\zeta}(\boldsymbol{A}, x; \boldsymbol{B}, y) - \sum_{j=0}^{D} \sum_{\lambda \in \mathrm{Ch}_j} c_{j,\lambda}(\boldsymbol{A}; x, o) S^{\zeta}(O_{j,\lambda}(o), \boldsymbol{B}(y))\,.$$

If $M_{\mathrm{geo}} r < \delta_{\mathrm{sp}}$, the concatenated positions are separated, $|x_i - y_j| \geq \delta_{\mathrm{sp}} - M_{\mathrm{geo}} r > 0$, so the remainder is defined in the small-radius range used below. A *contact extension* of $S_n^{\zeta}(\boldsymbol{A}; \cdot)$ is a distribution on $\mathbb{R}^{4n}$ whose value on every separated test f is $\int f(x) S_n^{\zeta}(\boldsymbol{A}; x)\, dx$; Theorem 6.21 constructs one by an explicit subtraction.

3 The main theorem

The main theorem concerns one Wilson family $\mathcal{W}$, fixed before any sequence of volumes is prescribed: the renormalized insertions of Definition 2.8 on the bare couplings of the multiscale theorem (Theorem 5.6), for a fixed block penalty. These couplings are tuned, volume by volume, so that the running coupling reaches one common endpoint coupling $\bar{\beta}$ at the unit scale (Section 5.4). Clause (ii) also packages the objects of clauses (iii)–(v) as hierarchy data satisfying the conditions Q2–Q6 of the formal statement (Section 8); there ν is the measure of the phase decomposition (Theorem 6.13), ξ_* the selected phase (Section 6.3) and Θ the centered stress trace (Definition 6.17), and Section 7 assembles the hierarchy data from these objects.

Theorem B (Main theorem)**.** *For every gauge group G of Definition 2.1 and $\mathsf{b} > 0$, the renormalized insertions (Definition 2.8) on the penalty-b trajectories of Theorem 5.6 form a regular Wilson family $\mathcal{W}$ with $a_N \downarrow 0$, $L_p \uparrow \infty$ and $\beta_{N,p}/\log(1/a_N) \to \kappa_G$ uniformly in p. Moreover:*

(i) *For every p, $v \in \mathcal{T}^n$ and p-admissible separated f, $C^{\mathrm{lat}}_{N,p}(v;f) \to \int f M^{\mathrm{UV}}_{p,v}$ as $N \to \infty$.*

(ii) *For every strictly increasing p there are a strictly increasing q, a table S satisfying E1–E7 and hierarchy data over $\mathcal{W}$ satisfying Q2–Q6 with $\mathcal{W} \to S$ and $\mathcal{W} \to S^{\mathrm{mix}} \circ \pi_{\mathrm{quo}}$ along $P = \mathsf{p} \circ \mathsf{q}$. In (iii)–(v), S is this table, $F_d = V_d/\mathcal{N}_d(S)$, and ν, ξ_*, Θ are those of these hierarchy data.*

(iii) *$S = \int_{\mathsf{X}} S^\xi \, d\nu(\xi)$ with ν a fully supported probability measure on a compact metric space; each S^ξ is the Euclidean continuation of a faithful Wightman theory with unique vacuum.*

(iv) *A conserved stress tensor is normalized by its shell fluxes (Theorem 6.10); in the selected phase ξ_* its centered trace Θ has*

$$r^8 \log^2(1/r) G_\Theta(r) \to 3 d_{\mathfrak{g}}/\pi^4,$$

and G_Θ agrees with no finite Wick profile on any interval $(0,\delta)$.

(v) *The fields F_d have an operator product expansion with spectator-independent coefficients and*

$$|\mathcal{R}^\zeta_D| \le r^{D+1-|\boldsymbol{A}|_{\deg}-\varepsilon},$$

with the quantifiers and uniformity of Theorem 6.20, and contact extensions from the same prescription.

The kernel $M^{\mathrm{UV}}_{p,v}$ of clause (i) is the fixed-volume limit of Theorem 5.9, the convergence $\mathcal{W} \to S$ in clause (ii) is that of Definition 2.6, and $\pi_{\mathrm{quo}}(d,\mathbf{c}) = (d,[\mathbf{c}])$ sends an insertion label to its field. Section 7 proves the theorem from the principal results of Sections 5 and 6.

Order of choices. The gauge group, its realization and the block penalty are fixed first. The lattice family, with its couplings and insertions, is then fixed before the volume sequence p. The refinement q, the table S, the field spaces, the phases, the selected phase, the relativistic theories, the product coefficients and the trace field may depend on p. One refinement serves every arity, every list of insertion labels and every separated test. The lattice limit in clause (ii) is the mixture S; the relativistic theories of clause (iii) belong to the phases, and the trace law of clause (iv) is stated in the selected phase.

What the theorem does not assert. It does not prove a mass gap, that is, a positive lower bound on the nonzero energies; uniqueness of the vacuum means only that the zero-energy space is one-dimensional, and no clause excludes nonzero energies accumulating at zero. It does not assert that the infinite-volume limit, the phase space or the selected phase is unique, nor that different block penalties give the same theory, and it does not take the spacing and volume limits simultaneously. It does not show that widely separated groups of observations become independent in the mixed state, or that any one phase is itself a limit of lattice states. The lattice correlations converge against separated tests; the contact extensions are constructed afterwards from the separated correlations, are not limits of unsubtracted lattice correlations at coincident points, and are not asserted to be unique, positive or to satisfy contact Ward identities. The product expansion is an estimate at each finite truncation degree, not the convergence of a series as that degree tends to infinity. The trace law is asserted in the selected phase only, and it excludes the finite Wick profiles of Definition 2.18; a generalized free field with the spectral measure of Θ has the same two-point function.

Section 8 states Theorem A, the existence statement in the exact form of the formal statement, with its conditions Q0–Q6, and deduces it from Theorem B with any $\mathsf{b} > 0$.

4 Strategy of the proof

The proof fixes one lattice family, takes its two limits and identifies the resulting fields; every estimate applies to this same family. Figure 2 shows the order in which the principal results use one another. This section explains the mechanisms that make the main steps work; Sections 5–6 state the results with the ideas of their proofs.

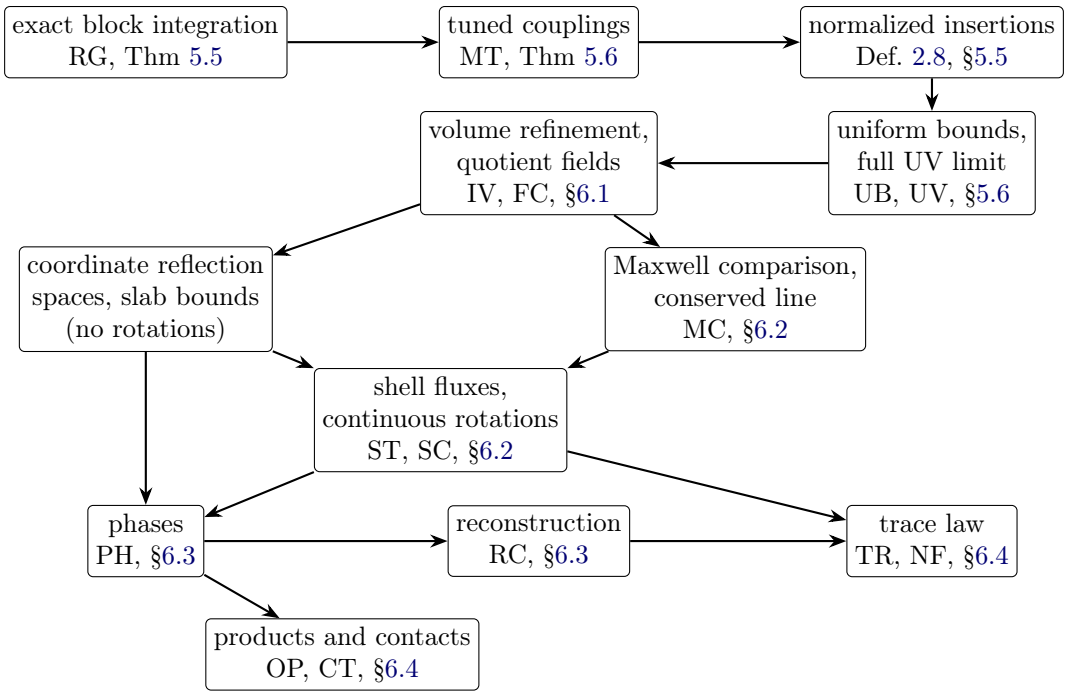


Figure 2: Dependency map. An arrow means that the later result uses the earlier one; the trace law and the products branch off the main route. The coordinate reflection spaces and slab bounds use only the volume limit and the field calculus; they enter the rotation argument before continuous rotations are available, so the order is not circular. The renormalization group runs only from the cutoff to the unit scale, and no mass gap is used.

4.1 Why the ultraviolet induction closes

Blocking integrates the fine links against a probability kernel and returns an exact density for the coarse links (Definition 5.1); it lowers the level from $m+1$ to m and multiplies the spacing by the odd block ratio ℓ_m (Figure 3). The density is written as a curvature action with the running coupling β_m, a scalar normalization, regular activities on small-field supports and forced activities covering every large-field site (Section 5.2). The equality is exact; the estimates control the coordinates in which it is written. Five points make the induction close.

One coupling to tune. In the local radial-gauge expansion of a gauge-invariant local term (Sections 12–13), the contributions of engineering weight at most four are a constant, terms linear in the curvature and its derivatives, and a quadratic curvature term. The constant goes into the scalar normalization. The linear terms vanish, since an $\mathrm{Ad}(G)$-invariant linear function on the simple algebra $\mathfrak{g}$ is zero; in particular there is no mass term. Simplicity of $\mathfrak{g}$ and the hypercubic

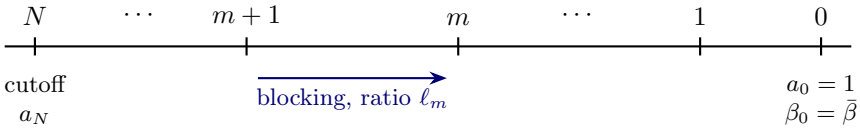


Figure 3: Levels and the direction of integration. Level m has spacing $a_m = D_m^{-1}$; blocking runs from the fine level $m+1$ to the coarse level m, so $a_{m+1} < a_m$, and the running coupling decreases by about $\kappa_G \log \ell_m$, so $\beta_{m+1} > \beta_m$ to leading order. The ratios grow: $\log \ell_m \to \infty$ and $\log \ell_m / \log D_m \to 0$. The endpoint coupling $\bar{\beta}$ is prescribed at level zero, the reference unit of length, which is not a measured mass scale.

and reflection symmetries leave one invariant quadratic curvature term, whose coefficient the construction moves into the coupling. The regular activities that remain vanish at the flat configuration and, on supports that do not wrap around the torus, have zero Hessian along affine directions (Definition 13.1). This subtraction plays the role of the localization operator of Bauerschmidt, Brydges and Slade [6, §5.2].

The scales. Write, at one level, $b = \widehat{\beta}_m$ for the reference coupling, $M = \exp((\log\log b)^2) = (\log b)^{\log\log b}$, which grows faster than every power of $\log b$ and slower than every power of b, the small-field threshold $t = b^{-1/2}M$ and the regular allowance $\varepsilon = b^{-11/10}$ (equation (5.2)). Plaquette fluctuations at coupling b have size $b^{-1/2}$, so t is that size enlarged by M, and $t \to 0$. The comparisons that fix these choices are, for $\beta \asymp b$:

Comparison	Consequence
$\beta t^5 \asymp b^{-3/2}M^5 = o(\varepsilon)$	the Wilson energy matched through fourth order leaves a regular error inside the allowance
$t^3/\varepsilon = b^{-2/5}M^3 \to 0$	a cubic term with bounded coefficient at the threshold fits inside the allowance
$\varepsilon/t^2 = b^{-1/10}M^{-2} \to 0$	a quadratic term at the threshold does not: residual quadratic coefficients go into the coupling
$\beta t^2 \asymp M^2$	a plaquette above the threshold costs a factor of order e^{-cM^2}, which is $o(b^{-k})$ for every fixed k

The forced allowance $e^{-M^{1/5}}$ is $o(b^{-k})$ for every fixed k but much larger than e^{-cM^2}; this leaves room for the combinatorial losses of the forced terms.

The cancellations. Two cancellations make the regular bound contract. First, an incoming regular activity, written in the coarse field, loses its constant, linear and affine quadratic terms by flat normalization, colour invariance and the affine Hessian condition; the remaining terms of its principal part carry at least ℓ_m^{-5} (on large supports after conversion of the support weight), against ℓ_m^4 fine anchors above a coarse anchor, a net factor V_j/ℓ_m with V_j fixed before the ratio, while the comparison remainder has relative size $b'^{-1/2+o(1)}$, where $b' = \widehat{\beta}_{m+1}$ is the reference coupling of the fine level (Lemma 13.6). Second, the new terms produced by the integration are connected local terms whose bound from the connected expansion is only $P_j(b')b'^{-7/16}$ (Lemma 13.5), where $j = J_{m+1}$ and P_j denotes functions of subpower growth fixed with the stage; this is far from the allowance $\widehat{\beta}_m^{-11/10}$. Most of these terms are already small enough: the connected expansion still converges when every elementary factor is weighted by $b'^{3/8}$, so terms with three or more factors, or

with a large-field overlap, are at most $P_j(b')b'^{-19/16} \le b'^{-23/20}$, and long paths, large supports and products with an incoming activity also stay below $b'^{-23/20}$. For the rest, the improvement comes from evaluating totals rather than separate terms. Section 13 selects a finite family of low-order connected terms (at most two connected products of at most two factors, short paths, no incoming activity), invariant under translations and H_4; the unselected low-order terms have weighted norm at most $\varepsilon_{\mathrm{tr}}$; every complete read support of these terms, relative to a site receiving a share, lies within distance R of that site. Let F_y be the sum of the shares at the site y, a connected term being shared equally among the sites of its support and a path term assigned to its anchor. For a radius $w > 4R$ fixed in Section 13, with $2(w+5)$ less than the side of Λ by the period requirements of Section 9, let Λ_i be the torus of side $2(w+i)$, $i = 0, \dots, 4$, let $V_{i,x}$ be the configuration on Λ_i obtained from the configuration near x by repeated reflection, let $F_y^{(i)}$ be the same local formula on Λ_i with its flat value subtracted, and put

$$\lambda_i = \frac{(-1)^i \binom{4}{i}}{2^4\, 4!}, \qquad G_x = \sum_{i=0}^{4} \lambda_i \sum_{y \in \Lambda_i} F_y^{(i)}(V_{i,x})\,. \tag{4.1}$$

Lemma 4.1 (Five-torus localization)**.** *For the selected low-order family, the function G_x depends only on the links within distance $w+5$ of x, and, in the local derivative seminorm of Definition 5.3,*

$$\|G_x\|_m \le P_j(b')\big(b'^{-23/20} + \varepsilon_{\mathrm{tr}}\big).$$

On configurations without large-field sites, $\sum_y F_y = c|\Lambda| + \sum_x G_x$ for a constant c.

This is Lemma 13.7, which also controls the correction near large-field sites. Two facts give it. On each auxiliary torus the total of the selected family is, up to the omitted terms, a normalized local integral, in which the flat value, the classical cubic and quartic terms and the one-loop quadratic term are subtracted exactly and Gaussian parity removes a lone cubic term; what remains has at least three factors of size $b'^{-1/2+o(1)}$ or a further Taylor gain. To recover a local function from totals, classify the sites of Λ_i by their distances from the reflection faces, evaluating the summand at y on the configuration $V_{i,x-\tilde{y}}$ reflected about $x - \tilde{y}$, with $\tilde{y}$ the integer representative of y. Site y of that cube then sees the configuration near x, so each class contributes a fixed local value, with a multiplicity that is a polynomial in i of degree at most four; degree four occurs only for the interior class, with leading coefficient 2^4. Since $\sum_{i=0}^{4}(-1)^i\binom{4}{i}i^k$ is 0 for $k < 4$ and 4! for $k = 4$, the weights λ_i cancel every boundary class and keep one copy of the interior value $F_x - F_x(1)$; summing over x, which turns $x - \tilde{y}$ back into x, gives the identity of Lemma 4.1. As $23/20 > 11/10$, the localized terms fit the allowance with room to spare.

Large fields. The estimates keep the configurations with large fields; they never replace the measure by its small-field restriction. A forced activity vanishes unless its support contains a large-field site, and the large-field lower bound on the quadratic energy (Lemma 11.12) gives a factor $e^{-c\beta t^2} \le e^{-cM^2/2}$ for each element of a selection (large fine plaquettes, mismatched coarse edges and flagged coarse sites), which pays for the polynomial and combinatorial losses of the forced terms. The resulting activity bounds (5.12) are local and uniform in the volume. They do not say that the whole torus is free of large fields with high probability: a large box may contain many flagged regions, each strongly suppressed, and the forced sum of Definition 5.2 keeps all of them; nor do they bound the density itself uniformly in the volume. The activities are real correction functions, not probabilities of their supports, and their norms use absolute values. Positivity of the density is inherited from the positive Wilson density through the exact block identity, since the block kernel is positive; it does not require any activity to be nonnegative.

Growing block ratios. The drift error of one step is bounded by a constant ω_j^{RG} depending only on the stage $j = J_{m+1}$ (Theorem 5.5). With a fixed block ratio ℓ, the accumulated drift bound $N\omega^{\mathrm{RG}}$ would be of the same order as the main term $\kappa_G \log D_N = N\kappa_G \log \ell$. The construction instead lets the ratios grow, $\log \ell_m \to \infty$ and $\log \ell_m / \log D_m \to 0$, and chooses them with $\omega_j^{\mathrm{RG}} \le j^{-2} \log \ell_m$; the summed errors are then $o(\log D_m)$ and the exact one-loop slope survives (equation (5.14)). Each level carries a stage J_m, with $4 \le J_m \le J_{m+1} \le J_m + 1$ and $J_m \to \infty$, which fixes the derivative order, the other estimate constants and the ratio $\ell_m = \ell^{\mathrm{st}}_{J_{m+1}}$. A stage may persist through many levels: entry into the next stage is delayed until the coupling and the period meet its requirements. Every fixed derivative order is therefore eventually available, while each step uses finitely many constants. All estimate constants of the induction are uniform in the volume; some are fixed before the block ratios and hence uniform in the ratio, the others may depend on a stage whose ratio is already fixed (Section 9).

4.2 Tuning, uniqueness and convergence

Three different tasks concern the cutoff, and they are not interchangeable.

Task	Mechanism	What it does not supply
Reach the endpoint coupling $\bar\beta$ at level zero	Continuous shooting and bracketing (Theorem 5.6)	Uniqueness of the shooting parameter
Identify the limits at a fixed level	Weighted contraction toward the common endpoint (Proposition 5.7)	Convergence of the scalar normalizations
Remove the cutoff at fixed volume	Fixed-level convergence and a uniform stopping error (Theorem 5.9)	An infinite-volume limit without refinement

Uniqueness of the shooting parameter is not needed: trajectories for different cutoffs are compared only through their common endpoint, and two fixed-level limits with that endpoint coincide by the contraction. Remark 7.2 records the uniqueness and analytic dependence that the recursion also gives; the proofs do not use it.

The ultraviolet limit is a Cauchy argument through a common level. For insertions at points of $a_m\mathbb{Z}^4$, which lies in every finer lattice, let M_N be the lattice moment and $C_{N,m}$ the normalized level-m integral of the unmarked density and the transferred one-insertion functions, with the components joining two or more insertions omitted. For cutoffs beyond a threshold depending on m,

$$\begin{aligned} |M_N - M_{N'}| &\le \underbrace{|M_N - C_{N,m}|}_{\text{stopping error}} + \underbrace{|C_{N,m} - C_{N',m}|}_{\text{fixed level}} + \underbrace{|C_{N',m} - M_{N'}|}_{\text{stopping error}} \\ &\le 4F_0 e^{-B_m/64} + |C_{N,m} - C_{N',m}|, \end{aligned}$$

where F_0 does not depend on m or on the cutoffs (Lemma 14.32). The reason is separation. At fine levels the sites of distinct insertions are at least B_m^2 apart, so every omitted component, which joins two or more insertions, has covering number at least $B_m/64$. Multiplying all these components by a complex z with $|z| < e^{B_m/64}$ therefore costs only two units of the support exponent, and the resulting normalized integral $F_m(z)$, carried back to a base level fixed before m, is bounded there by F_0. Since $F_m(1) = M_N$ and $F_m(0) = C_{N,m}$, Cauchy's estimate gives $|M_N - C_{N,m}| \le 2F_0 e^{-B_m/64}$. Choose m to make the stopping errors small, hold it fixed, and let $N, N' \to \infty$. The middle term tends to zero for three reasons. First, the scalar normalization cancels. Write $\mathrm{dens}^\circ \rho_{N,p,m}$ for the density of Definition 5.2 with v_{sc} replaced by zero, so that $\mathrm{dens}\, \rho_{N,p,m} = e^{v_{\mathrm{sc}}|\Lambda|}\, \mathrm{dens}^\circ \rho_{N,p,m}$

with $\Lambda = \Lambda_{L_p D_m}$, and $H^{\circ}_{N,m}$ for the integrand of the numerator of $C_{N,m}$ with v_{sc} replaced by zero, formed from the same coordinates and the transferred one-insertion functions, regular and forced. Then

$$C_{N,m} = \frac{\int e^{v_{\mathrm{sc}}|\Lambda|} H^{\circ}_{N,m}\, dU}{\int e^{v_{\mathrm{sc}}|\Lambda|}\, \mathrm{dens}^{\circ}\, \rho_{N,p,m}\, dU} = \frac{\int H^{\circ}_{N,m}\, dU}{\int \mathrm{dens}^{\circ}\, \rho_{N,p,m}\, dU}\,,$$

and the constant v_{sc} of $\rho_{N,p,m}$, whose convergence is not asserted, drops out. Second, the unmarked coordinates converge at each fixed level (Proposition 5.7), and the denominator stays above a positive constant for all large cutoffs, by restriction to small links in an axial gauge. Third, the marked coefficients converge because one step halves the distance $d_m(N, N')$ between two marked graphs at level m, so that

$$d_m(N, N') \le 2^{1-(h-m)} + \sum_{i=m}^{h-1} 2^{-(i-m)} \delta_i(N, N')$$

for every $h > m$, where each fixed-step discrepancy $\delta_i(N, N')$ tends to zero (Lemma 14.31). First choose h large, then the cutoffs. The bounds of Proposition 5.8 extend the convergence from $\bigcup_m a_m \mathbb{Z}^4$ to all separated positions and to derivatives.

4.3 Rotations

Continuous rotational covariance is a conclusion of the construction; the hypercubic lattice symmetry does not supply it. The ultraviolet comparison identifies each F_d with the Maxwell space through scaling frames (Propositions 6.4 and 6.7). In degree four, divergence maps the centered real symmetric tensorial families onto the corresponding vectors of degree five with a one-dimensional kernel (Proposition 6.9), which supplies a conserved tensor J. Its fluxes through small shells around one insertion define charge maps on the field spaces (Definition 6.8, Lemma 16.11). Conservation turns the sum of the angular charges of all insertions into the integral of J against the derivative of a vector field that equals the rotation field $y \mapsto Ky$ on a large anisotropic box; the sum does not depend on the size of the box. For configurations on a coordinate line this integral tends to zero by the coordinate reflection spaces of Definition 6.5, without a mass gap. The mechanism is a sign conflict (Lemmas 16.13, 16.14 and 16.15). Reflecting the integration variable in each axis replaces the spectator word by its projection that is odd under the reflections in the two axes of the rotation plane and even under the others. Because the word lies on a coordinate line, the transverse reflections fix its positions, so this projection is again a combination of words on the line. Such a combination has a positive spectral representation in each coordinate time axis, with the sign of its parity in that axis; one correlation represented with opposite signs bounds the low-energy spectral mass, and conservation with reflection positivity bounds that of the diagonal components of J. Every term of the integral is then bounded by a negative power of the box size. Finite Ward tests identify the translation charges with derivatives; analyticity then extends the identity to all separated configurations; finite tests remove the defects in the rotation algebra, and the charges integrate to continuous representations of $O(4)$ (Lemma 16.17).

4.4 Phases and reconstruction

Reconstruction starts from separated Euclidean correlations and quantitative slab bounds; it does not require a positive extension across collision diagonals. Three objects should be kept apart.

Object	Role
S_n^ξ on separated configurations	Euclidean correlations, reflection positive on positive-time words
Ordered-time continuations and boundary values	The Wightman functions and the relativistic fields
Contact extensions $\mathfrak{D}_{\boldsymbol{A}}^{\xi}$	A specified extension through collisions, with controlled local order

The infrared control available is a uniform low-energy estimate rather than a spectral gap. For a fixed finite combination A of compactly supported smeared field words, uniformly in the phase ξ,

$$\mu_{A\Omega_\xi}(\{0 < \mathbf{p}^4 \le u\}) \le M_A u^2 \qquad (0 < u \le 1),$$
$$0 \le \langle A\Omega_\xi, (e^{-sH_\xi} - P_{0,\xi})A\Omega_\xi\rangle \le M'_A s^{-2} \qquad (s \ge 1),$$

with $P_{0,\xi}$ the projection onto $\mathbb{C}\Omega_\xi$ (Lemma 17.9). Vacuum uniqueness first holds for almost every phase; continuity of the vacuum-subtracted heat pairings in ξ and full support of ν then extend it to every phase. The polynomial bound $M_A u^2$ is not a positive lower bound on the spectrum.

4.5 The trace

The free Maxwell stress tensor is traceless, so a qualitative Maxwell limit cannot determine the first nonzero trace: it requires the next coefficient, with a rate. Heuristically, at scale r the rescaled trace $r^4\Theta$ is close, in the comparison maps, to $(4d_2/\beta(r))\mathcal{O}_{F^2}$, where $\mathcal{O}_{F^2} = \frac{1}{4}\sum_{a,b,\alpha} :F^\alpha_{ab}F^\alpha_{ab}:$ is the free curvature square (Section 6.4), with $d_2 = -b_0/2$ and $\beta(r) \approx \kappa_G \log(1/r) = 2b_0 \log(1/r)$, so that

$$r^4\Theta \approx -\frac{\mathcal{O}_{F^2}}{\log(1/r)}, \qquad G_\Theta(r) \approx \frac{S_2^{\mathrm{Max}}(\mathcal{O}_{F^2}, \mathcal{O}_{F^2}; -\frac{r}{2}e_4, \frac{r}{2}e_4)}{\log^2(1/r)} = \frac{3d_{\mathfrak{g}}}{\pi^4}\, r^{-8} \log^{-2}(1/r)\,,$$

by homogeneity of degree -8. The amplitude does not depend on b_0: the trace coefficient and the running coupling both carry b_0, and only their ratio enters.

The coefficient d_2 is computed, not assumed, in five steps (Section 18). (1) The centered real symmetric H_4-tensorial free tensors of degree four are spanned by the Maxwell stress, the trace direction $\delta_{ab}\mathcal{O}_{F^2}$ and one further tensor, and a linear functional reads the trace of such a tensor from its divergence. (2) A lattice stress current built from plaquette curvatures has divergence equal to a variation of the action up to a small defect, and block transfer keeps its trace a lattice difference. (3) This lattice Ward identity has the regulated coefficient $-12\kappa_G$, the derivative $-\mu\, d/d\mu$ of a regulated one-loop determinant whose growth $12\kappa_G \log\mu$ comes from the computation that gives the running slope (Lemmas 12.9 and 18.5). Along suitable radii the trace functional is carried to the continuum frames with error $o(E^{-1})$ for a slowly growing scale E; the rate is needed because the cancelling coefficients of step (4) may grow like a smaller scale R, with $R/E \to 0$ (Lemmas 18.7 and 18.6). (4) Cancelling the divergence of the transferred current, multiplied by β, with multiples of the other two tensors gives a conserved tensor. By Proposition 6.9 it is a multiple of T, the shell normalization forces the multiple to tend to one, and its coefficient along $\delta_{ab}\mathcal{O}_{F^2}$ tends to $d_2 = -\kappa_G/4$; the trace $4\mathcal{O}_{F^2}$ of that direction gives $4d_2/\beta$ (Proposition 18.8). (5) The positive spectral representation makes G_Θ nonincreasing, which turns this comparison along subsequences of radii into a limit through all radii (Section 6.4). The product expansion, the last branch of Figure 2, is explained with its statement in Section 6.4.

5 The renormalization group

This section defines the block map, the coordinates of the effective densities and their norms, and states the inductive step, the multiscale theorem and the two ultraviolet results built on them. Fix $\mathsf{b} > 0$. Throughout we use the common choices of Sections 10–14, made once for the group realization and the penalty in the order of Section 9; constants may depend on these choices, and further dependence is stated where it occurs.

5.1 Scales and the block map

With odd ratios $\ell_m \geq 3$, set

$$D_0 = 1, \qquad D_{m+1} = \ell_m D_m, \qquad a_m = D_m^{-1}, \qquad s_m = \log D_m \,. \tag{5.1}$$

Level m has spacing a_m, and blocking passes from level $m+1$ to level m; the level-m torus has side $L_p D_m$ in lattice units and L_p in physical units. The cutoff sequence of the family is D_N. Each level carries the stage J_m of Section 4.1. The endpoint coupling $\bar\beta > 1$ is chosen large (item 7 of Table 3), and

$$\widehat\beta_m = \bar\beta + \kappa_G s_m, \qquad M_m^{\mathrm{cut}} = \exp\big((\log\log\widehat\beta_m)^2\big), \qquad t_m = \widehat\beta_m^{-1/2} M_m^{\mathrm{cut}}, \tag{5.2}$$

$$B_m = \lceil (\log\widehat\beta_m)^2 \rceil, \qquad \epsilon_m^{\mathrm{reg}} = \widehat\beta_m^{-11/10}, \qquad \epsilon_m^{\mathrm{for}} = e^{-(M_m^{\mathrm{cut}})^{1/5}} . \tag{5.3}$$

The reference coupling $\widehat\beta_m$ follows the leading running slope; t_m separates small from large plaquette fields; B_m is the radius of the support weights; and $\epsilon_m^{\mathrm{reg}}$, $\epsilon_m^{\mathrm{for}}$ are the allowances, the bounds kept by the induction for the regular and forced activity sums (Section 4.1 explains these sizes). They depend on the reference coupling, so they stay fixed as the actual coupling varies in the band $\mathfrak{B}_m = [\widehat\beta_m/2, 2\widehat\beta_m]$. The physical periods L_p are strictly increasing multiples of four, every positive integer divides L_p for infinitely many p, and $L_p D_m \geq 100(B_m + 1)$. At stage j, $\mathfrak{q}_j$ is the derivative order, ϖ_j the spatial power, σ_j the support exponent and $\omega_j^{\mathrm{RG}} \geq 0$ the coupling drift bound (Section 9).

At a step $m+1 \to m$, the fine links U and coarse links V live on $\Lambda_{L_p D_{m+1}}$ and $\Lambda_{L_p D_m}$; put $j = J_{m+1}$, $\ell = \ell_m$ and $\mathbb{B}_R = \{-R, \dots, R\}^4$. Fine transports between two coarse endpoints are compared after expressing them in the same frames. A frame is a gauge transformation $h_{y,z} \in G$ at each fine site $\ell y + z$ of the patch of radius $(j+3)\ell$ around ℓy, built from parallel transports in the patch (Definitions 11.1–11.3). For a coarse edge $e = (y, a)$ the framed straight connector is

$$Y_{e,z} = h_{y,z} \left(\prod_{r=0}^{\ell-1} U(\ell y + z + r e_a, a) \right) h_{y+e_a,z}^{-1} \,, \tag{5.4}$$

the fine transport from $\ell y + z$ to $\ell(y + e_a) + z$ in the frames at its ends. With Unif the uniform probability on a box and $*j$ the j-fold convolution, the averaging weights and the predictor are

$$\begin{aligned} w_z^{\mathrm{av}} &= \big(\mathrm{Unif}(\mathbb{B}_{(\ell-1)/2})\big)^{*j}(z), \qquad \overline{Y}_e = \sum_z w_z^{\mathrm{av}} Y_{e,z} \,, \\ Q_e^{\mathrm{pred}} &= \operatorname*{argmin}_{u \in G} \sum_z w_z^{\mathrm{av}} \,\mathrm{pr}(Y_{e,z} u^{-1}) = \operatorname*{argmin}_{u \in G} \|\overline{Y}_e - u\|_{\mathrm{F}} \,. \end{aligned} \tag{5.5}$$

The second form holds because $\mathrm{pr}(Y u^{-1}) = (2n_G)^{-1}\|Y - u\|_{\mathrm{F}}^2$ for unitary Y, u: the predictor is the point of G nearest to the weighted average of the connectors, an average of group variables

taken through the ambient matrices. On a good patch the connectors lie near one another, and this branch, on which the minimizer is unique, is used when the minimum lies below a fixed cutoff. Where the frame test or the predictor test fails, the kernel chooses at random, an axial frame from a finite list or a single connector $Y_{e,z}$ with probability w_z^{av}; these branches complete the kernel on the remaining configurations (Section 11 gives the decision rules, summarized in Table 4 and Figure 8). A tag α_{tag} records these discrete choices and nothing else: for each coarse site a frame label (z, o), an offset $z \in \mathbb{B}_{(j+3)\ell}$ and an order o of the four axes, and for each coarse edge an offset $z \in \mathbb{B}_{(j+3)\ell}$ selecting a connector; on the minimizing branches the label is drawn uniformly and does not affect the kernel. Tags are drawn with probabilities

$$\lambda^{\mathrm{tag}}(U, \alpha_{\mathrm{tag}}) \geq 0, \qquad \sum_{\alpha_{\mathrm{tag}}} \lambda^{\mathrm{tag}}(U, \alpha_{\mathrm{tag}}) = 1\,. \tag{5.6}$$

Given U and the tag, the coarse links are independent, each concentrated within about $(\mathsf{b}\beta)^{-1/2}$ of its predictor, with the normalized density $q_\gamma(u) = e^{-\gamma\,\mathrm{pr}(u)} / \int_G e^{-\gamma\,\mathrm{pr}}\,dh$; the block penalty plays the role of the smearing parameter of soft block-spin transformations.

Definition 5.1 (Exact block map)**.** For fine links U and coarse links V, define

$$k_\beta^{(m)}(U, V) = \sum_{\alpha_{\mathrm{tag}}} \lambda^{\mathrm{tag}}(U, \alpha_{\mathrm{tag}}) \prod_{e\ \mathrm{coarse}} q_{\mathsf{b}\beta}\big(V_e Q_e^{\mathrm{pred}}(U, \alpha_{\mathrm{tag}})^{-1}\big),$$
$$(\mathcal{B}_\beta^{(m)} f)(V) = \int k_\beta^{(m)}(U, V) f(U)\, dU\,,$$

with probability Haar measures on both link sets.

For fixed U and tag, right invariance of Haar measure makes each coarse factor integrate to one; summing the tag masses gives $\int k_\beta^{(m)}(U, V)\, dV = 1$. Hence, for integrable f,

$$\int (\mathcal{B}_\beta^{(m)} f)(V)\, dV = \int f(U)\, dU\ : \tag{5.7}$$

the map integrates one scale exactly and keeps the partition integral. Branches, measurability and symmetries of the kernel are proved in Section 11.

Probabilistically, a block step is the following experiment. Let U have law $f(U)\,dU / \int f$ for an integrable $f \geq 0$ with $\int f > 0$; draw a tag with probabilities $\lambda^{\mathrm{tag}}(U, \cdot)$; and put $V_e = R_e Q_e^{\mathrm{pred}}(U, \alpha_{\mathrm{tag}})$, where the R_e have density $q_{\mathsf{b}\beta}$ and are independent of each other and of $(U, \alpha_{\mathrm{tag}})$. The pair (U, V) then has law $f(U) k_\beta^{(m)}(U, V)\, dU\, dV / \int f$, and by Fubini the law of V has density $\mathcal{B}_\beta^{(m)} f / \int f$ with respect to Haar measure. Exactness refers to this identity, which includes every branch of the kernel. It does not mean that the coarse density is again a Wilson density; Section 5.2 gives the coordinates in which it is written.

5.2 The density and its norms

The density produced by the integrations records four things: the leading curvature action with its running coupling, the field-independent scalar accumulated so far, small local corrections on small-field supports, and the contributions of large-field regions. At level m write $\Lambda = \Lambda_{L_p D_m}$. The flagged sites $\mathrm{Flag}_m(U)$, or large-field sites, are the sites incident to a plaquette whose chord exceeds t_m; a support is healthy, or small-field in the sense of Bałaban, if it contains none. Flags

are gauge invariant, because plaquette chords are: the transform of the identity configuration in Section 2.1 has no flagged site, however far its links are from the identity. A regular activity E_i^{reg} is a Borel real gauge-invariant function, local on its support X_i, anchored at $o_i \in X_i$ and vanishing at the flat configuration; it is used only on healthy supports. A forced activity K_i^{for}, one of the large-field polymer activities, is local on its support Y_i and carries a nonempty inventory $I_i \subset Y_i$ of flagged sites; it vanishes off its eligibility event

$$\mathcal{E}_i^{\mathrm{elig}} = \{U : Y_i \cap \mathrm{Flag}_m(U) = I_i\} \cap \mathcal{E}_i^{\mathrm{extra}}, \qquad K_i^{\mathrm{for}}(U) = 0 \quad (U \notin \mathcal{E}_i^{\mathrm{elig}}) . \tag{5.8}$$

The first condition fixes exactly which sites of Y_i are flagged. The extra event is a fixed Borel gauge-invariant test supplied with the activity, which reads only links with both endpoints in Y_i and the flags in Y_i; it distinguishes local alternatives with the same support and inventory. The activity families are countable.

The path action carries the kinetic coefficient and the quadratic corrections generated by integration. At level m abbreviate the clipping constants and relaxation length of item 4 of Table 3,

$$c_{\mathrm{dir}} = c_{\mathrm{dir},J_m}, \qquad c_{\mathrm{lin}} = c_{\mathrm{lin},J_m}, \qquad 0 < 16c_{\mathrm{lin}} < c_{\mathrm{dir}} \le \min(\gamma_{J_m}, 1), \qquad R^{\mathrm{rel}} = R^{\mathrm{rel}}_{J_m} ,$$

where $\gamma_j > 0$ is half a uniform lower bound for the free effective plaquette forms of Lemma 11.5. Distances are periodic maximum-coordinate distances in lattice units: on a torus of side n, $\mathrm{dist}(x,y) = \min_{z \in n\mathbb{Z}^4} |\tilde{x} - \tilde{y} + z|_\infty$ for lifts $\tilde{x}, \tilde{y} \in \mathbb{Z}^4$. Set

$$\begin{aligned} \mathrm{clip}_t(r) &= \begin{cases} c_{\mathrm{dir}} r^2/2, & r \le t, \\ c_{\mathrm{lin}} t r, & r > t, \end{cases} \\ \chi^{\mathrm{rel}}_{m,x}(U) &= \mathbf{1}\Big\{ \mathrm{ch}(U_{y;ab}) \le t_m e^{\mathrm{dist}(x,y)/R^{\mathrm{rel}}} \text{ for every } y, a < b \Big\} . \end{aligned} \tag{5.9}$$

For each plaquette orientation $\mathfrak{p} = (a,b)$, let $\mathcal{F}_{\mathfrak{p}i}(U,x)$ be a curvature $\mathrm{proj}_{\mathfrak{g}}(U_\square - U_\square^{-1})/2$, with the orthogonal projection for the metric of Definition 2.1, transported along a specified finite path to x, and put $\mathcal{L}^{\mathrm{path}}_{\mathfrak{p}}(U,x) = \sum_i r_{\mathfrak{p}i} \mathcal{F}_{\mathfrak{p}i}(U,x)$, a smeared lattice curvature at x. The action splits the effective kinetic form into a clipped part, quadratic below t_m and linear beyond, and the remaining positive form, written as a sum of squares with one row per $(x,\mathfrak{p})$; a row is switched off once some plaquette exceeds its distance-relaxed threshold, that is, when the *relaxed threshold test* $\chi^{\mathrm{rel}}_{m,x}$ of (5.9) vanishes. With $\mathfrak{a}_i^{(q)}(U,x)$, $q = 2,3,4$, the fixed real invariant q-linear expressions in transported curvatures, each multiplied by the health indicator of its footprint (the union of its slot footprints and the anchor; Definition 12.1),

$$\begin{aligned} (P^{\mathrm{path}})^{\mathrm{raw}}_{N,m}(\beta,U) &= \beta \sum_{x,\mathfrak{p}} \Big[\mathrm{clip}_{t_m}(\mathrm{ch}(U_{x;\mathfrak{p}})) + \tfrac{1}{2}\chi^{\mathrm{rel}}_{m,x}(U) |\mathcal{L}^{\mathrm{path}}_{\mathfrak{p}}(U,x)|^2 \Big] \\ &\quad + \beta \sum_{x,i} (\mathfrak{a}_i^{(3)} + \mathfrak{a}_i^{(4)})(U,x) + \sum_{x,i} \mathfrak{a}_i^{(2)}(U,x), \\ P^{\mathrm{path}}_{N,m}(\beta,U) &= |H_4|^{-1} \sum_{h \in H_4} (P^{\mathrm{path}})^{\mathrm{raw}}_{N,m}(\beta, h^* U) , \end{aligned} \tag{5.10}$$

where h^*U transports links by the lattice isometry h. The path coefficients and multilinear expressions are independent of p (Definitions 12.1–12.2, Lemmas 12.3–12.4); the action is affine in β,

$$P^{\mathrm{path}}_{N,m}(\beta,U) = \beta\, \partial_\beta P^{\mathrm{path}}_{N,m}(U) + P^{\mathrm{path}}_{N,m}(0,U) ,$$

where, after the average over H_4, $\partial_\beta P^{\mathrm{path}}_{N,m}$ collects the clipped plaquette terms, the row squares with their switches $\chi^{\mathrm{rel}}_{m,x}$ and the cubic and quartic terms, and $P^{\mathrm{path}}_{N,m}(0,\cdot)$ the quadratic terms $\mathfrak{a}^{(2)}_i$. The clip drops at $r = t$ from $c_{\mathrm{dir}}t^2/2$ to $c_{\mathrm{lin}}t^2$. Accordingly the density coordinates are Borel functions; derivatives are taken only on healthy configurations with the masks held fixed, so coupling derivatives never differentiate the flag boundary. The positive linear branch beyond the threshold is what the local large-field energy bound (Lemma 11.12) uses.

For these fixed path actions and thresholds, a *representation* at level m is the datum

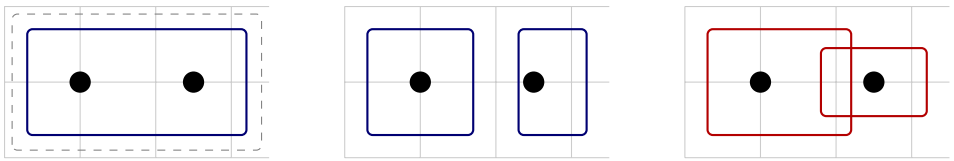
$$\rho = \big(\beta, v_{\mathrm{sc}}, (X_i, o_i, E^{\mathrm{reg}}_i)_i, (Y_i, I_i, \mathcal{E}^{\mathrm{extra}}_i, K^{\mathrm{for}}_i)_i\big)\,, \tag{5.11}$$

with scalar normalization per site v_{sc} (a constant in the effective density, not the vacuum vector of Section 6.3) and running coupling $\beta(\rho) = \beta$. The running coupling is thus a coordinate fixed by the choice of path action: the coefficient of the fixed function $\partial_\beta P^{\mathrm{path}}_{N,m}$, whose quadratic part is normalized so that β is the coefficient of the squared curvature at zero momentum at every level, while the quadratic terms $\mathfrak{a}^{(2)}_i$ carry no curvature-square term (Lemma 12.4). It is not read off from the whole density, which also contains v_{sc} and the activities. Let $\mathfrak{F}_m(U)$ be the set of finite $\mathfrak{a} \subset \mathbb{N}$ such that the supports Y_i, $i \in \mathfrak{a}$, are pairwise disjoint and $\bigcup_{i\in\mathfrak{a}} I_i = \mathrm{Flag}_m(U)$: the exact covers of the flagged sites.

Definition 5.2 (Effective density)**.** For a representation ρ, put

$$\mathcal{V}_\rho(U) = P^{\mathrm{path}}_{N,m}(\beta, U) + \sum_i \mathbf{1}_{X_i \text{ healthy}} E^{\mathrm{reg}}_i(U) - v_{\mathrm{sc}}|\Lambda|,$$
$$\operatorname{dens}\rho(U) = e^{-\mathcal{V}_\rho(U)} \sum_{\mathfrak{a}\in\mathfrak{F}_m(U)} \prod_{i\in\mathfrak{a}} K^{\mathrm{for}}_i(U)\,.$$

If there are no flagged sites, $\mathfrak{F}_m(U) = \{\varnothing\}$ and the forced sum is one. Otherwise every flagged site lies in exactly one selected support, but the sum runs over all exact covers: one support may carry two flags in one term while two disjoint supports carry one flag each in another (Figure 4). The local regularity of the activities and the restrictions on forced terms are Definitions 13.1–13.2.



(a) (b) (c)

Figure 4: Exact covers of two flagged sites (dots). (a) One support carries both flags; the dashed line marks the larger region its eligibility test inspects. (b) Two disjoint supports carry one flag each. Both selections occur in the sum of Definition 5.2, which does not choose a canonical cover. (c) Overlapping supports are not a cover.

For a link e let e^- be its initial site. The left derivative $D^L_{\dot{\mathcal{A}}}$ differentiates the increments $U_e \mapsto e^{t\dot{\mathcal{A}}_e}U_e$, with $\dot{\mathcal{A}}_e \in \mathfrak{g}$ normed by Definition 2.1.

Definition 5.3 (Local derivative seminorm)**.** For a regular activity anchored at o_i, set

$$\|E^{\mathrm{reg}}_i\|_m = \sup_{0\le q\le \mathfrak{q}_{J_m}} \sup_{U,\dot{\mathcal{A}}_1,\dots,\dot{\mathcal{A}}_q} \big|D^L_{\dot{\mathcal{A}}_1}\cdots D^L_{\dot{\mathcal{A}}_q} E^{\mathrm{reg}}_i(U)\big|\,,$$

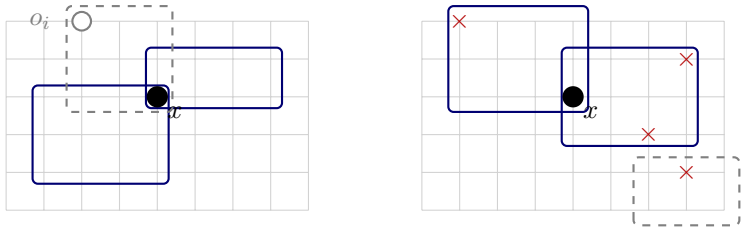
over configurations healthy on X_i and directions with $|(\dot{\mathcal{A}}_k)_e| \le t_m\big(1+\operatorname{dist}(e^-,o_i)\big)^{\varpi_{J_m}}$, $1 \le k \le q$.

The seminorm measures ordered local derivatives in units of the permitted field size. For a support X, let $\operatorname{cov}_{B_m}(X)$ be the least number of radius-B_m lattice balls with connected union covering X; the exponential weight below penalizes supports that need many such balls.

Definition 5.4 (Weighted activity sums)**.** For nonnegative majorants $e_i^{\mathrm{maj}} \ge \|E_i^{\mathrm{reg}}\|_m$ and $k_i^{\mathrm{maj}} \ge \sup_U |K_i^{\mathrm{for}}(U)|$, define

$$\mathcal{E}_m = \sup_x \sum_{o_i = x} e^{\sigma_{J_m} \operatorname{cov}_{B_m}(X_i)} e_i^{\mathrm{maj}}, \qquad \mathcal{K}_m = \sup_x \sum_{x \in Y_i} e^{\sigma_{J_m} \operatorname{cov}_{B_m}(Y_i)} k_i^{\mathrm{maj}}\,.$$

The two sums are formed differently (Figure 5). The regular sum at x counts the activities anchored at x, whatever their supports, so each regular activity is counted once, at the site from which its derivative directions are weighted (Definition 5.3). The forced sum at x counts every forced activity whose support contains x, so the weight $e^{\sigma_{J_m} \operatorname{cov}_{B_m}(Y_i)}$ charges a large support at each of its sites; if x is flagged, every activity that covers x in a term of Definition 5.2 is counted there. Section 13 converts anchored bounds into containing-site bounds where needed, at the cost of the number of possible anchors of a support.



(a) regular sum at x (b) forced sum at x

Figure 5: The weighted sums of Definition 5.4 at a site x. (a) The regular sum counts the activities anchored at x (solid supports), whatever their extent; the dashed support contains x but is anchored at $o_i \neq x$ and is counted at o_i. (b) The forced sum counts every activity whose support contains x (solid), whatever its inventory (crosses); overlapping supports both count, since the sum runs over activities, not covers. The dashed support does not contain x.

The induction keeps the bounds

$$\mathcal{E}_m \le \epsilon_m^{\mathrm{reg}}, \qquad \mathcal{K}_m \le \epsilon_m^{\mathrm{for}}\,. \tag{5.12}$$

These numerical bounds do not by themselves define the classes on which the step acts. The Wilson start is the continuous curve

$$\beta \longmapsto \rho_\beta^{\mathrm{W}}, \qquad \beta(\rho_\beta^{\mathrm{W}}) = \beta, \qquad \operatorname{dens}\rho_\beta^{\mathrm{W}} = e^{-\beta E^{\mathrm{W}}_{L_p D_N}} \quad (\beta \in \mathfrak{B}_N)\,, \tag{5.13}$$

with scalar normalization zero (Lemma 13.16). The *admissible classes* consist of everything reachable from it. Let $\mathcal{I}_{N,p} = \{\rho_\beta^{\mathrm{W}} : \beta \in \mathfrak{B}_N\}$ and $\mathcal{A}_{N,N,p} = \mathcal{I}_{N,p}$. For $m < N$, a representation at level m belongs to $\mathcal{A}_{N,m,p}$ if it is obtained from some $\rho' \in \mathcal{A}_{N,m+1,p}$ with coupling in $\mathfrak{B}_{m+1}$ by one block step of Section 13: its path is $P_{N,m}^{\mathrm{path}}$, its activities are produced by local subtraction, path replacement and fusion, its coupling is that of ρ' plus the one-loop coefficient $\alpha_{N,m}$ of Lemma 12.9 and a correction bounded by the first inequality of (13.24), and its density is the block integral

of the density of ρ' (the precise conditions precede Lemma 13.17). Membership is a property of a single representation; in particular the classes contain the outputs of the step construction on continuous curves and their restrictions to closed subintervals. Let $\mathcal{R}_{N,m,p} \subset \mathcal{A}_{N,m,p} \times \mathcal{A}_{N,m+1,p}$ be the set of the corresponding output–input pairs. Every member satisfies Definitions 13.1–13.2, the path bounds of Lemmas 12.3–12.4 and (5.12) (Lemma 13.16 at the Wilson start, Lemmas 13.12 and 13.17 at later levels). A curve $u \mapsto \rho_u$ in $\mathcal{A}_{N,m,p}$ is continuous (Definition 13.3) if its supports, anchors, inventories and extra tests do not depend on u, its coupling and scalar normalization are continuous, and, as $u \to u_0$, the differences of regular derivatives on healthy configurations and of forced values in supremum norm admit majorants whose weighted sums tend to zero.

5.3 The inductive step

Exact integration removes one scale. Extracting its marginal curvature-square term into the coupling leaves a bounded drift error and returns a density in the next admissible class.

Theorem 5.5 (RG. One step)**.** *For every N, p, $m < N$, reals $u_0 \le u_1$ and continuous $\rho'_u \in \mathcal{A}_{N,m+1,p}$ with $\beta'_u = \beta(\rho'_u) \in \mathfrak{B}_{m+1}$ for all $u \in [u_0, u_1]$, there is a continuous $\rho_u \in \mathcal{A}_{N,m,p}$ with $(\rho_u, \rho'_u) \in \mathcal{R}_{N,m,p}$ and, for every $u \in [u_0, u_1]$ and coarse configuration V,*

$$\operatorname{dens} \rho_u(V) = (\mathcal{B}^{(m)}_{\beta'_u} \operatorname{dens} \rho'_u)(V),$$
$$\beta(\rho_u) = \beta'_u - \kappa_G \log \ell_m + e_u, \qquad |e_u| \le \omega^{\mathrm{RG}}_{J_{m+1}} .$$

Idea of the proof (Sections 11–13; the proof follows Lemma 13.17).

Step 1. The density identity is exact because the output coordinates rewrite the block-integrated density on every configuration (Lemma 13.13); by (5.7) the partition integral is kept. Approximation enters only the coupling drift and the activity estimates.

Step 2. On small fields, gauge fixing in a rooted block-tree gauge and the stationary background turn the local integral into a Gaussian integral with its determinant and Haar-coordinate factors; the large-field lower bound (Lemma 11.12) pays for the forced terms.

Step 3. Classical fourth-order matching fixes the local terms, and the background curvature-square coefficient of the one-loop action, assembled from the inherited quadratic loop term, the fluctuation determinant and the Haar Jacobian, gives the change $-\kappa_G \log \ell_m$ up to a stage-bounded error (Lemma 12.9).

Step 4. The local expansion, the contraction of inherited terms and the five-torus localization (Lemma 4.1) give the regular bound; forced fusion gives the forced bound; the scalar coupling reset absorbs the residual kinetic coefficient (Lemma 13.14).

The bound ω^{RG}_j depends only on the stage, not on N, p, the interval or the curve. Since $\widehat{\beta}_m = \widehat{\beta}_{m+1} - \kappa_G \log \ell_m$, a step changes $\beta - \widehat{\beta}$ by e_u only.

5.4 Multiscale trajectories

A trajectory is a tuple $\rho_{N,p,m} \in \mathcal{A}_{N,m,p}$, $0 \le m \le N$, with $\rho_{N,p,N} \in \mathcal{I}_{N,p}$ and $(\rho_{N,p,m}, \rho_{N,p,m+1}) \in \mathcal{R}_{N,m,p}$ for $m < N$; write $\beta_{N,p,m} = \beta(\rho_{N,p,m})$. The bare coupling is selected by shooting. Put $\mathfrak{T}_m = [\bar\beta + \frac{\kappa_G}{2} s_m,\ \bar\beta + 2\kappa_G s_m] \subset \mathfrak{B}_m$, so that $\mathfrak{T}_0 = \{\bar\beta\}$. Start with the Wilson curve on $\mathfrak{T}_N$. Suppose a closed parameter interval has level-$(m+1)$ couplings in $\mathfrak{T}_{m+1}$, taking its end values at its endpoints. Since $s_{m+1} = s_m + \log \ell_m$ and $|e_u| \le \omega^{\mathrm{RG}}_{J_{m+1}} \le \frac{\kappa_G}{4} \log \ell_m$ (item 5 of Table 3), the output couplings at the two endpoints satisfy

$$\beta \le \bar\beta + \tfrac{\kappa_G}{2} s_m - \tfrac{\kappa_G}{4} \log \ell_m < \min \mathfrak{T}_m, \qquad \beta \ge \bar\beta + 2\kappa_G s_m + \tfrac{3\kappa_G}{4} \log \ell_m > \max \mathfrak{T}_m .$$

The intermediate value theorem gives a closed subinterval whose couplings lie in $\mathfrak{T}_m$ and take its end values at its endpoints. After N steps the level-zero coupling is $\bar\beta$. The couplings are chosen this way for all N, p before any sequence of volume indices is prescribed. Trajectories for different cutoffs thus share the coupling $\bar\beta$ at level zero; their scalar normalizations and activities need not agree (Figure 6), and Proposition 5.7 below compares them at a fixed level as the cutoff grows.

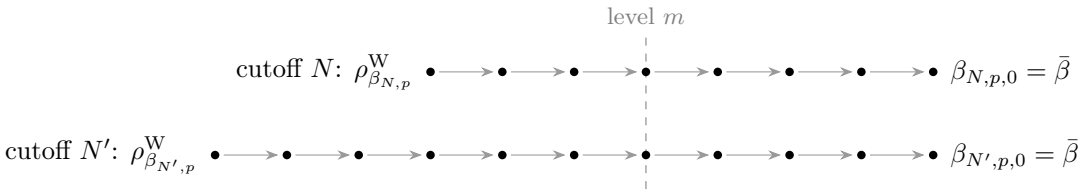


Figure 6: Trajectories for two cutoffs $N < N'$ at one volume, from the Wilson start at the cutoff (left) to level zero (right); each arrow is a block step, and levels are aligned vertically. Both reach the coupling $\bar\beta$ at level zero; their scalar normalizations and activities need not agree. At a fixed level m (dashed), Proposition 5.7 gives convergence of the couplings and activities as the cutoff tends to infinity.

Theorem 5.6 (MT. Multiscale trajectory)**.** *There are bare couplings $\beta_{N,p}$ and trajectories $(\rho_{N,p,m})_{m=0}^{N}$, for all $N, p \in \mathbb{N}$, with $\rho_{N,p,N} = \rho^{\mathrm{W}}_{\beta_{N,p}}$ and $\beta_{N,p,N} = \beta_{N,p}$, satisfying*

$$\beta_{N,p,0} = \bar\beta, \qquad \beta_{N,p,m} \in \mathfrak{B}_m \quad (0 \le m \le N),$$
$$\operatorname{dens} \rho_{N,p,m} = \mathcal{B}^{(m)}_{\beta_{N,p,m+1}} \operatorname{dens} \rho_{N,p,m+1} \quad (m < N)\,.$$

For every $\varepsilon > 0$, some $m_0(\varepsilon)$ satisfies

$$|\beta_{N,p,m} - \kappa_G s_m| \le \varepsilon s_m \qquad (N \ge m \ge m_0,\ p \in \mathbb{N})\,.$$

The proof is the shooting argument above (Section 14.1, Lemmas 14.2–14.3). Fix these trajectories for all later constructions. The drift bound of Theorem 5.5 holds on every adjacent pair, and all their integrals equal $Z_{L_p D_N, \beta_{N,p}}$ by (5.7) and the Wilson start. Summing the drift from the fixed endpoint gives

$$|\beta_{N,p,m} - \widehat{\beta}_m| \le \sum_{i=0}^{m-1} \omega^{\mathrm{RG}}_{J_{i+1}}\,. \tag{5.14}$$

Item 5 of Table 3 bounds the sum by $(\kappa_G/4)s_m$, which also keeps the couplings in $\mathfrak{B}_m$. Its stronger bound $\omega^{\mathrm{RG}}_{J_{i+1}} \le J_{i+1}^{-2} \log \ell_i$ makes the sum $o(s_m)$: split it at a fixed level and bound the tail by $\sup_{i \ge i_0} J_{i+1}^{-2}$ times s_m. Since $J_i \to \infty$ and $s_m \to \infty$, the finite initial sum and $\bar\beta$ disappear after division by s_m, uniformly in $N \ge m$ and p. As $\omega^{\mathrm{RG}}_j \ge 1$, the same bound forces $\log \ell_m \ge J^2_{m+1} \to \infty$: the block ratios must grow, and the level choices also give $\log \ell_m / s_m \to 0$. The step construction of Theorem 5.5 is fixed once on whole continuous curves (Lemma 13.17); the trajectories it produces from ρ^{W}_β, and their restrictions, form the *specified recursion*, to which the trajectories of Theorem 5.6 belong. Different cutoffs are compared through the following limit.

Proposition 5.7 (Fixed-level convergence)**.** *For fixed p and m, as $N \to \infty$ along the full sequence, the couplings $\beta_{N,p,m}$ of Theorem 5.6 converge, and the activity coordinates of $\rho_{N,p,m}$ converge in the weighted spaces $\mathcal{X}_{m,p}$ of (13.25). No convergence of the scalar normalizations is asserted.*

Idea of the proof (Lemmas 13.22 and 13.24). The argument uses geometric truncation, not compactness of a ball in an infinite-dimensional space. The scalar coordinates lie in fixed compact intervals, so a diagonal argument gives subsequential limits. At a fixed higher level k, replace the activity coordinate by zero and run the same maps down to level m: the error is at most $(8g_*)^{k-m}$ with $g_* = 1/128$, uniformly in the cutoff, while the truncated coordinates converge by the cutoff compatibility of finitely many step maps (Lemma 13.22). Letting $k \to \infty$ gives the activity limits. Two limits have the same endpoint $\bar\beta$, and their distance $\mathfrak{d}_\infty$ satisfies $\mathfrak{d}_\infty \le 8g_*\mathfrak{d}_\infty$, so they coincide.

5.5 Local fields under blocking

Blocking a density that carries one insertion returns a density of the same form: the unmarked density times a local insertion with new coefficients, plus regular remainders with vanishing Taylor coefficients through order d and forced remainders attached to large fields. The coefficients evolve linearly, and the insertions of Definition 2.8 are normalized by prescribing them at the reference level. Choose the reference level $m_{\mathrm{ref}}(d)$ with $J_{m_{\mathrm{ref}}(d)} \ge d+3$ and $\mathfrak{q}_{J_m} \ge d+1$, $\varpi_{J_m} \ge d$ for $m \ge m_{\mathrm{ref}}(d)$ (Lemma 14.7). Fix $N \ge m_{\mathrm{ref}}(d)$; starting from a fine-cutoff coefficient vector $\mathbf{v} \in V_d$, the marked density at the origin is initialized and blocked by

$$H^{\mathrm{ins}}_{N,p,N;d}(\mathbf{v};U) = \operatorname{dens}\rho_{N,p,N}(U)\, P^{\mathrm{loc}}_d[t_N^{-\mathsf{N}}\mathbf{v}](0;U), \tag{5.15}$$

$$H^{\mathrm{ins}}_{N,p,m;d}(\mathbf{v}) = \ell_m^d \mathcal{B}^{(m)}_{\beta_{N,p,m+1}} H^{\mathrm{ins}}_{N,p,m+1;d}(\mathbf{v}) \quad (m_{\mathrm{ref}}(d) \le m < N)\,. \tag{5.16}$$

For fixed N, p, d, the transfer construction chooses coefficient vectors $\mathbf{x}_m(\mathbf{v})$ and regular and forced remainders $R^{\mathrm{reg}}_{m,i}(\mathbf{v})$, $Q^{\mathrm{for}}_{m,i}(\mathbf{v})$, all linear in $\mathbf{v}$. Write $\mathbf{x}_m = x_{m,0}\mathbf{1} + \mathbf{x}^\circ_m$ and $\rho_m = \rho_{N,p,m}$. The regular remainders have supports $Z_{m,i}$ containing the origin. Their sum and the forced remainder are

$$\begin{aligned}
\mathcal{R}^{\mathrm{reg}}_m(\mathbf{v};U) &= \sum_i \mathbf{1}_{Z_{m,i}\,\mathrm{healthy}} R^{\mathrm{reg}}_{m,i}(\mathbf{v};U),\\
\mathcal{R}^{\mathrm{for}}_m(\mathbf{v};U) &= e^{-\mathcal{V}_{\rho_m}(U)} \sum_{\mathfrak{a}\in\mathfrak{F}_m(U)} \sum_{i\in\mathfrak{a}} Q^{\mathrm{for}}_{m,i}(\mathbf{v};U) \prod_{j\in\mathfrak{a}\setminus\{i\}} K^{\mathrm{for}}_j(U)\,,
\end{aligned} \tag{5.17}$$

with all unmarked quantities taken from ρ_m. The marked density has the exact decomposition

$$\begin{aligned}
H^{\mathrm{ins}}_{N,p,m;d}(\mathbf{v};U) = \operatorname{dens}\rho_m(U)\Big[x_{m,0}(\mathbf{v}) + \mathbf{1}_{\Sigma_d\,\mathrm{healthy}} P^{\mathrm{loc}}_d[t_m^{-\mathsf{N}}\mathbf{x}^\circ_m(\mathbf{v})](0;U)\Big]\\
+ \operatorname{dens}\rho_m(U)\mathcal{R}^{\mathrm{reg}}_m(\mathbf{v};U) + \mathcal{R}^{\mathrm{for}}_m(\mathbf{v};U)\,,
\end{aligned} \tag{5.18}$$

with the stencil Σ_d placed at the origin of the level-m torus. If $10\operatorname{dist}(x,0) < L_pD_m$ for every $x \in Z_{m,i}$, the regular remainder $R^{\mathrm{reg}}_{m,i}$ has zero radial-jet Taylor coefficients through order d; the forced remainders vanish outside their unmarked eligibility events (Lemmas 14.7–14.9). The fixed decomposition selected there gives the terminal map of Definition 2.8, invertible by Lemma 14.7:

$$\mathbf{x}_N(\mathbf{v}) = \mathbf{v}, \qquad \mathbf{x}_{m_{\mathrm{ref}}(d)}(\mathbf{v}) = \mathsf{Z}_{N,p,d}\mathbf{v}\,.$$

The factor $\ell_m^d = (a_m/a_{m+1})^d$ uses the degree of the whole space V_d, and the terminal map accounts for the mixing between its weights. The unit identity $\mathbf{x}_m(a_N^d\mathbf{1}) = a_m^d\mathbf{1}$, proved with the transfer symmetries (Lemma 14.8), explains why the coefficient of the constant insertion is $a^d_{m_{\mathrm{ref}}(d)}\mathbf{1}$, although V_d also contains polynomials of lower weight. With Theorem 5.6, these insertions specify one Wilson family; each insertion is continuous, gauge invariant and local on its fixed stencil, and Theorem 5.6 at $m = N$ gives the volume-uniform slope $\kappa_G > 0$.

5.6 Uniform bounds and the fixed-volume limit

For a finite list $\boldsymbol{l}$ of pairs (i,a), with i a slot and a an axis, the normalized differences and derivatives are

$$\Delta_{\boldsymbol{l}} = \prod_{(i,a)\in\boldsymbol{l}} a_N^{-1}(\tau_{i,a}-1), \qquad \partial_{\boldsymbol{l}} = \prod_{(i,a)\in\boldsymbol{l}} \partial_{x_i^a}\,,$$

where $\tau_{i,a}$ shifts slot i forward by one lattice link. The torus separation of a lattice tuple is $\min_{i\neq j}\min_{z\in L_p\mathbb{Z}^4}|\hat{y}_i-\hat{y}_j+z|_\infty$, with value $+\infty$ for $n\le 1$, and

$$\mathrm{Gen}_n = \{x : x_i^a \neq x_j^a \text{ for } i\neq j,\ a=1,\dots,4\}, \qquad \mathrm{NoHit}_{L,n} = \{x : x_i - x_j \notin L\mathbb{Z}^4 \text{ for } i\neq j\}\,;$$

the last set excludes collisions on a torus of physical side L. The first result bounds lattice correlations and their differences independently of the volume.

Proposition 5.8 (UB. Uniform difference bounds)**.** *For every $v\in\mathcal{T}^n$, finite difference list $\boldsymbol{l}$ and $r>0$, there are $B(v,\boldsymbol{l},r)\ge 0$ and $N_0(v,\boldsymbol{l},r)$ such that, for every $N\ge N_0$, every p and every lattice tuple of torus separation at least r,*

$$|\Delta_{\boldsymbol{l}} M^{\mathrm{lat}}_{N,p}(v;y)| \le B\,.$$

Idea of the proof (Section 14.2). Choose an intermediate level whose physical blocks fit below the separation, so the insertions lie in distinct blocks. By the decomposition (5.18), at that level each insertion is a bounded local function plus small remainders, with bounds independent of N and p; differences of insertions are matched in a higher degree, their equation directions are removed, and the remaining coefficients are tested against a finite free comparison matrix, whose injectivity (Lemma 14.22) rests on the free positivity of Lemma 16.1. The chessboard estimate then bounds the correlation by a product of reflected moments, schematically

$$\Big|\Big\langle\prod_i O_i\Big\rangle\Big| \le \prod_i \Big\langle\prod_{g\in\mathcal{G}} gO_i\Big\rangle^{1/|\mathcal{G}|}$$

for O_i localized in distinct blocks and $\mathcal{G}$ generated by the reflections in the block boundaries; each factor is independent of the volume. The reflected-moment constant is chosen before the intermediate level, and only the cutoff threshold depends on it.

The second result removes the spacing at each fixed volume along the full sequence; Section 4.2 displays the Cauchy argument.

Theorem 5.9 (UV. Fixed-volume limit)**.** *For every p,n and $v\in\mathcal{T}^n$, there is $M^{\mathrm{UV}}_{p,v}\in C^\infty(\mathrm{NoHit}_{L_p,n})$ such that, for every finite $\boldsymbol{l}$ and every p-admissible separated test f,*

$$\lim_{N\to\infty} a_N^{4n} \sum_{y\in\Lambda^n_{L_pD_N}} f(\hat{y})\Delta_{\boldsymbol{l}} M^{\mathrm{lat}}_{N,p}(v;y) = \int f(x)\partial_{\boldsymbol{l}} M^{\mathrm{UV}}_{p,v}(x)\,dx\,.$$

Idea of the proof (Lemmas 14.33–14.34). The Cauchy argument of Section 4.2 gives convergence on $\bigcup_m a_m\mathbb{Z}^4$: the transferred correlations at a fixed level converge, and the stopping error $2F_0e^{-B_m/64}$, with F_0 independent of m, compares them with the original ones. Proposition 5.8 extends convergence to all separated positions and identifies every derivative, also at moving points crossing the boundary of the chosen period cell. For $\boldsymbol{l}=\varnothing$ the left side is $\lim_N C^{\mathrm{lat}}_{N,p}(v;f)$, so the inner limits of Definition 2.6 exist at every admissible volume; admissibility places the support of f inside $\mathrm{NoHit}_{L_p,n}$. The limiting kernel is periodic in each position and smooth away from periodic collisions, including where positions share coordinates. The thresholds may depend on p.

6 The continuum limit, stress, phases, trace and products

The lattice family of Section 5 is now fixed; its couplings and insertions were chosen before any sequence of volume indices. This section takes the volume limit and identifies the continuum fields: their calculus, the stress tensor and the rotations, the phases and their reconstruction, the trace law and the products. Constants may depend on the group, its realization, the penalty and the construction choices of Section 9; further dependence is stated.

6.1 The volume limit and the field calculus

One volume refinement serves every field and all later constructions.

Theorem 6.1 (IV. Infinite-volume limit)**.** *For every strictly increasing* $\mathsf{p} : \mathbb{N} \to \mathbb{N}$*, there are a strictly increasing* $\mathsf{q} : \mathbb{N} \to \mathbb{N}$ *and a multilinear table* S *such that the Wilson family converges to* S *along* $P = \mathsf{p} \circ \mathsf{q}$ *in the sense of Definition 2.6. The table satisfies E1–E7.*

Remark 6.2 (Dependence on the volume sequence). The limiting table S may depend on the prescribed sequence p and the refinement q, hence on the retained volume indices $P = \mathsf{p} \circ \mathsf{q}$.

Fix p, q, P and S from now on.

Idea of the proof (Lemmas 15.1–15.6).

Step 1. Proposition 5.8 bounds the correlations and all their lattice differences uniformly in the volume; a diagonal extraction along p gives local smooth limits on Conf_n (E6). At each admissible volume the fixed-volume kernels are those of Theorem 5.9, so the full inner limit is kept; only the outer limit uses a subsequence.

Step 2. The chessboard seminorm, the cost of repeating one observation throughout the box, gives E7 at order zero, and the corrected derivative maps below give it at positive orders; lattice reflection positivity passes to the limit, giving E5 and then E4.

Step 3. A further refinement fixes finitely many coefficient maps in each degree. This auxiliary subsequence concerns finite-dimensional coefficients only; the limits of the correlations are the full ultraviolet limits already established.

The coefficient maps are needed because the normalization $a_N^{-d}\mathsf{Z}_{N,p,d}^{-1}$ depends on the degree. The same $c \in V_d \subset V_e$ defines different lattice fields as (d, c) and as (e, c), and a lattice difference of a degree-d insertion is a degree-$(d+1)$ object with yet another normalization; the naive inclusion $V_d \to V_e$ does not preserve correlations. Lemma 15.5 constructs linear *corrected coefficient maps* relating these normalizations, and they induce the inclusions and derivatives of the next proposition. Its dimension statement uses a Gaussian comparison theory, introduced by its covariance.

Definition 6.3 (Free Maxwell comparison)**.** For $d_{\mathfrak{g}}$ independent colours, let A_a^α be centred generalized Gaussian fields on $\mathbb{R}^4$ with

$$\mathbb{E}A_a^\alpha(x)A_b^\beta(y) = \delta_{\alpha\beta}\delta_{ab}\Gamma(x-y), \qquad \Gamma(x) = \frac{1}{4\pi^2|x|^2}, \qquad F_{ab} = \partial_a A_b - \partial_b A_a\,,$$

the derivatives taken in the sense of distributions. Let M_d^{Max} consist of the complex $\mathrm{Ad}(G)$-invariant Wick polynomials in curvature jets of weight at most d, modulo antisymmetry, the linear Bianchi identities and $\sum_a \partial_a F_{ab} = 0$. Their separated Wick correlations are S^{Max}.

Each component of A is a massless free field, with covariance $(-\Delta)^{-1}$ since $-\Delta\Gamma = \delta_0$, and F is the free Maxwell field. Wick polynomials in curvature jets at distinct points have smooth correlations, computed by Wick's theorem from derivatives of Γ at nonzero arguments. The

relations are of two kinds. Antisymmetry and the Bianchi identities $\partial_a F_{bc} + \partial_b F_{ca} + \partial_c F_{ab} = 0$ hold identically, since F is a derivative of A. The equation $\sum_a \partial_a F_{ab} = 0$ holds only in separated correlations: by $-\Delta\Gamma = \delta_0$,

$$\mathbb{E}\Big(\sum_a \partial_a F^{\alpha}_{ab}\Big)(x)\, F^{\beta}_{cd}(y) = \delta_{\alpha\beta}\big(\delta_{bd}\partial_{x^c} - \delta_{bc}\partial_{x^d}\big)\delta_0(x-y)\,,$$

which vanishes for $x \neq y$, so every separated Wick correlation with a factor of the divergence vanishes; but the divergence is not the zero distribution, since its covariance is $\delta_{\alpha\beta}(\partial_b\partial_d - \delta_{bd}\Delta)\delta_0(x-y)$. The quotient by this equation thus records a relation among separated correlations, not an identity of distributions. The comparison of Proposition 6.7 likewise concerns separated correlations at short distances; it does not identify the constructed fields with this Gaussian reference.

The constant one is the Maxwell unit, coefficient conjugation gives the real structure, and inclusions keep the same polynomial at a larger weight bound. Free differentiation is $\partial_a^{\mathrm{Max}}(:P:) =: \partial_a P :$, and the rotation action $R_d^{\mathrm{Max}}(h)(:P:) =: hP :$ is determined by $hF^{\alpha}_{ab} = \sum_{i,j} h_{ia}h_{jb}F^{\alpha}_{ij}$ and $h\partial_a = \sum_i h_{ia}\partial_i h$ for $h \in O(4)$; these operations preserve the defining relations.

Proposition 6.4 (FC. Field calculus)**.** *For the table S of Theorem 6.1, there are complex linear maps $\jmath_{d,e} : F_d \to F_e$ for $d \leq e$ and $\partial_a : F_d \to F_{d+1}$ for every d, a such that, for every $A \in F_d$,*

$$\iota_e \circ \jmath_{d,e} = \iota_d, \qquad \iota_{d+1}(\partial_a A) = \partial_{x_0^a}\iota_d(A),$$
$$\dim_{\mathbb{C}} F_d = \dim_{\mathbb{C}} M_d^{\mathrm{Max}}\,.$$

The proof is Lemma 15.5. The corrected coefficient maps supply representatives of inclusions and derivatives; the equation and rank comparisons of Lemmas 14.18–14.26 bound $\dim F_d$ from above, and the free tests of Lemma 14.22, with the free positivity of Lemma 16.1, bound it from below. Each ι_d is injective by Definition 2.12, so the inclusions are unique and injective, $\jmath_{e,f}\jmath_{d,e} = \jmath_{d,f}$, $\jmath_{d,d} = \mathrm{id}$, and by E1 they carry $\mathbf{1}_d$ to $\mathbf{1}_e$. Their images are nested, so $\mathcal{L} = \bigcup_d \iota_d F_d$. Permutation symmetry gives, for $\boldsymbol{A} \in \prod_i F_{d_i}$, $x \in \mathrm{Conf}_n$, slot i and axis a,

$$\partial_{x_i^a} S_n(\boldsymbol{A}; x) = S_n(A_1, \ldots, \partial_a A_i, \ldots, A_n; x)\,,$$

and E2 with injectivity of the responses makes inclusions and derivatives commute with conjugation, inclusions commute with H_4, and $h\partial_a = \sum_b h_{ba}\partial_b h$ for $h \in H_4$.

The table also defines reflection Hilbert spaces along the coordinate axes. They and their slab bounds use only the volume limit and the field calculus (Lemmas 15.6, 15.7 and Proposition 6.4), and they enter the rotation argument of Section 6.2 before continuous rotations are available.

Definition 6.5 (Coordinate reflection spaces)**.** For an axis a, let $\mathcal{H}_a^{\mathrm{OS}}$ be the completion of the complex span of words at distinct points in $\{x^a > 0\}$ modulo the null space of $\langle w, v\rangle_a = S(\theta_a(w)\, v)$, and put $e^{-tH_a^{\mathrm{OS}}}[v] = [\tau_{te_a} v]$ for $t \geq 0$. Let Ω_a^{OS} be the class of the empty word and P_a^0 the projection onto $\ker H_a^{\mathrm{OS}}$.

By E4 the form is positive semidefinite; with E7, the translations τ_{te_a}, $t \geq 0$, preserve its null space and define a strongly continuous self-adjoint contraction semigroup, whose generator is $H_a^{\mathrm{OS}} \geq 0$ (proof of Lemma 16.12). Inserting a word between positive-time vectors is bounded after a positive time translation (Lemma 16.12); these slab bounds are the coordinate estimates used before rotations.

6.2 Stress and rotations

The fields are compared with Maxwell fields at small distances, through bases that may vary with the radius to accommodate mixing, and a conserved tensor is then normalized by its shell fluxes. The volume refinement and S remain fixed; the subsequences below concern comparison radii only.

Definition 6.6 (Scaling frames)**.** A scaling sequence consists of $r_k > 0$, $r_k \to 0$, and complex linear isomorphisms $\mathsf{U}^{\mathrm{sc}}_{k,d} : M_d^{\mathrm{Max}} \to F_d$. It is stable if, for every $D \in \mathbb{N}$, these maps eventually preserve conjugation, H_4, units and inclusions exactly in all degrees $d \le e \le D$. Scaling dimensions are included in the maps: a comparison insertion is $\mathsf{U}^{\mathrm{sc}}_{k,d} u$ at $r_k x$, with no further factor r_k^d.

After rescaling positions the interacting correlations approach their Maxwell counterparts; one sequence serves every fixed list of fields.

Proposition 6.7 (MC. Maxwell comparison)**.** *There is one stable scaling sequence such that, for every arity n, degree list (d_i) and $u_i \in M_{d_i}^{\mathrm{Max}}$,*

$$S_n(\mathsf{U}^{\mathrm{sc}}_{k,d_i} u_i; r_k \cdot\,) \longrightarrow S_n^{\mathrm{Max}}(u; \cdot) \quad \text{in } L^1_{\mathrm{loc}}(\mathrm{Conf}_n)\,.$$

Idea of the proof (Lemma 16.8). The lattice comparison stops near the requested physical scale and compares the correlations with the free Gaussian ones. Finitely many separating correlations make the comparison maps injective, the upper bound $\dim_{\mathbb{C}} F_d \le \dim_{\mathbb{C}} M_d^{\mathrm{Max}}$ of Proposition 6.4 makes them isomorphisms, and finite-dimensional corrections impose the eventual equalities. Local L^1 convergence is equivalent, by duality and smooth approximation on slightly larger separated compact sets, to convergence uniform over smooth tests with $\|f\|_\infty \le 1$ and fixed compact support. This weaker form is used to construct and normalize the conserved current, before the stress tensor exists.

To read translation and angular charge, integrate a conserved symmetric tensor through a shell surrounding one insertion. In a free model the shell integral can be computed, and it differentiates the enclosed correlation. Let ϕ be a massless free scalar field with covariance Γ, write $\langle\cdot\rangle$ for its Wick correlations, and put $T^{\phi}_{ab} = -{:}\partial_a\phi\,\partial_b\phi{:} + \frac{1}{2}\delta_{ab}{:}|\nabla\phi|^2{:}$, the negative of the canonical tensor. With $\Gamma_0 = \Gamma(y - x_0)$ and $\Gamma_1 = \Gamma(y - x)$,

$$\langle T^{\phi}_{ab}(y)\phi(x_0)\phi(x)\rangle = -\partial_a\Gamma_0\,\partial_b\Gamma_1 - \partial_b\Gamma_0\,\partial_a\Gamma_1 + \delta_{ab}\nabla\Gamma_0\cdot\nabla\Gamma_1\,,$$

whose divergence in y vanishes away from x_0 and x, where $\Delta\Gamma_0 = \Delta\Gamma_1 = 0$. Take a smooth compactly supported χ equal to one on the ball of radius $\epsilon < |x - x_0|$ about x_0 and vanishing near x. The divergence theorem turns the flux $-\sum_a \int \partial_a\chi(y)\langle T^{\phi}_{ab}(y)\phi(x_0)\phi(x)\rangle\,dy$ into the integral of $\sum_a \langle T^{\phi}_{ab}(y)\phi(x_0)\phi(x)\rangle n_a$ over the boundary sphere of that ball, with outer normal n. There $\nabla\Gamma_0 = -n/(2\pi^2\epsilon^3)$, the last two terms cancel, and the integrand is $\partial_b\Gamma_1/(2\pi^2\epsilon^3)$. The sphere has area $2\pi^2\epsilon^3$ and $\partial_b\Gamma(\cdot - x)$ is harmonic on the ball, so the mean value property gives the flux $\partial_b\Gamma(x_0 - x) = \partial_{x_0^b}\langle\phi(x_0)\phi(x)\rangle$. This free calculation illustrates the identity that Theorem 6.10 establishes for the constructed fields by a different route.

We normalize the stress tensor in the same way, so that its translation flux acts as $+\partial_{x_0^b}$ on the enclosed insertion; with the minus sign in the flux below, this gives the negative of the usual Euclidean canonical Maxwell tensor (displayed after Theorem 6.10). Let $A \in F_d$, let $\boldsymbol{B}$ be a list of n spectator fields at x, and take $(x_0, x) \in \mathrm{Conf}_{n+1}$.

Definition 6.8 (Shell fluxes)**.** For $J_{ab} = J_{ba} \in F_4$ with $\sum_a \partial_a J_{ab} = 0$ and $K_{\mathrm{rot}} \in \mathfrak{so}(4)$, choose a real shell cutoff $\chi \in C_c^\infty(\mathbb{R}^4)$ equal to one near x_0 and supported away from the spectators. Define

$$\mathcal{P}_b^J = -\sum_a \int \partial_a \chi(y) S(J_{ab}(y), A(x_0), \boldsymbol{B}(x))\, dy,$$
$$\mathcal{Y}_{K_{\mathrm{rot}}}^J = -\sum_{a,b} \int \partial_a \chi(y)(K_{\mathrm{rot}}(y - x_0))_b S(J_{ab}(y), A(x_0), \boldsymbol{B}(x))\, dy\,.$$

The integrals avoid every insertion point, and they do not depend on the shell cutoff: two choices differ by a function vanishing near every insertion, integration by parts with Proposition 6.4 removes the divergence term, and the extra angular term is $\sum_{a,b}(K_{\mathrm{rot}})_{ba} J_{ab} = 0$ since K_{rot} is skew and J symmetric. Conservation makes the flux independent of the radius of the shell; it does not by itself control a flux through a distant surface. A field is real when fixed by conjugation; a family $J_{ab} \in F_4$ is H_4-tensorial when $hJ_{ab} = \sum_{i,j} h_{ia} h_{jb} J_{ij}$ for $h \in H_4$, conserved when $\sum_a \partial_a J_{ab} = 0$ in F_5 for every b, and centered when every component has zero one-point function.

Proposition 6.9 (Conserved line)**.** *The centered real symmetric H_4-tensorial conserved families (J_{ab}) in F_4 form a real vector space of dimension one.*

This is the first assertion of Lemma 16.10; its proof shows that divergence maps the centered real symmetric tensorial families onto the corresponding vectors in degree five, with one-dimensional kernel. Along a refinement of the sequence of Proposition 6.7, positive multiples of the rescaled nonzero member approach the free stress line in the degree-four frames. For a nonzero member J, Lemma 16.11 constructs complex linear charge maps $\mathsf{p}_b : F_d \to F_{d+1}$ and $\mathsf{Y}_{K_{\mathrm{rot}}} : F_d \to F_d$, the latter real linear in K_{rot}, with $\mathcal{P}_b^J = S((\mathsf{p}_b A)(x_0), \boldsymbol{B}(x))$ and $\mathcal{Y}_{K_{\mathrm{rot}}}^J = S((\mathsf{Y}_{K_{\mathrm{rot}}} A)(x_0), \boldsymbol{B}(x))$ for every degree, field, spectator list, separated configuration and shell cutoff.

These charges restore the rotations. Conservation of J turns the sum of the angular charges of the insertions, each combining $\mathsf{Y}_{K_{\mathrm{rot}}}$ at its own point with the translation charge weighted by the rotated position, into the integral of J against the derivative of a vector field equal to $y \mapsto K_{\mathrm{rot}} y$ on a large anisotropic box. For configurations on a coordinate line, the coordinate reflection spaces of Definition 6.5 show that this integral tends to zero as the box grows (Lemma 16.15; Section 4.3 explains the mechanism), so the angular sum vanishes there. Finite Ward tests then identify p_b with $c_\star \partial_b$, and analyticity extends the identity, infinitesimal rotation invariance, to all separated configurations. Lemma 16.17 thereby constructs continuous complex linear representations R_d of $O(4)$, extending H_4, fixing units and commuting with conjugation and inclusions; for a real $c_\star \neq 0$ common to every degree,

$$\mathsf{L}_d(K_{\mathrm{rot}}) = \frac{d}{dt} R_d(e^{tK_{\mathrm{rot}}})\Big|_{t=0}, \qquad \mathsf{p}_b = c_\star \partial_b, \qquad \mathsf{Y}_{K_{\mathrm{rot}}} = c_\star \mathsf{L}_d(K_{\mathrm{rot}})\,.$$

Integrating gives full covariance (Lemma 16.17): $S_n(R_{d_i}(h)A_i; hx) = S_n(\boldsymbol{A}; x)$ for every $\boldsymbol{A} \in \prod_i F_{d_i}$, $x \in \mathrm{Conf}_n$ and $h \in O(4)$. A family is $O(4)$-tensorial if $R_4(h) J_{ab} = \sum_{i,j} h_{ia} h_{jb} J_{ij}$ for every $h \in O(4)$. The common constant $c_\star$ fixes the normalization of the stress tensor by its action on fields.

Theorem 6.10 (ST. Stress tensor)**.** *There is a real symmetric $O(4)$-tensorial family $T_{ab} \in F_4$ with $\sum_a \partial_a T_{ab} = 0$ such that, for every degree, field, spectator list, separated configuration and shell cutoff of Definition 6.8, every b and every $K_{\mathrm{rot}} \in \mathfrak{so}(4)$,*

$$\mathcal{P}_b^T = \partial_{x_0^b} S(A(x_0), \boldsymbol{B}(x)), \qquad \mathcal{Y}_{K_{\mathrm{rot}}}^T = S((\mathsf{L}_d(K_{\mathrm{rot}}) A)(x_0), \boldsymbol{B}(x))\,.$$

The proof is in Lemmas 16.17–16.18: divide J by $c_\star$ and average its tensor transform over $O(4)$ with probability Haar measure. Covariance and the transformation laws of derivatives and generators give the same shell Ward identities for each transformed current, and integration preserves them, as well as symmetry, reality, centering and conservation. Fix this tensor; it is centered, since J is and R_4 preserves zero one-point values. The free stress T^{Max} is normalized by the same shell Ward identities, with $\partial_a^{\mathrm{Max}}$ and the free generators $\mathsf{L}_d^{\mathrm{Max}}$ (Section 16.1):

$$T_{ab}^{\mathrm{Max}} = -\sum_{c,\alpha} : F_{ac}^{\alpha} F_{bc}^{\alpha} : + \frac{\delta_{ab}}{4} \sum_{i,j,\alpha} : F_{ij}^{\alpha} F_{ij}^{\alpha} : \, .$$

The last comparison includes stress insertions. For a free list $u_1, \dots, u_n$, $u_i \in M_{d_i}^{\mathrm{Max}}$, a stress-slot choice is a subset $I \subset \{1, \dots, n\}$ with component pairs (a_i, b_i) for $i \in I$; put $A_i^0 = T_{a_i b_i}^{\mathrm{Max}}$ for $i \in I$ and $A_i^0 = u_i$ otherwise. The shell Ward identities fix the stress normalization, so its comparison insertion needs only the engineering factor r_k^4. Below, D_x^j differentiates the whole function $x \mapsto S_n(\boldsymbol{A}_k; r_k x)$.

Proposition 6.11 (SC. Differentiated short-distance comparison)**.** *Along the refinement of the scaling sequence of Proposition 6.7 chosen in Lemma 16.10, for every fixed free list and stress-slot choice, every derivative order j and compact $\mathcal{K} \subset \mathrm{Conf}_n$, with $A_{k,i} = r_k^4 T_{a_i b_i}$ on I for the fixed tensor of Theorem 6.10 and $A_{k,i} = \mathsf{U}_{k,d_i}^{\mathrm{sc}} u_i$ elsewhere,*

$$D_x^j S_n(\boldsymbol{A}_k; r_k x) \longrightarrow D_x^j S_n^{\mathrm{Max}}(\boldsymbol{A}^0; x) \quad \textit{uniformly on } \mathcal{K}\,.$$

Fix this refined sequence $(r_k, \mathsf{U}_{k,d}^{\mathrm{sc}})$ from now on. The proof follows Lemma 16.19; with $I = \varnothing$ and $j = 0$ it gives Proposition 6.7 in uniform form, but the earlier comparison is needed first, to construct the tensor. Along this sequence, for fixed d, a, K_{rot}, the transported infinitesimal actions converge in operator norm: $r_k (\mathsf{U}_{k,d+1}^{\mathrm{sc}})^{-1} \partial_a \mathsf{U}_{k,d}^{\mathrm{sc}} \to \partial_a^{\mathrm{Max}}$ and $(\mathsf{U}_{k,d}^{\mathrm{sc}})^{-1} \mathsf{L}_d(K_{\mathrm{rot}}) \mathsf{U}_{k,d}^{\mathrm{sc}} \to \mathsf{L}_d^{\mathrm{Max}}(K_{\mathrm{rot}})$.

6.3 Phases and reconstruction

Compression to zero energy separates the mixed correlations into phases. In the reflection space of the time axis, a long empty time interval removes the positive-energy components; the word insertions that survive commute on the zero-energy space, and a character of the algebra they generate assigns compatible scalar values to all of them. The phases are these characters. We now use the fourth coordinate as time.

Definition 6.12 (Compression)**.** For $h_{\mathrm{sl}} > 0$ and a word w with times in $(0, h_{\mathrm{sl}})$, let $\mathsf{B}_{h_{\mathrm{sl}}}(w)$ be the bounded extension to $\mathcal{H}_4^{\mathrm{OS}}$ (Lemma 16.12) of the slab insertion $A_w v = w\, \tau_{h_{\mathrm{sl}} e_4} v$ on positive-time words v, and put $M^0(w) = P_4^0 \mathsf{B}_{h_{\mathrm{sl}}}(w)|_{\ker H_4^{\mathrm{OS}}}$.

The slab insertion places w in the slab $(0, h_{\mathrm{sl}})$ below the translated state. A separated word is translated into a slab to define its compression; Lemma 17.1 proves independence of width and placement and that the compressed operators generate a separable commutative unital C*-algebra

$$\mathfrak{A}_0 = C^*(M^0(w) : w \text{ separated})$$

with cyclic and separating vector Ω_4^{OS}. Its vacuum state $a \mapsto \langle \Omega_4^{\mathrm{OS}}, a \Omega_4^{\mathrm{OS}} \rangle$ satisfies $S(w) = \langle \Omega_4^{\mathrm{OS}}, M^0(w) \Omega_4^{\mathrm{OS}} \rangle$ by translation invariance; the measure below represents this state on the character space, and the separating property gives full support. Call a character ξ a phase and S^ξ its table.

Theorem 6.13 (PH. Phase decomposition)**.** *The algebra $\mathfrak{A}_0$ has a compact metric character space X with a full-support probability measure ν such that, for every separated word w,*

$$S^{\xi}(w) = \xi(M^0(w)), \qquad S(w) = \int_{\mathsf{X}} S^{\xi}(w)\, d\nu(\xi)\,.$$

Each phase satisfies E1–E7 and has exactly the null spaces $\mathcal{N}_d(S^{\xi}) = \mathcal{N}_d(S)$. The identities of FC, $O(4)$ covariance and ST hold with S^{ξ} in place of S, for the same maps and tensor T. For fixed w, $\xi \mapsto S^{\xi}(w)$ is continuous.

Idea of the proof (Lemmas 16.12, 17.1 and 17.3). The matrix element $\langle [u], M^0(w_1)M^0(w_2)[v]\rangle$ is a limit of correlations of four words placed far apart in time; the long time intervals project onto $\ker H_4^{\mathrm{OS}}$. The same four words can instead be spread along a second positive coordinate direction, so that the two middle words occur in the opposite order. By $O(4)$ covariance (Lemma 16.17) both arrangements have the same limit, so $M^0(w_1)M^0(w_2) = M^0(w_2)M^0(w_1)$: rotation covariance is an input to the phase decomposition. Reflection positivity holds in each phase because the compressed reflected Gram matrices are positive operators and characters are positive; that no field becomes null in a phase uses the free tests at short distances. Commutativity alone gives neither.

For each phase apply Definitions 6.5 and 6.12 to S^{ξ}, obtaining $\mathcal{H}^{\mathrm{OS}}_{4,\xi}$, $H^{\mathrm{OS}}_{4,\xi}$ and $\Omega^{\mathrm{OS}}_{\xi}$. Fix ξ_* in the full-measure set with one-dimensional time-axis reflection ground space supplied by Lemma 17.3; Theorem 6.16 extends ground-space uniqueness to every phase. Two further notions enter the reconstruction and the trace law.

Definition 6.14 (Bounded weak commutant)**.** A bounded operator B belongs to the weak commutant if, for every smeared field Φ and all $u, v \in \mathcal{D}$, one has $u \in \operatorname{Dom}\Phi^*$ (automatic by W1) and $\langle \Phi^* u, Bv\rangle = \langle u, B\Phi v\rangle$, where Φ^* is the Hilbert-space adjoint.

Definition 6.15 (Hermitian scalar labels)**.** A label is Hermitian if $\ell^* = \ell$, and Lorentz scalar if $\rho_{\mathrm{lab}}(\Lambda, 0)\ell = \ell$ for every $\Lambda \in SO^+(1,3)$.

For the theory at a phase ξ write $\mathcal{H}_\xi$, $\mathcal{D}_\xi$, Ω_ξ, Φ_ξ, T_ξ, and define its energy by $\mathsf{T}_\xi(te_4) = e^{itH_\xi}$. Its reflection comparison map is the linear isometry $\mathsf{J}^{\mathrm{OS}}_\xi : \mathcal{H}_\xi \to \mathcal{H}^{\mathrm{OS}}_{4,\xi}$ with $\mathsf{J}^{\mathrm{OS}}_\xi \Omega_\xi = \Omega^{\mathrm{OS}}_\xi$ that sends each product of smeared fields applied to the vacuum to the norm limit of its heat-regularized positive-time reflection vectors as the ordered heights tend to zero (proofs of Lemmas 17.5 and 17.8).

Theorem 6.16 (RC. Reconstruction)**.** *For every $\xi \in \mathsf{X}$, there are a faithful Wightman theory (Definition 2.14) on $\mathcal{L}$ and functions F^{ξ}_n satisfying Definitions 2.16 and 2.17 with the response maps ι_d of Definition 2.12. Its OS comparison map $\mathsf{J}^{\mathrm{OS}}_\xi$ is unitary, intertwines spatial translations and satisfies $\mathsf{J}^{\mathrm{OS}}_\xi e^{-sH_\xi} = e^{-sH^{\mathrm{OS}}_{4,\xi}} \mathsf{J}^{\mathrm{OS}}_\xi$ for $s \geq 0$, and*

$$\ker H_\xi = \mathbb{C}\Omega_\xi, \qquad \ker H^{\mathrm{OS}}_{4,\xi} = \mathbb{C}\Omega^{\mathrm{OS}}_\xi\,.$$

Every bounded operator in the weak commutant of its fields is scalar.

Idea of the proof (Lemmas 17.5–17.11). Reflection positivity supplies the Hilbert spaces and the nonnegative energy; the slab bounds supply the growth estimates behind the boundary values; the restored rotations, continued to boosts, supply Lorentz covariance and, with the continuation, locality. The uniform low-energy estimate of Section 4.4, continuity of the vacuum-subtracted heat pairings and full support of ν extend vacuum uniqueness from almost every phase to every phase, so the choice of ξ_* stays valid. The same lemmas give joint Schwartz distributions, phase averages and stress continuation.

The label maps are common to all phases. The adjoint is induced by time reflection followed by conjugation, $(\iota_d A)^* = \iota_d(R_d(\mathsf{s}_4^{\mathrm{refl}})\overline{A})$ (Lemma 17.7). The Lorentz label action is the continuation of R_d [15]: spatial rotations keep their Euclidean action, and for $i = 1, 2, 3$ the boost along e_i has generator $-i\mathsf{L}_d(K_{4i})$, where $K_{4i} = e_4 e_i^{\mathsf{T}} - e_i e_4^{\mathsf{T}}$ generates rotation in the (e_4, e_i) plane. In particular, a real $O(4)$-invariant field gives a Hermitian Lorentz scalar label. Retain the theory at ξ_* and suppress its subscript in $\mathcal{H}, \mathcal{D}, \Omega, \Phi$ and the other relativistic data.

6.4 Trace and products

Several ultraviolet statements of different strength are proved, and they should not be confused.

Statement	What it asserts
Running (Theorem 5.6)	A volume-uniform leading logarithmic slope of the bare coupling of the tuned family
Maxwell comparison (Proposition 6.7)	Convergence along a sequence of radii, through scale-dependent frames that include field renormalization and mixing; not a perturbative expansion of each composite correlation
Stress comparison (Proposition 6.11)	The same, with derivatives and with the normalized stress insertions
Trace law (Theorem 6.18)	A specific asymptotic through all positive radii, in the selected phase

The trace is centered in the phase already selected.

Definition 6.17 (Centered trace)**.** For the tensor T fixed after Theorem 6.10 and $r > 0$, put

$$\tau = \sum_{a=1}^{4} T_{aa}, \qquad e_{\mathrm{tr}} = S^{\xi_*}(\tau), \qquad \Theta = \tau - e_{\mathrm{tr}}\mathbf{1}_4,$$
$$G_\Theta(r) = S_2^{\xi_*}(\Theta, \Theta; -re_4/2, re_4/2), \qquad A_G = \frac{3d_{\mathfrak{g}}}{\pi^4}\,.$$

The one-point value is position independent, and centering removes the vacuum contribution in the selected phase. Adding a real multiple of $\delta_{ab}\mathbf{1}_4$ to T preserves its shell Ward identities, the limit in Proposition 6.11 and Θ. The fixed stress tensor is centered in the mixture; the selected-phase centering subtracts its own one-point value.

Theorem 6.18 (TR. Trace law)**.** *The selected field Θ is nonzero, real and $O(4)$ invariant; its label $\iota_4\Theta$ is Hermitian and Lorentz scalar (Definition 6.15). For every $f \in \mathcal{S}$,*

$$\langle \Omega, \Phi(\iota_4\Theta, f)\Omega\rangle = 0\,.$$

As $r \downarrow 0$ through all positive radii,

$$\lim_{r\downarrow 0} r^8 \log^2(1/r) G_\Theta(r) = A_G > 0\,.$$

The amplitude comes from the free scalar curvature square, with $\alpha = 1, \dots, d_{\mathfrak{g}}$ indexing an orthonormal colour basis:

$$\mathcal{O}_{F^2} = \frac{1}{4}\sum_{a,b,\alpha} :F_{ab}^{\alpha}F_{ab}^{\alpha}:, \qquad S_2^{\mathrm{Max}}(\mathcal{O}_{F^2}, \mathcal{O}_{F^2}; -e_4/2, e_4/2) = \frac{3d_{\mathfrak{g}}}{\pi^4}\,.$$

At running coupling β the scalar trace coefficient is $4d_2/\beta$, with $d_2 = -b_0/2$, computed from a lattice Ward identity and the conserved line (Section 4.5, Proposition 18.8); since $\kappa_G = 2b_0$, its leading

value at radius r is $4d_2/(\kappa_G \log(1/r)) = -1/\log(1/r)$, and squaring it with the free correlation gives A_G (Section 4.5). The comparison needs fourth-order information beyond Proposition 6.7, with the rate that the slowly growing coefficients require; its estimates are uniform in the phase, although the theorem is stated at ξ_* only.

Idea of the proof (Section 18; Lemmas 18.1, 18.2, 18.3 and 18.7).

Step 1. Begin with any sequence of positive radii tending to zero.

Step 2. Adjustable stopping levels give the comparison along a subsequence, at generic configurations, not at the time-axis pair defining G_Θ.

Step 3. The positive spectral representation below makes G_Θ real and nonincreasing; the monotone function is G_Θ itself, not $r^8 \log^2(1/r) G_\Theta(r)$.

Step 4. By $O(4)$ invariance, generic separations slightly below and slightly above the target radius bound G_Θ there from both sides.

Step 5. Every starting sequence thus has a subsequence with the same normalized limit, which gives the limit through all radii.

The same proof identifies centering on the word domain, $\Phi(\iota_4\Theta, f) = \Phi(\iota_4\tau, f) - e_{\mathrm{tr}}(\int f)\mathrm{Id}_{\mathcal{H}}$, and constructs the unique positive tempered spectral measure μ_Θ, supported in $\overline{V}_+$, with

$$d\mu_{\Phi(\iota_4\Theta,f)\Omega} = |\widehat{f}_+|^2 \, d\mu_\Theta, \qquad G_\Theta(r) = \int e^{-r\mathbf{p}^4} \, d\mu_\Theta(\mathbf{p}) \quad (r > 0)\,,$$

where $\widehat{f}_+(\mathbf{p}) = \int e^{i\mathbf{p}\cdot x} f(x)\, dx$. Its Lorentz invariance and high-energy limits are Lemmas 18.2 and 18.9. Finite Wick profiles cannot reproduce the factor $r^{-8}\log^{-2}(1/r)$.

Proposition 6.19 (NF. Finite-profile exclusion)**.** *For every finite Wick profile W_{Wick} and every $\delta > 0$, there is $0 < r < \delta$ such that*

$$G_\Theta(r) \neq W_{\mathrm{Wick}}(r)\,.$$

The reason is the form of the expansions (Lemma 18.10). Each mass-shell kernel has an expansion to every finite order in integer powers of r and nonnegative powers of $\log r$, so a finite profile is, to bounded error, a finite sum of monomials $r^a(\log r)^k$ with $a \in \mathbb{Z}$, $k \in \mathbb{N}$. Multiplication by $r^8 \log^2(1/r)$ makes the terms with $a > -8$ vanish and the leading remaining nonzero term diverge, so a finite limit must be zero; the positive limit of Theorem 6.18 excludes equality on every interval $(0, \delta)$. Lemma 18.12 treats polynomial mass-shell comparisons, and Lemmas 18.15 and 8.10 give Fock countertests with their own comparison classes.

The phase quantifiers differ from statement to statement, and the following table records them.

Statement	Scope
Slab, comparison and low-energy estimates	Uniform in the phase
Vacuum uniqueness in the proof of Theorem 6.13	Almost every phase (an intermediate step)
Reconstruction, Theorem 6.16	Every phase
Trace law, Theorem 6.18	The selected phase ξ_*; uniform estimates in its proof do not change this
Product expansion, Theorem 6.20	Every phase and the mixture; degree cutoff and spectators fixed before the threshold, states and positions after it

The product expansion is a decoder, stated for the tables S^ζ, $\zeta \in \mathsf{X} \sqcup \{\mathrm{mix}\}$, on the common quotient fields. For each degree j, finitely many probe words placed away from the origin, with biorthogonal coordinate functionals, distinguish the fields of F_j. Applied to a cluster of fields

at positions x around a centre o, the probes read off the field with the same answers, averaged over $O(4)$ and conjugation; lower degrees are removed before the next one is read, and the coordinates in the basis $O_{j,\lambda}$ are the coefficients $c_{j,\lambda}(\boldsymbol{A}; x, o)$ (Definitions 19.3–19.5). For two fields, $A(x)B(0) \sim C^{\mathbf{1}}_{AB}(x)\mathbf{1} + \sum_\lambda C^\lambda_{AB}(x)O_\lambda(0) + \cdots$ is tested inside every correlation with distant spectators, and the error after degree D is at most $r^{D+1-d_A-d_B-\varepsilon}$: for small D it may still be singular, and what improves with D is its order. The probes, the bases and the coefficients are fixed once from the mixed table, before spectators, state indices and truncations; spectator and state independence hold by construction, and the content of the theorem is the size of the remainder of Definition 2.19. The coefficients depend on this prescription, the table and the probes; they are not a canonical expansion, and the theorem concerns this one coefficient system.

Theorem 6.20 (OP. Operator product expansion)**.** *For every finite list $\boldsymbol{A}$, $M_{\mathrm{geo}} \geq 0$, $\eta, \varepsilon > 0$, $j \in \mathbb{N}$ and $\lambda \in \mathrm{Ch}_j$, some $r_0 > 0$ gives, for $0 < r < r_0$ and every $(r, M_{\mathrm{geo}}, \eta)$-cluster,*

$$|c_{j,\lambda}(\boldsymbol{A}; x, o)| \leq r^{j-|\boldsymbol{A}|_{\mathrm{deg}}-\varepsilon}\,.$$

For every pair of finite lists $\boldsymbol{A}, \boldsymbol{B}$, $M_{\mathrm{geo}} \geq 0$, $\eta, \delta_{\mathrm{sp}}, \varepsilon > 0$ and $D \in \mathbb{N}$, some $r_1 > 0$ gives, for $0 < r < r_1$, such clusters, δ_{sp}-clear spectators and all state indices ζ,

$$|\mathcal{R}^\zeta_D| \leq r^{D+1-|\boldsymbol{A}|_{\mathrm{deg}}-\varepsilon}\,.$$

Idea of the proof (Lemmas 19.6–19.10). The discarded part is bounded in the compression algebra; evaluation at a character or at the vacuum state does not increase the bound, so it holds in every phase and in the mixture. The bound enters through moments (Lemma 17.2): lattice cluster replacement makes a group of insertions minus its matched field, one insertion at the centre, small in chessboard seminorm; the chessboard estimate applied to $2n$ widely translated, alternately conjugated copies bounds the nth vacuum moment of the compressed remainder U, so that $\int |\xi(U)|^{2n}\, d\nu(\xi) \leq B^{2n}$ for every n, and continuity in ξ with full support of ν gives $\sup_\xi |\xi(U)| \leq B$. Lattice estimates at three scales, the resolution of the insertions, the fusion scale and the exterior cell, give the degree count $D + 1 - |\boldsymbol{A}|_{\mathrm{deg}}$ up to factors $e^{o(\log(1/r))}$, which $r^{-\varepsilon}$ absorbs for small r. The two thresholds have the separate dependencies of the statement. The estimate holds at each finite truncation D; no series in D is claimed to converge. Lemmas 19.13–19.14 and Remark 19.15 give nested and integrated estimates.

Contact subtractions use the same original fields and correlations, with separate projections that extract fields through finite correlation probes; their clearance is a radius keeping the collapsed cluster separated from these probes, independent of the spectator clearance δ_{sp}. A collision forest is a family of subsets of slots, any two disjoint or nested. Since S^ζ_n satisfies E7, some contact extension exists by the standard extension theorem for kernels of polynomial growth at the diagonals; the content of the next theorem is that the explicit subtraction of normal Taylor jets along collision forests produces one, with the local order bounds of Lemma 19.21.

Theorem 6.21 (CT. Contact extension)**.** *For every state index ζ, arity n and list $\boldsymbol{A} \in \mathbb{F}^n$, the contact construction of Section 19.2, fixed on the underlying prescription of Lemma 19.6, gives a distribution $\mathfrak{D}^\zeta_{\boldsymbol{A}}$ on $\mathbb{R}^{4n}$ such that, for every separated test f,*

$$\mathfrak{D}^\zeta_{\boldsymbol{A}}(f) = \int f(x) S^\zeta_n(\boldsymbol{A}; x)\, dx\,.$$

The proof is in Lemmas 19.19–19.21. For two insertions, write the test in the centroid and one relative coordinate; for each contact degree w below the critical order L of the pair, subtract

the degree-w contact projection of the pair at the centroid, tested against the Taylor jet of order $L-1-w$ of the test in the relative coordinate, and the subtracted expression gains in each shell; three or more insertions bring partial collisions and nested or disjoint forests. A separated test extended by zero has vanishing normal jets on every collision diagonal, so only the correlation integral remains. For a general test, Lemma 19.20 controls the contact terms and the signed residuals after the child subtractions, and the shell induction of Lemma 19.21 gives a continuous functional; Lemma 19.19 supplies covariant contact projections with real bases. The extension is a later construction from the separated correlations, not a limit of unsubtracted lattice correlations at coincident points; neither uniqueness, positivity nor contact Ward identities are asserted.

7 Deduction of the main theorem

The principal results of Sections 5 and 6 concern one lattice family and one volume refinement; the deduction assembles them in the order of their choices.

Proof of Theorem B. 1. Fix G and $\mathsf{b} > 0$, and make the common choices in the order of Section 9. The Wilson start (5.13) begins the induction, Theorem 5.5 advances it, and Theorem 5.6 selects trajectories reaching the common endpoint. These choices precede any volume sequence.

2. Fix the local sections, reference levels and terminal maps on these trajectories; Definition 2.8 defines every insertion, also below its reference level. The local sections are polynomials in the links of their fixed stencils, hence continuous and local, and the terminal maps change only their coefficients, which preserves gauge invariance. Theorem 5.6 at $m = N$ gives the volume-uniform running law with $b_* = \kappa_G > 0$. Thus $\mathcal{W}$ is regular (Definition 2.5).

3. Theorem 5.9 gives the full fixed-volume limit for every field and difference list, so every admissible separated test has a full inner limit before any volume extraction. This proves (i).

4. Given a strictly increasing p, Theorem 6.1 chooses q and S with ordered convergence along $P = \mathsf{p} \circ \mathsf{q}$ and properties E1–E7; the same extraction fixes the corrected coefficient maps. This proves the table part of (ii).

5. Define $F_d = V_d/\mathcal{N}_d(S)$ and $\mathcal{L}$ by Definition 2.12. Proposition 6.4 gives the inclusions and derivatives, with injective response maps whose images cover $\mathcal{L}$, and Theorem 6.10 gives the conserved tensor, normalized by its shell fluxes, on these fields.

6. Theorem 6.13 gives X, ν, S^{ξ} on these quotient spaces. Choose ξ_* in the full-measure set with one-dimensional time-axis reflection ground space (Lemma 17.3). Theorem 6.16 reconstructs every phase with a unique vacuum, which proves (iii); retain the theory and continuation at ξ_*.

7. Set $\pi_{\mathrm{quo}}(d, \mathbf{c}) = (d, [\mathbf{c}])$; representatives make it onto. The unit response is nonzero since its one-point function is one, so faithfulness in Theorem 6.16 gives a nonzero acting field and, with Step 5, the label clauses of Q3. Theorem 6.13 gives its phase clauses, and the reconstruction gives Q2 and Q4. Assign arbitrary values to the quotient mixture and phase correlations on the collision diagonals; this supplies the function domains of Q0 and has no distributional content, unlike the contact extensions constructed in Step 9. Separated tests vanish there, so ordered convergence is unchanged, and representative independence gives $S_n^{\mathrm{mix}}(\pi_{\mathrm{quo}} v; x) = S_n(v; x)$ on Conf_n. This identifies the lattice limit $\mathcal{W} \to S^{\mathrm{mix}} \circ \pi_{\mathrm{quo}}$ of (ii).

8. Center the trace of the fixed tensor in the selected phase by Definition 6.17. Theorem 6.18 gives vacuum centering, the positive amplitude and the full-radius law, and Proposition 6.19 excludes every finite Wick profile; these give Q6 and (iv).

9. Choose the underlying prescription of Lemma 19.6 once, with its original fields and correlation table. Theorem 6.20 gives Q5 and its error bounds. Theorem 6.21 and Lemmas 19.19–19.21 use the same prescription, with separate contact projections, separating functionals and clearance, to construct real bases of the contact strata and the contact extensions; the contact proof uses the lattice-derived estimates of Lemma 19.9 and the signed transfer of Lemma 19.20. This proves (v) and completes (ii). □

Remark 7.1 (Further results). Further conclusions about these same objects are proved in Part II. Lemma 14.34 gives convergence at moving points, and Lemma 17.9 the uniform low-energy bounds. For the field Θ of Step 8, Lemma 18.1 gives the operator identity of centering, Lemma 18.2 the unique spectral measure and Lemma 18.9 its high-energy limits. Lemmas 18.12, 18.15 and 8.10 give further polynomial and Fock countertests in the selected phase, and Lemmas 19.13–19.14 and Remark 19.15 nested and integrated product estimates for the same prescription.

Remark 7.2 (Endpoint tuning). The specified recursion also varies the endpoint analytically. There is a width $w_{\mathrm{tun}} > 0$, independent of N, p, such that, for every N, p and $s \in [-1, 1]$, exactly one bare coupling $\beta_{N,p}(s)$ starts a trajectory of the specified recursion reaching $\bar{\beta} + w_{\mathrm{tun}} s$ at level zero. It satisfies $\beta_{N,p}(0) = \beta_{N,p}$ and has a holomorphic extension $b_{N,p}$ to $|z| < 1$ with

$$b_{N,p}(s) = \beta_{N,p}(s) \qquad (-1 < s < 1)\,. \tag{7.1}$$

This is Lemma 14.4; the proofs above do not use it.

8 Correspondence with the formal statement

This section states the existence theorem in the exact form of the formal statement and deduces it from Theorem B. The formal statement constrains abstract hierarchy data by the conditions Q0–Q6, of which Q1 is Definition 2.5. They do not impose every property established above: the correlations need not be multilinear, the logarithmic slope b_* is only required to be positive, and the degree-four field Θ is not identified as a stress trace. Section 8.2 records the conventions in which the formal statement differs from the main text.

8.1 Hierarchy data and the conditions Q0–Q6

Definition 8.1 (Q0. Hierarchy data)**.** Hierarchy data over a Wilson family $\mathcal{W}$ consist of:

- complex Hilbert and vector spaces $\mathcal{H}, \mathcal{L}$, and a relativistic datum with the group of Remark 8.8;
- finite-dimensional complex normed spaces F_d, linear maps $\iota_d : F_d \to \mathcal{L}$, and $\pi_{\mathrm{quo}} : \mathcal{T} \to \mathbb{F}$, where $\mathbb{F} = \bigsqcup_d F_d$;
- a compact metric space X, a Borel measure ν and $\xi_* \in \mathsf{X}$;
- functions $S_n^\zeta : \mathbb{F}^n \times (\mathbb{R}^4)^n \to \mathbb{C}$ for $\zeta \in \mathsf{X} \sqcup \{\mathrm{mix}\}$, and $\mathsf{F}_n : \mathbb{F}^n \times \mathbb{C}^n \times (\mathbb{R}^3)^n \to \mathbb{C}$;
- finite sets Ch_j, fields $O_{j,\lambda} \in F_j$, and coefficients $c_{j,\lambda} : \mathbb{F}^{I_{\mathrm{cl}}} \times (\mathbb{R}^4)^{I_{\mathrm{cl}}} \times \mathbb{R}^4 \to \mathbb{C}$ for every finite index set I_{cl};
- an element $\Theta \in F_4$.

The spaces, maps and functions of Q0 are arbitrary on the displayed domains; put $\deg(d, A) = d$ and $\iota(d, A) = \iota_d A$. The conditions below constrain the correlations at separated positions. In

Q2–Q6 all cited definitions and formulas are read with these hierarchy data and with the conventions of Section 8.2; in definitions indexed by a phase, read Ω_{ξ_*}, Φ_{ξ_*} and $\mathsf{F}_n^{\xi_*}$ as the Ω, Φ and F_n of these data.

Definition 8.2 (Q2. Wightman axioms)**.** The relativistic datum of Q0 is a Wightman theory in the sense of Definition 2.14.

Definition 8.3 (Q3. Labels and phases)**.** (a) Every nonzero $\ell \in \mathcal{L}$ has $\Phi(\ell, f) \neq 0$ for some $f \in \mathcal{S}$. (b) Each ι_d is injective, their images cover $\mathcal{L}$, and π_{quo} is onto. (c) $\Phi(\iota(A), f) \neq 0$ for some $A \in \mathbb{F}$ and $f \in \mathcal{S}$. (d) The measure ν is a probability measure of full support, and for every n, $\boldsymbol{A} \in \mathbb{F}^n$ and $x \in \mathrm{Conf}_n$, $\xi \mapsto S_n^{\xi}(\boldsymbol{A}; x)$ is continuous and $S_n^{\mathrm{mix}}(\boldsymbol{A}; x) = \int_{\mathsf{X}} S_n^{\xi}(\boldsymbol{A}; x)\, d\nu(\xi)$. (e) Each S_n^{ζ} is simultaneously permutation symmetric and smooth on Conf_n.

Definition 8.4 (Q4. Selected continuation)**.** The functions F_n satisfy the ordered-time continuation condition of Definition 2.16 and the boundary condition of Definition 2.17 at ξ_*, read as in Remark 8.9, using the relativistic datum and the maps ι_d of the hierarchy data.

Definition 8.5 (Q5. Products)**.** For the channels, fields, coefficients and tables S^{ζ} of Q0, form $\mathcal{R}_D^{\zeta}$ by Definition 2.19. These coefficients and remainders satisfy the two inequalities of Theorem 6.20: the coefficient threshold depends on $I_{\mathrm{cl}}, \boldsymbol{A}, M_{\mathrm{geo}}, \eta, \varepsilon, j, \lambda$; the remainder threshold depends on $I_{\mathrm{cl}}, I_{\mathrm{sp}}, \boldsymbol{A}, \boldsymbol{B}, M_{\mathrm{geo}}, \eta, \delta_{\mathrm{sp}}, \varepsilon, D$. Each threshold is uniform in positions and state indices.

Definition 8.6 (Q6. Trace test)**.** For the chosen $\Theta \in F_4$, with G_Θ its time-axis two-point function in the phase ξ_* (Section 2.7) and $A_G = 3d_{\mathfrak{g}}/\pi^4$, require $\langle \Omega, \Phi(\iota_4\Theta, f)\Omega\rangle = 0$ for every $f \in \mathcal{S}$,

$$\lim_{r\downarrow 0} r^8 \log^2(1/r) G_\Theta(r) = A_G > 0 \quad \text{in } \mathbb{C}\,,$$

and the finite-profile exclusion of Proposition 6.19.

For the lattice limit, $S^{\mathrm{mix}} \circ \pi_{\mathrm{quo}}$ denotes the kernels $K_n(v; x) = S_n^{\mathrm{mix}}(\pi_{\mathrm{quo}}(v_1), \dots, \pi_{\mathrm{quo}}(v_n); x)$.

Theorem A (Existence)**.** *For every gauge group of Definition 2.1, there is a regular* (Q1) *Wilson family $\mathcal{W}$ such that, for every strictly increasing $\mathsf{p} : \mathbb{N} \to \mathbb{N}$, there are a strictly increasing q and hierarchy data over $\mathcal{W}$* (Q0) *satisfying* Q2–Q6*, with*

$$\mathcal{W} \longrightarrow S^{\mathrm{mix}} \circ \pi_{\mathrm{quo}} \quad \textit{along } P = \mathsf{p} \circ \mathsf{q} \quad \textit{in the sense of Definition 2.6}.$$

Theorem A fixes the lattice theory before the volume sequence. Q2 gives relativistic fields; Q3 and Q4 express the limiting correlations as a phase average and continue the selected phase to those fields; Q5 approximates local products uniformly in the state and in separated spectators; Q6 fixes the vacuum centering and the short-distance law of the distinguished degree-four field.

Deduction of Theorem A. 10. Steps 1–9 of Section 7 give Theorem B and hierarchy data satisfying Q2–Q6 on the same family and volume refinement. Fix any $\mathsf{b} > 0$, for instance $\mathsf{b} = 1$. The regular family precedes p; its refinement and hierarchy data follow it, with the ordered convergence of Step 7. This proves Theorem A. □

As in the formal statement, Q0–Q3 do not require S_n^{ζ} to be multilinear in the fields; in the construction they are, since S and every S^{ξ} are tables (Theorems 6.1 and 6.13). Theorem A leaves $b_* > 0$ unspecified; Theorem 5.6 supplies $b_* = \kappa_G = 2b_0 = 11C_{\mathfrak{g}}/(24\pi^2)$.

Remark 8.7 (Nontriviality). Q3(c) requires a nonzero smeared field; given Q3(a)–(b) this is equivalent to $\mathcal{L} \neq 0$, and in the construction it is met by the unit field (Step 7). Nontriviality comes from the trace: $\Theta \neq 0$ (Theorem 6.18), so faithfulness gives $\Phi(\iota_4\Theta, f) \neq 0$ for some f, while its vacuum expectation vanishes; hence it is not a multiple of the identity. Q6 gives the positive limit $3d_{\mathfrak{g}}/\pi^4$ of $r^8 \log^2(1/r) G_\Theta(r)$, and on no interval $(0, \delta)$ does G_Θ agree with a finite Wick profile. Thus every Gaussian or finite free-field realization whose two-point profile for Θ belongs to the class of Definition 2.18 is excluded.

8.2 Formal conventions

The formal statement differs from the main text in the following conventions; none of them changes the content of the results.

Remark 8.8 (Abstract Poincaré group). In the formal statement a relativistic datum carries a set $\mathcal{P}'$ with a unit, a product and a topology, and a homeomorphism $\kappa_{\mathrm{P}} : \mathcal{P}' \to SO^+(1,3) \ltimes \mathbb{R}^4$ respecting the identity and the products (which makes $\mathcal{P}'$ a topological group), a strongly continuous unitary representation U of $\mathcal{P}'$ with $\mathsf{U}(\kappa_{\mathrm{P}}^{-1}(1,z)) = \mathsf{T}(z)$, label maps ρ_{lab} on $\mathcal{P}'$, and the action on tests through κ_{P}. Transport along κ_{P} turns such a datum into one for the ordinary group of Definition 2.13, and conversely, and it preserves W0–W3. Definition 2.14 is therefore equivalent to the formal one.

Remark 8.9 (Nonintegrable integrands). In the boundary condition of Q4 the formal statement does not require integrability: there a nonintegrable function has integral zero. For the continuations constructed in Theorem 6.16 the integrands are integrable for every $r > 0$ (by the bound (17.8) and rapid decay of the tests), so this convention is never used for them, and Definition 2.17 requires integrability.

The formal development also states the volume limit as local uniform convergence, on Gen_n, of the fixed-volume kernels $M^{\mathrm{UV}}_{P(k),v}$ restricted to the periodic generic domains $\{x : x_i^a - x_j^a \notin L_{P(k)}\mathbb{Z} \ (i \neq j,\ a = 1,\dots,4)\}$. This form follows from (15.1): a compact subset of Gen_n has bounded diameter and positive minimal coordinate separation, so it eventually lies in the periodic generic domain, and $\mathrm{Gen}_n \subset \mathrm{Conf}_n$.

8.3 An averaged countertest

The averaged Fock countertest of Section 18 belongs to this correspondence: it compares the trace field with free comparison vectors modified by arbitrary maps that preserve physical inner products and fix $U\mathcal{H}$. Its integrands are constant, so the convention of Remark 8.9 is not needed for them.

Let $n \geq 0$ and let U be a comparison (Definition 18.14), with $\mathcal{F}_n$ as in Definition 18.13. For a probability space (Z, λ), let each $P_z : \mathcal{F}_n \to \mathcal{F}_n$ preserve physical inner products and fix $U\mathcal{H}$ pointwise, without assuming linearity. The maps P_z model transformations of the free comparison vectors that fix every embedded interacting state in $U\mathcal{H}$. Because physical inner products are preserved, neither matrix element in Lemma 18.15 changes, so the averaged inequality states that averaging these transformations cannot restore the relative-locality identity.

Lemma 8.10 (Localized averaged countertest). *For every such $n, U, (Z,\lambda), (P_z)$ and $r > 0$, there are open A, B, with $\overline{A} \subset B(0,r)$ and $\mathsf{m}(x-y) < 0$ on $\overline{A} \times \overline{B}$, and tests f, h as in Lemma 18.15, supported respectively in A, B, such that*

$$\int_Z \langle P_z \mathcal{E}(-h), \psi_U(f)\rangle \, d\lambda(z) \neq \int_Z \langle \psi_U(\bar{f}), P_z \mathcal{E}(h)\rangle \, d\lambda(z) \,.$$

Proof. For physical v and $w \in U\mathcal{H}$ (Definition 18.14), $\langle P_z v, w\rangle = \langle P_z v, P_z w\rangle = \langle v, w\rangle$. Thus averaging preserves both integrands.

Apply Lemma 18.15 in $B(0, r/2)$. Compactness and strict spacelike separation give support neighbourhoods A, B with spacelike separated closures and $\overline{A} \subset B(0, r)$. □

9 Constants and order of choice

The choices come in four groups. First fix the gauge group, the block penalty, the local sections and, for each stage, the constants that do not involve the block ratio. Next choose the block ratios, and then delay the entry into each stage until the coupling and the period satisfy its remaining inequalities. These choices determine the lattice family, with its couplings and insertions, before any volume sequence is prescribed; only afterwards are the volume refinement and the continuum objects chosen. Table 3 records the individual requirements in this order. At each stage there are finitely many inequalities; a later coupling threshold or entry level may be increased without changing the earlier data.

Table 3: Choices in the order of dependence. Stage choices precede entry levels at each fixed $j \geq 4$.

	Choice	Dependence, requirements and purpose
1	Gauge normalization; $\mathsf{b} > 0$	Fix G, its realization, probability Haar measure and the metric, then $d_{\mathfrak{g}}, C_{\mathfrak{g}}, b_0, \kappa_G$ as in Definition 2.1; $\kappa_G > 0$. All later choices use these data.
2	Frame and predictor cutoffs; axial factor	Depend on G. The cutoffs select the unique-minimizer branches of the block kernel and are small enough for these branches to be valid on every patch (Lemma 11.4). On a radius-R patch, the total link chord in an eligible axial frame is at most $a_{\mathrm{ax}}(2R+1)$ times the total plaquette chord, with $a_{\mathrm{ax}} \geq 16$.
3	Bases, norms, local sections, stencils	Depend on G, d; Taylor normalization, reality and H_4 covariance (Lemma 10.2, Definitions 2.7–2.8).
4	Stage constants	Depend on items 1–3 and j. Derivative orders $\mathfrak{q}_j \geq 6$, spatial powers $\varpi_j \geq 2$, support exponents $\sigma_j = 100$; the path weight $w_j^{\mathrm{path}} > 0$ and the bound $A_j^{\mathrm{path}} > 0$ on the weighted row, cubic and quartic sums of Lemma 12.3. Fix the whole nondecreasing sequence $K_j \geq 1$ bounding the leading coefficient maps and their inverses, including the finite symmetry averages (before Lemma 14.2); then nondecreasing column bounds $C_j^{\mathrm{col}} \geq 64K_{j+1}^2$, including derivative factorials and basis-recombination constants (item 1 there), which bound the marked coefficient transfer (Lemma 14.7). With $\gamma_j > 0$ half a uniform lower bound for the free effective plaquette forms of Lemma 11.5, for the top form and both incoming stages, choose clipping coefficients $0 < 16c_{\mathrm{lin},j} < c_{\mathrm{dir},j} \leq \min(\gamma_j, 1)$, so that the remaining free form has the positive row factor of Definition 12.2; the threshold of the relaxed threshold test grows through $e^{\mathrm{dist}/R_j^{\mathrm{rel}}}$ with $R_j^{\mathrm{rel}} w_j^{\mathrm{path}} \geq 12$. With $d_j \geq 0$ the ratio-independent one-loop bound of Lemma 12.9, fix the drift bound $\omega_j^{\mathrm{RG}} \geq d_j + 1$ (Lemma 13.14). Orders, powers and support exponents are nondecreasing along levels, and orders and powers tend to infinity. All costs that may not depend on the ratio are now fixed.

Table 3 (continued)

	Choice	Dependence, requirements and purpose
5	Odd stage ratios ℓ_j^{st}	Depend on items 1–4; nondecreasing, $\ell_j^{\mathrm{st}} \geq 3$ and above every ratio threshold, including u_j of Lemma 11.9. At $j = J_{m+1}$ and $\ell_m = \ell_j^{\mathrm{st}}$, $$\frac{\omega_j^{\mathrm{RG}}}{\log \ell_m} \leq \min(\kappa_G/4, j^{-2}),$$ $$\frac{\log(1 + A_j^{\mathrm{path}} + (w_j^{\mathrm{path}})^{-1} + C_j^{\mathrm{col}} + R_j^{\mathrm{rel}} + \sigma_j)}{\log \ell_m} \leq j^{-2},$$ which control the drift and the logarithmic stage costs relative to $\log \ell_m$ (Theorems 5.5 and 5.6). Also the graph and matching requirements (14.8)–(14.9), the local marked error requirement of item 2 before Lemma 14.2, the ratio inequalities (19.2) and the source comparison constants through (14.11); among them $\ell^2 \geq C_{j,k}^{\mathrm{tr}} \zeta_j$ (Lemma 11.5) and $C_j^{\mathrm{pb}} \ell^{-2} \leq 1/256$ (proof of Lemma 13.22).
6	Exponents and coupling entry bounds	Depend on items 1–5. The positive antitone exponents θ_j control the quadratic loop bound of item 9. The bounds β_j^{ent} are lower bounds on the reference coupling whenever stage j is used. They meet the finite maximum of the remaining stage requirements, including those of Sections 16.3, 18.1 and 19, the comparison entry bounds (14.10) for each arity admitted at that stage, and the insertion-initialization bounds of item 6. They are chosen so that, at every level from stage entry on, $\ell_m \leq \log \widehat{\beta}_m$ (item 7 before Lemma 14.2) and $\max(\ell_{J_m}^{\mathrm{st}}, \ell_{J_m-1}^{\mathrm{st}}) \leq M_m^{\mathrm{cut}}$, with both adjacent stage ratios at stage transitions and the finite coefficient-error thresholds of Lemma 18.7 (used in Section 18.1 and (14.44)).
7	$\bar{\beta}$, entry levels m_j^{ent}, minimum periods n_j^{min}	Depend on items 1–6. The levels m_j^{ent} delay the first use of stage j; n_j^{min} are its minimum lattice periods. Choose $\bar{\beta} > 1$ large enough for the initial coupling requirements, including $\bar{\beta} \geq 2^{224}$, so that $\chi_m \leq 1/128$ in (13.26), and $t_m \leq 1$ at every level. Entry levels and periods satisfy the remaining level and geometric requirements, including the scale restrictions (14.3) and the comparison intervals (14.4). For each incoming stage $j = J_{m+1}$, fix integers $v_j^{\mathrm{per}}, h_j^{\mathrm{per}} \geq 1$, depending only on the stage data, and require at every step and every p $$L_p D_m \geq v_j^{\mathrm{per}} (1 + B_m + B_{m+1})^{h_j^{\mathrm{per}}},$$ which postponing stage entry achieves. The marked construction uses Lemma 19.7, with weight requirements (19.6) and entry requirements (19.7); also the support-enlargement and occupancy requirements in the proof of Lemma 19.8 and the regular marked weight inequalities before Lemma 19.7.
8	$J_m, \ell_m, D_m, a_m, L_p, m_{\mathrm{ref}}(d)$	Depend on items 1–7. $D_0 = 1$, $D_{m+1} = \ell_m D_m$, $a_m = D_m^{-1}$, $s_m = \log D_m$, $\ell_m = \ell_{J_{m+1}}^{\mathrm{st}}$, $4 \leq J_m \leq J_{m+1} \leq J_m + 1$, $J_m \to \infty$, $0 < \theta_{J_m} \leq J_m^{-1}$, $\log \ell_m \to \infty$, $\log \ell_m / s_m \to 0$. Strictly increasing L_p with $4 \mid L_p$ and every positive integer dividing infinitely many L_p; $L_p D_m \geq 100(B_m + 1)$ and $L_p D_{m+1} \geq 100((J_{m+1} + 3)\ell_m + 1)$. $J_{m_{\mathrm{ref}}(d)} \geq d + 3$, with $\mathfrak{q}_{J_m} \geq d + 1$ and $\varpi_{J_m} \geq d$ for $m \geq m_{\mathrm{ref}}(d)$.
9	Derived level scales	Determined by earlier choices: equation (5.2), $\mathfrak{B}_m = [\widehat{\beta}_m/2, 2\widehat{\beta}_m]$ and $\Lambda_m^{\mathrm{loop}} = 2(M_m^{\mathrm{cut}})^{\theta_{J_m}}$, which bounds the weighted quadratic loop sum of Lemma 12.3.

Table 3 (continued)

	Choice	Dependence, requirements and purpose
10	Actions, classes, recursion, trajectories, couplings; width w_{tun}	Depend on items 1–9: the path actions, the admissible classes, the specified recursion and its trajectories for every finite N, p, with the common endpoint (Theorem 5.6). Proposition 5.7 gives the fixed-level convergence; Remark 7.2 records the analytic dependence on the endpoint, not used later.
11	Marked transfers, terminal maps, insertions	Depend on items 1–10; reference normalization (Definition 2.8). Fixed before any volume sequence.
12	p, then $\mathsf{q}, P, S, F_d, \partial_a, \mathcal{L}$	An arbitrary strictly increasing p is prescribed first. Theorem 6.1 and Proposition 6.4 give $P = \mathsf{p} \circ \mathsf{q}$ and the continuum fields; Theorem 5.9 keeps the full inner limit.
13	Comparison radii, frames, J, charges, $c_\star, R_d, T$	Depend on item 12 and the lattice family: Proposition 6.7, then Theorem 6.10 and Proposition 6.11. Only comparison radii are refined.
14	Slab operators, X, ν, S^ξ, then ξ_*	Theorem 6.13; ξ_* is chosen in the full-measure set with one-dimensional time-axis reflection ground space (Lemma 17.3), before reconstruction.
15	Theories and continuations in every phase; Θ, μ_Θ	Theorem 6.16; Definition 6.17 centers the trace in the already selected phase; Theorem 6.18 and Lemma 18.2.
16	Product prescription; contact projections, distributions, bases, scales	Depend on items 12–14 and the lattice family. Theorems 6.20 and 6.21 use the same original fields and correlation table; the contact projections, separating functionals and clearance are chosen separately from those of the product coefficients.
17	Estimate constants and thresholds	Depend only on the data preceding them in each statement, uniformly in positions, volumes and states wherever asserted.

Index of notation

Two dictionaries. Geometric objects: a *site* is a point of Λ_s; a *plaquette* is an elementary square, with holonomy $U_{y;ab}$; a *path* is a finite chain of links along which curvatures are transported; the *stencil* $y + \Sigma_v$ is the finite site set an insertion depends on; the *support* X_i or Y_i is the set on which an activity is local; the *anchor* $o_i \in X_i$ is the site at which a regular activity is counted in its weighted sum; the *complete read support* consists of all links and flags a local term depends on, including those inspected by its tests; a *cover* is an exact cover $\mathfrak{a} \in \mathfrak{F}_m(U)$ of the flagged sites by disjoint forced supports; the *inventory* I_i is the set of flagged sites in a forced support; a *cluster* is a connected graph of polymers in a logarithmic expansion, or a group of nearby insertions in the product expansion. Field objects: a *label* is $c \in V_d$ and an *insertion label* is $(d, c) \in \mathcal{T}$; the *coefficient vector* $\mathbf{x}_m(\mathbf{v})$ is the label of a blocked insertion; the *lattice insertion* is $I_{N,p}(v, y; U)$; the *quotient field* is $[c] \in F_d$; its *response* is $\iota_d[c]$, its correlations against all spectators; a *field label* is $\ell \in \mathcal{L}$.

Conventions and gauge group

e_a coordinate vectors, time $a = 4$; Section 1.4.

m, $\overline{V}_+$ Minkowski square, closed forward cone; Section 1.4.

H_4, h signed permutation matrices; Section 1.4.

$\mathsf{s}_a^{\mathrm{refl}}$ reflection in coordinate a; Section 1.4.

G, $\mathfrak{g}$, n_G gauge group, Lie algebra; Definition 2.1.

$d_{\mathfrak{g}}$, $C_{\mathfrak{g}}$, b_0, κ_G dimension, adjoint Casimir, one-loop coefficient, slope; Definition 2.1.

pr, ch plaquette energy, normalized chord; equation (2.1).

Lattice theory and families

Λ_s periodic lattice; equation (1.1).

$U_{y;ab}$, E_s^{W}, $\mu_{s,\beta}$, $Z_{s,\beta}$ plaquette holonomy, Wilson energy, measure, normalization; Definition 2.3.

$\mathcal{W}$, D_N, L_p, $\beta_{N,p}$ Wilson family, cutoffs, periods, bare couplings; Section 2.2.

a_N, $\mu_{N,p}$, $\hat{y}$ spacing, measure, physical placement; equation (2.2).
$M^{\mathrm{lat}}_{N,p}$, $C^{\mathrm{lat}}_{N,p}$ lattice moment, smeared correlation; Definition 2.4.
Σ_v, b_* stencil, slope of a regular family; Definition 2.5.
Conf_n, sep_∞ distinct configurations, minimum separation; Section 2.2.
C^{UV}_p fixed-volume smeared limit; Definition 2.6.

Local fields

$\mathcal{J}$, V_d, $b_{d,i}$ radial jets, polynomial labels, bases; Definition 2.7.
N, t^{N} factor-count operator and scaling; Section 2.3.
$U(h_{\mathrm{lat}}, \mathsf{j})$, $\mathrm{sc}_{h_{\mathrm{lat}}}$ test holonomies, jet dilation; equation (2.3).
Tay_d Taylor map; Section 2.3.
P^{loc}_d, Σ_d local section, its stencil; equation (2.4).
$\mathcal{T}$ insertion labels; Section 2.3.
c_{F^2}, P^{cl} curvature-square label, clover reading; Section 2.3.
$I_{N,p}$, $\mathbf{u}_d$ renormalized insertion, unit coefficient; Definition 2.8.
$m_{\mathrm{ref}}(d)$, $\mathsf{Z}_{N,p,d}$ reference level, terminal map; Section 5.5.
$H^{\mathrm{ins}}_{N,p,m;d}$, $\mathbf{v}$ marked density, fine-cutoff coefficient vector; equation (5.15).
$\mathbf{x}_m$, $R^{\mathrm{reg}}_{m,i}$, $Q^{\mathrm{for}}_{m,i}$ coefficient vectors and remainders; equations (5.17)–(5.18).

Scales and blocking

ℓ_m, D_m, a_m, s_m block ratio, levels; equation (5.1).
J_m, ℓ^{st}_j stage, stage ratio; Sections 4.1 and 5.1.
$\bar{\beta}$ endpoint coupling; Section 5.1.
$\widehat{\beta}_m$, M^{cut}_m, t_m reference coupling, field multiplier, small-field threshold; equation (5.2).
B_m, $\epsilon^{\mathrm{reg}}_m$, $\epsilon^{\mathrm{for}}_m$ support radius, allowances (bounds kept by the induction); equation (5.3).
$\mathfrak{B}_m$, $\mathfrak{q}_j$, ϖ_j, σ_j, ω^{RG}_j coupling band and stage constants; Section 5.1.
b, q_γ block penalty, coarse-link density; Section 5.1.
$Y_{e,z}$, Q^{pred}_e, w^{av}_z framed connector, predictor, weights; equations (5.4)–(5.5).
λ^{tag} tag masses; equation (5.6).
$k^{(m)}_\beta$, $\mathcal{B}^{(m)}_\beta$ block kernel, exact block map; Definition 5.1.

Densities and classes

$\mathrm{Flag}_m(U)$ flagged (large-field) sites; Section 5.2.
E^{reg}_i, K^{for}_i, I_i regular and forced activities, inventory; equation (5.8).
clip_t, $\chi^{\mathrm{rel}}_{m,x}$ clip, relaxed threshold test; equation (5.9).
$P^{\mathrm{path}}_{N,m}$, $\mathcal{L}^{\mathrm{path}}_{\mathsf{p}}$ path action, smeared curvature; equation (5.10).
ρ, v_{sc} representation, scalar normalization; equation (5.11).
$\mathfrak{F}_m(U)$ exact covers of the flags; Section 5.2.
$\mathcal{V}_\rho$, $\mathrm{dens}\,\rho$ action, effective density; Definition 5.2.
$\mathrm{dens}^\circ\,\rho$ effective density with v_{sc} replaced by zero; Section 4.2.
$\|\cdot\|_m$, D^L local derivative seminorm, left derivative; Definition 5.3.
cov_{B_m}, $\mathcal{E}_m$, $\mathcal{K}_m$ ball-cover count, weighted sums; Definition 5.4.
ρ^{W}_β Wilson start; equation (5.13).
$\mathcal{A}_{N,m,p}$, $\mathcal{I}_{N,p}$, $\mathcal{R}_{N,m,p}$ admissible classes, initial classes, adjacent pairs; Section 5.2.
$\beta_{N,p,m}$, $\mathfrak{T}_m$ trajectory couplings, shooting intervals; Section 5.4.
λ_i, G_x five-torus weights, localized term; equation (4.1).

Limits and fields

$\Delta_{\boldsymbol{\iota}}$, $\partial_{\boldsymbol{\iota}}$ differences, derivatives; Section 5.6.
Gen_n, $\mathrm{NoHit}_{L,n}$ generic and periodically separated configurations; Section 5.6.
$M^{\mathrm{UV}}_{p,v}$ fixed-volume kernel; Theorem 5.9.
$P = \mathsf{p} \circ \mathsf{q}$, S retained volume indices, table; Theorem 6.1.
θ_a reflection of words; Section 2.4.
$\mathcal{N}_d(S)$, F_d, ι_d, $\mathcal{L}$ null space, fields, responses, field labels; Definition 2.12.
$\mathbb{F}$, deg, $\mathbf{1}_d$ degree-tagged fields, unit field; Section 2.5.
$\jmath_{d,e}$, ∂_a inclusion, field derivative; Proposition 6.4.
M^{Max}_d, S^{Max}, Γ Maxwell fields and correlations, free covariance; Definition 6.3.
$\mathcal{H}^{\mathrm{OS}}_a$, H^{OS}_a, Ω^{OS}_a, P^0_a coordinate reflection space, generator, vacuum, ground projection; Definition 6.5.

Stress and rotations

$\mathsf{U}^{\mathrm{sc}}_{k,d}$, r_k scaling frames, radii; Definition 6.6.
$\mathcal{P}^J_b$, $\mathcal{Y}^J_K$, χ shell fluxes, shell cutoff; Definition 6.8.
p_b, Y_K, R_d, L_d, $c_\star$ charges, rotation representations, generators, constant; Section 6.2.
T_{ab} stress tensor; Theorem 6.10.
T^{Max} free stress tensor; after Theorem 6.10.

Phases and relativistic fields

X, ν, S^ξ, ξ_* phases, measure, phase tables, selected phase; Sections 2.5 and 6.3.
$\mathsf{B}_{h_{\mathrm{sl}}}(w)$, $M^0(w)$, $\mathfrak{A}_0$ slab insertion, compression, compression algebra; Definition 6.12.
$\mathcal{S}_n$, $\mathcal{P}$ Schwartz spaces, Poincaré group; Section 2.6.
$\mathcal{H}$, $\mathcal{D}$, Ω, Φ, T, E, U, ρ_{lab} relativistic datum; Definition 2.13.
F^ξ_n ordered-time continuation; Definition 2.16.
$\mathsf{J}^{\mathrm{OS}}_\xi$, H_ξ comparison map, energy; before Theorem 6.16.
π_{quo} label projection; Theorem B.

Trace and products

Θ, G_Θ, A_G centered trace, its two-point function, amplitude; Definition 6.17.
$\mathcal{O}_{F^2}$, d_2, μ_Θ, $\widehat{f}_+$ curvature square, trace coefficient, spectral measure, Fourier transform; after Theorem 6.18.
W_{Wick} finite Wick profile; Definition 2.18.
ζ, Ch_j, $O_{j,\lambda}$, $c_{j,\lambda}$ state index, channels, fields, coefficients; Section 2.7.

$\mathcal{R}_D^{\zeta}$, $|\boldsymbol{A}|_{\deg}$ product remainder, cluster degree; Definition 2.19.

$\mathfrak{D}_{\boldsymbol{A}}^{\zeta}$ contact extension; Theorem 6.21.

Selected notation of Part II

$\mathcal{J}_j^{\mathrm{Max}}$, $\mathcal{I}_d$ on-shell jets, equation labels; Section 10.4.

$\mathbb{B}_r(y)$, F_{ext} closed ball of sites, exterior spectator; Section 10.4.

$\lambda_{N,p}$, $\mu_{N,p}$ terminal-map coefficients of the curvature square; Sections 2.3 and 10.5.

W_0 free interpolation; Section 11.2.

ψ, W_{ax}, $\|\cdot\|_{\mathrm{bl}}$ fine axial variable, axial tangent, block norm; Section 11.3.

C_μ, $\mathsf{C}_\mu^{\mathrm{rem}}$, $\mathsf{A}_\mu^{\mathrm{aux}}$, G_μ gauge-completed Hessian, the same without contact term, auxiliary fine operator, ghost operator; Section 11.4.

α_b curvature-square coefficient along transverse plane waves; Section 11.4.

upper rate of a quantity; Section 11.6.

$Q_{ab,ij}$, Q_{ab}, Q_{tot}, $Z^{(i)}$ free curvature quadratics, tensor basis; equation (16.2), Lemma 16.1.

c_k stress normalizers; Section 16.3.

$\mathfrak{B}$, $\mathcal{V}_n$ Borchers algebra, spectral support set; Section 17.2.

R_5, V_5 raw and covariant degree-five vector spaces; Section 18.1.

Conditions and constants

E1–E7, W0–W3, Q1 Definitions 2.9–2.11, 2.14, 2.5; Q0, Q2–Q6 Section 8; stage constants K_j, C_j^{col}, $c_{\mathrm{dir},j}$, $c_{\mathrm{lin},j}$, R_j^{rel}, β_j^{ent} Table 3.

Part II
Proofs

10 Local polynomial fields

The polynomial labels of Definition 2.7 prescribe continuum Taylor coefficients, whereas the Wilson measure integrates functions of group-valued links. Local sections connect the two: polynomials in the real and imaginary parts of finitely many link entries whose expansion on smooth test configurations reproduces a prescribed label. They initialize the renormalized insertions of Definition 2.8, hence the Wilson family of Theorem B.

To build them we recover the connection jet from finitely many transported plaquette curvatures, whose linearizations are unit-square circulations (Section 10.2); each curvature factor costs one power of the small-field threshold (Section 10.3). For the equation projections of Section 14, Section 10.4 constructs auxiliary sections whose regraded equation labels $\mathsf{R}_d(\mathcal{I}_d)$ have exactly vanishing moments against distant observables. Section 10.5 follows the curvature square through these constructions.

The lemmas of this section use only Part I; the idea paragraph and Section 10.5 also cite later results for illustration. The gauge metric and the Wilson measure of Definitions 2.1–2.3 fix the measurements and the one-link integration by parts, Definition 2.7 supplies the labels, and the derivative seminorm of Definition 5.3 measures the bounds. The sections are chosen once for each degree, and all constants of this section are fixed before the lattice volume and the stage block ratios, in the order of Section 9.

10.1 Labels, test holonomies and the Taylor map

Write $\mathcal{J}_j$ for the real space of the components $\mathcal{A}_j$ in Definition 2.7, the homogeneous radial potentials of degree $j+1$. Every label in V_d is a polynomial on the finite-dimensional space

$$\mathcal{J}_{\le d} = \bigoplus_{j+2\le d} \mathcal{J}_j \,.$$

Radial gauge and Euler's identity give $(j+2)\mathcal{A}_{j,a}(x) = \sum_b x^b F_{ba}^{(j)}(x)$, with $F^{(j)}$ the degree-j part of the curvature; so a coordinate of $\mathcal{A}_j$ has weight $j+2$, two for the curvature and one for each

derivative, and only finitely many monomials have weight at most d. Adjoint action and conjugation preserve both gradings. Real and imaginary parts span each invariant component over $\mathbb{C}$, giving bihomogeneous bases as in Definition 2.7. Conjugation in this real basis is coefficientwise and agrees with conjugation of polynomial values on real jets. Write k_i for the number of jet-coordinate factors in $b_{d,i}$. We identify a coefficient vector with its polynomial:

$$\mathbf{c}(\mathrm{j}) = \sum_i c_i b_{d,i}(\mathrm{j}), \qquad b_{d,0} = \mathbf{1}\,.$$

We evaluate lattice observables on the test holonomies (2.3), on the lattice $\mathbb{Z}^4$ covering the torus. With $\mathcal{A}^{\le d} = \sum_{j+2\le d} \mathcal{A}_j$, they are

$$U(h_{\mathrm{lat}}, \mathrm{j})(x, a) = u(1), \qquad u'(s) = u(s)\, h_{\mathrm{lat}} \mathcal{A}_a^{\le d}(h_{\mathrm{lat}}(x + s e_a)), \qquad u(0) = 1\,. \tag{10.1}$$

This linear equation with continuous $\mathfrak{g}$-valued coefficients has a unique global solution [32, Theorem 3.9]. It lies in the closed group G, being a uniform limit of products of exponentials of elements of $\mathfrak{g}$ by Gronwall's inequality [32, Lemma 2.7], and its Volterra series shows that it depends smoothly on $(h_{\mathrm{lat}}, \mathrm{j})$. Homogeneity and uniqueness give

$$U(h_{\mathrm{lat}}, \mathrm{j}) = U(1, \mathrm{sc}_{h_{\mathrm{lat}}} \mathrm{j}), \qquad \mathrm{sc}_{h_{\mathrm{lat}}} \mathrm{j} = (h_{\mathrm{lat}}^{j+2} \mathcal{A}_j)_j, \qquad U(0, \mathrm{j}) = 1\,. \tag{10.2}$$

The Taylor map of Section 2.3 reads off the continuum polynomial seen by a lattice observable. Recall that, for a smooth function O of finitely many cover links, $\mathrm{Tay}_d O$ is the Taylor polynomial at zero of $\mathrm{j} \mapsto O(U(1, \mathrm{j}))$ on $\mathcal{J}_{\le d}$, with the monomials of weight above d discarded. By (10.2), a monomial of weight w contributes h_{lat}^w to $O(U(h_{\mathrm{lat}}, \mathrm{j}))$. Hence the derivatives of $O(U(h_{\mathrm{lat}}, \mathrm{j}))$ at $h_{\mathrm{lat}} = 0$ through order d are those of $\mathbf{c}(\mathrm{sc}_{h_{\mathrm{lat}}} \mathrm{j})$ for every jet, where $\mathbf{c} \in V_d$, exactly when $\mathrm{Tay}_d O = \mathbf{c}$. A local section, constructed below, is a right inverse of the Taylor map, linear in the label: this is the sense of the word *section*.

For a cover plaquette $\square$ based at z and a path γ from 0 to z, write U_γ for the ordered product of successive oriented links, read from the initial point to the endpoint. Define the transported curvature (Figure 7) by

$$\sigma_{\gamma,\square}(U) = \mathrm{Ad}(U_\gamma) F_\square(U), \qquad F_\square(U) = \mathrm{proj}_{\mathfrak{g}} \frac{U_\square - U_\square^{-1}}{2}\,.$$

The projection is orthogonal for the normalized Frobenius metric of Definition 2.1. These measurements transform by the adjoint action at 0. They are real polynomial functions of the real and imaginary parts of link entries, since $U_e^{-1} = U_e^*$.

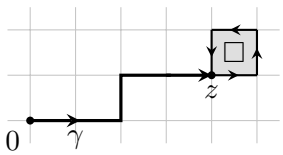


Figure 7: A transported curvature $\sigma_{\gamma,\square} = \mathrm{Ad}(U_\gamma) F_\square(U)$, drawn in one coordinate plane: the curvature of the plaquette $\square$ based at z is carried to the origin along the lattice path γ.

10.2 Taylor-normalized sections

Definition 10.1 (Local sections)**.** For a finite site stencil $\Sigma_d \subset \mathbb{Z}^4$, a local section has the form

$$P_d^{\mathrm{loc}}[\mathbf{c}](U) = \sum_{\nu,\mu} p_{\nu\mu}^{(d)}(\mathbf{c}) \prod_{e,\alpha,\delta} U_e(\alpha,\delta)^{\nu_{e\alpha\delta}} \overline{U_e(\alpha,\delta)}^{\mu_{e\alpha\delta}} .$$

The product uses positive links with both endpoints in Σ_d and matrix indices $1 \le \alpha, \delta \le n_G$. The sum is finite over nonnegative multi-indices; each coefficient is complex linear in $\mathbf{c}$. The construction below chooses such a map to represent each continuum label.

For a torus of side n_{tor} (the side s of Definition 2.3), write $[x]$ for reduction of $x \in \mathbb{Z}^4$ modulo n_{tor}. Placement at a torus site y means evaluation on the periodic extension $U^{[y]}(x,a) = U(y+[x],a)$; denote it by $P_d^{\mathrm{loc}}[\mathbf{c}](y;U)$. On the cover, $P_d^{\mathrm{loc}}[\mathbf{c}](0;U) = P_d^{\mathrm{loc}}[\mathbf{c}](U)$. For $h \in H_4$, define the configuration and jet actions by

$$(h^*U)(x,a) = \begin{cases} U(hx,b), & he_a = e_b, \\ U(hx-e_b,b)^{-1}, & he_a = -e_b, \end{cases}$$
$$(h\mathsf{j})_{j,a}(z) = \sum_b h_{ab}\mathcal{A}_{j,b}(h^{-1}z), \qquad (h\mathbf{c})(\mathsf{j}) = \mathbf{c}(h^{-1}\mathsf{j}) .$$

This action preserves homogeneity and radial gauge. Reality and covariance mean

$$P_d^{\mathrm{loc}}[\overline{\mathbf{c}}](U) = \overline{P_d^{\mathrm{loc}}[\mathbf{c}](U)}, \qquad P_d^{\mathrm{loc}}[\mathbf{c}](h^*U) = P_d^{\mathrm{loc}}[h\mathbf{c}](U) \quad (h \in H_4),$$

with conjugation as fixed above.

The following result supplies the sections specified in Section 2.3; the unit normalization is exact at every configuration.

Lemma 10.2 (Taylor-normalized sections)**.** *For every d, a box stencil $\Sigma_d = [-\varrho_d, \varrho_d]^4 \cap \mathbb{Z}^4$ and sections can be chosen gauge invariant, real and H_4 covariant, with $P_d^{\mathrm{loc}}[\mathbf{1}] = 1$, so that, for every $\mathbf{c} \in V_d$, radial jet j and $0 \le q \le d$,*

$$\frac{d^q}{dh_{\mathrm{lat}}^q} P_d^{\mathrm{loc}}[\mathbf{c}](U(h_{\mathrm{lat}},\mathsf{j}))\Big|_{h_{\mathrm{lat}}=0} = \frac{d^q}{dh_{\mathrm{lat}}^q} \mathbf{c}(\mathrm{sc}_{h_{\mathrm{lat}}}\mathsf{j})\Big|_{h_{\mathrm{lat}}=0} . \tag{10.3}$$

Equivalently, $\mathrm{Tay}_d P_d^{\mathrm{loc}}[\mathbf{c}] = \mathbf{c}$ for every $\mathbf{c} \in V_d$. The identity imposes no equality at positive spacing, and no compatibility between P_d^{loc} and P_{d+1}^{loc} is required. We fix the sections constructed in the following proof.

Proof. There are three steps: finitely many curvature measurements determine the truncated jet; an equivariant polynomial inverse of these measurements, composed with the label, gives a section with the right Taylor map; averaging over H_4 gives covariance and a common stencil. If $d < 2$, the labels are constant and their constant sections suffice. Fix $d \ge 2$.

Step 1: Curvatures determine the truncated jet. The measurements $\sigma_{\gamma,\square}(U(1,\mathsf{j}))$ are smooth in j, and all links equal 1 at $\mathsf{j} = 0$. The first variation of a link solution is the edge integral of the potential; that of a plaquette measurement is the circulation of the potential around the plaquette. Transport along γ contributes nothing at first order, because $F_\square(1) = 0$. By the fundamental theorem of calculus on each edge, the circulation around the unit square at x in the plane of the axes $a < b$ is

$$N_{ab}(x) = \int_0^1 \int_0^1 (\partial_a \mathcal{A}_b - \partial_b \mathcal{A}_a)(x + \xi e_a + \eta e_b)\, d\xi\, d\eta .$$

Suppose all these linearized measurements vanish at $\mathcal{A} = \sum_{j=0}^{j_*} \mathcal{A}_j$. Then the polynomial N_{ab} vanishes on $\mathbb{Z}^4$, hence identically: a polynomial vanishing on $\mathbb{Z}^4$ is zero, by induction on the number of variables. The averaging shifts only lower the degree, so the homogeneous part of N_{ab} of degree j_* is $(\partial_a \mathcal{A}_{j_*,b} - \partial_b \mathcal{A}_{j_*,a})(x)$. It vanishes, and antisymmetry covers the other ordered pairs. Differentiating the radial condition and using Euler's identity gives

$$(j_* + 2)\mathcal{A}_{j_*,b}(z) = \sum_a z^a (\partial_a \mathcal{A}_{j_*,b} - \partial_b \mathcal{A}_{j_*,a})(z) = 0\,.$$

Remove this component and descend in j_*: the common kernel is zero. Adding a measurement whenever the kernel is nonzero decreases its dimension, so finitely many measurements $\sigma_1, \dots, \sigma_{M_{\mathrm{meas}}}$ give a joint map

$$\mathsf{S} : \mathcal{J}_{\le d} \longrightarrow \mathfrak{g}^{M_{\mathrm{meas}}}, \qquad \mathsf{S}(\mathsf{j}) = (\sigma_l(U(1,\mathsf{j})))_{l=1}^{M_{\mathrm{meas}}}$$

with injective derivative $D\mathsf{S}(0)$.

Step 2: A polynomial inverse to the required order. By uniqueness of the link equation, a constant gauge transformation $v \in G$ takes the test links of j to those of $\mathrm{Ad}(v)\mathsf{j}$, so S is adjoint equivariant. The range of $D\mathsf{S}(0)$ is invariant; composing the inverse of $D\mathsf{S}(0)$ on it with orthogonal projection onto it gives an equivariant linear map $C : \mathfrak{g}^{M_{\mathrm{meas}}} \to \mathcal{J}_{\le d}$ with $CD\mathsf{S}(0) = I$. By the inverse function theorem [29, Theorem 9.24], $C\mathsf{S}$ has a local inverse φ near zero, smooth by differentiating $D\varphi = (D(C\mathsf{S}) \circ \varphi)^{-1}$. The adjoint actions are isometric, so φ can be taken on invariant balls, where injectivity of $C\mathsf{S}$ makes it equivariant. Put $f = \varphi \circ C$. Near zero, $f\mathsf{S}(\mathsf{j}) = \mathsf{j}$, $f(0) = 0$ and $f(\mathrm{Ad}(v)z) = \mathrm{Ad}(v)f(z)$ for $v \in G$. Consequently the Taylor polynomial $T_d f(z) = \sum_{k=1}^{d} D^k f(0)[z, \dots, z]/k!$ is adjoint equivariant and has no constant term. Define

$$Q_i(z) = b_{d,i}(T_d f(z)), \qquad \Psi_{\mathbf{c}}(U) = \sum_i c_i Q_i(\sigma_1(U), \dots, \sigma_{M_{\mathrm{meas}}}(U))\,.$$

The polynomials Q_i are invariant and real on real arguments, and $Q_0 = 1$. If $b_{d,i}$ has factor count k_i, every monomial of Q_i has at least k_i factors. Taylor's formula gives $T_d f(\mathsf{S}(\mathsf{j})) = \mathsf{j} + O(|\mathsf{j}|^{d+1})$ near zero, hence

$$\Psi_{\mathbf{c}}(U(1,\mathsf{j})) = \mathbf{c}(\mathsf{j}) + O_{\mathbf{c}}(|\mathsf{j}|^{d+1}), \qquad \mathrm{Tay}_d\, \Psi_{\mathbf{c}} = \mathbf{c}\,.$$

The polynomial Q_i itself need not have degree at most d in the curvature measurements.

Step 3: Covariance and a common stencil. Average the section by

$$P_d^{\mathrm{loc}}[\mathbf{c}](U) = \frac{1}{|H_4|} \sum_{h \in H_4} \Psi_{h^{-1}\mathbf{c}}(h^* U)\,. \tag{10.4}$$

Changing the summation variable to $h'h$ proves $P_d^{\mathrm{loc}}[\mathbf{c}](h'^* U) = P_d^{\mathrm{loc}}[h'\mathbf{c}](U)$. Simultaneous adjoint covariance of the measurements gives gauge invariance. The real basis and the signed-permutation action give reality; the constant column remains one. The action preserves factor count, so averaging retains the lower bound on the number of curvature factors.

For a positive image direction, the pulled-back link equation has the coefficient of $h^{-1}\mathsf{j}$, namely $h_{\mathrm{lat}} \sum_b h_{ba} \mathcal{A}_b^{\le d}(h_{\mathrm{lat}} h(x + s e_a))$. Reversing the parameter and inverting the transport covers a negative image direction. Uniqueness gives $h^* U(h_{\mathrm{lat}}, \mathsf{j}) = U(h_{\mathrm{lat}}, h^{-1}\mathsf{j})$. Each term of (10.4) therefore has Taylor map $\mathbf{c}$. A signed image of a transported curvature is again a transported curvature, up to sign: reversal changes $F_\square$ to $-F_\square$, and a change of base point gives conjugation by the intervening boundary transport. Enclose the finitely many resulting paths and plaquettes in a box $\Sigma_d = [-\varrho_d, \varrho_d]^4 \cap \mathbb{Z}^4$. This stencil contains zero and is signed-permutation invariant. Expanding the finite polynomial in matrix entries gives Definition 10.1. This proves (10.3), which is (2.4). □

10.3 Small-field bounds

We bound normalized initial insertions using the ordered left derivatives of Definition 5.3. At a placement y, set $Y = y + [\Sigma_d]$; being healthy on Y means that every plaquette meeting Y has chord at most t. A direction is permitted when

$$|\dot{\mathcal{A}}_e| \le t(1 + \operatorname{dist}(e^-, y))^{\varpi} .$$

Here $[\Sigma_d]$ is the torus image of the stencil, and $\operatorname{dist}(x, y) = \min_{z \in n_{\mathrm{tor}}\mathbb{Z}^4} |\tilde{x} - \tilde{y} + z|_\infty$ is the periodic maximum-coordinate distance, for any lifts $\tilde{x}, \tilde{y}$.

Lemma 10.3 (Section bounds)**.** *Each basis column is a polynomial in transported curvatures with at least k_i factors per monomial. For all $d, k, \varpi \in \mathbb{N}$ there are finite constants $C_{d,k,\varpi}$ and C_d such that, on every torus, for every placement y, $0 < t \le 1$, configuration healthy on Y, and permitted direction list of length $q \le k$,*

$$\left| D^L_{\dot{\mathcal{A}}_1} \cdots D^L_{\dot{\mathcal{A}}_q} P^{\mathrm{loc}}_d[b_{d,i}](y; U) \right| \le C_{d,k,\varpi} t^{k_i} . \tag{10.5}$$

For arbitrary configurations,

$$\sup_{i,y,U} |P^{\mathrm{loc}}_d[b_{d,i}](y; U)| \le C_d . \tag{10.6}$$

Each curvature factor is at most of the size t of the plaquettes it reads, so a column with k_i factors is of order t^{k_i} on healthy configurations. The proof uses the curvature polynomials constructed in Lemma 10.2.

Proof. First note that $|F_\square(U)| \le \mathrm{ch}(U_\square)$: orthogonal projection is contractive and $\|U_\square - U_\square^*\|_{\mathrm{F}} \le 2\|U_\square - 1\|_{\mathrm{F}}$. Adjoint transport is isometric, so an undifferentiated curvature measurement is bounded by t. An order-q left derivative of a fixed path product is a finite sum of products of unitary matrices and exactly q direction matrices. On the fixed stencil their norms are at most a constant times t, since periodic distance is bounded by cover distance. Thus each differentiated measurement is $O(t^q)$, and hence $O(t)$ when $q \ge 1$. This argument also applies when cover links coincide after placement: one first differentiates the cover expression and then assigns the same direction to coincident links.

Apply the product rule to the curvature polynomials of (10.4). Every differentiated monomial still has at least k_i factors, each bounded by a constant times t. There are finitely many terms; since $t \le 1$, their sum gives (10.5), including order zero. Each entry of a unitary matrix has absolute value at most one, so the finite matrix-entry polynomial gives (10.6). The formula also gives smoothness, measurability and locality. The constants depend on the sections, G, d, k, ϖ as applicable, and finite maxima make them nondecreasing in the integer parameters. They are fixed before the stage block ratios in Section 9. □

By linearity, Lemma 10.3 bounds permitted derivatives of $P^{\mathrm{loc}}_d[t^{-\mathsf{N}}\mathbf{c}]$ by $C_{d,k,\varpi} \sum_i |c_i|$ on the healthy set: the factor t^{-k_i} cancels t^{k_i}. Taking $t = t_N$, $k = \mathfrak{q}_{J_N}$ and $\varpi = \varpi_{J_N}$ bounds the initial marked density of (5.15) on the healthy set, in the sense of Definition 5.3. On unhealthy configurations the global bound gives $C_d \sum_i |c_i| t^{-k_i}$; the proof of Lemma 14.7 places this contribution in a merged forced remainder of (5.18).

10.4 Auxiliary equation sections

Exact lattice equation identities require different sections, still polynomial and local with the same small-field bounds. On $\mathcal{J}_j$ define the Maxwell subspace and the equation labels by

$$\mathcal{J}_j^{\mathrm{Max}} = \Big\{ \mathcal{A}_j : \sum_a \partial_a(\partial_a \mathcal{A}_{j,b} - \partial_b \mathcal{A}_{j,a}) = 0 \text{ for every } b \Big\}, \qquad \mathcal{I}_d = \Big\{ p \in V_d : p|_{\prod_j \mathcal{J}_j^{\mathrm{Max}}} = 0 \Big\}.$$

Here restriction uses only $j + 2 \le d$. Substitution into $\prod_j \mathcal{J}_j^{\mathrm{Max}}$ preserves bidegree, so $\mathcal{I}_d$ is the direct sum of its bihomogeneous parts. Order bidegrees (w, k) by the lexicographic order of (k, w). A triangular regrading (a unipotent change of labels) is a complex-linear map on V_d whose difference from the identity sends each bihomogeneous polynomial into strictly later bidegrees. There are finitely many bidegrees, so this difference is nilpotent and the map is invertible; the map and its inverse preserve the filtration by minimum factor count.

Idea: free equations in an interacting theory. Haar integration by parts in one link, a lattice Schwinger–Dyson identity, shows that for $\beta \neq 0$ the force $\mathcal{E}_e$ of (10.9), the derivative of the Wilson energy in the link e, has zero moments when multiplied by a function independent of U_e and by observables of distant links. Its linearization is the divergence of the linearized curvature, and the Maxwell equations enter only through this linearization. An equation label is a sum of products with one factor vanishing on Maxwell jets. Replacing that factor by forces and the others by curvature measurements gives an observable with exactly vanishing exterior moments, whose Taylor map is the label plus later terms, produced by the higher-weight linear terms of the lattice force and by nonlinear corrections. The labels with exact lattice equations therefore form $\mathsf{R}_d(\mathcal{I}_d)$, not $\mathcal{I}_d$: this is the lattice form of the vanishing of equation-of-motion operators at separated points. For example, put $F_{ab} = \partial_a \mathcal{A}_{0,b} - \partial_b \mathcal{A}_{0,a}$ and $(\partial \cdot F)_b = \sum_c \partial_c(\partial_c \mathcal{A}_{1,b} - \partial_b \mathcal{A}_{1,c})$, both constant. The label $q_{\mathrm{eq}} = \sum_b \langle (\partial \cdot F)_b, (\partial \cdot F)_b \rangle_{\mathfrak{g}}$ lies in $\mathcal{I}_6$ and has bidegree $(6, 2)$. The only later bidegree of weight at most six is $(6, 3)$, so $\mathsf{R}_6 q_{\mathrm{eq}} - q_{\mathrm{eq}}$ is a cubic invariant polynomial in the F_{ab}; nothing below requires it to vanish. The equation count is shared with the free theory, the separated correlations are not: Proposition 6.4 and (14.36) give $\dim \mathcal{N}_d(S) = \dim \mathcal{I}_d$, while in the selected phase G_Θ is asymptotic to $A_G r^{-8} \log^{-2}(1/r)$ (Theorem 6.18), and on no interval $(0, \delta)$ does G_Θ agree with a finite Wick profile (Proposition 6.19).

Lattice symmetries. We record the symmetries of the Wilson measure deferred from Definition 2.3. A site gauge transformation acts on plaquette holonomies by conjugation, and an affine isometry $h_{\mathrm{iso}}(x) = hx + b$ with $h \in H_4$ by conjugation, possibly after inversion; both preserve pr. Product probability Haar measure is invariant under fixed left and right multiplication, inversion and permutation [11, Corollary 2.28 and Proposition 2.31]. Hence, for every integer side $n_{\mathrm{tor}} \ge 1$, real β, site gauge transformation γ_{g} and bounded measurable F,

$$\int F(U^{\gamma_{\mathrm{g}}})\, d\mu_{n_{\mathrm{tor}},\beta}(U) = \int F(h_{\mathrm{iso}}^* U)\, d\mu_{n_{\mathrm{tor}},\beta}(U) = \int F(U)\, d\mu_{n_{\mathrm{tor}},\beta}(U). \tag{10.7}$$

For a torus site y, put $\mathbb{B}_r(y) = \{x : \mathrm{dist}(x, y) \le r\}$. An exterior spectator outside $\mathbb{B}_r(y)$ is a bounded measurable function of the positive links having both endpoints outside $\mathbb{B}_r(y)$; it need not be gauge invariant. The following identity supplies the auxiliary equation observables used in Section 14.

Lemma 10.4 (Auxiliary equation sections)**.** *For every d there are a local section V_d^{aux} as in Definition 10.1, an invertible triangular regrading R_d of V_d, and $r_d, M_d \in \mathbb{N}$ with the following properties.*

(i) *The section has its own box stencil, is gauge invariant, real and H_4 covariant, satisfies $V_d^{\mathrm{aux}}[\mathbf{1}] = 1$ and (10.3), and satisfies (10.5)–(10.6) with its own constants.*
(ii) *The map R_d fixes $\mathbf{1}$ and commutes with conjugation and H_4.*
(iii) *For every $n_{\mathrm{tor}} \geq M_d$, $\beta \neq 0$, placement y, $p \in \mathcal{I}_d$ and exterior spectator F_{ext} outside $\mathbb{B}_{r_d}(y)$,*

$$\int V_d^{\mathrm{aux}}[\mathsf{R}_d p](y;U) F_{\mathrm{ext}}(U)\, d\mu_{n_{\mathrm{tor}},\beta}(U) = 0\,. \tag{10.8}$$

The exact-equation subspace $\mathsf{R}_d(\mathcal{I}_d)$ is compared with the selected primary family in Section 14. All choices in the following proof depend only on G, d and the label basis.

Proof. There are three steps: one-link integration by parts gives zero moments; forces realize every linear functional vanishing on Maxwell jets; products of forces and curvature measurements, averaged, represent equation labels up to a triangular regrading, which is then inverted.

Step 1: One-link equations. Extend the curvature $F_\square$ of Section 10.1 to $F_{ab}(U;x)$ for ordered axes by antisymmetry, with $F_{aa} = 0$. For a positive link $e = (x,a)$ set

$$\mathcal{E}_e(U) = \sum_{b \neq a} \big(F_{ab}(U;x) - \mathrm{Ad}(U(x - e_b, b)^{-1}) F_{ab}(U; x - e_b) \big)\,.$$

On a torus of side at least two, direct differentiation gives

$$D^L_{e,A} E^{\mathrm{W}}(U) = \langle A, \mathcal{E}_e(U) \rangle_{\mathfrak{g}}\,. \tag{10.9}$$

Indeed, the forward plaquette contributes $-n_G^{-1} \operatorname{Re}\operatorname{tr}(A U_\square) = \langle A, F_\square \rangle_{\mathfrak{g}}$. The backward plaquette, rebased at x, contributes the negative of the transported curvature in the display. These are the two incidences for each transverse axis; antisymmetry gives the signs for the opposite ordering.

If a continuous multiplier g is independent of U_e, and a bounded measurable F_{ext} is also independent of U_e, left invariance of Haar measure gives

$$0 = \frac{d}{d\epsilon} \int g(U) F_{\mathrm{ext}}(U) e^{-\beta E^{\mathrm{W}}(U^{e,A,\epsilon})}\, dU \Big|_{\epsilon=0} = -\beta Z_{n_{\mathrm{tor}},\beta} \int \langle A, \mathcal{E}_e \rangle_{\mathfrak{g}} g F_{\mathrm{ext}}\, d\mu_{n_{\mathrm{tor}},\beta}\,.$$

Here $U_e^{e,A,\epsilon} = e^{\epsilon A} U_e$ and all other links are fixed. Compactness bounds the derivative of the smooth density uniformly, justifying differentiation even for measurable F_{ext}.

Choose a path γ_e from 0 to e^- avoiding e. If g also avoids e, then

$$\Upsilon_{e,\xi}(U) = \langle \xi, \mathrm{Ad}(U_{\gamma_e}) \mathcal{E}_e(U) \rangle_{\mathfrak{g}} g(U)$$

has zero moment for every $\xi \in \mathfrak{g}$ against every exterior spectator. To see this, expand $\mathrm{Ad}(U_{\gamma_e}^{-1})\xi$ in a fixed orthonormal basis of $\mathfrak{g}$ and apply the preceding identity to each coefficient times gF_{ext}. These multipliers are independent of U_e. There is one radius and one minimum torus side for any finite collection of such expressions: enlarge the radius to contain both endpoints of every varied link, and the side until distinct links in their finite cover supports remain distinct after wrapping. The same choices work at every placement.

Step 2: Enough measurements with avoidance. Any path can avoid one specified link: replace a traversal of that link by three sides of a square in a transverse direction. For each plaquette orientation only finitely many base points give a plaquette containing that link. Removing these points does not affect Step 1 of the proof of Lemma 10.2: if a polynomial vanishes at all but finitely many points of $\mathbb{Z}^4$, its restriction to each coordinate line through an integer point has infinitely

many zeros, so it vanishes on $\mathbb{Z}^4$ and hence identically. Thus curvature measurements whose paths and plaquettes avoid e still have injective joint linearization. Finite-dimensional duality therefore realizes every linear functional on $\mathcal{J}_{\le d}$ by finitely many components.

For every linear functional ℓ on $\mathcal{J}_j$ vanishing on $\mathcal{J}_j^{\mathrm{Max}}$, a finite combination of force linearizations equals $\mathsf{j} \mapsto \ell(\mathcal{A}_j)$ on $\mathcal{J}_{\le j+2} = \bigoplus_{i \le j} \mathcal{J}_i$. To see this, extend the circulations N_{ab} of Step 1 of the proof of Lemma 10.2 antisymmetrically. The force linearization at $e = (x, a)$ is

$$L_a(\mathsf{j}; x) = \sum_b \big(N_{ab}(x) - N_{ab}(x - e_b)\big) .$$

Suppose that every $L_a(\mathsf{j}; x)$, $x \in \mathbb{Z}^4$, vanishes for a jet $\mathcal{A} = \sum_{i \le j_*} \mathcal{A}_i$. Each $L_a(\mathsf{j}; \cdot)$ is a polynomial, hence zero. In $N_{ab}(x) - N_{ab}(x - e_b)$ a component $\mathcal{A}_i$ contributes a polynomial of degree at most $i - 1$, and $\mathcal{A}_{j_*}$ contributes $\partial_b(\partial_a \mathcal{A}_{j_*,b} - \partial_b \mathcal{A}_{j_*,a})(x)$ plus terms of lower degree. The part of degree $j_* - 1$ therefore gives $\sum_b \partial_b(\partial_a \mathcal{A}_{j_*,b} - \partial_b \mathcal{A}_{j_*,a}) = 0$, that is, $\mathcal{A}_{j_*} \in \mathcal{J}_{j_*}^{\mathrm{Max}}$. A functional of this top component annihilating $\mathcal{J}_{j_*}^{\mathrm{Max}}$ therefore annihilates the common kernel of the force measurements. Finite-dimensional duality puts it in their linear span. Transport paths do not enter the linearization and can be chosen to avoid their varied links. On $\mathcal{J}_{\le d}$ the same combination can have additional linear terms only at engineering weights above $j_* + 2$.

Step 3: Polynomial lifts and the triangular map. Choose linear projections $\pi_j : \mathcal{J}_j \to \mathcal{J}_j^{\mathrm{Max}}$ and put $\pi = \prod_j \pi_j$. A polynomial belongs to $\mathcal{I}_d$ precisely when $p \circ \pi = 0$. For a product of linear coordinates,

$$\prod_{i=1}^{k_{\mathrm{tot}}} \ell_i - \prod_{i=1}^{k_{\mathrm{tot}}} (\ell_i \circ \pi) = \sum_{i=1}^{k_{\mathrm{tot}}} \left[\prod_{l<i} (\ell_l \circ \pi)\right] (\ell_i - \ell_i \circ \pi) \prod_{l>i} \ell_l .$$

Consequently every bihomogeneous equation label is a finite sum of products with one linear factor vanishing on some $\mathcal{J}_j^{\mathrm{Max}}$, with the same engineering weight and factor count. The individual factors need not be invariant. We use the Taylor map Tay_d of Section 10.1; every coordinate has weight at least two. Call a lattice observable O a lift of a bihomogeneous polynomial q if $\mathrm{Tay}_d O - q$ consists of monomials later than q in the (k, w) order.

Lift the distinguished factor by a finite combination of transported forces from Step 2. For each summand, lift the other factors by curvature measurements avoiding its varied link, matching their full linear parts through engineering weight d. The resulting expression has the zero-moment property of Step 1 and at least k_{tot} curvature factors. Indeed, every force is a difference of two transported curvatures per transverse axis. The expression is a lift of the prescribed product: the additional linear terms of the force have greater engineering weight, and every nonlinear correction increases factor count. The square-measurement lifts have their prescribed full linear parts, so introduce only the latter kind of correction. Taylor polynomials exist by the smooth dependence of the test holonomies on the jet.

Average over constant gauge transformations. All factors transform by adjoint matrices at the origin, so the orbit lies in the finite-dimensional span of finitely many fixed-degree colour-coordinate monomials on the same measurements; each orbit function keeps zero moments, with colour vectors rotated by $\mathrm{Ad}(v^{-1})$. The Haar average is therefore a gauge-invariant finite combination of orbit functions; it keeps zero moments, minimum factor count and the lift property, and its Taylor map is invariant.

In each bidegree extend a basis of $\mathcal{I}_d$ to a basis of the invariant label space in that bidegree, lifting the complementary vectors by Lemma 10.2. Lift $\mathbf{1}$ to 1, since $\mathcal{I}_d$ contains no nonzero

constant. Summing gives a linear lift $p \mapsto \mathcal{Y}_p$. Replace this lift by the two successive averages

$$\mathcal{Y}'_p(U) = \frac{1}{|H_4|} \sum_{h \in H_4} \mathcal{Y}_{h^{-1}p}(h^* U),$$
$$\mathcal{Y}^{\mathrm{sym}}_p = \frac{1}{2} \left(\mathcal{Y}'_p + \overline{\mathcal{Y}'_{\bar{p}}} \right) .$$

The Taylor map commutes with both operations. The real Maxwell equations and the chain rule make $\mathcal{J}^{\mathrm{Max}}_j$, and hence $\mathcal{I}_d$, stable under signed permutations, adjoint transformations and conjugation. The zero-moment property survives the signed average by (10.7), changing the spectator by the inverse isometry; the exterior set is preserved. Conjugation uses $\overline{F_{\mathrm{ext}}}$. A reversed varied link uses a right derivative of the Wilson energy, whose Haar identity is identical. Thus the averaged equation representatives remain finite sums of one-link action derivatives with link-independent multipliers and avoiding transport paths.

Let $\mathsf{R}_d p = \mathrm{Tay}_d \mathcal{Y}^{\mathrm{sym}}_p$. Then $\mathsf{R}_d - I$ strictly raises (k, w). There are finitely many bidegrees of weight at most d, so $(\mathsf{R}_d - I)^{n_0} = 0$ for some integer n_0, and

$$\mathsf{R}_d^{-1} = \sum_{l=0}^{n_0 - 1} (-1)^l (\mathsf{R}_d - I)^l, \qquad V^{\mathrm{aux}}_d[\mathbf{c}] = \mathcal{Y}^{\mathrm{sym}}_{\mathsf{R}_d^{-1}\mathbf{c}} .$$

This gives invertibility and Taylor normalization; the averages give the symmetry and unit identities. The inverse preserves minimum factor count, so Lemma 10.3 applies to these sections with their own constants. A signed-invariant box enclosing all paths and plaquettes gives a common stencil. Take r_d, M_d as maxima for an ideal basis; linearity gives (10.8). □

10.5 Example: the curvature square

We follow one familiar observable through the construction. Sections 14 and 15 construct the terminal map and the quotient used at its end.

The label. With F_{ab} as in Section 10.4, radial gauge gives $\mathcal{A}_{0,a}(x) = \frac{1}{2} \sum_b x^b F_{ba}$; since $\mathcal{A}(0) = 0$, F_{ab} is the curvature of the jet at the origin. The curvature square is the label

$$c_{F^2}(\mathsf{j}) = \frac{1}{4} \sum_{a,b} \langle F_{ab}, F_{ab} \rangle_{\mathfrak{g}} = \frac{1}{4} \sum_{a,b,\alpha} (F^\alpha_{ab})^2 ,$$

with coordinates F^α_{ab} in an orthonormal basis of $\mathfrak{g}$: the Yang–Mills density at the origin. It is invariant, real and H_4 invariant, of bidegree $(4, 2)$, and lies in V_d for every $d \geq 4$.

Two lattice readings. For $a < b$ and $s, s' = \pm 1$, let $\square^{ss'}_{ab}$ be the plaquette with corner 0 spanned by $s e_a$ and $s' e_b$. The clover

$$P^{\mathrm{cl}}(U) = \frac{1}{4} \sum_{a<b} \sum_{s,s'=\pm 1} \mathrm{pr}(U_{\square^{ss'}_{ab}})$$

satisfies $\mathrm{Tay}_4 P^{\mathrm{cl}} = c_{F^2}$. Indeed, on test holonomies each of these plaquettes has $U_\square = e^X$ with $X = \pm h^2_{\mathrm{lat}} F_{ab} + O(h^3_{\mathrm{lat}}) \in \mathfrak{g}$, and $\mathrm{pr}(e^X) = \frac{1}{2} \langle X, X \rangle_{\mathfrak{g}} + O(|X|^3)$ because $\mathrm{Re}\,\mathrm{tr}\, X = 0$; hence $P^{\mathrm{cl}}(U(h_{\mathrm{lat}}, \mathsf{j})) = h^4_{\mathrm{lat}} c_{F^2}(\mathsf{j}) + O(h^5_{\mathrm{lat}})$. This familiar leading-order reading is gauge invariant, real and H_4 invariant, and at most $3t^2$ when its plaquettes have chord at most t: the size t^{k_i}, $k_i = 2$,

of Lemma 10.3. The column $P_4^{\mathrm{loc}}[c_{F^2}]$ of Lemma 10.2 is built differently, as the H_4 average of $c_{F^2}(T_4 f(\sigma_1, \dots, \sigma_{M_{\mathrm{meas}}}))$: it recovers the truncated jet from curvature measurements and evaluates the label on it. The two differ by an observable with vanishing Taylor map; the terminal map below refers to the section of Lemma 10.2.

The insertion. Take $d = 4$. A nonconstant monomial of weight at most four is a coordinate of $\mathcal{A}_0$, $\mathcal{A}_1$ or $\mathcal{A}_2$, or a product of two coordinates of $\mathcal{A}_0$, and invariant linear terms vanish because $\mathfrak{g}$ is simple. As the adjoint representation is absolutely irreducible, every label in V_4 is $m_0\mathbf{1} + \sum_{P,Q} m_{PQ}\langle F_P, F_Q\rangle_{\mathfrak{g}}$, summed over coordinate planes, with m symmetric. A reflection of one axis changes the sign of F_P exactly for the planes containing that axis. Hence H_4 invariance forces m to be diagonal and, by permutations, constant: the real H_4 invariant labels in V_4 are the real combinations of $\mathbf{1}$ and c_{F^2}. For $N \geq m_{\mathrm{ref}}(4)$, Lemma 14.8 makes $\mathsf{Z}_{N,p,4}$ commute with conjugation and H_4, and gives $\mathsf{Z}_{N,p,4}\mathbf{1} = (a_{m_{\mathrm{ref}}(4)}/a_N)^4\mathbf{1}$. The fine-cutoff coefficient vector and, by Definition 2.8, the insertion are therefore

$$\mathsf{Z}_{N,p,4}^{-1} c_{F^2} = \lambda_{N,p} c_{F^2} + \mu_{N,p}\mathbf{1}\,,$$
$$O_{N,p}^{(4)}(c_{F^2}, y; U) = a_N^{-4}\big(\lambda_{N,p} t_N^{-2} P_4^{\mathrm{loc}}[c_{F^2}](y; U) + \mu_{N,p}\big)\,,$$

with real $\mu_{N,p}$ and real $\lambda_{N,p} \neq 0$, since $\mathsf{Z}_{N,p,4}$ preserves $\mathbb{C}\mathbf{1}$. The renormalization of the curvature square is a multiplicative constant together with an additive one.

Mixing enters through the transfer: the representation (5.18) of the blocked marked density keeps a unit coordinate $x_{m,0}$, through which integration can turn part of a curvature insertion into a constant. The inverse terminal map makes the transported coefficient at the reference level exactly c_{F^2}, with unit coordinate zero, which specifies the label at the physical scale $a_{m_{\mathrm{ref}}(4)}$. A single factor ($\mu_{N,p} = 0$) would require the transfer to preserve the line through c_{F^2}, and nothing in the construction forces this. In degree six, $\mathsf{Z}_{N,p,6}^{-1} c_{F^2}$ may involve every real H_4 invariant label of weight at most six, including any that are not rotation invariant.

The degree tag acts separately from weight and factor count: it fixes ℓ_m^d, a_N^{-d} and the terminal map. Thus $(4, c_{F^2})$ and $(6, c_{F^2})$ are different insertion labels; the inclusion $\jmath_{4,6}$ of Proposition 6.4 maps the class of the first to the class in F_6 with the same correlations, which need not be the class of the second.

The field. Degree four has no invisible labels. Labels in V_4 depend only on $\mathcal{A}_0$, and $\mathcal{J}_0^{\mathrm{Max}} = \mathcal{J}_0$ because the curvature of $\mathcal{A}_0$ is constant; so $\mathcal{I}_4 = 0$, and (14.36) with Proposition 6.4 gives $\dim F_4 = \dim V_4$. Thus $\mathcal{N}_4(S) = 0$, the quotient map $V_4 \to F_4$ is bijective, and $[c_{F^2}] \neq 0$. In each phase ξ, Theorem 6.16 gives the smeared field $\Phi_\xi(\iota_4[c_{F^2}], f)$. Its free counterpart is the Maxwell curvature square $\mathcal{O}_{F^2}$ of Section 6.4, whose two-point function at separation r is $3d_{\mathfrak{g}}\pi^{-4}r^{-8}$; Section 18 compares the stress trace with it.

11 Gauges and block integration

The block map of Definition 5.1 averages fine parallel transports in local frames. A fixed axial frame would distinguish an ordering of the axes. We instead minimize a symmetric average of transport energies when the field is sufficiently small, and average over curvature-controlled axial frames elsewhere. Section 11.1 shows that these choices give a measurable, covariant probability kernel; consequently the block integration in Theorem 5.5 preserves the partition function.

The remaining subsections prepare the small-field integration used in Sections 12 and 13. Section 11.2 solves the free block problem: the free action at every level, the free interpolation W_0 from coarse to fine fields, and an exact square completion. Section 11.3 constructs the stationary fine field about a coarse field and integrates the Gaussian fluctuations about it; the determinant, the Haar Jacobian and the inherited loop action form the one-loop action T. Section 11.4 supplies what Section 12 needs to extract the curvature-square coefficient of T: inverse bounds uniform in the block ratio, vertex bounds whose input-dependent part carries a factor ℓ^{-1}, and a Ward identity showing that the predictor contact term (the chain-rule term containing the second derivative of the predictor) does not change that coefficient. Sections 11.5 and 11.6 give the local large-field energy (Lemma 11.12) and the simultaneous differentiated local integral (Lemma 11.14) used in Section 13.

Three uniformities require care.

(i) The frame and predictor cutoffs must work on patches with arbitrarily many sites. Centred frame minimizers satisfy a contraction whose constant does not depend on the patch (Lemma 11.4).

(ii) The free kernels must stay controlled after arbitrarily many blockings. We separate the principal Fourier pole before inverting (Lemma 11.5).

(iii) The determinant comparison must not see the ℓ^4 sites of a fine cell. The difference of the two compared operators either factors through coarse spaces or contains the regulator $\mathfrak{m}_\mu = \mu^2(P_\mu^3)^3$, and P_μ^3 is trace class uniformly in ℓ (Lemmas 11.8 and 12.8).

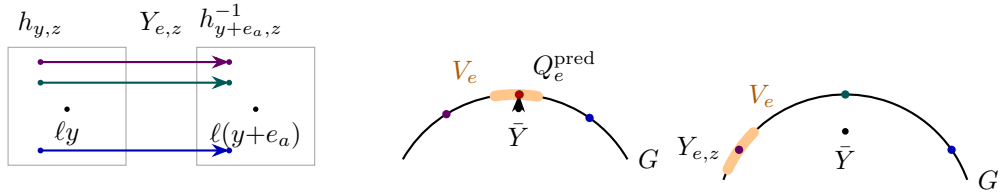
Induction scheme. Sections 11–13 prove one descending induction on m, started at $m = N$ by the Wilson representation of Lemma 13.16. The hypothesis of the step $m+1 \to m$ has two parts: the path bounds and cancellations of Lemmas 12.3–12.4 at level $m+1$, for the input path action of Definition 12.2; and an input curve satisfying Definitions 13.1–13.3 and (5.12). The stationary integration of this section and the determinant comparison of Section 12 first construct the level-m path arrays; the joint proof after Lemma 12.9 shows that they satisfy Lemmas 12.3–12.4 at level m. Lemmas 13.12 and 13.13 construct the output activities. Lemma 13.14 then fixes the output coupling; its proof uses Lemmas 13.9, 13.12, 13.13 and 13.15, none of which uses Lemma 13.14. Lemma 13.17 closes the step. Curl division (Lemma 12.5) and gauge completion (Lemma 12.7) do not use the induction.

Throughout, use the group and metric of Definition 2.1, the Wilson term (2.1), and the level choices (5.1). At a step $m+1 \to m$, put $j = J_{m+1}$, $\ell = \ell_m$ and $R = (j+3)\ell$. Unless stated otherwise, periods satisfy the minimum-period requirement in Section 9. All finite operators use counting inner products and are periodizations of the kernels described below. An adjoint at complex momentum means the analytic continuation of the real-momentum adjoint.

11.1 The block kernel

Before averaging transports, we express them in common endpoint frames. When the resulting connectors lie close to one another in G, their weighted average lies close to G, and its nearest point of G serves as a group-valued average. The other branches complete the kernel on the remaining configurations. Figure 8 shows the two cases, and Table 4 lists the tests.

For $o \in S_4$, let $T_U(y \to z; o)$ be the ordered product along the path that changes the four coordinates in the order o, using inverse links for negative steps.



(a) framed connectors (b) predictor (c) fallback

Figure 8: The predictor of a coarse link $e = (y, a)$. (a) Fine connectors between two blocks, each expressed in the frames at its two endpoints. (b) When the connectors lie close together in G, their weighted average $\bar{Y} = \sum_z w_z^{\mathrm{av}} Y_{e,z}$ is a matrix just off G; the predictor is its nearest point of G, and the coarse link V_e fluctuates about it with weight $e^{-\mathsf{b}\beta\,\mathrm{pr}(V_e Q_e^{-1})}$. (c) When the connectors are spread out, a single connector $Y_{e,z}$ is chosen with probability w_z^{av} and V_e fluctuates about it. This branch is part of the definition of the kernel on every configuration where the minimum test fails.

Definition 11.1 (Axial frames)**.** The frame tagged by (z, o), its normalization and its link coordinates are

$$g_y^{z,o} = T_U(z \to y; o), \qquad h_y^{z,o} = (g_0^{z,o})^{-1} g_y^{z,o}, \qquad U_{(y,a)}^{z,o} = g_y^{z,o} U(y, a)(g_{y+e_a}^{z,o})^{-1}.$$

These frames provide the finite alternatives for the symmetric gauge choice.

In the patch $\ell y + \mathbb{B}_R$ we abbreviate the frame $h_{y,r}$ of (5.4) at offset $r \in \mathbb{B}_R$ to h_r. Here $\delta_{\mathrm{fr}} > 0$ is the frame cutoff and $a_{\mathrm{ax}} \geq 16$ the axial factor of item 2 of Table 3; patch sums count internal positive links and plaquettes. The frame test below compares transports around loops through the patch centre: under a gauge transformation such a loop transport is conjugated, so its chord is gauge invariant.

Definition 11.2 (Symmetric frame choice)**.** For $x, y \in \mathbb{B}_R$ and $o, o_0 \in S_4$ let $\mathcal{T}_{xy;o}^{o_0} = g_x^{0,o_0} T_U(x \to y; o)(g_y^{0,o_0})^{-1}$, the transport from 0 to x in order o_0, then to y in order o, then back to 0 along the reversed route of order o_0. Tags lie in $\mathbb{B}_R \times S_4$.

(a) *Minimizing branch.* If $\mathrm{ch}(\mathcal{T}_{xy;o}^{o_0}) < \delta_{\mathrm{fr}}$ for all x, y, o, o_0, the frame $(h_r)_{r \in \mathbb{B}_R}$ is the minimizer with $h_0 = 1$ of

$$\frac{1}{24|\mathbb{B}_R|^2} \sum_{r,r' \in \mathbb{B}_R} \sum_{o \in S_4} \mathrm{pr}(h_r T_U(r \to r'; o) h_{r'}^{-1}),$$

and every tag has mass $|\mathbb{B}_R \times S_4|^{-1}$.

(b) *Axial branch.* Otherwise a tag (z, o) is *eligible* if

$$\sum_{e \text{ patch}} \mathrm{ch}(U_e^{z,o}) \leq a_{\mathrm{ax}}(2R + 1) \sum_{\square \text{ patch}} \mathrm{ch}(U_\square);$$

the frame is $h^{z,o}$, and eligible tags have equal mass.

On the minimizing branch the frame does not depend on its tag. For a coarse edge e use the framed connector $Y_{e,z}$ of (5.4) and the averaging weights w^{av} of (5.5); the weight is supported in $\mathbb{B}_{j(\ell-1)/2} \subset \mathbb{B}_R$.

Definition 11.3 (Coarse predictors)**.** If the minimum in (5.5) is below $\varepsilon_{\mathrm{pred}}$, then Q_e^{pred} is the minimizer there, and predictor tags have uniform mass on $\mathbb{B}_R$. Otherwise a tag z selects $Q_e^{\mathrm{pred}} = Y_{e,z}$, with mass w_z^{av}. The product of frame and predictor masses is $\lambda^{\mathrm{tag}}(U, \alpha_{\mathrm{tag}})$. These choices define every summand of Definition 5.1.

Choice	Test	If the test holds	Otherwise
Frame	$\mathrm{ch}(\mathcal{T}^{o_0}_{xy;o}) < \delta_{\mathrm{fr}}$ for all loops through the patch centre	the unique minimizing frame with $h_0 = 1$; all tags equally likely	an eligible axial frame $h^{z,o}$; eligible tags equally likely
Predictor	minimum in (5.5) below $\varepsilon_{\mathrm{pred}}$	the nearest point of G to $\bar{Y}$; tags uniform on $\mathbb{B}_R$	a connector $Y_{e,z}$, chosen with mass w_z^{av}

Result: for every fine configuration U, a probability law on coarse links, Borel in U and covariant (Lemma 11.4 and the paragraph after its proof). Analytic dependence on U is proved only on the minimizing branches near the flat field (Lemma 11.6).

Table 4: The two branch decisions of the block kernel.

The branch choices require constants independent of the patch radius. Their normalization then follows from finite probability sums and Haar invariance.

Lemma 11.4 (Branches and normalization)**.** *There are $\delta_{\mathrm{fr}}, \varepsilon_{\mathrm{pred}} > 0$, depending only on $G \subset U(n_G)$, such that for every patch $\mathbb{B}_R$, $R = (j+3)\ell$, and every $a_{\mathrm{ax}} \geq 16$:*

(i) on the minimizing branch of Definition 11.2, the frame with $h_0 = 1$ is unique;

(ii) on the minimizing branch of Definition 11.3, the predictor is unique;

(iii) for every $o \in S_4$ some tag (z, o) is eligible;

(iv) the masses and the kernel are normalized:

$$\sum_{\alpha_{\mathrm{tag}}} \lambda^{\mathrm{tag}}(U, \alpha_{\mathrm{tag}}) = 1, \qquad \int k_\beta^{(m)}(U, V)\, dV = 1\,.$$

The proof treats the predictor, the frame and the axial alternatives in turn, then the two normalizations.

Proof. Predictor branch. Work in the real Euclidean space of complex matrices with Frobenius norm. The tubular projection lemma of [12, p. 153, Lemma], applied to the compact smooth embedded submanifold G, gives a neighbourhood with unique smooth nearest-point projection π. At the identity its derivative is the orthogonal projection $P_{\mathfrak{g}}$ onto $\mathfrak{g}$. Consequently, for any $\epsilon > 0$, shrinking this neighbourhood gives

$$\|\pi(A) - \pi(B) - P_{\mathfrak{g}}(A - B)\|_{\mathrm{F}} \leq \epsilon \|A - B\|_{\mathrm{F}} \tag{11.1}$$

when A, B lie in a sufficiently small ball about 1. Integrate $D\pi - P_{\mathfrak{g}}$ along their joining segment.

Put $\bar{Y} = \sum_z w_z^{\mathrm{av}} Y_{e,z}$. Expansion of the square gives

$$2n_G \sum_z w_z^{\mathrm{av}}\, \mathrm{pr}(Y_{e,z} u^{-1}) = \sum_z w_z^{\mathrm{av}} \|Y_{e,z} - \bar{Y}\|_{\mathrm{F}}^2 + \|\bar{Y} - u\|_{\mathrm{F}}^2\,.$$

Thus the minimizers are precisely the nearest group points to $\bar{Y}$. If the minimum is below $\varepsilon_{\mathrm{pred}}$, then $\mathrm{dist}(\bar{Y}, G) < \sqrt{2n_G\varepsilon_{\mathrm{pred}}}$. Choose this smaller than a tubular radius. This choice uses neither the weights nor their number.

Frame branch: a contraction uniform in R. First make an origin-based axial change of gauge. Its route matrices, denoted $\mathcal{T}_{rr';o}$, satisfy $\mathcal{T}_{rr;o} = 1$ and $\mathcal{T}_{r'r;o'} = \mathcal{T}_{rr';o}^{-1}$ for the reversed order o'. On the minimizing branch choose δ_{fr} so that $\|\mathcal{T}_{rr';o} - 1\|_{\mathrm{F}} \le d$, where $d \le 1/4$ will be fixed below. Write av for the uniform averages over the indicated indices. A global minimizer h exists by compactness. Varying its coordinate h_r and pairing the two orientations shows that h_r is a nearest group point to

$$F_r(h) = \mathrm{av}_{r',o}\, h_{r'} \mathcal{T}_{rr';o}^{-1}\,.$$

The self term is harmless: replacing it by its value at a proposed competitor adds a nonnegative square. Comparison with the constant frame and the triangle inequality give

$$\mathrm{av}_{r,r'}\, \|h_r - h_{r'}\|_{\mathrm{F}}^2 \le 4d^2\,.$$

Left-multiply h so that 1 is a nearest point to its centroid $\bar{h} = \mathrm{av}_r\, h_r$. Jensen's inequality then gives $\|\bar{h} - 1\|_{\mathrm{F}} \le 2d$ and $\|F_r(h) - 1\|_{\mathrm{F}} \le 3d$.

Center two minimizers h, k in this way, and put $u = \max_r \|h_r - k_r\|_{\mathrm{F}}$. Their centroids are normal to G at 1, so $P_{\mathfrak{g}}(\bar{h} - \bar{k}) = 0$. Moreover,

$$\|F_r(h) - F_r(k)\|_{\mathrm{F}} \le u, \qquad \|F_r(h) - F_r(k) - (\bar{h} - \bar{k})\|_{\mathrm{F}} \le du\,.$$

Apply (11.1) with $\epsilon = 1/4$, choosing $3d$ inside its ball. At a coordinate attaining u it yields $u \le (d + 1/4)u \le u/2$. Hence the centered minimizers coincide. All minimizers differ by a common left multiple, which $h_0 = 1$ fixes uniquely. The contraction factor $d + 1/4$ comes from averages only, so neither d nor δ_{fr} depends on the number of sites.

Axial branch. For a rectangular slice X let $\Sigma_{\mathrm{link}}(X, U)$ and $\Sigma_{\mathrm{plaq}}(X, U)$ be the sums of link and plaquette chords internal to X. We show, by induction on the number r of varying axes of X, that some axial frame satisfies $\Sigma_{\mathrm{link}}(X, U^{\mathrm{ax}}) \le 4r(2R + 1)\Sigma_{\mathrm{plaq}}(X, U)$. Choose a layer X_0 perpendicular to the last axis whose plaquette sum is at most the slice average, so that $(2R + 1)\Sigma_{\mathrm{plaq}}(X_0, U) \le \Sigma_{\mathrm{plaq}}(X, U)$, and apply the induction hypothesis there. Extend its frame along the last axis, making those links equal to 1. A link in another layer differs from its transported link in X_0 by a product of intervening plaquettes. Each plaquette lies between at most $2R$ such pairs. The chord triangle inequality and conjugation invariance, summed over the links, give

$$\Sigma_{\mathrm{link}}(X, U^{\mathrm{ax}}) \le (2R + 1)\{\Sigma_{\mathrm{link}}(X_0, U^{\mathrm{ax}}) + \Sigma_{\mathrm{plaq}}(X, U)\} \le 4r(2R + 1)\Sigma_{\mathrm{plaq}}(X, U)\,.$$

The second inequality uses the induction bound for X_0 and the choice of the layer. At $r = 4$ this proves eligibility, for each axis ordering, with factor 16.

Normalization. Each frame mass therefore sums to one; each predictor mass sums to one since $w^{\mathrm{av}} \ge 0$ and $\sum_z w_z^{\mathrm{av}} = 1$. Successively summing the finite tag choices proves the first identity. For fixed tags, every coarse-link factor in Definition 5.1 integrates to one by right invariance of Haar measure. Their product and the tag sum prove the second identity. □

Measurability and covariance of the block kernel. The branch tests are Borel; unique compact minimizers depend continuously on their data. Axial eligibility is a closed inequality. Thus the finite tag sum is Borel, and it is bounded for fixed β since $0 \le \mathrm{pr} \le 2$ and the Haar denominator is positive. Gauge transformations act by conjugation on the rooted transports and transform both

ends of a connector; lattice isometries permute the roots, orders and offsets, including inversion of reversed transports. The objective functions, eligibility inequalities and tag masses are preserved. Uniqueness on the minimizing branches and reindexing on the other branches therefore give

$$k_\beta^{(m)}(U^\gamma, V^{\gamma\circ\iota}) = k_\beta^{(m)}(U,V), \qquad k_\beta^{(m)}(s_{\ell b}U, s_b V) = k_\beta^{(m)}(U,V), \qquad \iota(y) = \ell y\,.$$

Here s_b is the pullback by $x \mapsto Ox + b$, with $O \in H_4$; $s_{\ell b}$ uses the same O on fine sites. Fubini's theorem and Lemma 11.4 give (5.7). Thus the block kernel is measurable, covariant and normalized.

11.2 Free quadratic forms

Write d_r for the forward lattice exterior derivative on r-forms. Thus $(d_1 a)_{ab}(x) = a_b(x+e_a) - a_b(x) - a_a(x+e_b) + a_a(x)$. A history at level m is the finite sequence $(\ell_i, J_{i+1})_{m\le i<N}$, read from fine to coarse. Its stages decrease by at most one. Let $\mathcal{Q}_m$ be the derivative of the predictor at the flat field, and let

$$\mathcal{M}_m a(e) = \sum_z w_z^{\mathrm{av}} \sum_{r=0}^{\ell-1} a(\ell e^- + z + r e_{\mathrm{dir}(e)}, \mathrm{dir}(e))$$

be straight-link averaging.

The free problem is Gaussian. Let a free fine field a have diagonal covariance $\mathrm{diag}(c_a)$; at the top level $c_a = \lambda^{-1}$, the Feynman-gauge covariance of the Wilson Gaussian. Its blocked image $v = \mathcal{M}_m a + \mathsf{b}^{-1/2}\xi$, with independent standard noise ξ, again has a diagonal covariance. The frame changes $\mathcal{M}_m$ only by a gradient, so this recursion commutes with gauge reduction. Under blocking by ℓ, a coarse momentum p receives contributions from the fine momenta $\theta_z = (p+2\pi z)/\ell$, $z \in (\mathbb{Z}/\ell\mathbb{Z})^4$, its *aliases*. Near $p=0$ the alias $z=0$ carries the massless pole; the other aliases are regular. Aliasing the straight-link filter and adding b^{-1} gives the recursion (11.2) below. To write it, set

$$q_a(\theta) = \mathrm{e}^{i\theta_a} - 1, \quad \lambda(\theta) = \sum_a q_a(\theta) q_a(-\theta),$$

$$r_\ell(t) = \ell^{-1} \sum_{s=0}^{\ell-1} \mathrm{e}^{ist}, \quad \widehat{w}_\ell(\theta) = \prod_a \ell^{-1} \sum_{|s|\le(\ell-1)/2} \mathrm{e}^{is\theta_a}.$$

Here $\widehat{w}_\ell$ is the symbol of the uniform weight on $\mathbb{B}_{(\ell-1)/2}$, so $\widehat{w}^{\mathrm{av}} = \widehat{w}_\ell^{\,j}$. Starting from $c_a = \lambda^{-1}$, update these auxiliary diagonal covariances at each step $i+1 \to i$ of the history, with $\ell = \ell_i$ and $j = J_{i+1}$, by

$$c_a'(p) = \mathsf{b}^{-1} + \ell^{-2} \sum_{z \bmod \ell} \widehat{w}_\ell(\theta_z)^{2j} r_\ell((\theta_z)_a) r_\ell(-(\theta_z)_a) c_a(\theta_z), \qquad \theta_z = (p+2\pi z)/\ell. \tag{11.2}$$

Gauge reduction keeps only the precision of the curl: minimizing $\sum_a c_a^{-1}|a_a + q_a\phi|^2$ over gradients ϕ gives $\langle d_1 a, \mathsf{H}^0 d_1 a\rangle$ with the diagonal two-form metric below. For $p \neq 0$, define this unmodified metric and the chosen metric by

$$\widehat{\mathsf{H}}^0_{ab,cd}(p) = \frac{\mathbf{1}_{(a,b)=(c,d)}}{c_a(p)c_b(p)\sum_h q_h(p)q_h(-p)/c_h(p)}, \qquad \widehat{\mathsf{H}}^0(0) = I,$$
$$\mathsf{H}^{-1} = (\mathsf{H}^0)^{-1} + \zeta_{J_m}\big(d_2(\mathsf{H}^0)^{-1}\big)^* d_2 (\mathsf{H}^0)^{-1}.$$

Lemma 11.5 shows that the value at $p = 0$ is the analytic extension of the formula. Since $d_2 d_1 = 0$, the modification satisfies $\mathsf{H} d_1 = \mathsf{H}^0 d_1$: the free action on curls is unchanged. It changes H only off the range of d_1, where it makes the adjacent defect (11.3) positive, as the square completion of Lemma 11.12 requires. Fix $\zeta_k = \max(\zeta^0_{k,k}, \zeta^0_{k+1,k})$, where the positive thresholds for the two incoming stage pairs are chosen in the proof below. These constants are fixed before γ_k and the clipping coefficients. The ratio threshold also depends on the incoming ζ. The free action is

$$\mathsf{S}^{\mathrm{free}}_{N,m}(a) = \tfrac{1}{2}\langle d_1 a, \mathsf{H}_{N,m} d_1 a\rangle, \qquad \mathsf{K}_{N,m} = d_1^* \mathsf{H}_{N,m} d_1.$$

The smooth interpolation W_0 minimizes the fine free action plus $\mathsf{b}\|\mathcal{Q}_m a - v\|^2/2$, with the remaining gradient fixed to match the predictor; the tree representative is obtained afterwards.

Let T be the free two-form transfer (the lift of the straight-link block average): a periodized analytic alias kernel from coarse to fine two-forms which on curls satisfies

$$T^* d_1 = d_1 \mathcal{M}_m, \qquad T \mathsf{H}^0_c d_1 = \mathsf{H}^0_f d_1 W_0.$$

Its extension to all forms is given by the same alias formula, with a straight geometric-sum factor for each of the two indices. The two conditions are compatible: $\mathcal{Q}_m - \mathcal{M}_m$ takes values in gradients, $\mathsf{K}_c d_0 = 0$ and $\mathsf{K}_f W_0 = \mathcal{Q}_m^* \mathsf{K}_c$, so

$$\begin{aligned}\langle d_1 \mathcal{M}_m a, \mathsf{H}^0_c d_1 v\rangle &= \langle a, \mathcal{Q}_m^* \mathsf{K}_c v\rangle = \langle a, \mathsf{K}_f W_0 v\rangle \\ &= \langle d_1 a, \mathsf{H}^0_f d_1 W_0 v\rangle.\end{aligned}$$

Lemma 11.5 shows that the adjacent defect factors through d_2 on both sides,

$$\mathsf{H}_c(\mathsf{H}_c^{-1} - T^* \mathsf{H}_f^{-1} T - \mathsf{b}^{-1} d_1 d_1^*)\mathsf{H}_c = d_2^* \mathcal{W}_3 d_2, \tag{11.3}$$

with the three-form kernel $\mathcal{W}_3$ given in its proof. The factorization rests on one-leg exterior division. For a translation-invariant kernel f with coefficients f_x and a direction a,

$$\frac{f(p) - f(p|_{p_a=0})}{q_a(p)} = \sum_x f_x \mathrm{e}^{ip\cdot x} \frac{1 - \mathrm{e}^{-ip_a x_a}}{q_a(p)}, \tag{11.4}$$

and each quotient on the right is a finite geometric sum with $|x_a|$ terms. Applied in a leg in which the kernel is closed, this division writes it through d_2.

Write $k = J_m$, and let $C^{\mathrm{tr}}_{j,k}$ be the stage-pair transfer bound obtained below. Include the transfer inequality in the next lemma among the ratio requirements of item 5 of Table 3. At $m = N$ the free form is the identity on curls. This free integration fixes the normalization of the kinetic coefficient in Section 12. The coefficient arrays considered below are those of $\mathsf{H}_{N,m}$, $\mathsf{H}^{-1}_{N,m}$ and $\rho_{N,m}$, together with W_0 and T at the step $m+1 \to m$; we write $W_{0;N,m}$ and $T_{N,m}$ when the cutoff matters. For a translation-invariant kernel K between two-forms, links or sites put

$$\|K\|_c = \max \sum_{z\in\mathbb{Z}^4} e^{c|z|} |K_{\cdot\cdot}(z)|,$$

the maximum running over the finite component and residue indices. The constants $\gamma_k > 0$ below are uniform over histories, and $c_k > 0$ depends only on k; strip and coefficient bounds are uniform in N, m, p.

Lemma 11.5 (Free quadratic forms)**.** *For histories satisfying $\ell^2 \geq C^{\mathrm{tr}}_{j,k}\zeta_j$ and $c_{\mathrm{dir},k} \leq \gamma_k$:*

(i) $W_0^{*}\mathsf{K}_{N,m+1}W_0 + \mathsf{b}(\mathcal{Q}_m W_0 - I)^{*}(\mathcal{Q}_m W_0 - I) = \mathsf{K}_{N,m}$;

(ii) $\widehat{\mathsf{H}}_{N,m}(0) = I$ *and* $\mathsf{H}_{N,m} \geq 2\gamma_k I$;

(iii) $\mathsf{H}_{N,m} - c_{\mathrm{dir},k} I = \rho_{N,m}^{*}\rho_{N,m}$;

(iv) the adjacent defect (11.3) *has* $\mathcal{W}_3 \geq c_k I$;

(v) the coefficient arrays and their limits have uniform analytic strip bounds and exponential coefficient decay, with some rate $c > 0$*; at fixed* m*, for every* $0 < c' < c$*, as* $N \to \infty$*,*

$$\begin{aligned} &\|\mathsf{H}_{N,m} - \mathsf{H}_{\infty,m}\|_{c'} + \|\mathsf{H}_{N,m}^{-1} - \mathsf{H}_{\infty,m}^{-1}\|_{c'} + \|\rho_{N,m} - \rho_{\infty,m}\|_{c'} \\ &\qquad + \|W_{0;N,m} - W_{0;\infty,m}\|_{c'} + \|T_{N,m} - T_{\infty,m}\|_{c'} \longrightarrow 0. \end{aligned} \tag{11.5}$$

The proof separates the principal Fourier pole to obtain analytic, uniformly positive metrics, makes the defect positive off the curl range, and then proves the energy identity and the cutoff limit; the history bounds come first, so that inversion at zero momentum is justified.

Proof. Step 1: the covariance recursion and the two-form metric. The straight-link multiplier is $\ell\, \widehat{w}_\ell(\theta)^j r_\ell(\theta_a)$, and $\mathcal{Q}_m d_0 \phi = d_0(\phi \circ \iota)$, so gauge reduction commutes with the block step. Taking aliases of $\mathrm{Cov}(v) = \mathcal{M}_m \,\mathrm{diag}(c_a)\mathcal{M}_m^{*} + \mathsf{b}^{-1} I$ gives (11.2); $\ell^{-2} = \ell^{-4}\ell^2$ combines counting adjoints with the two straight connectors. For $w_a = 1/c_a$, minimization of the diagonal precision over gradients gives

$$\min_{\phi} \sum_a w_a |a_a + q_a \phi|^2 = \frac{\sum_{a<b} w_a w_b |q_a a_b - q_b a_a|^2}{\sum_d w_d |q_d|^2}.$$

Indeed the minimizer is $\phi = -\sum_a w_a \bar{q}_a a_a / \sum_a w_a |q_a|^2$; expanding its square proves the identity. This is precisely the two-form metric defined above.

Step 2: the cumulative filter. For total history dilation M at stage j, the cumulative averaging filter is

$$f(\theta) = \widehat{w}_M(\theta)^j \prod_{\text{stage increases at dilation } M'} \widehat{w}_{M'}(\theta)\,.$$

Successive dilations M' have ratio at least 3. On the enlarged coarse cube $|\operatorname{Re} p_a| < 5\pi/4$, $|\operatorname{Im} p_a| < \eta$, the geometric-sum formula gives

$$|f((p + 2\pi z)/M)| \leq C_j \prod_a (1 + |z_a|)^{-j}, \qquad |f(p/M)| \geq c_j > 0 \quad (p \text{ real})\,. \tag{11.6}$$

For the extra product, use $|\widehat{w}_{M'}(p/M)| \leq \mathrm{e}^{4\eta M'/M}$ and the corresponding real lower bound $\mathrm{e}^{-CM'/M}$. The sum of M'/M is bounded by a geometric series. The identities $\widehat{w}_\ell(\theta)\widehat{w}_{\ell'}(\ell\theta) = \widehat{w}_{\ell\ell'}(\theta)$ and $r_\ell(t) r_{\ell'}(\ell t) = r_{\ell\ell'}(t)$ follow by writing each integer in its quotient and residue classes. They give the cumulative filter. The basic $\widehat{w}_M^{\,j}$ factor supplies the displayed alias decay. It also bounds $r_M((p + 2\pi z)_a/M)$ by $C/(1 + |z_a|)$, with a positive principal lower bound. Thus every nonprincipal alias is damped by the j-th power of the filter, uniformly in the number of blockings.

Step 3: the principal pole. Separate the principal alias in the iterated (11.2). With $\lambda_M(p) = M^2 \lambda(p/M)$ this reads

$$c_a(p) = \frac{A_a(p)}{\lambda_M(p)} + c_a^{\mathrm{np}}(p), \qquad A_a(p) = f(p/M) f(-p/M) r_M(p_a/M) r_M(-p_a/M)\,.$$

The nonprincipal part c_a^{np} contains nonprincipal aliases, bounded by (11.6) and $|M^2\lambda((p+2\pi z)/M)| \geq c|z|^2$ for $z \neq 0$ on a smaller strip. Each earlier noise contributes $C_j(M')^{-2}$ by the squared filter bound. Summing these geometric contributions makes c_a^{np} analytic and uniformly bounded; it is nonnegative on real momenta.

Put $B_a = A_a + \lambda_M c_a^{\mathrm{np}}$ and $F = f(p/M)f(-p/M)$, so $c_a = B_a/\lambda_M$. The identity for a straight geometric sum implies

$$q_a(p)q_a(-p)F = M^2 q_a(p/M)q_a(-p/M)A_a\,.$$

It removes the apparent singularity in

$$\mathfrak{h}(p) = F^{-1} - \sum_a \frac{q_a(p)q_a(-p)c_a^{\mathrm{np}}}{A_a B_a}, \qquad \lambda_M \mathfrak{h} = \sum_a \frac{q_a(p)q_a(-p)}{B_a}\,.$$

On the real enlarged cube B_a is bounded above and below by positive constants. Since $\lambda(p)$ and $\lambda_M(p)$ are uniformly comparable there, the last identity bounds $\mathfrak{h}$ above and below, including $\mathfrak{h}(0) = 1$. Thin the strip using the uniform analytic bounds. The unmodified plaquette metric is then

$$\widehat{\mathsf{H}}^0_{ab,cd}(p) = \frac{\mathbf{1}_{(a,b)=(c,d)}}{\mathfrak{h}(p)B_a(p)B_b(p)} \qquad (a<b,\ c<d)\,.$$

Substitution of $w_a = \lambda_M/B_a$ in the minimization of Step 1 gives exactly $1/(\mathfrak{h}B_aB_b)$. Thus this formula extends the defined metric analytically to zero. It and its inverse are uniformly positive on real momenta, with $\widehat{\mathsf{H}}^0(0) = I$.

Step 4: the inverse modification and the defect. The inverse modification defining H has fixed stage coefficient ζ_j. This positive addition has uniform analytic bounds. Since $d_2d_1 = 0$, it gives $\mathsf{H}d_1 = \mathsf{H}^0 d_1$; it also leaves $\widehat{\mathsf{H}}(0) = I$.

The defect annihilates curls, for both the unmodified and the modified metrics: by the two conditions on T and $\mathsf{H}_f^{-1}\mathsf{H}_f^0 d_1 = d_1$, its value on $d_1 v$ is

$$\mathsf{H}_c d_1(v - \mathcal{M}_m W_0 v - \mathsf{b}^{-1}\mathsf{K}_c v) = 0,$$

since $\mathcal{Q}_m W_0 = I - \mathsf{b}^{-1}\mathsf{K}_c$ and $d_1 d_0 = 0$. For the unmodified metrics, the defect is closed in both legs, so one-leg division (11.4) in each leg writes it as $d_2^* F_0 d_2$. The two divisions lose two powers of path length; the exponential bounds leave a smaller strip. Symmetrize the resulting three-form kernel F_0. If S_0 is the corresponding lift of $d_2(\mathsf{H}_f^0)^{-1}T\mathsf{H}_c^0$, Woodbury inversion gives $d_2(\mathsf{H}_c^0)^{-1}\mathsf{H}_c = D_c d_2$. Substituting the two inverse modifications into the defect, the outgoing addition is $\zeta_k D_c^* D_c$ and the incoming subtraction is $-\zeta_j D_c^* S_0^* S_0 D_c$. Hence

$$\mathcal{W}_3 = D_c^*(F_0 + \zeta_k I - \zeta_j S_0^* S_0)D_c, \qquad D_c = (I + \zeta_k d_2(\mathsf{H}_c^0)^{-1} d_2^*)^{-1}.$$

The pole-free alias estimates give $\|F_0\| \leq C$ and $\|S_0\|^2 \leq C^{\mathrm{tr}}_{j,k}/\ell^2$ on real momenta: the lift has per-residue order ℓ^{-3}, whose square is summed over ℓ^4 residues. Choose $\zeta^0_{j,k} = C + 2$ and then $\ell^2 \geq \zeta_j C^{\mathrm{tr}}_{j,k}$ for the stage pair (j,k). The parenthesis is at least I. Both D_c and its inverse have uniform analytic bounds at the fixed stages. Consequently $cI \leq \mathcal{W}_3 \leq CI$ on real momenta, uniformly in the history. This proves (iv); it is the input for the square terms in Lemma 11.12.

Step 5: coercivity and the row factor. Choose $2\gamma_k$ below the lower bounds for the top and the two incoming possibilities at stage k. Then $\mathsf{H} - c_{\mathrm{dir},k} I \geq \gamma_k I$. Its positive square root ρ is analytic on a smaller common strip: expand the square root about a scalar exceeding the common upper bound. The resulting convergent matrix power series agrees with the positive root on the real torus.

Contour displacement in Fourier inversion gives exponential coefficient decay and the same bound for all periodizations. This proves (ii) and (iii).

Step 6: the normal equations and the energy identity. The normal equations for (11.2), with the gradient chosen to match the rooted predictor, give

$$\mathsf{K}_f W_0 = \mathcal{Q}_m^* \mathsf{K}_c, \qquad \mathcal{Q}_m W_0 = I - \mathsf{b}^{-1}\mathsf{K}_c\,.$$

At nonzero momentum write $\mathsf{C}_f = \operatorname{diag}(c_a)$ and $\mathsf{C}_c = \mathcal{M}_m \mathsf{C}_f \mathcal{M}_m^* + \mathsf{b}^{-1} I$. Gaussian conditioning gives $W_0 = \mathsf{C}_f \mathcal{M}_m^* \mathsf{C}_c^{-1}$ modulo gradients and $\mathcal{M}_m W_0 = I - \mathsf{b}^{-1}\mathsf{C}_c^{-1}$. Reduce these identities over gradients and select the gradient matching $\mathcal{Q}_m$. This gives the two normal equations. The pole-free formulas above extend them to zero; the constant fine potential is the coarse potential divided by ℓ. Expanding the square gives

$$\langle a, \mathsf{K}_f a\rangle + \mathsf{b}\|\mathcal{Q}_m a - v\|^2 = \langle v, \mathsf{K}_c v\rangle + \langle a - W_0 v, (\mathsf{K}_f + \mathsf{b}\mathcal{Q}_m^*\mathcal{Q}_m)(a - W_0 v)\rangle\,.$$

Set $a = W_0 v$ and polarize. This is (i). The empty history gives $\mathsf{H}^0 = I$ and hence the Wilson form on curls, also after the inverse modification.

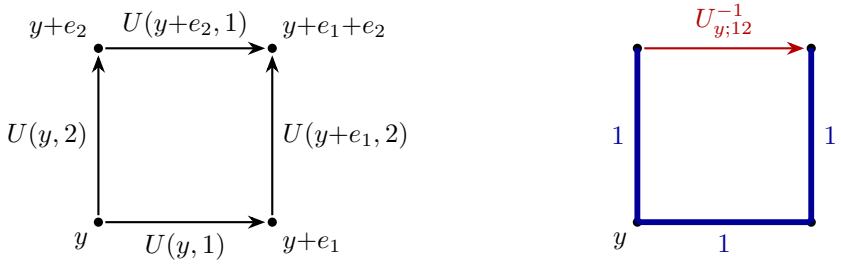
Step 7: the cutoff limit. Choose $c > 0$ below the common decay rate of the forms, inverses, row factors and the kernels W_0, T. For fixed m let $N \to \infty$, so the cumulative dilation $M \to \infty$. At each nonzero real momentum p, $\lambda_M(p) \to |p|^2$; $A_a(p)$ converges as a product of filter factors whose deviations from one are summable; the nonprincipal aliases converge by dominated convergence with (11.6); and the noise contributions form a convergent series since M at least doubles at each step. Hence the symbols of H^0, $\mathsf{H}^{\pm 1}$, W_0 and T converge at every such p, and so does that of ρ by continuity of the positive square root. The uniform strip bounds make the coefficient arrays precompact under diagonal extraction; an array with these bounds is determined by its symbol at nonzero real momenta, so all subsequential limits coincide, and dominated convergence gives convergence at every smaller width. This proves (11.5). Absolute summation also gives convergence of every fixed periodization. □

11.3 The stationary background and Gaussian coordinates

Tree gauge. Given a spanning tree of a set of sites, rooted at x_0, let g_x be the ordered product of the links along the tree path from x_0 to x, and apply the gauge transformation $U(x,a) \mapsto g_x U(x,a) g_{x+e_a}^{-1}$. Every tree link becomes 1. The axial frames of Definition 11.1 are of this form, for the tree of coordinate routes in the order o. Figure 9 shows one plaquette. With root y and tree links $(y,1)$, $(y+e_1,2)$, $(y,2)$, one has $g_y = 1$, $g_{y+e_1} = U(y,1)$, $g_{y+e_1+e_2} = U(y,1)U(y+e_1,2)$ and $g_{y+e_2} = U(y,2)$, so the remaining link becomes

$$g_{y+e_2} U(y+e_2,1) g_{y+e_1+e_2}^{-1} = U_{y;12}^{-1}, \qquad U_{y;12} = U(y,1)U(y+e_1,2)U(y+e_2,1)^{-1}U(y,2)^{-1}.$$

Its chord is the plaquette chord, and a change of frame k at the root replaces it by $kU_{y;12}^{-1}k^{-1}$. In a larger block, every non-tree link inside the block becomes the holonomy of a closed loop, a product of conjugated plaquette holonomies of a lattice surface spanning that loop; its chord is at most the sum of their chords. Lemma 11.4 (axial branch) proves the summed form of this bound, and the coercivity step in the proof of Lemma 11.6 is its linearization. A link joining two blocks becomes a transport between the two block roots and is compared with the coarse link. The gauge choice therefore does not make the remaining links small: their size is that of the curvature and of the coarse field.



(a) before (b) tree gauge

Figure 9: Tree gauge on one plaquette. After the gauge transformation the tree links (thick) equal 1, and the remaining link carries the inverse of the plaquette holonomy based at the root y.

In each block $\ell y + \mathbb{B}_{(\ell-1)/2}$ fix an axial spanning tree rooted at ℓy, and set its tree links to 1. (Section 11.5 allows every rooted block-tree gauge.) The remaining logarithmic link coordinates form the axial field ψ. For a fine edge $e = (x, a)$ put

$$(\operatorname{rep} V)_e = \begin{cases} V_{(y,a)}, & y = \operatorname{parent}(x),\ \operatorname{parent}(x + e_a) = y + e_a, \\ 1, & \operatorname{parent}(x + e_a) = \operatorname{parent}(x), \end{cases}$$
$$U(v, \psi)_e = \exp((\iota_{\mathrm{ax}}\psi)_e)(\operatorname{rep}\exp v)_e.$$

Here $\operatorname{dir}(e) = a$ for $e = (x, a)$, parent takes the centred block containing a site, $\operatorname{parent}(e) = \operatorname{parent}(x)$, and ι_{ax} extends by zero on tree edges. The block-constant lift $\operatorname{rep} V$ is one on every tree edge. The fine leading action is the hypercubic average of

$$\mathcal{A}_f(U) = c_{\mathrm{dir},j} \sum_{\square} \operatorname{pr}(U_\square) + \tfrac{1}{2} \sum_{x,\mathfrak{p}} |\mathcal{L}^{\mathrm{path}}_{\mathfrak{p}}(U, x)|^2 + \sum_{x,i} (\mathfrak{a}^{(3)}_i + \mathfrak{a}^{(4)}_i)(U, x).$$

The transported curvatures and coefficients are those of Definition 12.2, with masks and clipping removed. The action excludes the inherited quadratic loop action and the common inverse coupling. Let $S(v, \psi)$ be this action at $U(v, \psi)$ plus

$$\mathsf{b} \sum_{e\ \mathrm{coarse}} \operatorname{pr}\big((\exp v)_e Q^{\mathrm{pred}}_e(U(v, \psi))^{-1}\big),$$

again without the common inverse coupling, and put $\mathsf{Q} = D^2_\psi S(0, 0)$. The path bounds make these unclipped sums analytic near zero.

Block norms. A fine field that is smooth on the block scale has energy of order $\mu^2\|u\|^2$, with $\mu = \ell^{-1}$, so ordinary ℓ^2 norms would lose a factor of order ℓ^2 in inverse bounds. The block norm removes this loss by weighting each fine Fourier alias by $(\mu^2 + |q|^2)^{1/2}$. Put $\mu = \ell^{-1}$, $\theta_z(\xi) = (\xi + 2\pi z)/\ell$, with $x, z \in \{0, \dots, \ell - 1\}^4$, and

$$\omega_z(\xi) = \big(\mu^2 + \sum_a |q_a(\operatorname{Re}\theta_z(\xi))|^2\big)^{1/2},$$
$$\mathcal{U}_\ell(\xi)_{xz} = \ell^{-2} \mathrm{e}^{i\theta_z(\xi)\cdot x}, \quad T_\ell(\xi) = \mathcal{U}_\ell(\xi) \operatorname{diag}(\omega_z(\xi)^{-1}) \mathcal{U}_\ell(0)^{-1}.$$

A fine Bloch field at coarse momentum ξ obeys $u(x+\ell n)=\mathrm{e}^{i\xi\cdot n}u(x)$, and $\hat u=\mathcal U_\ell(\xi)^{-1}u$ are its alias coordinates on the fixed cell. For real ξ the block norm and its dual are

$$\|u\|_{\mathrm{bl}}^2=\sum_z \omega_z(\xi)^2|\hat u_z|^2,\qquad \|f\|_{\mathrm{bl},*}^2=\sum_z \omega_z(\xi)^{-2}|\hat f_z|^2,$$

the fine H^1 norm with mass μ and its dual, in Bloch coordinates; coarse coordinates carry the Euclidean norm. Since $\mathcal U_\ell(\xi)$ is unitary for real ξ, $\|u\|_{\mathrm{bl}}=\|T_\ell(\xi)^{-1}u\|$; the last factor of T_ℓ fixes the reference cell. On the strip $\mathcal S_\eta=\{\xi\in\mathbb C^4:|\operatorname{Im}\xi_a|<\eta\}$ we use this formula as the definition: T_ℓ on fine links or sites, the identity on coarse coordinates, direct sums on products, and $T_\ell(-\xi)^{-T}$ on dual fields. Axial variables inherit the norm of their slice embedding. For kernels set

$$\|K\|_\eta^{\mathrm{bl}}=\sup\|T_{\mathrm{out}}(\xi_{\mathrm{out}})^{-1}K(\xi_{\mathrm{out}},\xi_{\mathrm{in}})T_{\mathrm{in}}(\xi_{\mathrm{in}})\|_{\mathrm{op}}\,,$$

the supremum over momenta whose indicated shifts all lie in $\mathcal S_\eta$. Use tensor products for several inputs; field insertions are normalized coarse directions. The weights use $\operatorname{Re}\theta_z$ and need not be analytic. Write $W_{\mathrm{ax}}v=(v,D\psi(0)v)$ for the axial tangent, and retain W_0 for the smooth free interpolation before axial gauge. They are distinct coordinate representatives.

The next lemma supplies the stationary configuration and identifies its first derivative with the free solution. Let ∇_f^r denote any product of r forward fine differences. Fix an integer $r_0\ge 0$ with $r_0\le 2$ or $j\ge r_0+3$, and let x be any centred fine residue. The constants $c_j>0$ and $C_{j,r}$ depend on the stage and derivative order, but not on the ratio, history or period.

Lemma 11.6 (Stationary interpolation)**.** *(i) Near zero there is an analytic stationary branch with*

$$D_\psi S(v,\psi(v))=0,\quad \psi(0)=0,\quad \mathsf Q\ge\nu_{j,\ell}I\,,$$

where $\nu_{j,\ell}>0$ does not depend on the period.

(ii) $D\psi(0)$ is the axial representative of the free interpolation: $\operatorname{rep}v+\iota_{\mathrm{ax}}D\psi(0)v$ is the tree representative of W_0v.

(iii) The averaging symbol equals one at zero and vanishes to order at least j at every nonzero alias.

(iv) For $0\le r\le r_0$ and every bounded coarse field v,

$$|\nabla_f^r W_0(\ell y+x,y')|\le C_{j,r}\ell^{-1-r}e^{-c_j|y-y'|},$$
$$\|\nabla_f^r W_0 v\|_\infty\le C_{j,r}\ell^{-1-r}\|v\|_\infty.$$

The proof uses a local axial estimate for coercivity, a finite-dimensional implicit equation for the branch, and the alias decay of the averaging symbol for the kernel bounds.

Proof. The minimizing choices are analytic. At the flat field the frame objective has quadratic part $\operatorname{av}_{r,r'}|h_r-h_{r'}|^2/2$. On $h_0=0$ this is positive definite. The predictor objective has Hessian I in its group variable. Their chart equations are real analytic. Their local complexifications and the analytic implicit function theorem make both minimizing choices real analytic near zero. Apply [20, Theorem 1.3.8] to the complexified chart equations, which vanish at zero and have the displayed invertible Hessians. The neighbourhoods may depend on the period.

For a fine infinitesimal field a, let $I_a(x\to x';o)$ be its signed sum along the route. Differentiating the frame equation gives

$$\tau_y(r)=-\operatorname{av}_{r',o}I_a(\ell y+r\to\ell y+r';o),\quad h'_{y,r}=\tau_y(r)-\tau_y(0),\quad \mathcal Q_m a=\mathcal M_m a+d_0\mathcal F_m a\,.$$

Here $\mathcal{F}_m a(y) = \tau_y(0) - \sum_r w_r^{\mathrm{av}} \tau_y(r)$. In particular $\mathcal{Q}_m d_0 \phi = d_0(\phi \circ \iota)$.

Coercivity on the axial slice. Telescoping the boundary of each coordinate rectangle expresses a free link as a block-constant coarse-link value plus curls in its enlarged block. Averaging the straight connectors replaces the coarse-link value by $\mathcal{Q}_m a$; the frame differences are route sums of the same curls. For example, interchanging two positive steps gives

$$a_a(x) + a_b(x + e_a) - a_b(x) - a_a(x + e_b) = (d_1 a)_{ab}(x).$$

Reverse steps change the sign and translate this identity. Repeated interchanges compare the tree route, a straight connector and its averaged translates. Thus, at each free edge e,

$$|a_e| \le C_{j,\ell} \sum_{\square \in \ell \ \mathrm{parent}(e) + \mathbb{B}_{(j+7)\ell}} |(d_1 a)_\square| + |(\mathcal{Q}_m a)_{\mathrm{parent}(e),\mathrm{dir}(e)}| \,.$$

This is an identity followed by a triangle inequality: tree-edge coordinates are zero, and every interchange of two successive route steps contributes exactly one signed curl. Cauchy–Schwarz and the bounded number of enlarged blocks containing a given plaquette give constants $A_{j,\ell}, B_{j,\ell} < \infty$, independent of the period, with

$$\|a\|_2^2 \le A_{j,\ell} \|d_1 a\|_2^2 + B_{j,\ell} \|\mathcal{Q}_m a\|_2^2 \,.$$

The row construction of Section 12.2, at the finer input level, identifies the leading Hessian with $\mathsf{K}_f = d_1^* \mathsf{H}_f d_1$; Lemma 12.4 fixes its normalization at zero momentum. Cubic and quartic curvature atoms have zero quadratic derivative. Consequently, for the slice embedding ι_{ax},

$$\langle \psi, \mathsf{Q}\psi \rangle = \langle d_1 \iota_{\mathrm{ax}} \psi, \mathsf{H}_f d_1 \iota_{\mathrm{ax}} \psi \rangle + \mathsf{b} \|\mathcal{Q}_m \iota_{\mathrm{ax}} \psi\|^2 \ge \nu_{j,\ell} \|\psi\|^2 \,,$$

with $\nu_{j,\ell} = \min(2\gamma_j, \mathsf{b})/(1 + A_{j,\ell} + B_{j,\ell})$ (Lemma 11.5(ii)).

The branch and its tangent. The map $\mathcal{E}(v, \psi) = D_\psi S(v, \psi)$ is analytic, $\mathcal{E}(0, 0) = 0$, and $D_\psi \mathcal{E}(0, 0) = \mathsf{Q}$. The same implicit function theorem gives $\psi(v)$. Differentiation yields

$$D\psi(0) = -\mathsf{Q}^{-1} D_v D_\psi S(0, 0) \,.$$

To identify this tangent, note that at the flat field the quadratic part of $S(v, \cdot)$ in the fine field $a = \mathrm{rep}\, v + \iota_{\mathrm{ax}} \psi$ is one half of $\langle a, \mathsf{K}_f a \rangle + \mathsf{b} \|\mathcal{Q}_m a - v\|^2$. By the energy identity (Step 6 of the proof of Lemma 11.5), $W_0 v$ minimizes it over all fine fields; a gradient $d_0 \phi$ with ϕ vanishing at the block roots changes neither term, and exactly one such modification of $W_0 v$ has zero tree links. By coercivity it is the unique minimizer over the slice, which proves (ii). The implicit neighbourhood may depend on the volume; the coercivity constant does not.

Kernel bounds. Finally $\widehat{w}^{\mathrm{av}}(\theta) = \widehat{w}_\ell(\theta)^j$. At a nonzero alias $2\pi z/\ell$, at least one centred geometric sum has a simple zero, so the product has a zero of order at least $j \ge 4$. At zero it equals one. For a word of r fine differences the alias multiplier is a product of r factors $q_{a_h}(\theta_z)$, each bounded by $C\ell^{-1}(1 + |z|)$. This includes a difference crossing a fine cell boundary: its Bloch phase shifts the coarse base and gives exactly the same multiplier. After separating the principal pole as in Lemma 11.5, the interpolation alias has the bound $C_j \ell^{-1} \prod_{a=1}^4 (1 + |z_a|)^{-j}$ on a common smaller strip. The centred-residue exponential is bounded there. Consequently the sum of absolute alias terms after r differences is at most

$$C_{j,r} \ell^{-1-r} \sum_{z \in \mathbb{Z}^4} (1 + |z|)^r \prod_{a=1}^4 (1 + |z_a|)^{-j} \le C'_{j,r} \ell^{-1-r} \qquad (r \le j - 2).$$

Indeed $(1+|z|)^r \le C_r \prod_a (1+|z_a|)^r$ and each remaining exponent is at least two. The pole-free principal term has the same bound. This proves the required symbol estimate through order two for $j \ge 4$, and through order r_0 for $j \ge r_0 + 3$. There are finitely many difference words at a fixed stage; take a common smaller strip for them. Shifting the four coarse Fourier contours towards the sign of the displacement gives the first bound in (iv). It includes all link components and is uniform in the centred fine residue x. Summing over y' proves the second bound. Periodization preserves the row bound by the triangle inequality.

The same kernel bound permits polynomially growing inputs. For $0 \le c' < c_j$ and $h \ge 0$ put

$$w_{c',h}(y) = e^{c'|y|}(1+|y|)^h, \qquad \|v\|_{c',h} = \sup_y \frac{|v(y)|}{w_{c',h}(y)}.$$

The triangle inequality gives $w_{c',h}(y')/w_{c',h}(y) \le e^{c'|y-y'|}(1+|y-y'|)^h$. Thus absolute kernel summation yields

$$\sup_{y,x} \frac{|\nabla_f^r W_0 v(\ell y + x)|}{w_{c',h}(y)} \le C_{j,r,c',h} \ell^{-1-r} \|v\|_{c',h}. \tag{11.7}$$

The sum converges since $c' < c_j$. In particular (11.7) applies to polynomial jets of any fixed degree and to a cubic remainder when $j \ge 6$. □

Gaussian coordinates. Write $U_v = U(v, \psi(v))$ and $\mathsf{Q}(v) = D_\psi^2 S(v, \psi(v))$. Let s change axial logarithmic increments to left increments at U_v, and let J_{ax} be the product Haar density in these coordinates divided by its value at zero. For $x = \ell y + r$ in its centred block, set $\Psi(U, h_{\mathrm{fr}})(x) = h_{\mathrm{fr}}(y) h_{y,r}(U)$, using the patch frame of Definition 11.2. This extends a coarse frame h_{fr} over each fine block. The joint variable is (U, h_{fr}), with gauge action $(U, h_{\mathrm{fr}}) \mapsto (U^\gamma, h_{\mathrm{fr}}(\gamma \circ \iota)^{-1})$. Let $\mathcal{R}$ insert fine site gauge parameters, and let $\mathcal{I}$ embed free-edge increments, with zero coarse-frame component. Thus $(\mathcal{R}, \mathcal{I}) : \mathfrak{g}^{\text{fine sites}} \oplus \mathfrak{g}^{\text{free edges}} \to \mathfrak{g}^{\text{fine edges}} \oplus \mathfrak{g}^{\text{coarse sites}}$ is square: the number of tree edges is the number of fine sites minus the number of block roots. Write $\mathsf{J}(U, h_{\mathrm{fr}})$ for the joint left Hessian of the leading action plus block penalty, with the coarse field fixed. We evaluate it along the stationary background; its value at the flat field has fine kinetic part K_f. Put

$$X = D_{w=0}[-\log(\Psi(U, h_{\mathrm{fr}})^{-1} \Psi(\mathrm{e}^w (U, h_{\mathrm{fr}})))], \qquad \mathcal{L}_{m+1} = \sum_{x,i} \mathfrak{a}_i^{(2)}(\,\cdot\,, x)\,.$$

The last sum is the inherited quadratic loop action of Definition 12.2. The *one-loop action* is

$$\mathsf{T}(v) = \mathcal{L}_{m+1}(U_v) + \tfrac{1}{2} \log\det \mathsf{Q}(v) - \log J_{\mathrm{ax}}(v).$$

It is the negative logarithm of the field-dependent Gaussian factor.

The determinant and coordinate density must be compared in the same counting coordinates. Let n be the number of real axial coordinates.

Lemma 11.7 (Gaussian coordinates)**.** *Near $v = 0$ the Hessian $\mathsf{Q}(v)$ is positive,*

$$J_{\mathrm{ax}} = |\det Ds| > 0, \qquad J_{\mathrm{ax}}(0) = 1, \qquad X\mathcal{R} = I, \qquad \det(\mathcal{R}, \mathcal{I})^2 = 1,$$

and, for every $\beta > 0$,

$$J_{\mathrm{ax}}(v)\, \mathrm{e}^{-\mathcal{L}_{m+1}(U_v)} \int \mathrm{e}^{-\frac{\beta}{2}\langle w, \mathsf{Q}(v) w\rangle}\, dw = \Big(\frac{2\pi}{\beta}\Big)^{n/2} \mathrm{e}^{-\mathsf{T}(v)}.$$

Thus the stationary Gaussian integral, including the inherited loop action and the Haar density, has field-dependent factor $\mathrm{e}^{-\mathsf{T}(v)}$.

The proof computes the Haar density in axial coordinates at the stationary point. Lemma 12.7 then rewrites the determinant in gauge-completed form; (11.10) below collects the resulting normalization.

Proof. Tree gauge is an exact Haar change of variables. In tree order, each tree link is a Haar translate of its child gauge variable, whose integral is one. For fixed V, gauge transformations fixing every block root leave the kernel of Definition 5.1 invariant. Integration of these gauge variables reduces its integral to product Haar measure on free links; root transformations are not integrated here. In exponential coordinates write this measure as $J_*^{n_e}\prod_e J(\psi_e)\,d\psi$, with $J(0)=1$ and n_e the number of free links.

At a fixed stationary axial point $y=\psi(v)$ the coordinate change on each free link is

$$s_y(\psi)_e=\log(\exp\psi_e\exp(-y_e)),\qquad s_y(y)=0\,.$$

Right invariance of Haar measure gives $\prod_e J(\psi_e)=|\det Ds_y(\psi)|\prod_e J(s_y(\psi)_e)$. Evaluate at $\psi=y$ to obtain $J_{\mathrm{ax}}(v)=|\det Ds_y(y)|$. Since $Ds_0(0)=I$, its value at zero is one, and it stays positive nearby. Stationarity removes the second-derivative term in the Hessian chain rule, giving

$$\mathsf{Q}(v)=(Ds)^*(\mathcal{I}^*\mathsf{J}\mathcal{I})Ds \tag{11.8}$$

for the joint left Hessian J.

Gauge covariance of Ψ gives $X\mathcal{R}=I$. In tree-height order, each tree-edge row of $(\mathcal{R},\mathcal{I})$ has its child gauge coordinate $\pm I$ and earlier gauge coordinates; each coarse-frame row has its root gauge coordinate $-I$, and every free-edge slice column has its identity entry. Thus $\det(\mathcal{R},\mathcal{I})^2=1$. Positivity of $\mathsf{Q}(0)$ from Lemma 11.6 persists near zero.

The Gaussian integral of the positive form $\frac{\beta}{2}\langle w,\mathsf{Q}(v)w\rangle$ equals $(2\pi/\beta)^{n/2}(\det\mathsf{Q}(v))^{-1/2}$. Multiplying by $J_{\mathrm{ax}}(v)\mathrm{e}^{-\mathcal{L}_{m+1}(U_v)}$ gives the displayed identity by the definition of T. The factors $J_*^{n_e}$ and $(2\pi/\beta)^{n/2}$ do not depend on v; they enter the scalar normalization. □

11.4 Inverse, vertex and Ward estimates

Section 12 extracts the curvature-square coefficient of T by comparing it with an auxiliary fine determinant. This subsection supplies three inputs for that comparison: inverse bounds uniform in the block ratio (Lemma 11.8), vertex bounds (Lemma 11.9), and a Ward identity for the predictor contact term (Lemma 11.10).

The regulator. For a fine site field put

$$(d_U\phi)_e=\phi(e^-)-\mathrm{Ad}(U_e)\phi(e^+),\quad \Delta_U=d_U^*d_U,\quad P_\mu=\mu^2(\Delta_U+\mu^2 I)^{-1},\quad \mathfrak{m}_\mu=\mu^2P_\mu^9\,,$$

with $\mu=\ell^{-1}$. The regulator $\mathfrak{m}_\mu$ acts as a mass at the block scale: by (11.12) below, $\Delta_0+\mathfrak{m}_\mu\ge 2^{-10}(\Delta_0+\mu^2)$, so below momentum μ it is a mass of order μ^2, while above μ it decays like $\mu^2(\mu/|k|)^{18}$. The ninth power is chosen because P_μ^3 is trace class at each coarse momentum, uniformly in ℓ (Lemma 11.8(i)). After two field derivatives of $\mathfrak{m}_\mu=\mu^2(P_\mu^3)^3$ one undifferentiated factor P_μ^3 remains, and its trace norm keeps the ℓ^4 fine aliases out of the determinant comparison.

The predictor contact term. Let $p(w)$ be the logarithm of the framed predictor, in left coordinates of the joint fine-link and frame variables w, and $\mathrm{Pen}(p)=\mathsf{b}\sum_e\mathrm{pr}\big((\exp v)_e(\exp p_e)^{-1}\big)$. The second-derivative chain rule gives

$$D_w^2(\mathrm{Pen}\circ p)=H^*BH+\sum_\lambda d^\lambda C_\lambda,\qquad H=D_wp,\ B=D_p^2\mathrm{Pen},\ d=D_p\mathrm{Pen},\ C_\lambda=D_w^2p_\lambda. \tag{11.9}$$

The index λ comprises a coarse link and a colour index. The last sum is the *predictor contact term*; this use of the word is unrelated to the contact extensions of Theorem 6.21. Along the smooth interpolated field, $d(0) = 0$ and $Dd(0) = -\mathsf{K}_c$, by the normal equation $\mathsf{b}(I - \mathcal{Q}_m W_0) = \mathsf{K}_c$ (Step 6 of the proof of Lemma 11.5).

The compared operators. The gauge-completed Hessian, the ghost operator and the auxiliary fine operator are

$$\mathsf{C}_\mu = \mathsf{J} + F_\mu^* F_\mu, \quad F_\mu = d_U^* P_f + \mathfrak{m}_\mu X, \qquad \mathsf{G}_\mu = \Delta_U + \mathfrak{m}_\mu, \qquad \mathsf{A}_\mu^{\mathrm{aux}} = \mathsf{A} + d_U d_U^* + \mathsf{M}_\mu,$$

where P_f projects to fine links, $\mathsf{A}(U)$ is the fine leading Hessian and M_μ applies $\mathfrak{m}_\mu$ to each link component. Frame covariance gives $F_\mu \mathcal{R} = \mathsf{G}_\mu$. Write $\mathsf{C}_\mu^{\mathrm{rem}}$ for C_μ with the contact term of (11.9) removed; at the flat field $\mathsf{C}_\mu^{\mathrm{rem}} = \mathsf{C}_\mu$, since $d(0) = 0$.

These operators give the one-loop action a single normalization. The second line below follows from (11.8) and $J_{\mathrm{ax}} = |\det Ds|$, the third from Lemma 12.7. For every $\mu > 0$ and v in a neighbourhood of zero,

$$\boxed{\begin{aligned} \mathsf{T}(v) &= \mathcal{L}_{m+1}(U_v) + \tfrac{1}{2}\log\det \mathsf{Q}(v) - \log J_{\mathrm{ax}}(v) \\ &= \mathcal{L}_{m+1}(U_v) + \tfrac{1}{2}\log\det(\mathcal{I}^* \mathsf{J}(v)\mathcal{I}) \\ &= \mathcal{L}_{m+1}(U_v) + \tfrac{1}{2}\log\det \mathsf{C}_\mu(v) - \log\det \mathsf{G}_\mu(v). \end{aligned}} \tag{11.10}$$

The factor $\frac{1}{2}$ comes from the Gaussian integral; the Haar Jacobian converts the axial Hessian to left coordinates: $\det \mathsf{Q} = J_{\mathrm{ax}}^2 \det(\mathcal{I}^* \mathsf{J} \mathcal{I})$; the ghost determinant enters with power one because the gauge block of the completion contributes $(\det \mathsf{G}_\mu)^2$ to $\det \mathsf{C}_\mu$. The left side does not depend on μ. Field-independent factors, such as $(2\pi/\beta)^{n/2}$ and the Haar normalizations, form the scalar normalization; they cancel from every normalized expectation, so the convergence of correlations does not require them to converge. Only the dependence of T on v matters for correlations: Section 12 extracts the coupling coefficient from its quadratic kernel, and its higher Taylor terms enter the activities of Section 13.

Insertion bounds. For weighted bounds use the nearest projection π from Lemma 11.4 and centred coarse frames: $c_y(U) = \pi(\sum_r w_r^{\mathrm{av}} h_{y,r}(U))$, $\widetilde{h}_{\mathrm{fr}} = h_{\mathrm{fr}} c(U)$. Pull back the joint quadratic forms by this change. Its tangent is triangular with identity diagonal, so it preserves determinants. Lemma 12.8 compares $\mathsf{C}_\mu^{\mathrm{rem}}$ with $\mathsf{A}_\mu^{\mathrm{aux}}$; derivatives are at zero in free interpolated coarse directions. Let A_μ denote either of these two operators.

The uninserted inverse and the geometric insertions are controlled by one constant C_j^{ins}, chosen before the nonlinear path bounds; the derivatives of the full Hessian also contain the input path coefficients. For $r = 1, 2$ let M_r be the supremum of $\|D^r A_\mu\|$ from the primal block space to its dual, in normalized smooth coarse directions, on any common smaller strip for these inserted kernels. Define

$$Q_0(C) = C, \qquad Q_1(C) = C^2 M_1, \qquad Q_2(C) = 2C^3 M_1^2 + C^2 M_2.$$

The norms M_1, M_2 retain the actual cubic and quartic path contributions. The proof of the next lemma also shows, on a smaller common strip,

$$\sup_\ell M_r \le \mathcal{P}_{j,r}(A_j^{\mathrm{path}}), \qquad r = 1, 2, \tag{11.11}$$

where A_j^{path} is the stage path bound of item 4 of Table 3; the fixed polynomials also depend on the interpolation, free metric, frame and penalty bounds.

These estimates control the determinant comparison without a factor equal to the number of fine sites.

Lemma 11.8 (Scaled inverse insertions)**.** *There are $\eta_j^{\mathrm{inv}} > 0$ and $C_j^{\mathrm{ins}} < \infty$, fixed before the ratio and path bounds, such that for $A_\mu \in \{\mathsf{C}_\mu^{\mathrm{rem}}, \mathsf{A}_\mu^{\mathrm{aux}}\}$, $0 \le r \le 2$ and every odd ℓ allowed by item 5 of Table 3:*

(i) $\sup_{\xi \in \mathcal{S}_{\eta_j^{\mathrm{inv}}}} \|P_\mu(\xi)^3\|_1 \le C_j^{\mathrm{ins}}$ *and* $\|D^r P_\mu\|_{\eta_j^{\mathrm{inv}}}^{\mathrm{bl}} \le C_j^{\mathrm{ins}}$;

(ii) $\|D^r(A_\mu^{-1})\|_{\mathrm{dual}\to\mathrm{primal}} \le Q_r(C_j^{\mathrm{ins}})$ *and* $\|\mu^2 D^r(A_\mu^{-1})\|^{\mathrm{bl}} \le Q_r(C_j^{\mathrm{ins}})$;

(iii) on the common smaller strip, block-penalty and frame insertions have norms at most C_j^{ins}.

We first prove real coercivity in block units, then extend it to a strip and differentiate the inverse. Constants depend on the free bounds of Lemma 11.5, the stage and b.

Proof. Step 1: the mass floor. At zero background, Δ_0 has symbol $\lambda(\theta)$. With $x = \lambda(\theta) \ge 0$,

$$2^{-10}(x+\mu^2) \le x + \mu^2 \left(\frac{\mu^2}{x+\mu^2}\right)^9 \le x + \mu^2 . \tag{11.12}$$

For $x \ge \mu^2$ the first term supplies the lower bound; for $x < \mu^2$ the parenthesis is at least $1/2$. On links, the exterior identity $\|d_1 u\|^2 + \|d_0^* u\|^2 = \lambda(\theta)\|u\|^2$ and Lemma 11.5 give the same lower bound, with a fixed positive factor, for the auxiliary fine operator. Changing its quadratic form by T_ℓ therefore gives a positive lower bound independent of ℓ.

Step 2: coercivity modulo gauge. In centred coordinates fine gauge insertion has coarse component $-\sum_r w_r^{\mathrm{av}} \phi(\ell y + r)$: differentiate $h_{y,r}(U^\gamma) = \gamma(\ell y) h_{y,r}(U) \gamma(\ell y + r)^{-1}$ and the centroid. The uncompleted joint form is

$$\langle d_1 u, \mathsf{H}_f d_1 u\rangle + \mathsf{b}\|\mathcal{M}_m u - d_0 \vartheta\|^2 .$$

Modulo a joint gauge transformation choose $\vartheta = 0$ and make every nonprincipal fine alias transverse. This is possible by dividing its longitudinal part by $\lambda(\theta_z)$; the remaining principal gauge parameter fixes ϑ, since the principal averaging multiplier is bounded below by (11.6). Write u_z for the resulting normalized alias coordinates and $H_> = \sum_{z\neq 0} \omega_z^2 |u_z|^2$. The fine curvature energy bounds $cH_>$. The averaging multiplier on the principal alias is at least $c_j\mu$ in unitary cell coordinates; the weighted norm of all other alias columns is at most C_j. Therefore

$$(c_j\mu|u_0|)^2 \le 2\|\mathcal{M}_m u\|^2 + 2C_j^2 H_>, \qquad \omega_0 \le C\mu .$$

Combining these inequalities shows that the uncompleted form at w is at least $c\,\mathrm{dist}_{\mathrm{bl}}(w, \mathrm{ran}\,\mathcal{R})^2$, with c independent of ℓ.

Step 3: gauge completion. Let $\mathcal{R}\phi$ be joint gauge insertion. Frame covariance gives $F_\mu \mathcal{R} = \Delta_0 + \mathfrak{m}_\mu$ at zero background. By (11.12),

$$\|\mathcal{R}\phi\|_{\mathrm{bl}} \le C\|(\Delta_0 + \mathfrak{m}_\mu)\phi\|_2, \qquad \|F_\mu w\|_2 \le C\|w\|_{\mathrm{bl}} .$$

For the second bound one must also estimate $\mathfrak{m}_\mu X$ on arbitrary fine increments. In the centred frame, X is a sum of normalized averages of routes of length at most $C_j\ell$. Its row and column sums are at most $C_j\ell$, so

$$\|Xu\|_2 \le C_j\ell\|u\|_2 \le C_j\mu^{-2}\|u\|_{\mathrm{bl}}, \qquad \|\mathfrak{m}_\mu X u\|_2 \le C_j\|u\|_{\mathrm{bl}}.$$

The coarse-frame part satisfies the same inequality since its embedding has ℓ^2 as counting-norm factor and $\|\mathfrak{m}_\mu\| \le \mu^2$. Also $\|d_0^* u\|_2 \le C\|u\|_{\mathrm{bl}}$. For the first bound a fine gauge component contributes $\omega_z|q(\theta_z)\phi_z| \le \omega_z^2|\phi_z|$, and its averaged coarse component is bounded by the summable alias filter times $\omega_z^2|\phi_z|$. For $w = w_\perp + \mathcal{R}\phi$, the completed energy bounds $c\|w_\perp\|_{\mathrm{bl}}^2 + \|F_\mu w\|_2^2$. The last two inequalities give

$$\|w\|_{\mathrm{bl}} \le \|w_\perp\|_{\mathrm{bl}} + C\big(\|F_\mu w\|_2 + C\|w_\perp\|_{\mathrm{bl}}\big)\,.$$

Squaring proves real coercivity of the completed form with a constant independent of ℓ.

Step 4: the complex strip. The uninserted quadratic symbols and the geometric frame and penalty kernels have bounded weighted norms on a fixed coarse strip. To verify this, each fine derivative is bounded by $C\omega_z$, a free interpolated field by $C\mu$, and normalized frame or predictor averages by the summable filter (11.6). At the flat field these operators depend only on the free quadratic action, the interpolation, the frame and the penalty. Their defining inserted kernels use the smooth interpolation; its fine differences of order $s \le 2$ are $O(\ell^{-1-s})$ by the alias estimate (11.6). For complex momenta, freeze the weights at a real momentum: if the coarse displacement has norm at most one, the geometric-sum formula and $|q(\theta_z(\xi)) - q(\theta_z(\xi'))| \le C\mu|\xi - \xi'|$ make the two weights comparable by a fixed factor. Cauchy's estimate [20, Proposition 1.2.2] on a smaller disc bounds the change of the conjugated symbol by $C\eta$. Choose η_j^{inv} so this is less than half the real lower bound. A Neumann series gives the weighted inverse on that strip.

Step 5: insertions. For inserted directions v, w, write $A_v = DA_\mu[v]$ and $\mathsf{R} = A_\mu^{-1}$. Direct differentiation gives

$$D\mathsf{R}[v] = -\mathsf{R}A_v\mathsf{R}, \qquad D^2\mathsf{R}[v,w] = \mathsf{R}A_v\mathsf{R}A_w\mathsf{R} + \mathsf{R}A_w\mathsf{R}A_v\mathsf{R} - \mathsf{R}D^2A_\mu[v,w]\mathsf{R}\,.$$

The geometric insertions have norm at most C: each smooth field is $O(\mu)$, each derivative on it is $O(\mu^2)$, and $\mu \le \omega_z$. With two arbitrary fine legs, Cauchy–Schwarz accounts for their two factors ω_z^{-1}; a difference on either leg supplies its corresponding ω_z. This bounds the block penalty, frame and Δ_U insertions independently of the nonlinear path coefficients. For the path contribution, expand each transported curvature through total degree four. Its third derivative has one curl and its fourth has zero or more differences after two smooth insertions. In either case two arbitrary fine legs contribute at most μ^{-2}, while the two smooth insertions in the fourth derivative contribute μ^2. For the third derivative a smooth insertion and the remaining curl give the same bound. Summation by parts moves a difference off a smooth leg when necessary; its path-length loss is summable against the exponential path bound. Each coefficient is a product of at most two path-row coefficients, or one atom coefficient. The row and atom sums therefore give a fixed polynomial $\mathcal{P}_{j,r}(A_j^{\mathrm{path}})$, uniformly in ℓ. Adding the geometric insertions proves (11.11). The inverse identities therefore give dual-to-primal norms C^2M_1 and $2C^3M_1^2 + C^2M_2$. The inclusion of a primal source in the dual space has norm at most $C\mu^{-2}$; include its fixed factor in C_j^{ins}. Multiplication by μ^2 gives precisely Q_1, Q_2. Applying the same identities to $\Delta_U + \mu^2 I$, with the geometric insertion bounds, gives the stated bounds for $D^r P_\mu$. No derivative of a weight is taken.

Step 6: the trace bound. In centred alias representatives,

$$\left|\frac{\mu^2}{\lambda(\theta_z(\xi)) + \mu^2}\right| \le C(1 + |z|^2)^{-1}\,.$$

For real ξ this follows from $|q(k)| \ge 2|k|_\infty/\pi$ on the fine momentum cube and $|\xi + 2\pi z|_\infty \ge \pi|z|_\infty$ for centred aliases and $\xi \in [-\pi, \pi]^4$. The complex denominator differs from its real value by a small

multiple of ω_z^2, and the Bloch Fourier matrices have bounded norms. Hence

$$\|P_\mu^3\|_1 \le C d_{\mathfrak{g}} \sum_{z\in\mathbb{Z}^4} (1+|z|^2)^{-3} < \infty\,.$$

In $P_\mu^9 = (P_\mu^3)^3$, two insertions leave one undifferentiated cube, whose trace norm bounds the regulator products; the remaining factors have bounded operator norms. □

The inverse is thus bounded in block units with a constant that sees neither the ratio nor the nonlinear path coefficients; these enter only through the insertion norms M_1, M_2.

Vertices. For a translation-invariant Taylor kernel, momentum order at least s means that every coefficient of total momentum degree less than s vanishes at zero. Let V_r be the degree-r Taylor kernel of the fine leading action minus Wilson, for $r = 2, 3, 4$. We use:

(a) smooth free directions $\mathcal{W}v = (v, W_0 v, 0)$ in the coarse-link, fine-link and centred coarse-frame variables;

(b) the joint action

$$\mathcal{S}(v,u,\vartheta) = \mathcal{A}_f(\mathrm{e}^u) + \mathsf{b}\sum_e \mathrm{pr}\left((\exp v)_e \left[Q^{\mathrm{pred}}(\mathrm{e}^u)^{h_{\mathrm{fr}}}\right]_e^{-1}\right), \qquad h_{\mathrm{fr}} = \exp\vartheta\, c(\mathrm{e}^u)^{-1},$$

where the superscript is the coarse gauge action and c is the centring map above, and its third and fourth derivatives $\mathcal{V}_3, \mathcal{V}_4$ at zero;

(c) the vertex bounds C_3, C_4: the suprema on a smaller strip $\mathcal{S}_\eta$ of the absolute scalar Fourier entries of $\mathcal{V}_3, \mathcal{V}_4$ on momentum-conserving tuples whose legs are all $\mathcal{W}v_i$ with $|v_i| \le 1$, or normalized coarse legs and one fine leg of block norm at most one; these include the predictor and frame derivatives;

(d) $S_{\mathrm{stat}}(v) = S(v, \psi(v))$, and, for two mixed cubic functionals annihilating joint gauge directions,

$$b_h = \mathcal{V}_3[\mathcal{W}v_{h,1}, \mathcal{W}v_{h,2}, \cdot], \qquad h = 1, 2,$$

with axial restrictions $b_{1,\mathrm{ax}}, b_{2,\mathrm{ax}}$.

These pairings express the stationary fourth derivative in smooth coordinates. The constants a_3, c_4 below depend only on preceding stage data; a_3', c_4' may also depend on the input path bound. Put $E_{\mathrm{ins}} = \max(1, C_j^{\mathrm{ins}})$ with the constant of Lemma 11.8.

The extra momentum order of the difference makes the classical matching bounds independent of the input amplitude after enlarging ℓ.

Lemma 11.9 (Vertex bounds). *(i) The kernels V_2, V_3, V_4 have momentum orders at least $3, 2, 1$.*

(ii) There is $u_j \ge \max(a_3', c_4')$, depending on the input path data, such that for $\ell \ge u_j$

$$C_3 \le a_3 + \ell^{-1} a_3' \le a_3 + 1, \qquad C_4 \le c_4 + \ell^{-1} c_4' \le c_4 + 1\,.$$

(iii) In normalized coarse directions, $|D^4 S_{\mathrm{stat}}| \le C_4 + 3C_3^2 E_{\mathrm{ins}}$.

(iv) Gauge-annihilating mixed functionals satisfy $b_1 \mathsf{C}_\mu^{-1} b_2 = b_{1,\mathrm{ax}} \mathsf{Q}^{-1} b_{2,\mathrm{ax}}$, with C_μ at the flat field.

We first prove the low-momentum cancellations, using the input arrays of Lemmas 12.3–12.4, then count powers under the interpolation of Lemma 11.6, and finally pair the cubic vertices through the completed inverse.

Proof. Momentum orders. The quadratic difference is

$$\widehat{V}_2(p) = \tfrac{1}{2}\widehat{d}_1(-p)^T(\widehat{\mathsf{H}}_f(p) - I)\widehat{d}_1(p)\,.$$

Its middle factor vanishes at zero and its two curls each supply one momentum, proving order three.

To include the nonlinear transport terms, expand a connection with connection components of weight one and coordinate derivatives of weight one. A plaquette curvature starts at weight two, with leading term $\partial_a A_b - \partial_b A_a + [A_a, A_b]$. Changing its transport path changes this term only at higher weight. The row part at weight four is therefore its curvature square with coefficient $c_{\mathrm{dir}}\delta_{\gamma\delta} + \sum_\alpha s_{\alpha\gamma}s_{\alpha\delta} = \delta_{\gamma\delta}$ by Lemma 12.4 at the finer input level; $s_{\alpha\gamma}$ are the path-row coefficients of Definition 12.1. Cubic and quartic curvature atoms start at weights six and eight. The Wilson action has the same weight-four term. All contributions of weight at most four consequently cancel in their difference. Polynomial connection jets vary independently. Expand their finite parallel transports through weight four and sum using the exponential path bound. A degree-r field term of weight at least five has at least $5 - r$ coordinate derivatives, equivalently that many momentum powers. This proves the three claimed orders, including cubic and quartic terms arising inside a quadratic curvature row.

Power counting. An analytic coefficient f vanishing at zero factors as

$$f(p) = \sum_a p_a \int_0^1 \partial_a f(tp)\,dt\,.$$

Apply this to the vanishing Taylor coefficients, successively, on a smaller strip. Equivalently in position space, write $\mathrm{e}^{ip\cdot x} - 1$ as a sum of coordinate differences; the losses are fixed powers of the true path length. The exponential path norm bounds their sums. Wilson's degrees $r = 2, 3, 4$ have orders $4 - r$; the differences have orders $5 - r$.

Use the fine-difference bound for W_0 of Lemma 11.6. For one arbitrary fine leg use Cauchy–Schwarz over the block. If it carries $s = 0$ or 1 differences, its sum of absolute values over the block is at most $C\ell^{3-s}\|u\|_{\mathrm{bl}}$: there are ℓ^4 sites, $\|u\|_2 \le \mu^{-1}\|u\|_{\mathrm{bl}}$, and $\|\nabla_f u\|_2 \le C\|u\|_{\mathrm{bl}}$. Pair this with the other bounded legs. Summation by parts places at most one difference on the arbitrary leg. This gives the same bound in the dual weighted norm for a mixed vertex.

There are ℓ^4 sites per block. A term with r fields and s fine differences thus has scale ℓ^{4-r-s}. Wilson gives scale 1 at its leading orders, and the differences give ℓ^{-1}. The penalty has a fixed bound: normalized route and offset averages have mass one, straight sums cancel the factor ℓ^{-1} in a smooth input, and implicit differentiation of the centred frame and predictor equations bounds their derivatives through order four. In centroid-normalized frame variables the linear equation is identity minus the constant projection; every remaining term is a normalized average, so these bounds have no factor counting patch sites. These contributions define a_3, c_4; the path-dependent sums define a_3', c_4'. Choose $u_j \ge \max(a_3', c_4')$, so that $\ell^{-1}\max(a_3', c_4') \le 1$ for $\ell \ge u_j$. This gives (ii), independently of the input path bound.

The exchange term. In the joint coordinates, the exact nonlinear tree gauge change and the stationary chain rule give

$$D^4 S_{\mathrm{stat}} = \mathcal{V}_4[\mathcal{W}^4] - \sum_{\text{three pairings}} \mathcal{V}_3[\mathcal{W}, \mathcal{W}, \cdot]\,\mathsf{C}_\mu^{-1}\,\mathcal{V}_3[\cdot, \mathcal{W}, \mathcal{W}].$$

Lemma 11.6(ii) identifies the axial tangent with the tree representative of W_0: their difference is an infinitesimal joint gauge direction, including its coarse-frame component. A joint gauge transformation of the axial stationary curve remains stationary and can be chosen with tangent $\mathcal{W}v$. Differentiating this curve gives the displayed formula for D^4S_{stat}. Each paired functional annihilates the infinitesimal gauge directions by differentiated stationarity and gauge invariance. It may therefore be paired through the completed inverse of Lemma 11.8. If $b\mathcal{R} = 0$, the shift $x \mapsto x - \mathcal{R}(F_\mu\mathcal{R})^{-1}F_\mu x$ sets $F_\mu x$ to zero and preserves both the uncompleted energy and $b(x)$. Thus its pairing through the completed inverse equals its pairing through the slice inverse, which proves (iv). Lemma 11.8 gives the constant E_{ins} independently of ℓ and the input path bound. Duality then bounds each pairing by $C_3^2E_{\mathrm{ins}}$. The three pairings in a symmetric fourth derivative have identical bounds. This proves (iii). □

The stationary Ward identity. On the smooth interpolated field write $\mathsf{C}_\mu^{\mathrm{rem}}(v)$ for the contact-removed completed Hessian, and let $d(v)$ and $C_\lambda(v)$ be the factors of (11.9) there. Thus the full Hessian differs from $\mathsf{C}_\mu^{\mathrm{rem}}(v)$ by $\sum_\lambda d^\lambda(v)C_\lambda(v)$. To keep the derivatives of the contact insertion as well as those of the inverse, introduce an independent coarse link source u and put

$$\mathsf{R}(v,u) = \mathsf{C}_\mu^{\mathrm{rem}}(v)^{-1}\sum_\lambda u^\lambda C_\lambda(v), \qquad \mathsf{F}(v,u) = \operatorname{tr}\mathsf{R}(v,u), \qquad \mathsf{F}_2(v,u) = \operatorname{tr}\mathsf{R}(v,u)^2.$$

The traces sum colour and cell coordinates. Their translation-invariant kernels, with internal momentum integrated using $d\theta/(2\pi)^4$, are

$$\mathsf{b}_{i\lambda} = \partial_{v_i}\partial_{u_\lambda}\mathsf{F}(0,0), \qquad \mathsf{c}_{\lambda\rho} = \partial_{u_\lambda}\partial_{u_\rho}\mathsf{F}_2(0,0).$$

In particular $\mathsf{c}_{\lambda\rho}$ includes the two product-rule terms. Put $\mathsf{a}_i(\xi) = -\mathsf{K}_c(\xi)e_i$. The contact difference Y^{con} is the Hessian kernel of one half of the full minus contact-removed logarithmic determinant, evaluated on the stationary branch.

The coefficient to be preserved is read off by

$$[t^2]g(t) = \tfrac{1}{2}g''(0), \qquad \alpha_b(Y) = 2\operatorname{Re}[t^2]\,Y_{(2,b),(2,b)}(te_1),$$

for a quadratic Taylor kernel Y, including the factor $1/2!$, and a colour index b. The Wilson kinetic form $\frac{1}{2}\langle d_1a, d_1a\rangle$ has entry $\frac{1}{2}|q_1(t)|^2 = \frac{1}{2}t^2 + O(t^4)$, hence $\alpha_b = 1$; thus $\alpha_b(Y)$ is the multiple of the curvature square that Y carries along transverse plane waves of small momentum. Section 12 uses α_b to define the one-loop coefficient. The cancellation below is exactly the one needed in Lemma 12.8.

Lemma 11.10 (Stationary Ward estimates)**.** *The first penalty derivative has momentum order two;* b *vanishes at zero and* c *is analytic there. For a fixed colour* b *and the transverse link index* $i = (2,b)$*,*

$$Y_{ii}^{\mathrm{con}}(te_1) = O(t^3), \qquad \alpha_b(Y^{\mathrm{con}}) = 0.$$

The proof derives the trace contraction from the determinant expansion and then applies the stationary gauge identity, through the tangent identification of Lemma 11.6, to the mixed kernel.

Proof. The free normal equation gives $\mathsf{b}(I - \mathcal{Q}_mW_0) = \mathsf{K}_c$. Hence the first penalty derivative is $\mathsf{a} = -\mathsf{K}_c$, of order two because $\mathsf{K}_c = d_1^*\mathsf{H}_cd_1$.

The determinant difference along the stationary curve is

$$\tfrac{1}{2}\log\det(I + \mathsf{R}(v,d(v))) = \tfrac{1}{2}\mathsf{F}(v,d(v)) - \tfrac{1}{4}\mathsf{F}_2(v,d(v)) + O(|v|^3).$$

The first derivatives of each invariant scalar determinant vanish at the flat field: each is an adjoint-invariant linear functional on $\mathfrak{g}$, hence zero because $\mathfrak{g} = [\mathfrak{g}, \mathfrak{g}]$. In particular $\partial_u \mathsf{F}(0,0) = 0$, so the chain-rule term $\partial_u \mathsf{F}(0,0)\, D^2 d(0)[v,w]$ vanishes and the Hessian along the stationary curve equals that along its tangent. Since F is linear in u, its pure v and pure u Hessians vanish at $(0,0)$. The mixed derivative retains both contributions

$$\partial_{v_i} \operatorname{tr}(\mathsf{C}_\mu^{\mathrm{rem}\,-1} C_\lambda) = \operatorname{tr}\big(-\mathsf{C}_\mu^{\mathrm{rem}\,-1}(\partial_{v_i}\mathsf{C}_\mu^{\mathrm{rem}})\mathsf{C}_\mu^{\mathrm{rem}\,-1} C_\lambda + \mathsf{C}_\mu^{\mathrm{rem}\,-1}\partial_{v_i} C_\lambda\big).$$

The two mixed chain-rule terms of F and the pure u Hessian of F_2 now give

$$Y_{ii}^{\mathrm{con}}(\xi) = \tfrac{1}{2}\{W_i(\xi) + W_i(-\xi) - \tfrac{1}{2}V_i(\xi)\},$$
$$W_i(\xi) = \sum_\lambda \mathsf{a}_i^\lambda(\xi)\mathsf{b}_{i\lambda}(\xi), \qquad V_i(\xi) = \sum_{\lambda,\rho} \mathsf{a}_i^\lambda(-\xi)\mathsf{c}_{\lambda\rho}(\xi)\mathsf{a}_i^\rho(\xi).$$

These identities hold first for finite momentum sums. Lemma 11.8 bounds their traces on a common strip, so period limits give the indicated integrals and their derivatives.

Write $y_\lambda(v) = \operatorname{tr}(\mathsf{C}_\mu^{\mathrm{rem}}(v)^{-1} C_\lambda(v))$. The stationary gauge identity is $Dy(0)d_0\phi = 0$: gauge covariance of the stationary field gives $y((\exp 0)^\gamma) = \mathrm{Ad}(\gamma)y(0) = 0$, and differentiation gives this identity since $y(0) = 0$. Lemma 11.6 identifies the stationary tangent with the smooth one, so it applies to the kernels just defined. In Fourier coordinates it gives

$$\sum_a q_a(-\xi)\mathsf{b}_{(a,b'),\lambda}(\xi) = 0$$

for each colour b' and source index λ. Restrict to $\xi = te_a$. Only $q_a(-te_a)$ remains, hence the corresponding component is zero for $t \neq 0$. Continuity gives $\mathsf{b}(0) = 0$ for every component. Analyticity implies $\mathsf{b} = O(|\xi|)$ and $\mathsf{c} = O(1)$. On the tested ray $W_i = O(t^3)$ and $V_i = O(t^4)$. The displayed diagonal identity therefore has zero second Taylor coefficient, proving the assertion. □

Removing the predictor contact term therefore changes the tested entry of the quadratic kernel only at order t^3, and leaves the curvature-square coefficient unchanged.

11.5 The local large-field energy

The Gaussian integral is performed only on part of each field configuration. Its complement must retain a positive energy per selected large-field test. This energy is the local sum of completed squares and signed error terms, on a consistent selection of tests. The estimate is deterministic and keeps every configuration: a large field is not removed from the measure but charged the energy of Lemma 11.12, which Lemma 11.14 turns into a suppression factor.

Fix one hypercubic orientation. In Sections 11.5 and 11.6 write

$$t_{\mathrm{f}} = t_{m+1}, \qquad t_{\mathrm{c}} = t_m, \qquad B = B_m, \qquad R_* = \lfloor B/100000 \rfloor, \qquad t_{\mathrm{hi}} = \widehat{\beta}_{m+1}^{-49/100}$$

for the fine and coarse thresholds, the support radius, the path cutoff and the threshold of very large tests. Similarly $c_{\mathrm{dir,f}} = c_{\mathrm{dir},j}$ and $c_{\mathrm{dir,c}} = c_{\mathrm{dir},J_m}$, with the same convention for c_{lin}. Recall $\widehat{\beta}_{m+1} = \widehat{\beta}_m + \kappa_G \log \ell$. Increase the coupling threshold so that $B \geq j + 4$. In these two subsections the estimates hold on every coarse torus of period $n_{\mathrm{tor}} \geq 100(B+1)$, with fine period ℓn_{tor}. These periods also satisfy $\ell n_{\mathrm{tor}} \geq 100((j+3)\ell+1)$, so the patch-size requirement used in Lemmas 11.4–11.7 holds. Use the same periodized kernels and local sector conditions. By the formula for t_m in Section 5 and $M_m^{\mathrm{cut}} = \exp((\log\log \widehat{\beta}_m)^2)$ in Section 9,

$$t_{\mathrm{f}}/t_{\mathrm{c}} \to 1, \qquad B^n t_{\mathrm{f}}/t_{\mathrm{hi}} \to 0 \qquad (\widehat{\beta}_m \to \infty,\ \ell \text{ fixed})$$

for every fixed n. All limits below are taken in this way.

Square completion. The input and output row maps on two-forms are $\mathcal{B}_f = (\sqrt{c_{\mathrm{dir,f}}}I, \rho_f)^T$ and $\mathcal{B}_c = (\sqrt{c_{\mathrm{dir,c}}}I, \rho_c)^T$. They satisfy $\mathcal{B}_f^*\mathcal{B}_f = \mathsf{H}_f$ and $\mathcal{B}_c^*\mathcal{B}_c = \mathsf{H}_c$. The free square-completion maps are

$$M_f = \mathcal{B}_f\mathsf{H}_f^{-1}T\mathcal{B}_c^*, \qquad M_n = \mathsf{b}^{-1/2}d_1^*\mathcal{B}_c^*,$$
$$N = \begin{pmatrix} I - \mathcal{B}_c\mathsf{H}_c^{-1}\mathcal{B}_c^* \\ \mathcal{W}_3^{1/2}d_2\mathsf{H}_c^{-1}\mathcal{B}_c^* \end{pmatrix}.$$

They satisfy $M_f^*M_f + M_n^*M_n + N^*N = I$. Indeed $P_c = \mathcal{B}_c\mathsf{H}_c^{-1}\mathcal{B}_c^*$ is the orthogonal projection onto the range of $\mathcal{B}_c$, so the first block of N contributes $I - P_c$, while multiplying (11.3) by $\mathcal{B}_c\mathsf{H}_c^{-1}$ on the left and by its adjoint on the right gives

$$\mathcal{B}_c\mathsf{H}_c^{-1}d_2^*\mathcal{W}_3d_2\mathsf{H}_c^{-1}\mathcal{B}_c^* = P_c - M_f^*M_f - M_n^*M_n.$$

Thus a coarse row vector splits isometrically into a part transferred from the fine rows (M_f), a part carried by the penalty (M_n), and a remainder (N) seen only off the range of $\mathcal{B}_c$ and by the three-form defect. For an infinitesimal pair of fine and coarse fields, let $X_f^{\mathrm{lin}}, X_c^{\mathrm{lin}}$ be their curls followed by $\mathcal{B}_f, \mathcal{B}_c$, and let Z^{lin} be $\sqrt{\mathsf{b}}$ times their linear predictor discrepancy. The free interpolation identities give

$$M_f^*X_f^{\mathrm{lin}} + M_n^*Z^{\mathrm{lin}} = X_c^{\mathrm{lin}}. \tag{11.13}$$

These identities apply to the untruncated flat kernels. We retain them when estimating the errors below.

For the actual rows, transport every kernel coefficient along the fixed coordinate route joining its endpoints, using the block-constant lift of the coarse configuration V. Retain a coefficient only when that route, measured in fine units, has length at most R_*. For a fine-to-coarse coefficient also retain its residue and all the endpoints of that route. Denote the resulting maps, respectively truncating M_f, M_n, N, by $\Phi, \Phi_n, \mathcal{N}$. Write L_{f,R_*} and L_{c,R_*} for the path rows of Definition 12.2 truncated by their true lengths in fine units. At a plaquette p put

$$X_f(p) = \big(\sqrt{c_{\mathrm{dir,f}}}F_p(U), L_{f,R_*}(U,p)\big), \quad X_c(p) = \big(\sqrt{c_{\mathrm{dir,c}}}F_p(V), L_{c,R_*}(V,p)\big),$$
$$v_c(p) = \big(\min(1, t_{\mathrm{c}}/\operatorname{ch}(V_p))\sqrt{c_{\mathrm{dir,c}}}F_p(V), \chi_p(V)L_{c,R_*}(V,p)\big),$$
$$Z(e) = \sqrt{\mathsf{b}}\,F(V_eQ_e^{-1}), \qquad Q = Q^{\mathrm{pred}}(U).$$

Here $F(g) = P_{\mathfrak{g}}(g - g^{-1})/2$, the ratio in v_c is one at zero chord, and χ_p is the relaxed threshold test χ^{rel} of (5.9) at the plaquette base. At each plaquette X_f, X_c and v_c have two components, direct and path; norms and inner products are taken over both. The capped coarse coordinates v_c are bounded by Ct_{f} everywhere. Put

$$K_f^{\mathrm{loc}}(p) = \operatorname{clip}_{t_{\mathrm{f}}}(\operatorname{ch}(U_p)) + \tfrac{1}{2}\chi_p(U)|L_{f,R_*}(U,p)|^2, \quad K_c^{\mathrm{loc}}(p) = \operatorname{clip}_{t_{\mathrm{c}}}(\operatorname{ch}(V_p)) + \tfrac{1}{2}\chi_p(V)|L_{c,R_*}(V,p)|^2,$$

the clips and tests using the input and output coefficients, respectively. We split the local kinetic-energy difference into six terms, labelled on the right by what each accounts for:

$$\begin{aligned}
E_f(p) &= K_f^{\mathrm{loc}}(p) + \tfrac{1}{2}|\Phi v_c(p)|^2 - \langle \Phi v_c(p), X_f(p)\rangle && \text{fine energy,}\\
E_n(e) &= \tfrac{\mathsf{b}}{2}\operatorname{ch}(V_eQ_e^{-1})^2 + \tfrac{1}{2}|\Phi_n v_c(e)|^2 - \langle \Phi_n v_c(e), Z(e)\rangle && \text{mismatch penalty,}\\
E_c(p) &= \langle v_c(p), X_c(p)\rangle - \tfrac{1}{2}|v_c(p)|^2 - K_c^{\mathrm{loc}}(p) && \text{coarse subtraction,}\\
E_N(q) &= \tfrac{1}{2}|\mathcal{N}v_c(q)|^2 && \text{complement square,}\\
E_L(p) &= \langle v_c(p), \Phi^*X_f(p) + \Phi_n^*Z(p) - X_c(p)\rangle && \text{linear defect,}\\
E_D(p) &= \tfrac{1}{2}\langle v_c(p), (I - \Phi^*\Phi - \Phi_n^*\Phi_n - \mathcal{N}^*\mathcal{N})v_c(p)\rangle && \text{quadratic defect.}
\end{aligned}$$

The fine and mismatch terms supply the energy of selected large-field tests, and $E_N \geq 0$. The defects E_L, E_D vanish when $\Phi, \Phi_n, \mathcal{N}$ are replaced by M_f, M_n, N and the rows by their linearizations, by (11.13) and $M_f^* M_f + M_n^* M_n + N^* N = I$. The index q runs over the two-form complement and three-form components of N. Summing over every anchor, the cross terms of E_f, E_n, E_c cancel against E_L, and the squares $\frac{1}{2}|\Phi v_c|^2 + \frac{1}{2}|\Phi_n v_c|^2 + \frac{1}{2}|\mathcal{N} v_c|^2 - \frac{1}{2}|v_c|^2$ cancel against E_D:

$$\sum E_f + \sum E_n + \sum E_c + \sum E_N + \sum (E_L + E_D) = \sum K_f^{\mathrm{loc}} + \tfrac{\mathfrak{b}}{2} \sum_e \mathrm{ch}(V_e Q_e^{-1})^2 - \sum K_c^{\mathrm{loc}}. \quad (11.14)$$

The adjoints in this identity are those of the truncated transported matrices, not the flat matrices. The right side is the truncated clipped kinetic energy difference; the higher atoms and omitted tails are separate factors in Section 13. The identity concerns the sum over all anchors. Lemma 11.12 bounds instead the sum of the six terms over the anchors near a selection; on such a partial sum the cross terms need not cancel and individual terms can be negative, so the defects E_L, E_D must be estimated.

Selections. Fix a mismatch multiplier $W > 0$, chosen after ℓ below, and write $n_e = \mathrm{ch}(V_e Q_e^{-1})$. The fine field is taken in the tree slice of a rooted block tree, that is, in the tree gauge of Figure 9 for a rooted spanning tree of each block.

Definition 11.11 (Selections and allocation sectors)**.** A *selection* $A = (A_f, A_n, A_c)$ consists of fine plaquettes, coarse mismatch edges and coarse sites; $|A| = |A_f| + |A_n| + |A_c|$. Its location consists of every projected plaquette corner, both endpoints of each edge, and its sites; $A^{[r]}$ is the radius-r coarse neighbourhood of the location. The selection is *consistent* with (U, V) if every test meeting $A^{[5B]}$ obeys

(i) $\mathrm{ch}(U_p) \geq t_{\mathrm{f}}/4$ for $p \in A_f$, and $\mathrm{ch}(U_p) \leq t_{\mathrm{f}}/2$ for $p \notin A_f$;

(ii) $n_e \geq W t_{\mathrm{f}}$ for $e \in A_n$, and $n_e \leq 2W t_{\mathrm{f}}$ for $e \notin A_n$;

(iii) in $A^{[5B]}$, A_c is the set of coarse sites touched by an output plaquette with $\mathrm{ch}(V_p) > t_{\mathrm{c}}$, and every chosen frame tag has positive mass.

An *allocation sector* fixes these selected tests, their assignment to connected components of their neighbourhoods, a block tree, and the frame and predictor tags. Its *local energy* $\mathcal{E}_A$ sums the six local energy terms over anchors in $A^{[2B]}$, fine anchors being projected to coarse sites.

Let D select the fine or mismatch tests exceeding t_{hi}, and let Aff be the set of output anchors within $4B/5$ of D, the *affected* anchors. Put $r_p^{\mathrm{f}} = \mathrm{ch}(U_p)$ for fine plaquettes and $r_p^{\mathrm{c}} = \mathrm{ch}(V_p)$ for coarse ones. All sums below are restricted to the local anchors defining $\mathcal{E}_A$.

The positive fine and mismatch terms absorb the losses of the signed terms.

Lemma 11.12 (Local large-field energy)**.** *Fix the stage path bounds, free interpolation and penalty. There is ℓ_0 such that for every odd $\ell \geq \ell_0$ there is $W_*(\ell)$ such that for every $W \geq W_*(\ell)$ there are $\kappa > 0$ and a coupling threshold with the following property. For $\widehat{\beta}_m$ above the threshold and every coarse period $n_{\mathrm{tor}} \geq 100(B+1)$,*

$$\mathcal{E}_A(U, V) \geq \kappa t_{\mathrm{f}}^2 |A|$$

for every rooted block-tree gauge, allocation sector and consistent selection A. The constant κ and the threshold are uniform in the period and the sector.

The proof gives $\kappa = a/2$ with

$$a = \min(c_{\mathrm{lin,f}}/8,\ c_{\mathrm{dir,f}}/128,\ \mathsf{b}/32,\ c_{\mathrm{dir,c}}/64),$$

so κ does not in fact depend on ℓ or W; only the coupling threshold does. Table 5 lists which positive energy pays for which loss. We first bound the fine, mismatch and coarse terms, the last in the three ranges $r_p^{\mathrm{c}} \le t_{\mathrm{c}}$, $t_{\mathrm{c}} < r_p^{\mathrm{c}} \le \delta$ and $r_p^{\mathrm{c}} > \delta$ for a small fixed δ; then the signed terms on the affected anchors Aff and on the remaining small-field region; and finally absorb the losses by choosing ℓ, then W, then the coupling threshold.

Event charged	Energy cost	Loss that the cost pays
selected fine plaquette, $t_{\mathrm{f}}/4 \le r_p^{\mathrm{f}} \le t_{\mathrm{f}}$	$\ge c_{\mathrm{dir,f}} t_{\mathrm{f}}^2/128$	its share $a t_{\mathrm{f}}^2$ of the bound
selected fine plaquette, $r_p^{\mathrm{f}} > t_{\mathrm{f}}$	$\ge \frac{1}{2} c_{\mathrm{lin,f}} t_{\mathrm{f}} r_p^{\mathrm{f}}$, after the dropped path rows	signed-term loss $C t_{\mathrm{f}} \ell^{-2} r_p^{\mathrm{f}}$ (ℓ large)
selected mismatch, $n_e \ge W t_{\mathrm{f}}$	$\ge \frac{\mathsf{b}}{8} n_e^2$	signed-term loss $C t_{\mathrm{f}} n_e$ (W large)
test above t_{hi} (in D)	$\ge a t_{\mathrm{f}} t_{\mathrm{hi}}$	affected-region losses $(4aK + Z_{\mathrm{aff}} B^4) t_{\mathrm{f}}^2$ per test (coupling large)
coarse flag	$\ge 4 a t_{\mathrm{f}}^2$ per incident output plaquette with $t_{\mathrm{c}} < r_p^{\mathrm{c}} \le \delta$	its share $a t_{\mathrm{f}}^2$; at most four flags per plaquette
output plaquette, $r_p^{\mathrm{c}} > \delta$	none; $E_c \ge -c_{\mathrm{lin,c}} t_{\mathrm{c}} r_p^{\mathrm{c}}$	lies in Aff; cancelled by the positive signed term there
output plaquette, $r_p^{\mathrm{c}} \le t_{\mathrm{c}}$; signed terms off Aff	none	losses $C t_{\mathrm{c}}^4$ per local anchor and $\frac{a}{4} t_{\mathrm{f}}^2$ per selected test (coupling large)
Coupling threshold: $(4aK + Z_{\mathrm{aff}} B^4) t_{\mathrm{f}}^2 \le a t_{\mathrm{f}} t_{\mathrm{hi}}$ and $C t_{\mathrm{c}}^4 K B^4 \le a t_{\mathrm{f}}^2/4$. At least $a t_{\mathrm{f}}^2 \lvert A\rvert/2$ remains, so $\kappa = a/2$.		

Table 5: Gains and losses in Lemma 11.12. Every selected test is charged energy; no configuration is excluded. The losses are absorbed in the order listed: ℓ, then W, then the coupling threshold.

Proof. The path bounds of Lemmas 12.3–12.4 give the row and column sums of the transported maps, and the free kernels of Lemma 11.5 give the flat identities; both are taken at the input and output levels already fixed in the descending construction.

Step 1: fine, mismatch and coarse terms. Exponential row bounds, together with unitary parallel transport, give row and column sums for Φ bounded by $C\ell^{-2}$ and $C\ell^2$ respectively. Both sums for $\Phi_n, \mathcal{N}$ are at most C. Truncation restricts every coarse displacement to $B/50$. The capped coordinates give $\|\Phi v_c\|_\infty \le C\ell^{-2} t_{\mathrm{f}}$ and $\|\Phi_n v_c\|_\infty \le C t_{\mathrm{f}}$.

Write $R_f = \sum_{A_f : r_p^{\mathrm{f}} > t_{\mathrm{f}}} r_p^{\mathrm{f}}$ and $R_n = \sum_{A_n} n_e$. For the direct component, $|F(g)| \le \mathrm{ch}(g)$ gives

$$E_f^{\mathrm{dir}}(p) \ge \tfrac{1}{2}(\sqrt{c_{\mathrm{dir,f}}}\, r_p^{\mathrm{f}} - |\Phi v_c(p,0)|)^2 \quad (r_p^{\mathrm{f}} \le t_{\mathrm{f}}).$$

Choose ℓ so $|\Phi v_c| \le \epsilon_\ell t_{\mathrm{f}}$, with $\epsilon_\ell \le \sqrt{c_{\mathrm{dir,f}}}/8$. Each selected $t_{\mathrm{f}}/4 \le r_p^{\mathrm{f}} \le t_{\mathrm{f}}$ contributes $c_{\mathrm{dir,f}} t_{\mathrm{f}}^2/128$. For $r_p^{\mathrm{f}} > t_{\mathrm{f}}$, the linear clip and $\epsilon_\ell \sqrt{c_{\mathrm{dir,f}}} \le c_{\mathrm{lin,f}}/4$ leave $3 c_{\mathrm{lin,f}} t_{\mathrm{f}} r_p^{\mathrm{f}}/4$.

The path component is a square when its relaxed threshold test holds. When the test fails the term has a cross term of either sign, which must be summed. If a path row based at p is dropped, some plaquette q obeys $r_q^{\mathrm{f}} > t_{\mathrm{f}} \exp(\mathrm{dist}(p,q)/R_j^{\mathrm{rel}})$. For a fixed q, the number of possible p is at most $C_{R_j^{\mathrm{rel}}}(r_q^{\mathrm{f}}/t_{\mathrm{f}})$: its radius is $R_j^{\mathrm{rel}} \log(r_q^{\mathrm{f}}/t_{\mathrm{f}})$, and $(1 + R_j^{\mathrm{rel}} \log y)^4 \le C_{R_j^{\mathrm{rel}}} y$ for $y \ge 1$. Since $r_q^{\mathrm{f}} \le 2$,

each witness is within $R_j^{\mathrm{rel}} \log(2/t_{\mathrm{f}}) \le B/50$ for large $\widehat{\beta}_m$. It therefore lies in the neighbourhood where consistency applies and belongs to A_f. Split a path row into endpoints with chord at most t_{f} and those above t_{f}. The row bound controls the first part by Ct_{f} times the number of dropped rows; the column bound controls the second by CR_f. Hence

$$\sum_{\text{dropped } p} |L_{f,R_*}(U,p)| \le C(C_{R_j^{\mathrm{rel}}} + 1)R_f.$$

Multiplication by $|\Phi v_c| \le \epsilon_\ell t_{\mathrm{f}}$ loses at most $c_{\mathrm{lin,f}} t_{\mathrm{f}} R_f/4$ after increasing ℓ. Therefore

$$\sum E_f \ge \tfrac{1}{2} c_{\mathrm{lin,f}} t_{\mathrm{f}} R_f + \tfrac{c_{\mathrm{dir,f}}}{128} t_{\mathrm{f}}^2 |A_f \setminus \{r_p^{\mathrm{f}} > t_{\mathrm{f}}\}|.$$

For a selected mismatch $n_e \ge W t_{\mathrm{f}}$, $|\Phi_n v_c| \le C t_{\mathrm{f}}$. Choose W so this is at most $\sqrt{\mathsf{b}}\, n_e/2$. The same square inequality gives

$$\sum E_n \ge \tfrac{\mathsf{b}}{8} \sum_{A_n} n_e^2.$$

For E_c the path component cancels exactly, including its relaxed threshold test. Put $\alpha_p = \min(1, t_{\mathrm{c}}/r_p^{\mathrm{c}})$. The direct component is

$$E_c(p) = c_{\mathrm{dir,c}}(\alpha_p - \alpha_p^2/2)|F(V_p)|^2 - \mathrm{clip}_{t_{\mathrm{c}}}(r_p^{\mathrm{c}}).$$

The nearest-point Taylor expansion gives $|F(g)|^2 \ge \mathrm{ch}(g)^2 - C\,\mathrm{ch}(g)^4$ for small chord: the deficit is the squared normal displacement of $g - 1$, whose normal part is $O(\mathrm{ch}(g)^2)$. Fix δ small enough that also $|F(g)|^2 \ge 3\,\mathrm{ch}(g)^2/4$ below δ. Since $c_{\mathrm{dir,c}} > 16 c_{\mathrm{lin,c}}$, the three ranges yield

$$E_c(p) \ge \begin{cases} -C t_{\mathrm{c}}^4, & r_p^{\mathrm{c}} \le t_{\mathrm{c}}, \\ c_{\mathrm{dir,c}} t_{\mathrm{c}} r_p^{\mathrm{c}}/4, & t_{\mathrm{c}} < r_p^{\mathrm{c}} \le \delta, \\ -c_{\mathrm{lin,c}} t_{\mathrm{c}} r_p^{\mathrm{c}}, & r_p^{\mathrm{c}} > \delta. \end{cases}$$

The last range uses only nonnegativity of the first term. Fix the constant a above and set $\epsilon = a/4$.

Step 2: signed terms near very large tests. We first compare large output plaquettes with fine data. Let $\mathcal{P}_y(U) = \sum_q m_y(q)\,\mathrm{ch}(U_q)$ be the fine plaquette sum in the lifted macro-patch centred at ℓy, with $m_y(q)$ its actual multiplicity there. The macro radius is $(j+7)\ell$, so $\sum_y m_y(q) \le K_j$ independently of ℓ and the period. On its axial frame, denote a straight connector by Y_z and its two root adjustments by p_0, p_1. The averaging weight satisfies $\ell^4 w_z^{\mathrm{av}} \le 1$. Telescoping each length-ℓ connector and the two end routes, then summing their crossings, gives

$$\ell^4 \sum_z w_z^{\mathrm{av}}\,\mathrm{ch}(p_0^{-1} Y_z p_1) \le C_j \ell^2 \mathcal{P}_y(U).$$

For the straight segment, a fine edge occurs in at most ℓ translated segments, and the eligible axial link sum is at most $C_j \ell \mathcal{P}_y(U)$. The two end-frame sums obey the same $C_j \ell^2 \mathcal{P}_y(U)$ bound: write their coordinate routes, count each link at most $C_j \ell$ times after the normalized transverse average, and use the nearest-point comparison for the minimizing frame. The axial alternatives use their defining link-sum bound instead. The predictor branch comparison bounds its adjusted chord by C_{pred} times this weighted chord sum, on either branch. Around the four sides of an output plaquette the root adjustments cancel. Comparing its four links with Q gives

$$r_p^{\mathrm{c}} \le \sum_{e \in \partial p} n_e + C_j \ell^{-2} \mathcal{P}_y(U).$$

A coarse plaquette thus sees fine curvature only with weight ℓ^{-2}, and each fine plaquette enters boundedly many such comparisons. In an eligible axial frame a fine link differs from its replicated coarse link by at most C times the macro-patch plaquette sum, plus the mismatch on a boundary edge. Compare the four sides of a coarse plaquette, average the straight paths, and use the predictor equation. Each fine plaquette occurs with total coefficient at most $C\ell^{-2}$; mismatch chords enter linearly. Split each fine chord into its part at most t_{f} and its selected excess, and each mismatch into its part at most $2Wt_{\mathrm{f}}$ and its selected part. The resulting bound has the form

$$r_p^{\mathrm{c}} \le C_{j,\ell,W} t_{\mathrm{f}} + \Delta_p, \qquad \sum_p \Delta_p \le C\Big(\ell^{-2} \sum_{A_f : r_q^{\mathrm{f}} > t_{\mathrm{f}}} r_q^{\mathrm{f}} + \sum_{A_n} n_e\Big).$$

The excess constant depends on the fixed stage, predictor and penalty bounds and is fixed before ℓ, W; only the bounded-field term may depend on them. This follows by summing the actual route multiplicities before enlarging a macro-patch. Applying the same bound to $\Phi^* X_f$ and $\Phi_n^* Z$, with the column estimates above, bounds their sums on Aff by C times that excess sum plus $C_{j,\ell,W} t_{\mathrm{f}} |\mathrm{Aff}|$. Since $|v_c| \le C t_{\mathrm{f}}$, these estimates give

$$\sum_{\mathrm{Aff}} (E_L + E_D) \ge -C t_{\mathrm{f}} (\ell^{-2} R_f + R_n) - C_{j,\ell,W} t_{\mathrm{f}}^2 |\mathrm{Aff}|.$$

For E_D, expand its four operator terms. The bounds on $v_c, \Phi v_c, \Phi_n v_c, \mathcal{N} v_c$, together with both row and column sums of the three maps, bound each term by $C t_{\mathrm{f}}^2$ per anchor. The constants also depend on the predictor and penalty bounds.

Every output plaquette with $r_p^{\mathrm{c}} > \delta$ lies in Aff for large $\widehat{\beta}_m$: outside Aff the macro-patch comparison has no fine or mismatch test above t_{hi}, so $r_p^{\mathrm{c}} \le C_{j,\ell,W} t_{\mathrm{hi}} < \delta$. On Aff the same comparison and $t_{\mathrm{c}} \le 2t_{\mathrm{f}}$ give

$$c_{\mathrm{lin,c}} t_{\mathrm{c}} \sum_{r_p^{\mathrm{c}} > \delta} r_p^{\mathrm{c}} \le C_{j,\ell,W} t_{\mathrm{f}}^2 |\mathrm{Aff}| + C t_{\mathrm{f}} (\ell^{-2} R_f + R_n).$$

Add this inequality to the previous lower bound. There is a geometric constant K, fixed before the coupling threshold, with $|\mathrm{Aff}| \le K B^4 |D|$ and at most $K B^4 |A|$ local output anchors. Increasing the constants defines $Z_{\mathrm{aff}} = C_{j,\ell,W} K$ and gives

$$\sum_{\mathrm{Aff}} (E_L + E_D) \ge c_{\mathrm{lin,c}} t_{\mathrm{c}} \sum_{r_p^{\mathrm{c}} > \delta} r_p^{\mathrm{c}} - C t_{\mathrm{f}} (\ell^{-2} R_f + R_n) - Z_{\mathrm{aff}} t_{\mathrm{f}}^2 B^4 |D|.$$

The positive term cancels the third coarse-plaquette range of Step 1; the losses are paid by the fine excess (through ℓ), the mismatches (through W) and the tests in D.

Step 3: signed terms in the small-field region. For an anchor outside Aff, no fine chord or mismatch above t_{hi} occurs in the surrounding macro-patches. The preceding plaquette comparison gives coarse plaquette chords at most $C_{j,\ell,W} t_{\mathrm{hi}}$ on this region. Choose an axial gauge in a radius-$O(B)$ coarse cube. Lemma 11.4 bounds its link chords by CB times the sum of its $O(B^4)$ plaquette chords, hence by $C_{j,\ell,W} B^5 t_{\mathrm{hi}}$. Its block-constant extension acts on the fine field. Comparing a fine link with its replicated coarse link uses the local block-tree route; its fine plaquette sum is $O_{j,\ell}(t_{\mathrm{hi}})$ and its boundary mismatch is at most t_{hi}. Both fields are consequently small in this common gauge, with the convenient bound

$$\delta_{\mathrm{loc}} = C_{j,\ell,W} B^{10} t_{\mathrm{hi}}$$

through distance $2B/5$. In this region compare each transported kernel with its flat uncut kernel. A path of length n has transport error at most $Cn\delta_{\mathrm{loc}}$, by telescoping its product. Let $\tau_0 > 0$ be a

common exponential decay rate of the fixed-stage kernels. Their tails are at most $Ce^{-\tau_0 R_*}$. Each curvature or predictor differs from its linearization by $C\delta_{\mathrm{loc}}^2$. With $\epsilon_{\mathrm{loc}} = (1+R_*)\delta_{\mathrm{loc}} + \mathrm{e}^{-\tau_0 R_*}$, subtract (11.13) and $M_f^* M_f + M_n^* M_n + N^* N = I$ before applying the triangle inequality. This gives

$$\|\Phi^* X_f + \Phi_n^* Z - X_c\| \le C\delta_{\mathrm{loc}}\epsilon_{\mathrm{loc}}, \qquad \|(I - \Phi^*\Phi - \Phi_n^*\Phi_n - \mathcal{N}^*\mathcal{N})v_c\| \le Ct_{\mathrm{f}}\epsilon_{\mathrm{loc}}.$$

The assertion is local: extend the small logarithms by zero off the neighbourhood; the cut rows do not change there. Their range fits inside $2B/5$ for sufficiently large $\widehat{\beta}_m$. Thus each signed term is bounded below by $-C(t_{\mathrm{f}}\delta_{\mathrm{loc}} + t_{\mathrm{f}}^2)\epsilon_{\mathrm{loc}}$. Since $B^n t_{\mathrm{hi}}^2/t_{\mathrm{f}} \to 0$ and $B^n e^{-\tau_0 R_*}/t_{\mathrm{f}} \to 0$ for every fixed n, their total is at least $-\epsilon t_{\mathrm{f}}^2|A|$. Away from very large tests, the transported identities thus hold up to errors that are small compared with t_{f}^2 per selected test.

Step 4: absorption of the losses. The fine and mismatch bounds leave at_{f}^2 per selected fine or mismatch test, $at_{\mathrm{f}}t_{\mathrm{hi}}$ per test in D, and still retain the losses $Ct_{\mathrm{f}}(\ell^{-2}R_f + R_n)$ from the signed terms. First increase ℓ to handle the fine loss, then W to handle the mismatch loss. Coarse flags can be assigned to incident output plaquettes with chord above t_{c}, with at most four flags per plaquette. Plaquettes above δ number at most $K|D|$ by the macro-patch comparison; all local output anchors number at most $KB^4|A|$. Using $t_{\mathrm{f}} \le 2t_{\mathrm{c}}$, the middle range of Step 1 leaves $4at_{\mathrm{f}}^2$ per remaining such plaquette and so covers every other coarse flag.

The large output plaquette loss cancels against the positive term of Step 2; $E_N \ge 0$. The residual losses are at most $(4aK + Z_{\mathrm{aff}}B^4)t_{\mathrm{f}}^2|D| + Ct_{\mathrm{c}}^4 KB^4|A| + at_{\mathrm{f}}^2|A|/4$. Increase the coupling threshold, with ℓ, W, a fixed, until

$$(4aK + Z_{\mathrm{aff}}B^4)t_{\mathrm{f}}^2 \le at_{\mathrm{f}}t_{\mathrm{hi}}, \qquad Ct_{\mathrm{c}}^4 KB^4 \le at_{\mathrm{f}}^2/4.$$

The limits stated at the beginning of this subsection justify both choices. At least $at_{\mathrm{f}}^2|A|/2$ remains. Take $\kappa = a/2$. Every multiplicity used a fixed-radius local count or a row or column sum; none counts the period. This proves the asserted uniformity and the stated order of choices. □

11.6 Sector Gaussian integration

We now specify the Gaussian law for simultaneous local integration. It differs from the law of Lemma 11.7, whose covariance $\mathsf{Q}(v)^{-1}$ defines the one-loop action. The sector law is the reference law under which products of local factors are estimated, and its covariance is built from the linearized retained square terms of an allocation sector. Lemma 11.14 uses no identity between the two covariances: the difference between the sector integrand and the reference law is carried exactly by the determinant and Gaussian-density correction factors listed before Lemma 11.14.

Fix the allocation, its actual block trees, the selected tests, all tags, and the frozen fine links Y. The tree links equal 1 by the gauge choice of Section 11.3 and are not variables; the frozen links are free links, held fixed in the Gaussian calculation and integrated afterwards with Haar measure. Active coordinates are the remaining free-edge logarithms in the complement of the selected radius-$10B$ neighbourhoods. Linearize the retained fine and mismatch square terms at the block-constant lift of the coarse field, applying their actual cutoffs and stencils. Their stacked affine map is $a \mapsto \mathsf{D}a + b_Y$, so their quadratic energy is

$$Q_Y(a) = \tfrac{1}{2}|\mathsf{D}a + b_Y|^2, \qquad \mathsf{A}_Y = \mathsf{D}^*\mathsf{D}, \qquad f_Y = \mathsf{D}^* b_Y.$$

The stack includes the hypercubic average with its square-root normalization. A stencil for a term anchored at z contains every fine edge with projected distance at most $R_* + 5 + r_{\mathrm{pred}}$ from z; r_{pred}

is the maximum coarse distance read by a patch predictor. Stencils are thus fixed in advance and do not depend on the configuration.

Two windows are used, one for each calculation. The *sector window* gives the reference law of the exact Gaussian change of variables in an allocation-sector integral; it uses the largest distance between projected endpoints and the threshold $B/20$. The *local window* gives the reference law of the differentiated local integral of Lemma 11.14; it uses the smallest such distance and the threshold B. For an edge-coordinate i write E_i for the two endpoints projected to the coarse lattice. The two symmetric windows are

$$\Omega_i^{\mathrm{sec}} = \{i' : \max_{u\in E_i, v\in E_{i'}} \mathrm{dist}(u,v) \le B/20\},$$
$$\Omega_i^{\mathrm{loc}} = \{i' : \min_{u\in E_i, v\in E_{i'}} \mathrm{dist}(u,v) \le B\}.$$

Each window is restricted to the coordinates of its sector; unused coordinates may be completed by identity modes. For either prescription put

$$P_{i'i} = \mathbf{1}_{i'\in\Omega_i}(\mathsf{A}_Y|_{\Omega_i}^{-1})_{i'i}, \qquad \Gamma = \tfrac{1}{2}(P+P^*), \qquad m_Y = -P^* f_Y. \tag{11.15}$$

Table 6 lists the roles of these objects. The compressed columns make the covariance finite range: $\Gamma_{ii'} = 0$ unless $i' \in \Omega_i$ or $i \in \Omega_{i'}$. Factors whose dependence sets contain their windows and are disjoint are therefore exactly independent (Lemma 11.14(v)). For large B each window contains every coordinate within distance $B/60$ of i, since the two projected endpoints are at distance at most one; Lemma 11.13 uses only this property, so it applies to both prescriptions, with their own retained affine terms and mean. The Schur norm of a kernel is the maximum of its absolute row and column sums.

Object	Definition	Role
rows $\mathsf{D}a + b_Y$	retained fine and mismatch square terms, linearized	quadratic energy $Q_Y(a) = \frac{1}{2}\lvert\mathsf{D}a + b_Y\rvert^2$
precision A_Y	$\mathsf{D}^*\mathsf{D}$	coercive with exponential decay; its inverse is in general not finite range
columns Pe_i	inverse of A_Y restricted to Ω_i, zero outside	solve $\mathsf{A}_Y P e_i = e_i$ inside Ω_i (Figure 10)
covariance Γ	$\frac{1}{2}(P+P^*)$	finite range; $\Gamma \ge cI$ by comparison with A_Y^{-1}, not by symmetry
mean m_Y	$-P^* f_Y$	approximates the minimizer $-\mathsf{A}_Y^{-1} f_Y$ of Q_Y

The original integrand is not changed: the Gaussian substitution in Step 1 of the proof of Lemma 11.14 is exact for every mean and every positive covariance, and the differences from A_Y^{-1} and $-\mathsf{A}_Y^{-1} f_Y$ remain in its retained factor.

Table 6: The sector Gaussian law. Lemma 11.13 bounds the differences from the full inverse by $O(e^{-cB})$.

Fix a coarse root o for the derivative directions. The coarse flag tests remain the correct ones at the base field. Derivatives hold the selected index sets, Y and the tags fixed, and are taken in coarse directions of size at most $t_{\mathrm{c}}(1+\mathrm{dist}(o,e))^q$. Put $\upsilon_i = (1+\mathrm{dist}(o,i))^q$; each derivative moves one such factor to the coordinate i. Assume $0 < \beta \le 2\widehat{\beta}_{m+1}$, a weaker hypothesis than $\beta \in \mathfrak{B}_{m+1}$ that suffices here, and put $g = \beta^{-1/2}$. All bounds below hold for sufficiently large $\widehat{\beta}_m$.

For any principal coordinate set Ω, write $\mathsf{A}_\Omega = \mathsf{A}_Y[\Omega,\Omega]$ and $\Gamma_\Omega = \Gamma[\Omega,\Omega]$; their inverses are extended by zero outside Ω. They control the Gaussian marginals and the compressed columns in (11.15). In the next statement, K ranges over A_Y, A_Y^{-1}, Γ, A_Ω^{-1} and Γ_Ω^{-1}, and E over $\mathsf{A}_Y\Gamma - I$ and $\Gamma - \mathsf{A}_Y^{-1}$. At derivative order zero, exponential entry bounds give exponential Schur bounds by summing over coordinate locations.

The following estimate justifies differentiating the common Gaussian law and replacing its window covariance by the full inverse.

Lemma 11.13 (Window covariance bounds)**.** *For each fixed derivative order k and direction power q, there are $c > 0$ and $C_k < \infty$, depending only on the fixed stage and ratio data and k, q, such that, uniformly in the period, sector and Ω, for $0 \le r \le k$,*

$$\begin{aligned} \mathsf{A}_Y \ge cI, \qquad \Gamma \ge cI, \\ |D^r K_{ii'}| \le C_k \mathrm{e}^{-c\,\mathrm{dist}(i,i')} \upsilon_i^r, \qquad |D^r \mathsf{E}_{ii'}| \le C_k \mathrm{e}^{-cB - c\,\mathrm{dist}(i,i')} \upsilon_i^r, \\ |D^r m_Y(i)| \le C_k t_{\mathrm{f}} \upsilon_i^r, \qquad |D^r [g^{-1}(\mathsf{A}_Y m_Y + f_Y)]_i| \le C_k \mathrm{e}^{-cB} \upsilon_i^r. \end{aligned}$$

Coercivity comes from the axial estimate of Lemma 11.6, decay from a Combes–Thomas argument, and the window errors from the fact that a compressed column solves $\mathsf{A}_Y x = e_i$ exactly inside its window.

Proof. Coercivity. The linearized square terms and an eligible predictor control a fine link by those term values in a fixed macro-patch plus $C\delta_x$ times the surrounding fine increments, where $\delta_x = C_{j,\ell} t_{\mathrm{c}}$ tends to zero. Summing squares, using the bounded number of macro-patches meeting an edge, gives

$$\|a\|_2^2 \le C_{j,\ell} \|\mathsf{D} a\|_2^2 + C_{j,\ell} \delta_x^2 \|a\|_2^2.$$

For large $\widehat{\beta}_m$ the last coefficient is at most $1/2$, proving $\mathsf{A}_Y \ge cI$. All-frozen sectors have zero-dimensional active space and the assertion is vacuous there. Adding identity coordinates for inactive links preserves this lower bound.

Decay. Exponential path coefficients bound the weighted row and column sums of D. Products give the same Schur bound for A_Y. By the Combes–Thomas argument, for sufficiently small fixed $\theta > 0$ multiply the coordinates by $\mathrm{e}^{\theta\,\mathrm{dist}(i,\cdot)}$ on the left and by its inverse on the right. The resulting matrix differs from A_Y by at most $C\theta$ in operator norm: $|\mathrm{e}^{\theta d} - 1| \le \theta d \mathrm{e}^{\theta d}$ is summable against the exponential coefficient bound. Choose $C\theta < c/2$. The inverse Neumann series now gives exponential row and column bounds for A_Y^{-1} and every principal compression, with constants independent of the number of coordinates.

Windows. A compressed inverse column solves $\mathsf{A}_Y P e_i = e_i$ inside its window. Outside that window, its residual at i' is $\sum_{i'' \in \Omega_i} (\mathsf{A}_Y)_{i'i''} P_{i''i}$, a sum of products whose combined length $\mathrm{dist}(i,i'') + \mathrm{dist}(i'',i')$ crosses the boundary (Figure 10). Every window contains the coordinates within $B/60$ of i, and there are only $C_{j,\ell}$ coordinates per coarse site. The exponential Schur bounds therefore give $\|\mathsf{A}_Y P - I\|_{\mathrm{Schur}} \le C e^{-cB}$, and the identity $P - \mathsf{A}_Y^{-1} = \mathsf{A}_Y^{-1}(\mathsf{A}_Y P - I)$ gives

$$\|P - \mathsf{A}_Y^{-1}\|_{\mathrm{Schur}} \le C e^{-cB}, \qquad \|\Gamma - \mathsf{A}_Y^{-1}\|_{\mathrm{Schur}} \le C e^{-cB}.$$

The entrywise column and inverse bounds retain exponential decay in $\mathrm{dist}(i,i')$, while the preceding difference has the factor e^{-cB}. Taking the geometric mean of these bounds and decreasing c gives both factors together. For sufficiently large B, the uniform upper bound on A_Y and $\mathsf{A}_Y^{-1} \ge \|\mathsf{A}_Y\|^{-1} I$ give $\Gamma \ge c' I$ for a fixed $c' > 0$. Its row and column sums are bounded by a fixed constant. The

same estimate gives $\mathsf{A}_Y\Gamma - I = O(\mathrm{e}^{-cB})$. Apply the exponential conjugation argument to Γ and its principal compressions; their common lower bound gives the stated bounds for Γ_Ω^{-1}.

Derivatives. Differentiate A_Ω^{-1} by $D(\mathsf{A}_\Omega^{-1}) = -\mathsf{A}_\Omega^{-1}(D\mathsf{A}_\Omega)\mathsf{A}_\Omega^{-1}$ and iterate through order k. The index sets Ω stay fixed. Each derivative of a path coefficient is bounded by a fixed power of the path length and by the direction weight; $1 + \mathrm{dist}(o, i') \le (1 + \mathrm{dist}(o, i))(1 + \mathrm{dist}(i, i'))$ moves it to v_i, so r derivatives give v_i^r. A smaller exponential rate sums all remaining powers. In every differentiated residual, at least one chain still crosses the window boundary. This proves the displayed derivative bounds for A_Y and $\mathsf{A}_Y\Gamma - I$. For the inverse difference, use

$$\Gamma - \mathsf{A}_Y^{-1} = \mathsf{A}_Y^{-1}(\mathsf{A}_Y\Gamma - I).$$

Every differentiated product retains the exponentially small residual factor. The same inverse differentiation formula, applied to A_Y, Γ and their principal compressions, proves the remaining bounds for K. The spatial weights are moved to the first index as above; reducing the decay rate sums the intervening polynomial factors.

Mean. The affine source is formed from the same retained terms and from a bounded smooth extension of the frozen logarithms. Its derivatives are $O(t_{\mathrm{f}})$ with the same spatial weights. Multiplication by P^* proves the mean bound. Subtracting the exact inverse solution gives $\mathsf{A}_Y m_Y + f_Y = (I - \mathsf{A}_Y P^*) f_Y$, whose derivatives are $O(t_{\mathrm{f}} e^{-cB})$. Finally $g^{-1} \le C\widehat{\beta}_{m+1}^{1/2}$ and $B = \lceil (\log \widehat{\beta}_m)^2 \rceil$; this fixed power is absorbed by decreasing c. The estimates refer to the actual $\mathsf{A}_Y, \Gamma, m_Y$ of (11.15) throughout. □

The windowed law thus costs only e^{-cB} relative to the full Gaussian, and this bound persists through every fixed number of derivatives.

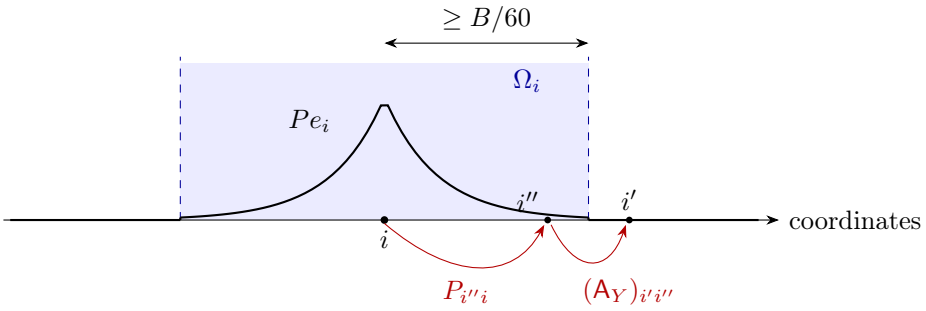


Figure 10: A compressed column. Inside the window Ω_i, $\mathsf{A}_Y P e_i = e_i$ exactly. At $i' \notin \Omega_i$ the residual is a sum of products $(\mathsf{A}_Y)_{i'i''} P_{i''i}$ with $i'' \in \Omega_i$; each decays exponentially in $\mathrm{dist}(i, i'') + \mathrm{dist}(i'', i') \ge B/60$.

The differentiated local integral. We state the local estimate for the finite products used in the expansion. In a fixed allocation sector, join selected large-field tests when their fixed enlarged neighbourhoods intersect; the connected components A_η are the *large-field components*. Write $A(M)$ for the union of the fine-plaquette, mismatch and coarse-site selections belonging to the components chosen in a product M. Fix the support enlargement h_{loc} once for the local factors. Compatibility means that the radius-$h_{\mathrm{loc}}B$ neighbourhoods of the distinct chosen components are disjoint.

A factor has a complete read support S_i containing the projected endpoints of every link it uses, every path and relaxed threshold test in its formula, every tag and frozen link, and the entire

window Ω_a^{loc} for each Gaussian coordinate a it reads. Thus a change outside S_i leaves its formula and its Gaussian marginal unchanged. A finite product M is a *polymer* when the intersection graph of its supports is connected. Choose o in their union. Its size s_M is the sum of the numbers of radius-B blocks covering these supports, including retained path lengths in the same units.

For an incoming forced activity retain its support Y_i and inventory I_i from Section 5. The admitted products satisfy

$$Y_i \cap Y_{i'} = \varnothing \quad (i \neq i'), \qquad I_i \subset \bigcup_{p \in A_f(M)} \mathrm{corners}(p),$$
$$\mathrm{Flag}_m(V) \cap \bigcup_{i \in M} S_i = \varnothing.$$

They also satisfy the component compatibility just specified. These conditions retain exactly the incidence and separation used in local integration.

Some incoming forced activities are only measurable, so the differentiated integral cannot be estimated by differentiating every factor in its link arguments. We instead change Gaussian coordinates as the coarse field varies, so that each physical fine link read by such an activity stays fixed. Derivatives then act on the smooth factors, the Jacobian and the Gaussian density. The nested chart cutoffs below ensure that this change is valid wherever a differentiated factor is nonzero.

Factors and their bounds. The retained square corrections are the exponential differences between the exact fine or mismatch terms and their affine squares $|\mathsf{D}a + b_Y|^2/2$. The other smooth factors are incoming regular activities, chart densities, and determinant and Gaussian-density corrections. Incoming forced activities are bounded by their measurable supremum envelopes.

For the chart cutoffs choose a smooth $\chi : [0,\infty) \to [0,1]$ equal to one on $[0,1]$ and zero on $[2,\infty)$. On an active logarithmic coordinate put $\chi_{a_*}(u_e) = \chi(|u_e|/(a_* t_{\mathrm{f}}))$. The fixed mean bound is $|m_Y(e)| \le c_* t_{\mathrm{f}}$, and choose $a_* > c_* + 1$. Fix a logarithm radius $r_{\log} > 0$ and take $\widehat{\beta}_m$ large enough that $4a_* t_{\mathrm{f}} < r_{\log}/2$. The smaller chart is $|u_e| < a_* t_{\mathrm{f}}$; the broader cutoff equals one for $|u_e| \le 2a_* t_{\mathrm{f}}$. A protected link is an active link accompanied by the broader cutoff χ_{2a_*} whenever an incoming measurable factor reads it. Nonactive free links are the frozen links Y; their physical values are held fixed. The nesting $\chi_{a_*} \neq 0 \Rightarrow \chi_{2a_*} = 1$ is what allows the physical-link change below on all derivative supports.

Cutoff remainders are the departures from one of the fine-plaquette and mismatch cutoffs, the frame and predictor tag likelihoods, the inventory tests, or the active chart cutoffs. The addresses are fine plaquettes for fine cutoffs, coarse edges for mismatches and predictor tags, coarse sites for frame tags, fine sites for inventory tests, and fine edges for chart cutoffs. A site's location is its coarse projection; an edge or plaquette is located at its projected initial vertex. Near a large-field component the corresponding departure is included in its component factor. Away from those components, each remaining cutoff remainder w and all its derivatives are supported on

$$E_w = \{z : \exists i,\ \mathrm{dist}(i, \mathrm{loc}(w)) \le 2B, \quad |z_i| \ge \xi t_{\mathrm{f}}/(2g)\}.$$

Indeed, if all these coordinates are below that threshold, then $|m_Y(e) + g z_e| \le (c_* + \xi/2) t_{\mathrm{f}} < a_* t_{\mathrm{f}}$ after reducing ξ, so both chart cutoffs equal one. The local predictor bound then makes the fine chord less than $t_{\mathrm{f}}/4$ and the mismatch less than $W t_{\mathrm{f}}$, and both tag likelihoods and the inventory test equal one. The strict margin persists on derivative supports. Choose the fixed $\xi > 0$ small enough for these finitely many local implications. Let $\mathcal{J}(M)$ be the set of distinct selected cutoff-remainder

addresses and $E_{\mathcal{J}(M)} = \bigcap_{w\in\mathcal{J}(M)} E_w$ their simultaneous event. Their amplitudes are bounded by the fixed stage-and-ratio constant

$$M_w = 2 + |\mathbb{B}_R \times S_4| + |\mathbb{B}_R|.$$

This includes the likelihood relative to uniform tags. Fix any $\tau_w > 0$ before the coupling threshold, and put $\varepsilon_w = \widehat{\beta}_{m+1}^{-\tau_w}$. For a factor i that is not a cutoff remainder, a_i is the product of its absolute coefficient bound (for a square correction, its deterministic envelope), its selected-test or path suppression, and the Gaussian exponential factor $\exp(C\Lambda_i)$ of Step 4 of the proof below. The weights are

$$f_i = a_i \quad (i \notin \mathcal{J}(M)), \qquad f_w = M_w\varepsilon_w \quad (w \in \mathcal{J}(M)).$$

Each distinct address contributes once.

The reference law and the transported integrand. Assume $\beta \in \mathfrak{B}_{m+1}$. Use the local prescription Ω^{loc} in (11.15) throughout this local integral. Conditional on the fixed sector and tags the reference law is

$$d\nu = dH_F(Y)\, d\gamma_\Gamma(z).$$

Here H_F is product Haar measure on the frozen tree-slice links and γ_Γ is the centred Gaussian. Reference tags are independent uniform tags; their actual likelihoods are included among the factors. All estimates below are uniform in those tags. Table 7 records which variables are fixed, integrated or changed.

Variable	In the Gaussian calculation	Afterwards, or as V varies
tree links	equal to 1 by the gauge choice	not variables
sector data	selected tests and block trees fixed	fixed; derivatives keep the selected index sets
tags	fixed; reference law uniform	summed; actual likelihoods are factors
frozen links Y	fixed	integrated under product Haar measure H_F
active coordinates	$u = m_Y + gz$, $z \sim \gamma_\Gamma$	replaced by z_V; protected physical links stay fixed
coarse field	base field V_0	differentiated at V_0

Table 7: Variables of the differentiated local integral.

A finite set of Gaussian reads I has marginal covariance $\Gamma_I = \Gamma[I, I]$. For a finite selection write its product as

$$F_M(V, Y, z) = \prod_{i\in M\setminus\mathcal{J}(M)} F_i(V, Y, z) \prod_{w\in\mathcal{J}(M)} W_w(V, Y, z),$$

where F_i are the factors other than cutoff remainders and W_w the cutoff remainders. The transported product uses one change of Gaussian coordinates on their common set of reads.

To differentiate while holding physical fine links fixed, fix a base coarse field V_0 and write $u = m_Y(V_0) + gz$. For a nearby V, set

$$u_V(e) = \log\left(\exp(u(e))(\operatorname{rep} V_0)_e(\operatorname{rep} V)_e^{-1}\right), \qquad z_V = g^{-1}(u_V - m_Y(V)).$$

Use a fixed smooth extension of this formula outside $|u(e)| < r_{\log}/2$, equal to the identity plus a bounded correction there. Its derivative is invertible for V sufficiently near V_0, uniformly over the

chart argument; the extension is used only with the stated protected cutoff. On active protected links,

$$\exp(u_V(e))(\operatorname{rep} V)_e = \exp(u(e))(\operatorname{rep} V_0)_e.$$

Every incoming measurable activity evaluated on those links is therefore constant under this differentiation. It is bounded by its supremum majorant and no field derivative is applied to it. For example, if V differs from V_0 only on the coarse link (y, a), then $\operatorname{rep} V$ differs from $\operatorname{rep} V_0$ only on the fine edges from block y to block $y + e_a$; on all other edges $u_V(e) = u(e)$ (inside the logarithm chart), while z_V may also change on other coordinates through the mean $m_Y(V)$; on those edges, $z_V(e) - z(e) = g^{-1}(m_Y(V_0)_e - m_Y(V)_e)$.

The density of the changed marginal, relative to the base one on the union I of all reads, is

$$\mathcal{R}_I(V, z) = |\det D_z z_{V,I}| \, \frac{(\det \Gamma_I(V))^{-1/2} \exp[-\frac{1}{2} z_{V,I}^T \Gamma_I(V)^{-1} z_{V,I}]}{(\det \Gamma_I(V_0))^{-1/2} \exp[-\frac{1}{2} z_I^T \Gamma_I(V_0)^{-1} z_I]}.$$

Define the transported integrand by

$$\widetilde{F}_M(V, Y, z) = \mathcal{R}_I(V, z) F_M(V, Y, z_V).$$

The Jacobian and the Gaussian density in $\mathcal{R}_I$ depend on all the reads of the product together, not factor by factor; this is why the whole product is estimated in one integral under the common law. The next lemma bounds its ordered derivatives at V_0 under the same base law, including products containing incoming measurable forced activities. It is the second local input to Section 13. Its bounds may depend on the fixed mismatch multiplier W and energy constant κ of Lemma 11.12.

For an admitted polymer put $A = A(M)$. For a large-field component with selection $A = (A_f, A_n, A_c)$ and $A_c \neq \varnothing$, let F_A contain its full local sector factor and every read meeting it. Adjoin every large-field component met by those reads. In this value estimate only, replace the healthy-support condition within this component by the consistency condition of Definition 11.11. Retain the incoming-support disjointness and inventory incidence. The other factors attached to F_A have the bounds f_i above; their product excludes the component's selected-test suppression and cutoff remainders already included in that component. Fix $0 < \kappa'_B < \kappa$ before the coupling threshold. Let s_A count the radius-B blocks of its complete read support, including retained path lengths in those units. The component itself is estimated in value; derivatives fall only on attached smooth factors and the common Gaussian density. Fix the orders k, q and take odd ℓ above the stage-data threshold. A function has *upper rate* r if it is at most $C_\epsilon \widehat{\beta}_{m+1}^{r+\epsilon}$ for every $\epsilon > 0$ once $\widehat{\beta}_m$ is large. Write $P = P_{j,\ell,k,q}$, of upper rate zero, and $d = d_{j,\ell,k,q}$; the large-field component bound may also depend on κ'_B. Directions and sector parameters have the restrictions of Lemma 11.13.

Lemma 11.14 (Simultaneous local integration). *(i) Admitted polymers M have integrable majorants G_M for their ordered derivatives through order k, with*

$$\int G_M \, d\nu \leq P(\widehat{\beta}_m)(1 + s_M)^d \prod_{i \in M} f_i.$$

(ii) If $A_c = \varnothing$, the selected-test envelopes satisfy, through order k,

$$\prod_{i \text{ selected test}} a_i \leq e^{-\kappa'_B \beta t_{\mathrm{f}}^2 |A|}.$$

(iii) A large-field component with $A_c \neq \varnothing$ has a value majorant G_A with

$$\int G_A \, d\nu \leq P(\widehat{\beta}_m)(1+s_A)^d e^{-\kappa'_B \beta t_{\mathrm{f}}^2 |A|} \prod_{i \text{ attached}} f_i.$$

(iv) Simultaneous cutoff remainders at r distinct addresses have probability at most ε_w^{2r}.

(v) Disjoint complete read supports factorize, including Haar and tag coordinates.

The window bounds of Lemma 11.13 control the Gaussian scores and exponentials, the energy of Lemma 11.12 supplies the suppression, and finite range gives the factorization.

Proof. Step 1: the affine Gaussian substitution. For any chart-supported F_Y,

$$\begin{aligned}\int \mathrm{e}^{-\beta Q_Y(a)} F_Y(a)\, da = (2\pi g^2)^{n/2} (\det \Gamma)^{1/2} \mathrm{e}^{-\beta Q_Y(m_Y)} \\ \times \int F_Y(m_Y + gz) \exp\left\{ \langle \eta_Y, z\rangle + \tfrac{1}{2}\langle z, (\Gamma^{-1} - \mathsf{A}_Y) z\rangle \right\} d\gamma_\Gamma(z), \\ \eta_Y = -g^{-1}(\mathsf{A}_Y m_Y + f_Y).\end{aligned}$$

Here n counts active real coordinates; the Haar density and chart cutoffs are included in F_Y. Expansion of $Q_Y(m_Y + gz)$ proves the identity, including its determinant and residual linear term. The identity holds for every vector m_Y and every positive Γ, in particular for either window prescription. The choices (11.15) only make the reference law finite range; the exponents of the retained factor vanish exactly when $m_Y = -\mathsf{A}_Y^{-1} f_Y$ and $\Gamma = \mathsf{A}_Y^{-1}$, and Step 4 estimates that factor. The local expansion in Section 13 uses Ω^{loc}.

Step 2: scores. For the common change of coordinates defined above, the first logarithmic derivative of the Gaussian density is

$$-\tfrac{1}{2}\operatorname{tr}(\Gamma_I^{-1}\dot{\Gamma}_I) + \tfrac{1}{2} z_V^T \Gamma_I^{-1} \dot{\Gamma}_I \Gamma_I^{-1} z_V - z_V^T \Gamma_I^{-1} \dot{z}_V.$$

Add the logarithmic Jacobian derivative. Repeated differentiation bounds every score through order k by a polynomial in the absolute values of the marginal Gaussian coordinates, with coefficients controlled by Lemma 11.13 and the fixed smooth chart extension. The number of placements is a fixed polynomial in $1 + s_M$, since k is fixed and $|I| \leq C_{j,\ell} B^4 s_M$. Connectivity and the chosen root bound all distances from o by $C_{j,\ell} B(1+s_M)$, so the spatial direction weights give another fixed polynomial.

The normalized coarse directions have size $O(t_{\mathrm{c}})$ with their stated spatial weights. On a fixed chart, derivatives of its multiplication, logarithm and bounded extension are bounded. The mean derivatives are $O(t_{\mathrm{f}})$, while $t_{\mathrm{c}}/t_{\mathrm{f}}$ stays bounded. In z_V the additional normalization is g^{-1}. Thus the score coefficients are bounded by a fixed power of $g^{-1}t_{\mathrm{c}}$, B and $1 + s_M$. Since

$$g^{-1} t_{\mathrm{c}} \leq C \exp\{(\log\log \widehat{\beta}_{m+1})^2\},$$

all fixed powers have upper rate zero. The small inverse and stationarity residuals in Lemma 11.13 have the same property after this normalization.

Step 3: square corrections. The chart-supported square corrections have deterministic envelopes: on their support all logarithms are $O(t_{\mathrm{f}})$, the affine remainder is $O(t_{\mathrm{f}}^2)$, and $|\frac{1}{2}|r+e|^2 - \frac{1}{2}|r|^2| \leq |r||e| + |e|^2/2$ gives an energy correction $O(\beta t_{\mathrm{f}}^3)$. The exponentially summable path coefficients

bound the same quantities with fixed path-length powers after differentiation. These bounds enter a_i.

Step 4: unbounded exponentials. The unbounded exponentials come from the residual Gaussian density and its local covariance comparisons. Write their upper exponents as $\eta_i \cdot z + z^T K_i z/2$, with $K_i \geq 0$. At the base field the Gaussian likelihood is one; its derivatives are the polynomial scores already estimated. Its exponent tends to zero with $V - V_0$, so on a sufficiently small neighbourhood its weighted row sums are at most $\delta_B = C_{j,k} B^d \mathrm{e}^{-cB}$. The residual Gaussian density has this bound by Lemma 11.13. The resolvent identity for each compressed inverse gives $C\delta_B$ for its local quadratic corrections. Each factor's coefficients decay exponentially from its own address, and each address has bounded multiplicity. Thus the sum at a coordinate is bounded by $C\delta_B \sum_y \mathrm{e}^{-c\,\mathrm{dist}(y,a)} \leq C'\delta_B$, uniformly in the period and number of factors. Here y ranges over address locations and a is the coordinate location. In the linear terms, weighted Cauchy–Schwarz at each coordinate gives

$$\left| \Gamma^{1/2} \sum_i \eta_i \right|^2 \leq C \sum_i \Lambda_i, \qquad \left\| \Gamma^{1/2} \sum_i K_i \Gamma^{1/2} \right\| \leq C\delta_B.$$

To specify the linear bound, let w_{ia} be the sum of the absolute path weights contributing to coordinate a of η_i; bounded endpoint multiplicity gives $\sup_a \sum_i w_{ia} \leq C$. Put

$$\Lambda_i = \sum_a \frac{|\eta_i(a)|^2}{w_{ia}} + \mathrm{tr}(\Gamma^{1/2} K_i \Gamma^{1/2}),$$

with zero summand if both numerator and denominator vanish. This quantity assigns the common linear exponential bound to the individual factors. To see the first bound, use $|\sum_r x_r|^2 \leq (\sum_r w_r) \sum_r |x_r|^2/w_r$ with the absolute path weights w_r; their sums are the preceding bounded row sums. The Schur bound for Γ completes the estimate. In particular, this bounds $|\Gamma^{1/2}\eta_0|^2$ with $\eta_0 = \sum_i \eta_i$, including all terms $\eta_i^T \Gamma \eta_{i'}$ with $i \neq i'$. This is the factor $\exp(C\Lambda_i)$ in a_i. Put $K = \sum_i K_i$. Since $0 \leq K_i \leq K$, each $\|\Gamma^{1/2} K_i \Gamma^{1/2}\|$ is at most $\|\Gamma^{1/2} K \Gamma^{1/2}\| < 1/8$ for large B, also after the fixed enlargement needed for two moments. Diagonalization and square completion, with $-\log(1-u) \leq 2u$ for $0 \leq u \leq 1/2$, give

$$\mathbb{E}_\Gamma \mathrm{e}^{2\eta_0 \cdot z + z^T K z} \leq \exp\left(C \sum_i \Lambda_i \right).$$

Under this tilt the covariance is at most 2Γ and its mean is $(I - 2\Gamma K)^{-1}(2\Gamma\eta_0)$. Polynomial moments are bounded by the preceding linear estimate; increasing the fixed constant in $\exp(C\Lambda_i)$ includes their mean contributions. Thus the differentiated factors and likelihood have, after removing the cutoff-remainder amplitudes, one L^2 majorant bounded by $P_{j,\ell,k,q}(\widehat{\beta}_m)(1 + s_M)^d \prod a_i$.

Step 5: cutoff remainders. We next bound the simultaneous event defined above, including all derivative supports. Let p_0 bound the number of Gaussian coordinates per coarse site and m_0 the number of distinct cutoff-remainder addresses per site. These are fixed once the stage and ratio are fixed. For r distinct addresses, choose one violating coordinate, a *witness*, for each. There are at most $(p_0|\mathbb{B}_{2B}|)^r$ choices. Distinct addresses may share a witness, but a single coordinate can serve at most $m_0|\mathbb{B}_{2B}|$ addresses, so the set J of distinct witnesses satisfies

$$r \leq m_0 |\mathbb{B}_{2B}|\, |J|, \qquad |J| \leq r.$$

The Schur bound gives covariance operator norm at most M_0. The exponential Markov inequality for each signed sum $\sum_{i\in J}\pm z_i$, and summation over the $2^{|J|}$ signs, give

$$\Pr\{|z_i|\ge T\ (i\in J)\}\le 2^{|J|}\exp\{(M_0/2-T)|J|\}.$$

Take $T=\xi t_{\mathrm f}/(2g)$ and choose $H>0$ with $M_0/2+Hm_0|\mathbb{B}_{2B}|\le T$. Then $(T-M_0/2)|J|\ge Hm_0|\mathbb{B}_{2B}||J|\ge Hr$: although witnesses are shared, the large deviations still gain $\mathrm e^{-H}$ per address. Summing over the witness choices gives

$$\Pr(E_{\mathcal J(M)})\le (p_0|\mathbb{B}_{2B}|)^r\,2^r\mathrm e^{-Hr}=\bigl(2p_0|\mathbb{B}_{2B}|\mathrm e^{-H}\bigr)^r.$$

The factor $t_{\mathrm f}/g$ grows faster than every fixed power of B and $\log\widehat\beta_{m+1}$. Consequently H can exceed $2\tau_w\log\widehat\beta_{m+1}+\log(2p_0|\mathbb{B}_{2B}|)$, proving (iv). A cutoff remainder is thus rare because its violating coordinate lies $t_{\mathrm f}/g\gg 1$ standard deviations out.

Step 6: the common majorant. Take the finite sum of the absolute differentiated smooth expressions and scores, multiplied by the fixed incoming measurable envelopes, the cutoff-remainder amplitudes $\prod_w M_w$, and the simultaneous indicator. The product rule bounds every derivative through order k by this same nonnegative function $G_M=G_M^0\mathbf 1_{E_{\mathcal J(M)}}$. By the Cauchy–Schwarz inequality, Step 4 and (iv),

$$\int G_M\,d\nu\le\|G_M^0\|_{L^2(\nu)}\Pr(E_{\mathcal J(M)})^{1/2}\le P(\widehat\beta_m)(1+s_M)^d\prod_{i\notin\mathcal J(M)}a_i\prod_{w\in\mathcal J(M)}M_w\,\varepsilon_w,$$

so the square root of the simultaneous bound leaves exactly one factor ε_w per address. Since $f_w=M_w\varepsilon_w$, this proves (i).

Step 7: selected-test suppression. Distribute the local energy of Lemma 11.12 among the selected-test factors before differentiating. When $A_c=\varnothing$ the admitted support is healthy. The physical-link change holds the fine links and selected test indices fixed; differentiated test cutoffs retain the selected-test inequalities on their supports. The exponential energy is retained in each differentiated envelope. Annular affine errors have size $E(\widehat\beta_m)|A|$ with E of upper rate $-3/2$, and the remaining per-test logarithmic corrections have a fixed-power bound $C_{j,\ell,k,q}(1+\log\widehat\beta_{m+1})^{d_0}|A|$. Derivatives of the smooth factors through order k multiply this bound by fixed powers of B, $g^{-1}t_{\mathrm c}$ and the number of selected tests. Their logarithms per test are $o(\beta t_{\mathrm f}^2)$, since $\beta t_{\mathrm f}^2$ is comparable to $(M_{m+1}^{\mathrm{cut}})^2$, which exceeds every fixed power of $\log\widehat\beta_{m+1}$. Also $E(\widehat\beta_m)/t_{\mathrm f}^2\to 0$. Increase the coupling threshold after fixing $\kappa_B'<\kappa$; absorption into $(\kappa-\kappa_B')\beta t_{\mathrm f}^2|A|$ gives (ii) through order k. The support factor $(1+s_M)^d$ remains outside this per-test estimate. The determinant and covariance corrections use the exponential bounds above. Integrating the independent frozen Haar variables and summing the finite tag probabilities preserve the bound.

Step 8: large-field components. Retain the full sector masks of the component and all frozen links read by it. The lower bound in Lemma 11.12 gives $\kappa t_{\mathrm f}^2|A|$ in its local energy. The enlarged neighbourhood used by the component adds annular affine remainders bounded by $E(\widehat\beta_m)|A|$, where E has upper rate $-3/2$: the term error is $O(t_{\mathrm f}^3)$, and the number of positions per test is a fixed power of B. The chart-density, tag, local determinant and normalized frozen-coordinate factors contribute at most $C_{j,\ell}(1+\log\widehat\beta_{m+1})^{d_0}|A|$ to the upper exponent, for a fixed d_0. Indeed, their read counts are at most $C_{j,\ell}B^4|A|$; the logarithm of each inverse Gaussian normalization is $O(1+\log\widehat\beta_{m+1})$, the chart logarithms are bounded on the protected chart, and the local determinant terms have the bounds of Lemma 11.13. The retained value envelope is thus at most

$$\exp\Bigl[-\kappa\beta t_{\mathrm f}^2|A|+\{\beta E(\widehat\beta_m)+C_{j,\ell}(1+\log\widehat\beta_{m+1})^{d_0}\}|A|\Bigr].$$

As in Step 7, both corrections are $o(\beta t_{\mathrm{f}}^2)$ per test. Choose κ_B'' strictly between κ_B' and κ and increase the coupling threshold. This bounds the component envelope by $e^{-\kappa_B''\beta t_{\mathrm{f}}^2|A|}$. Use this component envelope and the incoming measurable supremum envelopes. The attached factors and Gaussian scores have exactly the L^2 bound proved above. Cauchy–Schwarz with their simultaneous event retains every attached f_i. Polynomial factors in the number of component reads are bounded by a fixed polynomial in $1 + s_A$ times a fixed power of B, the latter entering P. Any remaining fixed per-test logarithmic loss is absorbed by $\kappa_B'' - \kappa_B'$. Frozen Haar integration has mass one. This proves (iii), without differentiating its flag masks or taking a supremum of a Gaussian coordinate.

Step 9: factorization. The Gaussian marginal on I has covariance Γ_I. Every complete read support contains Ω_i^{loc} for each coordinate i it reads. The windows are symmetric. Hence if i, i' belong to disjoint complete read supports, $i' \notin \Omega_i^{\mathrm{loc}}$ and $i \notin \Omega_{i'}^{\mathrm{loc}}$, so $\Gamma_{ii'} = 0$ by (11.15). The Gaussian characteristic function therefore factorizes over these supports. The frozen Haar and reference tag coordinates are disjoint as well, proving (v). □

12 Path actions and the one-loop coupling

After each integration the effective action keeps its curvature terms through fourth order in a path action. Its quadratic part is normalized so that β is the coefficient of the squared Wilson curvature at zero momentum at every level. The quadratic terms independent of β, the loop atoms, have vanishing zeroth and first displacement moments, so they carry no curvature-square term. This section constructs these arrays and computes the one-loop change of the coupling: the blocked determinant lowers the curvature-square coefficient by $\kappa_G \log \ell_m$, up to an error bounded independently of the block ratio (Lemma 12.9).

The path bounds enter Theorem 5.6, and the coefficient $\alpha_{N,m}$ enters Theorem 5.5. Section 11 supplies the inputs. The free forms and their row factor (Lemma 11.5) fix the quadratic part; the stationary interpolation and the vertex bounds (Lemmas 11.6 and 11.9) give the classical cubic and quartic terms; the Gaussian coordinates, the inverse insertions and the Ward estimates (Lemmas 11.7, 11.8 and 11.10) control the determinant. The scalar reset of the coupling uses the activities of Section 13 and is proved there, in Section 13.3.

Throughout, the group and metric are those of Definition 2.1, and the scales are those of Section 5. At a step $m + 1 \to m$ put $\ell = \ell_m$ and $j = J_{m+1}$. The path arrays (row coefficients and atoms of Definition 12.1) depend on N, m and the intervening block ratios, and are chosen on $\mathbb{Z}^4$ before periodic evaluation. They are independent of the volume index p and of the running coupling. Constants declared uniform in ℓ are chosen before ℓ.

The arrays are constructed by descending induction on m. Write (I_m) for the path bounds and cancellations of Lemmas 12.3–12.4 at level m, the path part of the induction scheme of Section 11. The top-level Wilson matching in the proof of Lemma 12.6 starts the induction. Under (I_{m+1}), Lemmas 12.6, 12.8 and 12.9 treat the step $m + 1 \to m$; the joint proof after Lemma 12.9 constructs the output arrays and proves (I_m). Curl division and the gauge determinant identity are independent of this induction.

12.1 Curvature coordinates and the path action

A path transports curvature to an anchor, where invariant colour contractions can be taken. Write U_γ for the ordered link product along γ, $\mathfrak{p} = (a, b)$, $a < b$, for an orientation, and $U_{x;\mathfrak{p}}$ for a plaquette holonomy. The footprint of a slot is the set of vertices visited by its path and plaquette boundary. The footprint of an atom is the union of its slot footprints and the origin, translated to

its anchor. Recall that a set is healthy if it avoids $\mathrm{Flag}_m(U)$ (Section 5); an atom is masked by the health of its footprint.

Definition 12.1 (Curvature slots)**.** A slot $s = (\gamma, \mathfrak{p})$, with $\gamma : 0 \to z_s$, has length $|\gamma| + 4$ and value

$$\mathcal{F}_s(U, x) = \mathrm{Ad}(U_{x+\gamma}) \operatorname{proj}_{\mathfrak{g}} \frac{U_{x+z_s;\mathfrak{p}} - U^{-1}_{x+z_s;\mathfrak{p}}}{2} .$$

Transport is from $x + z_s$ to x. Set $\mathcal{L}^{\mathrm{path}}_{\mathfrak{p}} = \sum_i r_{\mathfrak{p}i} \mathcal{F}_{\mathfrak{p}i}$. For $k = 2, 3, 4$, the atoms $\mathfrak{a}^{(k)}_i$ are real invariant k-linear expressions in slots, masked by health on their footprints. Atom length sums slot lengths; $|\mathfrak{a}^{(k)}_i|$ sums absolute colour coefficients. These arrays retain local terms through fourth order.

Here $r_{\mathfrak{p}i} \in \mathbb{R}$ and $\mathcal{F}_{\mathfrak{p}i}$ denotes the slot assigned to row $\mathfrak{p}$ and index $i \in \mathbb{N}$. In the orthonormal colour basis $(e^{\mathfrak{g}}_b)_b$ of Definition 2.1, an atom has the form

$$\mathfrak{a}^{(k)}_i(U, x) = \mathbf{1}_{X_i + x \text{ healthy}} \sum_{b_1, \dots, b_k} c_i(b_1, \dots, b_k) \prod_{\nu=1}^{k} \langle e^{\mathfrak{g}}_{b_\nu}, \mathcal{F}_{s_{i,\nu}}(U, x) \rangle_{\mathfrak{g}} .$$

The coefficients are invariant under the simultaneous adjoint action on all slots, and X_i is the footprint just defined.

For each stage k, take $\gamma_k > 0$ from Lemma 11.5, so the free plaquette forms for the top and both incoming stages are at least $2\gamma_k I$. Write $c_{\mathrm{dir}} = c_{\mathrm{dir},J_m}$ and $c_{\mathrm{lin}} = c_{\mathrm{lin},J_m}$. Whenever $c_{\mathrm{dir}} \le \gamma_{J_m}$, the form $\mathsf{H}_{N,m} - c_{\mathrm{dir}} I$ below is at least $\gamma_{J_m} I$ and has the positive row factor used below.

Definition 12.2 (Path action)**.** At level m the path action $P^{\mathrm{path}}_{N,m}$ is (5.10), with the clip and relaxed threshold test of (5.9), the rows and atoms of Definition 12.1, and clipping constants $0 < 16 c_{\mathrm{lin}} < c_{\mathrm{dir}} \le \min(\gamma_{J_m}, 1)$. Write

$$P^{\mathrm{path}}_{N,m}(\beta, U) = \beta \mathsf{K}^{\mathrm{kin}}_m(U) + \mathcal{L}_m(U),$$

where $\mathsf{K}^{\mathrm{kin}}_m$ is the hypercubic average of the clipped plaquette terms, the row squares and the cubic and quartic atoms, and $\mathcal{L}_m$ that of the quadratic atoms.

The kinetic part carries the coupling; the loop part collects the quadratic corrections that do not scale with β. The clip drops at $u = t$ from $c_{\mathrm{dir}} t^2/2$ to $c_{\mathrm{lin}} t^2$, and its linear branch gives $\mathrm{clip}_t(u) = c_{\mathrm{lin}} t u \ge c_{\mathrm{lin}} t^2$ for $u > t$. This positive linear large-field branch is the lower bound used in the local large-field energy estimate, Lemma 11.12, and through it in the forced bounds of Lemma 13.5. Because of the jump the path action is only Borel in the links. The derivatives used later are taken on healthy configurations, where every clip of the support is on its quadratic branch and every mask is fixed, and coupling derivatives hold the physical links and the masks fixed (Section 13.7). No derivative of a flag boundary or of the clip is ever taken.

Lemma 12.3 (Path bounds)**.** *For every N and $m \le N$ the chosen arrays are independent of p and β. There are widths $0 < w^{\mathrm{path}}_k < w'_k$ and amplitudes $A^{\mathrm{path}}_k > 0$, depending only on the stage k, such that, uniformly in N, with $w = w'_{J_m}$,*

$$\sum_i |r_{\mathfrak{p}i}| e^{w \operatorname{length}(\mathcal{F}_{\mathfrak{p}i})} \le A^{\mathrm{path}}_{J_m},$$

$$\sum_i |\mathfrak{a}^{(k)}_i| e^{w \operatorname{length}(\mathfrak{a}^{(k)}_i)} \le \begin{cases} \Lambda^{\mathrm{loop}}_m, & k = 2, \\ A^{\mathrm{path}}_{J_m}, & k = 3, 4, \end{cases}$$

where $\Lambda^{\mathrm{loop}}_m = 2(M^{\mathrm{cut}}_m)^{\theta_{J_m}}$ depends on the level. A fortiori these bounds hold at width $w^{\mathrm{path}}_{J_m}$.

The joint proof of Lemmas 12.3 and 12.4 follows Lemma 12.9. To state the cancellations, put

$$s_{\mathfrak{pq}} = \sum_{i:\,\mathrm{orientation}(\mathcal{F}_{\mathfrak{p}i})=\mathfrak{q}} r_{\mathfrak{p}i}\,.$$

For ordered orientations $\mathfrak{q},\mathfrak{r}$ let $I_{\mathfrak{qr}}$ be the set of quadratic atoms whose two slots have orientations $\mathfrak{q},\mathfrak{r}$, with colour coefficients $c_i(b_1,b_2)$ and slot offsets $z_{i,1}, z_{i,2}$.

The cancellations have a Fourier meaning. To quadratic order in the links, an atom in $I_{\mathfrak{qr}}$ pairs the curvatures at two points displaced by $z_{i,2}-z_{i,1}$, so the loop atoms contribute the plaquette-kernel symbol $\sum_{i\in I_{\mathfrak{qr}}} c_i(b_1,b_2)e^{i\xi\cdot(z_{i,2}-z_{i,1})}$ between two curvatures. A zero total coefficient removes its constant term and a zero first displacement moment removes its linear term, so the symbol is $O(|\xi|^2)$. Each curvature is a curl, of order $|\xi|$ in the link variables. Hence the loop atoms have a link kernel of order four and contain no curvature-square term: the marginal curvature square is carried by the coupling alone. Under blocking, the cell volume ℓ^4, the two interpolated fields ℓ^{-2} and the rescaled momentum ξ/ℓ turn an order-four link kernel into $\ell^4\ell^{-2}O(|\xi/\ell|^4)=\ell^{-2}O(|\xi|^4)$, an inverse-block-square gain (proof of Lemmas 12.3–12.4).

Lemma 12.4 (Kinetic and quadratic cancellations)**.** *For all orientations $\mathfrak{q},\mathfrak{r}$ and colours b_1,b_2,*

$$c_{\mathrm{dir},J_m}\delta_{\mathfrak{qr}} + \sum_{\mathfrak{p}} s_{\mathfrak{pq}}s_{\mathfrak{pr}} = \delta_{\mathfrak{qr}}, \qquad \sum_{i\in I_{\mathfrak{qr}}} c_i(b_1,b_2) = 0, \qquad \sum_{i\in I_{\mathfrak{qr}}} c_i(b_1,b_2)(z_{i,2}-z_{i,1}) = 0\,.$$

All these moment series converge absolutely.

Thus the kinetic part has the Wilson normalization $\widehat{\mathsf{H}}_{N,m}(0)=I$ below, and the loop atoms do not change the curvature-square coefficient.

12.2 Fourth-order matching

Near the flat field all relaxed threshold tests equal one. By $\mathrm{ch}^2 = 2\,\mathrm{pr}$ from (2.1), the leading analytic action is obtained from Definition 12.2 by replacing the clipped term by $c_{\mathrm{dir}}\,\mathrm{pr}$ and omitting the indicators and loop atoms. Denote this action, without its factor β, by $\mathsf{S}_{N,m}$. Its prescribed quadratic part is

$$\frac{1}{2}\langle d_1 a, \mathsf{H}_{N,m} d_1 a\rangle, \qquad \mathsf{H}_{N,m} = c_{\mathrm{dir},J_m} I + \rho^*_{N,m}\rho_{N,m}, \qquad \widehat{\mathsf{H}}_{N,m}(0) = I\,.$$

Here d_1 is the lattice curl. Lemma 11.5 gives the exact adjacent free relation and the row factor $\rho_{N,m}$, with $\mathsf{H}_{N,m} - c_{\mathrm{dir},J_m} I \geq \gamma_{J_m} I$. The rows are read off from $\rho_{N,m}$. Write $(\rho_{N,m}F)_{\mathfrak{p}}(x) = \sum_{\mathfrak{q}}\sum_{z\in\mathbb{Z}^4}\rho_{\mathfrak{pq}}(z)F_{\mathfrak{q}}(x+z)$. For each $\mathfrak{p},\mathfrak{q},z$ with $\rho_{\mathfrak{pq}}(z)\neq 0$ take one slot $(\gamma_z,\mathfrak{q})$, with γ_z a fixed coordinate path from 0 to z, and row coefficient $\rho_{\mathfrak{pq}}(z)$. At the flat field $\mathcal{L}^{\mathrm{path}}_{\mathfrak{p}}(U,x)$ linearizes to $(\rho_{N,m}d_1 a)_{\mathfrak{p}}(x)$, so the direct and row terms have quadratic part $\frac{1}{2}\langle d_1 a, \mathsf{H}_{N,m} d_1 a\rangle$; also $s_{\mathfrak{pq}} = \widehat{\rho}_{\mathfrak{pq}}(0)$, and the exponential decay of $\rho_{N,m}$ gives the row bound of Lemma 12.3. The cell volume contributes ℓ^4 and each curvature contributes ℓ^{-2}; hence $\ell^4(\ell^{-2})^2 = 1$, preserving the coefficient β.

We need a division that expresses a gauge-invariant Taylor difference in curvatures. For a momentum ξ put $q_a(\xi) = e^{i\xi_a} - 1$; its exterior action $d_1(\xi)a = q(\xi)\wedge a$ is the curl symbol. Momentum order s means vanishing of all Taylor coefficients of total momentum degree less than s.

Lemma 12.5 (Curl division)**.** *Let $k\geq 2$, $\sigma>0$, $A\geq 0$, and let f be a periodic analytic kernel in independent momenta $\xi_1,\dots,\xi_{k-1}$ ($\xi_0 = -\sum_\nu \xi_\nu$), valued in $\bigotimes_{\nu=0}^{k-1}\Lambda^1\mathbb{C}^4$, bounded coordinatewise*

by A on $|\operatorname{Im}\xi_\nu^a| < \sigma$, and transverse in every leg: $\delta_\nu f = 0$, $\delta_\nu = \iota_{q(\xi_\nu)}$. If $k = 2$ assume also momentum order $s + 2$. Then there are $0 < \eta \le \sigma$ and $K \ge 0$, depending only on k, σ, and a periodic analytic g valued in $\bigotimes_\nu \Lambda^2\mathbb{C}^4$, of order s if $k = 2$, with

$$f = \delta_0 \cdots \delta_{k-1} g, \qquad \sup_{\eta\text{-strip}} |g| \le KA\,.$$

The Fourier coefficients of g decay exponentially.

In words: a gauge-invariant multilinear kernel factors through the curls of its legs.

Proof. Dividing one leg by its curl is straightforward when its momentum is independent of the others. The difficulty is the leg with momentum ξ_0: a primitive in that leg may fail to remain transverse in another leg. We first construct the one-leg homotopy, then correct this transversality defect. Each correction reduces the number of independent momenta by one and raises the exterior degree in the dependent leg by two, and the iteration terminates when that degree exceeds four. The two-leg case instead uses the extra vanishing order at zero momentum.

Scalar division requires vanishing when the divided leg has zero momentum; positive exterior degree requires only closedness. For one leg let $\delta = \iota_q$ be contraction on exterior tensors. Split a δ-closed tensor as $f = e_a \wedge u + v$, with u, v containing no e_a. Define

$$D_a f(\xi) = \frac{f(\xi) - f(\xi)|_{\xi_a=0}}{q_a(\xi)}, \qquad h_a f = e_a \wedge D_a v, \qquad R_a f = v|_{\xi_a=0}\,.$$

Here D_a is the one-leg exterior division (11.4). The identity $\delta f = 0$ gives $\delta' v = -q_a u$, where δ' contracts the other axes. Since q_b is independent of ξ_a for $b \ne a$, D_a commutes with δ', and $D_a(q_a u) = u$. Consequently

$$\delta h_a f = q_a D_a v - e_a \wedge D_a \delta' v = v - v|_{\xi_a=0} + e_a \wedge u = f - R_a f\,.$$

The remainder $R_a f$ is again closed and has no component containing e_a. Applying this in the four axes gives the homotopy

$$f = \delta \left(\sum_{a=1}^{4} h_a R_{a-1} \cdots R_1 f \right) + R_4 R_3 R_2 R_1 f\,.$$

The final remainder vanishes in positive exterior degree because it contains none of the four axes. In degree zero it is the restriction to zero momentum of this leg, which must vanish for scalar division. In that case the formula reads $f = \sum_a q_a D_a (R_{a-1} \cdots R_1 f)$.

The quotients are analytic even at the zeros of q_a. On a Fourier monomial the identity

$$\frac{e^{in\xi_a} - 1}{e^{i\xi_a} - 1} = \begin{cases} \sum_{v=0}^{n-1} e^{iv\xi_a}, & n > 0, \\ -\sum_{v=n}^{-1} e^{iv\xi_a}, & n < 0, \\ 0, & n = 0 \end{cases}$$

bounds the quotient on a smaller strip by a constant times $|n|$ times its original exponential weight. The strip loss absorbs $|n|$, so the Fourier series converges absolutely. Each division loses at most one momentum order.

For independent legs, their remaining transverse subspaces do not depend on the divided momentum, so each restriction and division preserves them. The dependent-leg induction has

the following claim: with r independent momenta and dependent exterior degree $d \geq 1$, a tensor closed in the dependent leg and transverse in all independent legs has a dependent-leg primitive preserving those transverse conditions, provided $d + 2r > 4$. Analytic bounds hold after a strip reduction depending only on r, d and the original strip. For $r = 0$ the condition gives $d > 4$, so the tensor is zero. The induction step below reduces (r, d) to $(r - 1, d + 2)$, preserving the condition. Initially $r = k - 1, d = 1$, so it applies for $k \geq 3$.

For the dependent momentum $\xi_0 = -\sum_{i=1}^{k-1} \xi_i$, first shear the coordinates to replace ξ_1 by ξ_0. The other independent contractions remain unchanged. The one-leg formula gives $\delta_0 X = f$, retaining the contractions in legs $2, \dots, k-1$. Write c_1 for contraction in leg 1. Its defect obeys $\delta_0(c_1 X) = c_1 f = 0$. Solve $\delta_0 Y = c_1 X$ by the same formula. After c_1, leg 1 has exterior degree zero, so restoring that leg will require only scalar division.

On $\xi_1 = 0$, the restriction Y_0 is δ_0-closed, since $c_1 X$ vanishes there. The induction hypothesis with one fewer independent momentum and dependent degree two higher gives $\delta_0 Z = Y_0$, preserving the other transverse conditions. Extend Z independently of ξ_1. Then $Y' = Y - \delta_0 Z$ vanishes on $\xi_1 = 0$, and scalar division gives $c_1 D = Y'$. Thus

$$V = X - \delta_0 D, \qquad \delta_0 V = f, \qquad c_1 V = c_1 X - \delta_0 Y' = 0\,.$$

All corrections preserve the other independent contractions. The two successive solves $f \mapsto X$ and $c_1 X \mapsto Y$ each raise the dependent exterior degree by one; contraction c_1 changes only leg 1. Thus the restricted problem for Y_0 has one fewer independent momentum and dependent degree two higher. For three or more legs it terminates above exterior degree four, where the tensor is zero. The independent-leg divisions then finish the factorization.

For two legs, the restriction Y_0 is at zero external momentum. Starting from order $s+2$, the first division loses one order, contraction restores one, and the solve for Y leaves order $s + 1 \geq 1$. Hence $Y_0 = 0$ directly. The correction and final division leave order s. This verifies the scalar vanishing hypothesis where it is needed. For the Taylor differences of Definition 12.2, gauge invariance supplies transversality because all lower Taylor differences vanish. □

For $z = (z_1, \dots, z_{k-1})$, contour displacement gives $|\check{g}(z)| \leq C e^{-\zeta \sum_\nu |z_\nu|_1}$ for the Fourier coefficients $\check{g}$ of g. Choose the anchor at zero and coordinate paths to z_ν. Their total slot length is at most $4k + \sum_\nu |z_\nu|_1$. Every path width below ζ therefore gives a convergent weighted sum. Reality and simultaneous adjoint invariance pass to the coefficients by conjugation and invariant projection.

A finite-dimensional model. The classical matching below is the following calculation, with locality estimates attached. Let $S(v, z)$ be smooth near $0 \in \mathbb{R}^a \times \mathbb{R}^b$ with $DS(0, 0) = 0$ and $\mathsf{Q} = D_z^2 S(0, 0) > 0$, and let $z(v)$ solve $D_z S(v, z(v)) = 0$, $z(0) = 0$. Put $Wv = (v, Dz(0)v)$ and $\mathsf{B}(v, v)[u] = D^3 S(0)[Wv, Wv, (0, u)]$. Along $\gamma(t) = (tv, z(tv))$ one has $\gamma'(0) = Wv$ and $\gamma''(0) = (0, w)$ with $w = D^2 z(0)[v, v]$. Differentiating the equation for z once gives $D^2 S(0)[Wv, (0, u)] = 0$, and twice gives $w = -\mathsf{Q}^{-1}\mathsf{B}(v, v)$. In the third derivative of $S_{\mathrm{stat}}(v) = S(v, z(v))$ the only term containing w is $3D^2 S(0)[Wv, (0, w)] = 0$, and the term containing $D^3 z$ is multiplied by $DS(0) = 0$, so the cubic term is simple: $D^3 S_{\mathrm{stat}}(0)[v^3] = D^3 S(0)[(Wv)^3]$. In the fourth derivative the term $4D^2 S(0)[Wv, (0, D^3 z(0)[v^3])]$ vanishes by the first identity and the term containing $D^4 z$ is multiplied by $DS(0) = 0$; the terms containing w are

$$6\mathsf{B}(v, v)[w] + 3\langle w, \mathsf{Q}w\rangle = -3\mathsf{B}(v, v)\mathsf{Q}^{-1}\mathsf{B}(v, v)\,,$$

an exchange correction: the three ways of splitting four legs into two pairs, each pair joined through the fluctuation propagator Q^{-1}. Degree four is the stopping point. In the continuum the curvature

square $|da + \frac{1}{2}[a \wedge a]|^2$ has degree at most four in the connection, so matching through degree four captures the classical marginal term; the Taylor remainder of degree five is of order βt_m^5 per site and is absorbed by the activity bounds (proof of Lemma 13.7).

On the lattice the model applies with $z = \psi$: v is the coarse chart variable and ψ the fine axial variable; $S(v, \psi)$ is the fine leading action plus $\mathsf{b} \sum_e \mathrm{pr}$ of the predictor discrepancy, without the input coupling $\beta' = \beta'_u$, and it excludes the inherited loop action. Lemma 11.6 supplies the stationary branch $\psi(v)$ with $\mathsf{Q} \geq \nu_{j,\ell} I$, and we write $W_{\mathrm{ax}} v = (v, D\psi(0)v)$ for the axial tangent and $S_{\mathrm{stat}}(v) = S(v, \psi(v))$. Lemma 11.9 gives the vertex bounds $C_3 \leq a_3 + \ell^{-1} a'_3$, $C_4 \leq c_4 + \ell^{-1} c'_4$ and $|D^4 S_{\mathrm{stat}}| \leq C_4 + 3C_3^2 E_{\mathrm{ins}}$, with $E_{\mathrm{ins}} = \max(1, C_j^{\mathrm{ins}})$. The stationary identities hold for every block ratio of Table 3; only the uniform bound on the new arrays needs $\ell \geq u_j$.

Lemma 12.6 (Classical matching)**.** *With* $\mathsf{B}(v, v)[u] = D^3 S(0)[W_{\mathrm{ax}} v, W_{\mathrm{ax}} v, (0, u)]$,

$$D^3 S_{\mathrm{stat}}(0)[v^3] = D^3 S(0)[(W_{\mathrm{ax}} v)^3],$$
$$D^4 S_{\mathrm{stat}}(0)[v^4] = D^4 S(0)[(W_{\mathrm{ax}} v)^4] - 3\mathsf{B}(v, v)\mathsf{Q}^{-1}\mathsf{B}(v, v)\,.$$

For ℓ above a threshold depending on the input path data, cubic and quartic arrays satisfying the bounds of Lemma 12.3 make $\mathsf{S}_{N,m}$ agree with S_{stat} through degree four. Their bound depends only on fixed stage data.

Proof. The identities are the finite-dimensional calculation above applied to the analytic stationary branch of Lemma 11.6; in the joint coordinates of Lemma 11.9 the same identities hold with smooth interpolated legs. The kernel bounds are those of Lemma 11.9.

Increase ℓ until $\ell^{-1} a'_3, \ell^{-1} c'_4 \leq 1$. The output action built from c_{dir} and the rows of $\rho_{N,m}$ alone agrees with S_{stat} at degree two (Lemma 11.5). Both are gauge invariant, so their cubic Taylor difference is annihilated by linearized gauge transformations in each leg; that is, it is transverse. Lemma 12.5 with $k = 3$ writes it through curls; placing its coefficients on coordinate paths gives cubic atoms, bounded through $C_3 \leq a_3 + 1$. Adding these atoms, with their full nonlinear transport, changes the quartic Taylor coefficients. The remaining quartic difference is again transverse, and Lemma 12.5 with $k = 4$ gives quartic atoms bounded through $C_4 + 3C_3^2 E_{\mathrm{ins}}$. This makes the two actions agree through degree four. At $m = N$, the free form acts as the identity on curls, so the same construction matches the Wilson energy through degree four and starts with zero loop atoms. Lemma 13.16 places the remaining top-level difference, including clipping and unhealthy configurations, in the initial activities. □

The common factor β' in the penalty and fine leading action cancels from the stationary equation; hence the arrays are independent of β'. In $\log\det(\beta' \mathsf{Q})$, the term $(\dim \mathsf{Q}) \log \beta'$ is independent of the field and contributes only to the scalar normalization.

12.3 The determinant comparison

Lemma 11.7 gives the one-loop action

$$\mathsf{T}(v) = \mathcal{L}_{m+1}(U_v) + \tfrac{1}{2} \log\det \mathsf{Q}(v) - \log J_{\mathrm{ax}}(v)\,,$$

with the stationary fine configuration U_v, the inherited quadratic loop action $\mathcal{L}_{m+1}$, the axial Hessian $\mathsf{Q}(v)$ without β', and the Haar chart ratio J_{ax}, normalized to one at zero.

We measure the kinetic coefficient by a transverse plane wave: momentum te_1, link polarization e_2 and one fixed unit colour vector. For a quadratic Taylor kernel Y, including the Taylor factor $1/2!$,

recall from Section 11.4 the functional $\alpha_b(Y) = 2\,\mathrm{Re}[t^2]Y_{(2,b),(2,b)}(te_1)$, where $[t^2]g(t) = \frac{1}{2}g''(0)$, and define

$$\alpha_{N,m} = \alpha_b(Y^{\mathrm{blk}}_{N,m})\,.$$

For $a = e^{itx_1}e_2 \otimes e^{\mathfrak{g}}_b$ one has $|d_1 a|^2 = t^2 + O(t^4)$ per site, so $\alpha_b(Y)$ is the coefficient of $\frac{1}{2}|d_1 a|^2$ in Y, the curvature-square coefficient; the factor 2 compensates for the 1/2!. The kernel $Y^{\mathrm{blk}}_{N,m}$ is the infinite-volume quadratic kernel of T. The prescribed kinetic kernel has $\alpha_b = 1$. Simplicity makes $\alpha_{N,m}$ independent of b, as shown in the proof of Lemma 12.4; by construction it is independent of p and β'.

We use the joint left Hessian J, the gauge insertion $\mathcal{R}$, the slice embedding $\mathcal{I}$ and the frame derivative X of Lemma 11.7, and the covariant difference d_U, the Laplacian $\Delta_U = d_U^* d_U$, the regulator $\mathfrak{m}_\mu = \mu^2 P_\mu^9$ with $P_\mu = \mu^2(\Delta_U + \mu^2 I)^{-1}$, the fine-link projection P_f and the gauge condition $F_\mu = d_U^* P_f + \mathfrak{m}_\mu X$ defined before Lemma 11.8. Recall $X\mathcal{R} = I$ and $\det(\mathcal{R},\mathcal{I})^2 = 1$ (Lemma 11.7). Put

$$\mathsf{C}_\mu = \mathsf{J} + F_\mu^* F_\mu, \qquad \mathsf{G}_\mu = F_\mu \mathcal{R} = \Delta_U + \mathfrak{m}_\mu\,.$$

The next lemma is the Faddeev–Popov identity for the gauge condition F_μ: C_μ is the gauge-fixed Hessian, G_μ the ghost operator, and $-\log\det \mathsf{G}_\mu$ the ghost contribution.

Lemma 12.7 (Gauge completion)**.** *For v in a neighbourhood of zero,*

$$\mathsf{T}(v) = \mathcal{L}_{m+1}(U_v) + \tfrac{1}{2}\log\det \mathsf{C}_\mu(v) - \log\det \mathsf{G}_\mu(v)\,. \tag{12.1}$$

Proof. Gauge invariance and stationarity give $\mathsf{J}\mathcal{R} = 0$, and $F_\mu\mathcal{R} = \mathsf{G}_\mu$. Hence

$$(\mathcal{R},\mathcal{I})^*\mathsf{C}_\mu(\mathcal{R},\mathcal{I}) = \begin{pmatrix} \mathsf{G}_\mu & F_\mu\mathcal{I} \\ 0 & I \end{pmatrix}^* \begin{pmatrix} I & 0 \\ 0 & \mathcal{I}^*\mathsf{J}\mathcal{I} \end{pmatrix} \begin{pmatrix} \mathsf{G}_\mu & F_\mu\mathcal{I} \\ 0 & I \end{pmatrix}.$$

Since $\det(\mathcal{R},\mathcal{I})^2 = 1$, this gives $\det \mathsf{C}_\mu = (\det \mathsf{G}_\mu)^2 \det(\mathcal{I}^*\mathsf{J}\mathcal{I})$. The proof of Lemma 11.7 gives $\mathsf{Q} = (Ds)^*(\mathcal{I}^*\mathsf{J}\mathcal{I})Ds$ and $J_{\mathrm{ax}} = |\det Ds|$, where s changes axial coordinates to left increments at the stationary point. Hence $\frac{1}{2}\log\det\mathsf{Q} - \log J_{\mathrm{ax}} = \frac{1}{2}\log\det(\mathcal{I}^*\mathsf{J}\mathcal{I})$. Positivity of Q, J_{ax} and G_μ near zero permits the logarithms and proves (12.1); together these identities are the three lines of (11.10). □

The comparison uses the auxiliary fine operator and determinant

$$\mathsf{A}^{\mathrm{aux}}_\mu = \mathsf{A} + d_U d_U^* + \mathsf{M}_\mu, \qquad \Pi_\mu(U) = \tfrac{1}{2}\log\det \mathsf{A}^{\mathrm{aux}}_\mu(U) - \log\det(\Delta_U + \mathfrak{m}_\mu(U))\,,$$

where A is the left-coordinate Hessian of the fine leading action and M_μ applies $\mathfrak{m}_\mu$ to each link component. Let Y_μ be the infinite-volume quadratic Taylor kernel of Π_μ. Write u_j for the ratio threshold of Lemma 11.9; items 5 and 8 of Table 3 choose odd ratios above it.

Lemma 12.8 (Block-to-fine coefficient comparison)**.** *For every odd $\ell \geq u_j$ as in items 5 and 8 of Table 3, put $\mu = \ell^{-1}$. There is a constant $d_{\mathrm{cmp}} \geq 0$, depending only on the fixed stage bounds, such that*

$$|\alpha_{N,m} - \alpha_b(Y_\mu)| \leq d_{\mathrm{cmp}}\,.$$

The constant is independent of N, p, ℓ.

Proof. At one coarse momentum both operators act on the same joint space. Their difference either factors through a coarse space of dimension at most $4d_{\mathfrak{g}}$ per cell or contains the regulator $\mathfrak{m}_\mu = \mu^2(P_\mu^3)^3$. The first kind is bounded by finite rank, the second by the uniform trace bound

for P_μ^3, so the logarithmic Hessians differ by a trace-class amount bounded independently of ℓ; no trace is estimated by the fine-cell dimension ℓ^4. We (1) fold Π_μ to coarse momentum, which leaves α_b unchanged; (2) remove the contact term, which leaves α_b unchanged by Lemma 11.10; (3) bound the difference of logarithmic Hessians; (4) apply Cauchy's estimate. The input arrays satisfy Lemma 12.4 by (I_{m+1}).

Step 1: folding. Folding the auxiliary operator into coarse momentum does not alter α_b. The symbol of the weights w^{av} in (5.5) vanishes at every nonzero alias to order at least two in each external leg, so these aliases contribute $O(|\xi|^4)$. The zero alias in the tested entry is

$$\ell^4\ell^{-2}Y_\mu(\xi/\ell) + O(|\xi|^3)\,.$$

The factors are the cell volume and ℓ^{-1} per interpolated field. Extracting the quadratic coefficient supplies another ℓ^{-2}, so $\ell^4\ell^{-2}\ell^{-2} = 1$.

Step 2: the contact term. The full completed Hessian contains a contact term: the second derivative of the predictor multiplied by the first derivative of the penalty, the sum $\sum_\lambda d^\lambda C_\lambda$ in (11.9). Along the free interpolation, write $\mathsf{a}(\xi)$ for this penalty derivative and $\mathsf{b}(\xi)$ for the mixed fine/frame trace. Lemma 11.10 gives $\mathsf{a}(\xi) = O(|\xi|^2)$ and $\mathsf{b}(\xi) = O(|\xi|)$. The change in the tested entry on removing the contact term consists of ab, its reflected counterpart, and a term quadratic in a, with bounded analytic factors. It is $O(|\xi|^3) + O(|\xi|^4)$ and has zero α_b. Write $\mathsf{C}_\mu^{\mathrm{rem}}$ for C_μ with the contact term removed.

Step 3: the joint-space difference. At one coarse momentum let $\mathcal{V}_f$ be the fine-link fibre and $\mathcal{V}_0$ the coarse-frame fibre, of dimension $d_{\mathfrak{g}}$. Let $P_f : \mathcal{V}_f \oplus \mathcal{V}_0 \to \mathcal{V}_f$ and $Z : \mathcal{V}_f \oplus \mathcal{V}_0 \to \mathcal{V}_0$ be the coordinate projections, and put

$$\widetilde{\mathsf{A}}_\mu^{\mathrm{aux}} = P_f^*\mathsf{A}_\mu^{\mathrm{aux}}P_f + Z^*Z = \begin{pmatrix} \mathsf{A}_\mu^{\mathrm{aux}} & 0 \\ 0 & I \end{pmatrix}, \qquad \det \widetilde{\mathsf{A}}_\mu^{\mathrm{aux}} = \det \mathsf{A}_\mu^{\mathrm{aux}}\,.$$

Thus adjoining the frame identity leaves the auxiliary determinant unchanged. Let H be the differential of the logarithm of the framed predictor and B its coarse-link penalty Hessian; these are the two factors in the second-derivative chain rule after removal of the contact term. Put $K = \mathfrak{m}_\mu X$, and let J_a select direction a from fine links, so $\mathsf{M}_\mu = \sum_a J_a^*\mathfrak{m}_\mu J_a$. Then

$$\begin{aligned}
\mathsf{C}_\mu^{\mathrm{rem}} &= P_f^*\mathsf{A}P_f + H^*BH + (d_U^*P_f + K)^*(d_U^*P_f + K),\\
\widetilde{\mathsf{A}}_\mu^{\mathrm{aux}} &= P_f^*\mathsf{A}P_f + P_f^*d_Ud_U^*P_f + P_f^*\Big(\sum_a J_a^*\mathfrak{m}_\mu J_a\Big)P_f + Z^*Z,
\end{aligned}$$

and their difference is

$$\Theta = \mathsf{C}_\mu^{\mathrm{rem}} - \widetilde{\mathsf{A}}_\mu^{\mathrm{aux}} = H^*BH - Z^*Z + P_f^*d_UK + K^*d_U^*P_f + K^*K - P_f^*\Big(\sum_a J_a^*\mathfrak{m}_\mu J_a\Big)P_f\,.$$

The fine Hessians cancel exactly. The first term factors through coarse links, the second through coarse frames. For compatible maps V, W through a coarse space of dimension r,

$$\|VW\|_1 \le r\|V\|\|W\|, \qquad |\operatorname{tr}_{\mathrm{joint}}(VW)| = |\operatorname{tr}_{\mathrm{coarse}}(WV)| \le r\|W\|\|V\|\,.$$

Here $r = 4d_{\mathfrak{g}}$ or $d_{\mathfrak{g}}$, independently of ℓ. Differentiating these factorizations through order two preserves the intermediate coarse spaces. Lemma 11.8 bounds their factors in the mutually dual block norms.

The remaining terms keep a regulator factor. Lemma 11.8(i) bounds the trace norm of P_μ^3 on a complex strip and bounds D^rP_μ for $r \le 2$, independently of ℓ; Step 6 of its proof sums the fine

aliases $k_z = (\xi + 2\pi z)/\ell$ with weights $(1+|z|^2)^{-3}$. Two field derivatives of $\mathfrak{m}_\mu = \mu^2(P_\mu^3)^3$ leave one cube undifferentiated. Put that cube in trace norm and the other factors in operator norm; the bounds on $\mu^2 D^r(A^{-1})$ in Lemma 11.8 cancel the inverse scale before taking a trace. Applied to the factors of Θ, these estimates give a constant B_* for its trace norm through two field insertions, in the dual block norms, depending only on the fixed stage bounds. At each coarse momentum these bounds sum the fine aliases without a factor ℓ^4.

Let F_*, A_* bound the inverses and the operators through two normalized insertions in the compatible block norms (Lemmas 11.9 and 11.8). In $D_v D_w \log\det A = \operatorname{tr}(A^{-1}D_{vw}A - A^{-1}D_vA\,A^{-1}D_wA)$, telescoping the difference between $\mathsf{C}_\mu^{\mathrm{rem}}$ and $\widetilde{\mathsf{A}}_\mu^{\mathrm{aux}}$ with $(\mathsf{C}_\mu^{\mathrm{rem}})^{-1} - (\widetilde{\mathsf{A}}_\mu^{\mathrm{aux}})^{-1} = -(\mathsf{C}_\mu^{\mathrm{rem}})^{-1}\Theta(\widetilde{\mathsf{A}}_\mu^{\mathrm{aux}})^{-1}$ gives terms with exactly one factor Θ, or a derivative of it, in trace norm and only bounded factors otherwise; cyclically moving factors identifies the required dual norms. Hence the difference of the logarithmic Hessians is at most a fixed polynomial in A_*, B_*, F_*. The common ghost determinant cancels. Including the Taylor factors gives a bound C_* for the analytic difference kernel on a strip of width $\zeta_* > 0$, both independent of ℓ. The inherited loop kernel has order four by (I_{m+1}); pullback by the analytic interpolation W_0 preserves that order, so its α_b is zero.

Step 4: Cauchy's estimate. Restrict the tested entry g of the difference to $\xi = te_1$. Cauchy's coefficient inequality on $|t| = \zeta_*/2$ gives $|[t^2]g| \le 4C_*/\zeta_*^2$ and hence $|\alpha_b(g)| \le 8C_*/\zeta_*^2$. This gives d_{cmp}. □

12.4 The one-loop coefficient

The vertices of the fine action differ from Wilson's in momentum orders $3, 2, 1$ for the quadratic, cubic and quartic terms (Lemma 11.9); the input arrays satisfy its hypotheses by (I_{m+1}). The resulting logarithmic term is the part of the one-step drift which remains after the block ratio has been chosen.

Lemma 12.9 (One-loop coefficient)**.** *There is a constant $d_j \ge 0$, depending only on the fixed stage data, such that, for every step $m+1 \to m$ at stage j with odd $\ell_m \ge u_j$ as in Table 3,*

$$|\alpha_{N,m} + \kappa_G \log \ell_m| \le d_j\,.$$

It is independent of N, p, ℓ_m and the running coupling.

Thus one blocking step lowers the curvature-square coefficient by $\kappa_G \log \ell_m + O(1)$, with $\kappa_G = 2b_0$: the one-loop coefficient of asymptotic freedom. The proof is the background-field computation [1] on the lattice: gauge-field (vector) and ghost loops with two background insertions, where the vector vertex has an orbital part, equal to the ghost vertex, and a spin part.

Proof. The calculation pairs the one-loop vertices of Lemma 12.7, computes the leading bubble exactly, and compares the lattice integrand with it; Lemma 12.8 then transfers the result to $\alpha_{N,m}$. Choose a colour basis index b, external polarization e_2 and external momentum te_1. Signed permutations reduce other transverse axis pairs to this one. For nonzero loop momentum k the leading ghost vertex is $-2ik_2\,\mathrm{ad}(e_b^{\mathfrak{g}})$. The vector vertex has the same orbital term on its four components and the spin term

$$2it(\delta_{c2}\delta_{c'1} - \delta_{c1}\delta_{c'2})\,\mathrm{ad}(e_b^{\mathfrak{g}})\,,$$

with vector indices c, c'. Simplicity and the trace metric give $\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}(e_b^{\mathfrak{g}})^2) = -C_{\mathfrak{g}}$. The bubble pairs vertices at te_1 and $-te_1$. The orbital factor $-2ik_2$ is unchanged when $k \mapsto k + te_1$ and $te_1 \mapsto -te_1$.

Put $S_{\mathrm{sp}} = E_{21} - E_{12}$. Since $\operatorname{tr} S_{\mathrm{sp}} = 0$, the orbital–spin cross terms vanish; the spin product is $(2itS_{\mathrm{sp}})(-2itS_{\mathrm{sp}}) = 4t^2 S_{\mathrm{sp}}^2$, with trace $-8t^2$. The bubble factors are

contribution	directional trace	weight in $\frac{1}{2}\log\det \mathsf{C}_\mu - \log\det \mathsf{G}_\mu$
vector	$4(-2ik_2)^2 - 8t^2$	$-1/4$
ghost	$(-2ik_2)^2$	$+1/2$

Multiply by the colour trace, divide by $|k|^2|k+te_1|^2$, and take the coefficient of t^2; the extraction of α_b supplies the factor 2. Before taking the coefficient, ghost minus half vector is

$$-\frac{4C_{\mathfrak{g}}(k_2^2+t^2)}{|k|^2|k+te_1|^2}\,.$$

Its quadratic coefficient is

$$h(k) = 4C_{\mathfrak{g}}\left(\frac{k_2^2}{|k|^6} - \frac{4k_2^2k_1^2}{|k|^8} - \frac{1}{|k|^4}\right). \tag{12.2}$$

Expanding the denominator gives $|k|^{-4} - 2tk_1|k|^{-6} + t^2(4k_1^2|k|^{-8} - |k|^{-6}) + O(t^3)$; substitution yields (12.2) with its sign.

Let $I_{\mu,b}(k)$ be the integrand of $\alpha_b(Y_\mu)$, obtained with these weights and Taylor factors. Then

$$\alpha_b(Y_\mu) = (2\pi)^{-4}\int_{[-\pi,\pi]^4} \operatorname{Re} I_{\mu,b}(k)\,dk\,.$$

For constants $A_{\mathrm{int}}, D_{\mathrm{int}} \geq 0$ depending only on the fixed fine-stage bounds, uniformly in μ,

$$|I_{\mu,b}(k)| \leq \frac{A_{\mathrm{int}}}{(|k|+\mu)^4}, \qquad |I_{\mu,b}(k) - h(k)| \leq D_{\mathrm{int}}\left(|k|^{-3} + \mu^2|k|^{-6}\right) \quad (|k| \geq \mu)\,. \tag{12.3}$$

To obtain them, use $q_a(k) = ik_a + O(|k|^2)$ and the vertex orders above. For the free inverse $R_\mu(k)$, differentiation of $R_\mu \mathsf{A}_\mu^{\mathrm{aux}} = I$ gives

$$\|R_\mu\| \leq C(|k|+\mu)^{-2}, \quad \|\partial R_\mu\| \leq C(|k|+\mu)^{-3}, \quad \|\partial^2 R_\mu\| \leq C(|k|+\mu)^{-4}\,.$$

Each first-order vertex has order $|k|$, with bounded first external derivative. In a bubble, two momentum derivatives on one inverse give $|k|^2(|k|+\mu)^{-2}(|k|+\mu)^{-4} \leq (|k|+\mu)^{-4}$; moving derivatives to vertices gives the same bound. On $|k| \geq \mu$, a lattice-to-leading replacement gains one power of $|k|$, giving $C|k|^{-3}$. A regulator insertion contributes at most $C\mu^2|k|^{-6}$.

The terms with one two-insertion (seagull) vertex are absent from h. Their quadratic coefficients have bounds

$$C\left((|k|+\mu)^{-2} + \mu^2(|k|+\mu)^{-6}\right).$$

The first term comes from one inverse and a bounded seagull-vertex derivative; the second retains the regulator factor μ^2 with the differentiated inverse bounds. On the cube, $|k| \leq 2\pi$ and $\mu \leq 1$, so this is at most $C'(|k|+\mu)^{-4}$. On $|k| \geq \mu$, it is at most $C(2\pi|k|^{-3} + \mu^2|k|^{-6})$. Thus these terms satisfy both inequalities of (12.3). The product rule gives the same bounds for the ninth-power regulator. Summing proves (12.3).

On the spherical shell $\mu < |k| \leq 1$, rotational symmetry gives angular means $k_2^2/|k|^2 = 1/4$ and $k_2^2k_1^2/|k|^4 = 1/24$. For the latter, write the fourth moments as $m_4 = \mathbb{E}\theta_1^4$ and $m_{22} = \mathbb{E}\theta_1^2\theta_2^2$ on the

unit sphere: rotation by $\pi/4$ gives $m_4 = 3m_{22}$, and $(\sum_i \theta_i^2)^2 = 1$ gives $4m_4 + 12m_{22} = 1$. Since the sphere has area $2\pi^2$,

$$(2\pi)^{-4} \int_{\mu<|k|\le 1} h(k)\, dk = \frac{8\pi^2 C_{\mathfrak{g}}}{(2\pi)^4} \left(\frac{1}{4} - \frac{1}{6} - 1\right) \log \mu^{-1} \tag{12.4}$$
$$= -\frac{11 C_{\mathfrak{g}}}{24\pi^2} \log \mu^{-1} = -\kappa_G \log \mu^{-1} .$$

The errors in (12.3) have shell integrals at most $2\pi^2 D_{\mathrm{int}}$ and $\pi^2 D_{\mathrm{int}}$. On $|k| \le \mu$ the bound is $A_{\mathrm{int}}\mu^{-4}$ and the volume is $\pi^2\mu^4/2$; outside $|k| \le 1$ the bound is A_{int}. Hence

$$|\alpha_b(Y_\mu) + \kappa_G \log \mu^{-1}| \le (2\pi)^{-4}\{A_{\mathrm{int}}\pi^2/2 + A_{\mathrm{int}}(2\pi)^4 + 3\pi^2 D_{\mathrm{int}}\} .$$

Combine this bound with Lemma 12.8 and put $\mu = \ell^{-1}$. Their sum gives d_j. The computed slope is $\kappa_G = 2b_0$ of Definition 2.1, with its stated trace metric. □

Proof of Lemmas 12.3 and 12.4. The row normalization comes from the free forms of Lemma 11.5; the loop cancellations from the symmetry of the one-loop kernel and the two-leg curl division of Lemma 12.5; the bounds from the classical matching of Lemma 12.6 and the one-loop bound of Lemma 12.9. Start at $m = N$ and descend. The condition $c_{\mathrm{dir},k} \le \gamma_k$ supplies the positive row factor at every level. Its zero-momentum entries give

$$\widehat{\mathsf{H}}(0)_{\mathfrak{q}\mathfrak{r}} = c_{\mathrm{dir},J_m}\delta_{\mathfrak{q}\mathfrak{r}} + \sum_{\mathfrak{p}} s_{\mathfrak{p}\mathfrak{q}} s_{\mathfrak{p}\mathfrak{r}} = \delta_{\mathfrak{q}\mathfrak{r}} .$$

This proves the first cancellation and also forces $c_{\mathrm{dir},J_m} \le 1$ by taking a diagonal entry.

Subtract $\alpha_{N,m}$ times the kinetic kernel from the quadratic kernel of T. The residual is transverse. Its colour bilinear form is scalar: its self-adjoint representing operator commutes with the adjoint action, and its eigenspaces are ideals, so simplicity leaves one eigenvalue. Signed permutations identify the diagonal orientation coefficients. Reflection in an axis belonging to exactly one of two distinct orientations cancels the off-diagonal coefficient. Thus there is one degree-two coefficient, removed by the subtraction. Reflection through the origin removes degree three. The residual link kernel consequently vanishes to order four.

Apply the two-leg part of Lemma 12.5. Removing its two curl symbols leaves a plaquette kernel of order two. If its Fourier coefficients for a fixed orientation-colour pair are $a_{\mathrm{loop}}(z)$, then

$$\sum_z a_{\mathrm{loop}}(z) = 0, \qquad \sum_z z_a a_{\mathrm{loop}}(z) = 0 \quad (a = 1, \dots, 4) .$$

Exponential decay permits termwise differentiation. Put the coefficients on coordinate paths to obtain the quadratic atoms and their absolutely convergent moments. Inherited loop atoms already have these moments, hence a link kernel of order four. Pullback by the free interpolation preserves this order. Only the quadratic Taylor kernel of T is assigned to these atoms; its higher terms enter the preparatory expansion of Lemma 13.9.

For the bounds, contour displacement and the path-length estimate after Lemma 12.5 give a positive common decay rate for the top construction and the two possible incoming stages $k, k+1$. Choose $0 < w_k^{\mathrm{path}} < w_k'$ below that rate, and bound the series at w_k' when choosing A_k^{path}. Lemma 12.6 permits this choice before the corresponding ratios, by making the input-dependent terms at most one. Lowering the width gives the bounds at w_k^{path}. Make the following loop

estimates at w'_k as well, so that the quadratic bound retains the same strict inequality between widths.

The new loop contribution has a weighted bound $a_j^{\mathrm{lp}}\ell^4 + b_j^{\mathrm{lp}}\log\ell + c_j^{\mathrm{lp}}$. Here ℓ^4 bounds the trace dimension per coarse cell, and the logarithm comes from the subtracted coefficient. For an inherited order-four link kernel, rescaling gives

$$\ell^4\ell^{-2}O(|\xi/\ell|^4) = O(\ell^{-2}|\xi|^4)\,.$$

Its weighted pullback is therefore bounded by $C_j^{\mathrm{pb}}\ell^{-2}\Lambda_{m+1}^{\mathrm{loop}}$. Choose ℓ so that $C_j^{\mathrm{pb}}\ell^{-2} \le 1/4$, then raise the coupling entry threshold until

$$2(a_j^{\mathrm{lp}}\ell^4 + b_j^{\mathrm{lp}}\log\ell + c_j^{\mathrm{lp}}) \le \Lambda_m^{\mathrm{loop}}, \qquad \Lambda_{m+1}^{\mathrm{loop}} \le 2\Lambda_m^{\mathrm{loop}}\,.$$

The sum is at most $\frac{1}{2}\Lambda_m^{\mathrm{loop}} + \frac{1}{4}\cdot 2\Lambda_m^{\mathrm{loop}} = \Lambda_m^{\mathrm{loop}}$. These choices are in the order of Table 3: widths and amplitudes, ratios, then entry thresholds. The first inequality is possible because $2(M_m^{\mathrm{cut}})^{\theta_{J_m}} \to \infty$ as $\widehat{\beta}_m \to \infty$ with ℓ_m, J_m fixed. For the second, antitonicity of θ_k, $J_m \le J_{m+1} \le J_m + 1$, and $\theta_k \le 1$ give, once $M_m^{\mathrm{cut}} > 1$,

$$\frac{\Lambda_{m+1}^{\mathrm{loop}}}{\Lambda_m^{\mathrm{loop}}} \le \left(\frac{M_{m+1}^{\mathrm{cut}}}{M_m^{\mathrm{cut}}}\right)^{\theta_{J_m}} \le \frac{M_{m+1}^{\mathrm{cut}}}{M_m^{\mathrm{cut}}} \longrightarrow 1$$

as $\widehat{\beta}_m \to \infty$ with ℓ_m fixed. The descent starts with zero loops and the fourth-order Wilson matching of Lemma 12.6. The kernels and paths are chosen once for all periods. Finally $|\mathcal{F}_s| \le 2$ and $|\mathfrak{a}_i^{(k)}(U,x)| \le 2^k|\mathfrak{a}_i^{(k)}|$ give absolute convergence of the action. □

13 Activities and the inductive step

An exact block integral contains interactions on arbitrarily large supports. We express them as a sum of regular activities and a disjoint-support sum of forced activities, the large-field polymer activities. Regular terms are smooth where their supports are healthy; forced terms retain configurations with flagged plaquettes. Their norms serve different purposes: derivatives are required of the first family, whereas the second is estimated uniformly over configurations. The estimates and the exact density identity enter Theorem 5.5; the Wilson representation starts the recursion and supplies the initial curve of Theorem 5.6.

The step has five stages, and the sizes explain why each is needed. Write $\mathfrak{b} = \widehat{\beta}_{m+1}$; the output regular activities must be bounded by $\epsilon_m^{\mathrm{reg}} = \widehat{\beta}_m^{-11/10}$.

1. *Connected expansion* (Lemma 13.5). The block integral is a polymer gas whose elementary factors are $O(\mathfrak{b}^{-1/2+o(1)})$. Its logarithm has regular terms of total size $O(\mathfrak{b}^{-7/16})$, which is too large, and forced terms of size $O(e^{-r_j\beta'_u t_{m+1}^2})$.

2. *Linear terms* (Lemma 13.6). An incoming regular activity has zero value, gradient and affine Hessian, so after blocking its leading part has scaling weight five: ℓ_m^{-5} per fine anchor, a contraction ℓ_m^{-1} per coarse anchor.

3. *Low-order terms* (Lemmas 13.7–13.8). Terms with one or two elementary factors are only $O(\mathfrak{b}^{-1/2})$ individually. Their sum is the logarithm of a local Gaussian integral; after flat subtraction, parity and the classical and one-loop terms of Section 12, it is $O(\mathfrak{b}^{-23/20})$ up to subpower factors. We localize it by a fourth difference over five reflected tori. Terms with three or more factors are already $O(\mathfrak{b}^{-19/16})$ up to subpower factors.

4. *Curvature-square subtraction* (Section 13.2). The remaining affine Hessians are subtracted locally; lattice symmetry collects them into one coefficient Δ_u, added to the coupling.

5. *Fusion and reset* (Section 13.3). The resulting mask discrepancies vanish on healthy supports and are absorbed into the forced activities, giving the exact density (Lemma 13.13). The bound on Δ_u from the subtraction already gives the drift bound; the scalar reset (Lemma 13.14) reads Δ_u off at one test configuration, which identifies it from the density and gives its continuity.

Section 13.4 proves continuity at fixed volume, Section 13.5 treats the Wilson initial condition, and Section 13.6 fixes the step on whole curves and proves Theorem 5.5. Section 13.7 proves the analytic estimates used in Section 14 for endpoint tuning and the fixed-volume limit.

Throughout, the group and trace metric are those of Definition 2.1. Scales, flags, path actions and densities are those of Section 5 and Definitions 5.2, 12.1–12.2. At a step $m+1 \to m$ put $j = J_{m+1}$, $\Lambda = \Lambda_{L_p D_m}$, $t = t_m$, $B = B_m$ and $\sigma = \sigma_{J_m}$. Set $h_X(U) = \mathbf{1}\{X \cap \mathrm{Flag}_m(U) = \varnothing\}$. Constants may depend on the fixed stage data and, when indicated, on the chosen ratio, but are independent of the volume. Here C_j is a generic stage constant.

Local terms and their norms. An activity is a local function of the links, attached to a finite support, whose sums make up the effective density. Locality on X means dependence only on links with both endpoints in X. Write $\xi_o(y)$ for the centered lift of $y - o$ to $\mathbb{Z}^4$. All lattice distances in this section use the maximum norm. Set $N_R(X) = \{y : \mathrm{dist}(y, X) \le R\}$ and $N_R(o) = N_R(\{o\})$. A link internal to X has both endpoints in X. The nonwrapping condition is $10\,\mathrm{dist}(y, o) < L_p D_m$ for every $y \in X$; it permits local affine tests without crossing a choice of lift.

Definition 13.1 (Regular activities)**.** A regular activity is a Borel real gauge-invariant function E_i, local on X_i and anchored at $o_i \in X_i$. It satisfies $E_i(1) = 0$ and is $C^{\mathfrak{q}_{J_m}}$ in exponential left-link increments on healthy configurations, with all ordered derivatives. On nonwrapping supports the Hessian of $A \mapsto E_i(\exp A)$ at zero vanishes on every pair of affine directions

$$A_{(y,a)} = u_a + \sum_c \xi_{o_i}(y)^c v_{ac}, \qquad u_a, v_{ac} \in \mathfrak{g}\,.$$

Here $(\exp A)_e = \exp(A_e)$. This removes the affine quadratic contribution from the error.

Together with $E_i(1) = 0$ and the vanishing first derivative, which is automatic by gauge invariance (proof of Lemma 13.10), this says that E_i has no relevant or marginal part: its local constant and its curvature-square coefficient have been extracted, the analogue of the localization operator of [6].

For forced terms the inventory specifies exactly the flags inside the support. Additional tests distinguish local alternatives with the same support and inventory.

Definition 13.2 (Forced activities)**.** A forced activity with address $(Y_i, I_i, \mathcal{E}_i^{\mathrm{extra}})$, where $\varnothing \ne I_i \subset Y_i$, is a Borel real gauge-invariant function K_i that depends only on the links internal to Y_i and the flags in Y_i, and vanishes outside the eligibility event $\mathcal{E}_i^{\mathrm{elig}}$ of (5.8). Its extra test is Borel, gauge invariant, and reads the same links and flags. These conditions restrict the forced sum to the prescribed large-field inventory.

The address of a regular activity is the pair (X_i, o_i); that of a forced activity is the triple $(Y_i, I_i, \mathcal{E}_i^{\mathrm{extra}})$, and its eligibility indicator is $\mathbf{1}_{\mathcal{E}_i^{\mathrm{elig}}}$.

Three sets attached to the large-field sites, the flags, play different roles. The support Y_i carries the forced activity. The inventory $I_i \subset Y_i$ lists the flagged sites that the eligibility event requires in Y_i, exactly those and no others. The complete read support of a factor in the expansion (defined before Lemma 11.14) is the region that factor inspects; polymers are formed through overlaps of complete read supports, and a support Y_i need not be connected. Deciding the flags at the sites of Y_i reads plaquettes within distance one of Y_i, so the read regions of two disjoint supports may overlap although the supports do not. For example, let exactly two distant plaquettes have chord above t_m, so that the flags are their vertex sets F_1 and F_2. A selection may use one forced term whose support contains both, with inventory $F_1 \cup F_2$, or two terms with disjoint supports and inventories F_1 and F_2. Both alternatives occur in the exact-cover sum of Definition 5.2; in each selection every flag is covered exactly once, and no canonical covering is chosen. A term whose support contains $F_1 \cup F_2$ but whose inventory is F_1 is ineligible at this configuration and vanishes. Forced fusion (Lemma 13.11) groups overlap components together with the logarithmic factors meeting them and recovers every selection uniquely.

We use the ordered derivative seminorm and weighted sums of Definitions 5.3–5.4. Regular sums are bounded at their anchors; forced sums are bounded at every site of their supports, so the factor $e^{\sigma \operatorname{cov}_B(Y)}$ charges each support for its size. For a function f on Z anchored at o, $\|f\|_{m,Z}$ denotes the seminorm of Definition 5.3 with that support and anchor.

Continuity concerns the coefficients at fixed addresses. Health sets remain fixed as the curve parameter varies.

Definition 13.3 (Continuity of representations)**.** A continuous curve has fixed activity addresses and tests, and continuous coupling and scalar normalization. At each u_0, differences of regular derivatives and forced values admit nonnegative majorants $e_i(u), k_i(u)$ in the sense of Definition 5.3, with

$$\sum_i e^{\sigma \operatorname{cov}_B(X_i)} e_i(u) + \sum_i e^{\sigma \operatorname{cov}_B(Y_i)} k_i(u) \longrightarrow 0 \quad (u \to u_0)\,.$$

Both sums run over all addresses. This is continuity for coupling tuning, with no derivative of a flag indicator.

At fixed volume these sums control the density:

$$\sum_i \|E_i\|_m \le |\Lambda| \mathcal{E}_m, \qquad \sum_i \sup|K_i| \le |\Lambda| \mathcal{K}_m,$$
$$\sum_{\text{selections}} \prod_i \sup|K_i| \le \prod_i (1 + \sup|K_i|) \le e^{|\Lambda| \mathcal{K}_m}\,.$$

The first inequality sums the regular bounds over anchors, the second the forced bounds over sites, since each forced support is nonempty; the third sum runs over all finite selections. Restricting to disjoint supports and exact coverage preserves the third bound, so the density sums of Definition 5.2 converge. No volume-independent bound on the full density follows.

Coefficient algebras. Several estimates are applied to factors with coefficients in a commutative algebra; this is how observable insertions are carried through the expansion in Section 14. Let $z_1, \dots, z_n$ be commuting marks with $z_i^2 = 0$, the nilpotent external fields of [6]. An element of the coefficient algebra is $a = \sum_I a_I z_I$, where $I \subset \{1, \dots, n\}$, $z_I = \prod_{i \in I} z_i$ is a mode and a_I is real or complex. With mark weights $\mathsf{w}_i > 0$ and $\mathsf{w}_I = \prod_{i \in I} \mathsf{w}_i$, put $\|a\| = \sum_I |a_I| \mathsf{w}_I$. Since $z_I z_J = z_{I \cup J}$ for disjoint I, J and $z_I z_J = 0$ otherwise, $\|ab\| \le \|a\| \|b\|$. Taking the coefficient of a mode is a finite linear operation and commutes with integration; for one mark, the coefficient of z in $\prod_j (K_j + z L_j)$ is $\sum_j L_j \prod_{i \ne j} K_i$. The case $n = 0$ is the algebra of complex coefficients.

Support exponents. The output norms use $\sigma = \sigma_{J_m} = 100$. Intermediate estimates carry up to three spare units: Lemma 13.5 gives regular terms at $\sigma + 3$ and forced terms at $\sigma + 2$, and Lemma 13.8 keeps $\sigma + 3$ and $\sigma + 1$. The spare units pay for converting anchored sums to site sums (a support with covering number n has at most $C(1 + B)^4(1 + n)^4$ possible anchors; the factor $(1 + n)^4 \le Ce^n$ is absorbed by one spare unit, while $C(1 + B)^4$ is paid by the smallness of the activities), for collecting logarithmic corrections into the forced sum, and for fusion. Independently, the rooted clause of Lemma 13.5 holds with every exponent lowered by two; Section 14 uses this in its marked estimates.

13.1 The local expansion

Fix a compact parameter interval and a continuous input curve ρ'_u in the sense of Definition 13.3. Its prescribed path satisfies Lemmas 12.3–12.4; its regular and forced families satisfy Definitions 13.1–13.2 and the caps of Definitions 5.3–5.4; and $\beta'_u \in \mathfrak{B}_{m+1}$. Write $\mathcal{H}_u = \mathcal{B}^{(m)}_{\beta'_u} \operatorname{dens} \rho'_u$, and split the path action as in Definition 12.2. Write $\mathsf{K}^{\mathrm{kin}}_{2,m}$ for $\mathsf{K}^{\mathrm{kin}}_m$ without its cubic and quartic atoms; it is still nonlinear in the links. The coefficient $\alpha_{N,m}$ is the quadratic coefficient of the determinant comparison in Section 12. It is independent of the curve parameter and the period.

For $a < b$ define the clover curvature, the clover-leaf field strength of lattice gauge theory, by

$$F^{\mathrm{cl}}_{ab}(U, o) = \frac{1}{4} \sum_{\varepsilon,\eta=\pm1} \varepsilon\eta \operatorname{proj}_{\mathfrak{g}} \frac{U_{o;\varepsilon a,\eta b} - U^{-1}_{o;\varepsilon a,\eta b}}{2} .$$

The loop starts at o, takes steps $\varepsilon e_a, \eta e_b, -\varepsilon e_a, -\eta e_b$, and returns to o. Thus each term transforms by the adjoint action at o and uses links within distance two. For an affine field as in Definition 13.1, differentiating the four link factors gives

$$\left.\frac{d}{dz}\right|_{z=0} F^{\mathrm{cl}}_{ab}(e^{zA}, o) = v_{ba} - v_{ab} . \tag{13.1}$$

Each inverse contributes the negative link field; the four orientations have the same signed curl.

Expanding each row square $\frac{1}{2}|\mathcal{L}^{\mathrm{path}}_{\mathfrak{p}}|^2$ into the products $\frac{1}{2} r_{\mathfrak{p}i} r_{\mathfrak{p}k} \langle \mathcal{F}_{\mathfrak{p}i}, \mathcal{F}_{\mathfrak{p}k} \rangle$ and adding the direct plaquette terms gives a countable family of local energy terms indexed by λ; translations and the H_4 average, with weight $|H_4|^{-1}$, are included in the addresses. Each term has three versions: L_λ, the actual clipped or threshold-tested term; L^{raw}_λ, the same quadratic curvature function without clip or test; and L^{cl}_λ, the function L^{raw}_λ with each curvature replaced by the clover curvature of its orientation at the anchor. For a direct term $L^{\mathrm{raw}}_\lambda = c_{\mathrm{dir}} \operatorname{pr}(U_\square)$ and $L^{\mathrm{cl}}_\lambda = \frac{1}{2} c_{\mathrm{dir}} |F^{\mathrm{cl}}_q(U, o_\lambda)|^2$.

Choose Z_λ to contain the slot footprints and the radius-B neighborhood of the anchor needed to decide the relaxed threshold test. Queries outside that neighborhood are automatic once

$$t \exp((B - 2)/R^{\mathrm{rel}}_{J_m}) \ge 2 ,$$

since chords are at most two. Lemmas 12.3–12.4 give absolute convergence of $\mathsf{K}^{\mathrm{kin}}_m = \sum_\lambda L_\lambda + \mathcal{A}_m$, where $\mathcal{A}_m$ is the sum of the cubic and quartic atoms. Define the masked quadratic functional and record the clover identity:

$$\mathcal{Q}_m = \sum_\lambda h_{Z_\lambda} L^{\mathrm{raw}}_\lambda , \qquad \sum_\lambda h_{\{o_\lambda\}} L^{\mathrm{cl}}_\lambda = \frac{1}{2} \sum_o h_{\{o\}} \sum_q |F^{\mathrm{cl}}_q(U, o)|^2 . \tag{13.2}$$

A signed image in these formulas uses the pulled-back anchor, support and clover; the second equality follows from the coefficient identity of Lemma 12.4 and reindexing orientations. On Z_λ healthy, $L_\lambda = L^{\mathrm{raw}}_\lambda$; without flags, $\mathcal{Q}_m = \mathsf{K}^{\mathrm{kin}}_{2,m}$.

The local integral is expanded before any regular term is normalized. Its elementary factors F_a are: (i) the remainder of the stationary action; (ii) the variation $\mathrm{e}^{-[\mathcal{L}_{m+1}(U)-\mathcal{L}_{m+1}(U_v)]}$ of the inherited loop action about the stationary configuration U_v; (iii) the determinant and covariance ratios and the Haar Jacobian; (iv) the incoming regular and forced activities; (v) the cutoff remainders of Lemma 11.14, that is, the failures of the fluctuation, chart, tag and inventory tests. Put $W_a = F_a - 1$, except $W_a = K_a$ for an incoming forced activity. The support of a factor includes every field, flag and covariance window it reads. We join two factors when their complete read supports, as defined before Lemma 11.14, intersect. A polymer is a finite product of factors whose complete read supports have connected intersection graph; we use only the polymers allowed by Lemma 11.14, which have disjoint incoming forced supports, inventories covered by the corners of the selected fine plaquettes $A_f(M)$, and the component compatibility there. The selection retains these supports and inventories. A polymer is *ordinary* if its complete read supports also avoid the output flags.

For an ordinary polymer M on a healthy output support, its integrated weight and an absolute derivative majorant are

$$z_M(U) = \int \prod_{a\in M} W_a(U,\zeta)\, d\nu(\zeta), \qquad \int G_M\, d\nu \le P_j(\widehat{\beta}_m)(1+s_M)^{d_j} \prod_{a\in M} f_a\,. \tag{13.3}$$

Here ν is the single sector Gaussian/Haar law of Lemma 11.14, G_M majorizes the product and its required ordered derivatives, and f_a are that lemma's factor bounds. The integer s_M is the sum of the numbers of radius-B blocks covering the complete read supports, including retained path lengths in the same units. The function P_j has subpower growth after the ratio has been fixed. The formula is used only on compatible sectors; their sum retains the exact allocation indicators. Terms whose reads meet an output flag component remain attached to that component and have a nonempty forced inventory. The ordinary connected integrals are taken on the complement, where their read supports are disjoint from the output flags. Components with empty inventory are used in the regular logarithm on their healthy supports.

Ordinary or forced: an example. A cutoff remainder and an output flag are different objects. Let M consist of a stationary-action remainder and the cutoff remainder w of the chart cutoff on a fine edge e that it reads, with no selected large-field test near e. The factor W_w vanishes outside the event E_w that some Gaussian coordinate within distance $2B$ of the location of w is large; this is a property of the fluctuation, and the coarse configuration may be healthy. If the complete read supports of M avoid $\mathrm{Flag}_m(U)$, then M is ordinary: it enters the regular logarithm with the bound (13.3), in which w has its own factor $f_w = M_w\varepsilon_w$. If these supports meet a flag $x \in \mathrm{Flag}_m(U)$, the same product is attached to the large-field component of x and belongs to a forced term whose inventory contains x. That term keeps its eligibility indicator and the energy suppression of Lemma 11.14(iii), and its other attached factors keep their own bounds f_a.

The tree sums of Sections 13.1–13.3 are instances of one estimate. A support covered by n radius-B balls has at most $(2B+1)^4 n$ sites; this converts a site bound into an attachment bound.

Lemma 13.4 (Rooted polymer bound)**.** *Let $(Z_a, v_a)_a$ be a countable family of nonempty finite supports with weights $v_a \ge 0$, and let $\eta > 0$ satisfy*

$$(2B+1)^4 \sup_x \sum_{a:\, x\in Z_a} v_a e^{\eta\, \mathrm{cov}_B(Z_a)} \le \eta\,.$$

For a support $Z_\star$ with weight $v_\star \ge 0$, let $T_\star$ be the sum over finite rooted trees with root $(Z_\star, v_\star)$ and other vertices labelled by the family, repetitions allowed, in which every child support meets its

parent's; a tree contributes the product of its weights, and each set of k children of a vertex the factor $1/k!$. Then $T_\star \le v_\star e^{\eta \operatorname{cov}_B(Z_\star)}$.

Proof. Let $T^{(h)}$ denote the sum over trees of height at most h. The exponential formula for unordered children gives $T_\star^{(0)} = v_\star$ and $T_\star^{(h+1)} = v_\star \exp(\sum_{a: Z_a \cap Z_\star \neq \varnothing} T_a^{(h)})$. If $T_a^{(h)} \le v_a e^{\eta \operatorname{cov}_B(Z_a)}$ for every label, the exponent is at most $\sum_{x \in Z_\star} \sum_{a: x \in Z_a} v_a e^{\eta \operatorname{cov}_B(Z_a)} \le |Z_\star| \eta (2B+1)^{-4} \le \eta \operatorname{cov}_B(Z_\star)$. Induction on h and monotone convergence prove the bound. □

Every connected family of labels containing a distinguished label has a spanning tree rooted there, so the lemma dominates connected sums of nonnegative majorants. In applications the factor $e^{\eta \operatorname{cov}_B(Z_a)}$ is paid by one reserved unit of support exponent.

We now give the connected identity with its initial estimates. For a support exponent τ let $\mathcal{E}_\tau, \mathcal{K}_\tau$ be the sums of Definition 5.4 with σ replaced by τ. In the rooted clause below, $W_\star$ is a further elementary factor with complete read support $Z_\star$, $n_\star = \operatorname{cov}_B(Z_\star)$, and integrated majorant $v_\star$, including its support weight and required derivatives; any retained forced suppression of a selected large-field test is kept separately. The other factors have nonnegative majorants v_a on their complete read supports Z_a, including the reserved support weights. Lowering the support exponents in (d) includes the one in $v_\star$. Let $\mathcal{R}_{\mathrm{reg}}(W_\star)$ and $\mathcal{R}_{\mathrm{for}}(W_\star)$ be the absolute sums, at exponents $\sigma+3$ and $\sigma+2$, of the connected terms containing one distinguished occurrence of $W_\star$, counted once. The expansion produces a scalar τ_u^{old}, local functions f_a^0 on X_a, regular before flat-value and Hessian subtraction, and forced activities $k_{i,u}^{\mathrm{con}}$ with exact-cover sum $\mathcal{F}_u^{\mathrm{con}}$; weighted-continuous means continuous in the weighted norms of Definition 13.3. The stronger regular estimate needed for subtraction is obtained in Lemma 13.8.

Lemma 13.5 (Connected local expansion)**.** *There are $C_j, r_j, \eta > 0$, depending only on the stage and ratio, such that for every input curve:*

(a) These exist, with fixed countable addresses, equivariant and weighted-continuous, such that everywhere

$$\mathcal{H}_u = \exp\left\{ \tau_u^{\mathrm{old}} |\Lambda| - \beta_u' \mathsf{K}_m^{\mathrm{kin}} - \mathcal{L}_m - \alpha_{N,m} \mathcal{Q}_m - \sum_a h_{X_a} f_a^0 \right\} \mathcal{F}_u^{\mathrm{con}},$$

with absolutely convergent sums.

(b) $\mathcal{E}_{\sigma+3}(f^0) \le C_j \widehat{\beta}_{m+1}^{-7/16}$ and $\mathcal{K}_{\sigma+2}(k^{\mathrm{con}}) \le C_j e^{-r_j \beta_u' t_{m+1}^2}$.

(c) If $(2B+1)^4 \sup_x \sum_{a: x \in Z_a} v_a \le \eta$, then $\max\{\mathcal{R}_{\mathrm{reg}}(W_\star), e^{r_j \beta_u' t_{m+1}^2} \mathcal{R}_{\mathrm{for}}(W_\star)\} \le v_\star e^{\eta n_\star}$.

(d) Parts (b) and (c) hold coefficientwise for factors in a coefficient algebra, and with every support exponent lowered by two.

Proof. The block kernel of Definition 5.1 turns the density into a product of elementary factors; Lemma 11.14 integrates their products under one Gaussian law and factorizes disjoint supports, and the local energy of Lemma 11.12 pays for the large-field components.

Step 1: polymer gas. Expand finite products first. Fubini's theorem applies to each term by (13.3). Disjoint complete read supports use disjoint Gaussian marginal blocks and Haar variables, so their integrals multiply. This gives a gas of the connected weights z_M. Overlapping factors are always integrated together.

Step 2: elementary factors. Put $\mathfrak{b} = \widehat{\beta}_{m+1}$ and $g = (\beta_u')^{-1/2}$. For a healthy local chart write its link logarithm as $m_Y + gz$; the mean is $O(t_{m+1})$ by Lemma 11.13. The following bounds include

the prescribed finite number of ordered derivatives. A factor $P_j(\mathfrak{b})$ below absorbs fixed powers of B and M_{m+1}^{cut} after the ratio is fixed; read-count powers remain in $(1+s_M)^{d_j}$ as in (13.3):

factor	integrated majorant
stationary-action remainder	$P_j\beta'_u g^3 \le P'_j\mathfrak{b}^{-1/2}$
centered determinant remainder	$P_j(t_{m+1}+g) \le P'_j\mathfrak{b}^{-1/2}$
Gaussian-density and stationarity residuals	$P_j\mathfrak{b}^{c_j^{\mathrm{res}}}e^{-c_j^{\mathrm{tail}}B}$
Haar-Jacobian remainder	$P_j(t_{m+1}+g)^2$
inherited loop variation	$P_j\Lambda_{m+1}^{\mathrm{loop}}(t_{m+1}+g)g \le P'_j\mathfrak{b}^{-1/2}$
incoming regular sum	$P_j\epsilon_{m+1}^{\mathrm{reg}}$
incoming forced component	$P_j\epsilon_{m+1}^{\mathrm{for}}$
cutoff remainder	$M_w\mathfrak{b}^{-1/2}$

(13.4)

For the first row, Taylor's formula after the stationary quadratic part starts with $\beta'_u(m_Y+gz)^3$; substituting $m_Y = O(gM_{m+1}^{\mathrm{cut}})$ gives the displayed bound. For the determinant, subtract its flat value and the quadratic term already assigned to $\alpha_{N,m}\mathcal{Q}_m$. The inverse identity bounds the remaining variation by $P_j(t_{m+1}+g)$. The covariance and stationary residuals are exponentially small by Lemma 11.13; their rescaling costs only the displayed fixed power. The chart density has value one and zero first derivative at zero, giving its quadratic bound. The inherited loop action is quadratic in slot curvatures with weighted coefficient sum at most $\Lambda_{m+1}^{\mathrm{loop}}$ by Lemma 12.3; on the chart each slot is $O(t_{m+1}+g|z|)$ and moves by $O(g|z|)$, which gives its row. Incoming sums retain their own allowances. Chart-test failures, fine-plaquette and mismatch failures, and tag or inventory departures are precisely the cutoff remainders of Lemma 11.14.

Lemma 11.14 permits every fixed exponent $\tau_w > 0$ for the cutoff remainders. Take $\tau_w = 1/2$, so $\varepsilon_w = \mathfrak{b}^{-1/2}$ and the simultaneous probability bound is $\varepsilon_w^{2r} = \mathfrak{b}^{-r}$ for r distinct cutoff addresses, including the coordinate-choice multiplicity. The linear Chernoff bound there gives each cutoff remainder the exponent $c_j(t_{m+1}/g)/B^4$ for some $c_j > 0$. Since $t_{m+1}/g \ge M_{m+1}^{\mathrm{cut}}/\sqrt{2}$,

$$\frac{t_{m+1}/g}{B^4\log\mathfrak{b}} \longrightarrow \infty\,.$$

The fixed amplitudes M_w enter P_j. All rows of (13.4) are consequently at most $P'_j(\mathfrak{b})\mathfrak{b}^{-1/2}$ with P'_j of subpower growth. After the coupling threshold is raised, $P'_j(\mathfrak{b}) \le \mathfrak{b}^{1/16}$, so every factor majorant is at most $\mathfrak{b}^{-7/16}$. Path lengths are summed with the exponential bounds of Lemma 12.3. Every fixed polynomial in $1+s_M$ is bounded by $C_{j,\eta}e^{\eta s_M}$; choose η from the unused support exponent. These choices precede the coupling threshold.

Step 3: large-field components. Large-field components are the connected components of the neighbourhoods of the selected large-field tests in the allocation sector of Lemma 11.12, as defined before Lemma 11.14. Their selections $A = (A_f, A_n, A_c)$ consist of the selected fine plaquettes, mismatch edges and coarse flags; each retains $\exp(-\kappa_B\beta'_u t_{m+1}^2|A|)$ before integration, κ_B being the constant of Lemma 11.12. Fix the retained exponent $0 < \kappa'_B < \kappa_B$ in the large-field component bound of Lemma 11.14, and then choose $0 < r_j < \kappa'_B$. When $A_c \neq \varnothing$, keep all reads meeting this component in its local factor and apply that bound. Old measurable factors use their supremum envelopes, while the Gaussian variables are integrated. The frozen Haar integrals are probability integrals. After adjoining every large-field component met by the reads, $A_c = \varnothing$ means that these reads avoid output flags. In this case the selected-test clause of Lemma 11.14 retains $\prod_i a_i \le e^{-\kappa'_B\beta'_u t_{m+1}^2|A|}$ through the required derivative order; apply its ordinary polymer bound to

the remaining factors. With F_A denoting this component product, its integrable majorant satisfies

$$|F_A| \le g_A, \qquad \int g_A \, d\nu \le P_j(\mathfrak{b})(1+s_A)^{d_j} e^{-\kappa'_B \beta'_u t_{m+1}^2 |A|} \prod_{a \text{ attached}} f_a \,.$$

Here s_A counts its radius-B blocks, and κ'_B was fixed before r_j. The product contains the other attached factors' bounds. Its covariance scores and residual quadratic exponentials are estimated by square completion under the common Gaussian law there. The surrounding ordinary factors have supports disjoint from the output flags and use that same law. The read-count polynomial uses the unused support exponent: for every fixed $\delta > 0$, $(1+s_A)^{d_j} \le C_{j,\delta} e^{\delta s_A}$. Choose δ within that reserve, as for (13.3); this leaves the output support weights intact. The local sector multiplicity of the selected tests uses a fixed part of the energy exponent between κ'_B and r_j. The prescribed derivative costs are already included in the read-count polynomial and the subpower factor. A simultaneous set of r cutoff addresses contributes $\varepsilon_{\mathrm{w}}^{2r}$ to the Gaussian probability in Lemma 11.14; Cauchy–Schwarz leaves $\varepsilon_{\mathrm{w}}^{r}$ after the other factors have been estimated under that same law. Thus each cutoff remainder still has its own factor in $\prod f_a$, also when distinct ones share Gaussian coordinates.

Step 4: tree sums. The coupling threshold makes $(2B+1)^4 \sup_x \sum_{a: x \in Z_a} v_a$ at most η, with η smaller than the reserved exponent, which pays for the factors $e^{\eta \operatorname{cov}_B(Z_a)}$. Lemma 13.4 then bounds the rooted sums by $v_a e^{\eta n_a}$. This proves absolute summability, including all sectors and all differentiated terms. For a distinguished factor $W_\star$, the same lemma with root $(Z_\star, v_\star)$ gives $v_\star e^{\eta n_\star}$. Rooting there counts each distinguished occurrence once. A forced component keeps $\exp(-r_j \beta'_u t_{m+1}^2)$ in this sum, proving both forced bounds. For coefficient-valued factors take the norms of the coefficient algebra before integration; they are subadditive and submultiplicative, so every product and sum reduces to the same nonnegative tree calculation. Finite-product identities hold coefficient by coefficient, and absolute convergence permits passage to their limits. This proves the coefficientwise assertions without changing the health or inventory tests. Increasing the fine support exponent before choosing the ratio leaves the output exponents $\sigma + 3$ and $\sigma + 2$. The same tree calculation applies with two subtracted from every support exponent, including the exponent in $v_\star$: covering subadditivity still applies and the attachment sum decreases. This proves the bounds with the reduced weights.

Step 5: logarithm. There are two connectivity operations. First, elementary factors whose complete read supports intersect are integrated together; their connected products are the polymers M, with weights z_M. Second, the hard-core gas of these polymers is expanded logarithmically; its connected graphs, the clusters, have polymers as vertices and support overlaps as edges. In (13.5) connectedness refers to this second graph. The first operation handles dependence under the Gaussian law; the second produces the additive regular interaction. The finite range of the sector covariance enters only the first operation: by Lemma 11.14(v) it makes factors with disjoint complete read supports independent under the reference law. It neither replaces the second operation nor makes the effective density a product over regions. Write $\Xi(z) = \sum_{\mathcal{M}} \prod_{M \in \mathcal{M}} z_M$, where $\mathcal{M}$ runs over finite selections with disjoint supports, including the empty selection. This is the ordinary hard-core gas. Its logarithm has the connected coefficients

$$\begin{aligned} \log \Xi(z) &= \sum_{n \ge 1} \frac{1}{n!} \sum_{M_1, \dots, M_n} \phi^T(M_1, \dots, M_n) \prod_{h=1}^{n} z_{M_h}, \\ \phi^T &= \sum_{\Gamma \text{ connected}} \prod_{\{h,k\} \in \Gamma} \left[-\mathbf{1}_{Z_{M_h} \cap Z_{M_k} \neq \varnothing} \right]. \end{aligned} \tag{13.5}$$

The graph in this formula has vertex set $\{1,\dots,n\}$, and the one-vertex coefficient is one. Apply [34, Theorem 1, equations (3)–(4)] to the counting measure with masses z_M and interaction $\zeta(M,M') = -\mathbf{1}_{Z_M\cap Z_{M'}\neq\varnothing}$. Its hypotheses are $|1+\zeta|\le 1$, finite total variation, finite exponential mass, and

$$\sum_{M':Z_{M'}\cap Z_M\neq\varnothing} |z_{M'}|e^{\mathfrak{w}(M')} \le \mathfrak{w}(M), \qquad \mathfrak{w}(M) = \eta\,\mathrm{cov}_B(Z_M)\,.$$

The rooted estimate just proved verifies the last condition. At fixed volume the site sum gives finite total exponential mass. Equation (13.5) and its rooted estimate therefore apply. Fixing the distinguished occurrence changes the symmetry factor from $1/n!$ to $1/(n-1)!$, matching that theorem's rooted sum. Keep the components joined to forced polymers as coefficients of the exact-cover sum; take the logarithm only of the remaining ordinary gas. Expanding and regrouping finite selections proves this separation. Absolute convergence then proves the identity for the full integral. The deterministic part is exactly $\beta'_u\mathsf{K}_m^{\mathrm{kin}} + \mathcal{L}_m + \alpha_{N,m}\mathcal{Q}_m$: the stationary terms and one-loop quadratic term are those of Lemmas 12.6–12.9, with the masks in (13.2) retained.

Step 6: continuity and covariance. For two inputs at fixed addresses, telescope each product with one difference factor. Activity differences use their own difference norms under Lemma 11.14. For a coupling change, hold physical coordinates fixed and compare the Gaussian densities explicitly. On d active coordinates with fixed mean and positive precision A, put $Q(a) = \langle a-m, \mathsf{A}(a-m)\rangle/2$. The normalized laws satisfy

$$\frac{d\nu_{\beta_2}}{d\nu_{\beta_1}}(a) = \left(\frac{\beta_2}{\beta_1}\right)^{d/2} e^{-(\beta_2-\beta_1)Q(a)}\,.$$

Its derivative in β_2 is the same ratio times $d/(2\beta_2) - Q(a)$. On a small relative coupling interval its absolute value is dominated by a polynomial times a Gaussian exponential with precision strictly below that of the base law. Square completion bounds its integral. Mean and precision changes are treated by the inverse difference identity and the Schur bounds of Lemma 11.13; their density derivatives are again polynomial Gaussian scores. No old measurable factor is differentiated in this comparison. The tree bound with the same attachment margin controls the weighted discrepancy by a constant times the input activity and coupling discrepancies; this constant may depend on the fixed volume. Continuity holds locally and, by a finite covering of the compact interval, on fixed addresses throughout. Translation and signed permutations transport complete read supports, factors and sectors; their finite average gives equivariance. □

One factor followed to the output allowance. Lemma 13.5 and the lemmas that follow it combine into the output bound $\epsilon_m^{\mathrm{reg}}$, uniformly in the volume; this passage closes the induction. We follow one elementary factor through it, with $\mathfrak{b} = \widehat{\beta}_{m+1}$, $g = (\beta'_u)^{-1/2}$ and one law ν throughout.

One sector. On a healthy chart the link logarithm is $m_Y + gz$ with $m_Y = O(t_{m+1}) = O(gM_{m+1}^{\mathrm{cut}})$, since $\beta'_u \le 2\widehat{\beta}_{m+1}$. After the stationary quadratic part, the stationary-action remainder starts with $\beta'_u(m_Y+gz)^3$, and $\beta'_u g^2 = 1$ makes this $O(g(M_{m+1}^{\mathrm{cut}} + |z|)^3)$. Gaussian moments of z and $g \le \sqrt{2}\,\mathfrak{b}^{-1/2}$ give the first row of (13.4), an integrated majorant $P'_j\mathfrak{b}^{-1/2}$ with P'_j of subpower growth.

One law for the whole polymer. The factors of a polymer read common Gaussian coordinates, so the integral of their product is not the product of their integrals. Lemma 11.14(i) bounds it by $P_j(\mathfrak{b})(1+s_M)^{d_j}\prod_a f_a$ in a single estimate under ν; the square completions for covariance scores and the Cauchy–Schwarz step for cutoff remainders take place inside it (Step 3 above). No product in the expansion is bounded by multiplying separate integral bounds.

Reserve and supports. Raising the threshold until $P'_j(\mathfrak{b}) \le \mathfrak{b}^{1/16}$ makes every factor majorant at most $\mathfrak{b}^{-7/16}$; the exponent $1/16$ absorbs the fixed powers of B and M_{m+1}^{cut}. The read-count polynomial obeys $(1+s_M)^{d_j} \le C_{j,\eta} e^{\eta s_M}$ and is paid by a unit of support exponent that the output norm does not use. The threshold then makes the attachment parameter of Lemma 13.4 smaller than η, and the tree sums give regular terms of size $\mathfrak{b}^{-7/16}$ and forced terms of size $e^{-r_j \beta'_u t_{m+1}^2}$ (Lemma 13.5(b)).

Three factors. The size $\mathfrak{b}^{-7/16}$ is far above $\epsilon_m^{\mathrm{reg}}$. Multiply every ordinary factor majorant by $\mathfrak{h} = \mathfrak{b}^{3/8}$. The attachment parameter becomes $\mathfrak{h}\mathfrak{b}^{-7/16} = \mathfrak{b}^{-1/16}$, still small, so the weighted sums converge, with total a subpower factor times $\mathfrak{h}\mathfrak{b}^{-7/16}$. A term with at least three factors carries at least $\mathfrak{h}^3$ in this sum, and restoring the weights gives

$$P_j(\mathfrak{b})\, \mathfrak{h}^{-3} \cdot \mathfrak{h}\, \mathfrak{b}^{-7/16} = P_j(\mathfrak{b})\, \mathfrak{b}^{-19/16} \le \mathfrak{b}^{-23/20}\,,$$

the reserve $19/16 - 23/20 = 3/80$ absorbing P_j. Since $\widehat{\beta}_{m+1} \ge \widehat{\beta}_m$, this is at most $\widehat{\beta}_m^{-1/20} \epsilon_m^{\mathrm{reg}}$.

The branches. Terms with one or two factors lack this gain. The regular part of the connected logarithm is divided as follows; the proof of Lemma 13.8 spends at most $15\epsilon_m^{\mathrm{pre}}/16$ on the regular rows in total.

terms	bound	proof
linear in an incoming regular activity	$(V_j/\ell_m + \omega_{j,\ell_m}^{\mathrm{inh}})\epsilon_{m+1}^{\mathrm{reg}}$	Lemma 13.6
≥ 3 factors, large-field overlap, large support	$\mathfrak{b}^{-23/20}$	(13.8)
incoming regular factor and a further factor	$P_j \epsilon_{m+1}^{\mathrm{reg}} \mathfrak{b}^{-1/2}$	Lemma 13.5(c)
paths longer than B	$\mathfrak{b}^{-7/16} e^{-w_{J_m}^{\mathrm{path}} B/4}$	Lemma 12.3
selected low-order terms, localized	$P_j(\mathfrak{b}^{-23/20} + \varepsilon_{\mathrm{tr}})$	Lemma 13.7
unselected low-order terms	$\varepsilon_{\mathrm{tr}}$	Lemma 13.8

Forced components. Their gain is energy, spent in stages along

$$\kappa_B > \kappa'_B > r_j > r'_j\,.$$

The difference $\kappa_B - \kappa'_B$ absorbs, per selected test, the annular affine errors, the logarithmic corrections of chart densities, tags, determinants and frozen coordinates, and the derivative losses (Steps 7–8 of the proof of Lemma 11.14). The difference $\kappa'_B - r_j$ pays the local multiplicity of the selected tests (Step 3 above), and $r_j - r'_j$ the constants of collecting the corrections L_x into the forced sum, which leaves κ_j^{pre} (Lemma 13.8). In Lemma 13.12 half of the remaining exponent keeps the forced weights small in the attachment parameter of Lemma 13.11, and the factor $e^{-\kappa_{\mathrm{fus}}}$, $\kappa_{\mathrm{fus}} = \kappa_j^{\mathrm{pre}} (M_{m+1}^{\mathrm{cut}})^2/2$, is retained. The read-count polynomials $(1+s_A)^{d_j}$ are paid from support exponent, not from energy.

Two kinds of gain. A selected large-field test gains deterministically, by its energy. A cutoff remainder gains probabilistically: it vanishes unless a Gaussian coordinate within $2B$ of its address is large. Lemma 11.14(iv) bounds the probability of r distinct such events by ε_w^{2r}, and Cauchy–Schwarz under the same law leaves ε_w^r after the other factors are estimated: one factor $\varepsilon_w = \mathfrak{b}^{-1/2}$ for each distinct address, also when addresses share coordinates. Each address is counted once, and the selection retains the incoming forced supports and inventories.

Closure. The three linear operations of Sections 13.2–13.3 cost at most $C_j^0 \epsilon_m^{\mathrm{pre}}$, plus a term $C_j^1 |\alpha_{N,m}| t_m^3$ made small by the threshold. With $\epsilon_m^{\mathrm{pre}} = \theta c_j \epsilon_m^{\mathrm{reg}}$, Lemma 13.12 gives $\mathcal{E}_m \le \epsilon_m^{\mathrm{reg}}/1024$ and $\mathcal{K}_m \le \epsilon_m^{\mathrm{for}}/1024$, inside the input bounds required at level m. Every constant depends only on the stage and the ratio, and every threshold is independent of N, p and the period, so the same construction applies at the next step (Lemma 13.17).

The terms linear in an incoming regular activity must be treated separately from products of small factors. Their supports can be large, so they are not replaced by auxiliary tori. The function E_i and its derivative norm in this paragraph are at level $m+1$. For such an address i, let T_i be its contribution to the negative connected logarithm. The local Gaussian substitution of Lemma 11.14, restricted to the windows incident to X_i, reconstructs a fine configuration $U_i^0(V,z)$. With the same inner chart protector Γ_i, define

$$P_i(V) = \frac{1}{|\mathcal{T}_i|} \sum_{\theta \in \mathcal{T}_i} \mathrm{Ext}_{Y_i} \int \Gamma_{i,\theta}(z) E_i\big(U_{i,\theta}^0(V,z)\big)\, d\gamma_{C_{i,\theta}(V)}(z), \qquad R_i = T_i - P_i\,.$$

Here $\mathcal{T}_i$ is the finite set of rooted frames and tags in that local substitution. The operator Ext_{Y_i} extends the function from the healthy domain by its local Borel representative; on that domain it changes no value. The coarse support Y_i contains the projected fine support and every incident window. This principal Gaussian average retains the incoming function itself.

Lemma 13.6 is the contraction estimate for the linear part of the step. An incoming regular activity has zero value, zero gradient (by gauge invariance) and zero affine Hessian (Definition 13.1). Its Gaussian average P_i at fixed coarse field therefore sees only background terms of scaling weight at least five, each costing ℓ_m^{-5}; summing over the ℓ_m^4 fine anchors above a coarse anchor gives V_j/ℓ_m. The remainder R_i, the difference between the true local integral and this Gaussian average, is smaller by a factor $\widehat{\beta}_m^{-1/2+o(1)}$.

Lemma 13.6 (Inherited regular terms)**.** *The functions P_i, R_i are regular before flat-value and Hessian subtraction, on the same fixed coarse addresses as T_i. At exponent $\sigma+3$ their anchored derivative sums satisfy*

$$\mathcal{E}(P) \le \frac{V_j}{\ell_m} \epsilon_{m+1}^{\mathrm{reg}}, \qquad \mathcal{E}(R) \le \omega_{j,\ell_m}^{\mathrm{inh}}(\widehat{\beta}_m)\epsilon_{m+1}^{\mathrm{reg}},$$

where V_j is fixed before the ratio and $\omega_{j,\ell}^{\mathrm{inh}}$ has upper rate $-1/2$. The identity $T_i = P_i + R_i$ and the corresponding weighted comparisons hold on the entire input curve.

Proof. The principal average is estimated by a weight count in the radial gauge; the remainder by the Gaussian comparison of Lemma 11.14 and the unrefined bound of Lemma 13.5. For the principal average use the same normalized directions as in the incoming norm. The linear background lifts a unit coarse curl to fine curvature of size $O(\ell_m^{-2})$. Replace the background on X_i by its affine part at the anchor. In the Taylor formula for E_i the constant vanishes by Definition 13.1. The first derivative also vanishes: invariance under constant gauge transformations makes its coefficient on each link an $\mathrm{Ad}(G)$-invariant vector of $\mathfrak{g}$, which is zero since $\mathfrak{g}$ is simple and nonabelian (Definition 2.1). Its Hessian on two such affine fields vanishes by Definition 13.1. Since E_i is gauge invariant, evaluate it in the radial gauge at its fine anchor; at fine distance $\rho \le \ell_m$ the lifted field is then at most $C_j \ell_m^{-2}(t_m/t_{m+1})$ times a permitted fine direction, using $\varpi_{J_{m+1}} \ge \varpi_{J_m} \ge 2$. In the normalized curvature directions, the linear prediction P and its affine approximation $\mathsf{P}_{\mathrm{aff}}$ satisfy

$$\|\mathsf{P}\| \le C_j \ell_m^{-2}, \qquad \|\mathsf{P} - \mathsf{P}_{\mathrm{aff}}\| \le C_j \ell_m^{-3}\,.$$

Lemma 11.6 gives $\|\nabla_f^r W_0\|_{\infty\to\infty} \le C_j \ell_m^{-1-r}$ for $0 \le r \le 2$. Taking its curl and one further fine difference gives the two bounds above, with constants fixed before the ratio. For supports with $d_i^3 < \ell_m$ the weight count below shows that every surviving term carries ℓ_m^{-5}; larger supports are treated without affine subtraction.

For every support extending beyond the radius ℓ_m, use the incoming covering number $d_i = \operatorname{cov}_{B_{m+1}}(X_i)$. Split at $d_i^3 < \ell_m$. In this class $3d_i \le \ell_m$; together with $B_{m+1} \le 2B_m$, this bounds the inspected fine radius by $6(1 + B_m)\ell_m$. The support is nonwrapping because the coarse side is at least $100(1 + B_m)$. Perform the radial Taylor expansion on this whole support. In the incoming weighted direction norm, each differentiated lifted potential decomposes into a constant part, an affine part and a remainder with respective bounds

$$C\ell_m^{-1}, \qquad C\ell_m^{-2}, \qquad C\ell_m^{-3}.$$

Here C is bounded by a stage constant plus the normalized nonlinear and fluctuation errors. After the ratio is fixed these errors are made at most one by the entry choice: their bounds are fixed powers of ℓ_m, B_m times $t_{m+1} + g/t_{m+1}$, which tends to zero. The spatial weights are retained in this decomposition, using $\varpi_{J_{m+1}} \ge \varpi_{J_m} \ge 2$ from item 4 of Table 3.

Assign weights one, two and three to the three parts. In the local radial-gauge expansion, a site gauge removes the weight-one part of the potential and produces terms of weight at least two; commutators of constant parts first occur at weight two. Thus the first possible curvature has weight two. The terms through weight four are therefore a constant, linear terms in curvature and its spatial derivatives, and a quadratic curvature term. The first is zero by flat normalization, the second is zero by constant colour invariance, and the last is zero by the affine Hessian condition. Equivalently, multilinearity after the three-part decomposition cancels every selection of total weight at most four. A selection of larger weight has at least five inverse powers of ℓ_m. Taylor's integral formula along the healthy radial segment gives the same gain at the evaluation field: integrate the next derivative from the flat field, repeating up to the available order $\mathfrak{q}_{J_m} \ge 6$. The ordered chain rule has only stage-dependent many placements. The common Gaussian majorant of Lemma 11.14 bounds their moments.

For the complementary class $d_i^3 \ge \ell_m$, including winding supports, apply the derivative bound without an affine subtraction. Its direction and read-count losses are polynomial in $1 + d_i$. Both cases are expressed by

$$\delta_\ell(d) = \begin{cases} \ell^{-5}, & d^3 < \ell, \\ 1, & d^3 \ge \ell, \end{cases} \qquad \delta_\ell(d) \le \ell^{-5}(1 + d)^{15}.$$

Indeed $\ell^5 \le d^{15}$ in the second case. After the entry choice absorbs factors polynomial in B_m, if a_i is an incoming derivative majorant, the integrated principal bound is at most $C_j a_i \delta_{\ell_m}(d_i)(1 + d_i)^{q_j}$, for a fixed integer q_j; the polynomial comes from the ordered chain rule and the Gaussian moments just described.

For support compression, fix an integer r with $Y_i \subset N_{rB_m}(\operatorname{parent}(X_i))$ and put $T_r = (2r + 13)^4$. Choose the ratio to satisfy

$$20(\sigma + 3)T_r \le \ell_m \sigma_{J_{m+1}}.$$

Together with $B_{m+1} \le 2B_m$, this leaves half the incoming exponent:

$$(\sigma + 3)\operatorname{cov}_{B_m}(Y_i) \le (\sigma + 3)T_r + \tfrac{1}{2}\sigma_{J_{m+1}} d_i.$$

The remaining half bounds $(1 + d_i)^{q_j+15}$, since $\sup_{d\ge 0}(1 + d)^{q_j+15} e^{-\sigma_{J_{m+1}} d/2} < \infty$. There are ℓ_m^4 fine anchors above a coarse anchor. Multiplying their count by the per-address factor ℓ_m^{-5} proves the bound V_j/ℓ_m, with V_j fixed before the ratio.

For $\operatorname{cov}_{B_m}(Y_i) \le B_m$, compare the actual local integral with this principal Gaussian average before integrating. Subtract their means, covariances and transported kernels using the inverse

difference identity, and telescope the remaining factor product with one marked difference. The inverse difference identity $C^{-1} - \widetilde{C}^{-1} = C^{-1}(\widetilde{C} - C)\widetilde{C}^{-1}$ puts a reconstruction difference in one factor. Taylor's formula for the link exponential makes that factor at most $P_j(\mathfrak{b})(t_{m+1}+g)$ in each of the requested normalized derivatives; stationarity and window residuals contribute $P_j(\mathfrak{b})g^{-c_j}e^{-c'_jB}$. Consequently their common bound is

$$P_j(\mathfrak{b})\{t_{m+1} + g + g^{-c_j}e^{-c'_jB}\} = \mathfrak{b}^{-1/2+o(1)}, \qquad \mathfrak{b} = \widehat{\beta}_{m+1}\,.$$

A second occurrence of the incoming activity has bound $C_j(\epsilon^{\mathrm{reg}}_{m+1})^2$, which is also bounded by an upper-rate-$-1/2$ factor times $\epsilon^{\mathrm{reg}}_{m+1}$. In every term of the difference, one of these factors is retained before the common Gaussian majorant is applied. Lemma 11.14 bounds its remaining moments by $P_j(\mathfrak{b})(1+s_M)^{d_j}$, where P_j has upper rate zero after the ratio is fixed and s_M is the block count defined above. Summing with the unused incoming support exponent gives a function $\omega^{\mathrm{inh}}_{j,\ell}$ of rate $-1/2$. This proves the remainder estimate in the weighted norm. For the remaining supports use the unrefined bound for T_i from Lemma 13.5 and the same Gaussian majorant for P_i. Support compression implies that the incoming covering number is at least $cB_m - C_j$. Retaining a further fixed part of its exponent gives an extra e^{-cB_m} in the weighted sum, for some $c > 0$ fixed before the coupling threshold. Since $B_m \geq (\log \widehat{\beta}_m)^2$, this factor, times any fixed polynomial in B_m, is bounded by $\widehat{\beta}_m^{-1/2}$ after the threshold is raised. This supplies the stated remainder bound on large supports.

Local Gaussian marginals read only Y_i. A gauge transformation acts orthogonally on their colour coordinates and leaves Γ_i invariant; changing Gaussian variables proves gauge invariance of P_i. The frame average transports the support and anchor under lattice symmetries. The local extension preserves these identities, measurability and the healthy derivatives. The same properties hold for R_i by subtraction. For a parameter difference use that incoming difference in the displayed integral and in the marked comparison. The covariance score and protected-coordinate change are those of Lemma 11.14, so the same weighted estimates give continuity at fixed addresses. Finally choose the ratio so V_j/ℓ_m is less than the prescribed pre-ratio fraction, and then the threshold so $\omega^{\mathrm{inh}}_{j,\ell_m}$ is less than that fraction. □

The bound $\mathfrak{b}^{-7/16}$ of Lemma 13.5 does not fit $\epsilon^{\mathrm{reg}}_m = \widehat{\beta}_m^{-11/10}$. The obstruction comes from connected terms with one or two elementary factors, each of size $\mathfrak{b}^{-1/2}$. They improve only through cancellations in their sum, which is the logarithm of a local Gaussian integral: the lone cubic term vanishes by parity, the quartic terms belong to the classical action and the quadratic terms to the one-loop action. An estimate of each separate local term would miss these cancellations. We therefore evaluate the finite sum on small auxiliary tori, where the logarithm can be expanded directly, and recover a local density from the auxiliary sums. On the torus Λ_i of side $2(w+i)$, sum the local formula over all sites y, each evaluated at the configuration reflected about $x - \widetilde{y}$, so that y sits over x. The points of Λ_i fall into classes according to their distances from the reflection faces. When the configuration is healthy near x, the summand is the same on each class for every i, and the number of points in a class is a polynomial in i of degree at most four; only the interior class has degree four, with leading coefficient 2^4 (Figure 11). The fourth difference with the weights λ_i in (13.6) therefore returns the interior summand $F_x(V) - F_x(1)$ exactly, because

$$\sum_{i=0}^{4}(-1)^i\binom{4}{i}i^k = 0 \quad (0 \leq k < 4), \qquad \sum_{i=0}^{4}(-1)^i\binom{4}{i}i^4 = 4!\,.$$

The discrepancy of the health masks is supported near flags and will be included in the forced activities.

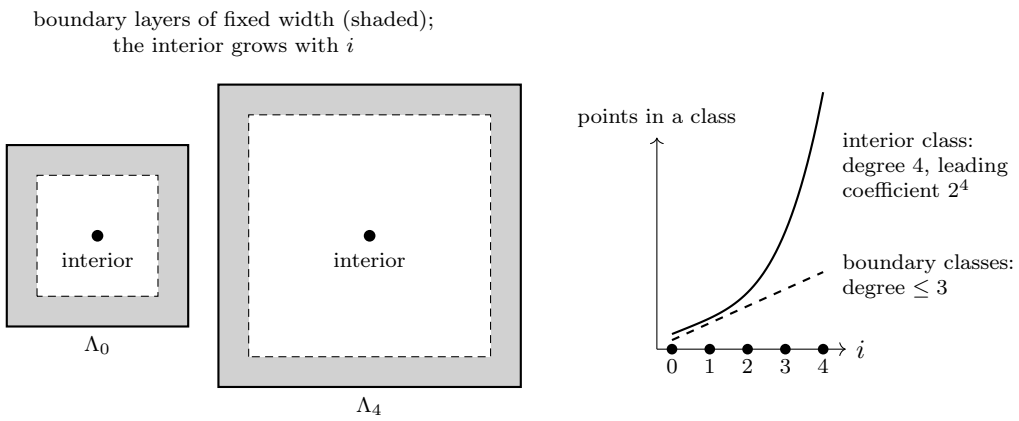


Figure 11: The five reflected tori Λ_i of Lemma 13.7, of sides $2(w+i)$, drawn for $i=0$ and $i=4$. A point y of Λ_i is classified by its distance, through the read radius R, from each reflection face; at the shifted configurations $V_{i,x-\widetilde{y}}$, healthy near x, the local formula takes the same value on a class for all i. The boundary layers have fixed width, so a boundary class has a number of points of degree at most three in i, while the interior class grows, with degree four and leading coefficient 2^4. The fourth difference $\sum_i (-1)^i \binom{4}{i}(\cdot)$ annihilates degrees below four, so it removes every boundary class from the total and keeps the interior one. The tori are auxiliary evaluation domains, not physical volumes, and the cancellation takes place in the sum over all points, not term by term.

Write f_a^0 for a regular summand of the negative logarithm in (13.5), including its coefficient. Keep the inherited terms of Lemma 13.6 separately. Call a remaining term *low-order* if it is a connected term with at most two polymers of at most two elementary factors each, every path atom having length at most B, with no incoming activity factor or large-field component, or a path term of length at most B. Such terms have complete read radius at most a fixed stage multiple of B. Low-order terms read only the prescribed path, the coupling and the addresses, not the values of the incoming activities. Fix an exhaustion of the low-order addresses by finite sets invariant under translations and signed permutations, and select the low-order terms in one of these sets. Its index and the truncation error $\varepsilon_{\mathrm{tr}}$ are fixed in the proof of Lemma 13.8. Give an ordinary term the shares $p_{xa} = \mathbf{1}_{X_a}(x)/|X_a|$, and a path term the share $\mathbf{1}_{\{o_a\}}(x)$; these shares sum to one. Put $F_x = \sum_{a \text{ selected}} p_{xa} f_a^0$, the selected low-order sum. The remaining terms form f^{tail}. Let R contain every complete read support of this finite sum relative to a site receiving a nonzero share. Its radius is a stage multiple of B. Set $w = 4R + 100(1+B)$, $\bar{w} = w+5$, and let Λ_i have side $2(w+i)$, $0 \le i \le 4$. A configuration on Λ gives a configuration $V_{i,x}$ on Λ_i by repeated reflection of its cube about x. The direction of a reflected link is reversed when necessary. Put

$$\lambda_i = \frac{(-1)^i \binom{4}{i}}{2^4\, 4!}, \qquad G_x(V) = \sum_{i=0}^{4} \lambda_i \sum_{y \in \Lambda_i} F_y^{(i)}(V_{i,x})\,. \tag{13.6}$$

Here $F^{(i)}$ is the same finite local formula on the auxiliary torus, with its value at 1 subtracted at each site. The mask discrepancy and the constant of the redistribution are

$$L_x = \sum_{a \text{ selected}} p_{xa} h_{X_a} f_a^0 - F_x(1) - \sum_{i=0}^{4} \lambda_i \sum_{y \in \Lambda_i} h_{N_{\bar{w}}(x-\widetilde{y})} F_y^{(i)}(V_{i,x-\widetilde{y}}), \qquad c = F_0(1)\,, \tag{13.7}$$

where $\widetilde{y}$ is the integer representative used in the reflection.

For the estimates below put $\mathfrak{b} = \widehat{\beta}_{m+1}$; P_j has subpower growth.

Lemma 13.7 (Finite-period localization)**.** *The function G_x is local on $N_{\bar{w}}(x)$, and*

$$\sum_a h_{X_a} f_a^0 = c|\Lambda| + \sum_a h_{X_a} f_a^{\mathrm{tail}} + \sum_x h_{N_{\bar{w}}(x)} G_x + \sum_x L_x \,.$$

Each L_x reads only links and flags in $N_{2\bar{w}}(x)$ and vanishes when that set is healthy, and

$$\|G_x\|_{m,N_{\bar{w}}(x)} \le P_j(\mathfrak{b})(\mathfrak{b}^{-23/20} + \varepsilon_{\mathrm{tr}}), \qquad \sup|L_x| \le P_j(\mathfrak{b})\mathfrak{b}^{-7/16}.$$

Proof. The identity is a rearrangement; the vanishing of L_x is the fourth-difference cancellation; the bound on G_x comes from the auxiliary logarithm, estimated with the classical and one-loop subtractions of Section 12. Summing (13.7) over x and changing x to $x - \widetilde{y}$ gives exactly the asserted identity. Translation symmetry makes $F_x(1)$ independent of x.

Use the polynomial period requirement in item 7 of Table 3, in addition to $100(B+1) \le L_p D_m$. Write $n = L_p D_m$. The radius R of the local construction is affine in B with stage-dependent coefficients, computed from the low-order radius; hence $\bar{w} = 4R + 100(1+B) + 5$ is affine in B with positive coefficients. Put $\bar{a} = 2\bar{w}|_{B=1}$. The chosen stage has integers $v_j^{\mathrm{per}}, h_j^{\mathrm{per}}$ with $v_j^{\mathrm{per}} \ge \bar{a}$, $h_j^{\mathrm{per}} \ge 1$ and

$$n \ge v_j^{\mathrm{per}} (1 + B_m + B_{m+1})^{h_j^{\mathrm{per}}} .$$

These are the minimum-period choices made after the ratios. Since $B \ge 1$,

$$2\bar{w} \le \bar{a}B < \bar{a}(1 + B_m + B_{m+1})^{h_j^{\mathrm{per}}} \le n, \qquad 4R + 16 < n.$$

The reflection cubes, whose side is at most $2(w+4) < 2\bar{w}$, therefore embed, as does every radius-R read set. The larger health set $N_{2\bar{w}}(x)$ is a set on the original torus and need not embed; the argument uses only its containment of the queried links and flags.

For the local comparison, suppose that the field is healthy on the radius-R neighbourhood. Translate its cube to $[0, 2R]^4$. On integer representatives $0 \le k < n$, fold each coordinate by $q(k) = k$ for $k \le 2R$, $q(k) = 4R - k$ for $2R < k \le 4R$, and $q(k) = 0$ thereafter. Pull back links along this map, reversing them on decreasing edges and setting them to the identity on collapsed edges. A plaquette becomes a plaquette of the original cube, with possible reversal and conjugation, or has identity holonomy. The field is therefore globally healthy and agrees with the original field on the cube. The bound $4R < n$ permits the construction without a divisibility assumption on n. Locality then permits this replacement when comparing the common local formulas on different periods.

We verify the vanishing of L_x on a healthy $N_{2\bar{w}}(x)$. The complete read supports then lie in that set, every mask in (13.7) is one, and every reflected configuration occurring there is healthy. Classify a point of Λ_i by its distance, through R, from each reflection face. In the y-summand of (13.7) the reflection cube is centred at $x - \widetilde{y}$, so its site y corresponds to x. For a fixed class the local formula is therefore independent of i. In one coordinate the number of points in an interior class is $2(w - 2R + 1) + 2i$; for a specified boundary class it is a fixed integer. Consequently each class has a multiplicity polynomial in i of degree at most four. Its degree is four only when all four coordinates are interior. The binomial identity displayed before (13.6) cancels every boundary class and retains the interior class with coefficient $2^4 4!/(2^4 4!) = 1$. Locality and gauge invariance identify its value with $F_x(V) - F_x(1)$. This proves $L_x = 0$. It also proves its stated read support: the largest displacement is at most $w + 4 < \bar{w}$, and each mask reads a further radius $\bar{w}$.

Classification of the terms. Here are the quantitative estimates on the sum before this localization. Put $\mathfrak{b} = \widehat{\beta}_{m+1}$ and $\mathfrak{h} = \mathfrak{b}^{3/8}$. In the convergent connected sums multiply the majorant of each ordinary elementary factor by $\mathfrak{h}$. The rooted attachment parameter still tends to zero, since $\mathfrak{h}\mathfrak{b}^{-7/16} = \mathfrak{b}^{-1/16}$. A term with at least three such factors across its polymers gains $\mathfrak{h}^{-3}$ upon restoring its original weights. A term with a retained large-field overlap has the same gain because its suppression is at least $\mathfrak{h}^{-3}$. The sum after multiplication has size at most a subpower factor times $\mathfrak{h}\mathfrak{b}^{-7/16}$. Thus the terms in these two classes have bound

$$P_j(\mathfrak{b})\mathfrak{h}^{-3}(\mathfrak{h}\mathfrak{b}^{-7/16}) = P_j(\mathfrak{b})\mathfrak{b}^{-19/16} \le \mathfrak{b}^{-23/20} \tag{13.8}$$

after the coupling threshold is raised. The reserve is $19/16 - 23/20 = 3/80$. This estimate also bounds fixed ordered derivatives: their placement polynomial was included before the rooted sum.

The connected sum separates into three parts. First retain the inherited terms using Lemma 13.6; their principal sum is made smaller than a prescribed multiple of $\epsilon_m^{\mathrm{pre}}$ by choosing the ratio, and their comparison remainder by raising the threshold. Among all other connected terms retain also those whose total support size exceeds the low-order radius: their support decay is at most $\mathfrak{h}^{-3}$ after the threshold is raised, so the estimate in (13.8) applies to them. A term with an incoming regular factor and at least one further factor is not low-order. The rooted clause of Lemma 13.5, with $W_\star$ that factor and at least one attached factor, bounds these terms by $P_j(\mathfrak{b})\epsilon_{m+1}^{\mathrm{reg}}\mathfrak{b}^{-1/2} \le \mathfrak{b}^{-23/20}$. The remaining terms are low-order: the selected connected contributions and path terms of length at most B, to which the reflected-period construction is applied, and the low-order tail, of weighted norm at most $\varepsilon_{\mathrm{tr}}$. Long paths have an additional factor $e^{-w_{J_m}^{\mathrm{path}}B/4}$ by Lemma 12.3; it converts their $\mathfrak{b}^{-7/16}$ bound to $\mathfrak{b}^{-23/20}$. Low-order connected terms with three or four elementary factors already obey (13.8); the remaining connected terms have at most two.

The auxiliary logarithm. For the finite sum use the stationary integral on each Λ_i. Their sides $2(w+i)$ satisfy $2(w+i) \ge 100(B+1)$. The entry choice $B \ge j+4$ also gives $\ell_m\, 2(w+i) \ge 100((j+3)\ell_m+1)$ for their fine periods. Thus the coarse-period form of Lemmas 11.12–11.14 applies, without requiring these auxiliary periods to belong to the chosen physical-volume sequence. Subtract its flat value and the classical cubic, quartic and quadratic one-loop terms of Lemmas 12.6–12.9. Differentiation of its logarithm gives connected expectations under the common Gaussian law. For the terms not involving incoming errors, write the normalized local integral as $\mathbb{E}\exp(V_3+V_4+V_{\ge 5})$, after the stationary action and its Gaussian determinant have been extracted. The subscripts refer to the Taylor degree in the link chart. Gaussian parity removes a lone odd cubic term. The fourth-order stationary terms are already in the classical path action, and the field-dependent quadratic part of the determinant is already in $\alpha\mathcal{Q}_m$ and the quadratic loop atoms. Subtracting the value at 1 removes the remaining vacuum terms. Expanding $\log \mathbb{E}e^V$ through two factors gives $\mathbb{E}V + \frac{1}{2}(\mathbb{E}V^2 - (\mathbb{E}V)^2)$; its remaining terms have at least three factors. In normalized fluctuation coordinates each nonquadratic factor starts at order $\mathfrak{b}^{-1/2}$. The common Gaussian majorant bounds the last terms by $P_j(\mathfrak{b})\mathfrak{b}^{-3/2}$. In the one- and two-factor terms that remain after the stated cancellations, Taylor's integral remainder and the healthy field bound give

$$P_j(\mathfrak{b})\{\mathfrak{b}^{-3/2} + \mathfrak{b}\, t_m^5 + \mathfrak{b}^{-1}t_m + t_m^3 + e^{-c_jB}\} \le \mathfrak{b}^{-23/20}\,.$$

Every radius and chart loss is included in the subpower factor. Each of the four power terms has exponent at most $-3/2$ before that factor; the path exponential decreases faster than any fixed power. Terms involving incoming activities are not low-order; apart from the inherited terms they obey (13.8) or the bound $P_j(\mathfrak{b})\epsilon_{m+1}^{\mathrm{reg}}\mathfrak{b}^{-1/2}$ above. Lemma 11.14 bounds these terms simultaneously, so the derivative bound for the centered total auxiliary sum is

$$C_j\mathfrak{b}^{-23/20} + 2|\Lambda_i|\{C_j\mathfrak{b}^{-23/20} + \varepsilon_{\mathrm{tr}}\}\,. \tag{13.9}$$

Here $\varepsilon_{\mathrm{tr}} > 0$ is the chosen truncation error. The first term is the centered local integral; the two volume terms bound the omitted regular sum at the field and at 1. Approaching a boundary point of the healthy domain by radial contraction preserves these estimates, so they hold on the closed domain used in Definition 5.3.

A direction of that definition, restricted to $N_{\bar w}(x)$ and reflected, is bounded by $2t(1+\bar w)^{\varpi_{J_m}}$. Rescaling each of at most $\mathfrak{q}_{J_m}$ directions in (13.9) therefore gives

$$\|G_x\|_{m,N_{\bar w}(x)} \le 2^{\mathfrak{q}_{J_m}}(1+\bar w)^{\varpi_{J_m}\mathfrak{q}_{J_m}} \sum_{i=0}^{4} |\lambda_i|[C_j\mathfrak{b}^{-23/20} + 2|\Lambda_i|(C_j\mathfrak{b}^{-23/20}+\varepsilon_{\mathrm{tr}})]\,.$$

All the factors outside the negative powers are fixed powers of $1+B$. Also $\mathrm{cov}_B(N_{\bar w}(x))$ is bounded by a stage constant. The regular bound divided by $\epsilon_m^{\mathrm{reg}}$ tends to zero, since $23/20 > 11/10$. Finally, (13.7) and the uncentered $\mathfrak{b}^{-7/16}$ bound give $\sup|L_x| \le P_j(\mathfrak{b})\mathfrak{b}^{-7/16}$, with the same polynomial read radius. These logarithmic corrections tend to zero in the site norm at exponent $\sigma+2$ and vanish on healthy supports. This proves all the estimates claimed for the localization. □

To specify the small coefficient, let C_j^{sub}, C_j^{low} and C_j^{high} be the operator bounds for local subtraction, low path replacement and the part of high path replacement proportional to $\epsilon_m^{\mathrm{pre}}$. These are the three linear operations in the following subsections; they are fixed from the output derivative norm and path bounds. Put $C_j^0 = C_j^{\mathrm{sub}} + C_j^{\mathrm{low}} + C_j^{\mathrm{high}}$. These constants use only derivative orders, direction powers, support exponents, and the output path bounds. For a prescribed $0 < \theta \le 1$, choose before the ratios

$$c_j = \frac{1}{64(1+C_j^0)}, \qquad \epsilon_m^{\mathrm{pre}} = \theta c_j \epsilon_m^{\mathrm{reg}}\,.$$

The quantity $\epsilon_m^{\mathrm{pre}}$ depends on m through $\epsilon_m^{\mathrm{reg}}$. The next assertion concerns the very same block integral and the masks of (13.2). Its forced sum $\mathcal{F}_u^0$ is the disjoint-support exact-cover sum of Definition 5.2 formed from the $k_{i,u}$, and $\mathfrak{e}_{i,u} \ge \|f_{i,u}\|_{m,X_i}$, $\mathfrak{k}_{i,u} \ge \sup_U |k_{i,u}(U)|$ are majorants.

Lemma 13.8 (Preparatory activities)**.** *For every input curve above there are a continuous scalar τ_u, local regular functions $f_{i,u}$, with the regularity of Definition 13.1 before flat-value and Hessian subtraction, and forced activities $k_{i,u}$ (Definition 13.2), all with fixed addresses, equivariant and continuous in the weighted norms with exponents $\sigma+3$ and $\sigma+1$, such that*

$$\mathcal{H}_u(U) = e^{\tau_u|\Lambda| - \beta_u'\mathsf{K}_m^{\mathrm{kin}}(U) - \mathcal{L}_m(U) - \alpha_{N,m}\mathcal{Q}_m(U) - \sum_i h_{X_i}(U) f_{i,u}(U)}\mathcal{F}_u^0(U), \tag{13.10}$$

$$\sup_x \sum_{o_i = x} e^{(\sigma+3)\,\mathrm{cov}_B(X_i)}\mathfrak{e}_{i,u} \le \epsilon_m^{\mathrm{pre}}, \tag{13.11}$$

$$\sup_x \sum_{i: x\in Y_i} e^{(\sigma+1)\,\mathrm{cov}_B(Y_i)}\mathfrak{k}_{i,u} \le e^{-\kappa_j^{\mathrm{pre}}(M_{m+1}^{\mathrm{cut}})^2}, \tag{13.12}$$

with $\kappa_j^{\mathrm{pre}} > 0$ fixed after the ratio. Restricting the interval preserves all this.

Proof. The connected expansion of Lemma 13.5 supplies the terms, Lemma 13.6 and the classification in the proof of Lemma 13.7 bound all but the low-order ones, and Lemma 13.7 localizes the latter; the choices below split $\epsilon_m^{\mathrm{pre}}$ among them. First choose the ratio so that the inherited principal terms of Lemma 13.6 use at most $\epsilon_m^{\mathrm{pre}}/32$, and then raise the threshold so that their comparison remainders use at most $\epsilon_m^{\mathrm{pre}}/32$. Bound the terms retained in (13.8) and the long paths

together by $\epsilon_m^{\mathrm{pre}}/16$. The localization bound is $o(\epsilon_m^{\mathrm{pre}})$ with the stronger support exponent. This fixes a coupling threshold independent of the period.

Next fix the truncation index. The low-order tail reads only the prescribed path, the coupling and the incoming addresses. For each coupling in $\mathfrak{B}_{m+1}$ summability gives a tail smaller than $\varepsilon_{\mathrm{tr}}/2$; weighted continuity in the coupling gives a neighbourhood on which the same tail is smaller than $\varepsilon_{\mathrm{tr}}$. Finitely many neighbourhoods cover the compact $\mathfrak{B}_{m+1}$; take the largest index. This index depends only on the incoming address lists, so it serves every input with these lists. Increasing the index preserves the tail bound. Choose the tail bound to be at most $\epsilon_m^{\mathrm{pre}}/16$. The regular terms outside the finite sum then have total at most $(1/16 + 2/16)\epsilon_m^{\mathrm{pre}}$. Choose $\varepsilon_{\mathrm{tr}}$ so that the two localization errors in (13.9), including the direction and support losses, have total at most $12\epsilon_m^{\mathrm{pre}}/16$. The resulting bound is $15\epsilon_m^{\mathrm{pre}}/16 < \epsilon_m^{\mathrm{pre}}$.

The change in the scalar is $\tau_u^{\mathrm{old}} - F_0(1)$, with the sign fixed by (13.7). Multiply the preliminary forced sum by $\exp(-\sum_x L_x)$. Expand each e^{-L_x} about one and collect intersecting supports, as in the proof of Lemma 13.11. The logarithmic factors vanish unless their supports contain flags, so each resulting connected component still contains a forced term. Their site bound at exponent $\sigma + 2$ is a subpower factor times $\mathfrak{b}^{-7/16}$, whereas each forced component retains $C_j e^{-r_j \beta'_u t_{m+1}^2}$. Choose a smaller positive $r'_j < r_j$. The site attachment parameter tends to zero and the remaining exponential gives

$$C'_j e^{-r'_j \beta'_u t_{m+1}^2} \le e^{-\kappa_j^{\mathrm{pre}} (M_{m+1}^{\mathrm{cut}})^2}$$

for some $\kappa_j^{\mathrm{pre}} > 0$, since $\beta'_u \ge \widehat{\beta}_{m+1}/2$. One extra support power is used in this collection, leaving $\sigma + 1$. Lemma 13.4 proves the asserted site bound and absolute convergence. No scalar depends on the output configuration, and all identities hold also when a chord is exactly at a threshold.

For continuity, the truncation and all reflected tori are now fixed. Finite linear combinations preserve the weighted comparison from Lemma 13.5. A difference of forced products has one marked difference factor; the rooted sum has the same strict attachment margin. Dominated convergence proves the weighted comparison after the collection. Flat evaluation gives continuity of the scalar. The support enlargement, inventories and conjunctions of tests are fixed. The same constructions commute with translations and signed permutations by reindexing. Every assertion is pointwise in the interval parameter or a convergence assertion within the interval, so restriction to a closed subinterval preserves it. □

The preparatory regular sum is $E_u^{\mathrm{pre}} = \sum_i f_{i,u}$ on the healthy domain. Its weighted bound is the anchored sum of Definition 5.4, before flat-value and Hessian subtraction. The next identity supplies the scalar reset.

Lemma 13.9 (Pre-reset expansion)**.** *For every input curve above and every u, there is a preparatory regular family with sum E_u^{pre} and weighted bound $\epsilon_m^{\mathrm{reg}}$ such that, when $\mathrm{Flag}_m(U) = \varnothing$,*

$$\mathcal{H}_u(U) = \exp\{\tau_u|\Lambda| - \beta'_u \mathsf{K}_m^{\mathrm{kin}}(U) - \alpha_{N,m} \mathsf{K}_{2,m}^{\mathrm{kin}}(U) - \mathcal{L}_m(U) - E_u^{\mathrm{pre}}(U)\}\,.$$

In particular $\mathcal{H}_u(1), \mathcal{H}_u(U) > 0$ on this set.

Proof. Restrict the identity (13.10) to configurations without flags. Set $E_u^{\mathrm{pre}} = \sum_i f_{i,u}$ on configurations without flags. Every health mask is one, and nonempty forced inventories give $\mathcal{F}_u^0 = 1$. Equation (13.2) gives $\mathcal{Q}_m = \mathsf{K}_{2,m}^{\mathrm{kin}}$. Dropping the extra three support powers gives the stated bound, since $\theta c_j \le 1/64$. Absolute convergence makes the exponent finite and real, proving positivity, including at 1. □

13.2 Local quadratic subtraction

The subtraction is local and uses no bound on the eventual coupling correction. It extracts from each preliminary regular function its response to constant curls, written with the clover curvature of (13.1).

For a preliminary f on X anchored at o, put $Z = X \cup N_2(o)$. For $q = (a, b)$ and $T \in \mathfrak{g}$, set $A^{q,T}_{(y,c)} = \mathbf{1}_{c=b}\xi_o(y)^a T$, and fix a unit vector $T_* \in \mathfrak{g}$. Define

$$q_f(q, r) = D^2[f(\exp \cdot)](0)[A^{q,T_*}, A^{r,T_*}],$$
$$Q_f(U) = \frac{1}{2}\sum_{q,r} q_f(q, r)\langle F^{\mathrm{cl}}_q(U, o), F^{\mathrm{cl}}_r(U, o)\rangle_{\mathfrak{g}}\,.$$

Set $q_f = 0$ when Z wraps. The matrix extracts the response to constant curls; its color independence is proved below. Define the regular remainder and the discrepancy of masks by

$$R_f = f - f(1) - Q_f,$$
$$\zeta_f = \mathbf{1}_{Z\,\text{unhealthy}}\{f(1) + h_{\{o\}}Q_f - h_X f\}\,. \tag{13.13}$$

The following identity is checked before any activities are summed.

Lemma 13.10 (Local subtraction)**.** *For every preliminary regular f,*

$$h_X f = f(1) + h_{\{o\}}Q_f - \zeta_f + h_Z R_f\,.$$

The function R_f is regular on Z, vanishes at 1, and has zero affine Hessian when Z is nonwrapping. The Borel gauge-invariant function ζ_f depends only on internal links and flags in Z and vanishes when Z is healthy.

Proof. Gauge invariance reduces the affine Hessian to its response to constant curls, Schur's lemma makes that response colour-independent, and the clover formula (13.1) reproduces it. The first derivative of f at the flat field vanishes: constant gauge transformations make its coefficient on each internal link an adjoint-invariant vector of $\mathfrak{g}$, hence a central vector, which is zero. Differentiating local gauge invariance again makes its Hessian annihilate an infinitesimal gradient in either argument. On a nonwrapping support an affine field with zero curl is a local gradient. Its constant and symmetric linear parts are differences of a quadratic site field, with a constant adjustment for the discrete difference. Multiply this site field by a periodic cutoff equal to one on Z. The internal links read by f are unchanged. The second differentiation of gauge invariance uses $Df(1) = 0$, as proved above, so no second derivative of the gauge orbit remains. Hence the affine Hessian depends only on the constant curls.

For each pair of orientations the colour form $(S, T) \mapsto D^2[f(\exp \cdot)](0)[A^{q,S}, A^{r,T}]$ is $\mathrm{Ad}(G)$-invariant. The complexified adjoint representation of the compact simple Lie algebra $\mathfrak{g}$ is irreducible, so every invariant bilinear form on $\mathfrak{g}$ is a multiple of the trace form (Schur's lemma); testing on T_* identifies the multiple.

The side satisfies $L_p D_m \geq 100(B_m + 1) > 20$, which permits the clover loops and affine tests without degeneracies. Equation (13.1) identifies the affine Hessian of Q_f with that of f. Subtracting gives zero diagonal affine Hessian for R_f; polarization gives zero on pairs. The value and first derivative of R_f vanish at 1. If Z wraps, no Hessian cancellation is required. The clover polynomial is smooth, Borel, gauge invariant and local on $N_2(o)$, which gives the other regularity assertions on Z.

All indicators in ζ_f read flags in Z. If Z is healthy, every mask in the identity is one and $\zeta_f = 0$; the identity reduces to the definition of R_f. If Z is unhealthy, $h_Z R_f = 0$ and the definition of ζ_f gives the identity directly. This proves the equality on every configuration and the stated locality and cancellations. □

The same subtraction has a quantitative bound. If a_f majorizes f, the diagonal tests tA^{q,T_*} and their half-sum are permitted directions in Definition 5.3. They give

$$t^2|q_f(q,q)| \le a_f, \qquad t^2|q_f(q,r)| \le 3a_f\,. \tag{13.14}$$

Indeed four times the half-sum's diagonal Hessian is $t^2(q_f(q,q) + q_f(r,r) + 2q_f(q,r))$. Its absolute value and the two diagonal bounds give the second estimate. Product differentiation of the four-link formula bounds every permitted derivative of F_q^{cl} by $C_j t$ at a healthy anchor. For order zero this follows from the plaquette chord bound; at positive order the direction factors supply t, and all undifferentiated group matrices are bounded. Hence

$$\|R_f\|_{m,Z} + \sup_U |\zeta_f(U)| \le C_j a_f\,. \tag{13.15}$$

For ζ_f, the terms Q_f and f occur only under their own relaxed threshold tests, so the estimates still apply when Z is unhealthy.

Enlargement costs at most $\mathrm{cov}_B(Z) \le \mathrm{cov}_B(X) + 1$: one radius-$B$ ball centered at $o \in X$ contains $N_2(o)$ and meets the original connected cover. Also $\mathrm{dist}(x,o) \le CB\,\mathrm{cov}_B(Z)$ for $x \in Z$, by following the connected cover. For a support of covering number n through x, its anchor lies within distance CBn of x, giving at most $C(1+B)^4(1+n)^4$ possible anchors. The inequality $(1+n)^4 \le Ce^n$ converts exponent $\sigma+3$ at the anchor to exponent $\sigma+1$ at a site: one support power bounds this polynomial and one remains unused. At exponent σ the enlargement itself costs only a stage constant. Without these extra powers the conversion would leave a polynomial in n at the original exponent.

Sum $q_{f_{i,u}}$ at each anchor. Translation covariance makes the result independent of the anchor. Reflection in an axis belonging to exactly one of two distinct orientations changes the sign of the off-diagonal entry, so that entry is zero. Permuting axes identifies the diagonal entries. Thus a single real Δ_u satisfies

$$\begin{aligned} \sum_{o_i = o} q_{f_{i,u}}(q,r) &= \Delta_u \delta_{qr}, \\ \sum_i h_{\{o_i\}} Q_{f_{i,u}} &= \frac{\Delta_u}{2} \sum_o h_{\{o\}} \sum_q |F_q^{\mathrm{cl}}(U,o)|^2\,. \end{aligned} \tag{13.16}$$

The coefficient sums converge absolutely by (13.14), and $t^2|\Delta_u| \le \epsilon_m^{\mathrm{pre}}$. In particular $|\Delta_u| \le \epsilon_m^{\mathrm{pre}}/t_m^2 = \theta c_j \widehat{\beta}_m^{-1/10}(M_m^{\mathrm{cut}})^{-2} < 1$, since $\theta c_j \le 1/64$, $\widehat{\beta}_m > 1$ and $M_m^{\mathrm{cut}} \ge 1$. This is already the numerical reset bound, and with Lemma 12.9 it gives the drift bound of Lemma 13.14 directly; no further restriction enters. The probe argument in the proof of Lemma 13.14 is a second proof of $|\Delta_u| \le 1$ with a separate role: it reads Δ_u off the two numbers $\mathcal{H}_u(1), \mathcal{H}_u(V)$ as the unique root of an affine equation, without the formula (13.16). Hence no other coupling correction fits the same activities, and Δ_u is continuous in u because the block integral and the activity sums are (Lemma 13.15).

13.3 Forced fusion, activity bounds and the reset

Put $\alpha = \alpha_{N,m}$. After Section 13.2 the exponent contains $\alpha\mathcal{Q}_m + \frac{\Delta_u}{2}\sum_o h_{\{o\}}\sum_q |F_q^{\mathrm{cl}}(U,o)|^2$, whereas the output density must contain $(\alpha+\Delta_u)\mathsf{K}_m^{\mathrm{kin}}$. The difference consists of clover versus plaquette curvature, the missing cubic and quartic atoms, and the clips and health masks. The first two become regular activities, $e_{\lambda,u}$ below and $-(\alpha+\Delta_u)\mathfrak{a}_i^{(k)}$; the third, $\ell_{\lambda,u}$, vanishes on healthy supports and, with the mask discrepancies ζ_f of Lemma 13.10, is absorbed into the forced activities by Lemma 13.11. Define

$$\begin{aligned} e_{\lambda,u} &= \Delta_u(L_\lambda^{\mathrm{cl}} - L_\lambda^{\mathrm{raw}}),\\ \ell_{\lambda,u} &= (\alpha+\Delta_u)(L_\lambda - h_{Z_\lambda}L_\lambda^{\mathrm{raw}}) + \Delta_u(h_{Z_\lambda} - h_{\{o_\lambda\}})L_\lambda^{\mathrm{cl}}\,. \end{aligned} \tag{13.17}$$

The regular function is used with mask h_{Z_λ}. Its affine Hessian vanishes because its two terms have the same constant-curl response. Add also the regular cubic and quartic functions $-(\alpha+\Delta_u)\mathfrak{a}_i^{(k)}$, with their prescribed masks; they have zero Hessian because $k \ge 3$. Every $\ell_{\lambda,u}$ vanishes on healthy Z_λ.

Write the logarithmic corrections $\zeta_{f_{i,u}}$ and $\ell_{\lambda,u}$ as ℓ_a on nonempty supports Z_a. A connected selection P consists of preliminary forced labels and logarithmic labels, contains at least one forced label, and has pairwise disjoint preliminary forced supports. Connected means connected in the graph joining intersecting supports. Its support Y_P is the union of all supports, including logarithmic supports; its inventory I_P is the union of the preliminary forced inventories. Define

$$K_P(U) = \mathbf{1}_{\{Y_P\cap \mathrm{Flag}_m(U)=I_P\}}\prod_{i\in P_{\mathrm{for}}} k_{i,u}(U)\prod_{a\in P_{\mathrm{log}}}(e^{\ell_a(U)}-1)\,. \tag{13.18}$$

The extra test is the conjunction of the preliminary forced tests. Selections and tests are fixed as u varies.

Lemma 13.11 (Forced fusion)**.** *Suppose* $\sum_a \sup_U|\ell_a(U)| < \infty$. *For* $\kappa_{\mathrm{fus}},\gamma \ge 0$ *assume nonnegative* v_a *for all forced and logarithmic labels with*

$$|k_i| \le e^{-\kappa_{\mathrm{fus}}}v_i, \quad |e^{\ell_a}-1| \le v_a, \quad \sup_x \sum_{a:x\in Z_a} e^{(\sigma+1)\,\mathrm{cov}_B(Z_a)}v_a \le \gamma, \quad (2B+1)^4\gamma \le 1\,.$$

Then the exact-cover gas of (13.18) *equals* $\mathcal{F}_u^0\exp(\sum_a \ell_a)$ *and its forced norm of Definition 5.4 is at most* $e^{-\kappa_{\mathrm{fus}}}(e^\gamma-1)$.

Proof. The identity is a regrouping of finite selections into overlap components; the bound is Lemma 13.4. Expand $\exp(\sum_a \ell_a) = \prod_a(1+(e^{\ell_a}-1))$. Decompose a finite selection, together with a selection from $\mathcal{F}_u^0$, into support-overlap components. A nonzero logarithmic factor meets a flag. Exact coverage assigns that flag to a preliminary forced label, whose support meets the logarithmic support. Every component therefore contains a forced label. Their union supports are disjoint; their union inventories cover all flags. Conversely the components recover the original selection uniquely. The eligibility indicator in (13.18) equals one on any such nonzero product: another component cannot supply a flag in Y_P, since its support is disjoint. This proves the identity for finite expansions.

For the absolute bound retain $e^{-\kappa_{\mathrm{fus}}}$ from one forced label per connected selection. Apply Lemma 13.4, with $\eta = 1$, to the weights $\mathfrak{v}_i = v_i e^{\sigma\,\mathrm{cov}_B(Z_i)}$; its hypothesis follows from $(2B+1)^4\gamma \le 1$ and the extra exponent in the hypothesis here. It bounds the rooted tree sums by $\mathfrak{v}_i e^{\mathrm{cov}_B(Z_i)}$,

and the sum of these bounds over roots whose supports contain a fixed site is at most γ. Every connected selection has a spanning tree rooted at a label whose support contains that site, so its sum is at most $\gamma \le e^{\gamma} - 1$. The covering number is subadditive on intersecting unions, which retains $e^{\sigma \operatorname{cov}_B(Y_P)}$ in this comparison. Restoring $e^{-\kappa_{\mathrm{fus}}}$ gives the claimed bound. The resulting summable envelope justifies passage from finite expansions to the infinite identity. □

Let E_u^{sum} be the masked sum of the $R_{f_{i,u}}$, the $e_{\lambda,u}$, and the cubic and quartic corrections. Let $\mathcal{F}_u$ be the fused gas. Hold these functions fixed while a trial parameter ϑ varies, and put

$$q_{u,\vartheta} = e^{-(\beta_u' + \alpha_{N,m} + \vartheta)\mathsf{K}_m^{\mathrm{kin}} - \mathcal{L}_m - E_u^{\mathrm{sum}}} \mathcal{F}_u \,.$$

The parameter ϑ is the free coupling correction of the scalar reset. Lemma 13.13 shows that $\vartheta = \Delta_u$ gives the density identity at every configuration; at one healthy test configuration V with $\mathsf{K}_m^{\mathrm{kin}}(V) > 0$, Lemma 13.14 uses the affine dependence on ϑ to read off Δ_u, which gives its continuity in u and, with Lemma 12.9, the drift bound. The output representation is the following single list: regular addresses (Z_i, o_i) with functions $R_{f_{i,u}}$, addresses (Z_λ, o_λ) with functions $e_{\lambda,u}$, the cubic and quartic addresses of Definition 12.1 with functions $-(\alpha + \Delta_u)\mathfrak{a}_i^{(k)}$, and forced addresses $(Y_P, I_P, \mathcal{E}_P^{\mathrm{extra}})$ with functions K_P. Each regular function carries its own health mask. We prove the inequalities

$$e^{-\kappa_{\mathrm{fus}}}(e^{\gamma} - 1) \le \epsilon_m^{\mathrm{for}}, \qquad \mathcal{E}_m \le \epsilon_m^{\mathrm{reg}} \,. \tag{13.19}$$

in Lemma 13.12 before using them for the density identity. They impose no condition on the trial coupling.

Lemma 13.12 (Activity bounds and covariance)**.** *For every coarse output constructed above, both activity families are countable and equivariant under translations and H_4, including supports, anchors, inventories and tests. There exist nonnegative majorants as in Definitions 5.3–5.4 whose weighted series converge and which satisfy*

$$\mathcal{E}_m \le \epsilon_m^{\mathrm{reg}}, \qquad \mathcal{K}_m \le \epsilon_m^{\mathrm{for}} \,.$$

Lemma 13.16 states the same bounds for the Wilson representation, and Lemma 13.17 extends them to all the chosen classes.

Proof for a coarse output. The regular bound adds the costs of the three linear operations to the preparatory bound of Lemma 13.8; the forced bound applies Lemma 13.11. Equation (13.15) controls each subtracted function. The low regular terms cost $C_j \epsilon_m^{\mathrm{pre}}$, since $t^2|\Delta_u| \le \epsilon_m^{\mathrm{pre}}$. The high terms cost $C_j(|\alpha|t^3 + \epsilon_m^{\mathrm{pre}})$: each has at least three curvature factors, and $|\Delta_u|t^3 \le \epsilon_m^{\mathrm{pre}} t \le \epsilon_m^{\mathrm{pre}}$. A derivative of a word of length l costs a fixed power of $1 + l$, by the product rule and the allowed spatial growth of the directions. The support obeys $\operatorname{cov}_B(Z_\lambda) \le C + l/B$. Raise the threshold until $\sigma/B \le w_{J_m}^{\mathrm{path}}/4$. A quarter of the path exponent then bounds the support weight, and another quarter bounds this polynomial, since $\sup_{l \ge 0}(1 + l)^k e^{-wl/2} < \infty$. Summing Lemma 12.3 gives the previously defined constant C_j^0 and a constant C_j^1 with

$$\mathcal{E}_m \le C_j^0 \epsilon_m^{\mathrm{pre}} + C_j^1 |\alpha| t^3 \,.$$

The constant $c_j = [64(1 + C_j^0)]^{-1}$ was chosen before the ratios in Lemma 13.8. After the ratio is fixed, Lemma 12.9 bounds $|\alpha|$; raise the entry threshold until $C_j^1 |\alpha| t^3 \le \epsilon_m^{\mathrm{reg}}/2048$. This is possible because $t^3/\epsilon_m^{\mathrm{reg}} = \widehat{\beta}_m^{-2/5} (M_m^{\mathrm{cut}})^3 \to 0$. Take $\theta \le 1/32$ before the ratio. Since $C_j^0 c_j \le 1/64$, the sum

is at most $\epsilon_m^{\mathrm{reg}}/1024$. The same argument allows any prescribed smaller coefficient by choosing θ first and raising the threshold for the second term.

For the logarithmic corrections, conversion from anchored sums to site sums costs at most $C_j(1+B)^4$, after the extra support powers pay the powers of the covering number. The local subtraction costs $C_j\epsilon_m^{\mathrm{pre}}$. The low terms in (13.17) cost

$$C_j(1+B)^4\{(|\alpha|+\epsilon_m^{\mathrm{pre}}/t^2)t+\epsilon_m^{\mathrm{pre}}\} =: m_j\,.$$

The linear clipped tail is bounded by $2c_{\mathrm{lin}}t$ times its coefficient, since chords are at most two. For a failed relaxed threshold test, the remote threshold supplies $e^{l/R^{\mathrm{rel}}_{J_m}}$ in a path of length l. It is paid by the path decay because $R^{\mathrm{rel}}_{J_m}w^{\mathrm{path}}_{J_m}\ge 12$. The common-anchor clover term costs $|\Delta_u|t^2\le\epsilon_m^{\mathrm{pre}}$. These estimates give exactly the displayed bound, including unhealthy configurations. Since $\epsilon_m^{\mathrm{pre}}/t = O_j(\widehat{\beta}_m^{-3/5}/M_m^{\mathrm{cut}})$ and $|\alpha|$ is fixed after the ratio, $(1+B)^4m_j\to 0$.

Let $\kappa_{\mathrm{fus}}=\kappa_j^{\mathrm{pre}}(M_{m+1}^{\mathrm{cut}})^2/2$. The majorants in (13.12), multiplied by $e^{\kappa_{\mathrm{fus}}}$, have rooted total at most $e^{-\kappa_{\mathrm{fus}}}$. For each logarithm use $|e^{\ell_a}-1|\le 2|\ell_a|$ once its individual majorant is at most $\log 2$; this follows by increasing the entry threshold until $m_j\le\log 2$. Their rooted total is at most $2m_j$. Increase the entry threshold so that $(2B+1)^4\gamma\le 1/2$ for $\gamma=e^{-\kappa_{\mathrm{fus}}}+2m_j$. Lemma 13.11 applies. Increase the threshold further so that $\gamma\le 1$ and $e^{-\kappa_{\mathrm{fus}}}(e^{\gamma}-1)\le\epsilon_m^{\mathrm{for}}/1024$. Indeed $e^{\gamma}-1\le 2\gamma$ and $\kappa_{\mathrm{fus}}\gg(M_m^{\mathrm{cut}})^{1/5}$. This proves the forced inequality and summability in (13.19).

Every address is a finite combination of countable preliminary and path addresses, so the families are countable. Subtraction and path replacement transport the entire address under translations. Signed permutations transform the curvature orientations with their signs; the finite group average includes every transformed term. These permutations act on supports, anchors and functions. On a connected selection they also transform the union inventory and conjunction of tests. Equation (13.18) therefore has the required covariance. For an inverse link, $(e^{zA}U)^{-1}=e^{-z\,\mathrm{Ad}_{U^{-1}}A}U^{-1}$. Differentiating this identity repeatedly gives only finitely many product-rule terms through order $\mathfrak{q}_{J_m}$; the adjoint-invariant norm bounds them by a constant depending on that order. This proves the complete address covariance and the convergent weighted bounds for a coarse output. The Wilson start is established below; induction in Lemma 13.17 then gives the class assertion. □

The identity below fixes the whole density before the scalar estimate of Lemma 13.14. Its proof uses the bounds just established.

Lemma 13.13 (Exact density representation)**.** *The final activities satisfy Lemma 13.12, vanish at the flat field in their regular part, and are independent of ϑ. For every u there is $\Delta_u\in\mathbb{R}$ such that*

$$q_{u,\Delta_u}(U)=\frac{\mathcal{H}_u(U)}{\mathcal{H}_u(1)}\qquad \text{for every } U\,.$$

Moreover $\mathcal{F}_u(U)=1$ when $\mathrm{Flag}_m(U)=\varnothing$.

Proof. Local subtraction rewrites the preliminary regular functions, path replacement turns the masked quadratic functional into the kinetic action, and fusion absorbs the logarithmic corrections; the bounds (13.19) make every sum converge. Choose Δ_u by (13.16) and put $c_u=\sum_i f_{i,u}(1)$. Local subtraction gives

$$\sum_i h_{X_i}f_{i,u}=c_u+\frac{\Delta_u}{2}\sum_o h_{\{o\}}\sum_q|F_q^{\mathrm{cl}}|^2-\sum_i\zeta_{f_{i,u}}+\sum_i h_{Z_i}R_{f_{i,u}}\,.$$

All sums converge absolutely. For each low term (13.17) gives

$$\begin{aligned}(\alpha+\Delta_u)L_\lambda+\Delta_u h_{Z_\lambda}(L_\lambda^{\mathrm{cl}}-L_\lambda^{\mathrm{raw}})-\ell_{\lambda,u}\\ =\alpha h_{Z_\lambda}L_\lambda^{\mathrm{raw}}+\Delta_u h_{\{o_\lambda\}}L_\lambda^{\mathrm{cl}}\,.\end{aligned}$$

Sum and apply (13.2). The high corrections cancel $(\alpha+\Delta_u)\mathcal{A}_m$. Thus the exponent in (13.10) becomes

$$\begin{aligned}&(\tau_u-c_u/|\Lambda|)|\Lambda|-(\beta_u'+\alpha+\Delta_u)\mathsf{K}_m^{\mathrm{kin}}-\mathcal{L}_m-E_u^{\mathrm{sum}}\\&\quad+\sum_i\zeta_{f_{i,u}}+\sum_\lambda\ell_{\lambda,u}\,.\end{aligned}$$

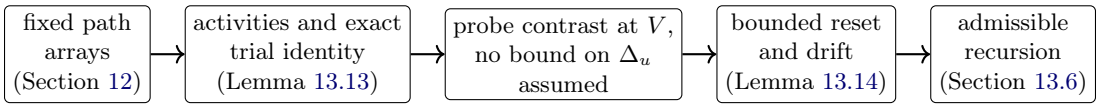
Lemma 13.11 incorporates the last two sums into $\mathcal{F}_u$. Hence, on every configuration,

$$\mathcal{H}_u(U)=e^{(\tau_u-c_u/|\Lambda|)|\Lambda|}q_{u,\Delta_u}(U)\,. \tag{13.20}$$

At 1 all path and final regular terms vanish. The only exact-cover forced selection is empty, so $q_{u,\vartheta}(1)=1$ for every ϑ. Equation (13.20) proves the quotient identity and positivity of its denominator. The same empty-selection argument gives $\mathcal{F}_u=1$ whenever there are no flags.

The regular cancellations were proved in Lemma 13.10 and after (13.17); the weighted bounds are those of Lemma 13.12. The value Δ_u used in the activity construction is already fixed by the preliminary functions. The trial variable ϑ occurs only in the exponential defining $q_{u,\vartheta}$. This proves the exact identity without using Lemma 13.14 or its reset estimate. □

The scalar reset. The density identity of Lemma 13.13 holds with a value Δ_u that the construction fixes before the bound on it proved here is used. The proof order is



Before the estimate below the following are available: the path arrays and $\alpha_{N,m}$ (Section 12), the activity families with their bounds (Lemma 13.12), and the exact identity $q_{u,\Delta_u}=\mathcal{H}_u/\mathcal{H}_u(1)$ (Lemma 13.13). Section 13.2 already gives $t_m^2|\Delta_u|\le\epsilon_m^{\mathrm{pre}}$, which the activity bounds use and which alone implies $|\Delta_u|<1$ and the drift bound. The estimate below is an alternative proof of $|\Delta_u|\le 1$ that uses the activity bounds but not the formula for Δ_u. The reset gives a second description of the same coefficient: evaluating the identity at one configuration with positive kinetic energy identifies Δ_u as the unique root of an affine scalar equation. This explicit formula, with the continuity of Lemma 13.15 in Section 13.4 (whose proof does not use Lemma 13.14), gives the continuity of Δ_u in u; with Lemma 12.9 it gives the drift bound.

For an input curve ρ_u' of Theorem 5.5, recall $\mathcal{H}_u=\mathcal{B}^{(m)}_{\beta_u'}\operatorname{dens}\rho_u'$, and put $b_u=\beta_u'+\alpha_{N,m}$. Lemma 13.13 supplies regular and forced sums $E_u^{\mathrm{sum}},\mathcal{F}_u$, independent of ϑ, with trial density

$$q_{u,\vartheta}(U)=e^{-(b_u+\vartheta)\mathsf{K}_m^{\mathrm{kin}}(U)-\mathcal{L}_m(U)-E_u^{\mathrm{sum}}(U)}\mathcal{F}_u(U)\,,$$

the flat-field normalizations $q_{u,\vartheta}(1)=1$, and $q_{u,\Delta_u}=\mathcal{H}_u/\mathcal{H}_u(1)$. Lemma 13.9 gives a preparatory sum E_u^{pre} with bound $c_{\mathrm{pre}}\epsilon_m^{\mathrm{reg}}/4$, for a stage constant $c_{\mathrm{pre}}\ge 4$, and Lemma 13.12 gives the final bound (5.12). The remaining scalar correction can be read from a single configuration with a definite kinetic energy.

Lemma 13.14 (Coupling reset)**.** *For each input curve of Theorem 5.5, take the families and real Δ_u supplied by Lemma 13.13, with no bound on Δ_u assumed. Then $|\Delta_u| \le 1$, and $\beta_u = \beta'_u + \alpha_{N,m} + \Delta_u$ is continuous and satisfies*

$$\beta_u = \beta'_u - \kappa_G \log \ell_m + e_u, \qquad |e_u| \le d_j + 1 \le \omega_j^{\mathrm{RG}} .$$

The bound is independent of N, p, u and the input curve.

Proof. We evaluate the exact identity of Lemma 13.13 at one configuration V with kinetic energy of order $|\Lambda| t_m^2$; the identity is affine in the reset, which is then read off. Recall $t = t_m \le 1$ from item 7 of Table 3. Put a group element of chord $h_V = \tau t$ on direction-1 links whose second coordinate is even, and the identity elsewhere. Continuity along a nonconstant one-parameter subgroup gives $\varepsilon_G > 0$ such that every chord in $[0, \varepsilon_G]$ occurs. Choose

$$0 < \tau \le \min\left(\varepsilon_G, \frac{1}{4}, \frac{c_{\mathrm{dir},J_m}}{512(1 + A_{J_m}^{\mathrm{path}})} \right) .$$

Every $(1,2)$ plaquette has chord h_V and all other plaquettes have chord zero. Thus this configuration V is everywhere healthy. All periods here are even because $4 \mid L_p$. Signed permutations permute the plaquettes and preserve their chords, so the direct contribution to $\mathsf{K}_m^{\mathrm{kin}}(V)$ is at least $|\Lambda| c_{\mathrm{dir},J_m} h_V^2/2$. Row squares are nonnegative. Since $|\operatorname{proj}_{\mathfrak{g}}(U - U^{-1})/2| \le \operatorname{ch}(U)$ and adjoint transport is isometric, every slot has norm at most $h_V \le 1/4$. Lemma 12.3 bounds the absolute sum of cubic and quartic terms by

$$2|\Lambda| A_{J_m}^{\mathrm{path}} h_V^3 \le |\Lambda| c_{\mathrm{dir},J_m} h_V^2/4 .$$

Each signed image of V has the same plaquette chords with a single nonzero orientation. Hence the bounds apply to every term of the hypercubic average. Consequently

$$K_V = \mathsf{K}_m^{\mathrm{kin}}(V) \ge |\Lambda| c_V t^2, \qquad c_V = c_{\mathrm{dir},J_m} \tau^2/4 > 0 .$$

Lemma 13.9 gives $\mathcal{H}_u(1), \mathcal{H}_u(V) > 0$. Put $v_u = |\Lambda|^{-1} \log \mathcal{H}_u(1)$. Taking the logarithmic contrast in Lemma 13.9 gives

$$\log \frac{\mathcal{H}_u(1)}{\mathcal{H}_u(V)} = \beta'_u \mathsf{K}_m^{\mathrm{kin}}(V) + \alpha_{N,m} \mathsf{K}_{2,m}^{\mathrm{kin}}(V) + \mathcal{L}_m(V) + E_u^{\mathrm{pre}}(V) - E_u^{\mathrm{pre}}(1) . \tag{13.21}$$

The pre-reset bound of Lemma 13.9, summed over anchors as in Definition 5.4, bounds each of the last two values by $|\Lambda| c_{\mathrm{pre}} \epsilon_m^{\mathrm{reg}}/4$. No vanishing condition at 1 is needed for this preparatory sum.

On everywhere-healthy configurations every curvature slot has norm at most t. Lemma 12.3 and $t \le 1$ therefore give

$$|\mathsf{K}_m^{\mathrm{kin}} - \mathsf{K}_{2,m}^{\mathrm{kin}}| \le 2|\Lambda| A_{J_m}^{\mathrm{path}} t^3 .$$

After ℓ is fixed, Lemma 12.9 bounds $|\alpha_{N,m}|$ by a stage constant. Raise the coupling entry threshold until $2|\alpha_{N,m}| A_{J_m}^{\mathrm{path}} t^3 \le c_{\mathrm{pre}} \epsilon_m^{\mathrm{reg}}/2$. This is possible since $t^3/\epsilon_m^{\mathrm{reg}} = \widehat{\beta}_m^{-2/5} (M_m^{\mathrm{cut}})^3 \to 0$. Together with (13.21) this yields

$$\left| b_u K_V + \mathcal{L}_m(V) - \log \frac{\mathcal{H}_u(1)}{\mathcal{H}_u(V)} \right| \le |\Lambda| c_{\mathrm{pre}} \epsilon_m^{\mathrm{reg}} .$$

For the final regular sum, Lemma 13.12 and (5.12) give

$$|E_u^{\mathrm{sum}}(V)| \le \sum_x \sum_{o_i=x} e_i^{\mathrm{maj}} \le |\Lambda| \mathcal{E}_m \le |\Lambda| \epsilon_m^{\mathrm{reg}} .$$

Since the activity families are fixed as ϑ varies, the scalar discrepancy is affine:

$$\begin{aligned} f(u,\vartheta) &= (b_u + \vartheta) K_V + \mathcal{L}_m(V) + E_u^{\mathrm{sum}}(V) - \log \frac{\mathcal{H}_u(1)}{\mathcal{H}_u(V)} \\ &= f(u,0) + \vartheta K_V . \end{aligned}$$

Since V has no flagged sites, $\mathcal{F}_u(V) = 1$. Evaluating the exact identity of Lemma 13.13 at V gives $f(u, \Delta_u) = 0$. As $K_V > 0$, this determines

$$\Delta_u = -f(u,0)/K_V, \qquad |\Delta_u| \le \frac{(c_{\mathrm{pre}} + 1)\epsilon_m^{\mathrm{reg}}}{c_V t_m^2} \le 1 .$$

Increase the entry threshold to obtain the final inequality, using $\epsilon_m^{\mathrm{reg}}/t_m^2 = \widehat{\beta}_m^{-1/10} (M_m^{\mathrm{cut}})^{-2} \to 0$. The block integral and activity sums are continuous in u by Lemma 13.15 of Section 13.4, whose proof does not use the present lemma; so Δ_u is continuous.

Multiplication by $\mathcal{H}_u(1) = e^{v_u|\Lambda|}$ gives the density identity at Δ_u. Its coupling is $\beta_u = \beta'_u + \alpha_{N,m} + \Delta_u$. Lemma 12.9 and $|\Delta_u| \le 1$ prove the claimed drift. □

13.4 Continuity at fixed volume

The scalar reset of Lemma 13.14 and the step on curves need continuity of the integral and both final sums. The next deduction uses the weighted comparison in Lemma 13.8 and does not use Lemma 13.14.

Lemma 13.15 (Continuity of the activity sums)**.** *For input curves of Theorem 5.5, the final activity families have fixed addresses and are continuous in the weighted sense of Definition 13.3. For every fixed configuration U,*

$$u \longmapsto \mathcal{H}_u(U), \qquad u \longmapsto E_u^{\mathrm{sum}}(U), \qquad u \longmapsto \mathcal{F}_u(U)$$

are continuous.

Proof. Every operation of the step is linear at fixed addresses or a convergent product with a strict attachment margin, so the weighted continuity of Lemma 13.8 carries over. The maps $f \mapsto f(1), q_f, R_f, \zeta_f$ are linear on functions with fixed address. Equation (13.14) applied to a difference bounds its projection coefficients by its derivative majorant divided by t^2. Weighted preliminary continuity therefore gives continuity of c_u, Δ_u, and the subtracted regular and logarithmic series. The low and high corrections are linear in Δ_u, with fixed summable path coefficients, so they retain this continuity. Every mask is a fixed function of U.

For fusion each coefficient is a finite product and hence continuous in supremum norm. To justify summing weighted differences, work first along subsequences with one summable majorant. Given $u_n \to u_0$, pass to a subsequence whose total weighted preliminary discrepancies are at most $2^{-n}\varepsilon$. Add their nonnegative majorants to those at u_0; the extra mass is at most ε. The margin $(2B+1)^4\gamma \le 1/2$ arranged in the proof of Lemma 13.12 permits this addition for small ε. Lemma 13.11 gives one summable envelope for the fused coefficients along this subsequence.

Dominated convergence gives convergence of their weighted differences. If the original sequence failed to converge, a subsequence with discrepancies bounded away from zero would admit such a further subsequence, a contradiction. This proves weighted continuity without assuming a common envelope on an arbitrary neighborhood. Constants here may depend on the fixed volume.

For the pointwise gas, put $M_i = \max(\sup|K_{i,u}|, \sup|K_{i,u_0}|)$. Telescoping finite products and summing selections gives

$$|\mathcal{F}_u(U) - \mathcal{F}_{u_0}(U)| \le e^{\sum_i M_i} \sum_i \sup|K_{i,u} - K_{i,u_0}|\,.$$

The restrictions on exact coverage use fixed inventories. Weighted continuity makes the right side tend to zero. The regular sum is continuous by the analogous sum of order-zero differences. The same bounds give continuity in u, uniform in U, for the input density at fixed volume, since its path action is affine in the continuous coupling.

Finally the kernel of Definition 5.1 is uniformly bounded on a compact coupling interval at this volume. The tag masses sum to one; its single-link normalizing integrals are positive and continuous in the coupling, and its link factors are bounded on the compact group. Dominated convergence against product probability Haar measure gives continuity of $\mathcal{H}_u(U)$. This proves all three pointwise continuities and weighted continuity, by Lemma 13.8. □

13.5 The Wilson start

At the finest level the identity free form permits a local choice. Put $\delta_{\mathrm{dir}} = c_{\mathrm{dir},J_N}$ and $F(u) = \mathrm{proj}_{\mathfrak{g}}(u - u^{-1})/2$. To define the quartic term, let ψ be the local inverse of $X \mapsto F(e^X)$ at zero and set

$$T_4(Y) = \frac{1}{4!} D^4[\mathrm{pr}(e^{\psi(\cdot)})](0)[Y,Y,Y,Y]\,.$$

The inverse exists since $DF(e^{\cdot})(0) = I$. This invariant quartic polynomial is chosen so that

$$R(u) = \mathrm{pr}(u) - \tfrac{1}{2}|F(u)|^2 - T_4(F(u)) \tag{13.22}$$

has zero Taylor terms through degree four at 1. This is the case $m = N$ in the proof of Lemma 12.6. The path action of Definition 12.2 at $m = N$ has row $\sqrt{1-\delta_{\mathrm{dir}}}\,F$, no cubic or loop terms, and quartic term $(1-\delta_{\mathrm{dir}})T_4(F)$ with its prescribed mask. These finite path sums satisfy Lemma 12.3.

The initial activities retain the difference between Wilson energy and this action also on flagged configurations. The proof constructs the full curve.

Lemma 13.16 (Wilson initialization)**.** *For every N, p there is a continuous curve $\beta \mapsto \rho_\beta^{\mathrm{W}}$ of representations at level N on $\mathfrak{B}_N$, with coupling β, scalar normalization zero, and*

$$\mathrm{dens}\,\rho_\beta^{\mathrm{W}} = e^{-\beta E^{\mathrm{W}}_{L_p D_N}}\,.$$

Its activity families are countable and equivariant, with majorants satisfying $\mathcal{E}_N \le \epsilon_N^{\mathrm{reg}}$ and $\mathcal{K}_N \le \epsilon_N^{\mathrm{for}}$, and the curve is continuous in the sense of Definition 13.3.

Proof of the initialization identity, bounds and continuity. The Wilson energy is split exactly into the path action of level N, a fifth-order regular remainder, and a nonnegative unassigned action that is charged to the flags. Everything here is at level N: $j = J_N$, $t = t_N$, $B = B_N$ and $\sigma = \sigma_{J_N}$.

For a plaquette q, write $k_q^{\mathrm{kin}} = (1-\delta_{\mathrm{dir}})|F(U_q)|^2/2$, $h_q = h_{\mathrm{vertices}(q)}$, and χ_q for its relaxed threshold test. Define its unassigned action by

$$\mathcal{W}_q(U) = \begin{cases} \mathrm{pr}(U_q) - c_{\mathrm{lin}} t\,\mathrm{ch}(U_q), & \mathrm{ch}(U_q) > t, \\ h_q k_q^{\mathrm{kin}} + (1-h_q)(1-\delta_{\mathrm{dir}})\,\mathrm{pr}(U_q) - \chi_q k_q^{\mathrm{kin}}, & \mathrm{ch}(U_q) \le t. \end{cases}$$

The relaxed threshold test $\chi_q = 1$ implies $\mathrm{ch}(U_q) \le t$. The identity $(u-u^{-1})/2 = (u-1)(1+u^{-1})/2$ and contraction of the projection give $|F(u)| \le \mathrm{ch}(u)$. Together with $\mathrm{ch}(u)^2 = 2\,\mathrm{pr}(u)$ this gives $\mathcal{W}_q \ge 0$. For $\mathrm{ch}(U_q) > t$, $c_{\mathrm{lin}} < \delta_{\mathrm{dir}}/16 \le 1/16$ gives $\mathcal{W}_q \ge \mathrm{ch}(U_q)^2/4 \ge t^2/4$. Substitution in the two cases gives

$$\mathrm{pr}(U_q) = L_q(U) + (1-\delta_{\mathrm{dir}})h_q T_4(F(U_q)) + (1-\delta_{\mathrm{dir}})h_q R(U_q) + \mathcal{W}_q(U)\,. \tag{13.23}$$

Here L_q is the direct clipped term plus the threshold-tested row square. The identity includes small plaquettes next to a flag and both sides of the clipping threshold.

Take the regular activities $\beta(1-\delta_{\mathrm{dir}})R(U_q)$ on their plaquette supports, anchored at the starting vertex, with the signed average of the path action. They vanish at 1 with their Hessians. Through the prescribed order their normalized healthy derivatives are bounded by $C_j\beta t^5$. To verify this without assuming small links, use the plaquette holonomy as one coordinate and the other three links as compact parameters. The remainder vanishes to order five in the holonomy chart. At derivative order $k \le 5$, Taylor's formula leaves order $5-k$ and the direction factors supply t^k. For $k > 5$, bounded derivatives and $t^k \le t^5$ give the same estimate. The four-link formula controls the remaining compact factors. There are finitely many plaquettes at each anchor, so the regular sum is at most $C_j\beta t^5$. On the top band,

$$\frac{\widehat{\beta}_N t_N^5}{\epsilon_N^{\mathrm{reg}}} = \widehat{\beta}_N^{-2/5}(M_N^{\mathrm{cut}})^5 \longrightarrow 0\,.$$

Increasing the entry threshold gives the regular bound $\epsilon_N^{\mathrm{reg}}/1024$ uniformly in volume.

Let $\mathfrak{F} = \mathrm{Flag}_N(U)$. Join two flags when their radius-$2B$ neighborhoods overlap. For each nonempty connected inventory I use $Y_I = N_{2B}(I)$ and

$$K_I(U) = \mathbf{1}_{\{Y_I \cap \mathfrak{F} = I\}} e^{-\beta d_I(U)}\,.$$

The charge d_I is the signed average of those $\mathcal{W}_q$ whose anchors lie within distance B of I, in the coordinates of that signed image. Every nonzero $\mathcal{W}_q$ is assigned: a failure of the relaxed threshold test has a witnessing flag within this distance, since $t\exp((B-2)/R_{J_N}^{\mathrm{rel}}) \ge 2$. The other cases already meet a flag at a plaquette vertex. The anchors lie within B of I, and each nonautomatic relaxed threshold test reads within $B-2$ of its anchor. Thus $N_{2B}(I)$ contains every link and flag used in the charge. Distinct actual components have disjoint supports, so each nonzero term is assigned once. Hence the sum of the component charges is the signed average of $\sum_q \mathcal{W}_q$.

Every flag touches a plaquette with $\mathrm{ch}(U_q) > t$, and a plaquette has at most four vertices. Retaining the identity orientation in the nonnegative signed average gives

$$d_I(U) \ge \frac{t^2|I|}{16|H_4|} \qquad (I \subset \mathfrak{F})\,.$$

A nonzero exact-cover product of K_I selects precisely the overlap components of $\mathfrak{F}$. Indeed any two flags in one selected connected inventory are connected in $\mathfrak{F}$. Conversely a flag joined to a

selected inventory cannot lie in another selected inventory, since their radius-$2B$ supports would intersect. Exact coverage then forces every component to be selected. The actual components satisfy eligibility. Thus the forced sum has exactly one nonzero term, equal to the exponential of minus β times the unassigned action. Equation (13.23), summed and averaged, proves the Wilson density identity for all configurations with scalar zero.

For the forced norm, a connected inventory of r sites is encoded by a rooted spanning walk of length at most $2(r-1)$ in the overlap graph. Each step has at most $C(1+B)^4$ choices, and $\mathrm{cov}_B(N_{2B}(I)) \le 5^4 r$. Counting possible roots for a support through a specified site costs another $C(1+B)^4$. With $c = 1/(16|H_4|)$, the weighted sum is therefore bounded by

$$C_j(1+B)^4 \sum_{r\ge 1} \Big[C_j(1+B)^8 e^{5^4\sigma - c\beta t^2} \Big]^r .$$

The top band gives $\beta t^2 \ge (M_N^{\mathrm{cut}})^2/2$. This dominates both $\log(1+B)$ and the exponent in $\epsilon_N^{\mathrm{for}}$, so the geometric sum is eventually at most $\epsilon_N^{\mathrm{for}}/1024$, uniformly in volume. Both normalized bounds can be made arbitrarily smaller by increasing the entry threshold.

Addresses and tests are independent of β. Regular differences have the same majorants with β replaced by $|\beta - \beta_0|$. For the forced functions $d_I \ge 0$ is bounded at fixed volume and

$$|e^{-\beta d_I} - e^{-\beta_0 d_I}| \le |\beta - \beta_0| \sup_U d_I(U) \qquad (\beta, \beta_0 \ge 0) .$$

There are finitely many subsets of this torus, so the weighted difference sums tend to zero. Translations and signed permutations transport complete plaquette terms, component supports, inventories and tests. This gives covariance as well as continuity. We have proved the Wilson representation, both bounds, and its full continuous curve with scalar zero. The one-step assertion for the later levels is proved in Section 13.6. □

13.6 The specified recursion

The shooting argument of Section 14 restricts parameter intervals repeatedly, so the step must be a fixed construction on curves that is compatible with restriction. We therefore fix, once for each continuous input curve, all choices of Sections 13.1–13.3: the tree gauge, the address identifications, the truncation index and the auxiliary tori; the last two depend only on the incoming address lists. The step on a closed subinterval is the restriction of the step on the larger interval. The estimates used to repeat the step are the following explicit ones. At level m the path is the fixed $P_{N,m}^{\mathrm{path}}$ of Definition 12.2, with the bounds of Lemmas 12.3–12.4. The activity functions satisfy Definitions 13.1–13.2 and Lemma 13.12. At an adjacent pair the output is exactly the functions in Lemma 13.13, its scalar is $|\Lambda|^{-1}\log \mathcal{H}(1)$, and its correction obeys

$$t_m^2|\Delta| \le \epsilon_m^{\mathrm{pre}}, \qquad \mathcal{E}_m \le \epsilon_m^{\mathrm{reg}}/1024, \qquad \mathcal{K}_m \le \epsilon_m^{\mathrm{for}}/1024, \qquad |\Delta| \le 1 . \tag{13.24}$$

The first three bounds precede the scalar reset; the last is Lemma 13.14, and it also follows from the first. The input coupling belongs to $\mathfrak{B}_{m+1}$; no output-band condition is imposed.

The classes are defined inductively from the Wilson start. Put $\mathcal{I}_{N,p} = \{\rho_\beta^{\mathrm{W}} : \beta \in \mathfrak{B}_N\}$, with the curve of Lemma 13.16, and $\mathcal{A}_{N,N,p} = \mathcal{I}_{N,p}$. For $m < N$, a representation ρ at level m belongs to $\mathcal{A}_{N,m,p}$ if there are $\rho' \in \mathcal{A}_{N,m+1,p}$ with $\beta' = \beta(\rho') \in \mathfrak{B}_{m+1}$ and a real Δ with $t_m^2|\Delta| \le \epsilon_m^{\mathrm{pre}}$ such that: the path of ρ is $P_{N,m}^{\mathrm{path}}$; its regular activities satisfy Definition 13.1 and are produced by local subtraction and path replacement, and its forced activities satisfy Definition 13.2 and are

produced by fusion (Lemma 13.11), from preliminary families with the properties of Lemma 13.8, including (13.11)–(13.12); its coupling is $\beta' + \alpha_{N,m} + \Delta$; its scalar normalization is $|\Lambda|^{-1} \log \mathcal{H}(1)$ with $\mathcal{H} = \mathcal{B}^{(m)}_{\beta'} \operatorname{dens} \rho'$; and its density equals $\mathcal{H}$. The relation $\mathcal{R}_{N,m,p}$ consists of the pairs (ρ, ρ') so obtained. Membership is a property of a single representation. The caps (5.12) are not imposed in this definition; Lemma 13.12 derives them. In particular the classes contain the outputs of the construction of Sections 13.1–13.3 on continuous curves and on their restrictions to closed subintervals. The analytic construction of Section 13.7 uses the same choices on the common address spaces; its step map and its restrictions are used for the selected recursion and endpoint tuning. Thus an adjacent pair always uses the actual input density and its own coupling in the kernel.

Lemma 13.17 (Specified recursion)**.** *For fixed G, b, every member of $\mathcal{A}_{N,m,p}$ satisfies Lemmas 12.3–12.4 and the activity bounds in (13.24). For every continuous curve in $\mathcal{A}_{N,m+1,p}$ with couplings in $\mathfrak{B}_{m+1}$, including curves assembled from earlier outputs, the step gives a continuous curve in $\mathcal{A}_{N,m,p}$, related to it pointwise by $\mathcal{R}_{N,m,p}$. Its restriction to a closed subinterval is the step on the restricted curve.*

Proof. The Wilson start (Lemma 13.16) supplies the initial curve and both caps; Sections 13.1–13.4 construct the step on a curve and prove its bounds, identity and continuity. At a step, Lemma 13.8 uses only the prescribed paths, the local regularity and forced eligibility, the bounds and the weighted continuity of its input, so the input curve need not coincide with an earlier chosen curve. Its truncation index depends only on the incoming address lists, so the same selection serves every input curve with these lists. Local subtraction, path replacement and fusion preserve the read supports, gauge invariance, cancellations and countability. Lemma 13.12 proves their numerical bounds, and Lemma 13.15 proves weighted continuity on the interval.

The normalized identity of Lemma 13.13 and the reset of Lemma 13.14 identify the same Δ. Lemma 13.14 identifies Δ with the continuous scalar root and gives $|\Delta| \le 1$ and the drift in Theorem 5.5. Defining the scalar normalization by flat evaluation gives the exact integral, since the normalized representation is one at 1. Hence every output is a member of the next class, (13.24) is inherited, and the prescribed paths are unchanged.

Restricting a chosen step to a closed parameter subinterval preserves its supports, tests, weighted continuity and estimates; weighted continuity becomes continuity in the relative topology of the smaller interval. We use this restricted step whenever the input curve is restricted. The construction applies to every continuous curve in the classes, regardless of how that curve was obtained. Thus the classes are closed under the step and under interval restriction, as required for the shooting argument in Section 14. Lemma 13.23 supplies the analytic estimates for this same recursion. □

Proof of Theorem 5.5. The step is the construction of Sections 13.1–13.3 applied to the input curve. Lemmas 13.9, 13.12, 13.13 and 13.15 give the final functions and their exact normalized identity. Put

$$v_u = |\Lambda|^{-1} \log \mathcal{H}_u(1), \qquad \beta_u = \beta'_u + \alpha_{N,m} + \Delta_u \,.$$

The denominator is positive by Lemma 13.9. Lemma 13.15 gives continuity of v_u and of the activity families. Lemma 13.14 identifies this same Δ_u with its continuous scalar root, and its drift estimate gives

$$\beta_u = \beta'_u - \kappa_G \log \ell_m + e_u, \qquad |e_u| \le \omega^{\mathrm{RG}}_{J_{m+1}} \,.$$

Define ρ_u by these scalars and the activities in Definition 5.2. Multiply the identity of Lemma 13.13 by $\mathcal{H}_u(1)$ to obtain $\operatorname{dens} \rho_u(U) = \mathcal{H}_u(U)$ for every configuration. Lemma 13.12 gives the activity

caps and Lemma 13.15 gives continuity. Lemma 13.17 gives $\rho_u \in \mathcal{A}_{N,m,p}$ and $(\rho_u, \rho'_u) \in \mathcal{R}_{N,m,p}$. The Wilson construction of Lemma 13.16 supplies the common initial curve with scalar zero. □

13.7 Weighted analytic estimates

This subsection writes the specified recursion in coordinates $(\mathfrak{x}_m, A_m)$, a normalized coupling and the activities, and proves the following. The step maps are real-analytic on balls of fixed radius, with multilinear coefficients of degree at least one bounded by $g_* = 1/128$ (Lemma 13.21). Hence the whole history is the unique fixed point of a contraction on power series in the endpoint coupling (Lemma 13.23), which gives endpoint tuning in Section 14. Each fixed-level coordinate converges along the full cutoff sequence (Lemmas 13.22 and 13.24), which drives the fixed-volume limit. Field differentiability does not give this parameter analyticity, because some activity functions are only measurable.

What is held fixed. A coupling derivative is never taken at fixed Gaussian coordinates. In each sector integral the physical link variables are held fixed, so every flag, allocation, tag and chart test, whose thresholds are t_{m+1} and fixed stage constants rather than $\beta^{-1/2}$, is independent of the coupling, and the supports, masks and local function domains of the activities are fixed. What varies with the coupling includes the Gaussian density in these fixed coordinates, its normalizing determinant, the stationary corrections and large-field energy factors such as $e^{-\beta(s)d}$, and these are analytic in the coupling (proof of Lemma 13.19). The fields themselves are not complexified, since the clips and threshold tests are not analytic in the links. Analyticity in the coupling and in the activity coordinates is therefore obtained from real evaluations. Lemma 13.19 bounds finite combinations $\sum_\nu c_\nu f(s_\nu)$ of values along real lines at a radius growing with $\widehat{\beta}_{m+1}$; Lemma 13.20 uses finite differences to bring these bounds to a fixed radius; and Lemma 13.18 recovers multilinear power series from them, on balls of radius $1/4$ centred at each point of the real domain of the recursion. In the fixed-level limit (Lemma 13.24) candidate limits are compared by geometric truncation of the activities; no compactness of a ball in the infinite-dimensional activity space is used.

We compare cutoffs using activity functions on fixed domains. At level m and period p, put all regular addresses into one countable list and all forced addresses into a second list. A missing term has coefficient zero. The lists contain every finite support, anchor and inventory used above, and every finite conjunction of their eligibility tests. The output of a fusion is indexed by its finite selection, so the lists are closed under the construction. Keep the same permutations for translations and signed permutations. This identifies different cutoffs without changing their health domains.

For a difference $A = (E, K)$ of activity families set

$$\|A\|_{m,p} = \max \left\{ \begin{array}{l} \dfrac{\sup_x \sum_{o_i = x} e^{\sigma_{J_m} \operatorname{cov}_{B_m}(X_i)} \|E_i\|_m}{\epsilon_m^{\mathrm{reg}}}, \\[2ex] \dfrac{\sup_x \sum_{x \in Y_i} e^{\sigma_{J_m} \operatorname{cov}_{B_m}(Y_i)} \sup_U |K_i(U)|}{\epsilon_m^{\mathrm{for}}} \end{array} \right\}. \tag{13.25}$$

The seminorms in this formula are the least majorants in Definitions 5.3–5.4. Quotient by zero seminorm and complete the space with the prescribed local cancellations, eligibility and permutations. Denote the resulting real Banach space by $\mathcal{X}_{m,p}$. This retains regular derivatives on healthy configurations and forced values on every configuration. The scalar normalization is excluded.

The coupling coordinate must allow relative changes of β while keeping a uniform analytic radius. Define

$$\chi_m = \widehat{\beta}_m^{-1/32}, \quad b_m = \chi_m \widehat{\beta}_m, \quad \mathfrak{x}_m = \chi_m^{-1}\sqrt{\beta_m/\widehat{\beta}_m}, \quad \beta_m = \widehat{\beta}_m(\chi_m \mathfrak{x}_m)^2 . \tag{13.26}$$

Thus b_m converts a change of $\mathfrak{x}_m$ into a change of β_m, up to the factor $2\chi_m\mathfrak{x}_m$. Monotonicity gives $b_m \geq \chi_0\widehat{\beta}_0 > 0$. Item 7 of Table 3 gives $\bar{\beta} \geq 2^{224}$; then $\chi_m \leq 1/128$. In the following paragraphs write $b_{\min} = \chi_0\widehat{\beta}_0$, $g_* = 1/128$ and $\lambda_* = 1/16$.

Let $F_{N,m}$ denote the selected activity map and $\mathcal{D}_{N,m}$ its scalar correction, so that

$$A_m = F_{N,m}(\mathfrak{x}_{m+1}, A_{m+1}), \qquad \beta_m = \beta_{m+1} + \alpha_{N,m} + \mathcal{D}_{N,m}(\mathfrak{x}_{m+1}, A_{m+1}) .$$

The top map is denoted by $F_N^{\mathrm{W}}(\mathfrak{x}_N)$. The maps are constructed for $\|A\| \leq 1$ and $33/40 \leq \chi_m\mathfrak{x} \leq 47/40$. All inputs in this domain have the common address lists, so the truncation index and auxiliary tori of Lemma 13.8 are the same at every point. Hence $F_{N,m}$ and $\mathcal{D}_{N,m}$ are single-valued, and their restriction to a line is the construction on that line. We use expansions centred in the smaller domain

$$\|A\| \leq 1/8, \qquad 17/20 \leq \chi_m\mathfrak{x} \leq 23/20 . \tag{13.27}$$

A ball of radius $1/4$ about such a centre stays in the larger domain: its activity norm is at most $3/8$, and the change of $\chi_m\mathfrak{x}$ is at most $1/512 < 1/40$. For a retained history with $|\beta_0 - \bar{\beta}| \leq \bar{\beta}/4$, summing the drift and the error bound in Section 9 gives $|\beta_m - \widehat{\beta}_m| \leq \widehat{\beta}_m/4$. Its square root therefore lies in $[17/20, 23/20]$. Every activity in the history, including the Wilson start, has norm at most $1/1024$ by the proofs of Lemmas 13.12 and 13.16. Thus the entire history lies in (13.27).

For a finite parameter evaluation formula with real weights c_ν and nodes $s_\nu \in [-1,1]$, define its moment bound

$$\mathfrak{s}(c,s;R) = \sup_{k\geq 0}\left|\sum_\nu c_\nu (s_\nu/R)^k\right| , \qquad R > 1 . \tag{13.28}$$

It expresses analyticity through real evaluations: if $f(s) = \sum_k a_k s^k$ with $\sum_k \|a_k\|R^k \leq M$, then $\|\sum_\nu c_\nu f(s_\nu)\| \leq M\,\mathfrak{s}(c,s;R)$, and Lemma 13.18 below proves a converse, up to halving the radius. The nodes lie on a parameter interval; they are unrelated to the spatial stencils of local fields.

Uniform line estimates at nearby centres control mixed finite differences; this supplies the multilinear coefficients for the endpoint equations.

Lemma 13.18 (Analyticity from real parameter estimates)**.** *Let X be a real normed space, Y a real Banach space, $r > 0$, $R > 1$, $C \geq 0$, and let f be defined on a set containing $x' + [-1,1]w$ for all x' near x and all $\|w\| \leq r$. Assume $\|\sum_\nu c_\nu f(x' + s_\nu w)\| \leq C\,\mathfrak{s}(c,s;R)$ for all such x', w and all parameter evaluation formulas with nodes in $[-1,1]$. Then there are continuous symmetric k-linear maps $T_k : X^k \to Y$ with*

$$f(x+h) = \sum_{k\geq 0} T_k(h,\dots,h), \qquad \|T_k\| \leq 2C\left(\frac{4e}{rR}\right)^k$$

whenever $\|h\| \leq r$ and $4e\|h\| < rR$. The T_k extend with the same bounds to the completion of X.

Proof. Associate to a parameter evaluation formula the sequence $(\sum_\nu c_\nu(s_\nu/R)^k)_{k\geq 0}$ with supremum norm. The assumed inequality defines a bounded linear map, of norm at most C, from their span to Y. Extend it to the closure. The kth coordinate vector lies in that closure: the kth finite difference at 0, divided by $k!(a/R)^k$, has zero moments below k, moment one at k, and higher

moments tending uniformly to zero as its mesh $a \to 0$. To see the uniformity, split the higher moments at any fixed order and use $|ja/R|^n \le (ka/R)^n$ for the remaining tail. Evaluation on a single point then gives the exact line series. Its coefficient sum at radius $R/2$ is at most $2C$. Uniqueness follows by successively dividing a vanishing series by its first possible power.

Write $P_k(h)$ for the coefficient of s^k in $f(x+sh)$. Rescaling s gives $P_k(ah) = a^k P_k(h)$ and

$$\|P_k(h)\| \le 2C \left(\frac{2\|h\|}{rR}\right)^k .$$

We must prove the polynomial identity in the direction h. Put $\delta_h g(a) = g(a+h) - g(a)$. Uniform line remainders at nearby centres imply

$$\delta_{sh}^n f(x+sa) = O(s^n) \quad (s \to 0).$$

Indeed insert the line series at $x+sa$ into the nth forward difference. The coefficients below degree n cancel by $\sum_{j=0}^n (-1)^{n-j}\binom{n}{j} j^k = 0$ for $k < n$; the uniformly bounded tail is $O(s^n)$.

For $h_0, \dots, h_{n-1}$ and $T \subset \{0, \dots, n-1\}$ set

$$v_T = \sum_{i \in T} \frac{h_i}{n-i}, \qquad w_T = -\sum_{i \in T} \frac{i h_i}{n-i}.$$

There is the exact mixed-difference identity

$$\delta_{sh_0} \cdots \delta_{sh_{n-1}} f(x+sa) = \sum_T (-1)^{n-|T|} \delta_{sv_T}^n f(x + s(a + w_T)) \,. \tag{13.29}$$

Expand the right side as a sum over its $n+1$ forward-difference points. At the point indexed by $j < n$, pairing sets T differing by the index $i = j$ cancels the terms, since that index has coefficient $(j-i)/(n-i) = 0$. At $j = n$ the coefficient of each h_i is one, leaving precisely the left side. Hence every mixed nth difference above is $O(s^n)$.

Expand each of its finitely many values in its line series at x and compare powers of s. It follows that

$$\delta_{h_0} \cdots \delta_{h_{n-1}} P_k(a) = 0 \qquad (k < n).$$

For $n = k+1$ this makes the kth difference of P_k independent of its base point. Define

$$T_k(h_1, \dots, h_k) = \frac{1}{k!} \delta_{h_1} \cdots \delta_{h_k} P_k(0).$$

The identity $\delta_{u+v} = \delta_u + \delta_v + \delta_u \delta_v$ proves additivity in each slot, since the extra $(k+1)$st difference vanishes. The displayed bound on P_k makes these additive maps locally bounded, hence continuous and real linear in each slot. The operators commute, giving symmetry. Homogeneity gives $\delta_h^k P_k(0) = k! P_k(h)$, so their diagonal is the original line coefficient.

There are 2^k terms in this polarization sum, and the argument in each has norm at most k on unit directions. Thus $k^k/k! \le e^k$ gives the stated operator bound. The diagonal series equals $f(x+h)$ by the line recovery; the norm bound proves its convergence on the stated domain. Continuous multilinear maps extend uniquely through the dense inclusion into the completion, with unchanged norm. □

We evaluate the maps along lines $(\mathfrak{x}(s), A(s)) = (\mathfrak{x} + sv, A_0 + sA_1)$, with $|v| \le 1$ and $\|A_0 + sA_1\| \le 1$ for $|s| \le 1$. Supports, flag and threshold tests and local function domains are fixed along the

line. Write $F(s) = F_{N,m}(\mathfrak{x}(s), A(s))$ and $\mathcal{D}(s) = \mathcal{D}_{N,m}(\mathfrak{x}(s), A(s))$. Let $\mathcal{N}_{\mathrm{pre}}$ be the anchored regular norm at exponent $\sigma + 3$, before division by $\epsilon_m^{\mathrm{reg}}$ and before subtraction. Let $\mathcal{N}_{\mathrm{for}}$ be the corresponding forced site norm at exponent $\sigma + 1$.

The coupling varies through a square of an affine function. The next estimate controls that dependence at a radius growing with the reference coupling; a bound at one fixed field-derivative order would not suffice.

Lemma 13.19 (Estimates on coupling lines)**.** *Put* $\mathfrak{b} = \widehat{\beta}_{m+1}$ *and* $\mathfrak{R} = \mathfrak{b}^{1/256}$*. For every line* $s \mapsto (\mathfrak{x} + sv, A_0 + sA_1)$*,* $|s| \le 1$*, with* $|v| \le 1$*,* $\|A_0 + sA_1\| \le 1$ *and* $33/40 \le \chi_{m+1}\mathfrak{x} \le 47/40$*, the preparatory families satisfy*

$$\mathcal{N}_{\mathrm{pre}}\left(\sum_{\nu} c_\nu f(s_\nu)\right) \le \mathfrak{b}^{-1/4}\mathfrak{s}(c, s; \mathfrak{R}/2),$$
$$\mathcal{N}_{\mathrm{for}}\left(\sum_{\nu} c_\nu k(s_\nu)\right) \le C_j e^{-\kappa_j^{\mathrm{pre}}(M_{m+1}^{\mathrm{cut}})^2}\mathfrak{s}(c, s; \mathfrak{R}/4).$$

For any prescribed $\theta_{\mathrm{W}} > 0$*, the Wilson map satisfies the analogous normalized estimate with radius* $\mathfrak{R}$ *and bound* $4\theta_{\mathrm{W}}$ *after the entry threshold is raised.*

The proof keeps physical fields fixed and then applies the connected and localization operations.

Proof. The parameter coefficients of the connected expansion of Lemma 13.5 are bounded by the same rooted estimate, with the Gaussian rescaling expanded in convergent Neumann series; the five tori of Lemma 13.7 then act on these coefficients. Put $\upsilon = \mathfrak{b}^{-1/128}$ and raise the threshold until

$$128\mathfrak{R}\chi_{m+1} \le \upsilon \le 1\,. \tag{13.30}$$

This is possible since $1/256 - 1/32 + 1/128 = -5/256 < 0$. Set $\zeta = \chi_{m+1}\mathfrak{x}$ and $L = \zeta - 6\mathfrak{R}\chi_{m+1}$. For $|s| \le 6\mathfrak{R}$,

$$\beta(s) = \mathfrak{b}L^2\left(\frac{\zeta}{L} + \frac{\chi_{m+1}\upsilon}{L}s\right)^2, \qquad \frac{3}{4} \le \chi_{m+1}(\mathfrak{x} + sv) \le \frac{4}{3}\,.$$

The displayed margins follow from (13.30) and $33/40 \le \zeta \le 47/40$. In particular the real couplings lie in $\mathfrak{B}_{m+1}$. The relative scale change is at most a fixed multiple of υ.

To estimate parameter coefficients, use physical link variables in each sector integral. All flag, allocation, tag and chart tests are then independent of s: their thresholds are t_{m+1} and the prescribed stage constants, not $\beta(s)^{-1/2}$. Old measurable forced functions occur as $K_0 + sK_1$ at the same physical links. Gaussian rescaling is used to estimate the coefficients, after these functions have been held fixed. Thus parameter coefficients do not differentiate their field arguments or any test.

Here is the coefficient bound for this change of Gaussian scale. For a positive local precision matrix A and affine source b_0, put $m_0 = -\mathsf{A}^{-1}\mathsf{b}_0$ and $Q(a) = \langle a - m_0, \mathsf{A}(a - m_0)\rangle/2$. This is the centred quadratic energy in the sector Gaussian density. The parameter occurs through $\beta(s)Q$, its normalizing determinant, and the local stationary corrections. For a precision perturbation H,

$$(\mathsf{A} + H)^{-1} = \sum_{n\ge 0}(-\mathsf{A}^{-1}H)^n\mathsf{A}^{-1}, \qquad \log\det(\mathsf{A} + H) - \log\det\mathsf{A} = \sum_{n\ge 1}\frac{(-1)^{n+1}}{n}\operatorname{Tr}\big((\mathsf{A}^{-1}H)^n\big)\,.$$

Both series are estimated through the same dimensionless perturbation $\mathsf{A}^{-1}H$. The inverse bound and exponential Schur sums of Lemma 11.13, together with (13.30), make their radius-$\mathfrak{R}$ coefficient

sums less than a fixed constant. For the Gaussian density, expansion of its exponential is dominated by $e^{c|z|^2}$ with c smaller than half its precision lower bound. Indeed the coefficient of $|z|^2$ changes by $O(\mathfrak{R}\chi_{m+1})$. Integrating this majorant bounds every parameter degree at once. The local number of coordinates is a fixed power of $1+B$; hence its additional loss is bounded because $(1+B)^C\mathfrak{R}\chi_{m+1}\to 0$. The same Neumann series solves each stationary correction with the uniformly bounded inverse. Ordered field derivatives only add the fixed read polynomials already present in Lemma 11.14.

Each incoming factor is affine in s. For a regular occurrence its exponential has radius-$\mathfrak{R}$ majorant $\exp(\mathfrak{R}\|E_1\|)$; for a forced occurrence the majorant is $|K_0|+\mathfrak{R}|K_1|$. Products use Cauchy products of coefficients. Reserve the support powers before summing these expressions. The initial small factor in (13.4) loses at most one factor $\mathfrak{R}$ for an incoming affine occurrence; the change of Gaussian scale has the bounded sums just proved. Thus the weighted attachment parameter is bounded by

$$P_j(\mathfrak{b})\mathfrak{R}\mathfrak{b}^{-7/16}=P_j(\mathfrak{b})\mathfrak{b}^{-111/256}\longrightarrow 0\,.$$

The connected logarithm therefore has an absolutely summable parameter series by the same rooted estimate as in Lemma 13.5. A forced component retains a smaller fixed portion of the energy in Lemma 11.12; coefficient multiplication and sector counting use the rest. Its coefficient majorant is $C_j e^{-\kappa_j^{\mathrm{pre}}(M_{m+1}^{\mathrm{cut}})^2}$.

Apply the five reflected tori of Lemma 13.7 to these same parameter coefficients. A parameter evaluation formula commutes with their finite linear combinations. The direction rescaling, auxiliary volumes, support conversion and flat subtraction cost a subpower factor. The resulting regular coefficient sum has bound $P_j(\mathfrak{b})\mathfrak{b}^{-3/8}$: the preceding attachment bound is smaller, since $111/256>3/8$. Raise the threshold until this is at most $\mathfrak{b}^{-1/4}$. The regular logarithm and the forced collection may use radii $\mathfrak{R}/2$ and $\mathfrak{R}/4$, respectively; their Cauchy products then have geometric reserve. Multiplying degree n by $\sum_\nu c_\nu s_\nu^n$ proves the asserted bounds.

For the Wilson map use the formulas in the proof of Lemma 13.16. Its regular term is $\beta(s)$ times a fixed fifth-order remainder. Its radius-$\mathfrak{R}$ coefficient sum is at most $C_j\mathfrak{b}t_N^5/\epsilon_N^{\mathrm{reg}}$, which tends to zero. For a forced charge $d_I\ge 0$, write $\beta(s)=\beta(0)+b_1s+b_2s^2$. Its exponential has coefficient sum at most

$$\exp\{-\beta(0)d_I+(|b_1|\mathfrak{R}+|b_2|\mathfrak{R}^2)d_I\}\le e^{-\beta(0)d_I/2}$$

by (13.30). The inventory count in that proof is still summable with this half exponent. Both normalized coefficient sums can be made at most θ_{W}; enlarging the bound to $4\theta_{\mathrm{W}}$ gives the stated estimate. All bounds were taken before the period was counted, so they are uniform in the period. □

We now reduce the growing radius to one fixed radius chosen before the ratios. Choose a radius $\mathfrak{R}_0>1$, and then positive $\delta_{\mathrm{map}}\le g_*/2$ and $\eta_{\mathrm{par}}\le\delta_{\mathrm{map}}$. Put $\varphi_m=\eta_{\mathrm{par}}\epsilon_m^{\mathrm{pre}}/t_m^2$ and $\bar\varphi_m=2\varphi_m$; here $\epsilon_m^{\mathrm{pre}}=\theta c_{J_{m+1}}\epsilon_m^{\mathrm{reg}}$ is the coefficient for this step. The small fraction θ is chosen below before the ratios.

Lemma 13.20 (Real parameter estimates)**.** *For every line as in Lemma 13.19, the selected local maps satisfy*

$$\left\|\sum_\nu c_\nu F(s_\nu)\right\|\le\delta_{\mathrm{map}}\mathfrak{s}(c,s;\mathfrak{R}_0/2),\qquad\left|\sum_\nu c_\nu\mathcal{D}(s_\nu)\right|\le\varphi_m\mathfrak{s}(c,s;\mathfrak{R}_0),\tag{13.31}$$

with $\sum_{i<m}\bar\varphi_i\le g_*b_m$. The radius $\mathfrak{R}_0$ can be arbitrarily large. The Wilson map satisfies the first estimate with the same δ_{map}.

Proof. Finite differences recover the low line coefficients from the bound of Lemma 13.19, as in Lemma 13.18; the high ones are paid by the growing radius; the three linear conversions and fusion then pass the bound to the output. Recover the regular line coefficients by finite differences at zero. Put $\mathfrak{Q} = \mathfrak{b}^{1/256}/2$. Choose an integer K with $(K+1)/256 + 1/4 > 11/10$; we take $K = 300$. Split after degree K and interpolate the finite part at $K+1$ fixed distinct points of $[-1,1]$. The norm of this finite interpolation matrix gives a constant L_0, depending only on $\mathfrak{R}_0$ and K, independent of the coupling and volume. The radius-$\mathfrak{R}_0$ coefficient sum is bounded by

$$L_0 \epsilon_m^{\mathrm{pre}} + (1+L_0)(\mathfrak{R}_0/\mathfrak{Q})^{301} \mathfrak{b}^{-1/4}\,.$$

For the tail each degree $n \geq 301$ contributes $(\mathfrak{R}_0/\mathfrak{Q})^n$; its geometric denominator is bounded once $\mathfrak{Q} \geq 2\mathfrak{R}_0$. The tail has rate

$$-301/256 - 1/4 = -365/256 < -11/10\,.$$

Construct the preparatory families with θ replaced by $\theta\eta_{\mathrm{par}}/[4(1+L_0)]$, so that (13.11) holds with $\epsilon_m^{\mathrm{aux}} = \eta_{\mathrm{par}}\epsilon_m^{\mathrm{pre}}/[4(1+L_0)]$ in place of $\epsilon_m^{\mathrm{pre}}$. Since this only decreases the bound, Lemma 13.8 holds as stated, and no identity changes. Choose this fraction before the ratio, and raise the threshold until the tail is at most $\eta_{\mathrm{par}}\epsilon_m^{\mathrm{pre}}/4$. The radius-$\mathfrak{R}_0$ bound is then at most $\eta_{\mathrm{par}}\epsilon_m^{\mathrm{pre}}/2$. Projection by (13.14) gives a scalar bound at most $\eta_{\mathrm{par}}\epsilon_m^{\mathrm{pre}}/t_m^2$. The scalar reset in the proof of Lemma 13.14 is exactly this extracted coefficient.

For the regular output, apply its three linear conversions with total bound C_j^0. Choose θ so the radius-$\mathfrak{R}_0$ preparatory contribution is at most $\epsilon_m^{\mathrm{reg}}/1024$ after this conversion, and raise the threshold so the high path term contributes at most another $\epsilon_m^{\mathrm{reg}}/1024$. Explicitly it suffices to choose the real-interval coefficient so that $C_j^0 L_0 \theta c_j \leq 1/2048$ and then to use the tail reserve just proved. For the forced output, retain its coefficient majorant from Lemma 13.19; subtraction adds logarithmic factors with the bounds in the proof of Lemma 13.12. At fixed $\mathfrak{R}_0$, their radius-$\mathfrak{R}_0$ sums have the same small attachment parameter, with a constant depending on $\mathfrak{R}_0$. Fusion therefore gives

$$\mathcal{K}_m(K)_{\mathfrak{R}_0/2} \leq C_{j,0} \exp\{-\kappa_j^{\mathrm{out}} (M_{m+1}^{\mathrm{cut}})^2\} \leq \epsilon_m^{\mathrm{for}}/1024$$

for a smaller $\kappa_j^{\mathrm{out}} > 0$ and a larger threshold. Here $\mathcal{K}_m$ is the forced norm at the output exponent σ, and the subscript denotes the parameter coefficient sum at that radius. The exponent $\sigma + 1$ of the preparatory bound supplies the extra power used in fusion. The maximum norm (13.25) is thus at most $1/512$, and can be made smaller than the prescribed δ_{map}. This derives both the order-zero bound and the full bound on parameter evaluation formulas. For Wilson choose $4\theta_{\mathrm{W}} \leq \delta_{\mathrm{map}}$ in Lemma 13.19 and reduce its radius to $\mathfrak{R}_0/2$.

Since $\theta \leq 1$ and $c_j \leq 1/64$, the scalar normalization gives

$$\bar{\varphi}_m \leq (\eta_{\mathrm{par}}/32)\widehat{\beta}_m^{-1/10}, \qquad \sum_{i<m} \bar{\varphi}_i \leq C_G \eta_{\mathrm{par}} \widehat{\beta}_m^{9/10}\,.$$

The second inequality follows by comparison with an integral, since the reference-coupling increments have a fixed positive lower bound. Divide by $b_m = \widehat{\beta}_m^{31/32}$ and raise the entry threshold until $C_G \eta_{\mathrm{par}} \bar{\beta}^{-11/160} \leq g_*$. This proves the weighted scalar bound. □

The real parameter estimates yield norm-convergent multilinear expansions in the common completed spaces.

Lemma 13.21 (Local analytic estimates)**.** *At every point of the real domain above, the centered maps $F_{N,m}$, $\mathcal{D}_{N,m}$ and F_N^{W} have convergent real multilinear expansions on the ball of radius $1/4$. Their degree-k coefficients, $k \geq 1$, have norms at most g_*, $\bar{\varphi}_m$ and g_*, respectively, where $\bar{\varphi}_m \geq 0$ is independent of N, p and*

$$\sum_{i<m} \bar{\varphi}_i \leq g_* b_m \,.$$

Proof. Lemma 13.18 turns the bounds of Lemma 13.20 into coefficient bounds. The larger real domain preceding (13.27) supplies the neighbouring centres required in the recovery lemma. Apply it with $r = 1/4$ and $R = \mathfrak{R}_0/2$ for the activity map, and $R = \mathfrak{R}_0$ for the scalar correction. Their coefficient bounds are respectively

$$2\delta_{\mathrm{map}}(32e/\mathfrak{R}_0)^k, \qquad 2\varphi_m(16e/\mathfrak{R}_0)^k \,.$$

Choose $\mathfrak{R}_0 \geq 128e$ before the ratios. Since $2\delta_{\mathrm{map}} \leq g_*$, these are the asserted bounds. The same argument applies to Wilson. Completion preserves each coefficient, its norm and its diagonal identity. The fixed permutation subspace preserves equivariance. □

The strict path reserve also permits a comparison across cutoffs at a fixed level.

Lemma 13.22 (Continuity in the cutoff)**.** *Fix m, p. If $(\mathfrak{x}^{(N)}, A^{(N)}) \to (\mathfrak{x}, A)$ in $\mathbb{R} \times \mathcal{X}_{m+1,p}$ within the domain of the maps, then $F_{N,m}(\mathfrak{x}^{(N)}, A^{(N)})$ converges in $\mathcal{X}_{m,p}$ and $\mathcal{D}_{N,m}(\mathfrak{x}^{(N)}, A^{(N)})$ converges in $\mathbb{R}$. The path coefficients of $P_{N,m}^{\mathrm{path}}$ converge in the weighted path norm with exponent $w_{J_m}^{\mathrm{path}}$, being bounded uniformly in N at the strictly larger exponent of Lemma 12.3.*

Proof. The path coefficients converge degree by degree through the affine coefficient maps below; the connected weights converge by dominated convergence, and the remaining operations are linear or have a strict attachment margin. Compare first the quadratic rows and then the cubic, quartic and quadratic loop kernels in that order. The free forms, their row factors and the kernels W_0, T converge by Lemma 11.5; placing the coefficients of $\rho_{N,m}$ on coordinate paths (Section 12) gives convergence of the quadratic rows. The resolvent identity and the uniform inverse bound of Lemma 11.13 give convergence of the remaining fixed-window inverses. In each subsequent comparison fix these rows and all preceding kernels. Let $\mathcal{V}_a$ be the affine coefficient space for the kernel of the current degree, with its prescribed symmetries and cancellation identities. Divide its weighted coefficient norm by the stage allowance. The corresponding coefficient map, denoted $\mathcal{T}_m^a$, obeys

$$\mathcal{T}_m^a(K + \lambda(H - K)) = \mathcal{T}_m^a(K) + \lambda(\mathcal{T}_m^a(H) - \mathcal{T}_m^a(K)) \,. \tag{13.32}$$

For the cubic coefficient this follows by retaining degree three in the stationary substitution: it is linear in the incoming cubic kernel. At degree four, terms quadratic in the cubic kernel are fixed, and the incoming quartic kernel occurs once. The quadratic loop coefficient is affine in its incoming loop kernel, with the preceding classical kernels fixed. These are the degree extractions in Lemmas 12.6–12.9.

Take the ratio threshold of Lemma 12.6 for the enlarged input path bound $64A_{J_{m+1}}^{\mathrm{path}}$, and, for the enlarged quadratic loop bound $64\Lambda_{m+1}^{\mathrm{loop}} \leq 128\Lambda_m^{\mathrm{loop}}$, the threshold in the proof of Lemmas 12.3–12.4 with $C_j^{\mathrm{pb}}\ell^{-2} \leq 1/256$, so that $\frac{1}{2}\Lambda_m^{\mathrm{loop}} + \frac{1}{256} \cdot 128\Lambda_m^{\mathrm{loop}} = \Lambda_m^{\mathrm{loop}}$. These choices precede the coupling threshold. With the preceding kernels fixed, the coefficient constructions then accept normalized input size 64 and give normalized output size at most one. The enlarged input must still belong to $\mathcal{V}_a$ and satisfy these bounds. For inputs $K, H \in \mathcal{V}_a$ of normalized size at most one

put $\delta_K = \|H - K\|$. If $\delta_K > 0$, the extrapolated input $K_* = K + (32/\delta_K)(H - K)$ belongs to the same affine space and has norm at most $33 < 64$. Its linear cancellation and symmetry conditions hold because their homogeneous differences form a linear subspace. Hence its stage bound applies. Equation (13.32) gives

$$\frac{32}{\delta_K}\|\mathcal{T}_m^a(H) - \mathcal{T}_m^a(K)\| \le \|\mathcal{T}_m^a(K_*)\| + \|\mathcal{T}_m^a(K)\| \le 2\,.$$

The output difference is at most $\delta_K/16 \le \delta_K/4$; at $\delta_K = 0$ it is zero by the same affine identity. This calculation concerns one kernel, with all preceding kernels fixed.

For two cutoffs, the change of those fixed preceding kernels contributes an error $e_{N,N',i}$ tending to zero at each fixed level i, once their preceding comparisons have been made. The normalized difference of the current kernels therefore satisfies $\delta_i \le \delta_{i+1}/4 + e_{N,N',i}$. Truncate this coefficient recursion at level k, retaining any terminal kernel in $\mathcal{V}_a$ of normalized size at most one there. Its terminal difference is at most two, and iteration gives

$$\delta_m \le 2(1/4)^{k-m} + \sum_{i=m}^{k-1}(1/4)^{i-m} e_{N,N',i}\,.$$

At fixed k the sum tends to zero; then $k \to \infty$ proves the Cauchy property, successively for each degree.

Retain now finitely many addresses and finite path lengths. Window covariances and means on these sets converge with their derivatives by the uniform inverse bound and the resolvent identity. Stationary corrections use this same inverse, and determinant differences are integrals of traces of resolvent differences. Lemmas 12.6–12.9 express the path coefficients by these kernels. The stronger-weight clause of Lemma 12.3 bounds these coefficients uniformly in N at an exponent $w' > w$, with $w = w_{J_m}^{\mathrm{path}}$. A tail of length at least l at exponent w is bounded by its uniform w' norm times $e^{-(w'-w)l}$. Thus finite-coefficient convergence and uniformly small tails prove path convergence with a strict reserve.

For the connected weights, write both integrals in the same fixed physical coordinates and subtract their densities. The converging means and precisions make these Gaussian densities converge pointwise. Their strict positive lower bounds give a common integrable Gaussian exponential majorant, including the fixed derivative scores; combine it with G_M from Lemma 11.14 and the weighted input discrepancy. Dominated convergence therefore gives convergence for a finite collection. The rooted estimates bound the discarded collections uniformly. Localization uses only five fixed auxiliary tori, finite linear combinations and such uniformly bounded tails. Subtraction is linear, and fusion has a strict attachment margin. Consequently input coordinates converging in (13.25) and converging normalized couplings give outputs and corrections converging in the same common spaces. □

For two histories at the same N, p, let $\mathfrak{x}_m, A_m$ and $\mathfrak{x}'_m, A'_m$ be these coordinates and set

$$\mathfrak{d} = \max_{0\le m\le N} \max\{|\mathfrak{x}'_m - \mathfrak{x}_m|, \|A'_m - A_m\|_{m,p}\}, \qquad \delta = \beta'_0 - \beta_0\,. \tag{13.33}$$

The parameter series below belong to $\mathfrak{S}(X) = \{\sum_{k\ge0} z^k X_k : \sum_k \|X_k\| < \infty\}$, with norm $\sum_k \|X_k\|$. Evaluation has norm at most one for real $|z| \le 1$. This space is complete, and its Cauchy product is bounded by the product of the norms.

For the series equations fix a history reaching $\bar\beta$, whose existence is proved in Lemma 14.3, and set $z_m = \chi_m \mathfrak{x}_m$. Write $\mathfrak{u}_m$ for the perturbation of $\mathfrak{x}_m$, W_m for that of A_m, and $\mathfrak{e}$ for the endpoint series. For any map F let $\widehat F$ be its centered series. The simultaneous equations are

$$\begin{aligned} W_N &= \widehat F^{\mathrm W}_N(\mathfrak{u}_N), \\ W_m &= \widehat F_{N,m}(\mathfrak{u}_{m+1}, W_{m+1}) \quad (m < N), \\ \mathfrak{u}_m &= (2z_m)^{-1}\left[b_m^{-1}\left(\mathfrak{e} - \sum_{i<m} \widehat{\mathcal D}_{N,i}(\mathfrak{u}_{i+1}, W_{i+1}) \right) - \chi_m \mathfrak{u}_m^2 \right] . \end{aligned} \tag{13.34}$$

The last line is obtained by expanding the square in (13.26); it retains its quadratic term.

These right sides define a map on the product of the spaces $\mathfrak S(\mathbb R)$ and $\mathfrak S(\mathcal X_{m,p})$, with the maximum of their series norms. The real endpoint interval is

$$\mathfrak J = [\bar\beta - \lambda_* b_{\min}/4, \bar\beta + \lambda_* b_{\min}/4] .$$

The following is the estimate used by Section 14. It applies to the selected step maps, their actual Wilson initial family and the same fixed address spaces.

Lemma 13.23 (Analytic endpoint equations)**.** *The histories above satisfy*

$$\mathfrak d \le |\delta|/b_{\min} + 8 g_* \mathfrak d .$$

For $\|\mathfrak e\|_{\mathfrak S(\mathbb R)} \le \lambda_* b_{\min}/4$, equations (13.34) preserve the closed radius-λ_* ball in the product of the spaces $\mathfrak S(\mathbb R)$ and $\mathfrak S(\mathcal X_{m,p})$ and contract by at most $1/2$. Their unique solution evaluates to the selected real recursion.

Proof. The small coefficients of Lemma 13.21 make every step map nearly constant, and summing the scalar recursion from the endpoint cancels the coefficients $\alpha_{N,i}$. The centered expansions of Lemma 13.21 give

$$\begin{aligned} \|A'_m - A_m\| &\le 8 g_* \max\{|\mathfrak x'_{m+1} - \mathfrak x_{m+1}|, \|A'_{m+1} - A_{m+1}\|\}, \\ |\mathcal D'_{N,m} - \mathcal D_{N,m}| &\le 4 \bar\varphi_m \max\{|\mathfrak x'_{m+1} - \mathfrak x_{m+1}|, \|A'_{m+1} - A_{m+1}\|\} . \end{aligned} \tag{13.35}$$

For distances at most $1/4$ these follow by subtracting the series and summing $k(1/4)^{k-1}$. For larger distances the separate bounds $\|A_m\| \le g_*/2$ and $|\mathcal D_{N,m}| \le \bar\varphi_m/2$ give the displayed, weaker constants. The Wilson top map has the first bound as well.

Write $z_m = \chi_m \mathfrak x_m$ and $z'_m = \chi_m \mathfrak x'_m$ in this proof. Both are in $[17/20, 23/20]$. Equation (13.26) gives the exact identity

$$\beta'_m - \beta_m = b_m(\mathfrak x'_m - \mathfrak x_m)(z'_m + z_m) .$$

Summing the scalar recursion from the endpoint cancels $\alpha_{N,i}$ between the two histories, and hence

$$b_m|\mathfrak x'_m - \mathfrak x_m| \le |\delta| + \sum_{i<m} |\mathcal D'_{N,i} - \mathcal D_{N,i}| \le |\delta| + 4 g_* b_m \mathfrak d .$$

Take the maximum together with the activity estimates and use $b_m \ge b_{\min}$. This proves the asserted inequality before absorbing its last term. In particular $\delta = 0$ gives $\mathfrak d = 0$, and otherwise $\mathfrak d \le 2|\delta|/b_{\min}$. Thus real evaluation can be identified by uniqueness.

On the product ball of radius λ_* the series pair $(\mathfrak u_m, W_m)$ has norm at most $2\lambda_* < 1/4$. A centered series with coefficients bounded by c consequently has norm at most $c/2$, and the difference

of its two evaluations is bounded by $2c$ times the difference of its arguments. For the scalar prefix the latter gives $4\sum_{i<m}\bar{\varphi}_i$ times the product distance, because the two coordinates contribute at most twice that distance. Also

$$\|\mathfrak{u}_m^2 - \mathfrak{v}_m^2\| \le 2\lambda_*\|\mathfrak{u}_m - \mathfrak{v}_m\|\,.$$

Since $2z_m \ge 1$ and $\chi_m \le 1$, the scalar part of (13.34) has norm at most

$$\lambda_*/4 + g_* + \lambda_*^2 < \lambda_*$$

and Lipschitz constant at most $4g_* + 2\lambda_* = 5/32 < 1/2$. The activity part has norm at most $g_*/2 < \lambda_*$ and Lipschitz constant at most $4g_* < 1/2$. Thus the map preserves the closed ball and contracts there. Starting at zero, its successive differences are bounded by a geometric series with ratio $1/2$; completeness gives its unique fixed point.

Evaluation commutes with sums, products and each absolutely convergent multilinear series. It turns the fixed point into the real recursion and Wilson start. The comparison already proved identifies it with the selected history. The endpoint width $\lambda_* b_{\min}/4$ is smaller than the real shooting width: $b_{\min} = \chi_0\bar{\beta} \le \bar{\beta}/128$. Hence the whole closed real interval stays in that shooting interval. □

Only the endpoint coupling is compact in the next argument; candidate limits are compared by geometric truncation of the activities, and no compactness of a ball in the activity Banach space is used.

Lemma 13.24 (Fixed-level limits of couplings and activities)**.** *For fixed p, m, the selected scalar couplings and activity coordinates in $\mathcal{X}_{m,p}$ converge along every full cutoff sequence with convergent endpoints in the interval of Lemma 13.23. The prescribed paths converge with the strict reserve of Lemma 13.22. The limits are independent of subsequences. The additive scalar normalizations are excluded from this convergence statement.*

Proof. Compactness is used only for the scalar coordinates; the activities are controlled by truncating the recursion at a fixed level, which costs a geometric factor by (13.35), and by Lemma 13.22 for the truncated orbit. First extract a subsequence on which every scalar coordinate $\mathfrak{x}_i$ converges. Each fixed coordinate is in a compact real interval by $17/20 \le \chi_i\mathfrak{x}_i \le 23/20$; a diagonal selection gives the simultaneous subsequence.

Fix a stopping index $k > m$. At level k replace the activity coordinate by zero and run the same selected maps downward using the unchanged scalar coordinates. Write $A_m^{[k]}$ for the resulting coordinate. The first inequality in (13.35), applied successively at equal scalar inputs, gives

$$\|A_m - A_m^{[k]}\|_{m,p} \le (8g_*)^{k-m}\,. \tag{13.36}$$

The terminal difference is at most one. For fixed k the truncated coordinates converge by a finite downward induction, using Lemma 13.22. The geometric bound is uniform in the cutoff and tends to zero as $k \to \infty$. Given $\varepsilon > 0$, choose k to make both truncation errors smaller than $\varepsilon/3$, then choose the cutoffs so that the truncated difference is smaller than $\varepsilon/3$. Thus the actual activity coordinates are Cauchy in their common completed spaces on the extracted subsequence. The scalar corrections converge by the second inequality in (13.35) and Lemma 13.22.

It remains to identify two possible subsequential limits. Denote them by $\mathfrak{x}_i, A_i, \mathcal{D}_i$ and $\mathfrak{x}_i', A_i', \mathcal{D}_i'$. Their common endpoint is β_0. Passing to the limit in the finite scalar recursion gives

$$\beta_i - \beta_i' = -\sum_{j<i}(\mathcal{D}_j - \mathcal{D}_j')\,. \tag{13.37}$$

The limits of $\alpha_{N,j}$ cancel because the prescribed paths have one limit. Put

$$\mathfrak{d}_\infty = \sup_{i\geq 0} \max\{|\mathfrak{x}_i - \mathfrak{x}'_i|, \|A_i - A'_i\|_{i,p}\}\,.$$

This supremum is finite with bound $1/4$. Indeed the activity difference is at most $1/4$, and (13.37) together with $|\mathcal{D}_j|, |\mathcal{D}'_j| \leq \bar{\varphi}_j/2$ gives $|\mathfrak{x}_i - \mathfrak{x}'_i| \leq g_*$, after division by b_i as in the previous proof. Passing the one-step estimates to the limit gives $\|A_i - A'_i\| \leq 8g_*\mathfrak{d}_\infty$. Equations (13.37) and (13.35) give $|\mathfrak{x}_i - \mathfrak{x}'_i| \leq 4g_*\mathfrak{d}_\infty$. Hence $\mathfrak{d}_\infty \leq 8g_*\mathfrak{d}_\infty$, so $\mathfrak{d}_\infty = 0$.

Every subsequence therefore has a further subsequence with these same scalar and activity limits. If the full sequence failed to converge at some fixed level, a subsequence staying a positive distance from the limit would contradict the preceding construction. This proves full convergence. The continuous map $x \mapsto \widehat{\beta}_m(\chi_m x)^2$ gives convergence of β_m. The path assertion is already part of the cutoff comparison, with a strictly larger exponent than the one defining the local charts. None of these arguments compares the scalar normalizations, which disappear from the normalized activity and coupling equations. □

14 Trajectories, marked fields and ultraviolet convergence

This section constructs the trajectories and the renormalized insertions used in Part I, and proves their volume-uniform bounds and full fixed-volume limits: Theorem 5.6, Proposition 5.8 and Theorem 5.9. It also supplies the cutoff equation identities, coefficient corrections and finite-rank comparisons from which Section 15 builds the quotient fields. All these results concern the Wilson measures of Definition 2.3 and the insertions of Definition 2.8.

Three different tasks are involved, and each has its own mechanism. A continuous shooting argument chooses the bare coupling so that the trajectory reaches one prescribed coarse coupling (Section 14.1). Weighted estimates for the specified recursion give uniqueness and analytic dependence near that endpoint. Finally, at fixed volume, convergence of the effective coordinates at each fixed level is combined with a bound on the error made by stopping the insertion transfer there (Section 14.3). The additive scalar normalization cancels from normalized expectations, so its convergence is never needed.

The main difficulty in the bounds of Section 14.2 is differentiation in position. Each lattice difference introduces an inverse spacing, so a bound for undifferentiated insertions does not suffice. Local comparison replaces a differentiated insertion by one of higher degree, with a negligible chessboard error. Exact equation identities remove its equation component. A finite matrix of free Maxwell correlation tests bounds the remaining coefficient. This argument takes place at finite cutoff, before any continuum field space has been constructed.

Throughout, the group, metric, penalty, scales and admissible classes are those of Section 5 and Section 9. The trajectories are built from the path actions of Lemmas 12.3–12.4, which fix the classical and one-loop terms, the Wilson initialization of Lemma 13.16, and the inductive step of Theorem 5.5, which carries an admissible representation one level down. Additional dependence of constants is indicated. Write $M_{N,p} = M^{\mathrm{lat}}_{N,p}$ and $\langle F\rangle_{N,p} = \int F\, d\mu_{N,p}$.

For an insertion label $v = (d, c)$ put $O_{N,p}(v, y) = O^{(d)}_{N,p}(c, y)$. Physical points x in a moment are read as lattice sites x/a_N, modulo $L_p D_N$. We use the torus separation

$$\mathrm{sep}_{\mathrm{tor}}(y) = \min_{i\neq j} \min_{z\in L_p\mathbb{Z}^4} |a_N(y_i - y_j) + z|_\infty,$$

with value $+\infty$ for at most one slot. Multilinear interpolation on the physical grid means the affine interpolation in each coordinate of each grid cube, using its vertex values.

14.1 Trajectories and tuning

The entry levels allow the ratios and stage constants to increase sufficiently slowly. Besides the conditions of Section 9, we use the following quantitative choices. Let $c_{\mathrm{dir},j}$ be the direct clipping coefficient at stage j, and let ν_G^{mat} be the real dimension of the ambient matrix space. Put

$$Q_j(b) = 2(2+4A_j^{\mathrm{path}})+1+6(c_{\mathrm{dir},j}+(A_j^{\mathrm{path}})^2)+4\nu_G^{\mathrm{mat}}(\log 16+\tfrac{1}{2}\log b), \qquad q_m = Q_{J_m}(\widehat{\beta}_m). \tag{14.1}$$

Here A_j^{path} is the path bound of Lemma 12.3. For an admissible unmarked representation ρ_m on V sites write $\Phi_m = v_{\mathrm{sc},m}V - P_{N,m}^{\mathrm{path}} - \sum_i \mathbf{1}_{X_i\,\mathrm{healthy}} E_i^{\mathrm{reg}}$ and $K_{m,Y}$ for its forced activities. The next estimate bounds what it costs, per site, to replace the positive density by absolute values in a normalized integral.

Lemma 14.1 (Normalization cost)**.** *For every sufficiently late level m and every admissible unmarked representation ρ_m on V sites,*

$$\frac{\int e^{\Phi_m(U)} \sum_{\mathcal{A}} \prod_{Y\in\mathcal{A}} |K_{m,Y}(U)|\, dU}{\int \mathrm{dens}\,\rho_m\, dU} \le e^{Vq_m}. \tag{14.2}$$

Proof. Put $\widehat{g}_m = \widehat{\beta}_m^{-1/2}$. The scalar-free unmarked exponent is at most VE_m, and its absolute forced sum is at most $e^{V\epsilon_m^{\mathrm{for}}}$ with $\epsilon_m^{\mathrm{for}} \le 1$; at every sufficiently late level the path bounds give $E_m \le 2 + 4A_{J_m}^{\mathrm{path}}$. Restrict the denominator to links of chord at most $\widehat{g}_m/4$. There are no flags there, the forced sum is one, and its scalar-free exponent is at least $-V(E_m + 6(c_{\mathrm{dir},J_m} + (A_{J_m}^{\mathrm{path}})^2))$. The elementary Haar-ball bound gives this set measure at least $(\widehat{g}_m/16)^{4V\nu_G^{\mathrm{mat}}}$. The common scalar normalization in numerator and denominator cancels, so the ratio is at most

$$\exp\Big\{V\Big[2E_m + \epsilon_m^{\mathrm{for}} + 6(c_{\mathrm{dir},J_m} + (A_{J_m}^{\mathrm{path}})^2) + 4\nu_G^{\mathrm{mat}}\log(16/\widehat{g}_m)\Big]\Big\} \le e^{Vq_m}. \qquad \square$$

The per-site cost $q_m = O(1 + \log\widehat{\beta}_m)$ comes from the Haar-ball loss $\widehat{g}_m^{-\nu_G^{\mathrm{mat}}}$ on every link. Lemma 14.25 improves it to a constant depending only on the stage: in an axial gauge the denominator pays the Haar-ball loss on $3V+1$ links and the numerator recovers it on $3V-3$ one-link integrals, so only a loss of order $\log\widehat{\beta}_k$ for the whole torus remains. The source comparison needs that improvement, because its gain per cell is only a power of M_t^{cut}. Enlarge Q_j at the finitely many initial levels, so that $q_m \ge 1$ and (14.2) hold at every level. The choices include

$$B_{i+1}^2 \le 2B_i^2, \qquad (\ell_i B_i^2)^4 q_{i+1} = o(s_{i+1}), \qquad \varpi_{J_i} = o(s_i). \tag{14.3}$$

Define $k(m)$ to be the first $k \ge m$ with $B_k^2 \le a_m/a_k$.

The short-distance comparison uses the following comparison windows of levels. For $b > 0$ and a level t in stage j, write $L = \ell_j^{\mathrm{st}}$ and

$$h_t(b) = \left\lceil \frac{(b+2)\log M_t^{\mathrm{cut}}}{\log L} \right\rceil, \quad v_t = \left\lceil \frac{10\log B_t}{\log L} \right\rceil, \quad k_t = t - h_t(b) - v_t. \tag{14.4}$$

The numerical requirements of the marked proofs serve three purposes: contraction of one-insertion remainders (items 1–3), control of replicated sources and comparison with the free field (items 4–5), and the initialization of insertions (item 6). At each stage the finite-dimensional and support constants are fixed first, then the block ratio, and finally the coupling entry. The constants

named here are introduced with their estimates; we list them so that the order of choice is visible in one place.

For each stage j, let $K_j \geq 1$ be a nondecreasing bound for the leading coefficient maps and their inverses through that stage's finitely many degrees and symmetry averages. These bounds are fixed before the column bounds C_j^{col}. Let A_d^{sym} bound the finite basis recombination in degree d, and let C_j^{loc} bound the local marked error of the one-step estimate before the ratio is chosen (proof of Lemma 14.6). Each step below has $j = J_{m+1}$ and $L = \ell_m$; in items 2–3 and in the proofs of Lemmas 14.6, 14.17 and 14.29 we write $K = K_j$, $S = C_j^{\mathrm{col}}$ and $\varepsilon_L = L^{-3/8}$. For the retained incoming regular term, and for the unsubtracted incoming contribution in Section 19, let $R_j^{\mathrm{supp}} \in \mathbb{N}$ bound the enlargement of its projected incoming read support in units of B_m, over the finitely many adjacent stage pairs with fine stage j. Put

$$\mathcal{A}_m = \exp\Big\{(\sigma_{J_m} + 2)(2R_{J_{m+1}}^{\mathrm{supp}} + 13)^4\Big\}. \tag{14.5}$$

This factor converts an incoming support weight into the outgoing weight. Indeed, for fine and coarse covering numbers s_f, s_c, the connected-cover bound gives

$$\begin{aligned} Ls_c &\leq (L + 10s_f)(2R_j^{\mathrm{supp}} + 13)^4, \\ (\sigma_{J_m} + 2)s_c &\leq \log \mathcal{A}_m + \tfrac{1}{2}\sigma_{J_{m+1}} s_f, \end{aligned}$$

provided $20(\sigma_{J_m} + 2)(2R_j^{\mathrm{supp}} + 13)^4 \leq L\sigma_{J_{m+1}}$. Thus half the incoming exponent remains for the other support factors. The constant $\mathcal{C}_j \geq 2$ includes these bounds and the finite source comparison constants. In item 6, C_d is the section bound in (10.6). For the source estimates, choose an integer q_d^{ht} bounding the powers of the inverse health threshold in degree d, and put

$$\zeta_{d,i} = \widehat{\beta}_i^{-(q_d^{\mathrm{ht}}+4)}.$$

This allowance controls attachments of other marked factors to a distinguished mark. Let $A_{j,d}$ bound the finite section, derivative and support-conversion constants in these attachments at stage j, and let $\eta_i^{\mathrm{att}} > 0$ be the attachment threshold in Lemma 13.5 for the step at level i. Decrease it so that $\eta_i^{\mathrm{att}} \leq 1/4$. All these constants are fixed before the coupling entry. Lemma 14.13 uses

$$2A_{j,d}(2B_i + 1)^4(1 + B_i)^4 \widehat{\beta}_i^{q_d^{\mathrm{ht}}} \zeta_{d,i} \leq \eta_i^{\mathrm{att}}/2, \qquad j = J_{i+1}, \quad d \leq J_i. \tag{14.6}$$

The factor $(1 + B_i)^4$ bounds the spatial sum from marked anchors to a site; it is derived in the proof of that lemma. For transport to finer levels we also require

$$\widehat{\beta}_{i+1}^{-1} \ell_i^d C_{J_{i+1}}^{\mathrm{col}} \zeta_{d,i} \leq \zeta_{d,i+1}. \tag{14.7}$$

1. *Coefficient and support bounds.* Choose nondecreasing columns with $C_j^{\mathrm{col}} \geq 64K_{j+1}^2$, above the derivative factorials and finite basis-recombination constants at stage j. The support exponent is $\sigma_j = 100$; we also retain the estimates at every exponent in $[98, 100]$.

2. *One-insertion contraction* (Lemma 14.6). Choose $L^{3/5} > 2C_j^{\mathrm{loc}}$, $\mathcal{A}_m \leq L$ and $20(\sigma_{J_m} + 2)(2R_j^{\mathrm{supp}} + 13)^4 \leq L\sigma_{J_{m+1}}$. The local marked error ϵ_m^{mk} satisfies $\epsilon_m^{\mathrm{mk}} \leq L^{-2/5}$, and

$$\epsilon_m^{\mathrm{mk}}(1 + S)A_d^{\mathrm{sym}} \leq \varepsilon_L, \quad 2K\varepsilon_L(1 + S) \leq 1, \quad 4S\varepsilon_L \leq 1, \quad 2K \leq S, \quad 4K^2 \leq C_{J_m}^{\mathrm{col}}. \tag{14.8}$$

3. *Backward matching* (Lemmas 14.17 and 14.29). Require

$$4K\,32(1+S)^2\varepsilon_L \le 1, \qquad (4K^2+1)L^{1/8}\varepsilon_L \le 1. \tag{14.9}$$

For each auxiliary section through the stage's degrees, require also $2(1+C_j^{\mathrm{col}})H_{j,d}A_d^{\mathrm{sym}}L^{-1/8} \le 1/4$, where $H_{j,d}$ bounds its normalized healthy derivatives.

4. *Accumulated transport losses.* Choose ratios to satisfy (14.11), including every constant in $\mathcal{C}_j$.

5. *Gaussian comparison and source bounds* (Lemmas 14.23–14.25). On a comparison window, for an arity bound $n \ge 2$, put $\mathsf{w}_t = (M_t^{\mathrm{cut}})^{1/(8n)}$. Let $\mathcal{E}_t^{\mathrm{G}}$ be the sum of the covariance, rounding, Taylor and localization errors in the fixed finite source comparison. Require

$$\mathsf{w}_t^n \mathcal{C}_j^{h_t(b)+v_t+2}\mathcal{E}_t^{\mathrm{G}} \le 1, \qquad Q_j^{\mathrm{cmp}} \le \log M_t^{\mathrm{cut}}/(16n), \qquad \widehat{\beta}_i \in [\widehat{\beta}_t/2, \widehat{\beta}_t] \quad (k_t \le i \le t). \tag{14.10}$$

Here Q_j^{cmp} is the stage normalization constant in (14.48), enlarged to cover the source modes and amplitudes of the finitely many arity bounds $n \ge 2$ used at stage j, and the second bound is required for each of these n. These constants are fixed first; the entry is then raised until $\log M_t^{\mathrm{cut}} \ge 16n\,Q_j^{\mathrm{cmp}}$. The first bound is imposed with the stronger smallness used in (14.46). The entry also meets the local marked bound in item 2, the forced-source bounds at every exponent between 98 and 100, and (14.6)–(14.7). The unmarked attachment sum in Lemma 13.5 is at most $\eta_i^{\mathrm{att}}/2$, and $64\log\widehat{\beta}_{i+1}/B_i \le 1/4$.

6. *Initialization of insertions* (Lemma 14.7). At every level m and $d \le J_m$, require B_m at least the radius of Σ_d and $C_d t_m^{-d} e^{\sigma_{J_m}}(e^{|\Sigma_d|\epsilon_m^{\mathrm{for}}} - 1) \le (\epsilon_m^{\mathrm{for}})^{1/2}$.

7. *Ratio below the logarithmic coupling* (Lemma 18.3). At every level m, $\ell_m \le \log\widehat{\beta}_m$.

The finite comparisons in item 5 exhaust bounded configurations with positive point separation, fixed arities and finite derivative orders as j increases. Their errors and constants are calculated in Lemma 14.24; none depends on the period.

The following choice gives both the sparse insertion grids and the windows used for free comparison.

Lemma 14.2 (Scale and period choices)**.** *All requirements 1–7 and* (14.3) *can be met with*

$$\ell_m \ge 3 \text{ odd}, \qquad \log\ell_m \to \infty, \qquad \log\ell_m/s_m \to 0, \qquad 4 \mid L_p,$$
$$(a_m/a_{k(m)})^4 q_{k(m)} = o(s_{k(m)}).$$

The sides L_p increase strictly; every positive integer divides infinitely many. For every finite set of exponents $b \ge 1$ and finite comparison requests, arbitrarily late terminal levels t meet the comparison requirements simultaneously on their windows (14.4)*, with $k_t \to \infty$ and $J_i = J_t$ throughout each window.*

Proof. The choices follow the order of Table 3: stage constants, then ratios, then entries. First fix the entire sequence K_j and the finite section bounds. Choose $C_j^{\mathrm{col}} \ge 64K_{j+1}^2$, and then fix $\mathcal{C}_j$ to include these columns and all source constants. Since $J_{m+1} \le J_m + 1$, this gives $4K_{J_{m+1}}^2 \le C_{J_m}^{\mathrm{col}}$ even at a stage transition. Next choose nondecreasing odd stage ratios $\ell_j^{\mathrm{st}} \ge \max(3, j)$ large enough that

$$\frac{\omega_j^{\mathrm{RG}}}{\log\ell_j^{\mathrm{st}}} \le \min(\kappa_G/4, j^{-2}), \qquad \frac{\log\mathcal{C}_j}{\log\ell_j^{\mathrm{st}}} \le j^{-2}. \tag{14.11}$$

Each denominator can be increased after its numerator is fixed. The ratio requirements in items 2 and 3 are lower bounds on this same later choice: $L^{-2/5}/L^{-3/8} = L^{-1/40} \to 0$ and $L^{1/8}L^{-3/8} = L^{-1/4} \to 0$. Choose also $L^{3/5} > 2C_j^{\mathrm{loc}}$, so $C_j^{\mathrm{loc}}/L < \frac{1}{2}L^{-2/5}$. For this fixed L, raise the coupling entry until the absolute value of the $o_{\widehat{\beta}_m}(1)$ term in the one-step estimate is at most $\frac{1}{2}L^{-2/5}$. The local error $C_j^{\mathrm{loc}}/L + o_{\widehat{\beta}_m}(1)$ is then at most $L^{-2/5}$. The column and ratio choices also give the fixed auxiliary-section inequality. The support radii and exponents in (14.5) are fixed before L, so its compression condition and $\mathcal{A}_m \le L$ are further fixed lower bounds on the ratio. Thus no column is chosen in response to a ratio or to a later-stage column.

Choose consecutive constant-stage intervals, postponing each new stage until all its fixed coupling requirements hold. On a fixed stage s_i grows linearly, while $B_i = O((\log(2+s_i))^2)$ and $q_i = O(1+\log(2+s_i))$. Thus

$$(\ell_i B_i^2)^4 q_{i+1}/s_{i+1} \longrightarrow 0 \quad \text{on that stage as its entry is postponed.}$$

For the initial stage, choose $\bar{\beta}$ after its fixed ratio L_0. For $B(b) = \lceil (\log b)^2 \rceil$, the quotient $B(b + \kappa_G \log L_0)^2/B(b)^2$ tends to one as $b \to \infty$. Choose $\bar{\beta}$ beyond a threshold where it is at most two for every $b \ge \bar{\beta}$. This covers every step of the first stage, including $i = 0$. At a transition to a later ratio, impose the same bound at its entry and thereafter, with that ratio fixed. Impose the second condition of (14.3) with bound $1/j$ after entry to stage j. Impose also $\varpi_j \le s_i/j$ and $\log \ell_j^{\mathrm{st}} \le s_i/j$ at its entry. These are lower bounds on that entry. By item 9 of Table 3, $t_m^{-1} \le \widehat{\beta}_m$ and $(\epsilon_m^{\mathrm{for}})^{1/2}\widehat{\beta}_m^{d+1} \to 0$, so item 6, including the finite stencil-radius requirement, is another entry bound. Since $\log \widehat{\beta}_m \to \infty$ at a fixed ratio, item 7 is also an entry bound, imposed for both ratios that occur at the levels of a stage. At each fixed ratio, the same threshold formula gives $t_m/t_{m+1} \to 1$ as $\widehat{\beta}_m \to \infty$; impose $\frac{1}{2} \le t_m/t_{m+1} \le 2$ at entry and thereafter, with $t_m \le 1$ supplied by item 7 of Table 3. The preceding entry restrictions give both little-oh assertions and $\log \ell_i/s_i \to 0$; $s_i \ge i \log 3$ at every level.

For each stage, the length $h_t(b) + v_t$ in (14.4) is $O_j((\log\log(2+s_t))^2 + \log\log(2+s_t)) = o(t)$. Lengthen that constant-stage interval to contain a terminal point with the whole window (14.4) in the same stage and with its left endpoint beyond all earlier requested levels. Do this with the longest window $b = j$, all arities $n \le j$ and the finitely many comparison orders at stage j, on the compact sets of test configurations with point separations at least $1/j$ and coordinates of absolute value at most j. Every $1 \le b \le j$ uses a subinterval of this window. The length and error bounds are uniform for b in any fixed compact subinterval of $[1, j]$; take the maximum of the finite comparison constants before postponing the common terminal level. Their finitely many derivative bounds are uniform on each such compact set. The estimates (14.44)–(14.46) show that the first requirement in (14.10) holds once that entry is late enough. The second is a lower bound on $\log M_t^{\mathrm{cut}}$ after Q_j^{cmp} is fixed. The total window length is $o(t)$, so its coupling variation is $o(\widehat{\beta}_t)$ and gives the third requirement. These are lower bounds on a later entry or interval length. Each fixed request is then met at arbitrarily large stages. The source requirements are also entry bounds. On a fixed stage, the left side of (14.6) is a fixed constant times $(2B_i+1)^4(1+B_i)^4\widehat{\beta}_i^{-4}$ and tends to zero. In (14.7), division by $\zeta_{d,i+1}$ leaves

$$\ell_i^d C_{J_{i+1}}^{\mathrm{col}} \widehat{\beta}_{i+1}^{-1} (\widehat{\beta}_{i+1}/\widehat{\beta}_i)^{q_d^{\mathrm{ht}}+4},$$

which tends to zero at fixed ratio. Include both adjacent stages at a transition. There are finitely many degrees at each entry. The unmarked attachment sum tends to zero by the factor rates in the proof of Lemma 13.5, so its half-threshold requirement is another entry bound. These choices do not alter earlier intervals. Taking $\ell_i = \ell_{J_{i+1}}^{\mathrm{st}}$ gives odd ratios, and $\ell_j^{\mathrm{st}} \ge j$ gives $\log \ell_i \to \infty$.

The level $k(m)$ exists: $a_i B_i^2$ decreases by at least the factor $2/3$ at each step. If $k(m) = u+1 > m$, minimality gives

$$a_m/a_{k(m)} \le \ell_u B_u^2. \tag{14.12}$$

Consequently (14.3) gives $(a_m/a_{k(m)})^4 q_{k(m)} = o(s_{k(m)})$. The case $k(m) = m$ satisfies the same conclusion directly from $q_m = o(s_m)$.

Choose $L_p = 4(M_{\mathrm{per}} + p + 1)$. For $j = J_{m+1}$, the required period inequalities, including item 7 of Table 3, are

$$L_p e^{s_m} \ge 100(B_m + 1), \qquad L_p e^{s_{m+1}} \ge 100((j+3)\ell_m + 1),$$
$$L_p e^{s_m} \ge v_j^{\mathrm{per}} (1 + B_m + B_{m+1})^{h_j^{\mathrm{per}}}.$$

The first follows by choosing M_{per} large, since $(B_m + 1)e^{-s_m}$ is bounded. After canceling ℓ_m in the second, postpone each stage until $e^{s_m} \ge 100(j+4)$. For the third, v_j^{per} and h_j^{per} are fixed before entry to stage j, and $v_j^{\mathrm{per}}(1 + B_m + B_{m+1})^{h_j^{\mathrm{per}}} e^{-s_m} \to 0$ on that stage. Postpone each later stage until this expression is at most one thereafter, and enlarge M_{per} for the finite initial interval. This proves all three inequalities. The third supplies the period bound for the five-torus localization in the proof of Lemma 13.7. The periods are strictly increasing, divisible by four, and every integer divides infinitely many of them. □

The next identities retain the normalization of the original Wilson integral through every integration.

Lemma 14.3 (Exact trajectory identities)**.** *For the trajectories constructed below with endpoint $\bar\beta$ and every $m < N$,*

$$\rho_{N,p,N} = \rho^{\mathrm{W}}_{\beta_{N,p}}, \qquad |\beta_{N,p,m+1} - \beta_{N,p,m} - \kappa_G \log \ell_m| \le \omega^{\mathrm{RG}}_{J_{m+1}},$$
$$\int \mathrm{dens}\, \rho_{N,p,m} = Z_{L_p D_N, \beta_{N,p}}.$$

All representations are admissible, and the terminal integral has the same value.

The following proof also proves Theorem 5.6.

Proof of Lemma 14.3 and Theorem 5.6. The trajectory is obtained by bracketing, level by level, along the continuous Wilson curve of Lemma 13.16, carried down by the continuous step of the specified recursion in Lemma 13.17; the drift margin comes from (14.11). Set

$$b_m^- = \bar\beta + \tfrac{1}{2}\kappa_G s_m, \qquad b_m^+ = \bar\beta + 2\kappa_G s_m.$$

These intervals lie in the coupling bands $\mathfrak{B}_m$. Suppose the fine endpoint couplings are b_{m+1}^- and b_{m+1}^+. The drift estimate and (14.11) give coarse endpoint bounds

$$\beta_m^- \le b_m^- - \tfrac{1}{4}\kappa_G \log \ell_m < b_m^-, \qquad \beta_m^+ \ge b_m^+ + \tfrac{3}{4}\kappa_G \log \ell_m > b_m^+.$$

For a continuous function whose endpoints bracket $[b_m^-, b_m^+]$, restrict between its last lower hit before its first upper hit. Its values then remain in this interval. Restrict the parameter interval accordingly. Every preceding representation is restricted along with the new one, so all already obtained identities persist.

Start at level N with the Wilson curve restricted to $[b_N^-, b_N^+]$ and repeat this operation down to zero. Since $b_0^- = b_0^+ = \bar\beta$, it produces a trajectory with the required endpoint. The case $N = 0$

is the Wilson representation at $\bar{\beta}$. All retained representations and adjacent pairs belong to the chosen classes and relation because Theorem 5.5 returns admissible representations whose adjacent pairs lie in the chosen relation.

The exact density identity at each step is the one in Theorem 5.5, now applied only on the retained interval. By Lemma 11.4, integration of the block kernel of Definition 5.1 in its coarse argument is one. Fubini therefore preserves the integral, and the Wilson start gives its displayed value at every level. The path arrays were fixed before p in Lemma 12.3; restricting the parameter interval does not change them.

Summing the drift from zero gives

$$|\beta_{N,p,m} - \bar{\beta} - \kappa_G s_m| \le \sum_{i<m} \omega^{\mathrm{RG}}_{J_{i+1}}. \tag{14.13}$$

For a fixed splitting level k, the tail is at most $\sup_{i\ge k} J_{i+1}^{-2} s_m$ by (14.11). The initial sum is fixed. Divide by s_m, let $m \to \infty$ and then $k \to \infty$. Since $\bar{\beta}/s_m \to 0$, this proves the uniform running law in Theorem 5.6; neither bound depends on $N \ge m$ or p. □

The uniqueness estimate concerns the specified recursion, rather than arbitrary representations of a density. Lemma 13.23 supplies its weighted analytic coordinates and their real power-series extension. Section 14.3 uses the fixed-level convergence of Lemma 13.24.

Lemma 14.4 (Analytic endpoint tuning)**.** *There is $w_{\mathrm{tun}} > 0$, depending only on the construction, such that, for every N, p and $s \in [-1, 1]$, exactly one bare coupling $\beta_{N,p}(s)$ starts a trajectory of the specified recursion ending at $\bar{\beta} + w_{\mathrm{tun}} s$. Moreover,*

$$\beta_{N,p}(0) = \beta_{N,p}, \qquad \beta_{N,p}(s) = b_{N,p}(s) \quad (-1 < s < 1),$$

where $b_{N,p}$ is holomorphic on $|z| < 1$.

Proof. Uniqueness and the analytic extension both come from the weighted contraction of Lemma 13.23, applied to histories of the specified recursion (Definitions 5.2 and 13.3). Let $\mathfrak{d}$ be the history distance of two histories in the domain of that lemma, including the coupling coordinate, and δ their coarse endpoint difference. With the constants $b_{\min}, \lambda_* > 0$ fixed before that lemma, independent of N, p, it gives

$$\mathfrak{d} \le |\delta|/b_{\min} + 8 g_* \mathfrak{d}, \qquad g_* = 1/128. \tag{14.14}$$

For equal endpoints, $\mathfrak{d} \le \mathfrak{d}/16$ implies equality of the histories and hence of their bare couplings.

Take $w_{\mathrm{tun}} = \lambda_* b_{\min}/4$, where $\lambda_* = 1/16$ is fixed before Lemma 13.23. The local action is affine in the varying coupling; the masks and sector domains are fixed in its analytic coordinates. Its power-series map preserves the radius-λ_* ball and contracts by $1/2$ for endpoint series $w_{\mathrm{tun}} z$. Successive substitution converges in the norm $\sum_j \|u_j\|$ of absolutely summable series. The map preserves real coefficients, so evaluation gives a real trajectory in that domain for every $s \in [-1, 1]$, pointwise in s. This supplies existence on the whole stated interval, including its endpoints. The selected coefficient at $s = 0$ is the trajectory of Lemma 14.3, by uniqueness. The bare-coupling series is holomorphic in $|z| < 1$ and has the claimed real values. All clipping and health conditions are covered by the analytic neighbourhood in Lemma 13.23; they are not differentiated across their boundaries. □

14.2 Marked fields and volume-uniform estimates

Proposition 5.8 needs bounds on differentiated insertions, uniform in the cutoff and the volume. The argument has four parts. (i) One marked insertion is carried to coarser levels as a polynomial coordinate plus remainders, with a two-sided bound on the coordinate that loses no factor ℓ_m^d in the inverse direction (Section 14.2.1). (ii) Placing one insertion in every reflected cell of a tiling of side a_m and integrating down to a level k with $(a_m/a_k)^4$ sites per cell gives the chessboard bound $Aa_m^{-d-\varepsilon}$ (Section 14.2.2). (iii) A lattice difference of a degree-d insertion is a degree-$(d+1)$ insertion up to a negligible error; its coefficient splits into an equation part, which vanishes in separated correlations, and a part in a space of dimension at most $\dim_{\mathbb{C}} M_{d+1}^{\mathrm{Max}}$ (Sections 14.2.3–14.2.4). (iv) That part is bounded through a nondegenerate matrix of free Maxwell tests (Section 14.2.5). Lemma 14.28 combines (ii)–(iv) and proves Proposition 5.8 (Section 14.2.6). The bounded approximations of Section 14.2.7 serve the phase argument of Section 17.

14.2.1 One marked insertion

Fix a degree d and suppress N, p, d while describing one transfer. We use the marked densities of (5.15), the local section $P_d = P_d^{\mathrm{loc}}$ of Lemma 10.2, and the factor-count scaling t^{N} of Definition 2.8. Write ρ_m for the unmarked representation, $v_{\mathrm{sc},m}$ for its scalar normalization and $\Lambda_{n_{p,m}}$ for its site torus. An anchor is the lattice site at which the insertion is placed. Denote the marked regular supports by Z_i, to distinguish them from the unmarked supports X_i. For a bare polynomial coordinate $\mathbf{v} \in V_d$,

$$H_N(\mathbf{v}) = \operatorname{dens} \rho_N\, P_d[t_N^{-\mathsf{N}}\mathbf{v}], \qquad H_m(\mathbf{v}) = \ell_m^d \mathcal{B}^{(m)}_{\beta_{N,p,m+1}} H_{m+1}(\mathbf{v}), \tag{14.15}$$

so that $H_m(\mathbf{v}) = H^{\mathrm{ins}}_{N,p,m;d}(\mathbf{v})$. These densities keep track of one insertion under exact integration. The equality is exact; the estimates below control the coordinates in which it is written.

Definition 14.5 (Marked density)**.** A representation of $H_m(\mathbf{v})$ at level m consists of a coordinate $\mathbf{x} = x_0\mathbf{1} + \mathbf{x}^\circ \in V_d$, complex regular remainders R_i on supports Z_i, and forced remainders Q_i on forced addresses of ρ_m, such that the exact decomposition (5.18) holds with $\mathbf{x}_m = \mathbf{x}$, $R^{\mathrm{reg}}_{m,i} = R_i$ and $Q^{\mathrm{for}}_{m,i} = Q_i$. The section and the stencil Σ_d are translated to the anchor.

The remainders restore the exact density; eligibility is that of Definition 5.2. The marked regular norm below is anchored at the insertion site, taken to be zero; it is not anchored separately at each support. For selected majorants $e_i \ge \|R_i\|_m$, $k_i \ge \sup|Q_i|$, set

$$\begin{aligned} \mathcal{E}_m^{\mathrm{mk}} &= \sum_i e^{\sigma_{J_m} \operatorname{cov}_{B_m}(Z_i)} e_i, \qquad \mathcal{K}_m^{\mathrm{mk}} = \sum_i e^{\sigma_{J_m} \operatorname{cov}_{B_m}(Y_i)} k_i, \\ \|g_m\|_{\mathrm{rem}} &= \mathcal{E}_m^{\mathrm{mk}} + (\epsilon_m^{\mathrm{for}})^{-1/2} \mathcal{K}_m^{\mathrm{mk}}. \end{aligned} \tag{14.16}$$

The derivative seminorm is Definition 5.3. A coefficient graph at level m is a linear choice $\mathbf{x} \mapsto g_m(\mathbf{x}) = (R(\mathbf{x}), Q(\mathbf{x}))$ of such remainders, with the supports fixed and $\|g_m(\mathbf{x})\|_{\mathrm{rem}} \le C^{\mathrm{col}}_{J_m} |\mathbf{x}^\circ|_1$. As in the graph transform for invariant manifolds, the marked step maps coefficient graphs at level $m+1$ to graphs at level m and halves the distance between two of them (Lemma 14.6); the current polynomial coordinate thus bounds the whole inserted density.

The following step retains the complete inserted density while reexpressing its remainders in the new polynomial coordinate. The inverse estimate is essential: a bound in the original bare coordinate would accumulate at every step.

To leading order a step multiplies a homogeneous coordinate of engineering weight w by ℓ_m^{d-w}: the factor ℓ_m^d comes from the normalization in (14.15) and ℓ_m^{-w} from blocking. These factors form the diagonal map

$$(\mathsf{S}_L\mathbf{x})_i = L^{d-w_i}x_i, \qquad 0 \le w_i \le d,$$

where (w_i, k_i) are the engineering weight and factor count of the basis polynomial $b_{d,i}$; the unit has $w_0 = 0$. What remains is a change of basis T_m, bounded with its inverse uniformly in N, p and m within a stage, obtained by reexpressing the fine section, evaluated on the free interpolation of a coarse jet, in the coarse basis. Precisely, for a coarse radial jet j, let $U^{W_0}(\mathsf{j})$ be the fine stencil configuration obtained by exponentiating the free interpolation W_0 of Lemma 11.6 of the coarse potential of j, the degree-d Taylor polynomial, in the jet coordinates, of the coarse-link logarithms. Define $\mathsf{R}_m^{\mathrm{lead}}$ by

$$\frac{d^q}{dh_{\mathrm{lat}}^q}(\mathsf{R}_m^{\mathrm{lead}}b)(\mathrm{sc}_{h_{\mathrm{lat}}}\mathsf{j})\Big|_{h_{\mathrm{lat}}=0} = \frac{d^q}{dh_{\mathrm{lat}}^q}P_d[b]\big(0; U^{W_0}(\mathrm{sc}_{h_{\mathrm{lat}}}\mathsf{j})\big)\Big|_{h_{\mathrm{lat}}=0} \qquad (0 \le q \le d),$$

put $(\mathsf{D}_L^{\mathrm{eng}}\mathbf{x})_i = L^{w_i}x_i$, and set

$$T_m = t_m^{\mathsf{N}}\mathsf{R}_m^{\mathrm{lead}}\mathsf{D}_{\ell_m}^{\mathrm{eng}}t_{m+1}^{-\mathsf{N}}.$$

Taylor matching gives $(\mathsf{R}_m^{\mathrm{lead}})_{bi} = \delta_{bi}\ell_m^{-w_i}$ for $w_b \le w_i$ and $= 0$ for $k_b < k_i$. To bound the other entries, use the kernel form of Lemma 11.6: at averaging order $j \ge d+3$,

$$|\nabla_f^r W_0(\ell_m y + a, y')| \le C_{j,r}\ell_m^{-1-r}e^{-c_j|y-y'|}, \qquad 0 \le r \le d.$$

The choice of the reference level ensures this order. A jet potential grows polynomially, so the weighted bound (11.7) with $c' = 0$ and $h = d$ applies to it. Thus its kernel sum and the required differences converge absolutely. Applying the displayed bound to each factor of a homogeneous jet monomial gives $|(\mathsf{R}_m^{\mathrm{lead}})_{bi}| \le C\ell_m^{-w_i}$, with C depending only on d and the stage. Periodization gives the same bound on a torus by summing over its translates. The threshold formula in item 9 of Table 3 and the entry choices of Lemma 14.2 give $0 < t_m \le 1$ and $\frac{1}{2} \le t_m/t_{m+1} \le 2$. Hence T_m sends each basis vector to its own and higher weights, with nonzero diagonal $(T_m)_{ii} = (t_m/t_{m+1})^{k_i}$, and back substitution in increasing weight bounds $T_m^{\pm 1}$ uniformly in N, p and m within a stage.

The marked step constructs, from a coefficient graph at level $m+1$, the coarse coordinate and a coefficient graph at level m for which the marked identity holds again. Its statement is in the coordinate $\mathbf{u} = \mathsf{M}_m\mathbf{x}$ extracted by the step.

Lemma 14.6 (One marked step)**.** *For $m \ge m_{\mathrm{ref}}(d)$ and each coefficient graph g_{m+1} at level $m+1$, there are an invertible linear M_m on V_d and a coefficient graph g_m at level m such that every representation of $H_{m+1}(\mathbf{v})$ with coordinate $\mathbf{x}$ and remainders $g_{m+1}(\mathbf{x})$ yields one of $H_m(\mathbf{v})$ with coordinate $\mathbf{u} = \mathsf{M}_m\mathbf{x}$ and remainders $g_m(\mathbf{u})$. Moreover:*

(i) $|\mathbf{u}|_1 \le C_{J_{m+1}}^{\mathrm{col}}\ell_m^d|\mathbf{x}|_1$ *and* $|\mathbf{x}|_1 \le C_{J_{m+1}}^{\mathrm{col}}|\mathbf{u}|_1$*;*

(ii) M_m *and* g_m *respect conjugation and signed permutations, and* $\mathsf{M}_m\mathbf{1} = \ell_m^d\mathbf{1}$ *with zero remainders;*

(iii) fine graphs differing by at most $\tau C_{J_{m+1}}^{\mathrm{col}}|\mathbf{x}|_1$ *give coarse graphs differing by at most* $\frac{\tau}{2}C_{J_m}^{\mathrm{col}}|\mathbf{u}|_1$*.*

The proof gives the construction and the bounds, including the contraction between two choices of incoming remainders.

Proof. The new coordinate is read off by Taylor extraction after one exact block integration (Definition 5.1) in the local sections of Definition 2.7. The connected expansion of Lemma 13.5, in a nilpotent algebra, controls the marked terms; the pre-reset estimates of Lemma 13.9, the subtraction of Lemma 13.10 and the density identity of Lemma 13.13 provide the unmarked part of the step on which the marked terms ride. Put $L = \ell_m$, $S = C^{\mathrm{col}}_{J_{m+1}}$, $K = K_{J_{m+1}}$ and $\kappa_m = (\epsilon^{\mathrm{for}}_m)^{1/2}$. All norms in this proof are coefficient sums or the anchored norm (14.16).

Step 1: Marked terms in the local integration. Introduce $z^2 = 0$. Replace the regular exponential factor by its product with $1 + z(x_0 + \mathbf{1}_{\Sigma_d \text{ healthy}} P_d[t_{m+1}^{-\mathsf{N}} \mathbf{x}^\circ] + \sum_i \mathbf{1}_{Z_i \text{ healthy}} R_i)$ and replace a forced factor K_i by $K_i + zQ_i$. The coefficient of z in the eligible covering sum is exactly the representation of Definition 14.5. Apply the coefficientwise rooted clause of Lemma 13.5 in this nilpotent algebra. Each connected term now contains one distinguished factor. For a term with ordinary factors $a_1, \dots, a_h$, its marked coefficient is

$$[z] \prod_{i=1}^{h} (a_i + z b_i) = \sum_{i=1}^{h} b_i \prod_{j \neq i} a_j.$$

Root its connected support at this distinguished factor. The sums over all other factors are bounded by the rooted clause of Lemma 13.5; there is no sum over possible translates of the marked anchor. Absolute convergence justifies the displayed extraction and integration term by term. A regular mark has its derivative majorant e_i; a forced mark has its amplitude majorant k_i. The latter retains its original eligibility event, so the large-field estimate used in the forced expansion still applies.

The three contributions to a basis input are bounded as follows. The map T_m fixes the unit and preserves the nonconstant subspace. It is triangular in the homogeneous weights: $(T_m)_{ij} \neq 0$ and $i \neq j$ imply $w_j < w_i$. Its diagonal entries are nonzero, so inversion proceeds in increasing weight, with the finite stage bounds included in K. Let $C_{\mathrm{loc}} \geq 1$ bound its two operator norms and the following local remainder estimates before symmetry averaging; the leading stage bound was chosen with $K \geq 2A_d^{\mathrm{sym}} C_{\mathrm{loc}}$. Write $\epsilon = \epsilon^{\mathrm{mk}}_m$ for the local marked error of item 2. Taylor subtraction of a primary mark leaves coefficient error at most $\epsilon |\mathbf{x}^\circ|_1$ and remainder at most $C_{\mathrm{loc}} |\mathsf{S}_L \mathbf{x}^\circ|_1$. A regular remainder of mass $e^{\sigma \operatorname{cov}(Z)} e_i$ contributes at most $\epsilon e^{\sigma \operatorname{cov}(Z)} e_i$ to each of these two bounds. A forced remainder contributes at most $\epsilon e^{\sigma \operatorname{cov}(Y)} k_i / \kappa_{m+1}$.

For the primary mark, the local polynomial is subtracted through degree d before rescaling. Each remaining term has at least one additional scaled jet weight, and the block rescaling gives the inverse block factor in its norm. The finite connected sums and the ordered derivative product rule introduce only stage constants.

An incoming regular mark is never moved to the forced family. A forced remainder must vanish off an eligibility event with a nonempty inventory of flagged sites (Definition 13.2), whereas the support of a regular mark may wrap around the torus while every plaquette in it is healthy. The integrated contribution of a regular mark consists of regular coarse remainders and, from those terms of its connected expansion that carry a flagged inventory at level m, of forced coarse remainders on their own eligibility events. Wrapping plays no role in this assignment; it only decides whether the radial cancellation can be used. Fix such a mark R_i on Z_i, with derivative majorant e_i, let $\widehat{Z}_i$ be the coarse support of its contribution, and put

$$d_i = \operatorname{cov}_{B_{m+1}}(Z_i), \qquad \widehat{Z}_i \subset N_{R_j^{\mathrm{supp}} B_m}(\mathrm{parent}(Z_i)).$$

Call Z_i short if $d_i^3 < \ell_m$. Three classes of supports are estimated separately.

(i) If $\operatorname{cov}_{B_m}(\widehat{Z}_i) > B_m$, the contribution is bounded at a coarse support exponent at least $\sigma_{J_m}+3$, one more than the exponent $\sigma_{J_m}+2$ in (14.5). Half of this unit excess absorbs the derivative losses, which are polynomial in $\operatorname{cov}_{B_m}(\widehat{Z}_i)$. The other half gives a factor $e^{-B_m/2}$, since $\operatorname{cov}_{B_m}(\widehat{Z}_i) > B_m$. As for the large supports in the proof of Lemma 13.6, the resulting factor tends to zero as the coupling entry is raised at fixed stage and ratio; it is part of the term $o_{\widehat{\beta}_m}(1)$ below.

(ii) If $\operatorname{cov}_{B_m}(\widehat{Z}_i) \le B_m$ and Z_i is short, then Z_i is nonwrapping about the anchor. Indeed, as in the proof of Lemma 13.6, $3d_i \le \ell_m$ and $B_{m+1} \le 2B_m$ bound the fine radius of Z_i by $6(1+B_m)\ell_m$, while the coarse side is at least $100(1+B_m)$. The radial-jet coefficients of R_i through degree d therefore vanish: this is the format of the incoming remainders, recorded in Lemma 14.9 at level $m+1$. Taylor's formula along the healthy radial segment gives the factor $\ell_m^{-(d+1)}$, with derivative losses polynomial in $1+d_i$.

(iii) If $\operatorname{cov}_{B_m}(\widehat{Z}_i) \le B_m$ and $d_i^3 \ge \ell_m$, no radial cancellation is used. By (ii), this class contains every wrapping support with $\operatorname{cov}_{B_m}(\widehat{Z}_i) \le B_m$; it also contains nonwrapping supports that are not short. The derivative bound without subtraction has losses polynomial in $1+d_i$, and

$$1 \le \ell_m^{-(d+1)}(1+d_i)^{3(d+1)}$$

gives the same factor $\ell_m^{-(d+1)}$ at the cost of a further polynomial in $1+d_i$.

In classes (ii) and (iii), the support conversion (14.5) gives $(\sigma_{J_m}+2)\operatorname{cov}_{B_m}(\widehat{Z}_i) \le \log \mathcal{A}_m + \frac{1}{2}\sigma_{J_{m+1}} d_i$. The remaining half of the incoming exponent bounds the polynomial in $1+d_i$, since $\sup_{s\ge 0}(1+s)^q e^{-\sigma_{J_{m+1}} s/2} < \infty$ for each fixed q. After multiplication by ℓ_m^d, the contribution of the mark is at most $C\ell_m^{-1} e^{\sigma_{J_{m+1}} d_i} e_i$, with C fixed before the ratio. Thus the smallness of a healthy wrapping mark comes from its unused support weight, not from a Taylor cancellation.

An incoming forced mark retains its own eligibility event, whether or not its support wraps. Its support sum keeps half the original forced suppression: its normalization is κ_{m+1}, not $\epsilon_{m+1}^{\mathrm{for}}$. The remaining suppression absorbs the same derivative and support factors. Thus the bounds before choosing parameters have the form $C_j^{\mathrm{loc}}/L + o_{\widehat{\beta}_m}(1)$, with C_j^{loc} fixed before L and the second term tending to zero at fixed stage and block factor; the ratio and entry choices of item 2, made in the proof of Lemma 14.2, make it at most $\epsilon_m^{\mathrm{mk}} \le L^{-2/5}$. The degree-$d$ Taylor extraction is on healthy charts; the indicators themselves are never differentiated.

Let $\mathbf{u}$ be the extracted coarse polynomial coordinate, r the incoming anchored remainder norm and r' the resulting coarse remainder norm, including its forced normalization. Summing the contributions gives

$$|\mathbf{u} - T_m \mathsf{S}_L \mathbf{x}|_1 \le \epsilon(|\mathbf{x}^\circ|_1 + r), \qquad r' \le C_{\mathrm{loc}}|\mathsf{S}_L \mathbf{x}^\circ|_1 + \epsilon r, \qquad r = \|g_{m+1}(\mathbf{x})\|_{\mathrm{rem}}. \tag{14.17}$$

The regular-remainder terms in (14.17) use the vanishing Taylor coefficients through degree d on short supports, and the unused support weight on the other supports. Retained incoming regular terms use the support conversion factor (14.5). The same sums with the primary mark canceled give the Lipschitz bound $\epsilon\|g-\widetilde{g}\|_{\mathrm{rem}}$ for the coefficient and remainder differences; this uses the entire forced marked family at once.

Step 2: Linear choice and inverse bound. Perform the construction on each homogeneous basis vector and on all its signed and conjugated images. Take their union of supports, with zero entries where necessary, and recombine using the basis coefficients. Average over the finite signed-permutation group and over conjugation. This gives a complex-linear family with the required symmetries and exact density identity. Let $A_d^{\mathrm{sym}} \ge 1$ bound the sum of absolute recombination coefficients in this finite averaging. After averaging, the error in (14.17) is at most $\epsilon(1+S)A_d^{\mathrm{sym}}|\mathbf{x}^\circ|_1$, and the remainder is at most $(C_{\mathrm{loc}} + \epsilon S)A_d^{\mathrm{sym}}|\mathsf{S}_L\mathbf{x}^\circ|_1$.

Use (14.8). The cross-stage inequality follows from item 1, fixed before the ratios. Since $\epsilon S \le 1 \le C_{\mathrm{loc}}$, the averaged remainder constant is at most K. In the notation $\mathbf{y} = \mathsf{S}_L\mathbf{x}$, the resulting inequalities are

$$|\mathbf{u} - T_m\mathbf{y}|_1 \le \varepsilon_L(|\mathbf{x}^\circ|_1 + r), \qquad r' \le K|\mathbf{y}^\circ|_1 + \varepsilon_L r, \qquad r \le S|\mathbf{x}^\circ|_1.$$

Since $|\mathbf{x}|_1 \le |\mathsf{S}_L\mathbf{x}|_1 \le L^d|\mathbf{x}|_1$, the first two conditions in (14.8) imply

$$|\mathbf{y}|_1 \le 2K|\mathbf{u}|_1, \qquad |\mathbf{u}|_1 \le 2K|\mathbf{y}|_1, \qquad |\mathbf{y}^\circ|_1 \le 2K|\mathbf{u}^\circ|_1.$$

For example, apply T_m^{-1} and use $2K|\mathbf{u} - T_m\mathbf{y}|_1 \le |\mathbf{y}^\circ|_1$ to absorb the error; the same argument on the nonconstant subspace gives the last bound. Moreover $\varepsilon_L r \le K|\mathbf{y}^\circ|_1$, so

$$r' \le 2K|\mathbf{y}^\circ|_1 \le 4K^2|\mathbf{u}^\circ|_1 \le C^{\mathrm{col}}_{J_m}|\mathbf{u}^\circ|_1.$$

The linear map $\mathbf{x} \mapsto \mathbf{u} = \mathsf{M}_m\mathbf{x}$ is injective by $|\mathbf{x}|_1 \le S|\mathbf{u}|_1$, hence invertible in finite dimension. This defines the new graph by $g_m(\mathbf{u}) = g_m(\mathsf{M}_m\mathbf{x})$; the last display is the graph bound required of a coefficient graph at level m. The two coefficient estimates of (i) now follow; property (ii) is inherited from the symmetric construction. In particular the inverse estimate has no factor L^d: S_L^{-1} has norm at most one.

For the unit, the primary extraction is exact, T_m fixes it and its nonconstant coordinate is zero. All the preceding remainder bounds are then zero, and the unit is sent to $L^d\mathbf{1}$ with zero remainders. Finally, when two fine graphs differ by at most $\tau S|\mathbf{x}|_1$ in the remainder norm, the Lipschitz estimate above and rebasing by the inverse bound of (i) give a coarse difference at most $\frac{1}{2}\tau C^{\mathrm{col}}_{J_m}|\mathbf{u}|_1$: the two contributions are the changed remainder at fixed input and the changed input coordinate, each bounded by $\tau/4$ using (14.8). This is the graph contraction used in Lemma 14.31. □

Choose the reference level $m_{\mathrm{ref}}(d)$ with $J_{m_{\mathrm{ref}}(d)} \ge d+3$, $\mathfrak{q}_{J_m} \ge d+1$ and $\varpi_{J_m} \ge d$ for $m \ge m_{\mathrm{ref}}(d)$. Apply Lemma 14.6 along the trajectory, starting from the representation at the cutoff constructed in the proof of Lemma 14.7, and let M_m be its coefficient maps. Define, before normalizing the lattice fields,

$$\mathsf{Z}_{N,p,d} = \mathsf{M}_{m_{\mathrm{ref}}(d)} \cdots \mathsf{M}_{N-1} \quad (N \ge m_{\mathrm{ref}}(d)). \tag{14.18}$$

An empty product is the identity. For $N < m_{\mathrm{ref}}(d)$ choose an arbitrary invertible map, as allowed in Definition 2.8.

The curvature square at the terminal map. We continue the example of Section 10.5. The curvature square c_{F^2} is a label of degree $d = 4$ with engineering weight four and factor count two, the top weight in V_4. Its column of the leading map is therefore diagonal, $T_m c_{F^2} = (t_m/t_{m+1})^2 c_{F^2}$, and S_{ℓ_m} fixes it. With the leading maps alone, the terminal map would multiply c_{F^2} by the single factor $(t_{m_{\mathrm{ref}}(4)}/t_N)^2$. The coefficient maps, however, are controlled only up to an error in the full coefficient norm, $|\mathsf{M}_m\mathbf{x} - T_m\mathsf{S}_{\ell_m}\mathbf{x}|_1 \le \varepsilon_{\ell_m}(|\mathbf{x}^\circ|_1 + \|g_{m+1}(\mathbf{x})\|_{\mathrm{rem}})$ (Step 2 in the proof of Lemma 14.6), so at each step the curvature-square input may acquire a component along the unit $\mathbf{1}$, the only other real H_4-invariant label in V_4 (Section 10.5), for instance from integrated remainders. The estimates therefore do not force the unit coefficient $\mu_{N,p}$ of Section 10.5 to vanish. The reference normalization instead prescribes the whole bare coefficient vector $\mathsf{Z}^{-1}_{N,p,4}c_{F^2}$; Lemma 14.7 bounds it by $|c_{F^2}|_1 \prod_{m_{\mathrm{ref}}(4)\le i<N} C^{\mathrm{col}}_{J_{i+1}} = a_N^{-o(1)}|c_{F^2}|_1$, by (14.11). Section 15 continues the example with the quotient field.

The next lemma carries one insertion from the cutoff to the reference level. Its content is the representation at the cutoff, whose forced part needs the initialization requirement, and the transport bounds obtained by iterating the marked step.

Lemma 14.7 (Marked trajectory)**.** *Let $N \geq m_{\mathrm{ref}}(d)$ and $m_{\mathrm{ref}}(d) \leq m \leq N$. The densities (14.15) have representations at level m with coordinates $\mathbf{x}_m(\mathbf{v})$ and remainder families linear in $\mathbf{v}$, with fixed supports, such that $\mathbf{x}_N(\mathbf{v}) = \mathbf{v}$, $\mathbf{x}_m = \mathsf{M}_m \mathbf{x}_{m+1}$ for $m < N$, $\mathbf{x}_{m_{\mathrm{ref}}(d)}(\mathbf{v}) = \mathsf{Z}_{N,p,d}\mathbf{v}$ with $\mathsf{Z}_{N,p,d}$ invertible, and*

$$\mathcal{E}_m^{\mathrm{mk}}(\mathbf{v}) + (\epsilon_m^{\mathrm{for}})^{-1/2}\mathcal{K}_m^{\mathrm{mk}}(\mathbf{v}) \leq C_{J_m}^{\mathrm{col}} |\mathbf{x}_m(\mathbf{v})^\circ|_1,$$

also with σ_{J_m} replaced by any exponent in $[98, 100]$. For $m < N$,

$$|\mathbf{x}_{m+1}(\mathbf{v})|_1 \leq C_{J_{m+1}}^{\mathrm{col}} |\mathbf{x}_m(\mathbf{v})|_1, \qquad |\mathbf{x}_m(\mathbf{v})|_1 \leq C_{J_{m+1}}^{\mathrm{col}} \ell_m^d |\mathbf{x}_{m+1}(\mathbf{v})|_1.$$

Proof. At level N, split the unmasked section into its healthy part and its complement $f_{\mathbf{v}} = \mathbf{1}_{\Sigma_d \text{ unhealthy}} P_d[t_N^{-\mathsf{N}}\mathbf{v}^\circ]$. The healthy part is the polynomial term of (5.18); the complement must be carried by the forced family, whose eligibility events (Definition 5.2) cover the flagged configurations. For an eligible family $\mathcal{A}$ let $\mathcal{A}_\Sigma$ be its forced terms whose supports meet Σ_d. If $\mathcal{A}_\Sigma = \varnothing$, exact coverage leaves no flag in Σ_d and $f_{\mathbf{v}} = 0$. Otherwise replace $\mathcal{A}_\Sigma$ by one new marked forced address with support $\Sigma_d \cup \bigcup_{i \in \mathcal{A}_\Sigma} Y_i$, inventory $\bigcup_{i \in \mathcal{A}_\Sigma} I_i$, zero unmarked activity, and $Q = f_{\mathbf{v}} \prod_{i \in \mathcal{A}_\Sigma} K_i^{\mathrm{for}}$ on the event that the flags in that support are exactly this inventory, $Q = 0$ otherwise. An index of the new marked family records the cluster $\mathcal{A}_\Sigma$, even when two clusters have the same support and inventory; retain all original forced factors in Q. The remaining supports are disjoint from the merged support, since they avoid Σ_d and were disjoint from every merged support. This is a bijection between eligible families meeting Σ_d and eligible marked families containing one new address, so the forced sum restores $f_{\mathbf{v}} \operatorname{dens} \rho_N$ exactly. The regular remainder is zero and $\mathbf{x}_N = \mathbf{v}$.

The support weight of the merged address is at most $e^{\sigma_{J_N} \operatorname{cov}_{B_N}(\Sigma_d)}$ times the product of the old support weights: their connected covers all meet the stencil cover. The activity caps bound the sum of the weighted majorants of addresses meeting Σ_d by $|\Sigma_d|\epsilon_N^{\mathrm{for}}$. Dropping disjointness and summing their products over nonempty subsets gives at most $e^{|\Sigma_d|\epsilon_N^{\mathrm{for}}} - 1$. By (10.6), $|f_{\mathbf{v}}| \leq C_d t_N^{-d} |\mathbf{v}^\circ|_1$; with (5.12) and $\operatorname{cov}_{B_N}(\Sigma_d) \leq 1$,

$$\begin{aligned}(\epsilon_N^{\mathrm{for}})^{-1/2}\mathcal{K}_N^{\mathrm{mk}} &\leq (\epsilon_N^{\mathrm{for}})^{-1/2} C_d t_N^{-d} e^{\sigma_{J_N}} (e^{|\Sigma_d|\epsilon_N^{\mathrm{for}}} - 1) |\mathbf{v}^\circ|_1 \\ &\leq |\mathbf{v}^\circ|_1\end{aligned}$$

by item 6. This is the bound of the lemma at $m = N$, since $C_{J_N}^{\mathrm{col}} \geq 1$.

Iterate the marked step on the same unmarked trajectory. Lemma 14.6 gives $\mathbf{x}_m = \mathsf{M}_m \mathbf{x}_{m+1}$, the exact equality in (14.15), the graph bound and the two transport bounds; the factor ℓ_m^d agrees with $a_m^d = \ell_m^d a_{m+1}^d$. Each M_m is invertible, so their product is invertible and $\mathbf{x}_{m_{\mathrm{ref}}(d)} = \mathsf{Z}_{N,p,d}\mathbf{v}$. No choice of a different unmarked representation is made. For fixed supports, take the union of the addresses selected for the finitely many basis inputs and fill unused addresses by zero; the basis construction in Lemma 14.6 then makes the coefficients and both remainders linear in $\mathbf{v}$. Finally, for $s \geq 0$ and $98 \leq \tau \leq 100$ we have $e^{\tau s} \leq e^{100 s}$, so lowering the support exponent in (14.16) only decreases both sums. □

The normalization retains the symmetries of the local sections. Its unit includes the degree-dependent spacing factor.

Lemma 14.8 (Symmetry and the unit)**.** *For every d, $N \geq m_{\mathrm{ref}}(d)$ and p, the maps $\mathsf{Z}_{N,p,d}$ commute with conjugation and H_4. For $m_{\mathrm{ref}}(d) \leq m \leq N$,*

$$\mathbf{x}_m(a_N^d \mathbf{1}) = a_m^d \mathbf{1}.$$

Consequently $\mathbf{u}_d = a^d_{m_{\mathrm{ref}}(d)}\mathbf{1}$ has insertion one.

Proof. Complex conjugation and signed permutations commute with integration (Definition 5.1), with Taylor extraction in the symmetric sections of Lemma 10.2, and with the equivariant coefficient graph of Lemma 14.6. Each operation therefore commutes with every M_m and their product.

For a unit input, the marked density is $a_N^d \operatorname{dens} \rho_N$. Exact integration and multiplication by ℓ_m^d give $a_m^d \operatorname{dens} \rho_m$ at level m. The selected extraction has zero nonconstant coordinate and zero remainder for this density. Induction proves the displayed identity. In particular $\mathsf{Z}^{-1}_{N,p,d}\mathbf{u}_d = a_N^d \mathbf{1}$. Substitute into Definition 2.8; factor scaling fixes the polynomial unit, so its insertion equals $a_N^{-d} a_N^d = 1$. □

The remainders have the cancellations of the marked format (5.18) of Section 5.

Lemma 14.9 (Marked local format)**.** *For $N \geq m_{\mathrm{ref}}(d)$ and $m_{\mathrm{ref}}(d) \leq m \leq N$, the complex regular remainders are Borel, gauge invariant, local and smooth as in Definition 13.1. Their supports contain the anchor. Their values at the identity, and their radial-jet coefficients through degree d on nonwrapping supports, vanish. The forced remainders satisfy Definition 13.2, and*

$$Q_i \neq 0 \quad \Longrightarrow \quad \text{the anchor belongs to } Y_i.$$

Proof. The anchored extraction in Lemma 14.6 subtracts the value at the identity and the radial Taylor polynomial; this gives exactly the two stated regular cancellations. On wrapping supports no radial Taylor cancellation is imposed. In the connected expansion every marked component contains the original insertion anchor. The support is the union of the supports in that component, so it still contains that anchor after enlargement. A component reaches a forced address only through terms that carry a flagged inventory at that level, and then only on the eligibility event of that address (Definition 13.2); an incoming regular mark is never moved to the forced family as a whole (Step 1 in the proof of Lemma 14.6). The anchor argument applies to these terms as well. Locality, gauge invariance and Borel measurability are preserved by local integration and by these tests. The regular derivatives are taken on healthy configurations, as in Definition 13.1; no derivative of a relaxed threshold test is used. The fixed basis construction and its support unions retain these properties under iteration. □

14.2.2 Chessboard seminorms and replicated sources

A bond tiling has cell boundaries through link midpoints. Its reflected copies place an observable in every cell, with conjugation on odd reflections. The copy $(F)_C$ is the pullback of F under the composition of bond reflections taking the base cell to C, conjugated when that composition has odd parity.

Definition 14.10 (Chessboard seminorm)**.** For a tiling $\mathcal{C}^{\mathrm{tile}}$ into congruent reflected cells and an observable F in its base cell, set

$$\|F\|_{\mathcal{C}^{\mathrm{tile}}} = \left| \left\langle \prod_{C \in \mathcal{C}^{\mathrm{tile}}} (F)_C \right\rangle_{N,p} \right|^{1/|\mathcal{C}^{\mathrm{tile}}|} .$$

Interior means dependence only on links with both endpoints strictly inside the base cell. The centered tiling $\mathcal{C}_m^{\mathrm{tile}}(y)$ has physical cell side a_m and base cell centered at y.

The seminorm measures the cost of repeating one observation throughout the box; it converts reflected-product estimates into bounds on one insertion. For interior complex observables these quantities are seminorms and satisfy

$$\left| \left\langle \prod_{C \in \mathcal{C}^{\mathrm{tile}}} (F_C)_C \right\rangle_{N,p} \right| \le \prod_{C \in \mathcal{C}^{\mathrm{tile}}} \|F_C\|_{\mathcal{C}^{\mathrm{tile}}} . \tag{14.19}$$

Apply the abstract chessboard theorem, [14, Theorem 4.1], successively in the four axes. The real vector space is the space of complex cell functions, and the reflection includes conjugation. Translation invariance gives cyclicity; reflection positivity gives the required Schwarz inequality. For the Wilson measure this positivity is [31, Theorem 2.1, pp. 14–15], with compact matrix group, positive coupling and observables on one side of the bond planes. For $\mathcal{C}_m^{\mathrm{tile}}$, there are $L_p D_m$ cells per axis, an even integer. The cell side $a_m/a_N = D_N/D_m$ is odd, so the centered boundaries pass through link midpoints. Bounded Borel cell functions follow by approximation in the finite-dimensional configuration measure.

The centered seminorm estimate must also control anchors near arbitrary, separated configurations. Write $v = (d, c)$ for the insertion label. A signed image means the image under H_4; conjugation is also allowed. Two tilings are aligned when their cells have the same axes and every coarse boundary is a fine boundary.

Lemma 14.11 (Interior seminorm bound)**.** *Fix $m \le k \le N$, $N \ge m_{\mathrm{ref}}(d)$, and $B_{\mathrm{rt}} \ge 0$. Suppose, for every anchor y', every signed image or image under conjugation of $O_{N,p}(v, y')$ has seminorm at most B_{rt} in $\mathcal{C}_k^{\mathrm{tile}}(y')$ and stencil within a_k of y'. In every translated bond tiling $\mathcal{T}$ of side a_m, an anchor at distance at least $3a_k$ from its cell boundary satisfies*

$$O_{N,p}(v, y) \text{ is interior}, \qquad \|O_{N,p}(v, y)\|_{\mathcal{T}} \le B_{\mathrm{rt}} . \tag{14.20}$$

Proof. At an aligned fine-cell center the bound follows from (14.19); between centers it follows from log-convexity in each axis, using the symmetry of Lemma 14.8. Since a_k/a_N is odd, the fine-cell centers are lattice sites. At an aligned fine-cell center, reflect all coarse copies into distinct fine cells. Their signs and conjugations are among the assumed images. Apply (14.19) with 1 in every unoccupied fine cell. Taking the coarse seminorm bounds it by B_{rt}.

For intermediate anchors, let $H(z)$ be the coarse seminorm after z lattice translations in one axis, with the other coordinates fixed. Translation invariance and reflection give

$$H(z)^2 \le H(z-1)H(z+1) . \tag{14.21}$$

Indeed, translate the reflected product by one lattice step. Alternate cells then contain shifts $z-1$ and $z+1$; apply (14.19). On a finite interval a nonnegative sequence satisfying this inequality

cannot exceed its endpoint maximum: an interior positive maximum extends to both neighbours and hence to an endpoint. This includes zero endpoints.

Bracket each coordinate of y by its two neighbouring fine-center coordinates. All anchors in the resulting rectangle have margin at least $2a_k$, so their stencils are interior. Induct on $i = 0, \dots, 4$: the seminorm is at most B_{rt} when the first i coordinates are arbitrary lattice coordinates in these intervals and the remaining coordinates are endpoints. The case $i = 0$ is the aligned-center bound. In the next axis, the endpoint bounds hold uniformly in the coordinates already moved; (14.21) supplies the intervening bounds. The case $i = 4$ proves (14.20). □

Replicated sources. Several insertions are carried at once by giving each a nilpotent mark. This is the device of the observable field in [5, Section 2.3], where only terms of order at most one in each external field are retained. For a finite set of slots I_0, work in the commutative algebra $\mathbb{C}[z_i : i \in I_0]/(z_i^2 : i \in I_0)$. Write $z_I = \prod_{i \in I} z_i$ for the mode I. The exact source density starts at

$$\mathcal{H}_N(z; U) = \operatorname{dens} \rho_N(U) \prod_{i \in I_0} (1 + z_i F_i(U)), \qquad \mathcal{H}_m = \mathcal{B}^{(m)}_{\beta_{N,p,m+1}} \mathcal{H}_{m+1}. \tag{14.22}$$

Thus its normalized z_I integral is the cutoff moment of the corresponding physical insertions. At level m write its local representation as

$$e^{\Phi_m(U) + \sum_i z_i f_{i,m}(U) + \sum_X H_{m,X}(z;U)} \cdot \sum_{\mathcal{A}} \prod_{\text{eligible } Y \in \mathcal{A}} \left(K_{m,Y}(U) + \sum_i z_i k_{i,m,Y}(U) + K^{\geq 2}_{m,Y}(z; U) \right). \tag{14.23}$$

Here Φ_m is the unmarked exponent, $f_{i,m}$ and $k_{i,m,Y}$ are the complete one-insertion exponential and forced terms, including their physical spacing factors. The polynomials $H_{m,X}$ and $K^{\geq 2}_{m,Y}$ contain only modes z_I with $|I| \geq 2$. Their support contains every anchor indexed by a nonzero mode. The regular terms include their health masks, and each forced term retains its eligibility test. For a single mark, $z_1^2 = 0$ gives $e^{\Phi_m + z_1 f_{1,m}} = e^{\Phi_m}(1 + z_1 f_{1,m})$, and (14.23) is the representation of Definition 14.5 with

$$f_{1,m} = a_m^{-d} \Big\{ x_0 + \mathbf{1}_{\Sigma_d \text{ healthy}} P_d[t_m^{-\mathsf{N}} \mathbf{x}^\circ_m] + \sum_i \mathbf{1}_{Z_i \text{ healthy}} R_i \Big\}, \qquad k_{1,m,Y_i} = a_m^{-d} Q_i,$$

at $\mathbf{v} = \mathsf{Z}^{-1}_{N,p,d} c$. The higher modes record the connected terms that integration creates between different marks. Rescaling mark h by $\mathsf{w}_h > 0$, its *mark weight*, multiplies the coefficient of z_I by $\prod_{h \in I} \mathsf{w}_h$; a bound of order one on the rescaled source therefore bounds that coefficient by $\prod_{h \in I} \mathsf{w}_h^{-1}$.

Connected terms with at least two marks need large supports, because distinct marked anchors are far apart. We record this once, since the stopping estimate uses it again.

Lemma 14.12 (Marked covering estimate)**.** *Let $B \geq 8$. If a support X contains $n \geq 2$ anchors at pairwise periodic max-distance at least B^2, then*

$$Bn \leq 64 \operatorname{cov}_B(X). \tag{14.24}$$

Proof. Take a minimal cover of X by $M = \operatorname{cov}_B(X)$ radius-B balls with connected union. Join their centers by a tree with edges of length at most $2B + 1$. Its doubled traversal has length at most $6B(M - 1)$. Assign each anchor to one covering center. Distinct anchors have distinct centers,

and the distance between any two assigned centers is at least $B^2 - 2B$. Visit these centers in their cyclic order in the traversal. The triangle inequality gives $n(B^2 - 2B) \le 6B(M-1)$, so $n(B-2) \le 6M$. For $B \ge 8$ we have $B - 2 \ge \frac{3}{4}B$, hence $Bn \le 8M \le 64M$. The argument uses the periodic max-distance and needs no nonwrapping assumption. □

The estimate permits each source weight to be enlarged by $e^{\delta B/64}$ per mark when δ units of the support exponent are removed.

The rooted expansion needs a bound on all the other marked factors. We verify it for the primary insertions of Definition 2.8, with their selected coefficient graphs, before taking any replicated integral. For mark h at level i let $\mathbf{x}_{h,i}$ be its dimensionless singleton coefficient, let $b_{h,i} = a_i^{-d}|\mathbf{x}_{h,i}|_1$, and let $v_{a,i}^{\mathrm{src}}$ be the weighted integrated majorants of the other marked factors on their complete read supports $Z_{a,i}$, with derivative and support weights as in Lemma 13.5.

Lemma 14.13 (Marked attachments)**.** *Let $m_{\mathrm{ref}}(d) \le k \le N$. Consider finitely many degree-d primary insertions whose anchors have level-i ancestors pairwise at periodic distance at least B_i^2 for $k \le i \le N$. Give mark h weights with $\mathsf{w}_{h,i+1} = \widehat{\beta}_{i+1}^{-1}\mathsf{w}_{h,i}$ and $\mathsf{w}_{h,k}b_{h,k} \le \zeta_{d,k}$. Then, for $k \le i \le N$,*

$$(2B_i+1)^4 \sup_x \sum_{a: x \in Z_{a,i}} v_{a,i}^{\mathrm{src}} \le \eta_i^{\mathrm{att}}/2,$$

uniformly in the cutoff, the period and the number of marks.

Proof. The proof is an induction down the levels: the rooted clause of Lemma 13.5 bounds each step, the covering estimate pays for the changing mark weights, and the allowance (14.6) closes the bound. The singleton coefficients are controlled by Lemma 14.6 and their format by Lemma 14.9.

For the local expansion, give a mode z_I the weight $\prod_{i\in I} w_i$, with $w_i > 0$. The resulting coefficient norm is submultiplicative: multiplication sums over disjoint splittings of I, exactly the splittings in the product of the two weighted norms. The coefficientwise rooted clause of Lemma 13.5 applies in this algebra, with the attachment bound at the preceding stage of the construction. Derivative and support weights are included in the separate majorants $v_{a,i}$; no submultiplicativity is asserted for them. Connected terms with at least two marks obey (14.24) with $B = B_k$, because distinct anchors are separated by B_k^2 and $B_k \ge 8$.

The weight change from $\mathsf{w}_{h,i+1}$ to $\mathsf{w}_{h,i}$ costs $\widehat{\beta}_{i+1}^{|I|}$ on a mode I. For $|I| \ge 2$, Lemma 14.12 at level i gives

$$\widehat{\beta}_{i+1}^{|I|} \le \exp\left\{ \frac{64 \log \widehat{\beta}_{i+1}}{B_i} \operatorname{cov}_{B_i}(X) \right\}.$$

The coefficient of the covering number tends to zero at fixed stage, since $B_i \asymp (\log \widehat{\beta}_i)^2$ and $\widehat{\beta}_{i+1}/\widehat{\beta}_i \to 1$. The source entry bounds therefore pay this increase from the support reserve at each step. The singleton terms use the coefficient and graph bounds established below. This pays for the varying weights on all modes with at least two marks.

The weights $\mathsf{w}_{h,i}$ estimate the intermediate source polynomials; the weight in the final integral remains $\mathsf{w}_{h,k}$. The inverse estimate of Lemma 14.6 gives $b_{h,i+1} \le \ell_i^d C_{J_{i+1}}^{\mathrm{col}} b_{h,i}$. Therefore (14.7) proves by induction that

$$\mathsf{w}_{h,i}b_{h,i} \le \zeta_{d,i} \qquad (k \le i \le N).$$

The graph norm has the same bound with its column constant included in $A_{j,d}$. The normalized forced terms retain their original eligibility events. Their remaining large-field suppression pays the fixed derivative factors before the forced normalization is divided out.

Here is the conversion from a marked-anchor bound to the site sum required by Lemma 13.5. Each support contains its marked anchor. If it also contains a site x at distance R from that anchor, its connected covering number is at least $R/(2B_i+1)$. One reserved unit of support exponent thus gives a factor at most $e^{-R/(2B_i+1)}$. Distinct anchors have occupancy one, so summing this factor over them is bounded by

$$\sum_{z\in\mathbb{Z}^4} e^{-|z|_\infty/(2B_i+1)} \le C(1+B_i)^4.$$

This estimate also bounds the periodic sum, by lifting the distinct anchors to nearest representatives. The finite derivative products and polynomial evaluations cost at most $A_{j,d}\widehat{\beta}_i^{q_d^{\mathrm{ht}}}$. Rooted attachment sums use the other reserved support unit to absorb $e^{\eta_i^{\mathrm{att}} n_\star}$; summing their convergent tree majorants enlarges the bound by at most two, after decreasing η_i^{att} if necessary. For the weighted majorants $v_{a,i}^{\mathrm{src}}$ of all other marked factors, including the connected source terms already formed, this gives

$$(2B_i+1)^4 \sup_x \sum_{a:x\in Z_{a,i}} v_{a,i}^{\mathrm{src}} \le 2A_{j,d}(2B_i+1)^4(1+B_i)^4\widehat{\beta}_i^{q_d^{\mathrm{ht}}}\zeta_{d,i} \le \eta_i^{\mathrm{att}}/2.$$

Start with finite source products at level N, where there are no connected higher modes. In each descending step, apply the rooted bound to the input majorants, change the mark weights using the covering estimate, and use the displayed site bound for the output majorants. This closes the induction, with a bound independent of the number of marks and of the truncation. The unmarked factors contribute at most the other $\eta_i^{\mathrm{att}}/2$, by item 5. Thus the complete attachment sum is at most η_i^{att} at every level $i\in[k,N]$. Passing to the absolutely convergent limit preserves the bound. In particular the rooted clause applies to the other marked factors as well as to the ordinary factors. □

The next estimate is the source of all volume-uniform bounds. Place one copy of the insertion in every cell of the level-m tiling, give each copy a nilpotent mark, and integrate down to a level k at which every cell contains $(a_m/a_k)^4$ sites. The marks stay at distance at least B_i^2 at every intermediate level, so the attachment condition of the rooted clause holds uniformly in the number of marks (Lemma 14.13). Each mark then costs e^6/w_k, where $\mathsf{w}_k^{-1} = a_k^{-d}e^{o(s_k)}$ is the size of the transported insertion, and each level-k site costs the unmarked normalization e^{q_k} of Lemma 14.1. Taking the $|\mathcal{C}_m^{\mathrm{tile}}|$-th root gives the per-insertion bound $\mathsf{w}_k^{-1}e^{6+(a_m/a_k)^4 q_k}$.

We also use the estimate for a residual observable E with an exact source representation. Call $\mathsf{w}E$ controlled at level k when its weighted singleton exponential bound is at most 3, its forced singleton bound is at most 1, and the connected regular and forced bounds contribute at most one each per mark, with the original eligibility tests. The source evolution from its initial level down to k must retain these bounds and satisfy the attachment bounds (14.6) at every intervening level. This condition concerns the complete source, including its forced terms. Fix d and a bounded coefficient ball. A stopping pair is a pair of levels $m_{\mathrm{ref}}(d)\le m\le k$ with $B_k^2\le a_m/a_k$.

Lemma 14.14 (Replicated source bound)**.** *There are mark weights $\mathsf{w}_k>0$, independent of p, such that, for each stopping pair, some $N_0(d,\mathrm{ball},m,k)$ gives, for all $N\ge N_0$ and all p,*

$$\mathsf{w}_k^{|\mathcal{C}_m^{\mathrm{tile}}|}\left|\left\langle \prod_{C\in\mathcal{C}_m^{\mathrm{tile}}} (O(c))_C \right\rangle_{N,p}\right| \le e^{|\Lambda_{n_{p,k}}|q_k+6|\mathcal{C}_m^{\mathrm{tile}}|}.$$

For every $\varepsilon'>0$, there is $k_0(\varepsilon')$ with $6-\log\mathsf{w}_k\le(d+\varepsilon')s_k$ for all $k\ge k_0(\varepsilon')$. The same integral bound holds with $O(c),\mathsf{w}_k$ replaced by E,w whenever $\mathsf{w}E$ is controlled at k.

The proof bounds the complete source before taking its replicated integral.

Proof. The marks are attached through Lemma 14.13, the expansion of Lemma 13.5 bounds the higher modes, and Lemma 14.1 bounds the unmarked ratio; Lemmas 14.6 and 14.9 control the singleton coefficients and their format. Enlarge the fixed coefficient radius to include its signed and conjugated images. Put

$$E_k = R' C^{\mathrm{col}}_{J_k} \prod_{i<k} C^{\mathrm{col}}_{J_{i+1}}, \qquad \mathsf{w}_k = a_k^d \widehat{\beta}_k^{-w_d} / E_k,$$

where $R' > 0$ depends on the coefficient ball and w_d is a fixed nonnegative integer large enough for its finite polynomial evaluations and forced counterparts. The transport inequality of Lemma 14.7, iterated from $m_{\mathrm{ref}}(d)$, bounds the level-k coefficients by the product in E_k. Threshold powers introduce only a fixed power of $\widehat{\beta}_k$; enlarging w_d includes them. Increase w_d to at least $q_d^{\mathrm{ht}} + 4$. Multiplying every mark by w_k then bounds its full singleton size by $\widehat{\beta}_k^{-w_d} \le \zeta_{d,k}$.

The reflected anchors at level k have spacing $a_m/a_k \ge B_k^2$, and their parents agree under each block map. Since $B_{i+1}^2 \le 2B_i^2$ and $\ell_i \ge 3$, their spacing is at least B_i^2 at every finer level. Apply Lemma 14.13 with initial weight w_k. Its full-interval bound verifies the hypothesis of the rooted clause of Lemma 13.5. Use the mode-weighted coefficient norm of that proof; derivative and support weights remain in the separate majorants.

Singleton terms are bounded directly by their one-mark norms. Taylor's product rule introduces at most the factorial of the derivative order; the stage column constant was chosen above this factorial. With $j = J_{k+1}$, the rooted clause of Lemma 13.5 supplies the derivative bound at support exponent $\sigma_{J_k} + 3$ and the forced bound $v_\star e^{\eta_k^{\mathrm{att}} n_\star} e^{-r_j \beta'_u t_{k+1}^2}$ at exponent $\sigma_{J_k} + 2$, where $v_\star$ is the integrated majorant of the distinguished factor and $n_\star$ its covering number. Multiplying by a mark weight changes only the distinguished factor in its tree sum. Item 1 pays the derivative factorial. The weighted sums therefore retain those bounds; decreasing the final exponent from 100 to any value in $[98, 100]$ only decreases the sums. The forced normalization uses half the suppression, leaving the other half for the marked derivative and attachment losses. Their spatial anchor is always one of the marks; a nonzero mode is assigned only to a support containing all its marked anchors. This proves the support assertion in (14.23), as well as its absolute convergence. Restricting the set of marks commutes with the construction, since every operation takes disjoint unions of their modes.

At the stopping level, the weighted supremum of each complete singleton exponential term is at most 3, and that of its forced profile is at most 1. Indeed, its polynomial part is at most its coefficient bound, its marked regular sum is at most that bound, and the normalization of the forced part is at most one; Lemma 14.7 and the definition of E_k bound the sum by 3. The forced profile follows from its κ_k normalization. For (14.23), the local absolute sums give an additional two per mark: exponentiation of the connected regular terms and the eligible forced sum each contribute at most one. Integrating the remaining scalar-free unmarked exponent and dividing by the unmarked partition function gives at most $e^{|\Lambda_{n_{p,k}}| q_k}$, by Lemma 14.1. Thus coefficient extraction in the weighted source norm yields

$$\mathsf{w}_k^{|I_0|} \left| \left\langle \prod_{i \in I_0} F_i \right\rangle_{N,p} \right| \le \exp\{ |\Lambda_{n_{p,k}}| q_k + (3+1+2)|I_0| \}.$$

Exact integration preserves these normalized coefficients, proving the first assertion. The ratio of site and cell counts is $(a_m/a_k)^4$. For a residual source controlled at weight w, the same coefficient

extraction uses w in place of w_k; only the bounds $3+1+2$ and the unmarked ratio enter this calculation. Taking the cell root gives

$$\|E\|_{\mathcal{C}_m^{\mathrm{tile}}} \le \mathsf{w}^{-1} \exp\{(a_m/a_k)^4 q_k + 6\}. \tag{14.25}$$

In particular $\mathsf{w} = c\mathsf{w}_0$ gives a bound $C_{m,k,\mathsf{w}_0}/c$, uniformly in the cutoff and period. This proves the residual clause, for which E need not be a primary insertion.

Finally (14.11), summed as in (14.13), gives

$$\sum_{i<k} \log C_{J_{i+1}}^{\mathrm{col}} = o(s_k), \qquad \log \widehat{\beta}_k = o(s_k).$$

The separate final column is included in this sum for $k > 0$. Hence, for every $\varepsilon' > 0$, there is $k_0(\varepsilon')$ with $6 - \log \mathsf{w}_k \le (d+\varepsilon')s_k$ for all $k \ge k_0(\varepsilon')$. All choices precede the period, and all the estimates are on full cutoff tails. □

The centered estimate fixes its constant before the physical cell size. A sufficiently fine cutoff then places the fixed insertion stencil strictly inside that cell.

Lemma 14.15 (Centered bound)**.** *For every d and $R, \varepsilon > 0$, there is $A(d,R,\varepsilon) \ge 0$ such that, for every $m \ge m_{\mathrm{ref}}(d)$, some $N_0(d,R,\varepsilon,m) \ge m$ gives, for $N \ge N_0$, every p, y and $\|c\| \le R$,*

$$O_{N,p}^{(d)}(c,y) \text{ is interior}, \qquad \|O_{N,p}^{(d)}(c,y)\|_{\mathcal{C}_m^{\mathrm{tile}}(y)} \le A a_m^{-d-\varepsilon}.$$

Proof. The replicated source bound of Lemma 14.14 gives the seminorm at the stopping level $k = k(m)$ of Lemma 14.2; the choice of $k(m)$ makes the per-cell site cost negligible against s_k. For $0 < \varepsilon' < 1$,

$$\begin{aligned} \|O(c,0)\|_{\mathcal{C}_m^{\mathrm{tile}}(0)} &\le \exp\{(a_m/a_k)^4 q_k + 6 - \log \mathsf{w}_k\}, \\ 6 - \log \mathsf{w}_k &\le (d+\varepsilon')s_k \quad (k \ge k_0(\varepsilon')), \\ (a_m/a_{k(m)})^4 q_{k(m)} &\le \varepsilon' s_{k(m)} \quad (m \ge m_0(\varepsilon')). \end{aligned} \tag{14.26}$$

The cutoff threshold is uniform in the coefficient ball and in p.

Put $\tau = s_{k(m)}$, $\upsilon = s_m$ and $W = (a_m/a_k)^4$. Since $q_k \ge 1$,

$$\tau - \upsilon = \log(a_m/a_k) \le W \le W q_k \le \varepsilon' \tau.$$

The exponent in the first line of (14.26) is therefore at most $(d+2\varepsilon')\tau \le (d+2\varepsilon')\upsilon/(1-\varepsilon')$. Choose ε' so that $(d+2\varepsilon')/(1-\varepsilon') \le d+\varepsilon$. This proves the required estimate with constant one for all sufficiently large m. For the finitely many remaining $m \ge m_{\mathrm{ref}}(d)$, use the first line of (14.26) at the selected fixed $k(m)$ and take the maximum of its ratio to $a_m^{-d-\varepsilon}$ and one. This fixes A before m.

Finally the stencil of Definition 2.8 has fixed lattice radius. For fixed m its physical radius tends to zero as $N \to \infty$, uniformly in p. Translation covariance moves the centered calculation to any y. □

14.2.3 Negligible errors and local matching

The following estimates concern the insertions of Definition 2.8, with the terminal maps $\mathsf{Z}_{N,p,d}$ of Lemma 14.7. They are used to identify the quotient fields in Section 15. A statement holds on full

cutoff tails if it holds for all $N \ge N_0$ and every p. A nonnegative family $b_{N,p}$ has subpower growth when, for every $\varepsilon > 0$, $b_{N,p} \le a_N^{-\varepsilon}$ on such a tail. A coefficient family is tame if

$$\forall \varepsilon > 0\ \exists N_0\ \forall N \ge N_0\ \forall p : \|\mathsf{Z}_{N,p,d}^{-1} c_{N,p}\|_\infty \le a_N^{-\varepsilon}. \tag{14.27}$$

For later volume extraction we also use strictly increasing subsequences $N = \mathsf{n}(j)$, $p = P(k)$ and the ordered limit

$$\operatorname{olim} X_{N,p} = x \quad \Longleftrightarrow \quad \forall \varepsilon > 0\ \exists k_0\ \forall k \ge k_0\ \exists j_0(k)\ \forall j \ge j_0(k) : |X_{\mathsf{n}(j),P(k)} - x| < \varepsilon. \tag{14.28}$$

The ordered versions of tameness and negligibility replace full cutoff tails by these quantifiers. We prove the full-tail versions of the bounds below; the same constructions work for ordered-tame families, with their ordered thresholds, and this is the form used in Section 15.

An anchor is the lattice site at which an insertion is placed; translates of a bond tiling are by lattice vectors. In a base cell of side a_k, let z^a be the anchor's integer coordinate measured from the first site in direction a, and put

$$M_{n_*,k,N} = \left\lfloor \frac{a_k}{8(n_*+1)a_N} \right\rfloor, \qquad M_{n_*,k,N} \le z^a, \quad z^a + M_{n_*,k,N} + 1 \le a_k/a_N\,. \tag{14.29}$$

These inequalities specify a relative interior margin.

Definition 14.16 (Negligible errors)**.** An observable family $E_{N,p}(y)$ is negligible if there are $\eta_{N,p} \ge 0$ with $\sup_p \eta_{N,p} \to 0$ as $N \to \infty$ such that, for every arity n_* and level m, there are $k \ge m$ and $N_0 \ge k$ with the following property: for all $N \ge N_0$, all p, every translated bond tiling $\mathcal{C}^{\mathrm{tile}}$ of side a_k and every base-cell anchor y satisfying (14.29),

$$E_{N,p}(y) \text{ is interior}, \qquad \|E_{N,p}(y)\|_{\mathcal{C}^{\mathrm{tile}}} \le \eta_{N,p}\,.$$

The tolerance is chosen before n_* and m; the level and the cutoff tail may depend on these two requests. Here n_* is the number of insertions in a later product test, and the lower level m requires the chosen cell scale to be arbitrarily fine. All residuals below are translation covariant, and their signed and conjugated images obey the same estimates. To pass from centered to arbitrary interior seminorms, choose $k' \ge k$ with $24(n_*+1)a_{k'} \le a_k$ and apply the finite reflection argument of Lemma 14.11. It uses only fixed stencils and the chessboard inequality, so it also applies to these residuals on full cutoff tails.

Local matching converts a local observable of controlled size into an insertion of the same degree. It makes no assumption on the size of the physical coefficient it produces; it bounds its inverse terminal coordinate, which is the quantity in (14.27). The derivative bound below controls Taylor extraction, and the supremum bound controls the forced complement.

Lemma 14.17 (Local comparison)**.** *(i)* Local matching. *Let $F_{N,p}(y)$ be local observables such that $a_N^{d'} F_{N,p}$ is continuous on a fixed stencil and regular on healthy configurations in the sense of Definition 5.3, with derivative bounds and a supremum bound over all configurations of subpower growth. Then there are tame $b_{N,p} \in V_{d'}$ with $O^{(d')}(b) - F$ negligible.*

(ii) Auxiliary equation insertions. *For the auxiliary section of Lemma 10.4, there are invertible $P_{N,p}$, with $\|P_{N,p}^{-1}\|_1 \le 2$ eventually uniformly in p, such that, with $B_{d,N,p} = \mathsf{Z}_{N,p,d} P_{N,p} t_N^{\mathsf{N}} \mathsf{R}_d$, the comparison (14.30) holds for every equation label $\pi_{N,p} \in \mathcal{I}_d$ with tame $B_{d,N,p}\pi_{N,p}$.*

Here $\mathcal{I}_d$ is the equation ideal described in the proof of Lemma 14.18, and the comparison is

$$O^{(d)}_{N,p}(B_{d,N,p}\pi_{N,p}, y) - a_N^{-d} V_d^{\mathrm{aux}}[\mathsf{R}_d \pi_{N,p}](y) \quad \text{is negligible.} \tag{14.30}$$

Both parts use the same backward coefficient correction.

Proof. We represent the observable as a one-mark source at the cutoff, carry the difference with the selected primary insertion backwards with the marked step (Lemma 14.6), and bound its replicated seminorm by Lemma 14.14. Write $h_N = a_N^{d'} F_{N,p}(0)$. Enlarge its stencil by the section stencil. The radial Taylor coefficients through degree d' (in the sections of Lemma 10.2) give a coefficient v_N and a remainder. They are bounded by a fixed multiple of the derivative bound: it is a linear map on a finite jet space. Subtract the section with that coefficient on the healthy part. Its regular remainder has zero value and zero radial jet through degree d'. Assign the complementary observable to the forced supports meeting the enlarged stencil. This gives an exact one-mark representation of $\operatorname{dens} \rho_N h_N$, with

$$|v_N|_1 + r_N \le E_{N,p}, \qquad \text{forced majorant} \le E'_{N,p}, \tag{14.31}$$

where E, E' have subpower growth. The support weight of this fixed enlarged stencil is bounded by $e^{81\sigma_{J_N}}$ eventually; its B_N covering number is at most 81. Threshold powers, the Taylor-coefficient bound and this support weight are included in E, E'. The entry conditions make each of them subpower. No division by the forced allowance is made on the uncontrolled complement: it is retained with its original eligibility tests and the supremum bound E'.

To compare with the selected graph, carry this representation backwards and at the reference level choose the primary coefficient with the same polynomial coordinate. Subtract the corresponding selected inserted density. Its polynomial coordinate at the reference level is zero. The following calculation carries this matching condition towards the fine scale.

For one step, let u, v be the coarse and fine coordinates, R, T their remainder bounds, $Y = |\mathsf{S}_L v|_1$, K the norm bound for the leading map and its inverse, $\varepsilon_L = L^{-3/8}$ and $\vartheta_L = L^{1/8}$. The marked-step inequalities give

$$Y \le K\big(|u|_1 + \varepsilon_L(Y+T)\big), \qquad R \le KY + \varepsilon_L T. \tag{14.32}$$

Suppose $a|u|_1 \le R$, with $a = 32(1 + C^{\mathrm{col}}_{J_i})^2 \ge 4K^2$. Then $4K|u|_1 \le Y + \varepsilon_L T$. Substitution in the first inequality, using $4K\varepsilon_L \le 1$, gives $Y \le 4K\varepsilon_L T$. Item 3 in Lemma 14.2 gives

$$4K\, 32(1 + C^{\mathrm{col}}_{J_{i+1}})^2 \varepsilon_L \le 1, \qquad (4K^2+1)\vartheta_L \varepsilon_L \le 1.$$

Consequently $32(1 + C^{\mathrm{col}}_{J_{i+1}})^2 |v|_1 \le T$ and $\vartheta_L R \le T$. The matching condition starts this induction with $u = 0$ at the reference level. Multiplying the second inequality through the interval gives the inverse-scale gain, while the finest-level bound is closed by absorbing the coefficient contribution into its remainder bound. In particular, with subpower E'' absorbing E, E',

$$|\Delta_N|_1 \le \frac{2(1 + C^{\mathrm{col}}_{J_N}) E''_{N,p}}{\ell_{N-1}^{1/8}}, \qquad R_k \le 2(1 + C^{\mathrm{col}}_{J_N}) E''_{N,p} e^{-(s_N - s_k)/8}. \tag{14.33}$$

Here Δ_N is the difference from the selected fine polynomial coordinate. The same bounds hold when a local source is first represented at any level $n \le N$, with its polynomial coordinate matched to the selected graph at a lower reference level. If $E''_{n;N,p}$ bounds its starting data after the coefficient contribution is absorbed, then

$$|\Delta_n|_1 \le \frac{2(1 + C^{\mathrm{col}}_{J_n}) E''_{n;N,p}}{\ell_{n-1}^{1/8}}, \qquad R_k \le 2(1 + C^{\mathrm{col}}_{J_n}) E''_{n;N,p} e^{-(s_n - s_k)/8} \quad (m_{\mathrm{ref}}(d') \le k < n). \tag{14.34}$$

Indeed the cone starts at the same zero matched coordinate, and its induction uses only the steps through n. The boundary absorption uses the starting bound there. The product $\prod_{i=k}^{n-1} \ell_i^{-1/8}$ is exactly $e^{-(s_n-s_k)/8}$. Taking $n = N$ recovers (14.33). The factor $e^{-(s_n-s_k)/8}$ is the inverse-scale gain: each matched step contracts the remainder by $\ell^{-1/8}$.

To estimate the replicated residual, for a source of size E started at N, use at level k the mark weight

$$\mathsf{w} = \frac{\zeta_{d',k}/4}{a_k^{-d'} E \exp\{-(s_N - s_k)/8\}},$$

where $\zeta_{d',k} = \widehat{\beta}_k^{-(q_{d'}^{\mathrm{ht}}+4)}$ is the source allowance fixed before N, p in (14.6). Its reciprocal is a fixed power of the reference coupling. At every finer step multiply this weight by the source contraction factor of that step. Its product with the dimensionless singleton size is at most one quarter of the available source allowance; the other three quarters bound the graph remainder, the forced complement and their sum. This follows inductively from the two inequalities in (14.17). For the first step the forced complement is bounded by E' in (14.31); the large-field suppression absorbs its fixed support and derivative factors before division by the new forced normalization. The radius transport (14.7) and the site-sum calculation in the proof of Lemma 14.13 verify the attachment bound at every finer level. Thus the residual is controlled at level k in the sense of Lemma 14.14. The singleton polynomial coordinate at the reference level is zero, so the representation constructed this way is exactly the difference with the selected primary coefficient.

Here m labels the tiling and k the finer stopping level at which its replicated sources are integrated. Replicate the difference at a stopping pair $m \le k$, and use the weighted source representation (14.23). Coefficient extraction contributes w^{-1} per mark; the complete singleton bounds and the normalized unmarked ratio (14.2) contribute a constant depending on m, k. Consequently, for every fixed such pair,

$$\|O^{(d')}(b) - F\|_{\mathcal{C}_m^{\mathrm{tile}}(0)} \le C_{m,k} \mathfrak{e}_{N,p} a_N^{1/8}, \qquad \|\mathsf{Z}_{N,p,d'}^{-1} b_{N,p}\|_\infty \le 2E'''_{N,p}, \tag{14.35}$$

where $\mathfrak{e}, E'''$ have subpower growth. The factor $e^{s_k/8}$ and the fixed weight factor $\zeta_{d',k}^{-1}$ belong to $C_{m,k}$. The same argument applies after each signed permutation and conjugation, with translation covariance and the union of their fixed stencils. Absorbing two factors $a_N^{-1/32}$ bounds the first expression by $a_N^{1/16}$ on every fixed pair. The interior-seminorm comparison after Definition 14.16 gives the common tolerance $a_N^{1/32}$, chosen before all cell requests. This proves negligibility and tameness on full cutoff tails, and part (i).

For part (ii), perform the construction on all its basis polynomials together, with the original Taylor normalization. Let u_j be the primary bare coefficient matching its jth basis vector, and set $Px = \sum_j x_j u_j$, after the finite symmetry averaging. For this fixed section the regular derivative bound $H_{j,d}$ at stage j is fixed before its ratio, independently of the later coupling. The sharper coefficient estimate in (14.33) therefore has numerator bounded by $2(1 + C_j^{\mathrm{col}})H_{j,d}$ on each basis input. The ratio condition, including the finite basis-recombination factor, gives

$$|x - Px|_1 \le \tfrac{1}{4}|x|_1.$$

Indeed, before summing over the fixed basis the right side is the last-step factor $\ell_{N-1}^{-1/8}$ times the stage coefficient bound; the stage ratio was chosen after that bound. The matrix is therefore invertible and $|P^{-1}x|_1 \le 2|x|_1$. The construction is common to all later cell requests. Multiplication by $t_N^{\mathsf{N}} \mathsf{R}_d$ changes a polynomial label into its auxiliary bare insertion coordinate; multiplication by

Z changes that coordinate to the reference normalization. These are exactly the maps B in the statement. The inverse bound preserves tameness, and (14.35) compares the two sections on its transported equation ideal. Finite combinations use the same subpower factors. The argument only replaces eventual bounds on coefficients, so ordered-tame families obey the corresponding ordered version as well. □

14.2.4 Equation identities and coefficient corrections

The equation projections separate the coefficient directions controlled by exact cutoff identities. The free equation space consists of polynomial labels that vanish on curvature jets satisfying the Maxwell equations. Its transported image is the kernel of the projections chosen below, and the free quotient has dimension $\dim_{\mathbb{C}} M_d^{\mathrm{Max}}$. The projections and the equation observables in Lemmas 14.18–14.19 are chosen together, before the spacing and volume subsequences.

Lemma 14.18 (Equation projections)**.** *For every d, there are complex linear idempotents $\Pi_{d,N,p} : V_d \to V_d$ and $C_d^{\Pi} < \infty$ such that*

$$\|\Pi_{d,N,p}\| \le C_d^{\Pi}, \qquad \dim \operatorname{ran} \Pi_{d,N,p} \le \dim_{\mathbb{C}} M_d^{\mathrm{Max}}, \qquad \ker \Pi_{d,N,p} = B_{d,N,p}\mathcal{I}_d,$$

with the maps B of Lemma 14.17(ii).

Proof. The kernel is prescribed by the transported equation ideal, and the bound comes from choosing orthogonal projections in a fixed norm. Let $\mathcal{I}_d \subset V_d$ be the polynomials of Definition 2.7 belonging to the ideal generated by the linear Maxwell equations in radial jet coordinates. Radial coordinates already satisfy the linear Bianchi identities. Taking invariant polynomials commutes with the on-shell quotient: average any representative over the compact adjoint group to obtain an invariant representative. Thus the on-shell map is onto M_d^{Max} (Definition 6.3) and

$$\dim V_d - \dim \mathcal{I}_d = \dim_{\mathbb{C}} M_d^{\mathrm{Max}}. \tag{14.36}$$

Lemma 14.17(ii) constructs invertible coefficient maps $B_{d,N,p} : V_d \to V_d$. They transport $\mathcal{I}_d$ to the coefficient equations for auxiliary local representatives, including their change to the selected insertion coordinates. Set $U_{d,N,p} = B_{d,N,p}\mathcal{I}_d$. In the Euclidean norm on the fixed basis take the orthogonal projection $Q_{d,N,p}$ onto $U_{d,N,p}$ and set $\Pi_{d,N,p} = 1 - Q_{d,N,p}$. Then $\Pi^2 = \Pi$ and $\ker \Pi = U_{d,N,p}$. Equivalence with the fixed coefficient norm bounds $\|\Pi\|$ independently of N, p, even when B is poorly conditioned. Since B is invertible, $\dim \ker \Pi = \dim \mathcal{I}_d$, and (14.36) gives the dimension bound.

The same local comparison maps will be used in Lemma 14.19. They and these projections are chosen for all cutoffs before any subsequences. □

The next estimate converts a coefficient equation into an observable with exact separated moment identities.

Lemma 14.19 (Equation observables)**.** *For these same projections, every tame family $c_{N,p} \in \ker \Pi_{d,N,p}$ admits a measurable observable family $\mathcal{Q}_{N,p}(y)$, bounded at each anchor and cutoff on a full cutoff tail, for which $O_{N,p}^{(d)}(c_{N,p}, y) - \mathcal{Q}_{N,p}(y)$ is negligible. For every fixed spectator list and $\delta > 0$, on a full cutoff tail,*

$$\int \mathcal{Q}_{N,p}(y_i) \prod_{h \ne i} O_{N,p}(v_h, y_h)\, d\mu_{N,p} = 0 \qquad (\mathrm{sep}_{\mathrm{tor}}(y) \ge \delta)\,.$$

Proof. The observable is the auxiliary equation section of Lemma 10.4, whose separated moments vanish exactly; Lemma 14.17(ii) compares it with the insertion. Lemma 10.4 supplies $\widetilde{P}_d = V_d^{\mathrm{aux}}$ and R_d. Using the kernel identity of Lemma 14.18, write $c_{N,p} = B_{d,N,p}\pi_{N,p}$ with $\pi_{N,p} \in \mathcal{I}_d$, and set

$$\mathcal{Q}_{N,p}(y) = a_N^{-d} V_d^{\mathrm{aux}}[\mathsf{R}_d \pi_{N,p}](y). \tag{14.37}$$

Lemma 14.17, specifically (14.30), proves that $O^{(d)}(c) - \mathcal{Q}$ is negligible. Its inverse change has subpower growth for tame c. The common tolerance and interior-seminorm comparison were proved there and apply to this family.

The exact zero identity is (10.8) in Lemma 10.4: every auxiliary equation term is a Wilson action derivative whose multiplier and transport avoid the varied link, so Haar invariance makes its exterior moment zero. The auxiliary section has one finite equation radius. At prescribed positive physical separation, every fixed spectator stencil is exterior to it for all sufficiently fine cutoffs, uniformly in p. Its product is bounded and measurable, as required by that lemma. Apply the identity with that product as spectator. Finally $\widetilde{P}_d$ is continuous on a finite product of compact groups. At each fixed cutoff its evaluation with the chosen finite coefficients is bounded. □

Changing the degree changes the insertion normalization. The next estimate corrects that change before taking limits.

Lemma 14.20 (Corrected inclusion)**.** *For every $d \le d'$ and $c \in V_d$, there is a tame family $b_{N,p} \in V_{d'}$ such that*

$$O_{N,p}^{(d')}(b_{N,p}, y) - O_{N,p}^{(d)}(c, y) \quad \textit{is negligible}.$$

Proof. The target observable satisfies the hypotheses of local matching (Lemma 14.17(i)) in degree d'. After multiplication by $a_N^{d'}$, it is

$$a_N^{d'-d} P_d[t_N^{-\mathsf{N}} \mathsf{Z}_{N,p,d}^{-1} c](y).$$

It is a local continuous function on the fixed degree-d stencil. Its ordered derivatives in the allowed link directions and its supremum have subpower bounds. Indeed, the inverse terminal coordinate has this growth by the first transport inequality of Lemma 14.7 and (14.11); threshold powers have subpower growth because $\log \widehat{\beta}_N = o(s_N)$. The factor $a_N^{d'-d} \le 1$ can only decrease these bounds.

Local matching extracts a degree-d' coefficient $b_{N,p}$ whose inverse terminal coordinate still has subpower growth. Its residual seminorm is bounded by $C_{m,k}\mathfrak{e}_{N,p} a_N^{1/8}$ at every fixed stopping pair of levels, with $\mathfrak{e}$ of subpower growth. The absorption and interior-seminorm argument in the proof of Lemma 14.17 gives one tolerance $a_N^{1/32}$ for all the later cell requests. The residual is therefore negligible. For $d = d'$ one may simply take $b_{N,p} = c$, with zero residual. □

The physical difference quotient requires a correction in the next degree.

Lemma 14.21 (Corrected difference)**.** *For every degree d, $c \in V_d$ and axis a, there is a tame family $b_{N,p} \in V_{d+1}$ such that*

$$O_{N,p}^{(d+1)}(b_{N,p}, y) - \frac{O_{N,p}^{(d)}(c, y + e_a) - O_{N,p}^{(d)}(c, y)}{a_N} \quad \textit{is negligible}.$$

Proof. Multiplying the target difference by a_N^{d+1} removes the inverse spacing and gives exactly

$$P_d[t_N^{-\mathsf{N}} \mathsf{Z}_{N,p,d}^{-1} c](y + e_a) - P_d[t_N^{-\mathsf{N}} \mathsf{Z}_{N,p,d}^{-1} c](y),$$

to which local matching (Lemma 14.17(i)) applies in degree $d+1$. Its stencil is the union of one fixed stencil and its unit translate. It is continuous and local. Moving an ordered derivative anchor by one lattice site multiplies its spatial weight by at most a stage-dependent factor. The factor is at most $2^{\varpi_{J_N}}$ per derivative. The logarithmic condition $\varpi_{J_N} = o(s_N)$ in Lemma 14.2 makes each fixed product of these factors subpower. The other bounds are those in the preceding proof. Local matching in degree $d+1$ now supplies a tame coefficient family and the same negligible residual. The inverse spacing has been removed before matching, so it causes no further loss in that residual. The four axes use translates of this one computation and their fixed spatial-weight bounds. □

14.2.5 Comparison with free fields

Fix d and a real basis $\mathfrak{m}_1 = 1, \mathfrak{m}_2, \dots, \mathfrak{m}_{\dim_{\mathbb{C}} M_d^{\mathrm{Max}}}$ of M_d^{Max}. Choose finitely many one- or two-point tests. In test i, the first label is the variable $\mathfrak{m}_j$; the optional second label is a fixed basis element. Let $y^i \in \mathrm{Gen}_{n_i}$, $n_i \in \{1,2\}$, be the test's configuration. The free test matrix is

$$\mathcal{G}_{ij} = S_{n_i}^{\mathrm{Max}}(\mathfrak{m}_j,\ \text{test } i \text{ spectator}; y^i)\,. \tag{14.38}$$

The next lemma supplies a finite family that distinguishes every free field of degree at most d.

Lemma 14.22 (Free tests)**.** *For every d and real basis $\mathfrak{m}_1 = 1, \mathfrak{m}_2, \dots, \mathfrak{m}_{\dim_{\mathbb{C}} M_d^{\mathrm{Max}}}$, there are finitely many one- or two-point tests at generic configurations, with every second label a basis element, for which the matrix* (14.38) *has rank* $\dim_{\mathbb{C}} M_d^{\mathrm{Max}}$.

The proof uses only the Maxwell correlations of Definition 6.3 and the free positivity in Lemma 16.1, independently of Section 15.

Proof. Let $c \neq 0$ in M_d^{Max}. Lemma 16.1 gives a strictly positive reflected two-point value for c and the reflection of $\bar{c}$. Expand that second label in the basis $(\mathfrak{m}_j)$. At least one of the resulting two-point functionals is nonzero on c. Each free Wick correlation is a finite sum of products of derivatives of $\Gamma(x) = (4\pi^2|x|^2)^{-1}$ between distinct sites, hence is smooth there. This permits a perturbation of the pair to Gen_2 preserving this nonzero value. Thus the common kernel of all permitted test functionals is zero. Starting with M_d^{Max}, choose a test nonzero on a nonzero vector in the current common kernel. Each choice strictly lowers its dimension. After finitely many choices the common kernel is zero, so $\mathcal{G}$ has rank $\dim_{\mathbb{C}} M_d^{\mathrm{Max}}$. □

Fix such a family for the rank comparison below.

For an index ν of the comparison sequence, a common coefficient list $b^{\nu}_{j,N,p} \in V_d$ and lattice-compatible configuration $x^{\nu,i} = r_\nu y^{\nu,i}$ define the test insertions $F_h^{\mathrm{tst}} = O^{(d)}_{N,p}(b^{\nu}_{j_h,N,p}, x_h^{\nu,i}/a_N)$, with $j_1 = j$ and the spectator index fixed by test i. Physical points are read modulo the lattice period. Let $\mathcal{G}^{\nu}_J$ be the Maxwell moment of the labels $(\mathfrak{m}_{j_h})_{h\in J}$ at $(y_h^{\nu,i})_{h\in J}$, with $\mathcal{G}^{\nu}_{\varnothing} = 1$. Define the relative errors by

$$R_{\varnothing} = 1, \qquad R_I = \prod_{h\in I} F_h^{\mathrm{tst}} - \sum_{\varnothing \neq J \subseteq I} \mathcal{G}^{\nu}_J R_{I\setminus J}\,. \tag{14.39}$$

The rank argument uses only the singleton and pair cases. Suppressing the superscript ν, these are

$$\begin{aligned} R_{\{1\}} &= F_1^{\mathrm{tst}} - \mathcal{G}_{\{1\}}, \\ R_{\{1,2\}} &= (F_1^{\mathrm{tst}} - \mathcal{G}_{\{1\}})(F_2^{\mathrm{tst}} - \mathcal{G}_{\{2\}}) - (\mathcal{G}_{\{1,2\}} - \mathcal{G}_{\{1\}}\mathcal{G}_{\{2\}}). \end{aligned}$$

Thus the pair error subtracts the connected free two-point value from the product of the centered insertions. The lifts b^{ν} include the scale normalization, so F_h^{tst} carries no additional power of r_ν. A

lift is the reference coefficient whose Wick-ordered leading polynomial at scale r_ν is the specified basis element $\mathfrak{m}_j$. The construction uses $r_\nu = a_{t_\nu}(M_{t_\nu}^{\mathrm{cut}})^b$ for one fixed $b > 0$ chosen from the degree and the finite tests, before ν varies.

The source estimates below also apply to tests with any fixed arity bound $n \geq 2$. For a slot set I_0 with $|I_0| \leq n$, define the free generating polynomial and its logarithm, the free cumulant generating function,

$$\mathsf{G}^{\mathrm{Max}}(z) = \sum_{J \subseteq I_0} \mathcal{G}_J z_J, \qquad \mathsf{C}^{\mathrm{Max}}(z) = \log \mathsf{G}^{\mathrm{Max}}(z), \qquad z_i^2 = 0 \quad (i \in I_0).$$

The logarithm is a finite polynomial because the constant term of $\mathsf{G}^{\mathrm{Max}}$ is one and its nonconstant part is nilpotent. Its coefficient on a nonempty I is

$$[z_I]\mathsf{C}^{\mathrm{Max}} = \sum_{\mathcal{P} \text{ partition of } I} (-1)^{|\mathcal{P}|-1}(|\mathcal{P}|-1)! \prod_{J \in \mathcal{P}} \mathcal{G}_J.$$

For two slots these are the two singleton means and the connected pair moment; the rank comparison uses just this case. In the source estimates, the test positions may range over any fixed compact set $K \subset \mathrm{Conf}_{|I_0|}$ with $\mathrm{sep}_\infty \geq \delta > 0$. Constants may depend on K, δ, n and the finite degree and derivative orders; the lifts are common to these positions. The coefficients to be estimated are

$$\mathfrak{c}_I = [z_I]\left(e^{-\mathsf{C}^{\mathrm{Max}}(z)} \prod_h (1 + z_h F_h^{\mathrm{tst}})\right). \tag{14.40}$$

For the source comparison, use the intervals (14.4). Fix $b > 0$ large enough for the degree and the finite tests, put $l = t - h_t(b)$, $k = l - v_t$ and $r = a_t(M_t^{\mathrm{cut}})^b$. The interval from t to l separates the reading points from the last fluctuation scale; the interval from l to k lets the mark weights of the corrected sources increase while their remainders decrease. With $L = \ell_{J_t}^{\mathrm{st}}$, these intervals satisfy

$$L^{h_t(b)} \geq (M_t^{\mathrm{cut}})^{b+2}, \qquad L^{v_t} \geq B_t^{10}, \qquad \widehat{\beta}_i \in [\widehat{\beta}_t/2, \widehat{\beta}_t] \quad (k \leq i \leq t) \tag{14.41}$$

on sufficiently late intervals. The last assertion follows from $h_t(b) + v_t = o(t)$ at fixed stage and the entry choices. All rounded test points lie strictly inside the level-k cell: their distance from the origin is $O(r)$, while $a_l/r \geq (M_t^{\mathrm{cut}})^2$. Rounding changes their rescaled coordinates by $O((M_t^{\mathrm{cut}})^{-b})$.

Fix the arity bound $n \geq 2$ and the compact separated position set specified above. Put $\mathcal{M} = M_t^{\mathrm{cut}}$. Let q_{der} bound the required derivative orders, and let $c, \delta_{\mathrm{G}} > 0$ be the localization and healthy-chart exponents in the estimates below. Define

$$\mathcal{E}_t^{\mathrm{G}} = \mathcal{M}^{-b} + C_j^{\mathrm{G}} \mathcal{M}^{q_{\mathrm{der}}}(\widehat{\beta}_t^{-\delta_{\mathrm{G}}} + e^{-cB_t}). \tag{14.42}$$

Comparison windows. A comparison window is a triple of levels $k < l < t$; put $\mathcal{M} = M_t^{\mathrm{cut}}$. The proofs of Lemmas 14.23–14.25 use the window only through the following conditions:

(Win1) $\log(a_l/a_t) \geq (b+2)\log \mathcal{M}$, $\log(a_k/a_l) \geq 10 \log B_t$, and $\widehat{\beta}_i \in [\widehat{\beta}_t/2, \widehat{\beta}_t]$ for $k \leq i \leq t$;

(Win2) $\mathcal{C}_{J_t} \prod_{k \leq i \leq t} \mathcal{C}_{J_i} \leq \mathcal{M}^{1/64}$;

(Win3) every level of $[k, t]$ is past the entries of item 5 for the finitely many requests, and $Q_{J_k}^{\mathrm{cmp}} \leq \log \mathcal{M}/(16n)$.

Late windows (14.4) satisfy (Win1)–(Win3) by (14.41), (14.44) and Lemma 14.2. A window need not lie in one stage: under (Win1)–(Win3) the three lemmas hold with $\mathcal{C}_j^{h_t(b)+2}$ replaced by $\mathcal{C}_{J_t}\prod_{l\le i\le t}\mathcal{C}_{J_i}$.

The polynomial comparison supplies the initial error for the subsequent source iteration.

Lemma 14.23 (Middle-level source comparison)**.** *For each fixed finite label family, sufficiently late intervals admit common real lifts, fixing the unit and bounded in N, p for that interval. For every comparison window satisfying (Win1)–(Win3) and uniformly over the prescribed compact position set, the coefficients of z_I, $I \neq \varnothing$, of $e^{-\mathsf{C}^{\mathrm{Max}}}\prod_h(1+z_hF_h^{\mathrm{tst}})$, represented at level l, have total regular and normalized forced bound*

$$\mathcal{C}_j^{h_t(b)+2}\mathcal{E}_t^{\mathrm{G}}.$$

The assertion holds on a full cutoff tail, uniformly in p.

The proof gives the Wick correction and the inverse-separation gain.

Proof. The free covariance is compared with the lattice covariance of Lemmas 11.5 and 11.13 at the reading scale, the internal contractions are removed by Wick ordering, and the connected expansion of Lemma 13.5 carries the corrected sources from t to l, with the marked step of Lemma 14.6 supplying the singleton coefficients. The sources have at most n marks in each cell; the number of cells is unrestricted. Use the coefficientwise rooted clause of Lemma 13.5 in their weighted nilpotent algebra. Only point separation is needed for the covariance comparison; coincident coordinates of distinct points are allowed.

On a fixed healthy chart, subtract the radial Taylor polynomial of each local insertion. The coefficient norm of the remainder is bounded by its next ordered derivative times the corresponding power of the coordinate radius. All derivatives needed here are among the finitely many stage orders. Express each polynomial in curvature and differentiated-curvature variables, carrying along their powers of r. In the finite jet coordinates $\xi_1,\dots,\xi_{n_\xi}$ write the covariance as $C=(C_{ab})$ and put $\Delta_C=\sum_{a,b}C_{ab}\,\partial^2/\partial\xi_a\partial\xi_b$. Gaussian contraction and Wick ordering are the mutually inverse operators

$$e^{\Delta_C/2}\pi=\sum_{j\ge0}\frac{1}{2^j j!}\Delta_C^j\pi,\qquad :\pi:_C=e^{-\Delta_C/2}\pi.$$

The series terminate on the chosen polynomial space, since each contraction removes two factors. We Wick-order each insertion with respect to its internal contractions; contractions between distinct insertions remain. The leading coefficient change T_m of Lemma 14.6 and Wick ordering, in the jet normalization of Definition 2.7, are invertible and preserve the unit and real form. The error in replacing the selected coefficient map by this leading map has norm less than $1/4$ after its inverse, by the marked estimate (14.17) and the entry condition. Solving this finite linear system gives one lift for each free basis polynomial, common to every row. Transport only between the fixed levels $m_{\mathrm{ref}}(d)$ and t bounds these reference coefficients independently of N, p for that interval. All matrices are real on the real form, and the unit equation has the exact solution $\mathbf{u}_d$.

The covariance calculation involves finitely many derivatives of $\Gamma(x)=(4\pi^2|x|^2)^{-1}$. For each pair of coloured plaquette stencils the rescaled Fourier chart $K(\xi)$ is supported in $|\xi|\le2$. Its leading symbol is the Maxwell curvature symbol $H(\xi)$. For every fixed required order q_{der}, the quadratic-form calculation of Lemma 11.5, differentiated in frequency, gives

$$|\partial^j K(\xi)|\le C|\xi|^{-j},\qquad |\partial^j(K-H)(\xi)|\le C|\xi|^{1-j}\quad(0<|\xi|\le1,\ j\le q_{\mathrm{der}}).$$

These bounds follow by differentiating the inverse quadratic symbol: its curl factors remove the inverse-square singularity; replacing each lattice difference by its linear term leaves one extra power of $|\xi|$. The regulator and its derivatives obey the same fixed bounds; different colours have zero cross covariance. To convert the second estimate to position space, decompose a cutoff of $K - H$ into annuli $|\xi| \asymp 2^{-v}$. A derivative of order s in position inserts s powers of ξ. After $q_{\mathrm{der}} > 5 + s$ integrations by parts on each annulus its transform at Rx, with x in a compact separated set, is bounded by

$$C2^{-v(5+s)} \min\{1, (R2^{-v})^{-q_{\mathrm{der}}}\}.$$

Splitting the sum at $2^v \asymp R$ gives CR^{-5-s}. The smooth part away from zero decreases faster than this. The leading homogeneous symbol is the corresponding curvature derivative of Γ away from the origin, with the Fourier normalization of Part I. Including the curvature scaling powers therefore gives relative error $O(R^{-1})$.

For the finite jet samples, terms of the matching homogeneity give this leading derivative exactly. Every other term has at least one extra inverse power of R; summing their finitely many absolute coefficients preserves the $O(R^{-1})$ bound. Here $R = (M_t^{\mathrm{cut}})^b$, and rounding in the rescaled variables gives an additional $O(R^{-1})$. Localizing the sector covariance introduces $C(M_t^{\mathrm{cut}})^{q_{\mathrm{der}}} e^{-cB_t}$ by the inverse-covariance bound in Lemma 11.13, with smaller c, since $\widehat{\beta}_i \geq \widehat{\beta}_t/2$ throughout the interval. Its nonlinear healthy-chart error is $C(M_t^{\mathrm{cut}})^{q_{\mathrm{der}}} \widehat{\beta}_t^{-\delta_{\mathrm{G}}}$ for some fixed $\delta_{\mathrm{G}} > 0$. The derivative orders and powers q_{der} are fixed by the degree and the finite row family; their stage constants are fixed before entry. Since $\log M_t^{\mathrm{cut}} = o(\log \widehat{\beta}_t)$ and $B_t \asymp (\log \widehat{\beta}_t)^2$, both errors are smaller than any prescribed fixed negative power of M_t^{cut} after entry. Choose b so that the inverse-separation error is smaller than the finite power required by the mark weight below. The chart and inspected supports lie inside the nonwrapping comparison interval in (14.41), so these local estimates hold on every period, with the same constants.

The finite-polynomial estimate combining these errors is as follows. Let U bound the Gaussian comparison polynomials, T the one-factor error, and Z the two-factor error in their coefficient norm. For the connected two-point term, subtract the product of the one-point terms. Expanding that product gives the bound

$$T + Z + (U + T)T + TU. \tag{14.43}$$

For example, if a_i and a_i^0 are the two one-point terms, then $a_1 a_2 - a_1^0 a_2^0 = (a_1 - a_1^0)a_2 + a_1^0(a_2 - a_2^0)$, whose norm is at most $(U + T)T + UT$. The terms T, Z are the errors in the remaining one- and two-mark polynomial terms. Gaussian contraction expands into pairings, so the bound for any fixed polynomial is a finite polynomial in the bounds for its contractions. After b has been fixed sufficiently large, the preceding covariance and Taylor estimates make (14.43) smaller than every source bound required at l.

For more slots, let $\mathcal{G}'_J$ be the Gaussian polynomial moment before replacing its covariance by the Maxwell covariance. On the compact separated position set, all these moments are bounded by a fixed $U \geq 1$, and their discrepancies are at most ε, a constant times the covariance and Taylor error above. For a partition with q blocks, telescoping its product bounds the discrepancy by $q\varepsilon(U + \varepsilon)^{q-1}$. The partition formula for $\mathsf{C}^{\mathrm{Max}}$ therefore bounds the sum of all nonempty corrected coefficients by

$$C_n \varepsilon (1 + U + \varepsilon)^{n-1}.$$

There are finitely many subsets and partitions for fixed n. Their constants are included in $\mathcal{C}_j$. Subtracting $\mathsf{C}^{\mathrm{Max}}$ cancels every leading free coefficient in the logarithm; for a pair this is precisely the mean-product subtraction in (14.43). A support carries a mode only when it contains all its

anchors. The support-independent scalar correction is retained separately per cell. Thus no local term is assigned to a support lacking one of its marks.

For terms meeting a flagged region, keep that region's complete source support and its original eligibility test. At a step $i+1 \to i$ in stage j, let $v_\star$ be the integrated majorant of the distinguished marked factor and $n_\star$ its covering number, as in Lemma 13.5. Its rooted forced bound is $v_\star e^{\eta_i^{\mathrm{att}} n_\star} e^{-r_j \beta'_u t_{i+1}^2}$ before division by the forced source allowance. Half this suppression pays the derivative and support factors of the finite marks; the other half bounds their normalized forced remainder by the error above after entry. The finite polynomial contractions, Taylor subtractions and connected attachments cost at most $\mathcal{C}_j$ at each of the $h_t(b)$ steps. Their product is the factor in the statement. At each step the mode-weighted coefficient norm is submultiplicative; the derivative and support weights remain in the separate integrated majorants. The rooted sums are anchored at marks, so their bounds have no factor counting the replicated cells. □

The next estimate increases the mark weight while retaining the full regular and forced representation.

Lemma 14.24 (Corrected Gaussian sources)**.** *Uniformly on that set, arbitrarily late intervals admit common real lifts of the free basis, bounded in N, p for each fixed interval and fixing the unit, such that the corrected source $e^{-\mathsf{C}^{\mathrm{Max}}(z)} \prod_h (1 + z_h F_h^{\mathrm{tst}})$, carried to level k, has a representation (14.23) there with mark weight*

$$\mathsf{w}_t = \exp(\log M_t^{\mathrm{cut}}/(8n)).$$

At this weight its regular and forced source sums are each at most one, with scalar source bound at most one per cell. For each such interval there is N_0 such that the assertion holds for all $N \geq N_0$ and all p.

Its proof iterates the middle-level error over the second part of the interval.

Proof. Starting from the error of Lemma 14.23 at l, a quadratic recurrence from the connected expansion of Lemma 13.5 carries the bound to k; the extra length v_t of the window absorbs the per-step losses. Work in the submultiplicative mode-weighted coefficient norm defined in the proof of Lemma 14.13. Keep derivative and support weights in separate integrated majorants and retain the forced eligibility tests.

We now bound the loss over all steps; their number grows with t. Use (14.42). Increasing the fixed power q_{der} includes the finite Taylor and rounding bounds above. The mark weight contributes at most $\mathsf{w}_t^n = \mathcal{M}^{1/8}$ to any mode in one cell. Enlarge $\mathcal{C}_j$ to include all per-step finite polynomial and support constants. From (14.11) and the two ceilings in (14.4), on sufficiently late intervals with fixed b,

$$\mathcal{C}_j^{h_t(b)+v_t+2} \leq \mathcal{M}^{(b+6)/j^2} B_t^{10/j^2} \leq \mathcal{M}^{1/64}. \tag{14.44}$$

Indeed $\log B_t = 2 \log\log \widehat{\beta}_t + o(1)$ and $\log \mathcal{M} = (\log\log \widehat{\beta}_t)^2$, so $B_t = \mathcal{M}^{o(1)}$. The two ceilings and the $+2$ contribute at most $L^{4/j^2} \leq \mathcal{M}^{4/j^2}$, since $\ell_{J_t}^{\mathrm{st}} \leq M_t^{\mathrm{cut}}$ by item 6 of Table 3; the final inequality holds at all sufficiently large stages for each fixed b.

We use the following one-step source estimate. For the fixed lifts of Lemma 14.23, let S_i majorize the corrected singleton terms, and P_i the corrected terms with at least two marks, in the weighted regular norm at mark weight w_t, including their scalar terms per cell. Use the same quantities for the forced norm after division by its positive source allowance, and take the larger values. For

$k \le i < l$, the connected product expansion gives

$$S_i \le K_j^{\mathrm{src}} S_{i+1} + \mathcal{M}^{1/2} S_{i+1}^2,$$
$$P_i \le K_j^{\mathrm{src}} P_{i+1} + \mathcal{M}^{1/2} (S_{i+1} + P_{i+1})^2. \tag{14.45}$$

Here $K_j^{\mathrm{src}} \ge 1$ is fixed at the stage, and $\mathcal{C}_j \ge K_j^{\mathrm{src}} + 2$. To obtain these bounds, separate connected terms linear in the incoming corrected sources from those with at least two source factors. The former use the one-mark integration bound. In the latter, the product rule leaves a product of two source norms; summing the additional attachments uses the rooted clause of Lemma 13.5. Its finite derivative and localization losses are bounded by $\mathcal{M}^{1/2}$ after entry. Every higher product is absorbed in the same quadratic bound while the source norm is at most $\mathcal{M}^{-1/2}$. For the forced terms the retained eligibility event supplies the forced suppression before its allowance is divided out. The entry requirement in item 5 includes this normalized forced bound. Disjoint cell modes use the multiplicative weights of Lemma 14.14; their connecting supports keep the same estimate. No factor counts the number of cells.

Set $\mathfrak{a}_i = S_i + P_i$. Equations (14.45) imply $\mathfrak{a}_i \le K_j^{\mathrm{src}} \mathfrak{a}_{i+1} + 2\mathcal{M}^{1/2} \mathfrak{a}_{i+1}^2$. The first part of the interval gives $\mathfrak{a}_l \le \mathsf{w}_t^n \mathcal{C}_j^{h_t(b)+2} \mathcal{E}_t^{\mathrm{G}}$. By (14.44), for $b \ge 1$ and sufficiently late entry,

$$\mathsf{w}_t^n \mathcal{C}_j^{h_t(b)+v_t+2} \mathcal{E}_t^{\mathrm{G}} \le \mathcal{M}^{-3/4}, \qquad \mathfrak{a}_i \le \mathcal{C}_j^{l-i} \mathfrak{a}_l \le \mathcal{M}^{-3/4} < \mathcal{M}^{-1/2}. \tag{14.46}$$

The second inequality follows by backward induction: below $\mathcal{M}^{-1/2}$ the quadratic term in the recurrence is at most $2\mathfrak{a}_{i+1}$. For the first, the covariance contribution is at most $\mathcal{M}^{1/8+1/64-b}$, whose exponent is at most $-55/64$. The other contributions tend to zero faster than every fixed negative power of $\mathcal{M}$, since $\log \mathcal{M} = o(\log \widehat{\beta}_t)$ and $B_t \asymp (\log \widehat{\beta}_t)^2$. Thus the initial-error condition is compatible with the whole interval, although its length grows with t.

The extra length v_t also provides the support margin: a level-l neighborhood of radius B_t^2 has level-k radius at most $B_t^2 / L^{v_t} \le B_t^{-8}$. It therefore fits inside the protected cell around its rounded reading points. The successive kernel neighborhoods add a bounded geometric sum in those units; their fixed margin was included in the choice of the comparison cell. Thus the inspection and attachment tests used in the iteration stay inside the stated interval. The unit has no remainder by Lemma 14.6.

The support margin in (14.41) makes every replicated inspected stencil fit its own cell. Terms meeting flagged regions keep their original forced eligibility and the suppression used in Lemma 13.5; their source amplitude replaces only the amplitude of the distinguished factor. Thus the local source sums at the enlarged mark weight are at most one in both regular and forced norms. The constant source contribution is at most one per cell. These estimates prove the lemma on arbitrarily late intervals, with all thresholds independent of p. □

The remaining normalization uses the complete positive unmarked density. In particular it cannot discard the forced sum or replace it by its healthy part.

Lemma 14.25 (Normalized source integral)**.** *For the corrected sources of Lemma 14.24, put* $u_t = \log M_t^{\mathrm{cut}} / (16n)$*. For each such interval there is* N_0 *such that, for all* $N \ge N_0$ *and all* p*, for every nonempty mode* I*,*

$$\left| \left\langle \prod_{C \in \mathcal{C}_k^{\mathrm{tile}}} (\mathfrak{c}_I)_C \right\rangle_{N,p} \right| \le e^{|\mathcal{C}_k^{\mathrm{tile}}| q_t^{\mathrm{cmp}}} \mathsf{w}_t^{-|I||\mathcal{C}_k^{\mathrm{tile}}|}, \qquad q_t^{\mathrm{cmp}} \le u_t.$$

Proof. The bound compares the source integral with the complete positive unmarked density of Definition 5.2, whose positivity follows from the density identity of Theorem 5.5, the positive Wilson start and the probability kernels of Lemma 11.4; the source sizes come from Lemma 14.24. Let ρ be the unmarked representation at the width level k, $V = |\Lambda_{n_{p,k}}| = |\mathcal{C}_k^{\mathrm{tile}}|$, $g = \beta(\rho)^{-1/2}$, $H_{\mathrm{path}} = A_{J_k}^{\mathrm{path}}$, and let $\delta, \Lambda^{\mathrm{loop}}$ bound its regular and loop terms. The admissibility inequalities give $0 < g \le t_k$, $\delta \le 1$, $\Lambda^{\mathrm{loop}} \le \Lambda_k^{\mathrm{loop}}$ and $\beta(\rho) \le 4\widehat{\beta}_k$.

Choose constants $s_H \in (0,1]$, $b_H \ge 1$ such that small Haar balls have measure at least $s_H r^{d_{\mathfrak{g}}}$ and the clipped one-link integrals are at most $b_H(g/\sigma_*)^{d_{\mathfrak{g}}}$, where σ_* is the smaller of one and the linear clip coefficient. These follow from the nonsingular exponential chart and compactness of the group, with constants chosen before all levels. Write

$$\mathfrak{h} = \log b_H - d_{\mathfrak{g}} \log \sigma_*, \qquad \mathcal{P}(\rho, X, g) = 2E_{\mathrm{ex}} + 6 + \beta(\rho)E_{\mathrm{base}} + X + 3\mathfrak{h} - 4\log s_H.$$

Here E_{ex} bounds the loop and regular exponent per site, including the exponent discrepancy δ in the clipped comparison, and on links of size at most $g/4$, $\beta(\rho)E_{\mathrm{base}} = 3(c_{\mathrm{dir},J_k} + H_{\mathrm{path}}^2)$. For any integrable complex F with $z|F(U)| \le e^{\Phi_\rho(U)+VX}$, $z \ge 0$, the ratio estimate is

$$\frac{z\left|\int F\, dU\right|}{\int \mathrm{dens}\,\rho\, dU} \le e^{V\mathcal{P}(\rho,X,g) - 4d_{\mathfrak{g}}\log g}. \tag{14.47}$$

The gauge fixing cancels the small-coupling powers in this ratio. The denominator has $3V+1$ free links and the numerator will integrate $3V-3$ selected plaquette factors, so their Haar-ball powers differ only by $g^{d_{\mathfrak{g}}(3V-3)}/g^{d_{\mathfrak{g}}(3V+1)} = g^{-4d_{\mathfrak{g}}}$, a cost that is not extensive in V; all remaining constants are exponential in V. For the denominator, fix an axial spanning tree with $V-1$ links. Gauge invariance removes those links, leaving $3V+1$ free links. Restrict each free link to the ball of chord $g/4$. All plaquettes are then healthy, so the forced covering sum equals one. The lower action bound and the Haar-ball bound give

$$\int \mathrm{dens}\,\rho\, dU \ge e^{v_{\mathrm{sc}}V - V(E_{\mathrm{ex}}+\beta(\rho)E_{\mathrm{base}})}\left(s_H g^{d_{\mathfrak{g}}}\right)^{3V+1}.$$

For the numerator, the clipped plaquette inequality is

$$\beta(\rho)E_{\mathrm{plaq}}(U) \ge \frac{\mathrm{ch}(U_\square)}{g/\sigma_*} - 1.$$

It follows in the quadratic branch from $(a/g-1)^2 \ge 0$ and $\sigma_* \le \min(1, c^{\mathrm{dir}})$, and in the linear branch from $\sigma_* \le c^{\mathrm{lin}}$ and g below the threshold. Summing over the six plaquettes per site retains the product of $e^{-\mathrm{ch}(U_\square)/(g/\sigma_*)}$ in the upper integrand.

Keep the $3V-3$ plaquette factors selected as follows and discard the others, whose values lie in $[0,1]$. The axial tree returns each site towards zero in its highest nonzero coordinate. Order free links by that coordinate, treating wrap links first in each coordinate, except for the four links at zero, and then transverse links in decreasing height. Associate to each treated link its plaquette towards the tree. That plaquette contains the link once; every other remaining selected plaquette is independent of that link in this order. Haar invariance therefore replaces its factor by the one-link integral. The four untreated links account for the difference $(3V+1) - 4 = 3V-3$. All discarded plaquette factors lie in $[0,1]$, including at the periodic boundary. Thus

$$z\left|\int F\, dU\right| \le e^{v_{\mathrm{sc}}V + V(E_{\mathrm{ex}}+6+X)}\left(b_H(g/\sigma_*)^{d_{\mathfrak{g}}}\right)^{3V-3}.$$

Divide these inequalities. The powers of g differ by $-4d_{\mathfrak{g}}$. Using $3V-3\le 3V$, $3V+1\le 4V$, $b_H\ge 1$ and $s_H\le 1$ gives (14.47).

One may use the clipped comparison exponent in this calculation even if it is not the exponent used for the source bound. Their difference has absolute value at most $V\delta$. Multiplying the upper source bound by $e^{V\delta}$ and bounding the two positive partition integrals in the opposite direction by the same factor accounts for this change. These two factors are included in $2E_{\mathrm{ex}}$ because each occurrence of E_{ex} includes δ. Thus (14.47) applies to the actual source integral on the original exact history.

The enlarged-weight bounds of Lemma 14.24 give $X\le 1+n$: one is the scalar bound, and each of at most n marks contributes at most one. The local coefficient estimate before integration includes $z=\mathsf{w}_t^{|I|V}$. The entry inequalities imply

$$E_{\mathrm{ex}}\le 8H_{\mathrm{path}}+2,\qquad -4d_{\mathfrak{g}}\log g/V\le 1.$$

For the second, $-4d_{\mathfrak{g}}\log g=2d_{\mathfrak{g}}\log\beta(\rho)$ is $O(\log\widehat{\beta}_k)$, whereas $V\ge D_k=e^{s_k}$. The first follows from $\widehat{\beta}_k t_k^3\le 1$, $\widehat{\beta}_k t_k^4\le 1$ and $\Lambda_k^{\mathrm{loop}}t_k^2\le 1$ on sufficiently late levels. Hence choose a stage constant

$$Q_j^{\mathrm{cmp}}\ge 2(8A_j^{\mathrm{path}}+2)+6+3(c_{\mathrm{dir},j}+(A_j^{\mathrm{path}})^2)+(1+n)+3\mathfrak{h}_j-4\log s_H+1 \tag{14.48}$$

for each arity used at stage j. It bounds the exponent per cell in (14.47). All its quantities are fixed before the entry to stage j. Item 5 in Lemma 14.2 requires $Q_j^{\mathrm{cmp}}\le \log M_t^{\mathrm{cut}}/(16n)$ on the chosen intervals. This proves $q_t^{\mathrm{cmp}}\le u_t$ and the asserted coefficient estimate. In particular the volume appears only through the number of cells, so the cutoff threshold is independent of p. □

The next estimate uses the corrected source comparison in Lemma 14.24. Each relative error is interior whenever its insertions are interior, since (14.39) is a polynomial in those insertions.

Lemma 14.26 (Relative rank comparison)**.** *For every such finite test family, there are $r_\nu\downarrow 0$, $y^{\nu,i}\to y^i$, levels $k_\nu\le t_\nu$ tending to infinity, $\gamma_\nu\ge 0$ tending to zero, and real lifts $b^\nu_{j,N,p}$ of the free basis $(\mathfrak{m}_j)$, bounded in N,p for fixed ν and common to all tests. They satisfy $b^\nu_{1,N,p}=\mathbf{u}_d$. Each $x^{\nu,i}$ is generic on the spacing-a_{t_ν} grid. Eventually uniformly in p, all test insertions are interior to $\mathcal{C}^{\mathrm{tile}}_{k_\nu}(0)$, and, for every nonempty I,*

$$\|R_I\|_{\mathcal{C}^{\mathrm{tile}}_{k_\nu}(0)}\le\gamma_\nu\,.$$

Proof of Lemma 14.26. The relative errors are the corrected coefficients (14.40), and Lemmas 14.24–14.25, with the free correlations of Definition 6.3, bound their seminorms by a negative power of the cutoff size. Apply those lemmas with arity bound $n=2$, so $\mathsf{w}_t=(M_t^{\mathrm{cut}})^{1/16}=e^{2u_t}$ and $u_t=\log M_t^{\mathrm{cut}}/32$. The normalized source lemma gives

$$\left|\left\langle\prod_{C\in\mathcal{C}^{\mathrm{tile}}}(\mathfrak{c}_I)_C\right\rangle_{N,p}\right|\le\exp\{|\mathcal{C}^{\mathrm{tile}}|q_t^{\mathrm{cmp}}\}\mathsf{w}_t^{-|I||\mathcal{C}^{\mathrm{tile}}|},\qquad q_t^{\mathrm{cmp}}\le u_t. \tag{14.49}$$

The cutoff threshold is independent of p.

Taking the $|\mathcal{C}^{\mathrm{tile}}|$-th root in (14.49) gives

$$\|\mathfrak{c}_I\|_{\mathcal{C}^{\mathrm{tile}}}\le e^{u_t-2|I|u_t}\le e^{-u_t}=(M_t^{\mathrm{cut}})^{-1/32}\longrightarrow 0.$$

Since $e^{\mathsf{C}^{\mathrm{Max}}}$ has coefficient $\mathcal{G}_J$ at z_J, multiplication by $e^{\mathsf{C}^{\mathrm{Max}}}$ shows that $\mathfrak{c}_I$ is exactly R_I in (14.39). Choose successive comparison windows with both levels increasing, reading radii strictly decreasing

to zero, and e^{-u_t} decreasing to zero. The windows of Lemma 14.2 allow these finitely many requests at each step. Call the resulting levels $k_\nu \le t_\nu$, radii r_ν and errors $\gamma_\nu = e^{-u_{t_\nu}}$. Rounding on the finer grid preserves genericity for each fixed ν once that grid is fine enough. Interior insertion supports imply interiority of every R_I, by its polynomial formula. □

14.2.6 Bounds for lattice differences

Positional differences are a separate problem. A difference quotient multiplies the insertion by the inverse spacing,

$$\frac{O^{(d)}_{N,p}(c, y+e_a) - O^{(d)}_{N,p}(c,y)}{a_N} = a_N^{-(d+1)}\Big(P_d[t_N^{-\mathsf{N}}\mathsf{Z}^{-1}_{N,p,d}c](y+e_a) - P_d[t_N^{-\mathsf{N}}\mathsf{Z}^{-1}_{N,p,d}c](y)\Big), \tag{14.50}$$

so Lemma 14.15, applied to the two terms on the left, bounds its seminorm only by $2Aa_m^{-d-\varepsilon}/a_N$, which diverges as $N \to \infty$. The remedy has three steps. Local matching represents the difference quotient as a degree-$(d+1)$ insertion up to a negligible error (Lemma 14.21); the factor $a_N^{-(d+1)}$ becomes the normalization of the new insertion, but its coefficient is only tame, not bounded. The equation projection removes the directions of that coefficient whose insertions vanish in separated correlations (Lemmas 14.18–14.19). The remaining coefficient lies in a space of dimension at most $\dim_{\mathbb{C}} M^{\mathrm{Max}}_{d+1}$ and is bounded by testing it against the finite free matrix (Lemmas 14.22 and 14.26).

The tests of the difference must be bounded without any bound on differentiated insertions. Averaging the difference quotient over $Q_N \approx h_{\mathrm{av}}/a_N$ consecutive translates telescopes to a difference over the fixed physical distance $Q_N a_N$, which is bounded by undifferentiated insertions alone; this is why the argument is not circular. Comparing averaged tests at nearby positions uses the following displacement estimate, which also involves only undifferentiated insertions. For a grid configuration x define its minimum coordinate separation on the physical torus by

$$g_p(x) = \min_{i\neq j}\ \min_{a=1,\dots,4}\ \min_{z\in L_p\mathbb{Z}} |x_i^a - x_j^a + z|,$$

with value $+\infty$ for one slot. This separation supplies the empty slabs for reflection positivity. A coordinate hybrid of x, y chooses each coordinate from either configuration. Assume the insertions and their single-axis reflections are interior at a centered level $D \le N$, with all their seminorms at most $A \ge 0$.

Lemma 14.27 (Displacement of correlation tests)**.** *For every $r > 0$ with $a_D < r$, a one-coordinate grid move $x \mapsto y$ of size at most r from $g_p(x) \ge 4r$ satisfies*

$$|M_{N,p}(v; x/a_N) - M_{N,p}(v; y/a_N)| \le A^n |x-y|/r.$$

If $|x-y| \le r$ and every coordinate hybrid has coordinate separation at least $4r$, then

$$|M_{N,p}(v; x/a_N) - M_{N,p}(v; y/a_N)| \le 4nA^n |x-y|/r.$$

Proof. A move changes the lengths of the two empty slabs next to the moved insertion, and the spectral theorem for the transfer operator bounds the effect of that change; the reflection inequality (14.19) bounds what remains. Lemma 14.15 supplies the assumed seminorms for bounded coefficient lists on full cutoff tails. The separation hypothesis gives every directed coordinate distance at least $4r$, at any period. No fixed test neighbourhood is used.

Fix an axis and all other coordinates. Reflection positivity and translation invariance give a positive transfer operator, rescaled to $0 \le T \le 1$. On the periodic axis the expectation is a

normalized trace. If X denotes the insertion being moved, its two adjacent empty slabs have lengths $n_1^{\mathrm{sl}}, n_2^{\mathrm{sl}}$ in fine sites. A move by $l \geq 0$ changes their factor by

$$\begin{aligned} T^{n_1^{\mathrm{sl}}+l} X T^{n_2^{\mathrm{sl}}-l} - T^{n_1^{\mathrm{sl}}} X T^{n_2^{\mathrm{sl}}} &= T^{n_1^{\mathrm{sl}}}(T^l X - X T^l) T^{n_2^{\mathrm{sl}}-l} \\ &= -T^{n_1^{\mathrm{sl}}}(1-T^l) X T^{n_2^{\mathrm{sl}}-l} + T^{n_1^{\mathrm{sl}}} X (1-T^l) T^{n_2^{\mathrm{sl}}-l}. \end{aligned} \tag{14.51}$$

Both expressions are inserted in the same remaining cyclic product and divided by the same partition trace. The stencil is contained in a centered cell of lattice side $d_{\mathrm{cell}} = a_D/a_N$. Since $a_D < r$, the directed separations exceed $2d_{\mathrm{cell}} + 2r/a_N$, and $la_N \leq r$ by assumption. Splitting the two surrounding empty slabs at their reflection planes leaves $n_{\mathrm{sl}} + 1 \geq 2r/a_N$ fine sites next to each occurrence of $1 - T^l$. The spectral theorem gives

$$\|(1-T^l)T^{n_{\mathrm{sl}}}\| \leq \sup_{0\leq t\leq 1} l(1-t)t^{n_{\mathrm{sl}}} \leq l/(n_{\mathrm{sl}}+1) \leq la_N/(2r).$$

For each normalized trace in (14.51), reflection Schwarz at the slab ends and the chessboard inequality bound the remaining factors by the product of their centered seminorms, at most A^n. One applies this inequality to the complete cyclic product, so no supremum norm of X is required. The two terms together are bounded by $A^n la_N/r$. An arity-one moment is translation invariant and has change zero.

For a negative move reverse the chosen axis. Coordinate separation is unchanged and the assumed reflected seminorms have the same bound, so the same slab estimate applies. Replace the $4n$ coordinates one at a time. Each starting configuration is a coordinate hybrid and hence has the required separation. Each move is at most $|x - y| \leq r$. Summing proves the second bound. The constants depend only on the seminorms, arity and slab margin, uniformly in N, p. □

All the preceding comparison lemmas hold on full cutoff tails, uniformly in p. Local matching also applies to any fixed iterated difference: after multiplying by a_N^{d+h}, a difference of order h is a finite sum of translates of the degree-d section, with subpower derivative bounds. Matching in degree $d + h$ leaves a seminorm error tending to zero at every fixed cell size, uniformly in p. This finite-cutoff argument supplies the difference bounds required in Part I.

Lemma 14.28 (Difference estimate)**.** *For every arity n, insertion-label list v, difference list $\boldsymbol{l}$ and $r > 0$, the constants in Proposition 5.8 can be chosen independently of p:*

$$\sup_p \sup_{\mathrm{sep}_{\mathrm{tor}}(y)\geq r} |\Delta_{\boldsymbol{l}} M_{N,p}(v;y)| \leq B(n,v,\boldsymbol{l},r) \qquad (N \geq N_0(n,v,\boldsymbol{l},r)).$$

Proof. We first bound the chessboard seminorm of one differentiated insertion at a fixed fine cell size, uniformly in N, p (Steps 1–4), and then apply (14.19) (Step 5). Let $F = O^{(d)}(c)$, let $\Delta_{\boldsymbol{a}} F$ be a fixed iterated normalized difference in its anchor, and put $d' = d + |\boldsymbol{a}|$.

Step 1: Matching and the equation part. Local matching gives tame coefficients $c^{\boldsymbol{a}}$ in degree d' with $O^{(d')}(c^{\boldsymbol{a}}) - \Delta_{\boldsymbol{a}} F$ of vanishing seminorm. Write $c^{\mathrm{pr}} = \Pi_{d',N,p} c^{\boldsymbol{a}}$. For a tame coefficient in $\ker \Pi_{d'}$, the auxiliary equation section has seminorm zero. To see this, take all the other reflected copies as the spectator in the exact Haar identity in Lemma 10.4. For a fixed physical cell, its side in lattice units eventually exceeds twice the equation radius plus two; all those copies are then exterior. The section-comparison residual of Lemma 14.19 tends to zero in seminorm. In particular it applies to $z = (c^{\boldsymbol{a}} - c^{\mathrm{pr}})/(1 + \|c^{\mathrm{pr}}\|)$. Indeed, $c^{\boldsymbol{a}}/(1 + \|c^{\mathrm{pr}}\|)$ is tame, and $c^{\mathrm{pr}}/(1 + \|c^{\mathrm{pr}}\|)$ has norm at most one. Every bounded coefficient family is tame, because Lemma 14.7 gives

$$\|\mathsf{Z}_{N,p,d'}^{-1}\| \leq C_{d'} \prod_{i=m_{\mathrm{ref}}(d')}^{N-1} C_{J_{i+1}}^{\mathrm{col}} = a_N^{-o(1)}$$

by (14.11). Subtracting these two tame families proves tameness of z before c^{pr} is bounded.

Step 2: The averaged tests are bounded. It remains to bound c^{pr}. Use a fixed sufficiently short-distance element of the comparison sequence of Lemma 14.26 for degree d'. Let $\delta_0 > 0$ be the minimum coordinate separation of its configuration on the physical torus. Fix a slab margin $0 < b < \delta_0/8$ and a centered level D with $a_D < b$. Average the variable slot of every test over $Q_N = \lceil h_{\mathrm{av}}/a_N \rceil$ sites in each direction occurring in $\boldsymbol{a}$, where $0 < h_{\mathrm{av}} < b/[2(|\boldsymbol{a}|+1)]$. Eventually $Q_N a_N \le 2h_{\mathrm{av}}$, so the total shift in each coordinate is less than b and every coordinate hybrid used in the averaging has coordinate separation at least $4b$. Denote this linear map on coefficients by $\mathcal{P}_{N,p}$. Telescoping in one direction gives

$$\frac{1}{Q_N}\sum_{j=0}^{Q_N-1}\frac{f(z+(j+1)e_a)-f(z+je_a)}{a_N}=\frac{f(z+Q_Ne_a)-f(z)}{Q_Na_N}.$$

Since $Q_N a_N \ge h_{\mathrm{av}}$, iterating bounds the tests of $\Delta_{\boldsymbol{a}}F$ by $(2/h_{\mathrm{av}})^{|\boldsymbol{a}|}$ times the supremum of tests of F. The latter is bounded by Lemma 14.15, the interior-seminorm comparison and (14.19). Matching errors tend to zero in these tests: put the error and the bounded spectator insertions in distinct cells with fixed margins, use Lemma 14.11 for their interior seminorms, and apply (14.19). All these bounds hold on full cutoff tails and use no positional difference estimate. Thus $\|\mathcal{P}_{N,p}c^{\boldsymbol{a}}\| \le C_{\mathrm{test}}$ eventually, uniformly in p.

Step 3: Invertibility of the tests. The finite-dimensional lower bound is obtained before any cutoff limit. The free matrix has a positive lower bound on the unit sphere. The relative-error estimate and the equation-seminorm bound put the tests of the projected lifts within $O(\gamma_\nu) + o_{N\to\infty}(1)$ of the free matrix, uniformly in p. Choose the fixed ν with this error smaller than its matrix lower bound. Averaging over sufficiently small h_{av} preserves the lower bound: the undifferentiated correlation tests have uniform positional continuity by Lemma 14.27, with margin b, the centered level D and the coordinate-hybrid separations just chosen. The level and its seminorm bounds are fixed before reducing h_{av}, so the displacement bound tends to zero with h_{av}. The projected lifts are therefore linearly independent. Their number equals $\dim_{\mathbb{C}} M_{d'}^{\mathrm{Max}}$, whereas Lemma 14.18 bounds $\dim \operatorname{ran}\Pi_{d'}$ by that number. They consequently span the range. Their uniform coefficient bounds give

$$\|c^{\mathrm{pr}}\| \le C_{\mathrm{det}}\|\mathcal{P}_{N,p}c^{\mathrm{pr}}\| \tag{14.52}$$

with C_{det} independent of N, p on the retained tails. Choose an invertible square minor H_0 of the free matrix and the corresponding interacting minor H. For $\|H - H_0\| \le (2\|H_0^{-1}\|)^{-1}$, the inverse bound $\|H^{-1}\| \le 2\|H_0^{-1}\|$ follows.

Step 4: The bound on c^{pr}. The equation coefficient $z = (c^{\boldsymbol{a}} - c^{\mathrm{pr}})/(1 + \|c^{\mathrm{pr}}\|)$ is tame. Its tests tend to zero, so eventually $C_{\mathrm{det}}\|\mathcal{P}_{N,p}z\| \le 1/2$. Equation (14.52) yields

$$\|c^{\mathrm{pr}}\| \le C_{\mathrm{det}}C_{\mathrm{test}} + \tfrac{1}{2}(1 + \|c^{\mathrm{pr}}\|), \qquad \|c^{\mathrm{pr}}\| \le 2C_{\mathrm{det}}C_{\mathrm{test}} + 1.$$

Lemma 14.15 now bounds the seminorms of $O^{(d')}(c^{\mathrm{pr}})$ at each fixed sufficiently fine scale. Now $c^{\boldsymbol{a}} - c^{\mathrm{pr}}$ is tame, so its equation residual has vanishing seminorm. Together with the matching residual, the triangle inequality for seminorms gives the same type of bound for $\Delta_{\boldsymbol{a}}F$. All thresholds and constants in this argument are independent of p.

Step 5: Products at separated points. Distribute the differences in $\boldsymbol{l}$ among their indicated slots. A difference in slot i acts only on that insertion, so the resulting moment is exactly the expectation of the product of these differentiated insertions. Choose a fixed tiling side smaller than r and translate its boundaries away from all anchors. In each axis the excluded boundary

strips occupy less than one full cell length if their widths are chosen below $a_m/[4(n+1)]$. For sufficiently large N a lattice translation with these margins exists. Distinct anchors occupy distinct cells, since their torus separation exceeds the cell side. The interior-seminorm comparison, with a finer centered scale, bounds all the factors uniformly. Apply (14.19) and put one in every unused cell. Their product gives the asserted B. Empty difference lists and arities at most one obey the same argument. This proves the lemma and Proposition 5.8. □

14.2.7 Bounded approximations

The later phase argument needs bounded approximations with a bound independent of the cutoff. Compactness at each cutoff gives no such bound. We obtain it by stopping at one fixed coarse lattice and pulling bounded observables back through the probability block kernels.

In the next two lemmas, a statement holds eventually in the ordered limits when it holds for all sufficiently large indices of the chosen volume sequence and, at each such volume, all sufficiently fine cutoffs of the chosen spacing sequence. No common cutoff threshold in other subsequently fixed parameters is asserted. The chessboard seminorms used below are defined in Definition 14.10.

Lemma 14.29 (Bounded coarse approximations)**.** *For every d, $k \geq m_{\mathrm{ref}}(d)$, real $c \in V_d$ and $\delta > 0$, there are measurable real observables $\mathcal{B}_{N,p}(c,y)$ and $M_{\mathrm{bnd}} < \infty$, independent of N, p, y, with $|\mathcal{B}_{N,p}(c,y)| \leq M_{\mathrm{bnd}}$ and*

$$\|\mathcal{B}_{N,p}(c,y) - O^{(d)}_{N,p}(c,y)\|_{C_k^{\mathrm{tile}}(y)} \leq \delta$$

eventually in the ordered limits. Both observables are interior to that cell, and the same holds for their signed and reflected images.

Proof. We stop at a fixed coarse level h, replace the insertion by a bounded coarse polynomial pulled back through the probability block kernels (Definition 5.1), and bound the discrepancy by replicating it in every cell (Lemma 14.14); the marked step of Lemma 14.6 matches the coefficients. The construction is independent of the phase decomposition. Choose a fixed $m \geq k$ with $B_m^2 \leq a_k/a_m$, using Lemma 14.2. At a finer level $h \geq m$, the bounded coarse polynomial with coordinate v is

$$\mathsf{V}_h(v;U) = a_h^{-d}\left(v_0 + \mathbf{1}_{\Sigma_d\,\mathrm{healthy}} P_d[t_h^{-\mathsf{N}} v^\circ](U)\right).$$

Its supremum for each basis input is bounded by a constant depending on d, h, uniformly in N, p: the healthy polynomial uses a fixed compact coordinate range, and the unit term is constant. Its inspected stencil has a fixed radius, including the links needed for the relaxed threshold test.

Let $\mathcal{K}_{N,h}(U, dV)$ be the probability kernel obtained by composing the block kernels from N to h, and set

$$(\mathcal{P}_{N,h}F)(U) = \int F(V)\,\mathcal{K}_{N,h}(U, dV).$$

It preserves real values and does not increase the supremum norm. The locality of the block kernels confines its dependence to the fine preimage of the inspected stencil. For reflected cells these ancestor cones are disjoint once h has the stated margin. At each intermediate level the block kernels are products in the coarse-link variables given the preceding field; integrating disjoint ancestor cones therefore gives conditional independence, and hence $\prod_C \mathcal{P}_{N,h}F_C = \mathcal{P}_{N,h}(\prod_C F_C)$. Choose h sufficiently large that this preimage lies strictly inside the centered level-k cell, including its boundary-link margin.

To compare with the actual insertion, choose auxiliary fine coefficients u_j whose selected level-h polynomial coordinates differ by at most $1/4$ from the basis vectors. They are obtained by

carrying the pure coarse polynomial back through the inverse graph construction in Lemma 14.6. Apply (14.32) to the difference with this primary coarse polynomial. Write $S_h = C^{\mathrm{col}}_{J_h}$, and let $\mathcal{J}_h$ be the matrix of the selected level-h coordinates of these trials. For a basis input, the cone and the finest-level boundary estimate give

$$32(1+S_h)^2|e_j - \mathcal{J}_h e_j|_1 \le R_h, \qquad R_h \le 2(1+S_h), \qquad |e_j - \mathcal{J}_h e_j|_1 \le \frac{1}{16(1+S_h)} \le \frac{1}{4}. \tag{14.53}$$

Taking the maximum over columns gives $\|\mathcal{J}_h - 1\|_1 \le 1/4$, which implies $\|\mathcal{J}_h^{-1}\|_1 \le 2$. If c is the required reference coefficient, linear recombination therefore gives

$$O(c) = \sum_j t_j O(\mathsf{Z}_{N,p,d} u_j), \qquad \sum_j |t_j| \le 2e^{\sum_{i<h} \log C^{\mathrm{col}}_{J_{i+1}}} |c|_1.$$

Replace each term on the right by $\mathcal{P}_{N,h}\mathsf{V}_h(e_j)$. This defines a bounded complex observable $\mathcal{B}'_{N,p}$; its bound depends on h, d, c, and hence may depend on δ, but is independent of N, p.

The discrepancy is controlled by replicating all its reflected copies before taking the marginal kernel. Formula (14.22) applies to the fine insertions; at level h multiply the source density by the corresponding coarse factors. The conditional factorization above identifies the coefficient at each mark with $\mathcal{P}_{N,h}\mathsf{V}_h(e_j) - O(\mathsf{Z} u_j)$. The probability kernel supplies no additional supremum factor. Weighted coefficient extraction and (14.2) then give, after recombination,

$$\|\mathcal{B}'_{N,p} - O(c)\|_{\mathcal{C}^{\mathrm{tile}}_k} \le C_{k,m,d,c}\, e^{\sum_{i<h} \log C^{\mathrm{col}}_{J_{i+1}}} (1 + C^{\mathrm{col}}_{J_h}) e^{-(s_h - s_m)/8}. \tag{14.54}$$

The prefactor has subpower growth in a_h^{-1} by (14.11); the last factor tends to zero. Choose h so that this expression is at most δ, before choosing the cutoff. This also meets the stencil-fit condition.

For real c, the original insertion is real by Lemmas 10.2 and 14.8. Set $\mathcal{B} = \operatorname{Re}\mathcal{B}'$. In the centered cell,

$$\mathcal{B} - O(c) = \tfrac{1}{2}(\mathcal{B}' - O(c)) + \tfrac{1}{2}\overline{\mathcal{B}' - O(c)}.$$

The chessboard seminorm and its invariance under conjugation show that this does not increase (14.54); the supremum bound also does not increase. Translation, signed permutation and conjugation use the same finite collection of stencils and bounds. This proves the lemma on the stated ordered tails. □

Fix a degree bound d, a lower level m and an ordered extraction. A real grid word is a list $w = ((d_i, c_i), x_i)_{i=1}^n$ with $d_i \le d$, $\bar{c}_i = c_i$ and distinct spacing-a_k positions. We assume its configuration is generic. Let ϑ denote linear time reflection on positions and labels, without complex conjugation of scalar coefficients. For an ordered word w, write w^* for conjugation and reversal of order, and $\theta w = \vartheta w^*$ for the antilinear reflection of Osterwalder and Schrader. Put

$$F_{N,p}(w) = \prod_i O^{(d_i)}_{N,p}(c_i, x_i/a_N), \qquad z_j(T) = T j a_k(1,1,1,1),$$

$$f_{j,N,p,T} = F_{N,p}(\tau_{z_j(T)} w) - F_{N,p}(\tau_{z_j(T)} \vartheta w).$$

These are the real word differences used in the off-diagonal variance estimate in Section 17.

Lemma 14.30 (Bounded real word approximations)**.** *There is $k \geq m$, chosen before w, such that every such word has, for each $\varepsilon > 0$, measurable real approximations $g_{j,N,p,T}$ and $R_\varepsilon < \infty$ with $|g_{j,N,p,T}| \leq R_\varepsilon$ for all N, p, T, j. For each $i < j$ and every sufficiently large integer T, eventually in the ordered limits,*

$$|\langle g_{i,N,p,T} g_{j,N,p,T}\rangle_{N,p} - \mathrm{Re}\langle f_{i,N,p,T} f_{j,N,p,T}\rangle_{N,p}| \leq \varepsilon.$$

Proof. Each insertion of the word is replaced by its bounded approximation from Lemma 14.29; the chessboard inequality (14.19), with the seminorm bounds of Lemmas 14.15 and 14.11, controls the replacement error by seminorms alone. Choose $k \geq \max(m, \max_{e \leq d} m_{\mathrm{ref}}(e))$. It depends only on d, m, before the length, labels and positions of w. All its finer comparison levels remain available by Lemma 14.2.

Include every label in w and its signed and conjugated images in a finite list. Lemma 14.15 gives a bound C for their centered seminorms at any one fixed sufficiently fine level. Choose that level so that its cells and the margins required by Lemma 14.11 fit between distinct spacing-a_k points. Apply Lemma 14.29 with error δ at a still finer centered level, preserving these margins. Denote its real bounded approximations by $\mathcal{B}_i$. The same family, translated to any anchor, is used for every copy of the word. Its seminorms are at most $C + \delta$, independently of its possibly much larger supremum bound.

For a product of n_{word} such insertions at distinct grid sites, telescope the products, replacing one factor at a time. Put all n_{word} factors into distinct cells with the chosen margins. The interior-seminorm comparison and (14.19) give

$$\left| \left\langle \prod_{a=1}^{n_{\mathrm{word}}} \mathcal{B}_a - \prod_{a=1}^{n_{\mathrm{word}}} O_a \right\rangle_{N,p} \right| \leq n_{\mathrm{word}} \delta (C + \delta)^{n_{\mathrm{word}} - 1}. \tag{14.55}$$

Only the seminorms, not the supremum bounds on $\mathcal{B}_a$, occur on the right. For each fixed finite configuration, sufficiently large periods avoid wrapping identifications; its stencils then lie in the prescribed cells on the inner cutoff tail. This explains the order of the two eventualities.

Let $G_{j,N,p,T}$ be the product of the $\mathcal{B}_i$ for the translated word, and let $G^\theta_{j,N,p,T}$ be the corresponding product for its reflected word. Set $g_j = G_j - G^\theta_j$. Choose M_i^{bnd} to bound both the original and reflected approximations in slot i. Then

$$|g_j| \leq 2 \prod_i M_i^{\mathrm{bnd}} =: R_\varepsilon,$$

independently of N, p, T, j. Every g_j is real and measurable. Set $g_j = 0$ outside the ordered tails on which the coarse approximations are defined, so this supremum bound holds for all N, p, T, j. For fixed $i < j$, sufficiently large T separates the two translated finite lists, for each of the four combinations of a word and a reflected word. Indeed their first coordinates differ by $T(j - i)a_k$ plus a fixed finite difference. Each list is itself distinct because the original word is generic and reflection is injective.

Expand $g_i g_j$ and $f_i f_j$ into their four product terms and apply (14.55) with $n_{\mathrm{word}} = 2n$. Choose $0 < \delta \leq 1$ so that $4(2n)\delta(C + 1)^{2n-1} \leq \varepsilon$. Taking real parts cannot increase the error. Intersecting the four eventual sets proves the assertion. The empty word has zero difference and may be assigned $g_j = 0$. There is no common cutoff threshold in T, i, j; a finite list of pairs is handled by intersecting its eventual sets. The choice gives such a grid level above every prescribed lower level m, which is the cofinality required in Section 17. □

14.3 Full ultraviolet convergence

We prove convergence as the ultraviolet cutoff increases, at fixed volume. The argument is a triangle inequality through a common fixed level m. At a separated configuration x on a fixed common grid,

$$|M_{N,p} - M_{N',p}| \le \underbrace{|M_{N,p} - C_{N,p,m}|}_{\text{stopping error}} + \underbrace{|C_{N,p,m} - C_{N',p,m}|}_{\text{fixed-level difference}} + \underbrace{|C_{N',p,m} - M_{N',p}|}_{\text{stopping error}}, \tag{14.56}$$

where $M_{N,p} = M_{N,p}(v; x/a_N)$ is the cutoff moment and $C_{N,p,m} = C_{N,p,m}(v; x)$ is its approximation computed at level m from the unmarked representation and the one-mark families (14.58). Lemma 14.32 bounds each stopping error by $2F_0 e^{-B_m/64}$, with F_0 independent of m. Lemma 14.31 shows that the fixed-level difference tends to zero as $N, N' \to \infty$ at fixed m. So choose m to make both stopping errors small, hold it fixed, and then send both cutoffs to infinity.

The middle term compares two cutoff histories at the same level m. This differs from the contraction of the marked step, which compares two coefficient graphs carried by one unmarked history. Table 8 lists the discrepancies in the middle term and the estimate that controls each. The third row compares with the limiting unmarked history. This history exists: Lemma 13.24

Discrepancy between cutoffs N, N'	Controlled by
unmarked couplings, activities and paths at a fixed level	convergence along the full cutoff sequence (Lemma 13.24)
two marked graphs on one unmarked history	one marked step halves their distance (Lemma 14.6(iii))
one marked step on the history at cutoff N and on the limiting history	continuity in the unmarked coordinates; the discrepancy $\delta_i(N, N')$ tends to zero at fixed i
scalar normalization $e^{v_{\mathrm{sc},m}\lvert\Lambda_{n_{p,m}}\rvert}$	not compared: it cancels in the normalized ratios (14.57) and (14.59)
scalar-free partition integral	Cauchy by unmarked convergence; bounded below on a cutoff tail by restriction to small links in an axial gauge

Table 8: The discrepancies in the fixed-level difference $C_{N,p,m} - C_{N',p,m}$.

constructs subsequential limits of the coupling and activity histories and proves their uniqueness, hence convergence along the full cutoff sequence; that is why the unmarked convergence is proved first. Fix p. In the normalized density at a fixed level, the scalar normalization appears as the same field-independent factor in numerator and denominator:

$$\frac{\operatorname{dens}\rho_{N,p,m}}{\int \operatorname{dens}\rho_{N,p,m}} = \frac{e^{v_{\mathrm{sc},m}\lvert\Lambda_{n_{p,m}}\rvert}\, e^{-\mathcal{V}^{\flat}_{N,p,m}}\mathcal{F}_{N,p,m}}{e^{v_{\mathrm{sc},m}\lvert\Lambda_{n_{p,m}}\rvert}\int e^{-\mathcal{V}^{\flat}_{N,p,m}}\mathcal{F}_{N,p,m}} = \frac{e^{-\mathcal{V}^{\flat}_{N,p,m}}\mathcal{F}_{N,p,m}}{\int e^{-\mathcal{V}^{\flat}_{N,p,m}}\mathcal{F}_{N,p,m}}, \tag{14.57}$$

where $\mathcal{V}^{\flat}_{N,p,m} = P^{\mathrm{path}}_{N,m} + \sum_i \mathbf{1}_{X_i\,\mathrm{healthy}} E^{\mathrm{reg}}_i = \mathcal{V}_{\rho_{N,p,m}} + v_{\mathrm{sc},m}\lvert\Lambda_{n_{p,m}}\rvert$ is the scalar-free part of the potential, and $\mathcal{F}_{N,p,m}$ is the forced sum, both as in Definition 5.2. The factor $e^{v_{\mathrm{sc},m}\lvert\Lambda_{n_{p,m}}\rvert}$ depends on the cutoff, but it has canceled, so no convergence of that factor is proved or used.

For a fixed finite insertion-label list and a separated common-grid configuration x, let $o_{i,m} = x_i/a_m$ be its sites at all sufficiently fine levels. They are read modulo the period; their parents agree at consecutive levels. In (14.23), form $\mathcal{H}^0_{N,p,m}$ by setting $H_{m,X} = K^{\ge 2}_{m,Y} = 0$, keeping the complete singleton functions $f_{i,m}$ and $k_{i,m,Y}$. Define

$$C_{N,p,m}(v; x) = \frac{\int [z_{\{1,\dots,n\}}]\mathcal{H}^0_{N,p,m}(z; U)\, dU}{\int \operatorname{dens}\rho_{N,p,m}\, dU}. \tag{14.58}$$

The numerator is specified by the unmarked representation and the one-mark families alone. Both integrands contain the factor $e^{v_{\mathrm{sc},m}|\Lambda_{n_{p,m}}|}$, the numerator through Φ_m, and it cancels as in (14.57):

$$\begin{aligned} C_{N,p,m}(v;x) &= \frac{e^{v_{\mathrm{sc},m}|\Lambda_{n_{p,m}}|}\int [z_{\{1,\dots,n\}}]e^{-v_{\mathrm{sc},m}|\Lambda_{n_{p,m}}|}\mathcal{H}^0_{N,p,m}\,dU}{e^{v_{\mathrm{sc},m}|\Lambda_{n_{p,m}}|}\int e^{-v_{\mathrm{sc},m}|\Lambda_{n_{p,m}}|}\,\mathrm{dens}\,\rho_{N,p,m}\,dU} \\ &= \frac{\int [z_{\{1,\dots,n\}}]e^{-v_{\mathrm{sc},m}|\Lambda_{n_{p,m}}|}\mathcal{H}^0_{N,p,m}\,dU}{\int e^{-v_{\mathrm{sc},m}|\Lambda_{n_{p,m}}|}\,\mathrm{dens}\,\rho_{N,p,m}\,dU}. \end{aligned} \tag{14.59}$$

After the cancellation both integrands are scalar-free; they are formed from the scalar-free density and the singleton functions, whose convergence is proved below. The difference from the exact source is the stopping error, estimated in Lemma 14.32.

The first lemma treats one fixed coarse level. The unmarked trajectories converge at that level; contraction of the marked recursion then gives convergence of the singleton insertions.

Lemma 14.31 (Fixed-level marked convergence)**.** *For every fixed p, v, x as above and every sufficiently fine fixed level m, the scalar-free unmarked density and the complete singleton functions in (14.23) are Cauchy in their common function spaces as $N \to \infty$. Their normalized integrals satisfy*

$$C_{N,p,m}(v;x) \text{ is Cauchy as } N \to \infty. \tag{14.60}$$

The scalar-free partition integral is bounded below by a positive constant depending on p, m.

Proof. The proof establishes the rows of Table 8. Lemma 13.24 gives full-sequence convergence of the scalar couplings and scalar-free activities in fixed weighted spaces at each fixed p, m, and includes the prescribed paths with the strict reserve. Thus the cutoff-dependent path arrays converge in the same charts. The scalar normalization is excluded, as in (14.57).

Use $\kappa_i = (\epsilon_i^{\mathrm{for}})^{1/2}$ in the remainder norm (14.16). For coefficient graphs $g, \widetilde{g}$ at level i put

$$\mathrm{dist}_i^{\mathrm{gr}}(g,\widetilde{g}) = \sup_{\mathbf{x}\neq 0} \frac{\|g(\mathbf{x}) - \widetilde{g}(\mathbf{x})\|_{\mathrm{rem}}}{C_{J_i}^{\mathrm{col}}|\mathbf{x}|_1},$$

and let $\mathrm{dist}_i^{\mathrm{gr}}(N, N')$ be this distance between the graphs selected at cutoffs N, N', after the fixed-level identification of their unmarked coordinates. The graph bound in the definition of a coefficient graph gives $\mathrm{dist}_i^{\mathrm{gr}} \le 2$; by Lemma 14.6(iii), one step on a common unmarked representation halves it. For two different cutoffs, first compare both steps with a step on their limiting unmarked coordinates. The one-mark local integrals, their Taylor coefficients and the forced regrouping are continuous in these coordinates. To check this continuity, retain the common summable majorants in (14.17); each finite local integral is continuous in its coefficients and chart, and the remaining tails are bounded by these same summable majorants. The inverse coefficient maps are continuous because their inverses have the common bound already proved. Hence for each fixed i the extra discrepancy $\delta_i(N, N')$ tends to zero as both cutoffs tend to infinity.

Iterating from a remote fixed level h down to m gives the estimate

$$\mathrm{dist}_m^{\mathrm{gr}}(N, N') \le 2^{1-(h-m)} + \sum_{i=m}^{h-1} 2^{-(i-m)}\delta_i(N, N'). \tag{14.61}$$

First choose $h - m$ large enough to make the geometric term small. Keeping this h fixed, let $N, N' \to \infty$; the finite sum of discrepancies then tends to zero. This proves the Cauchy property at level m. The finitely many coefficient maps between the reference level and m converge as well.

Solving their fixed-dimensional linear systems therefore gives Cauchy polynomial coordinates for each fixed reference label. Substituting in the graph gives the complete singleton functions.

On the fixed finite torus, weighted convergence bounds the sums of the regular terms uniformly and bounds the forced terms after grouping identical supports and inventories. There are only finitely many such supports and inventories. The resulting scalar-free density and the singleton functions are uniformly Cauchy on their fixed health domains. The health threshold at level m is fixed; no moving indicator is being differentiated or replaced. Formula (14.23), with its higher modes set to zero, is a finite polynomial in the grouped functions and their exponentials. Its scalar-free coefficients are consequently uniformly Cauchy and integrable against Haar probability measure.

The denominator has a common positive lower bound. Restrict to an axial tree gauge with all free links of chord at most $g/4$, choosing $g > 0$ below every admitted $\beta_{N,p,m}^{-1/2}$ on the retained fixed-level tail. The forced sum on this set is one, and the scalar-free action is uniformly bounded above there. Its product Haar measure is positive. The same lower bound applies to all cutoffs on that tail. Dividing the Cauchy numerator integrals by these positive Cauchy denominators proves (14.60). □

The source density transfers exactly, but its singleton part does not: connected components with two or more marks remain. Two marks show what the stopping removes and what it keeps. Write $H_{m,X} = z_1 z_2 h_X$ and $K^{\geq 2}_{m,Y} = z_1 z_2 \kappa_Y$, put $K^{\mathcal{B}} = \prod_{Y\in\mathcal{B}} K_{m,Y}$, and suppress m and U on the singleton functions. Since $z_1^2 = z_2^2 = 0$, expanding the exponential and the product in (14.23) gives

$$\begin{aligned}[z_1 z_2]\mathcal{H}_m = {} & e^{\Phi_m} \sum_{\mathcal{A}} \Big[f_1 f_2 K^{\mathcal{A}} + \sum_{Y\in\mathcal{A}} (f_1 k_{2,Y} + f_2 k_{1,Y}) K^{\mathcal{A}\setminus\{Y\}} + \sum_{Y\neq Y'} k_{1,Y} k_{2,Y'} K^{\mathcal{A}\setminus\{Y,Y'\}} \Big] \\ & + e^{\Phi_m} \sum_{\mathcal{A}} \Big[\Big(\sum_X h_X \Big) K^{\mathcal{A}} + \sum_{Y\in\mathcal{A}} \kappa_Y K^{\mathcal{A}\setminus\{Y\}} \Big],\end{aligned} \tag{14.62}$$

with sums over eligible families and ordered pairs of distinct $Y, Y' \in \mathcal{A}$. The first line is $[z_1 z_2]\mathcal{H}^0_{N,p,m}$. It keeps the marked forced terms $k_{i,Y}$, which enter through the same eligible families as the unmarked activities. The second line is removed, and its normalized integral is the stopping error for two marks. On a configuration without flags only the empty family is eligible, and the coefficient reduces to $e^{\Phi_m}(f_1 f_2 + \sum_X h_X)$.

The next estimate bounds the effect of omitting all components with two or more marks. Its quantifiers are those needed in (14.56), where m is chosen before the cutoffs; so the constant F_0 below must not depend on m or N. Convergence at each fixed level does not supply such a bound, since it concerns $C_{N,p,m}$ and not its distance from the cutoff moment.

Lemma 14.32 (Stopping estimate)**.** *For fixed p, v, x, there are $F_0 < \infty$ and m_0 such that, for each $m \geq m_0$ and every sufficiently large N,*

$$|M_{N,p}(v; x/a_N) - C_{N,p,m}(v; x)| \leq 2F_0 e^{-B_m/64}. \tag{14.63}$$

The constant F_0 is independent of m, N; the cutoff threshold may depend on m.

Every mode with at least two marks has support covering number at least $B_m/64$, by Lemma 14.12. Multiply all such modes, and only those, by a complex z. The complete singleton functions, the original forced eligibility tests and the unmarked representation are unchanged. Two units of support exponent pay for $|z| < r_m = e^{B_m/64}$, so the normalized integral $F_m(z)$ is holomorphic on this disk, with $F_m(1)$ the cutoff moment and $F_m(0) = C_{N,p,m}$. Carried back to one

base level k_0 chosen before m, it is bounded by a constant F_0 computed at that level. Cauchy's estimate on circles of radius $r_m/2$ centered on $[0,1]$ gives

$$|F_m(1) - F_m(0)| \le \frac{F_0}{r_m/2} = 2F_0 e^{-B_m/64}.$$

The error is the bound on the disk divided by half its radius; it is small because the radius grows with m while the bound does not.

Proof. The expansion is that of (14.23) for the exact source evolution (14.22); the bounds at the base level come from Lemma 14.14 and the singleton bounds of Lemma 14.31. Choose a base level k_0 above the reference levels and the grid containing x. At all sufficiently fine levels, distinct ancestral sites have separation at least B_i^2: their separation is a fixed positive physical number divided by a_i, while B_i^2 grows only as a power of $\log \widehat{\beta}_i$. Their occupancy is one. Enlarge k_0 so these assertions and all required positive source allowances hold thereafter.

Choose mark weights $\mathsf{w}_i > 0$ at this base level so the weighted complete singleton sizes lie below those allowances. This choice is independent of N: Lemma 14.31 bounds each singleton uniformly along the fixed-level cutoff tail. Let $s_0^{\mathrm{reg}}, s_0^{\mathrm{for}}$ bound the weighted regular singleton supremum and the sum of forced singleton suprema at k_0. Set

$$F_0 = \frac{\exp\{|\Lambda_{n_{p,k_0}}| q_{k_0} + n(s_0^{\mathrm{reg}} + s_0^{\mathrm{for}} + 2)\}}{\prod_i \mathsf{w}_i}. \tag{14.64}$$

The numerator is the normalized-integral bound of the several-mark expansion, now with these fixed base weights. It depends on the base level, not on the later stopping level.

At level $m > k_0$, every nonzero mode of $H_{m,X}$ or $K_{m,Y}^{\ge 2}$ contains at least two marks and its support contains their anchors. The covering estimate of Lemma 14.12 gives

$$\mathrm{cov}_{B_m}(X) \ge B_m/64.$$

Multiply every one of these higher-mode functions by a common complex variable z. For $|z| \le e^{B_m/64}$, its amplitude increase is absorbed by decreasing the support exponent by two:

$$|z| e^{(\sigma-2)\,\mathrm{cov}_{B_m}(X)} \le e^{\sigma\,\mathrm{cov}_{B_m}(X)}.$$

Singleton functions are not multiplied by z. The source representation therefore remains integrable on $|z| < e^{B_m/64}$, with its prescribed singleton functions, original forced eligibility tests and two fewer units of support exponent. Here $\sigma = \sigma_{J_m} = 100$. For $s \ge 0$ and $98 \le \tau \le 100$, $e^{\tau s} \le e^{100 s}$, so every regular or forced weighted sum decreases when its exponent decreases. The coefficientwise rooted clause of Lemma 13.5 bounds the connected regular terms at exponent $\sigma + 3$ and the forced terms at $\sigma + 2$. The marked product factors are paid at the entry chosen in item 5. Applying these bounds in the weighted source norm of (14.23) leaves the two units used here, with final exponent 98.

Carry this modified density back to the same base level k_0, using the rooted clause of Lemma 13.5 with all support exponents lowered by two, as permitted in its statement. Every nonzero mode contains only marks whose anchors lie in its support, so multiplying modes retains their positive weights without adding a mark outside that support. At the base level, the singleton exponential and forced factors contribute at most $n(s_0^{\mathrm{reg}} + s_0^{\mathrm{for}})$ to the logarithmic majorant; the remaining connected regular and forced sums contribute at most $2n$. Equation (14.2) contributes $|\Lambda_{n_{p,k_0}}| q_{k_0}$. Coefficient

extraction divides by $\prod_i \mathsf{w}_i$, giving exactly (14.64). Absolute convergence on $|z| < e^{B_m/64}$ makes the normalized full-mode integral $F_m(z)$ holomorphic there, with

$$|F_m(z)| \le F_0.$$

This avoids the level-m volume factor that would arise from estimating the normalized integral only where it was stopped.

For $e^{B_m/64} > 2$, the Cauchy derivative estimate on circles of radius $e^{B_m/64}/2$ centered on $[0,1]$ gives

$$|F_m(1) - F_m(0)| \le 2F_0 e^{-B_m/64}.$$

At $z = 1$ this is the exact multiple-source integral. Its equality to the original cutoff moment follows from Fubini and the probability normalization of every block kernel in (14.22). At $z = 0$ it is exactly (14.58). Substitution proves the estimate. For each fixed m, the required bound on its singleton functions holds on a cutoff tail by Lemma 14.31; it changes the threshold but leaves F_0 fixed. □

The fixed-level and stopping estimates, together with the positional bounds already proved, give the full limit.

Lemma 14.33 (Lattice-point convergence)**.** *For every p, n, v, there is $f_{p,v} \in C^\infty(\mathrm{NoHit}_{L_p,n})$ such that, for every $\boldsymbol{l}$, compact $K \subset \mathrm{NoHit}_{L_p,n}$ and $\varepsilon > 0$, some $N_0(p, n, v, \boldsymbol{l}, K, \varepsilon)$ gives*

$$|\Delta_{\boldsymbol{l}} M_{N,p}(v; x/a_N) - \partial_{\boldsymbol{l}} f_{p,v}(x)| < \varepsilon$$

at every spacing-a_N point of K in the periodic lift, for every $N \ge N_0$.

Proof. Pointwise convergence on the common grids is (14.56); the difference bounds of Lemma 14.28 extend it to uniform convergence of all derivatives. By Lemmas 14.31–14.32, the right side of (14.56) is at most $4F_0 e^{-B_m/64} + |C_{N,p,m} - C_{N',p,m}|$; taking first m and then N, N' large proves the Cauchy property at every separated configuration on a fixed common grid. Since every ratio is an integer, each coarser grid is contained in every finer grid. The union of these grids is dense in the periodic noncollision domain.

We next extend this limit and identify its derivatives. Choose a compact neighbourhood K' of K contained in that domain, and a positive lower bound for its torus separation. Lemma 14.28 bounds the lattice functions and all their fixed differences there. One more difference gives, by telescoping along coordinate lattice paths,

$$|G_N(z) - G_N(z')| \le B\,|z - z'|_1$$

for nearby physical grid points, where $G_N = \Delta_{\boldsymbol{l}} M_{N,p}$. For order zero, a finite dense grid in K' and this estimate extend the pointwise Cauchy property to uniform convergence on K. The multilinear interpolants obey the same bound, with an additional $O(a_N)$ displacement to their vertices. The resulting limit $f_{p,v}$ is continuous and periodic.

Suppose G_N already converges locally uniformly to g. Bounds for two more differences compare its first forward difference with a difference over a fixed physical interval. Put $h_N = Q_N a_N \to h > 0$, with $Q_N \in \mathbb{N}$. Summing successive first differences and bounding their variation by the second difference gives

$$\left| \frac{G_N(z + h_N e) - G_N(z)}{h_N} - \Delta_e G_N(z) \right| \le \tfrac{1}{2} B h_N. \tag{14.65}$$

Choose h small and then let $N \to \infty$. The endpoint values converge uniformly, so (14.65) proves that $\Delta_e G_N$ is uniformly Cauchy on K. Call its continuous limit g_e. The exact telescoping identity on grid segments, followed by Riemann summation of the uniformly convergent first differences, gives

$$g(z + te) - g(z) = \int_0^t g_e(z + ue)\, du.$$

Thus $\partial_e g = g_e$. Induct on the length of the difference list. Every required additional order is available from Lemma 14.28, so all derivative limits exist and $f_{p,v}$ is smooth. Finite covers by such neighbourhoods give the asserted uniform statement on K. □

The same limit applies when observation points move across periodic representative boundaries. Smearing then follows from a compact-support Riemann sum.

Lemma 14.34 (Interpolation and smearing)**.** *For fixed $p, n, v, \boldsymbol{l}$ and $x_N \to x \in \mathrm{NoHit}_{L_p,n}$, the multilinear interpolants of $\Delta_{\boldsymbol{l}} M_{N,p}$ at x_N converge to $\partial_{\boldsymbol{l}} f_{p,v}(x)$. For every p-admissible separated test t,*

$$t\partial_{\boldsymbol{l}} f_{p,v} \in L^1, \qquad a_N^{4n} \sum_y t(\hat{y}) \Delta_{\boldsymbol{l}} M_{N,p}(v; y) \longrightarrow \int t\partial_{\boldsymbol{l}} f_{p,v}.$$

Proof. Both assertions follow from the locally uniform convergence of Lemma 14.33, with the placement and moment conventions of Definition 2.4 and admissibility from Section 2.2. Choose a compact neighbourhood of x inside the periodic noncollision domain. Every vertex used by the interpolant at x_N lies there eventually and tends to x. The interpolant is a convex combination of the vertex values. Uniform convergence at these vertices and continuity of the limiting derivative give the moving-point assertion. Work on the periodic lift throughout; no continuity of the choice of centered representatives is needed.

The closed support of t is a compact subset of that domain, strictly inside the chosen period box. The smooth limiting derivative is bounded there, so its product with t is integrable. On a fixed compact neighbourhood, Lemma 14.33 makes the supremum of the difference between the lattice values and the limiting derivative tend to zero. Its contribution to the sum is at most that supremum times $\|t\|_\infty$ times the total volume of the grid cubes meeting the support. This volume is bounded independently of N. The remaining sum is the Riemann sum for the continuous compactly supported function $t\partial_{\boldsymbol{l}} f_{p,v}$ and converges to its integral. □

Proof of Theorem 5.9. Set $M_{p,v}^{\mathrm{UV}} = f_{p,v}$. Lemma 14.33 gives its smoothness on the full periodic noncollision domain, and Lemma 14.34 gives the asserted interpolation and smearing limits for every fixed derivative list. Thus every inner limit in Definition 2.6 exists at every admissible fixed volume along the full cutoff sequence. □

15 Infinite volume and field calculus

We pass from the fixed-volume kernels of Theorem 5.9 (fixed-volume limit) to correlations on $\mathbb{R}^4$. The order of limits is that of Definition 2.6 and Figure 1: at every fixed admissible volume the spacing tends to zero along the full cutoff sequence; only then does the volume grow, along one refinement of the prescribed volume sequence that serves every field list, derivative and test. Section 15.1 extracts this volume limit, with all derivatives, from the uniform bounds of Proposition 5.8 (Lemma 15.1). Section 15.2 bounds separated correlations cell by cell and shows that negligible errors vanish in them (Lemma 15.3). Section 15.3 identifies the quotient fields: it constructs coefficient maps

for inclusions and positional derivatives and proves $\dim_{\mathbb{C}} F_d = \dim_{\mathbb{C}} M_d^{\mathrm{Max}}$ (Lemma 15.5, which proves Proposition 6.4). Section 15.4 passes reflection positivity to the limit and proves the power bounds of E7 (Lemma 15.6, which completes Theorem 6.1). Section 15.5 bounds repeated reflected insertions, the input for the slab operators of Lemma 16.12.

The Wilson family and separated tests are those of Section 2.2, the insertion labels those of Section 2.3, and the differences those of Section 5.6. Write $M_{p,v} = M_{p,v}^{\mathrm{UV}}$ and $\langle F\rangle_{N,p} = \int F\, d\mu_{N,p}$. Constants may depend on the fixed construction; further dependence is indicated.

15.1 Volume limits

Fix a strictly increasing $\mathsf{p} : \mathbb{N} \to \mathbb{N}$. Proposition 5.8 bounds the lattice kernels and their differences on separated sets, uniformly in the volume; one further difference makes these bounds equicontinuous.

Lemma 15.1 (Volume limits)**.** *There are a strictly increasing* q *and a complex multilinear table* S *such that, with* $P = \mathsf{p} \circ \mathsf{q}$*, for every* n*, label list* v *and difference list* $\boldsymbol{l}$*,*

$$S_n(v;\cdot) \in C^\infty(\mathrm{Conf}_n), \qquad \partial_{\boldsymbol{l}} M_{P(k),v} \longrightarrow \partial_{\boldsymbol{l}} S_n(v;\cdot) \quad \text{locally uniformly on } \mathrm{Conf}_n\,. \tag{15.1}$$

Every compact $K \subset \mathrm{Conf}_n$ *eventually lies in* $\mathrm{NoHit}_{L_{P(k)},n}$*, and for every separated test* f*,*

$$C^{\mathrm{UV}}_{P(k)}(v;f) \longrightarrow \int f(x) S_n(v;x)\, dx. \tag{15.2}$$

Both convergences persist along every further refinement of q*.*

Proof. Fix compact sets $K \subset K' \subset \mathrm{Conf}_n$, with K' a neighbourhood of K. For large p, K' lies in $\mathrm{NoHit}_{L_p,n}$ with torus separation at least some $2r > 0$ independent of p. Proposition 5.8 bounds $\Delta_{\boldsymbol{l}} M^{\mathrm{lat}}_{N,p}$ by $B(v,\boldsymbol{l},r)$ near K' for large N, uniformly in p, and by Lemma 14.34 the multilinear interpolants converge at each fixed p to $\partial_{\boldsymbol{l}} M_{p,v}$. Hence

$$\sup_{p\ge p_0}\sup_{x\in K'} |\partial_{\boldsymbol{l}} M_{p,v}(x)| \le B(v,\boldsymbol{l},r), \tag{15.3}$$

although the cutoff threshold may depend on p. One more derivative gives equicontinuity on K. Arzelà–Ascoli [29, Theorem 7.25] and a diagonal extraction over compact exhaustions of Conf_n, arities, degree lists, basis coefficient lists and difference lists, a countable family, give q. Multilinearity extends the limits to every label list, and integration by parts against compactly supported tests identifies the limit of each derivative with the derivative of the limit. This proves (15.1).

A compact separated set has bounded diameter, so it eventually lies in $\mathrm{NoHit}_{L_{P(k)},n}$. Theorem 5.9 gives, at every admissible p, the full inner limit $C^{\mathrm{UV}}_p(v;f) = \int f\, M_{p,v}$; the support of f is eventually admissible along P, and local uniform convergence proves (15.2). Both properties pass to subsequences. □

Lemma 15.5 refines q once more without changing S; from then on P denotes the final refinement. At each volume the kernel limits remain the full ultraviolet limits of Theorem 5.9: no spacing subsequence enters (15.2).

Symmetries of the limit. Permutation symmetry, the empty correlation, deletion of the distinguished unit, conjugation and signed-permutation covariance hold on the lattice once N exceeds the reference levels (Definition 2.8, Lemma 14.8). Approximating separated configurations by lattice points, Lemma 14.34 and (15.1) pass them to S. For translations, approximate the translation vector on the same lattice; translation invariance and (15.3) give arbitrary real translations. This proves E1–E3, and (15.1) gives E6.

15.2 Separated cells and negligible errors

Chessboard seminorms, interior observables, signed images, aligned tilings, anchors and margins are those of Definition 14.10 and of the paragraphs before Lemma 14.11 and Definition 14.16. For observables $F_1, \dots, F_n$ in distinct cells $C_1, \dots, C_n$ of a translated bond tiling $\mathcal{T}$, each interior to its cell, write $\|F_h\|_{\mathcal{T}} = \|G_h\|_{\mathcal{T}}$, where G_h is the base-cell observable with $(G_h)_{C_h} = F_h$. The chessboard inequality (14.19), with the constant one in the remaining cells, gives

$$\left| \left\langle \prod_{h=1}^{n} F_h \right\rangle_{N,p} \right| \le \prod_{h=1}^{n} \|F_h\|_{\mathcal{T}} . \tag{15.4}$$

The seminorm measures the cost of repeating one observation throughout the box, so this bound is independent of the volume and of the distances between the cells.

Separated anchors must therefore be placed in distinct cells, away from the boundaries. In each axis, the translations of a tiling of side a that bring one of n anchors within distance w of a cell boundary form, modulo a, at most n intervals of total length $2nw$. If $2nw < a$, some lattice translation avoids them once a_N is small. We call this the strip count. Anchors at torus sup-distance greater than a never share a cell.

Lemma 15.2 (Separated cells)**.** *Let $m \le k$ and $n \ge 1$ with $6na_k < a_m$, and suppose the insertions $O_{N,p}(v_h, \cdot)$, $h \le n$, satisfy the hypothesis of Lemma 14.11 at level k with a common bound B. For all sufficiently large N and all anchors with pairwise torus sup-distance greater than a_m,*

$$\left| \left\langle \prod_{h=1}^{n} O_{N,p}(v_h, y_h) \right\rangle_{N,p} \right| \le B^n .$$

Proof. The strip count with $a = a_m$ and $w = 3a_k$ gives a tiling of side a_m with the anchors in distinct cells, at distance at least $3a_k$ from their boundaries. Lemma 14.11 bounds each seminorm by B, and (15.4) gives the claim. □

A negligible error has a tolerance chosen before the cell scale (Definition 14.16), and this order makes it vanish against each fixed list of spectators.

Lemma 15.3 (Vanishing errors)**.** *For every negligible error E of Definition 14.16, fixed label list v, distinguished slot i and $\delta > 0$, there is $C(v, \delta) < \infty$ such that, eventually in (14.28),*

$$\left| \left\langle E_{N,p}(y_i) \prod_{h \ne i} O_{N,p}(v_h, y_h) \right\rangle_{N,p} \right| \le C(v, \delta)\eta_{N,p} \qquad (\mathrm{sep}_{\mathrm{tor}}(y) \ge \delta) . \tag{15.5}$$

Consequently its smear against any fixed separated test tends to zero in that order.

The error and the spectators receive distinct cells; the error pays its tolerance $\eta_{N,p}$, and each spectator a constant.

Proof. Let n be the arity. Take m with $a_m < \delta$, and let $k \ge m$ be the level that Definition 14.16 assigns to the arity n and the lower level m; the error tiling has side a_k. Choose $k_{\mathrm{sp}} \ge k$, beyond the reference levels of the spectators, with $3a_{k_{\mathrm{sp}}} \le a_k/[8(n+1)]$. Lemma 14.15 bounds the centered seminorms of all signed and conjugated images of the spectators at level k_{sp} by some B_{sp}. The

strip count with $a = a_k$ and $w = a_k/[4(n+1)]$, and $\delta > a_k$, give a tiling of side a_k with the anchors in distinct cells at distance at least w from their boundaries. This margin contains the integer margin (14.29), including its extra lattice unit, and exceeds $3a_{k_{\mathrm{sp}}}$. Translating by a multiple of a_k, we may take the error's cell to be the base cell. Definition 14.16 bounds the seminorm of the error by $\eta_{N,p}$, and Lemma 14.11, with tiling side a_k and centered level k_{sp}, bounds those of the spectators by B_{sp}. Now (15.4) gives (15.5) with $C = B_{\mathrm{sp}}^{n-1}$. On a compact test support the number of lattice tuples times a_N^{4n} is bounded independently of N, p, which proves the smeared assertion. □

15.3 Quotient fields

Two labels represent the same field when every separated correlation gives the same answer for them (Definition 2.12). For a degree d, a slot, fixed spectators v and a separated test f, the one-slot functionals on V_d are

$$L_{N,p}(c) = C^{\mathrm{lat}}_{N,p}\big((d,c),v;f\big), \qquad L(c) = \int f(x_0,x)\, S_{n+1}\big(c(x_0),v(x)\big)\, dx_0\, dx.$$

As S is smooth, $\mathcal{N}_d(S)$ is the intersection of the kernels of all such L: it consists exactly of the labels that no separated smear detects. Four inputs from Section 14.2 identify it, each with one task. (C1) Lemma 14.20 represents a degree-d insertion in a degree $e \geq d$, and (C2) Lemma 14.21 its lattice difference quotient in degree $d+1$, up to negligible errors. (C3) Lemma 14.19 replaces a tame family in $\ker \Pi_{d,N,p}$ by observables with vanishing separated moments, up to a negligible error; *tame* means the subpower bound (14.27). (C4) The free test rows of Lemma 14.22 and the relative rank comparison of Lemma 14.26 keep the quotient from losing dimension. The families supplied by (C1)–(C3) may have unbounded coefficients; Lemma 15.4 turns them into fixed coefficient vectors.

The coefficient limits are taken along a common spacing subsequence n at every volume and a further volume refinement, in the order (14.28). They do not affect the kernel limits, which at each admissible volume remain the full ultraviolet limits required by Definition 2.6. A family in a finite-dimensional space is *bounded* if its norm is bounded eventually in (14.28); an *ordered limit point* of countably many bounded families is their joint limit along an ultrafilter containing the ordered tails. It exists by compactness, and families that converge along the ordered tails keep their limits, so passing to ordered limit points changes neither S nor P.

The next lemma, finite-dimensional linear algebra, converts approximate representatives, which need not be bounded, into one exact representative.

Lemma 15.4 (Representation of limits)**.** *Let V be a finite-dimensional complex normed space and $\mathcal{F}$ a proper filter on an index set. Let Π_u, Π be idempotents on V with $\Pi_u \to \Pi$, and $L_{i,u}, L_i$, $i \in I$, linear functionals with $L_{i,u} \to L_i$ and $\ker \Pi = \bigcap_i \ker L_i$. Suppose $x_u \in V$ satisfy, for every i, along $\mathcal{F}$,*

$$L_{i,u}(x_u) \longrightarrow t_i, \qquad L_{i,u}\big((x_u - \Pi_u x_u)/(1 + \|\Pi_u x_u\|)\big) \longrightarrow 0. \tag{15.6}$$

Then some $z \in V$ satisfies $\Pi z = z$ and $L_i(z) = t_i$ for every i.

The second condition makes the null part of x_u negligible relative to its retained part. If the retained parts $\Pi_u x_u$ escaped to infinity, normalizing them would produce a unit vector in the range of Π annihilated by every L_i: a null vector in the retained range, which $\ker \Pi = \bigcap_i \ker L_i$ excludes.

Proof. Put $b_u = (1 + \|\Pi_u x_u\|)^{-1}$ and $z_u = b_u \Pi_u x_u$, so $b_u + \|z_u\| = 1$ and $\Pi_u z_u = z_u$. The pairs (b_u, z_u) have a cluster point (λ, z) along $\mathcal{F}$ in $[0,1] \times \{\|z\| \leq 1\}$. As $L_{i,u}(z_u) = b_u L_{i,u}(x_u) - L_{i,u}(b_u(x_u - \Pi_u x_u))$,

$$L_i(z_u) - b_u t_i = b_u\big(L_{i,u}(x_u) - t_i\big) - L_{i,u}\big(b_u(x_u - \Pi_u x_u)\big) + (L_i - L_{i,u})(z_u) \longrightarrow 0,$$

and $\Pi z_u - z_u = (\Pi - \Pi_u) z_u \to 0$. Hence $L_i(z) = \lambda t_i$, $\Pi z = z$ and $\lambda + \|z\| = 1$. If $\lambda = 0$, then $z \in \bigcap_i \ker L_i = \ker \Pi$, so $z = \Pi z = 0$, contradicting $\|z\| = 1$. Thus $\lambda > 0$, and z/λ has the required properties. □

Lemma 15.5 (Corrected coefficient maps)**.** *After a strictly increasing refinement of* $\mathfrak{q}$*, which does not change* S*, there are complex linear maps* $\hat\jmath_{d,e} : V_d \to V_e$ *for* $d \le e$ *and* $\hat\partial_{d,a} : V_d \to V_{d+1}$ *such that, for every* $c \in V_d$*, spectator word* $v(x)$ *and* $(x_0, x) \in \mathrm{Conf}_{n+1}$*,*

$$\begin{aligned} S_{n+1}\big(\hat\jmath_{d,e} c(x_0), v(x)\big) &= S_{n+1}\big(c(x_0), v(x)\big), \\ S_{n+1}\big(\hat\partial_{d,a} c(x_0), v(x)\big) &= \partial_{x_0^a} S_{n+1}\big(c(x_0), v(x)\big). \end{aligned} \tag{15.7}$$

Moreover $\dim_{\mathbb{C}} F_d = \dim_{\mathbb{C}} M_d^{\mathrm{Max}}$*, and the induced maps* $\jmath_{d,e}[c] = [\hat\jmath_{d,e} c]$ *and* $\partial_a[c] = [\hat\partial_{d,a} c]$ *satisfy Proposition 6.4.*

The proof identifies the null space with the kernel of a limiting projection, then applies Lemma 15.4 to the families of (C1) and (C2).

Proof. Limiting projections. Bounded coefficient families are tame: Lemma 14.7 bounds their inverse terminal coordinates by a fixed factor times $\prod_{m<N} C^{\mathrm{col}}_{J_{m+1}}$, and $\log C^{\mathrm{col}}_{J_{m+1}} \le J_{m+1}^{-2} \log \ell_m$ (Section 9) with $J_m \to \infty$ makes this product grow more slowly than every power of a_N^{-1}. A countable diagonal extraction, first in N at every p, then in the volume, retaining (15.1), makes the following converge: the projections of Lemma 14.18, which are bounded by C_d^{Π}, the comparison lifts of Lemma 14.26, and the couplings $\beta_{N,p,k} \in \mathfrak{B}_k$, $k \in \mathbb{N}$, of Theorem 5.6. In operator norm,

$$\Pi_{d,\mathfrak{n}(j),p} \longrightarrow \Pi_d^{(p)}, \qquad \Pi_d^{(P(k))} \longrightarrow \Pi_d, \qquad \Pi_d^2 = \Pi_d\,, \tag{15.8}$$

and by (14.28) this iterated limit is an ordered limit.

The null space. By finite dimensionality, the full fixed-volume limits and (15.2), $L_{N,p} \to L$ in operator norm in the ordered limit. For fixed c, the bounded family $c - \Pi_{d,N,p} c$ is tame and lies in $\ker \Pi_{d,N,p}$, so (C3) and Lemma 15.3 give $L_{N,p}(c - \Pi_{d,N,p} c) \to 0$. Thus $L((1 - \Pi_d)c) = 0$ for every L:

$$\ker \Pi_d \subset \mathcal{N}_d(S). \tag{15.9}$$

For the reverse inequality, use (C4). Rearranging (14.39) isolates $\mathcal{G}_I^\nu$; every other term contains a nonempty R_J, whose expectation is at most its seminorm γ_ν by (15.4) with one occupied cell. Thus

$$\left| \left\langle \prod_{h \in I} F_h^{\mathrm{tst}} \right\rangle_{N,p} - \mathcal{G}_I^\nu \right| \le \gamma_\nu \sum_{J \subsetneq I} |\mathcal{G}_J^\nu|, \tag{15.10}$$

and the sum stays bounded as $y^{\nu,i} \to y^i$. Pass to the coefficient and kernel limits. Each entry of the resulting matrix is a fixed spectator functional, vanishing on $\mathcal{N}_d(S)$, applied to the limiting lift of $\mathfrak{m}_j$; so the matrix factors through F_d and has rank at most $\dim F_d$. As $\nu \to \infty$ these matrices tend to the rank-$\dim_{\mathbb{C}} M_d^{\mathrm{Max}}$ matrix $\mathcal{G}$, so $\dim F_d \ge \dim_{\mathbb{C}} M_d^{\mathrm{Max}}$. Ranks of converging idempotents are eventually constant (if $\|P - Q\| < 1$, then P is injective on $\operatorname{ran} Q$), so the dimension inequality of Lemma 14.18 passes to Π_d, and with (15.9), $\dim F_d \le \dim V_d - \dim \ker \Pi_d \le \dim_{\mathbb{C}} M_d^{\mathrm{Max}}$. Hence

$$\mathcal{N}_d(S) = \ker \Pi_d, \qquad \dim_{\mathbb{C}} F_d = \dim_{\mathbb{C}} M_d^{\mathrm{Max}}. \tag{15.11}$$

Targets. For the inclusion of $c \in V_d$, (C1) and Lemma 15.3 give a tame family $w_{N,p} \in V_e$ whose one-slot functionals tend to $t_L = \int fS(c,\dots)$. For the derivative in slot i and axis a, (C2) gives $w_{N,p} \in V_{d+1}$, and torus summation by parts gives exactly

$$a_N^{4n} \sum_y f(\hat{y})\Delta_{i,a} M^{\mathrm{lat}}_{N,p}(v;y) = a_N^{4n} \sum_y \frac{f(\hat{y} - a_N e_{i,a}) - f(\hat{y})}{a_N} M^{\mathrm{lat}}_{N,p}(v;y), \tag{15.12}$$

with $e_{i,a}$ the coordinate vector of slot i, axis a. The backward quotient tends to $-\partial_{i,a} f$ with residual at most $a_N \|\partial_{i,a}^2 f\|_\infty/2$, which Proposition 5.8 at order zero controls, so the target is $t_L = -\int (\partial_{i,a} f) S(c,\dots)$.

Representation. Apply Lemma 15.4 in V_e, with $\mathcal{F}$ the filter of ordered tails $\{(j,k) : k \geq k_0,\ j \geq j_0(k)\}$, $\Pi_u = \Pi_{e,\mathsf{n}(j),P(k)}$ and the one-slot functionals of degree e. They converge as shown above, and $\ker \Pi_e = \bigcap \ker L$ by (15.11). With $\Pi = \Pi_{e,N,p}$, the family $(w - \Pi w)/(1 + \|\Pi w\|)$ lies in $\ker \Pi$ and is tame, being a bounded multiple of w minus a family of norm at most one; (C3) and Lemma 15.3 give the second condition of (15.6). The lemma gives one vector representing the target against every spectator and test at once; no further subsequence is chosen for these uncountably many tests. Choose such vectors on a basis of V_d and extend linearly; (15.1) turns the distributional identities into (15.7).

Induced maps. By (15.7), $\hat{\jmath}_{d,e}$ and $\hat{\partial}_{d,a}$ carry $\mathcal{N}_d(S)$ into null spaces, so they descend to F_d and satisfy the identities of Proposition 6.4; with (15.11) this proves that proposition. The further properties listed after it (injective responses, units, conjugation, H_4 covariance, the positional identity in every slot) follow from these identities, E1 and E2, as explained there. □

Example: the curvature square (continued). Section 10.5 follows the label $c_{F^2} \in V_4$ of $\frac{1}{4}\sum_{a,b,\alpha} F^\alpha_{ab} F^\alpha_{ab}$ to its insertion; in degree four the quotient removes nothing. Every jet coordinate has weight at least two, and since $\mathfrak{g}$ is simple no nonzero $\mathrm{Ad}(G)$-invariant polynomial is linear in a $\mathfrak{g}$-valued jet; so an element of V_4 is a constant plus an invariant quadratic form in the lowest jet $\mathcal{A}_0$, that is, in the curvature at the origin. The Maxwell equations do not constrain the curvature at one point, so no nonzero such polynomial vanishes on every Maxwell solution: the equation space $\mathcal{I}_4$ of Lemma 14.18 is zero. That lemma gives $\Pi_{4,N,p} = 1$, and (15.11) gives $\mathcal{N}_4(S) = 0$. Distinct degree-four labels give distinct fields, and $[c_{F^2}] \neq 0$ in F_4. The quotient first acts on derivatives: the lattice difference quotient of the insertion of c_{F^2} is not a degree-five insertion, (C2) represents it by a tame family in V_5, and Lemma 15.4 replaces that family by one vector, with $\partial_a[c_{F^2}] = [\hat{\partial}_{4,a} c_{F^2}]$. In degree five, $\mathcal{I}_5$ contains the invariant contraction $\sum_{a,b,c,\alpha} F^\alpha_{ab}\, \partial_c F^\alpha_{cb}$ of a curvature with the Maxwell expression. By Lemma 14.18 and (15.11), $\mathcal{N}_5(S) = \ker \Pi_5$ is therefore nonzero, and $\hat{\partial}_{4,a} c_{F^2}$ is determined only modulo these limits of transported equation directions.

15.4 Euclidean properties

Lattice symmetries and positivity pass through the ordered limits; Lemma 15.2 controls separated points as they approach one another, uniformly in their diameter.

Lemma 15.6 (Euclidean properties)**.** *The table S of Lemma 15.1 satisfies E1–E7.*

Proof. E1–E3 and E6 were proved after Lemma 15.1; the positive derivative orders of E7 use (15.7).

Reflection positivity. We prove E5 and deduce E4. Fix joint smooth tests f_i with compact support on positive-side separated configurations for an axis a. Use the bond reflection $x^a \mapsto a_N - x^a$, a lattice symmetry with planes $x^a = a_N/2$ and $x^a = (a_N + L_p)/2$. For large p, N the lattice smears F_i are supported strictly inside the positive half-torus, with stencils avoiding both planes, since the

supports keep a positive distance from $x^a = 0$ and the stencils have physical radius tending to zero. Periodic Wilson reflection positivity [31, Theorem 2.1, pp. 14–15] gives

$$\sum_{i,j} \bar{z}_i z_j \langle \overline{F_i \circ \mathsf{r}_{a,N}} F_j \rangle_{N,p} \geq 0,$$

where $\mathsf{r}_{a,N}$ reflects configurations. Insertion covariance reflects the positions and applies $\mathsf{s}_a^{\mathrm{refl}}$ to the labels; conjugation acts on the labels of the first word. The reflected joint tests converge with all derivatives to $\bar{f}_i \otimes f_j$ with the first group reflected through $x^a = 0$, and all supports lie in one compact separated set. Proposition 5.8 at order zero bounds the effect of the moving tests by their uniform difference times a bounded weighted site count. Lemma 14.34 and (15.2) then give a positive semidefinite limiting matrix; smoothness of S makes every pairing absolutely integrable. This is E5. Concentrating normalized bumps at positive-side words gives E4 by continuity.

Power bound at order zero. Take a nonempty label list and put $d_{\mathrm{pow}} = 1 + \max_i d_i$. Lemma 14.15 with $\varepsilon = 1$, for the finitely many signed and conjugated images of the labels, gives k_0 and C_{ctr} such that their centered seminorms at levels $k \geq k_0$ are at most $C_{\mathrm{ctr}} a_k^{-d_{\mathrm{pow}}}$; these constants precede the separation. The block ratios grow slowly: $\log \ell_u / s_u \to 0$ (Lemma 14.2) gives $\ell_u \leq a_u^{-1}$ eventually, so for some $c_{\mathrm{mesh}} > 0$ and all u

$$a_{u+1} \geq c_{\mathrm{mesh}} a_u^2. \tag{15.13}$$

Thus the first $u \geq k_0$ with $a_u < \delta \leq 1$ satisfies $a_u \geq c_0 \delta^2$, where $c_0 = \min(c_{\mathrm{mesh}}, a_{k_0})$: a discrete scale lies within a fixed power of any prescribed length. Fix $0 < r \leq 1$ and x with $\mathrm{sep}_\infty(x) \geq r$. Choose the first $m \geq k_0$ with $a_m < r/2$, then the first $k \geq k_0$ with $a_k < a_m/[6(n+1)]$; then $k \geq m$ and $a_k \geq c'(n) r^4$. Lattice approximations of x in large volumes have torus separation greater than a_m, so Lemma 15.2 bounds the lattice correlation by $(C_{\mathrm{ctr}} a_k^{-d_{\mathrm{pow}}})^n$. Lemma 14.34 and (15.1) give

$$|S_n(v; x)| \leq C(n, d, c) r^{-4 d_{\mathrm{pow}} n}, \tag{15.14}$$

with constants independent of the positions and the volume. Each insertion costs its centered seminorm at a cell scale comparable to r^4, and the chessboard bound makes the total independent of the diameter of x.

Higher orders. By (15.7), every coordinate derivative is a correlation with fixed degree-raised labels, finitely many at each order. Apply (15.14) to each, take the largest exponent, and bound the Fréchet norm by the sum of the coordinate coefficients in an orthonormal basis of $\mathbb{R}^{4n}$. This proves E7; for arity zero, $S_0 = 1$. □

Proof of Theorem 6.1. Compose the volume refinements of Lemmas 15.1 and 15.5. Equation (15.2) proves ordered convergence, with the full inner limit of Theorem 5.9 at every admissible volume, and Lemma 15.6 proves E1–E7. The coefficient corrections change no kernel limit. □

15.5 Repeated reflected words

Use the OS space of Definition 6.5. For a word w, let w^* reverse it and apply conjugation to its labels, and let $\theta w = \mathsf{s}_4^{\mathrm{refl}} w^*$ be its antilinear OS reflection. For $s > 0$ and a word w with times in $(0, s)$ set

$$A_w v = w \, \tau_{s e_4} v, \qquad w^\dagger = \tau_{s e_4} \theta w;$$

A_w acts on words, and boundedness on OS classes is not assumed. A word has buffer r in a slab of width h_{sl} if its points have pairwise sup-distance at least r and times in $[r, h_{\mathrm{sl}} - r]$. Fix a degree bound d and a coefficient bound $R \geq 0$, and put $Q = 4(d+1)$. The next estimate bounds repeated insertion before any operator extension.

Lemma 15.7 (Repeated reflected words)**.** *There is $C_0 = C_0(d, R) > 0$ such that every nonempty length-n word w with these bounds and buffer $0 < r \le 1$ in a slab of width h_{sl} admits a width $0 < s \le h_{\mathrm{sl}}$ in which w has buffer $r/2$ and, for every separated exterior word v with positive times,*

$$\|[(A_{w^\dagger} A_w)^j v]\|_{\mathrm{OS}}^2 \le B_v [C_0^n (n/r)^{Qn}]^{4j} \qquad (j \ge 0),$$

where $B_v = B_v(w, r, s) < \infty$ is independent of j.

The operators use the chosen width s, and s, C_0 and Q precede v. The main insertions occupy cells of a size chosen from r alone and the exterior insertions a finer aligned tiling, so the cost of the exterior does not grow with the number j of repetitions.

Proof. Enlarge the coefficient ball to contain all signed and conjugated images in degrees at most d. Lemma 14.15 with $\varepsilon = 1$ gives k_* and $B_* > 0$ such that their centered seminorms at levels $k_{\mathrm{w}} \ge k_*$ are at most $B_* a_{k_{\mathrm{w}}}^{-d_{\mathrm{pow}}}$, $d_{\mathrm{pow}} = d + 1$; this precedes n, r, w. Choose the first $m \ge k_*$ with $a_m < r/32$ and put $s = a_m \lfloor h_{\mathrm{sl}}/a_m \rfloor$. As w is nonempty with buffer r, $h_{\mathrm{sl}} \ge 2r$, so $0 < s \le h_{\mathrm{sl}}$ and $h_{\mathrm{sl}} - s < a_m$, and w has buffer $r/2$ in width s. By (15.13), $a_m \ge c_1 r^2$ with $c_1 > 0$ depending on k_*. Fix $0 < \alpha \le 1$ with $20\alpha < c_1$, and choose the first $k_{\mathrm{w}} \ge k_*$ with $a_{k_{\mathrm{w}}} < \alpha (r/n)^2$. Then

$$20 n a_{k_{\mathrm{w}}} < a_m, \qquad a_{k_{\mathrm{w}}} \ge c_2 (r/n)^4, \qquad (B_* a_{k_{\mathrm{w}}}^{-d_{\mathrm{pow}}})^n \le (B_* c_2^{-d_{\mathrm{pow}}})^n (n/r)^{4 d_{\mathrm{pow}} n}. \tag{15.15}$$

Expanding the OS norm gives a correlation of $4jn$ main insertions and two copies of v. The main insertions are separated from one another and from the exterior by at least $r/2 > a_m$. Translations by multiples of s preserve residues modulo $a_m \mathbb{Z}^4$, so the main anchors have at most $2n$ residue patterns, those of w and θw, for every j. For nonempty v, let $t_{\min} > 0$ be its minimum time and choose $m_{\mathrm{v}} \ge k_{\mathrm{w}}$ with $a_{m_{\mathrm{v}}} < 2t_{\min}$ and $a_{m_{\mathrm{v}}}$ below every distance between distinct anchors of v. The two exterior copies have times $\pm(2js + t)$, t a time of v, so they are more than $2t_{\min} > a_{m_{\mathrm{v}}}$ apart for every j. Choose $k_{\mathrm{v}} \ge m_{\mathrm{v}}$, beyond the exterior reference levels, with $12|v| a_{k_{\mathrm{v}}} < a_{m_{\mathrm{v}}}$, and let $X \ge 0$ bound the centered seminorms of the exterior labels and their images at level k_{v} (Lemma 14.15). Neither m_{v} nor k_{v} depends on j.

The coarse tiling has side a_m and separates the main insertions; the fine tiling has side $a_{m_{\mathrm{v}}}$, shares its origin and separates the exterior insertions. Their side ratio is odd, so coarse reflections preserve the fine grid and its interior margins. The strip count for the $2n$ main residues modulo a_m with $w = 5a_{k_{\mathrm{w}}}$, possible by (15.15), keeps the main anchors at distance at least $5a_{k_{\mathrm{w}}}$ from the coarse boundaries. A further translation by less than $a_{m_{\mathrm{v}}} \le a_{k_{\mathrm{w}}}$ puts the $2|v|$ exterior anchors at distance at least $3a_{k_{\mathrm{v}}}$ from the fine boundaries, since $12|v| a_{k_{\mathrm{v}}} < a_{m_{\mathrm{v}}}$, and leaves the main anchors margin at least $3a_{k_{\mathrm{w}}}$. The main anchors occupy distinct coarse cells, none containing an exterior anchor.

Apply (15.4) to the coarse tiling, treating the product of the exterior factors in one coarse cell as one observable. Lemma 14.11 bounds each main factor by $B_* a_{k_{\mathrm{w}}}^{-d_{\mathrm{pow}}}$. For a coarse cell with n_C exterior factors, reflect their product into every coarse cell; in each copy the exterior anchors are more than $a_{m_{\mathrm{v}}}$ apart and occupy distinct fine cells, so (15.4) for the fine tiling and Lemma 14.11 bound the coarse seminorm of that product by X^{n_C}. Therefore

$$\|[(A_{w^\dagger} A_w)^j v]\|_{\mathrm{OS}}^2 \le X^{2|v|} (B_* a_{k_{\mathrm{w}}}^{-d_{\mathrm{pow}}})^{4jn}.$$

For each fixed j, lattice approximations of the anchors preserving these strict margins, Lemma 14.34 and (15.1) pass this bound to the limit; no cutoff threshold uniform in j is needed. Take $B_v = \max(1, X^{2|v|})$, $C_0 = B_* c_2^{-d_{\mathrm{pow}}}$ and $Q = 4d_{\mathrm{pow}}$, and use (15.15). For empty v omit its factors and take $B_v = 1$. □

16 Free fields, Ward identities and rotations

The only rotations that preserve the lattice, the Wilson action and the block-spin transformation of Sections 11–15 are those of the hypercubic group H_4. Continuous rotational covariance of the limit is therefore an output of the construction. This section obtains it from a conserved stress tensor and proves Proposition 6.7, Theorem 6.10 and Proposition 6.11. The argument has three tasks.

(1) *Find the conserved line.* The Maxwell comparison identifies the fields of degrees four and five with their free counterparts. A three-dimensional space of candidate tensors maps by divergence onto a two-dimensional space of vectors, so conserved tensors form a line (Lemma 16.10).

(2) *Make the translation shell act as differentiation.* Shell fluxes of a member of that line define charge maps on the field spaces (Lemma 16.11). Finite free Ward tests identify the translation charge with a multiple of differentiation (Lemma 16.17).

(3) *Identify the angular action and recover rotations.* Coordinate reflection positivity controls the angular flux at infinity for collinear configurations, with no lower bound on the energy spectrum (Lemma 16.15). Analyticity extends the angular identity to all configurations; finite internal tests remove the defect in the bracket relations, and the angular charges integrate to an $O(4)$ action (Lemma 16.17).

Section 16.1 treats the free field, Sections 16.2–16.3 the comparison and the conserved line, Section 16.4 the coordinate reflection spaces, and Section 16.5 the Ward identities. Section 16.4 uses no continuous rotation; continuous rotations are first available in Lemma 16.17. The free calculation of Lemma 16.1 also supplies the positivity used in Lemma 14.22; its proof uses no interacting input.

Two estimates go beyond convergence at fixed separated points. The lattice comparison must control tests uniformly in their supremum norm, including near configurations with a shared coordinate; narrow strips around those configurations are removed, with different scales for the kernel and for its variation. Later, conservation removes the inner boundaries in the angular identity but not the boundary at infinity, which an anisotropic cutoff and the coordinate spectral estimates remove.

Except in the free calculation, fix the table, volume refinement, quotient fields and calculus of Definitions 2.6, 2.12 and 6.3 and Lemmas 15.1–15.6. Subsequences below change only comparison radii. Write $\jmath_{d,e}$ for the inclusions and ∂_a for field differentiation.

16.1 Free fields

Put $z = x - y$ and $\mathsf{h}_{ab}(z) = \partial_a \partial_b \Gamma(z)$. The curvature covariance in Definition 6.3 is

$$C_{ab,ij}(z) = -\delta_{bj}\mathsf{h}_{ai} + \delta_{bi}\mathsf{h}_{aj} + \delta_{aj}\mathsf{h}_{bi} - \delta_{ai}\mathsf{h}_{bj}, \qquad \mathsf{h}_{ab} = \frac{8 z_a z_b - 2\delta_{ab}|z|^2}{4\pi^2 |z|^6}. \tag{16.1}$$

Different colours have zero covariance. Wick correlations sum products of covariances over the pairings that join factors of different insertions; for instance $S_2^{\mathrm{Max}}(:F_1F_2:, :F_3F_4:) = C_{13}C_{24} + C_{14}C_{23}$, and no covariance at zero separation enters. Set

$$\begin{aligned} Q_{ab,ij} &= \sum_{\alpha} :F^{\alpha}_{ab}F^{\alpha}_{ij}:, \qquad Q_{ab} = \sum_{c} Q_{ac,bc}, \qquad Q_{\mathrm{tot}} = \sum_{a} Q_{aa}, \\ T^{\mathrm{Max}}_{ab} &= -Q_{ab} + \tfrac{1}{4}\delta_{ab}Q_{\mathrm{tot}}, \qquad \mathcal{O}_{F^2} = \tfrac{1}{4}Q_{\mathrm{tot}}. \end{aligned} \tag{16.2}$$

The minus sign is fixed by the shell convention in Definition 6.8.

For the coordinate reflection s_a, strict reflected positivity of $u \in M_d^{\mathrm{Max}}$ means

$$S_2^{\mathrm{Max}}(R^{\mathrm{Max}}(s_a)\bar{u}, u; s_a x, x) > 0 \qquad (x = te_a,\ t > 0).$$

Write $\mathsf{L}^{\mathrm{Max}}(K)$ for the derivative at zero of $R^{\mathrm{Max}}(e^{tK})$. The two free shell Ward identities, with the fluxes of Definition 6.8, are

$$\mathcal{P}_b^{T^{\mathrm{Max}}} = S^{\mathrm{Max}}((\partial_b^{\mathrm{Max}} u)(x_0), \boldsymbol{v}(x)), \qquad \mathcal{Y}_K^{T^{\mathrm{Max}}} = S^{\mathrm{Max}}((\mathsf{L}^{\mathrm{Max}}(K)u)(x_0), \boldsymbol{v}(x)). \tag{16.3}$$

The next lemma gives the nondegeneracy of free correlation tests and the normalization used in the interacting comparison.

Lemma 16.1 (Free positivity and stress)**.** *(i) The Maxwell table satisfies E1–E7, the field calculus and $O(4)$ covariance.*

(ii) Every $u \in M_d^{\mathrm{Max}} \setminus \{0\}$ has strict reflected positivity.

(iii) The centered real symmetric H_4-tensorial conserved degree-four tensors form the line $\mathbb{R}T^{\mathrm{Max}}$, and T^{Max} satisfies (16.3).

(iv) For $x \neq y$,

$$S_2^{\mathrm{Max}}(\mathcal{O}_{F^2}, \mathcal{O}_{F^2}; x, y) = \frac{3d_{\mathfrak{g}}}{\pi^4 |x-y|^8}.$$

Proof. (i) Formula (16.1) satisfies Bianchi and $\sum_a \partial_a C_{ab,ij} = 0$ off zero, so Wick correlations descend to the on-shell quotient. It also gives permutation symmetry, reality, units, translations and orthogonal covariance. Each differentiated contraction is homogeneous of a fixed negative degree and bounded on the unit sphere; a finite product of these bounds proves E6–E7. Differentiating the pairings proves the derivative law; inclusions retain the same polynomial. Reflection positivity E4–E5 is proved after (ii).

(ii) Kelvin inversion $z \mapsto z^\vee = z/|z|^2$ makes the reflected two-point pairing of curvature jets triangular with positive pivots (16.5); strict positivity then follows from Wick's theorem. Let Harm_j be the real space of $\mathfrak{g}$-valued homogeneous polynomial two-forms of degree j satisfying $df = \delta f = 0$. Its Frobenius inner product identifies it with the dual of the order-j on-shell curvature jets. Use Taylor coefficients $D^j/j!$ and the inner product summing both ordered form indices and ordered derivative indices; $p(\partial)$ denotes contraction with this normalized derivative. For $z \neq 0$ put $\mathcal{R}_z = \mathbf{1}_4 - 2zz^{\mathsf{t}}/|z|^2$. Let $R_p(y)$ be the covariance of $F(y)$ with the jet probe p at zero. Then

$$R_p(y) = \varsigma_j |y|^{-4} (\mathcal{R}_y \otimes \mathcal{R}_y) p(y^\vee), \qquad \varsigma_j = \frac{(j+1)(j+2)(-2)^j}{\pi^2}. \tag{16.4}$$

Indeed $p(\partial)\Gamma = (-2)^j |y|^{-2j} p(y)\Gamma$; induction proves this, since traces of the coefficient tensor vanish by harmonicity. In the two curls, terms containing $\sum_a \partial_a p_{ab}$ and $\partial_a p_{bc} + \partial_b p_{ca} + \partial_c p_{ab}$ vanish. The remaining radial derivatives use $\sum_a y_a \partial_a p = jp$ and $\partial_a |y|^{-2j-2} = -(2j+2) y_a |y|^{-2j-4}$, and give $\varsigma_0 = 2/\pi^2$, $(j+1)\varsigma_{j+1} = -2(j+3)\varsigma_j$.

For towers $p = (p_j)$ and $q = (q_j)$, let $B_z(p, q)$ be their two-site covariance at separation z, with $\mathcal{R}_z$ acting on all tensor indices of the second probe; this is the reflected jet pairing at $\pm z/2$. Define the triangular map σ_z by replacing a single order-j probe p with the tower whose order-k component is $(\varsigma_j/\varsigma_k) T_k[p(\cdot - z^\vee)]$, $k \leq j$, where T_k is the homogeneous Taylor term at zero. Its generated field is the right side of (16.4) with $p(y^\vee - z^\vee)$. Since $D(z^\vee) = |z|^{-2}\mathcal{R}_z$, its Taylor terms

at z below j vanish, and its top term is $\varsigma_j|z|^{-2j-4}\mathcal{R}_z p$. Symmetry removes the entries above the diagonal. Thus

$$B_z(\sigma_z p, \sigma_z q) = \sum_j \frac{(j+1)(j+2)2^j}{\pi^2|z|^{2j+4}}\langle p_j, q_j\rangle. \tag{16.5}$$

Evaluating the second order-j probe supplies $(-1)^j$, since its positional derivatives are opposite to derivatives of the separation; this cancels the sign in ς_j. In coordinates diagonalizing this form, Wick pairing between two sites gives $\langle X^\mu, X^\nu\rangle = \mathbf{1}_{\mu=\nu}\mu!\prod_i b_i^{\mu_i}$ with $b_i > 0$. Every polynomial uses a finite tower, proving strict positivity.

For several positive-side sites the reflected curvature covariance is positive semidefinite: spatial Fourier transformation of Γ gives $e^{-|p|(t+s)}/(2|p|)$, and its two curls, with reflection of the form indices, give the sum of squares of the two transverse polarizations. Jet derivatives differentiate the same factors. Include the contractions within each positive-side word in its Fock vector; its reflected pairing is then the symmetric Fock pairing of this positive semidefinite form, so finite sums of words have nonnegative squared norm. On compact separated supports every pairing is continuous and bounded; integration gives E5 and point evaluation gives E4.

(iii) Simplicity makes every invariant linear colour form zero and every invariant bilinear form a multiple of the fixed metric. Through weight four only the unit and the quadratics Q remain, and sign changes and permutations give the basis

$$Z^{(1)}_{ab} = \delta_{ab}Q_{aa}, \qquad Z^{(2)}_{ab} = \delta_{ab}(Q_{\mathrm{tot}} - 2Q_{aa}), \qquad Z^{(3)}_{ab} = (1-\delta_{ab})Q_{ab}$$

of centered real symmetric tensors. Sign parity reduces their divergences to contractions of $F\partial F$ with the free index b. The equations remove $\sum_d \partial_d F_{dc}$, and Bianchi gives $2\sum_{c,d} F_{cd}\partial_c F_{bd} = \sum_{c,d} F_{cd}\partial_b F_{cd}$. So the divergences lie in the plane spanned by

$$\sum_c : F_{bc}\partial_b F_{bc} := \tfrac{1}{2}\partial_b Q_{bb}, \qquad \sum_{c<d,\ c,d\neq b} : F_{cd}\partial_b F_{cd} := \tfrac{1}{4}\partial_b(Q_{\mathrm{tot}} - 2Q_{bb}),$$

with colour sums understood. Test the first components of the divergences of $Z^{(1)}, Z^{(2)}, Z^{(3)}$ (columns) at e_1 against $Q_{12,12}(0)$ and $Q_{23,23}(0)$ (rows), dividing by $d_{\mathfrak{g}}/(4\pi^2)^2$. For example $S_2^{\mathrm{Max}}(Q_{11}(te_1), Q_{12,12}(0)) = 2d_{\mathfrak{g}}/(\pi^4 t^8)$, whose derivative gives the first entry; differentiating (16.1) gives the others:

$$\begin{pmatrix} -256 & 0 & 128 \\ 0 & -512 & -128 \end{pmatrix}. \tag{16.6}$$

Its first two columns are independent, so divergence maps the span of the $Z^{(i)}$ onto this plane. The test of $Z^{(3)}_{12}$ against $Q_{13,23}$ gives -16, and zero for $Z^{(1)}, Z^{(2)}$, so the three tensors are independent. The kernel is the line of $\frac{1}{2}Z^{(1)} - \frac{1}{4}Z^{(2)} + Z^{(3)} = -T^{\mathrm{Max}}$.

To compute its charge, polarize T^{Max} between a smooth Maxwell wave f and the field g generated by a probe at x_0. Replace Γ by Γ_ε with $-\Delta\Gamma_\varepsilon = \rho_\varepsilon$, a smooth approximate identity. For a probe F_{ij} the divergence of g is $\delta_{bj}\partial_i\rho_\varepsilon - \delta_{bi}\partial_j\rho_\varepsilon$. The polarized tensor is $t_{ab} = -\sum_c(f_{ac}g_{bc} + g_{ac}f_{bc}) + \frac{1}{2}\delta_{ab}\sum_{i,j} f_{ij}g_{ij}$, and Bianchi gives $\sum_a \partial_a t_{ab} = -\sum_c f_{bc}\sum_a \partial_a g_{ac}$. With the minus sign of the shell integral, $\int \chi v_b \partial_a t_{ab} = -\int \chi v_b f_{bc}(\delta_{cj}\partial_i\rho_\varepsilon - \delta_{ci}\partial_j\rho_\varepsilon)$, and a second integration by parts gives

$$\partial_i(v_b f_{bj})(x_0) - \partial_j(v_b f_{bi})(x_0) = (\mathrm{Lie}_v f)_{ij}(x_0).$$

No derivative of χ appears at x_0, and the contraction with Dv vanishes by symmetry. For $v = K(\cdot - x_0)$ this is $\sum_c(K_{ci}f_{cj} + K_{cj}f_{ic})(x_0)$, the generator of the free action on curvature; for

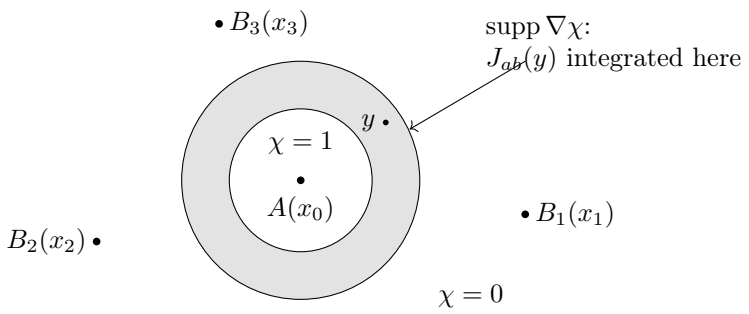


Figure 12: The shell flux of Definition 6.8. The cutoff χ equals one near the insertion $A(x_0)$ and vanishes near the spectators $B_i(x_i)$; the current is integrated against $\nabla\chi$ on the shaded annulus. Conservation makes the flux independent of χ, so the annulus may shrink towards x_0 or grow until it meets a spectator. It says nothing about flux through large spheres. The vanishing of the total angular flux at infinity is proved separately: on coordinate lines in Lemma 16.15, and for all configurations in Lemma 16.17.

constant v it is $v \cdot \nabla f$. Higher probes follow by differentiating in x_0. In a Wick contraction, the terms in which both stress legs end at x_0 contribute zero to the shell flux: dilation by 2 multiplies their shell integral by $2^{3+p-8-j-l} \neq 1$ (probe orders j, l, weight degree $p = 0, 1$), while the flux is independent of the cutoff. Terms with both legs outside vanish by conservation. The remaining contractions give the derivation Lie_v, proving (16.3).

(iv) At $z = re_1$ the six unordered curvature components have diagonal covariances $\pm(\pi^2 r^4)^{-1}$. Since $\mathcal{O}_{F^2} = \frac{1}{2}\sum_{a<b,\alpha} :(F^\alpha_{ab})^2:$, its two-point Wick sum is $\frac{1}{4} \cdot 2 \cdot 6 d_{\mathfrak{g}}/(\pi^4 r^8)$. Orthogonal covariance gives the formula for every $z \neq 0$. □

The sign convention can be seen in one free example (Figure 12). By Lemma 16.1(iv), $S_2^{\mathrm{Max}}(\mathcal{O}_{F^2}(x), \mathcal{O}_{F^2}(y)) = c|x-y|^{-8}$ with $c = 3d_{\mathfrak{g}}/\pi^4$. The translation shell Ward identity (16.3) and the free derivative law give, for a shell around x,

$$\mathcal{P}_b^{T^{\mathrm{Max}}} = \partial_{x_b}\big(c|x-y|^{-8}\big) = -8c\,(x_b - y_b)|x-y|^{-10}.$$

Replacing T^{Max} by λT^{Max} multiplies this flux by λ, so the requirement that the flux be the derivative fixes both the normalization and the sign of the stress tensor. This is a free-field calculation; nothing is asserted here about the form of interacting correlations.

The next lemmas concern only the invariant polynomial algebra of Definition 6.3, realized by the Taylor sections of Lemma 10.2. A curvature jet of order j has weight $j+2$. Let $\mathfrak{P}$ be the complex polynomial algebra on all on-shell jets, $\mathfrak{P}_{\le j}$ its subalgebra using orders at most j, and $\mathfrak{P}^G$ the invariant subalgebra. The invariant polynomials of weight at most d identify with M_d^{Max}. For a spectator degree d put

$$\mathfrak{N}_d = \{P \in \mathfrak{P} : S^{\mathrm{Max}}(P(0), u_1(x_1), \dots, u_n(x_n)) = 0 \text{ for all } n,\ u_i \in M_d^{\mathrm{Max}},\ (0, x) \in \mathrm{Conf}_{n+1}\},$$

with every polynomial Wick ordered at its own site. The degree of P is unrestricted. The next result says that spectators of one bounded degree detect every invariant polynomial, however high its jet order.

Lemma 16.2 (Detection by bounded spectators)**.** *There is $d_0 \in \mathbb{N}$ such that*

$$\mathfrak{N}_{d_0} \cap \mathfrak{P}^G = \{0\}. \tag{16.7}$$

The proof has two stages. At a fixed jet order, finite generation and reflected positivity give a bound on the spectator degree. To make the bound independent of the jet order, a quadratic spectator isolates a highest-order jet derivative of P by its harmonic pole, and induction on weight removes that jet order.

The extraction rests on the difference between a single harmonic pole and a product of poles. In four dimensions $K(z) = \partial_1\partial_2|z|^{-2} = 8z_1z_2|z|^{-6}$ is harmonic and homogeneous of degree -4; multiplication by $|z|^4$ leaves $8z_1z_2|z|^{-2}$, which is not analytic at zero, and only $|z|^6$ clears it. The product of two poles $|z|^{-2}\cdot|z|^{-2} = |z|^{-4}$ has the same degree but is already cleared by $|z|^4$. In general a product of poles of orders $r_1, r_2 \geq 2$ is cleared by $|z|^{2(r_1+r_2-2)}$, while a harmonic pole of order $r_1 + r_2$ needs $|z|^{2(r_1+r_2-1)}$.

Proof of Lemma 16.2. Step 1: bounded jet order. Fix j. The algebra $\mathfrak{P}_{\leq j}$ is a polynomial algebra on finitely many variables, hence Noetherian by the Hilbert basis theorem [33, Lemma 10.31.1]. Its ideal generated by invariant polynomials of positive ordinary degree therefore has finitely many invariant, ordinary-homogeneous generators f_α. Every invariant ordinary-homogeneous polynomial is a sum of products of invariant homogeneous polynomials of weight at most some b_j: write $P = \sum a_\alpha f_\alpha$, keep only terms of total degree $\deg P$, and replace a_α by its gauge average, whose degree is smaller; induction on the degree concludes.

Spectators of degree b_j detect: $\mathfrak{N}_{b_j} \cap \mathfrak{P}^G \cap \mathfrak{P}_{\leq j} = \{0\}$. Otherwise take its lowest nonzero ordinary-homogeneous part P_r and split a product of homogeneous factors of total ordinary degree r into distinct nearby spectator sites. Every leg of P must pair with a spectator leg, so terms with internal spectator contractions, or with higher-degree parts of P, vanish. As the spectator sites coalesce, Wick's rule gives the two-site pairing of P_r with their ordinary product. Each factor has weight at most b_j, so for $P \in \mathfrak{N}_{b_j}$ this limit is zero. Choose the product sum to be the reflection of $\overline{P_r}$; Lemma 16.1(ii) makes this pairing strictly positive, a contradiction.

Step 2: pole extraction. Put $g = d_{\mathfrak{g}}$ and choose $d_0 \geq \max\{b_g, 2g+5\}$. In a real colour basis write $X_{j,\alpha}(p)$ for the jet coordinate tested by a scalar order-j probe p. For scalar probes p, q of orders k, l, the invariant quadratic $\sum_\alpha {:}X_{k,\alpha}(p)X_{l,\alpha}(q){:}$ has weight $k+l+4$. Its collision with a polynomial using jets through order j has, in the coefficient where just the order-k leg contracts at order j, the polynomial

$$\sum_\alpha X_{l,\alpha}(q) \sum_\mu C_{k,j}(p, e_\mu; z) \frac{\partial P}{\partial X_{j,\alpha}(e_\mu)}, \tag{16.8}$$

where (e_μ) is a basis of order-j probes and $C_{k,j}$ their differentiated curvature covariance.

Claim. If $P \in \mathfrak{N}_{d_0}$, $l < k$ and $k + l + 4 \leq d_0$, then (16.8) lies in $\mathfrak{N}_{d_0}$. Put $r = 4 + k + j$. Tested against fixed exterior spectators, the term (16.8) is harmonic and homogeneous of degree $-r$ in z, so $|z|^{2(r-1)}$ times it is analytic at zero. The other terms, expanded in Taylor series at each singular order, give at order r homogeneous functions that become analytic after multiplication by $|z|^{2(r-2)}$: a double contraction is a product of poles of orders $r_1, r_2 \geq 2$, cleared by $|z|^{2(r_1+r_2-2)}$, and Taylor terms keep this improvement. The contraction of the order-l leg alone has order at most $4 + l + j < r$. The claim now follows from:

> a harmonic K, homogeneous of degree $-r \leq -2$, with $|z|^{2(r-2)}K$ analytic at 0, vanishes.

Indeed $\Delta(|z|^{2s}K) = 4s(s+1-r)|z|^{2s-2}K$ in four dimensions; successive Laplacians show that K extends analytically, and negative homogeneity makes it zero. Expand the collision through an order exceeding $8 + k + l + 2j$. Since its correlation with the quadratic test and further bounded-degree spectators vanishes identically, comparing successive radial homogeneities makes the order-r harmonic pole zero, so (16.8) belongs to $\mathfrak{N}_{d_0}$.

Step 3: induction on weight. Take $k = g + 1$ and $l = 0, \dots, g - 1$. For each l choose a probe q_l and a jet value v_l with $q_l(v_l) = 1$. The matrix $(X_{l,\alpha}(q_l))_{l,\alpha}$ has nonzero polynomial determinant: assigning v_l to colour l and zero to the other colours in order l makes it the identity. For $j > g$ the polynomial (16.8) replaces an order-j jet by an order-l jet, so its weight is smaller, and the induction hypothesis makes it zero. Multiplication by the transposed cofactor matrix shows that every one-colour contraction in (16.8) vanishes. As z, p vary, the forms $C_{k,j}(p, \cdot; z)$ span the dual of the order-j jets: their common annihilator generates a Maxwell wave whose order-k jets vanish everywhere, a polynomial of degree below k with negative homogeneity, hence zero, and (16.4) makes the probe zero. Hence every order-j partial derivative of P vanishes. Descending from the top order removes all orders above g; Step 1 with spectator bound b_g treats the rest. □

Let $\mathcal{C}_d$ be the real space of complex linear tuples $\mathsf{X}_a : M_d^{\mathrm{Max}} \to M_{d+1}^{\mathrm{Max}}$ that commute with conjugation, kill the unit, are vector-covariant under H_4, and send M_e^{Max} into M_{e+1}^{Max} for $e \le d$. For $\mathsf{X} \in \mathcal{C}_d$ the Ward sum is

$$\mathcal{E}_b(\mathsf{X}; \boldsymbol{u}, x) = \sum_i S_n^{\mathrm{Max}}(u_1, \dots, \mathsf{X}_b u_i, \dots, u_n; x).$$

Finitely many tests on lines determine its kernel.

Lemma 16.3 (Finite vector Ward tests)**.** *For all sufficiently large d, finitely many positional derivatives of $\mathcal{E}_b(\mathsf{X}; \boldsymbol{u}, x)$, with $u_i \in M_d^{\mathrm{Max}}$ and x on a coordinate line perpendicular to e_b, define a real linear map $\mathcal{T}_d$ with*

$$\ker \mathcal{T}_d = \mathbb{R}\partial^{\mathrm{Max}} \quad \text{in } \mathcal{C}_d. \tag{16.9}$$

Proof. Take $d \ge d_0$ from Lemma 16.2. Suppose all the stated jets of the Ward sums of X vanish.

Step 1: continuation off the lines. Free correlations are real analytic off the diagonals by (16.1), so every Ward sum vanishes near each testing line. For a separated configuration choose distinct ordered line coordinates respecting the strict order of its coordinates in that axis. The straight segment to the line configuration stays separated: strict inequalities persist, and coordinate ties are separated by the line coordinates. Analytic continuation along it gives $\mathcal{E}_b = 0$ on Conf_n.

Step 2: weight four. Let J_j be the scalar on-shell order-j jet space. For $p \in \mathsf{J}_0^*$ put $U(p) = \sum_\alpha {:}X_{0,\alpha}(p)^2{:}$. Simplicity, the filtration and inversion parity force $\mathsf{X}_b U(p)$ to be a colour contraction of one order-zero and one order-one jet, represented by a linear map $A_p : \mathsf{J}_0 \to \mathsf{J}_1^*$. In the Ward identity of $U(p), U(q), U(w)$ let the first two points coalesce along tu and multiply by t^5. The term with X_b at the distant point tends to zero, and the two remaining triangular Wick contractions give, for every $X \in \mathsf{J}_0$ and with $C_{1,0}$ taken in one colour,

$$C_{1,0}(A_p X, q; u)\, q(X) = C_{1,0}(A_q X, p; u)\, p(X).$$

Arbitrary X are reached by choosing the distant probe w through the invertible order-zero covariance (16.4). If $p(X) = 0$, choose $q(X) \neq 0$; varying q, u gives $A_p X = 0$. Thus $A_p X = p(X) f_b p$, and the same identity shows that $f_b : \mathsf{J}_0^* \to \mathsf{J}_1^*$ is real linear with $C_{1,0}(f_b p, q; u) = C_{1,0}(f_b q, p; u)$.

Calculation of f_b. Write e^{ij} for the curvature coordinate and $\partial_h e^{ij}$ for its first jet. For $f_1 e^{12}$, sign parity permits only $\partial_1 e^{12}, \partial_3 e^{23}, \partial_4 e^{24}$; interchange of axes $3, 4$ and $\sum_i \partial_i e^{i2} = 0$ reduce their span to $\mathbb{R}\partial_1 e^{12}$. For $f_1 e^{23}$ the candidates $\partial_1 e^{23}, \partial_2 e^{13}, \partial_3 e^{12}$ reduce, by antisymmetry in axes $2, 3$ and Bianchi, to $\mathbb{R}\partial_1 e^{23}$. Permutations leave two constants, which agree because $C_{1,0}(\partial_1 e^{12}, e^{23}; e_1 + e_3) = -1/(2\pi^2) \neq 0$ by (16.1). Therefore $f_b = c_{\mathrm{jet}} \partial_b$, and since $\partial_b U(p)$ has two identical terms, $\mathsf{X}_b = c\, \partial_b^{\mathrm{Max}}$ on weight four with $c = c_{\mathrm{jet}}/2$.

Step 3: higher weights. Set $\mathsf{X}_b^\circ = \mathsf{X}_b - c\,\partial_b^{\mathrm{Max}}$. Translation invariance makes its Ward sums zero, and it kills every free stress component. Insert into such a sum a free stress shell for the dilation field $v(y) = y - x_0$, surrounding only the distinguished point. The term where X_b° acts on the stress is zero. The other shell terms are given by the polarized calculation in Lemma 16.1, now with this v; the extra derivative of v contracts with the trace of the free stress, which is zero by (16.2). The local Lie derivative of a two-form under dilation is $y \cdot \nabla + 2$, and on jets it gives the weight operator $\mathsf{D}^{\mathrm{wt}} X_{j,\alpha} = (j+2) X_{j,\alpha}$. Subtracting the Ward sum with distinguished label $\mathsf{D}^{\mathrm{wt}} u$ shows $[\mathsf{D}^{\mathrm{wt}}, \mathsf{X}_b^\circ] u \in \mathfrak{N}_d$, and Lemma 16.2 makes it zero. So X_b° preserves each weight. Inversion acts there by $(-1)^{\mathrm{weight}}$, whereas vector covariance makes X_b° odd under inversion; hence $\mathsf{X}_b^\circ = 0$.

Step 4: finite selection. Conversely translation invariance annihilates every test on ∂^{Max}. The real space $\mathcal{C}_d$ is finite-dimensional; successively choose a test nonzero on the current kernel modulo this line. Each choice lowers its dimension, so finitely many tests have precisely the asserted kernel. □

The internal tests rest on one elementary fact about the plane generators $K_{ab} = E_{ba} - E_{ab}$ of $\mathfrak{so}(4)$. Call a coordinate quarter turn the rotation $e_a \mapsto e_b$, $e_b \mapsto -e_a$ in one coordinate plane.

Lemma 16.4 (Alternating planes)**.** *Let $\beta : \mathfrak{so}(4) \times \mathfrak{so}(4) \to V$ be real bilinear and alternating, with $\beta(hKh^{-1}, hLh^{-1}) = \beta(K, L)$ for every coordinate quarter turn h. Then $\beta = 0$.*

Proof. The quarter turns generate the determinant-one signed permutations: they project onto the transpositions of S_4, and squares of quarter turns give the diagonal sign changes with two minus signs. Conjugation by a diagonal sign matrix s multiplies K_{ab} by $s_a s_b$; conjugation by a permutation matrix permutes the planes. *Overlapping planes:* $s = \mathrm{diag}(1, 1, -1, -1)$ fixes K_{12} and reverses the sign of K_{13}, so $\beta(K_{12}, K_{13}) = -\beta(K_{12}, K_{13}) = 0$. *Disjoint planes:* the even permutation $(13)(24)$ interchanges K_{12} and K_{34}, so $\beta(K_{12}, K_{34}) = \beta(K_{34}, K_{12}) = -\beta(K_{12}, K_{34}) = 0$. Every pair of distinct planes is carried to one of these two pairs by a signed permutation of determinant one, and $\beta(K, K) = 0$; bilinearity gives $\beta = 0$. □

Let $\mathcal{B}_d$ consist of alternating real-bilinear families $\Delta : \mathfrak{so}(4)^2 \to \mathrm{End}_{\mathbb{C}}\, M_d^{\mathrm{Max}}$ which commute with conjugation, kill the unit, preserve every lower-degree subspace, and satisfy

$$R^{\mathrm{Max}}(h)\Delta(K, L) = \Delta(hKh^{-1}, hLh^{-1}) R^{\mathrm{Max}}(h) \qquad (h \in H_4,\ \det h = 1).$$

The next result removes the defect in the rotation commutator.

Lemma 16.5 (Finite internal Ward tests)**.** *For every $d \geq 4$, finitely many sums*

$$\sum_i S_n^{\mathrm{Max}}(u_1, \dots, \Delta(K, L) u_i, \dots, u_n; x), \qquad u_i \in M_d^{\mathrm{Max}}, \quad x \in \mathrm{Conf}_n, \tag{16.10}$$

separate $\mathcal{B}_d$.

Proof. Suppose all these sums vanish and fix $\mathsf{Z} = \Delta(K, L)$. On the weight-four space let $b_x(u, v) = S_2^{\mathrm{Max}}(u, v; x, 0)$. The order-zero Kelvin formula and Wick's rule give $b_x(u, v) = b_y(R^{\mathrm{Max}}(\mathcal{R}_y \mathcal{R}_x) u, v)$ for $|x| = |y| \neq 0$. Both Z and the orthogonal action preserve this space. Applying $b_x(\mathsf{Z}u, v) + b_x(u, \mathsf{Z}v) = 0$ at x and y, and nondegeneracy of b_y, shows that Z commutes with $R^{\mathrm{Max}}(\mathcal{R}_y \mathcal{R}_x)$, in particular with every coordinate quarter turn. With the signed covariance of Δ, the function $(K, L) \mapsto \Delta(K, L) u$ satisfies the hypothesis of Lemma 16.4 for each weight-four u, so Z vanishes on weight four, including the stress.

Insert a free angular stress shell around the distinguished point only, as in Step 3 of the proof of Lemma 16.3. Subtracting the Ward sum with the angularly differentiated label gives $[\mathsf{L}^{\mathrm{Max}}(M), \mathsf{Z}]u \in \mathfrak{N}_d$. This polynomial has weight at most d, so Lemma 16.1(ii) makes it zero. Hence Z commutes with all infinitesimal rotations, and differentiating $e^{t\mathsf{L}(M)}\mathsf{Z} - \mathsf{Z}e^{t\mathsf{L}(M)}$ gives commutation with coordinate quarter turns in every degree. Lemma 16.4 again gives $\Delta = 0$. Finite-dimensionality selects a finite separating subfamily by successively reducing the common kernel. □

16.2 Lattice inputs

The insertions are those of Definition 2.8, with coefficient polynomials from Definition 10.1 and Lemma 10.2. The inverse coefficient estimate of Lemma 14.6 inverts the marked coefficient map; the corrected sources and normalized integral of Lemmas 14.24 and 14.25 bound the remainders; the uniform displacement estimate of Lemma 14.27 controls positions, with coordinate separations at least four times the displacement margin. Chessboard seminorms are those of Definition 14.10. Cutoff limits are taken in the order (14.28), followed by the fixed extraction of Lemma 15.1; an ordered limit point of a bounded coefficient family is its limit along an ultrafilter containing the ordered tails.

Fix arity and degree bounds $n \geq 2, D \geq 1$ and $R \geq 1$. A basis word is a list of members of fixed real bases of M_d^{Max}, $d \leq D$. Put

$$K_{R,m} = \{x : |x| \leq R,\ \mathrm{sep}_\infty(x) \geq R^{-1}\}, \qquad E_\delta = \{x : |x_i^a - x_j^a| > \delta\ (i \neq j,\ a = 1, \dots, 4)\}.$$

The comparison exponents $\alpha_0, \alpha_1, \alpha_2 > 0$ and $c < 0$ have the following roles: α_0 sets the comparison radius, α_1 the cells used for moment bounds, α_2 the smaller cells used for displacement bounds, and c the width of the excluded coordinate strips. They satisfy

$$\alpha_1 + 6 < \alpha_0, \quad \alpha_2 + 5 < \alpha_0 + c, \quad (\alpha_0 - \alpha_1)Dn + c < 0, \quad (\alpha_0 - \alpha_2)Dn - \alpha_0 - c < 0. \tag{16.11}$$

The first two inequalities place the chessboard cells; the last two make the excluded strips negligible. One explicit choice, with $n_{\max} = \max(n, D)$, is $\kappa_1 = 7n_{\max}(n_{\max} + 2) + 1$, $\kappa_2 = 6 + \kappa_1$, $\alpha_0 = 17 + n_{\max} + \kappa_1 + \kappa_2 n_{\max}(n_{\max} + 2) + \kappa_2$, $\alpha_1 = \alpha_0 - 7$, $\alpha_2 = \alpha_0 - \kappa_2$ and $c = -\kappa_1$. Write $M_k = M_k^{\mathrm{cut}}$ and set

$$r_k = a_k M_k^{\alpha_0}, \qquad \mathfrak{s}_k = M_k^{-\alpha_0}, \qquad \delta_k = M_k^c, \qquad R_k^{\mathrm{cell}} = C M_k^{(\alpha_0 - \alpha_1)D}, \quad R_k^{\mathrm{var}} = C M_k^{(\alpha_0 - \alpha_2)D}, \tag{16.12}$$

where $C \geq 1$ depends only on n, D, R and the construction and is fixed in the proof of Lemma 16.6. Thus $r_k \mathfrak{s}_k = a_k$: the normalized grid is the physical construction grid. For an ordered limit point $c_{k,d} : M_d^{\mathrm{Max}} \to V_d$ of the lifts in Lemma 16.6, and the quotient $\pi_d : V_d \to F_d$ of Definition 2.12, define the rescaled correlation

$$S_k^{\mathrm{sc}}(\boldsymbol{u}; x) = S_m(\pi_{d_i} c_{k,d_i} u_i; r_k x).$$

The following comparison uses one coefficient family for every word within the fixed bounds. Its constants may depend on these bounds, the word and the construction. The arity, degree and radius bounds are fixed first, then the level and the lifts; the estimates then hold for every basis word within these bounds.

Lemma 16.6 (Lattice comparison at small radii)**.** *There are arbitrarily large levels k, real H_4-equivariant lifts $\Lambda_{k,N,p,d} : M_d^{\mathrm{Max}} \to V_d$ preserving the degree-zero unit, and $\varepsilon_k \to 0$ such that every ordered limit point of these lifts satisfies, for every basis word with $m \le n$,*

$$\begin{aligned} |S_k^{\mathrm{sc}}| &\le (R_k^{\mathrm{cell}})^m && \text{on } K_{R,m}, \\ \|\nabla S_k^{\mathrm{sc}}\| &\le 4m(R_k^{\mathrm{var}})^m/\delta_k && \text{on } K_{2R,m} \cap E_{4\delta_k}, \\ |S_k^{\mathrm{sc}} - S_m^{\mathrm{Max}}| &\le \varepsilon_k && \text{on } (\mathfrak{s}_k\mathbb{Z})^{4m} \cap K_{2R,m} \cap E_0. \end{aligned} \tag{16.13}$$

At fixed k the lifts are bounded on ordered tails.

The lifts are chosen so that the rescaled insertions have the free Wick polynomials as Gaussian leading part (16.14). A subset recursion removes the continuum Gaussian moments; for exact Gaussian variables each remainder has mean zero, and the source estimates of Section 14 make each remainder small in chessboard seminorm (16.16). The chessboard inequality then gives the comparison on the grid, and the coarser normalizations α_1, α_2 give the size and gradient bounds off the grid.

Proof. Step 1: lifts. Choose a real H_4-equivariant polynomial representative map ι_d^{free} for the on-shell quotient, using the Taylor sections of Lemma 10.2 and averaging a real splitting over H_4. Let $\mathsf{C}_{k,N,p,d}$ be the marked coefficient map from level k down to $m_{\mathrm{ref}}(d)$, after the Wick-ordering correction of Lemma 14.23. Write δ_t^{wt} for multiplication of a weight-j monomial by t^j; the factor-count scaling t^{N} is that of Definition 2.8. The Gaussian field divided by the small-field threshold has scale

$$\mathfrak{t}_{k,N,p} = (t_k\sqrt{\beta_{N,p,k}})^{-1}, \qquad 0 < \mathfrak{t}_{k,N,p}^{-1} \le 2M_k,$$

since $t_k = M_k/\sqrt{\widehat{\beta}_k}$ is the small-field threshold of Part I and $\beta_{N,p,k} \le 2\widehat{\beta}_k$. Solve

$$\mathsf{C}_{k,N,p,d}\Lambda_{k,N,p,d}u = \mathfrak{t}_{k,N,p}^{-\mathsf{N}}\delta_{M_k^{\alpha_0}}^{\mathrm{wt}}\iota_d^{\mathrm{free}}u. \tag{16.14}$$

The solution exists and obeys

$$|v|_1 \le K_{k,d}a_k^d|\mathsf{C}_{k,N,p,d}v|_1, \tag{16.15}$$

with $K_{k,d}$ independent of N, p at fixed k. Indeed, apply the inverse estimate of Lemma 14.6 at each step $m+1 \to m$ from level k down to $m_{\mathrm{ref}}(d)$, and the coefficient comparison in the proof of Lemma 14.23, whose relative error is less than $1/4$, so its inverse costs at most 2. One may take $K_{k,d}$ to be twice the product of the finitely many column and block factors up to k, enlarged by the fixed basis norms. The coefficient map is injective, hence invertible on its finite coefficient space, and its unit, conjugation and signed laws pass to the inverse. The bound on $\mathfrak{t}^{-1}$ bounds the right side at fixed k, so the lifts are bounded on cutoff tails.

Step 2: subset recursion. Lemmas 14.23–14.25 hold for an arity bound $n \ge 2$, at most n marks per cell and any compact position set with positive point separation; apply them to the interval (14.4) with exponent α_0 and position set $K_{2R,m}$. Their covariance and Taylor bounds are uniform on $K_{2R,m}$, whose point separations are at least $(2R)^{-1}$, and need no lower bound on coordinate differences; only the reflected placement below requires E_0. Write $I_i = O_{N,p}^{(d_i)}(\Lambda_{k,N,p,d_i}u_i, r_k x_i/a_N)$ at compatible lattice points. For a subset J put $G_J = S_{|J|}^{\mathrm{Max}}((u_i)_{i\in J}; (x_i)_{i\in J})$ and define

$$R_\varnothing = 1, \qquad R_I = \prod_{i\in I} I_i - \sum_{\varnothing \ne J \subseteq I} G_J R_{I\setminus J}.$$

This subtracts the full continuum Gaussian moment on every subset of slots; for exact continuum Gaussian variables its nonempty coefficients have expectation zero, by induction on $|I|$. The Wick-ordering correction and (16.14) identify the leading Gaussian correlations with these continuum moments at the normalized grid points, as calculated in Lemma 14.23. Covariance, Taylor and localization errors remain in the regular and forced source terms.

Step 3: chessboard seminorm of the remainders. The finite-set clause of Lemma 14.2 gives arbitrarily late common terminal levels k for the three exponents $\alpha_0, \alpha_1, \alpha_2$. Use the interval (14.4) with exponent α_0, its width level $w_{\alpha_0}(k) = k - h_k(\alpha_0) - v_k$ and its centered tiling $\mathcal{C}^{\mathrm{tile}}_{w_{\alpha_0}(k)}(0)$. Put $q_k = \log M_k/(16n)$ and give each mark weight $e^{2q_k} \le M_k^{1/16}$. The weighted mode norm in the proof of Lemma 14.14 multiplies mark weights over subsets, so coefficient extraction for a nonempty I contributes $e^{-2q_k|I|}$ in each reflected cell, and the corrected-source iteration of Lemma 14.24 keeps this mark weight on the selected comparison intervals. Work in $\mathbb{C}[z_1, \dots, z_m]/(z_i^2)$ and multiply by the reciprocal of the continuum free moment polynomial; its coefficient of z_I is exactly R_I. After the finite Gaussian contractions on every subset, the remaining source modes have zero leading Gaussian coefficient. In the source iteration, terms linear in these modes cost the finite stage constant, and every other connected term contains at least two residual modes and is bounded by their squared norm times a fixed power of M_k; this is (14.45). The covariance and localization errors of Lemma 14.23 are below the corresponding source tolerance at entry (14.46). Lemma 14.24, applied to the coefficients (14.40) of these finitely many subset modes, gives this mark weight with constants depending only on n, D, R and the construction; its sums are rooted at a mark and do not count the replicated cells.

The normalized positive integral costs, per cell, at most the stage constant $Q^{\mathrm{cmp}}_{J_k}$ of (14.48) (Lemma 14.25), which contains the term $1 + n$ for at most n marks. Item 5 imposes $Q^{\mathrm{cmp}}_j \le \log M_t/(16n)$ for each arity bound, hence $Q^{\mathrm{cmp}}_{J_k} \le q_k$; the entry is chosen after the finitely many source constants. Taking the root over the reflected cells gives

$$\|R_I\|_{\mathcal{C}^{\mathrm{tile}}_{w_{\alpha_0}(k)}(0)} \le e^{q_k - 2q_k|I|} \le M_k^{-1/(16n)} \qquad (\varnothing \ne I \subseteq \{1, \dots, m\}). \tag{16.16}$$

The smallness comes from the corrected density at large coupling: its source terms stay bounded at a growing mark weight, while the unmarked normalization grows more slowly than that weight. The estimate concerns the complete reflected array, before absolute values are taken, and needs no independence of the insertions.

Step 4: comparison on the grid. Distinct coordinate differences at normalized grid points are at least $\mathfrak{s}_k$, so their physical differences are at least a_k. The reading map assigns the corresponding level-k sites without rounding, and the interval margin (14.41) contains the inspected stencils and their reflected copies. Expanding the recursion, applying (14.19) to each R_I and bounding the free submoments on $K_{2R,m}$ gives $C_{R,\boldsymbol{u}} M_k^{-1/(16n)}$, the last line of (16.13).

Step 5: size and variation off the grid. At its own normalization the singleton comparison gives a chessboard seminorm at most $1 + |S_1^{\mathrm{Max}}(u)|$. For exponent α_1, changing normalization to $M_k^{\alpha_0}$ multiplies a weight-j coefficient by $M_k^{(\alpha_0 - \alpha_1)j}$, so the finite coefficient sum bounds these seminorms by R_k^{cell} for C large; exponent α_2 gives R_k^{var}. Include every reflected basis label and its complex conjugation in these maxima. The two interval lengths of (14.4) give, with the stage block ratio L,

$$\frac{a_{w_b(k)}}{a_k} = L^{h_k(b) + v_k} \le L^2 M_k^{b+2} B_k^{10} \le M_k^{b+5} \qquad (b = \alpha_1, \alpha_2), \tag{16.17}$$

after the stage entry, where L is fixed and $B_k = M_k^{o(1)}$. So the α_1 cell is at most $a_k M_k^{\alpha_1 + 6} = o(r_k)$ and the α_2 cell is $o(r_k \delta_k)$ by (16.11). The chessboard inequality bounds separated continuum

moments by $(R_k^{\mathrm{cell}})^m$. The physical diameter on $r_k K_{2R,m}$ is at most $4Rr_k < L_p/2$ on the ordered tails, so the torus separation g_p of Lemma 14.27 is the minimum ordinary coordinate separation, which exceeds $4r_k\delta_k$ on $E_{4\delta_k}$. The uniform clause of Lemma 14.27, with margin $r_k\delta_k$ and the reflected seminorms included in R_k^{var}, gives $4m(R_k^{\mathrm{var}})^m/\delta_k$ after rescaling. The three normalizations use the same coefficient map and lifts and the same terminal levels; the finitely many cutoff tails can be intersected.

Step 6: limits. Take any ordered limit point of the bounded lifts. Lemma 14.33 and (15.1) pass the moment estimates to the fixed table, and Proposition 6.4 with Lemma 15.5 passes the difference bound. No extraction of the table or the volume sequence is made. □

For the shell estimate take $\varkappa \in \{0,1\}$, $A \in F_d$, $B \in F_4$ and $D = d+1-\varkappa$. The shell factor has scaling degree $3+\varkappa$: integration over four dimensions contributes four powers of the radius, the differentiated cutoff removes one, and the constant or linear vector field contributes $\varkappa$. Since the two inserted fields have total degree $d+4$, the expected output degree is $(d+4)-(3+\varkappa) = D$. A grid test is a separated configuration $(0, x_i)$ on $a_{j_0}\mathbb{Z}^4$ for some level j_0, generic in each coordinate, with a fixed spectator list $\boldsymbol{B}_i$. Its response functional and marked response are

$$\mathfrak{r}_i(v) = S(v(0), \boldsymbol{B}_i(x_i)), \qquad \mathfrak{h}_i(y) = S(B(y), A(0), \boldsymbol{B}_i(x_i)).$$

Finite collections may use different grids, and the union of all these configurations is dense. The next lemma shows that every finite combination of tests annihilating F_D also annihilates the limiting shell response.

Lemma 16.7 (Annular annihilator estimate)**.** *For every finite list of grid tests and coefficients $g_i \in \mathbb{C}$ with $\sum_i g_i \mathfrak{r}_i = 0$ on F_D, there are $\rho_l \downarrow 0$ such that*

$$\sup_{1/4\le|z|\le 2}\left|\sum_i g_i \rho_l^{3+\varkappa}\mathfrak{h}_i(\rho_l z)\right| \longrightarrow 0. \tag{16.18}$$

Proof. The local sections of Lemma 10.2 and Lemmas 14.7–14.15 represent the product of two nearby marked insertions by one marked insertion of degree D; Proposition 6.4 and Lemma 15.5 pass to the quotient. Put $h = 3+\varkappa$, so $D+h = d+4$. Recall $s_m = \log(a_m^{-1})$ from (5.1). Use the support radius $B_m = \lceil(\log\widehat{\beta}_m)^2\rceil$ of Part I and take $\rho_m = 128B_{m+1}^2 a_{m+1}$, so $\log\rho_m = -s_m + o(s_m)$.

Step 1: two-mark representation. Fix a centered cell around the distinguished point, disjoint from all spectator stencils, and a finer stopping level $k_0 \ge m_{\mathrm{ref}}(D)$ at which its replicated source estimate applies. Choose levels $j(m)$ with

$$\frac{s_{j(m)}}{s_m} \longrightarrow t = 1 + \frac{1}{64(d+12)};$$

they exist since $s_{m+1}/s_m \to 1$ (Lemma 14.2). At a grid point $y \in a_{j(m)}\mathbb{Z}^4$ with $\rho_m/16 < |y| < 4\rho_m$, choose representatives $a \in V_d, b \in V_4$ of A, B, block their two marked insertions at $y, 0$ together, and retain the Taylor terms of weight at most D. Their coefficient vector $c_{N,p}(y) \in V_D$ is chosen by inverting the marked coefficient map. The exact difference is

$$\rho_m^h O_{N,p}^{(4)}(b, y/a_N) O_{N,p}^{(d)}(a, 0) - O_{N,p}^{(D)}(c_{N,p}(y), 0).$$

Let $u_{m,1}, u_{m,2}$ be the available mark weights of the degree-d and rescaled degree-four marks at level $j(m)$. The one-mark bounds, including ρ_m^h, give $\log u_{m,1} = (-dt + o(1))s_m$ and $\log u_{m,2} = (-4t + h + o(1))s_m$; support weights, coefficient columns and threshold powers have logarithm $o(s_m)$

by (14.3) and (14.11). Transporting each source to level $m+1$ adds the volume factor $(a_{j(m)}/a_{m+1})^4$ to its weight. So the joint weight v_m and the dimensionless coefficient bound e_m of its Taylor remainder satisfy

$$\log v_m = [-D-(d+12)(t-1)+o(1)]s_m, \qquad e_m \le \exp\{[(d+12)(t-1)+o(1)]s_m\}. \tag{16.19}$$

Indeed the transported exponents sum to $-(d+4)t+h+8(1-t) = -D-(d+12)(t-1)$, and Taylor extraction costs the reciprocal joint weight times a_{m+1}^D, which cancels the D term. This is the two-mark use of the source representation (14.23), retaining its forced terms and their eligibility tests.

Step 2: backward matching. Apply (14.34) from $m+1$ to the fixed stopping level k_0. The matched remainder gains $\prod_{i=k_0}^m \ell_i^{1/8} = e^{(s_{m+1}-s_{k_0})/8} \ge e^{(s_m-s_{k_0})/8}$ in its inverse bound. Put $c_m = e^{s_m/32}$ and multiply the source by c_m; its size is then at most

$$c_m e_m e^{-(s_m-s_{k_0})/8} \le \exp\{[-5/64+o(1)]s_m\}, \qquad \tfrac{1}{32}+(d+12)(t-1)-\tfrac{1}{8} = -\tfrac{5}{64},$$

which eventually fits every fixed source allowance at k_0. The residual-source estimate (14.25) at the fixed cell and level k_0 then bounds the chessboard seminorm of the unamplified residual by $2C/c_m$, with C independent of m, N, p. The inverse estimate (16.15) in degree D bounds $c_{N,p}(y)$ on cutoff tails for each fixed m, y. Interior support follows from $32\rho_m \to 0$ and the support diameter $a_{m+1}(B_{m+1}^2+1+2048B_{m+1}^2) \to 0$.

Step 3: cancellation at grid points. Pass to an ordered limit point $c(y)$ and project it to F_D. The chessboard inequality (14.19) for the residual and the fixed exterior stencils gives, simultaneously for the finite tests,

$$|\rho_m^h \mathfrak{h}_i(y) - \mathsf{r}_i(\pi_D c(y))| \le C_i c_m^{-1},$$

with the same coefficient vector for every spectator. Multiplication by g_i and summation cancel the response term, so $f_m(y) = \sum_i g_i \rho_m^h \mathfrak{h}_i(y)$ is $O(c_m^{-1})$ at all grid points of the enlarged annulus whose four coordinates are nonzero.

Step 4: interpolation across coordinate planes. For a derivative order q, Proposition 6.4 replaces the moving label by one of degree $4+q$. The centered and reflected bounds of Lemmas 14.15 and 14.14, with (14.19), bound a field of fixed degree e by $\exp[(e+2+o(1))s_m]$, with a field-dependent constant of logarithm $o(s_m)$. The finitely many later levels required for the fixed spectator word have logarithmic scale $s_m+o(s_m)$. Choose degree bounds e_{ij} for the spectator labels and put $C_0 = 8+d+\max_i \sum_j (e_{ij}+2)$. Since $\rho_m^h \le 1$,

$$|\partial_a^q f_m(y)| \le \exp[(q+C_0+o(1))s_m] \qquad (\rho_m/16 < |y| < 4\rho_m), \tag{16.20}$$

with C_0 independent of q. Choose q with $q(t-1) > C_0$; since $a_{j(m)} = \exp[-(t+o(1))s_m]$, the qth derivatives along grid-sized segments tend to zero. On each coordinate segment of length $4(2^q+2)a_{j(m)}$ choose q grid nodes with nonzero coordinates, mutually separated by at least $a_{j(m)}$. Lagrange interpolation gives $|g| \le C_q(\max_{\text{nodes}}|g| + a_{j(m)}^q \sup|\partial^q g|)$ on the segment, with C_q independent of m. Apply it to $\operatorname{Re} f_m$ and $\operatorname{Im} f_m$ in each of the four axes; the segments stay in the enlarged annulus since $4(2^q+2)a_{j(m)} = o(\rho_m)$. Four successive interpolations give $\sup_{\rho_m/4\le|y|\le 2\rho_m}|f_m(y)| \to 0$. Select a decreasing subsequence of ρ_m; it depends on the fixed tests and g, as allowed. □

16.3 Free comparison and the conserved line

The next lemma converts Lemma 16.6 into the comparison of Part I, measured uniformly over the supremum-norm unit ball of tests. No volume refinement is needed.

Lemma 16.8 (Stable weak comparison)**.** *There is one stable scaling sequence such that, for every arity n, degree list, $u_i \in M_{d_i}^{\mathrm{Max}}$ and compact $\mathcal{K} \subset \mathrm{Conf}_n$,*

$$\sup_{\|f\|_\infty \le 1,\ \operatorname{supp} f \subset \mathcal{K}} \left| \int f(x) \big[S_n(\mathsf{U}^{\mathrm{sc}}_{k,d_i} u_i; r_k x) - S_n^{\mathrm{Max}}(\boldsymbol{u}; x) \big] dx \right| \longrightarrow 0.$$

Proof. Lemma 16.6 supplies the three estimates, Lemma 16.1 the free modulus of continuity and the free separating tests, and Proposition 6.4 the dimension bound. The set $K_{R,m} \setminus E_{8\delta_k}$ lies in finitely many strips about coordinate coincidence hyperplanes and has volume at most $C_{m,R}\delta_k$; its contribution to a tested difference is at most $C_{m,R}\|f\|_\infty((R_k^{\mathrm{cell}})^m + \|S_m^{\mathrm{Max}}\|_{K_{R,m}})\delta_k$. Outside these strips, rounding to the grid moves a point by at most $4m\mathfrak{s}_k \le \delta_k$, so the segment stays in $K_{2R,m} \cap E_{4\delta_k}$. The mean value bound and the free modulus of continuity ϖ give total tested error at most

$$C_{m,R}\|f\|_\infty \big[\varepsilon_k + ((R_k^{\mathrm{cell}})^m + \|S_m^{\mathrm{Max}}\|_{K_{R,m}})\delta_k + (R_k^{\mathrm{var}})^m \mathfrak{s}_k/\delta_k + \varpi(4m\mathfrak{s}_k) \big]. \tag{16.21}$$

Every term tends to zero by the strict inequalities (16.11): the strips cost $(R_k^{\mathrm{cell}})^m\delta_k$, the rounding costs $(R_k^{\mathrm{var}})^m\mathfrak{s}_k/\delta_k$, and both are negative powers of M_k. Arbitrary fields expand in the finitely many basis words, with coefficient sums bounded by the product of their norms.

At a fixed comparison level the lifts are bounded in a finite-dimensional space. Take an ordered limit point and project to F_d; ordered convergence holds also for convergent coefficients, by multilinearity. This proves (16.21) for linear maps $U_{k,d} : M_d^{\mathrm{Max}} \to F_d$, for every ordered limit point and the same table (Lemma 15.1).

Choose successively arity, degree and radius bounds tending to infinity, tolerances tending to zero, and strictly decreasing radii tending to zero; this is possible because $\log M_k = o(\log a_k^{-1})$ by (5.2) and (14.13), whence $a_k M_k^{\alpha_0} \to 0$. The sequence is selected before any field list. To impose inclusions exactly, choose real H_4-equivariant left inverses L_d of the free inclusions i_d and replace the maps recursively by

$$\widetilde{U}_{k,0} = U_{k,0}, \qquad \widetilde{U}_{k,d+1} = \jmath_{d,d+1}\widetilde{U}_{k,d}L_d + U_{k,d+1}(I - i_d L_d). \tag{16.22}$$

The first summand is exact on included fields and the second agrees with $U_{k,d+1}$ on a complement; since free and interacting inclusions preserve correlations, the comparison is preserved, as are reality and signed covariance, and the degree-zero unit gives all units. Strict free positivity supplies finitely many separating tested correlations in each degree; their matrices converge, so a nonzero full-rank minor makes $\widetilde{U}_{k,d}$ eventually injective, and Proposition 6.4 makes it an isomorphism. Fill the finitely many earlier indices in each degree with any of these isomorphisms. □

Proof of Proposition 6.7. By the equivalence after Proposition 6.7, uniform tested convergence on each compact set is local L^1 convergence; Lemma 16.8 gives it for one stable sequence. □

The comparison upgrades to smooth convergence, which the identification of the current needs.

Corollary 16.9 (Smooth comparison)**.** *For the stable frames of Lemma 16.8,*

$$\mathsf{d}_{k,a} = r_k(\mathsf{U}^{\mathrm{sc}}_{k,d+1})^{-1}\partial_a \mathsf{U}^{\mathrm{sc}}_{k,d} \longrightarrow \partial_a^{\mathrm{Max}} \tag{16.23}$$

in operator norm, and, whenever $u_{k,i} \to u_i$,

$$S_n(\mathsf{U}^{\mathrm{sc}}_{k,d_i} u_{k,i}; r_k\,\cdot\,) \longrightarrow S_n^{\mathrm{Max}}(u_i; \cdot) \quad \text{in } C^\infty_{\mathrm{loc}}(\mathrm{Conf}_n). \tag{16.24}$$

Proof. Choose finitely many tested free responses injective on M_{d+1}^{Max}; their interacting matrices converge to this injective matrix by Lemma 16.8. Testing $\mathsf{d}_{k,a}u$ gives, by integration by parts, minus the correlation tested against $\partial_{x_0^a} f$, which converges to the test of $\partial_a^{\mathrm{Max}} u$. An injective finite matrix has a bounded left inverse on all small perturbations, proving (16.23) on a basis.

The chain rule and (16.23) make every positional derivative converge locally in L^1, also for convergent label lists. For $g = \chi\cdot\big(S_n(\mathsf{U}^{\mathrm{sc}}_{k,d_i}u_{k,i}; r_k\,\cdot) - S_n^{\mathrm{Max}}(u_i;\cdot)\big)$, with $\chi \in C_c^\infty(\mathrm{Conf}_n)$ equal to one on a compact $\mathcal{K}$, and $M > 4n+j$, integration by parts gives $|\widehat{g}(p)| \le C_M(1+|p|)^{-M}\sum_{|\alpha|\le M}\|\partial^\alpha g\|_{L^1}$. Fourier inversion [10, Theorem 8.26] applies since g and $\widehat{g}$ are integrable; multiplying by p through order j and integrating proves uniform convergence of those derivatives on $\mathcal{K}$. □

The rank calculation (16.6) transfers to the mixed fields; the proof also fixes the sign of one current.

Lemma 16.10 (Conserved line)**.** *The real space of centered real symmetric H_4-tensorial tensors $J_{ab} \in F_4$ with $\sum_a \partial_a J_{ab} = 0$ is one-dimensional. For some nonzero J, a refinement of the sequence of Proposition 6.7 and positive c_k,*

$$(\mathsf{U}^{\mathrm{sc}}_{k,4})^{-1}(c_k^{-1} r_k^4 J_{ab}) \longrightarrow T^{\mathrm{Max}}_{ab}$$

componentwise.

Proof. Transport a free tensor by $\mathsf{U}^{\mathrm{sc}}_{k,4}$ and subtract its mixed one-point value times the unit. On centered tensors this is a real isomorphism C_k; the inverse transports back and subtracts the free expectation, and unit preservation makes the two composites identities. Let V_k be transport by $\mathsf{U}^{\mathrm{sc}}_{k,5}$ on real covariant vectors. By (16.23),

$$V_k^{-1}(r_k\,\mathrm{div})C_k \longrightarrow \mathrm{div}^{\mathrm{Max}},$$

the centering term disappearing since derivatives kill the unit. Composition with a right inverse of free divergence tends to the identity, so mixed divergence is onto. The domain and target have dimensions three and two by these isomorphisms and (16.6), so the kernel is a real line.

Choose a nonzero B on this line and set $c_k = \|(\mathsf{U}^{\mathrm{sc}}_{k,4})^{-1}(r_k^4 B)\|/\|T^{\mathrm{Max}}\|$, with norms on whole tensors. The normalized preimages have a convergent subsequence on this sphere. Stability preserves symmetry, reality and signed tensoriality; (16.24) at arity one gives centering of the limit, and (16.23) gives zero free divergence. Lemma 16.1(iii) makes the limit T^{Max} or $-T^{\mathrm{Max}}$; choose $J = B$ or $-B$ once to obtain the stated limit with positive c_k. □

Lemma 16.7 now applies to each component of the current, and the resulting fields represent both fluxes of Definition 6.8. The proof combines it with finite-dimensional duality, conservation, translation covariance and continuity of separated responses.

Lemma 16.11 (Raw shell charges)**.** *For the chosen J there are complex linear maps $\mathsf{p}_b : F_d \to F_{d+1}$ and $\mathsf{Y}_K : F_d \to F_d$, the latter real linear in K, such that, for every shell cutoff, field, spectator list and separated configuration,*

$$\mathcal{P}_b^J = S((\mathsf{p}_b A)(x_0), \boldsymbol{B}(x)), \qquad \mathcal{Y}_K^J = S((\mathsf{Y}_K A)(x_0), \boldsymbol{B}(x)).$$

Proof. Use a constant Killing field v ($\varkappa = 0$) or $v(y) = Ky$ ($\varkappa = 1$). Independence of the shell cutoff follows by integration by parts: the divergence vanishes, and $\sum_{a,b}(\partial_a v_b)J_{ab} = 0$ by symmetry.

Fix finitely many grid response tests r_i and coefficients g_i annihilating F_D, $D = d + 1 - \varkappa$. Shrink a common shell cutoff to $\chi(y/\rho_l)$, with $\chi = 1$ near zero and $\nabla\chi$ supported in $1/2 \le |y| \le 1$. Changing variables gives the sum of their fluxes as

$$-\sum_{a,b} \int (\partial_a \chi)(z) v_b(z) \sum_i g_i \rho_l^{3+\varkappa} S(J_{ab}(\rho_l z), A(0), \boldsymbol{B}_i(x_i))\, dz,$$

which is independent of l and tends to zero by (16.18). So every linear relation between the response tests is a relation between their flux values, and finite-dimensional duality gives a field in F_D matching these finitely many values: the annihilator of the image of the response matrix annihilates the flux vector.

Choose a finite separating family of grid tests; it exists because the common kernel of all separated responses is zero, grid configurations are dense, and responses are continuous. A field matching this family is unique, and adjoining any further grid test, the same field matches it. Continuity of the responses and of shell integrals, with the cutoff fixed locally, extends the identity to all separated configurations with distinguished point zero; translation covariance moves that point to arbitrary x_0. Linearity of the integral and injectivity of responses give complex linearity in A and real linearity in K. Taking $v = e_b$ and $v = K(\cdot - x_0)$ gives the two raw shell identities. □

16.4 Coordinate reflection spaces and spectral estimates

Coordinate reflection positivity alone gives a reflection Hilbert space for each coordinate axis. This subsection constructs these spaces, bounds the slab insertion operators, and bounds the low-energy spectral mass of collinear words of mixed reflection parity and of the components of a conserved tensor. Section 16.5 uses these bounds to remove the boundary at infinity before continuous rotations are known. Nothing here uses rotations or Sections 16.1–16.3: the inputs are E1–E7, Proposition 6.4 and Lemmas 15.6–15.7.

A word is $w = \prod_{i=1}^n A_i(x_i)$ with distinct points, and $\tau_z w = \prod_i A_i(x_i + z)$. Fix a norm $\|\cdot\|_{V_d}$ on each coefficient space; insertion-label bounds refer to representatives in the spaces V_d of Definition 2.8, and the resulting operators depend only on their classes in F_d. Write w^* for the word with reversed order and conjugated labels, extended antilinearly to word combinations, ϑ for linear time reflection, acting on labels through the hypercubic action of $\mathsf{s}_4^{\mathrm{refl}}$, and $\theta w = \vartheta w^*$; thus θ is the antilinear reflection in the form of Definition 6.5. The insertion rule and buffer are those preceding Lemma 15.7: $A_w v = w\, \tau_{he_4} v$, and an r-buffered word in width h has mutual sup-distance at least r and times in $[r, h - r]$. Write $\partial_{i,a} w$ for the word with A_i replaced by $\partial_a A_i$.

Repeated reflected insertions bound the operator that inserts a word into a time slab. Taking arbitrarily high powers removes the dependence of the bound on the exterior vector.

Lemma 16.12 (Slab operators)**.** *For $d \in \mathbb{N}$ and $R \ge 0$ there are $C > 0$ and $Q \in \mathbb{N}$ such that every r-buffered word w of length n, with degrees at most d and coefficient norms at most R, in a slab of width $h > 0$, has a bounded insertion operator $\mathsf{B}_h(w)$ extending A_w, with $\|\mathsf{B}_h(w)\| \le C^n (n/r)^{Qn}$ for $0 < r \le 1$. For widths h, k in which the words are buffered,*

$$\mathsf{B}_h(w)^* = \mathsf{B}_h(\tau_{he_4}\theta w), \qquad \mathsf{B}_h(w)\mathsf{B}_k(v) = \mathsf{B}_{h+k}(w\, \tau_{he_4} v), \tag{16.25}$$

$$\partial_{x_i^a} \mathsf{B}_h(w) = \mathsf{B}_h(\partial_{i,a} w), \tag{16.26}$$

with derivatives in operator norm; the empty word gives $e^{-hH_4^{\mathrm{OS}}}$.

Proof. Replace R by $\max(1, R)$. *The heat semigroup.* On the span of positive words, $\|\tau_{te_4} v\|^2 = \langle v, \tau_{2te_4} v\rangle \le \|v\|\, \|\tau_{2te_4} v\|$. E7 bounds the separated orbit, so iteration gives $\|\tau_{te_4} v\| \le \|v\|$ and

null vectors are preserved. Translation invariance gives symmetry and commuting unitary spatial translations; Gram continuity gives strong continuity. Since $e^{-H}\psi = 0$ implies $e^{-2^{-j}H}\psi = 0$, strong continuity excludes spectral mass at zero of the heat operator. Spectral calculus [19, Theorem 2.5.1 and Proposition 4.2.3] defines $H_4^{\mathrm{OS}} = \int_{(0,1]} -\log v \, dP(v)$ from the spectral measure P of the time-one heat operator; positive roots and continuity give $e^{-tH_4^{\mathrm{OS}}} \to P_4^0$ strongly.

The power argument. For a nonempty word, Lemma 15.7 supplies a width s and $D = C_0^n (n/r)^{Qn}$. On the word span put $A = A_w$, $B = A_{w^\dagger}$ with $w^\dagger = \tau_{se_4}\theta w$, and $T = BA$. The OS form gives $\langle Av, u\rangle = \langle v, Bu\rangle$, and Lemma 15.7 gives $\|T^j v\| \le K_v D^{2j}$. Set

$$d_j = D^{-2j}\|T^j v\|, \qquad M_v = \sup_{j\ge 1} d_j < \infty.$$

Symmetry of T gives $d_j^2 \le \|v\| d_{2j}$, hence $M_v^2 \le \|v\| M_v$ and $M_v \le \|v\|$. In particular $\|Av\|^2 = \langle v, Tv\rangle \le \|v\|\,\|Tv\| \le D^2\|v\|^2$. The bound is independent of K_v, and density gives the unique bounded extension. Right multiplication by $e^{-(h-s)H_4^{\mathrm{OS}}}$ gives the required width.

Identities. Positive-word matrix entries prove (16.25). Proposition 6.4 bounds differentiated-label operators in a smaller buffer; integrating their matrix entries twice along segments gives an operator-norm Taylor remainder, and induction proves (16.26) and norm smoothness. □

With e_μ as time, the same construction gives $\mathcal{H}_\mu^{\mathrm{OS}}$, $e^{-tH_\mu^{\mathrm{OS}}}$ and commuting transverse translations $\mathsf{T}_\perp(y)$, $y^\mu = 0$, using only coordinate reflection positivity and signed permutations. Their joint projection-valued measure P_μ lives on $[0,\infty)\times\mathbb{R}^3$, with variables (E,p): apply Gelfand theory to the commutative unital C*-algebra generated by $e^{-H_\mu^{\mathrm{OS}}}$ and the translation representation of $C_0(\mathbb{R}^3)$ [19, Theorems 2.4.1, 3.7.11, equation (3.89) and Proposition 4.2.3], then take the logarithm of the heat coordinate. For a word combination U, let $v_{\mu,u}(U)$ be its OS vector translated by ue_μ into positive times, and undo the damping by

$$d\nu_\mu^U(E,p) = e^{2uE}\, d\langle v_{\mu,u}(U), P_\mu(E,p) v_{\mu,u}(U)\rangle. \tag{16.27}$$

This measure is independent of u and finite on $E \le M$; point-field measures may have infinite total mass before damping. Polarize to ν_μ^{UV} on integrable tests. For nonnegative integrable g_1, g_2, spectral Cauchy–Schwarz gives

$$\left|\int f\, d\nu_\mu^{UV}\right|^2 \le \int g_1\, d\nu_\mu^U \int g_2\, d\nu_\mu^V \quad \text{if } |f|^2 \le g_1 g_2. \tag{16.28}$$

If $\vartheta_j U = \varepsilon_j U$ with $\varepsilon_j \in \{1,-1\}$, then, for $x^\mu = 0$ and t separating the words,

$$S\big(U^*\,\tau_{te_\mu + x} U\big) = \varepsilon_\mu \int e^{-tE + ip\cdot x}\, d\nu_\mu^U(E,p)\,.$$

Both point fields and collinear words are represented.

Opposite reflection signs compare coordinate spectra. Fix a set of axes $\mathcal{A} \subset \{1,2,3,4\}$ and a finite word combination U whose offsets have zero coordinates in $\mathcal{A}$, with $\vartheta_j U = \varepsilon_j U$ for $j \in \mathcal{A}$, and assume that both signs occur. For even nonnegative $f_j \in C_c^\infty(\mathbb{R})$ set $f_{j,v}(p) = v^{-1} f_j(p/v)$. Throughout, λ ranges over $e^{\mathbb{N}}$.

Lemma 16.13 (Opposite reflection signs)**.** *For every $i \in \mathcal{A}$, $q < 0$, $\beta_j \le 0$ and $\delta > 0$,*

$$\int_{E\le 1} \frac{\prod_{j\in\mathcal{A}\setminus\{i\}} f_{j,\lambda^{\beta_j}}(p_j)}{E + \lambda^q}\, d\nu_i^U = O_\delta(\lambda^\delta).$$

Constants may depend on $U, \mathcal{A}, q, \beta, f, \delta$.

Tested against a product of even profiles, the reflected correlation of U has a positive spectral representation in each time axis $i \in \mathcal{A}$, with sign ε_i. For two axes of opposite parity, these representations of the same number differ by a sign; after the annular corrections below, this gives the transfer inequality (16.33). A narrow momentum profile is replaced by an annular profile with the same low moments; the residual has a nonnegative Laplace response, so the low-energy response in axis i is bounded by two responses, one with each of the profiles in i and in the next axis j replaced by its annular correction, which the moment bounds control. Each transfer replaces one narrow profile by one whose width is bounded below; once every axis has been treated, the moment bound applies directly.

Proof. Moments. For a placement of margin $r > 0$, E7 bounds the $2m$th transverse derivative of the reflected-pair correlation by Cr^{-D}. Its centered difference has spectral integrand $[2\sin(tp_j/2)/t]^{2m}e^{-2uE}$, and the integral remainder formula and Fatou give

$$\int e^{-2uE}|p_j|^{2m}\,d\nu_\mu^U \le Cr^{-D}. \tag{16.29}$$

Extra heat supplies energy powers, so for every damping u as in (16.27) and $m \in \mathbb{N}$,

$$\int e^{-2uE}(1+E+|p|)^m\,d\nu_\mu^U < \infty\,. \tag{16.30}$$

For offsets perpendicular to μ take $u = t/2$, $r \asymp t$; integrating Ct^{-D} against $t^D\,dt$ on $(0,1)$ gives, after enlarging D, $\int(1+E)^{-D}|p_j|^{2m}\,d\nu_\mu^U < \infty$.

Annular correction. Choose even $\sigma \ge 0$ of integral one supported in $1 \le |v| \le 2$. Let L_j, $0 \le j < m$, be the Lagrange polynomials of degree $m-1$ with $L_j((i+2)^2) = \delta_{ij}$. For radius $\varrho > 0$ define

$$\int \mathcal{C}_{m,\varrho}f(q)g(q)\,dq = \int \sigma(v)\int f(p)\sum_{j<m} L_j((p/(\varrho v))^2)g(\varrho v(j+2))\,dp\,dv.$$

Since $\sum_j L_j(x)(j+2)^{2i} = x^i$ for $i < m$, the residual $f - \mathcal{C}_{m,\varrho}f$ has vanishing moments $\int p^{2i}(\cdot)\,dp$ for $i < m$, hence $2m$ vanishing moments when f is even; the correction is an annular profile whose coefficients contain ϱ^{-2r}. Put $\operatorname{Lap} f(E) = \int e^{-E|t|}\widehat{f}_+(t)\,dt$. Interpolating $(p^2+E^2)^{-1}$ gives the residual response

$$\mathcal{R}_{m,\varrho}f(E) = 2E\int \frac{f(p)}{p^2+E^2}\int \sigma(v)\prod_{j<m}\frac{(\varrho v(j+2))^2 - p^2}{(\varrho v(j+2))^2 + E^2}\,dv\,dp = \operatorname{Lap}(f - \mathcal{C}_{m,\varrho}f)(E). \tag{16.31}$$

Its value at $E = 0$ is $2\pi f(0)$, and for $f \ge 0$ supported in $[-\varrho,\varrho]$ all factors lie in $[0,1]$. Let $f \ge 0$ be even, supported in $[-1,1]$ and equal to one on $[-\frac12,\frac12]$, and let $0 < v \le 1 \le \varrho$. Then

$$\begin{aligned} \frac{c_m}{E+v} \le \mathcal{R}_{m,\varrho}f_v(E) &\le 2\pi\|f_v\|_\infty && (0 \le E \le 1),\\ \mathcal{R}_{m,\varrho}f_v(E) &\le C(\varrho/E)^D && (E \ge 1,\ m \ge D/2), \end{aligned}$$

with $C = C(m, D, \|f\|_{L^1})$: the upper bounds follow from $2E\int(p^2+E^2)^{-1}dp = 2\pi$ and from bounding the product by $\prod_j(2\varrho(j+2)/E)^2$; for the lower bound the product is at least $2^{-m}(1+E)^{-2m}$ for $|p| \le v$, and one integrates over $0 < p < vE/(2(E+v))$. With the moment bound, for $\varrho \ge 1$, $\operatorname{supp} f_v \subset [-\varrho,\varrho]$ and $m \ge D/2$ chosen after the moment exponent D,

$$\int \mathcal{R}_{m,\varrho}f_v(E)|p_j|^{2r}\,d\nu_\mu^U \le C(v^{-1} + \varrho^D). \tag{16.32}$$

Transfer between two axes. For $i, j \in \mathcal{A}$ write $\operatorname{Resp}_i(g) = \int \operatorname{Lap} g_i(E) \prod_{k \in \mathcal{A}\setminus\{i\}} g_k(p_k)\, d\nu_i^U$. For $\mathcal{A}$ containing all axes, the same product test gives

$$(2\pi)^{-3} \int S(U^* \tau_x U) \prod_k \widehat{g}_{k+}(x_k)\, dx = \varepsilon_i \operatorname{Resp}_i(g) = \varepsilon_j \operatorname{Resp}_j(g),$$

since $\int e^{ipx} \widehat{g}_+(x)\, dx = 2\pi g(p)$ for even profiles, while the time integral is $\operatorname{Lap} g(E)$; both temporal profiles have the zero moments required at collisions. For smaller $\mathcal{A}$, put $g_k(p) = h(p/T)$ in omitted axes, with $h \in C_c^\infty(\mathbb{R})$ even and equal to one near zero, and let $T \to \infty$ by (16.32). Hence $\operatorname{Resp}_i(g) = -\operatorname{Resp}_j(g)$ when $\varepsilon_i = -\varepsilon_j$. Subtract annular corrections a_i, a_j from the two profiles in the current axes; positivity of the residual responses gives

$$\operatorname{Resp}_i(g_i - a_i; g_{\neq i}) \le \operatorname{Resp}_i(g_i - a_i; g_j \mapsto a_j) + \operatorname{Resp}_j(g_j - a_j; g_i \mapsto a_i), \tag{16.33}$$

after expanding the signed identity and discarding $\operatorname{Resp}_j(g_j - a_j; g_{\neq j}) \ge 0$.

The walk. Choose a finite walk $i_0, \dots, i_L$ in $\mathcal{A}$ with opposite signs at consecutive entries, $i_0 = i$, every axis among $i_0, \dots, i_{L-1}$, and i_L already visited; it exists because both signs occur (insert an axis of the opposite sign between two entries of the same sign). Put $V_s = \{i_0, \dots, i_{s-1}\}$. Fix $K \ge 0$ and $\delta > 0$. The induction claim is that there is $A_s > 0$ such that, for every fixed family of even nonnegative compactly supported profiles f_j, all exponents with $0 \le \omega_j \le A_s$ $(j \in V_s)$ and $-K \le \omega_j \le A_s$ $(j \notin V_s)$, every $0 \le \eta \le A_s$ with $\omega_{i_s} \le \eta$, and $R_0 \ge 1$ such that the profile in axis i_s is supported in $[-R_0\lambda^\eta, R_0\lambda^\eta]$, all large subtraction orders m satisfy

$$0 \le \int \mathcal{R}_{m, R_0\lambda^\eta} f_{i_s, \lambda^{\omega_{i_s}}}(E) \prod_{j \in \mathcal{A}\setminus\{i_s\}} f_{j,\lambda^{\omega_j}}(p_j)\, d\nu_{i_s}^U \le C\lambda^\delta, \tag{H_s}$$

with C and the threshold in λ depending on the fixed data. Small positive exponents on unvisited axes permit later visits.

At $s = L$ every width is at least one, so all transverse suprema are bounded, and (16.32) with $r = 0$ bounds the integral by $C(\lambda^{-\omega_{i_L}} + \lambda^{\eta D})$; $A_L D < \delta$ gives (H_s). For the step from $s+1$ to s put $i = i_s$, $j = i_{s+1}$ and $p_* = K|\mathcal{A} \setminus \{i, j\}|$. The parameters are chosen in this order:

	choice	depends on
1	$\alpha' = A_{s+1}$	the case $s+1$ of (H_s)
2	$r \in \mathbb{N}$ with $2r\alpha' > p_* + \max(K, \delta) + 1$	$\alpha', K, \delta, \lvert\mathcal{A}\rvert$
3	moment exponent D of (16.32) for $\lvert p_j\rvert^{2r}$	r, U
4	$0 < A_s < \min(\alpha', \delta/(D+1))$	α', δ, D
5	subtraction orders m	D and the collision orders

Correct the profile in axis i at radius $R_0\lambda^\eta$ and the one in axis j at radius $\lambda^{\alpha'}$. The correction in i is bounded by a fixed even nonnegative annular profile of width λ^η, since its coefficients contain only factors $\lambda^{2r'(\omega_i - \eta)} \le 1$; so the second term of (16.33) is bounded by the case $s+1$, the exponent in axis i having become nonnegative. In the first term the correction in j is bounded by $C_r \lambda^{-2r\alpha'} |p_j|^{2r}$, with bounded coefficients since $\omega_j < \alpha'$, and the other transverse suprema cost at most λ^{p_*}. Equation (16.32) bounds this term by

$$C\lambda^{p_* - 2r\alpha' + \max(-\omega_i, \eta D)} \le C\lambda^{p_* - 2r\alpha' + \max(K, A_s D)} = O(\lambda^\delta).$$

Both residual responses are nonnegative, so (16.33) with the positive annular majorants proves the case s.

Conclusion. Apply the case $s = 0$ with $K \ge \max(-q, \max_j(-\beta_j))$, $\omega_i = q$, $\omega_j = \beta_j$ for $j \ne i$, and $\eta = 0$, the profile in axis i being the bump of the lower response bound. On $E \le 1$ that bound makes the stated integral at most c_m^{-1} times the integral in (H_s). □

Fix $a \ne b$ and a line axis l. Set $\omega_a = \omega_b = -100$, $\omega_l = -150$ when $l \notin \{a, b\}$, and $\omega_j = -300$ for $j \notin \{a, b, l\}$. For a fixed even nonnegative $\psi \in C_c^\infty(\mathbb{R})$ equal to one on $[-1, 1]$ put

$$w_j = \lambda^{1+\omega_j}, \qquad \mathcal{K}_\lambda(p) = \prod_{j \ne \mu} w_j^{-1} \psi(p_j / w_j), \qquad I_\mu(U; v) = \int_{E \le 1} \frac{\mathcal{K}_\lambda(p)}{E + \lambda^v} \, d\nu_\mu^U .$$

These are the scales of the box on which the vector field of Lemma 16.15 equals Ky: $|y_a|, |y_b| \le \lambda^{100}$, $|y_l| \le \lambda^{150}$, $|y_j| \le \lambda^{300}$. Set $B = 2^{-4} \prod_j (1 + \varepsilon_j \vartheta_j) U$ for a word combination U with offsets on $\mathbb{R} e_l$, with $\varepsilon_a = \varepsilon_b = -1$ and the other signs positive. Let $J_{ij} \in F_4$ be real, symmetric, H_4-tensorial and conserved; no Ward normalization is assumed. The index $J_{\mu b}$ in a measure denotes the one-point word $J_{\mu b}(0)$. Consider also tests $\Psi_\lambda(E, p)$ such that, for some fixed K and every $L \ge 0$,

$$|\Psi_\lambda(E, p)| \le C_L \lambda^K e^{-\lambda^{-\omega_\mu} E} \prod_{j \ne \mu} (1 + \lambda^{-\omega_j} |p_j|)^{-L} , \tag{16.34}$$

and put $X_\lambda = \{E > 1\} \cup \bigcup_{j \ne \mu} \{|p_j| > w_j\}$. The scaled Fourier–Laplace tests of Lemma 16.15, with fixed polynomial prefactors, have this bound. Constants below may depend on the fixed words, tensor, profiles and exponents.

Lemma 16.14 (Coordinate spectral estimates)**.** *For every projected collinear B, $\delta > 0$ and $M \ge 0$, with $\gamma_\mu = 1$ if $\mu = l$ and $\gamma_\mu = -\omega_l$ otherwise, and $W = \lambda^{-99}$,*

$$I_\mu(B; -3000) = O(\lambda^{\gamma_\mu}), \qquad I_\mu(J_{\mu b}; \omega_\mu) = O_\delta(\lambda^\delta) \quad (\mu \ne b), \tag{16.35}$$

$$\int_{E \le 1} \frac{E W^4}{(E + W)^4} \mathcal{K}_\lambda \, d\nu_\mu^{J_{\mu\mu}} = O_\delta(\lambda^\delta W^4), \tag{16.36}$$

and $I_\mu(J_{\mu\mu}; 0) = O(\lambda^{900})$ and $\int_{X_\lambda} \Psi_\lambda \, d\nu_\mu^{UV} = O_M(\lambda^{-M})$.

The sign estimates come from Lemma 16.13; the diagonal estimate comes from conservation. Recall also (16.28), (16.30), and the holomorphy for $\operatorname{Re} t > 0$ of one-insertion heat matrix elements, which follows from Lemma 16.12 and spectral calculus.

Proof. Sign estimates. In direction l, choose $2u$ separating the offsets and use $R_v(E) = \int e^{-E|t|}(1 - \chi(t))\phi(vt)\, dt$, where $0 \le \chi \le 1$ is compactly supported and equals one on $[-2u, 2u]$, and $\phi = \widehat{(\sigma * \sigma)}_+ \ge 0$, $\phi(0) = 1$. It satisfies $C/(E + v) \le R_v(E) \le C' v^{-1} e^{-2uE}$ for $E \le 1$ and small v: integrate over $R < t < R + c/(E + v)$ beyond $\operatorname{supp} \chi$ for the lower bound, and drop $1 - \chi$ and scale for the upper. The removed profile has bounded supremum, and (16.29) permits the transfer (16.33) through an opposite transverse sign; $\delta = 1$ gives $O(\lambda)$ when $\mu = l$. For $\mu \ne l$, omit the l profile, apply Lemma 16.13 on $\mathcal{A} = \{j : j \ne l\}$, where both signs occur for the specified parity, and restore its supremum $O(\lambda^{-1-\omega_l})$; $\delta = 1$ gives $O(\lambda^{-\omega_l})$. The same lemma for the point word $J_{\mu b}$, $\mu \ne b$, gives the second estimate.

Diagonal tensor: moment-subtracted extension. Write $C_{ab,cd}(x) = S(J_{ab}(0) J_{cd}(x))$, $x \ne 0$. Choose D_{col} bounding the collision powers of these pairs and their first derivatives. For even $\chi = 1$ near zero with compact support put $\Pi_{D_{\mathrm{col}}} f(t) = \chi(t) \sum_{r < D_{\mathrm{col}}} f^{(r)}(0) t^r / r!$, and define

$$R^0_{ab,cd}(f) = \int C_{ab,cd}(x) \left[\prod_j f_j(x_j) - \prod_j \Pi_{D_{\mathrm{col}}} f_j(x_j) \right] dx.$$

Taylor vanishing gives integrability, and a zero Taylor factor recovers the unsubtracted integral. Average under signed symmetries and pair exchange to obtain R. Conservation makes $\sum_i \partial_i R_{ib,cd}$ a jet at zero, and inversion $x \mapsto -x$ makes R even, eliminating even divergence orders. To remove the linear orders, put $\chi_0(x) = \prod_j \chi(x_j)$ and $q_{ab,cd} = \sum_i R_{ib,cd}(\partial_i(x_a\chi_0))$, and replace R by $R_{ab,cd} - q_{ab,cd}\delta_0$; reflections kill coefficients unless indices occur in pairs, and permuting them gives the pair symmetries. Write $\widehat{R}_{ab,cd}$ for the Fourier transform of this corrected extension. Its divergence is the polynomial $F_{b,cd}(p) = \sum_i p_i \widehat{R}_{ib,cd}(p)$, with no terms of degree below three. Symmetry gives

$$\sum_{j,k\neq\mu} p_j p_k \widehat{R}_{k\mu,j\mu} = p_\mu^2 \widehat{R}_{\mu\mu,\mu\mu} + \sum_{j\neq\mu} p_j F_{\mu,j\mu} - p_\mu F_{\mu,\mu\mu},$$

and the same two contractions on the other index pair give

$$\begin{aligned} \mathcal{Q}_0 = \widehat{R}_{\mu\mu,\mu\mu}, \qquad \mathcal{Q}_1 = \sum_{j,k\neq\mu} p_j p_k \widehat{R}_{k\mu,j\mu} = p_\mu^2 \mathcal{Q}_0 + D_1, \\ \mathcal{Q}_2 = \sum_{j,k,r,s\neq\mu} p_j p_k p_r p_s \widehat{R}_{sk,rj} = p_\mu^2 \mathcal{Q}_1 + D_2, \end{aligned} \tag{16.37}$$

where multiplication by p_j acts on the test and D_1, D_2 are polynomial remainders.

Diagonal tensor: annular bounds. On a fixed temporal annulus, Lemma 16.13 and (16.28) bound each $\widehat{R}_{ab,cd}$ by $O_\delta(\lambda^\delta)$. For a diagonal entry, divide the annular profile by p_μ and move the derivative to the label, which then has both signs; moment subtraction justifies this at the collision. To remove the remaining zero-moment restrictions, expand one transverse profile into its residual and its fixed annular correction. For an odd profile f supported in $[-r_0, r_0]$ use $p^{-1}\mathcal{C}_{m,\varrho}(pf)$ as correction; for $f_v(p) = v^{-1}f(p/v)$, splitting at $E = v$ gives $\int_{|p|\le r_0 v} |pf_v(p)|/(p^2+E^2)\,dp \le C/(E+v)$ from $|f(p)| \le C|p|$. The factor $(1+E)^{-2m}$ makes the even lower bound dominate this residual, including at zero, and (16.28) with (16.33) controls it. Each correction fixes one more transverse annulus with bounded coefficients $\lambda^{2r\beta}$, $\beta \le 0$; the induction ends after at most three coordinates, and finite Taylor terms are bounded on normalized shrinking profiles. Since each transverse momentum costs at most W, the annular bounds for $\mathcal{Q}_0, \mathcal{Q}_1, \mathcal{Q}_2$ are $O_\delta(\lambda^\delta)$, $O_\delta(\lambda^\delta W^2)$ and $O_\delta(\lambda^\delta W^4)$. The errors cost $O(W^4)$ and $O(W^6)$, since divergence has degree at least three and the contractions add one and three powers; one subtraction order m above these degrees annihilates them by its zero temporal moments.

Diagonal tensor: positivity. Put $g_{v,r}(p) = v^{-1}(p/v)^{-2r}\sigma(p/v)$, so $p^2 g_{v,r+1} = v^2 g_{v,r}$. Keep $\mathcal{K}_\lambda$ fixed and set

$$c_r(t) = W^{-2r} \operatorname{Re} \mathcal{Q}_r(g_{tW,r}), \qquad d_r = W^{-2r} \operatorname{Re} \mathcal{Q}_r\big((1 - \mathcal{C}_{m,8})(g_{W,r} - g_{2W,r})\big).$$

Conservation gives $c_{r+1}(t) = t^2 c_r(t) + O_\delta(\lambda^\delta W^2)$ for $t = 1, 2$, $r = 0, 1$. The correction of $g_{W,r} - g_{2W,r}$ has zero zeroth moment and other even moments $O(W^2)$, so $d_r = c_r(1) - c_r(2) + O_\delta(\lambda^\delta W^2)$. Reflection positivity for $J_{\mu\mu}$ makes the spectral representation of $\mathcal{Q}_0$ positive; on corrected profiles it gives $d_0 \ge 0$, $d_1 \le 0$, $d_2 \ge 0$, since multiplication by p_μ^2 becomes $-E^2$. Consequently $4d_0 = 5d_1 - d_2 + O_\delta(\lambda^\delta W^2)$ and $d_0 = O_\delta(\lambda^\delta W^2)$. Subtracting $(E^2 + 4W^2v^2)^{-1}$ from $(E^2 + W^2v^2)^{-1}$ in (16.31), with $1 \le |v| \le 2$, bounds the response difference below by $c_m EW^2/(E+W)^4$ for $0 < E \le 1$; both vanish at zero. Since the response difference and $\mathcal{K}_\lambda$ are nonnegative,

$$c_m \int_{E\le 1} \frac{EW^2}{(E+W)^4} \mathcal{K}_\lambda \, d\nu_\mu^{J_{\mu\mu}} \le d_0 = O_\delta(\lambda^\delta W^2),$$

and multiplying by W^2/c_m gives the third estimate.

Crude and far-region bounds. Finiteness of $\nu_\mu^{J_{\mu\mu}}(E \le 1)$ and $\|\mathcal{K}_\lambda\|_\infty = O(\lambda^{897})$ give the crude bound. On $E > 1$, (16.34) gives $C_M \lambda^{-M} e^{-t_0 E}$ for a fixed $t_0 > 0$; on $|p_j| > w_j$, the inequality $\lambda^{-\omega_j}|p_j| > \lambda$ gives arbitrary decay. Integrate by (16.30) and use (16.28) for mixed terms. □

16.5 Ward identities and rotations

Write $\boldsymbol{q} = (A_i(x_i))_i$ and define the angular sum

$$\mathrm{Ang}_K[\mathsf{Y}, \mathsf{p}](\boldsymbol{q}; x) = \sum_i \left[S(\dots, \mathsf{Y}_K A_i, \dots; x) + \sum_b (Kx_i)_b S(\dots, \mathsf{p}_b A_i, \dots; x) \right]. \tag{16.38}$$

It is the total angular flux of the raw charges of Lemma 16.11. Fix distinct axes a, b, a line axis l, and the exponents ω_j, profile ψ and weights $\mathcal{K}_\lambda$ of Section 16.4. For a word B on $\mathbb{R}e_l$ projected to odd parity in a, b and even parity in the other axes, Lemma 16.14 gives the sign estimates (16.35), the diagonal bound (16.36), the crude bound $I_\mu(J_{\mu\mu}; 0) = O(\lambda^{900})$, and $O_M(\lambda^{-M})$ from the region X_λ. These estimates use coordinate reflection positivity, signed permutations and conservation only; no continuous rotation enters them, and they precede the phase decomposition of Section 17, which uses the rotations obtained here.

The next lemma removes the angular boundary at infinity before continuous rotations are known. Its mechanism is as follows. The shell identities and conservation write the angular sum as the integral of the conserved tensor against the derivative of a vector field equal to Ky on a large box; the sum does not depend on the box, so it suffices that this integral tends to zero. No decay of correlations is assumed, so the smallness must come from positivity. After a translation the configuration lies on a line $\mathbb{R}e_l$. A coordinate reflection in an axis other than l fixes every point of this line, and the reflection in l maps it to itself; so reflecting the integration variable replaces the spectator word by its parity projections, which are again combinations of words on the line, odd in a, b and even in the other axes. Such a word has positive spectral representations of opposite signs in two coordinate time axes, and Lemma 16.13 turns this into a bound on its low-energy spectral mass; the diagonal components of the tensor are controlled by conservation instead (Lemma 16.14). The box has the large anisotropic extents $\lambda^{-\omega_j}$ of Section 16.4. In the terms that survive two exact cancellations, differentiating the cutoff in the time axis μ leaves a profile of bounded L^1 norm supported where $|y_\mu| \ge \lambda^{-\omega_\mu}$, whose Laplace transform gains the factor $\lambda^{\omega_\mu}/(E + \lambda^{\omega_\mu})$. On the momentum side the cutoff is concentrated where $|p_j| \lesssim \lambda^{\omega_j}$; the windows $|p_j| \le w_j = \lambda^{1+\omega_j}$ are wider by the factor λ, so outside them every power of λ is gained. The exponents are chosen so that every term is $O(\lambda^{-101})$ before the prefactor λ^{100}, the size of $|Ky|$ on the box.

Lemma 16.15 (Collinear angular identity)**.** *For every separated configuration x on a coordinate line, every field list $\boldsymbol{q}$ and every $K \in \mathfrak{so}(4)$, the raw charges satisfy*

$$\mathrm{Ang}_K[\mathsf{Y}, \mathsf{p}](\boldsymbol{q}; x) = 0. \tag{16.39}$$

Proof. It suffices to treat $K = E_{ba} - E_{ab}$, $a \ne b$, for which $(Ky)_b = y_a$, $(Ky)_a = -y_b$ and the other components vanish. Choose even $\eta \in C_c^\infty(\mathbb{R})$ equal to one on $[-1, 1]$ and put $\ell(t) = t\eta(t)$. Set

$$f_{ab}(y) = \lambda^{100} \ell(\lambda^{-100} y_a) \prod_{j \ne a} \ell'(\lambda^{\omega_j} y_j), \qquad V_b = f_{ab}, \quad V_a = -f_{ba}, \quad V_j = 0 \ (j \ne a, b).$$

The vector field V equals Ky on the box $|y_a|, |y_b| \le \lambda^{100}$, $|y_l| \le \lambda^{150}$, $|y_j| \le \lambda^{300}$, hence near every insertion for large λ. Disjoint shell cutoffs, the raw shell identities of Lemma 16.11 and conservation give $\mathrm{Ang}_K = -\int \sum_{\mu,c} \partial_\mu V_c \, S(J_{\mu c}(y), \boldsymbol{q})\, dy$.

Two exact cancellations. Only the columns $c = a, b$ occur, since $V_c = 0$ otherwise. Since $\omega_a = \omega_b$, both $\partial_a f_{ab}$ and $\partial_b f_{ba}$ equal $\prod_j \ell'(\lambda^{\omega_j} y_j)$, and $J_{ab} = J_{ba}$; so the term $\mu = a$ of the column b cancels the term $\mu = b$ of the column a:

$$\sum_{\mu,c} \partial_\mu V_c \, J_{\mu c} = \sum_{\mu \ne a} \partial_\mu f_{ab} \, J_{\mu b} - \sum_{\mu \ne b} \partial_\mu f_{ba} \, J_{\mu a} \,. \tag{16.40}$$

These are identities for every λ; no estimate enters them. The second sum is the first with a, b exchanged, and $\omega_a = \omega_b$, so the bounds below for the first sum apply to it; this includes its diagonal component J_{aa}. Since the angular sum is independent of λ, it suffices to show that each term $\mu \ne a$ of the first sum is $O(\lambda^{-101})$ before the prefactor λ^{100}.

Spectral form of one term. Fix $\mu \ne a$. Reflecting the integration variable in each axis, including the tensor signs, replaces the spectator word by its odd projection in a, b and its even projections in the other axes; if the line axis is reflected, the projection is a finite linear combination of collinear words. The term becomes a pairing against the measure $\nu_\mu^{B, J_{\mu b}}$, the polarization of the coordinate spectral measures (16.27) of B and $J_{\mu b}$ in the time axis μ; it lives on $\{E \ge 0\}$, including $E = 0$. The hypotheses of Lemma 16.14 hold: B is a combination of words on $\mathbb{R}e_l$, odd in a, b and even in the other axes, and J is real, symmetric, H_4-tensorial and conserved. The differentiated profile is supported where $|y_\mu| \ge \lambda^{-\omega_\mu}$, so its Laplace transform has the factor $e^{-E/\lambda^{\omega_\mu}} \le \lambda^{\omega_\mu}/(E + \lambda^{\omega_\mu})$. For each of the three spatial factors $f_j = \ell$ or ℓ', Fourier scaling gives, on $|p_j| \le w_j$,

$$|\widehat{f_j(\lambda^{\omega_j} \cdot)}(p_j)| \le \lambda^{-\omega_j} \|f_j\|_{L^1} \le C \lambda w_j^{-1} \psi(p_j / w_j).$$

Their product gives $\lambda^3 \mathcal{K}_\lambda$. For an off-diagonal component, (16.28) bounds the integral over $E \le 1$ and the windows by $C\lambda^{3+\omega_\mu}[I_\mu(B; -3000) I_\mu(J_{\mu b}; \omega_\mu)]^{1/2}$, with $\delta = 1$. For $\mu = b$ split at $E = \lambda^{-3000}$. Above it use $e^{-E/\lambda^{-100}} \le 4[W/(E+W)]^2$ and (16.36); below it, where $E = 0$ lies, $E + \lambda^{-3000} \le 2\lambda^{-3000}$ supplies λ^{-1500} in Cauchy–Schwarz, and the crude bound $I_\mu(J_{\mu\mu}; 0) = O(\lambda^{900})$ completes the estimate.

One term in detail. Take $a = 1$, $b = 2$, $l = 3$, so $(\omega_1, \omega_2, \omega_3, \omega_4) = (-100, -100, -150, -300)$, and the term $\mu = 3$, with component J_{32}. Its four factors are the prefactor λ^{100}; the Fourier factors in the axes $1, 2, 4$, giving $\lambda^3 \mathcal{K}_\lambda$; the Laplace factor $\lambda^{-150}/(E + \lambda^{-150})$ in the time axis 3; and the Cauchy–Schwarz factor $[I_3(B; -3000)\, I_3(J_{32}; -150)]^{1/2} = O\big((\lambda \cdot \lambda)^{1/2}\big)$. Together:

$$\lambda^{100} \cdot \lambda^3 \cdot \lambda^{-150} \cdot \lambda^{(1+1)/2} = \lambda^{-46}.$$

This is one term of the estimate, not the whole of it; the table lists all of them.

All terms. The table lists the power of λ in each case, before and after the prefactor; the line-axis cost 150 is the largest value of γ_μ in (16.35) for $\mu \ne l$.

term and region	bounds used	before	after
$\mu = l \notin \{a, b\}$	$\gamma_\mu = 1$, $\delta = 1$	$3 - 150 + (1+1)/2 = -146$	-46
$\mu \notin \{a, b, l\}$	$\gamma_\mu \le 150$, $\delta = 1$	$3 - 300 + (150+1)/2 = -221.5$	-121.5
$\mu = b$, $E \ge \lambda^{-3000}$	$\gamma_\mu \le 150$, (16.36)	$3 + (150 + 1 - 396)/2 = -119.5$	-19.5
$\mu = b$, $E < \lambda^{-3000}$	$\gamma_\mu \le 150$, crude bound	$3 - 1500 + (150 + 900)/2 = -972$	-872
outside $E \le 1$ and the windows	region X_λ	$-M$	$100 - M$

Every row before the prefactor is at most -101, the last for $M > 201$. The angular sum is therefore $O(\lambda^{-1})$, which is more than needed: any negative power would do, since the sum is independent of λ. So it is zero, and linearity in K proves (16.39). □

To pass from lines to arbitrary configurations, the rotation argument needs analyticity of a vector-valued function, obtained from its Gram kernel.

Lemma 16.16 (Analyticity from a stationary Gram kernel)**.** *Let $v : \mathbb{R} \to \mathcal{H}$ be a smooth curve in a Hilbert space with $\langle v(s), v(s')\rangle = A(s'-s)$, and suppose that the derivatives of A at zero coincide with those of a function analytic at zero. Then v is real analytic at every point of $\mathbb{R}$.*

Proof. Differentiating the Gram identity gives $\langle v^{(m)}(s), v^{(k)}(s')\rangle = (-1)^m A^{(k+m)}(s'-s)$. Let $\rho > 0$ be smaller than the radius of the analytic function; Cauchy estimates give $|A^{(2m)}(0)| \le C\rho^{-2m}(2m)!$. Hence

$$\|v^{(m)}(s)\|^2 = |A^{(2m)}(0)| \le C\rho^{-2m}(2m)! \le C(2/\rho)^{2m}(m!)^2$$

for every s, using $(2m)! \le 4^m(m!)^2$. Taylor's remainder in the Hilbert space then tends to zero on the interval of radius $\rho/2$ around every point. □

For a finite-dimensional rotation action write $\mathsf{L}_d(K) = \frac{d}{dt} R_d(e^{tK})\big|_{t=0}$ for its generator. The shell charges are now identified without assuming rotation covariance; the proof fixes their common scale and then constructs an orthogonal-group action.

Lemma 16.17 (Rotation and charge identification)**.** *There are $c_\star \in \mathbb{R} \setminus \{0\}$ and continuous complex linear $O(4)$ representations R_d extending H_4, fixing units and commuting with conjugation and inclusions, with*

$$\mathsf{p}_b = c_\star \partial_b, \qquad \mathsf{Y}_K = c_\star \mathsf{L}_d(K), \qquad R_{d+1}(h)\partial_a = \sum_i h_{ia}\partial_i R_d(h),$$

$$S_n(R_{d_i}(h)A_i; hx) = S_n(\boldsymbol{A}; x) \qquad (h \in O(4)).$$

For every d, an isomorphism $\mathcal{V}_d : M_d^{\mathrm{Max}} \to F_d$ intertwines the free and interacting $O(4)$ actions.

Proof. We first identify the translation charge up to one common scalar. After this normalization the angular charge satisfies the derivative commutator needed to extend the collinear identity. We then prove that identity on all separated configurations, remove the bracket defect by finite free tests, and construct an intertwiner with the free rotation action.

Step 1: shell commutators. Fix a shell cutoff and move its distinguished point. Differentiating the two shell identities of Lemma 16.11 under the compact integral gives

$$\partial_b \mathsf{p}_a = \mathsf{p}_a \partial_b, \qquad \partial_b \mathsf{Y}_K + \sum_i K_{ib}\mathsf{p}_i = \mathsf{Y}_K \partial_b, \tag{16.41}$$

the second term coming from the derivative of $K(y - x_0)$. Testing the unit, applying conjugation or a signed permutation, and including a field show that the charges kill units, are real, have vector or angular covariance, and respect inclusions; injectivity of responses turns these tested identities into field identities.

Step 2: the common normalization. Translate a collinear configuration by e_l in (16.39), with $K = E_{bl} - E_{lb}$; subtraction gives $\sum_i S(\dots, \mathsf{p}_b A_i, \dots; x) = 0$ for $b \neq l$, and (16.41) gives every positional jet of these identities. Translation invariance gives the same identities for ∂. Apply the finite free vector tests (16.9) to the transported mixed vector maps; their test matrices converge

by (16.24). Adjoining a real functional nonzero on ∂^{Max}, the free matrix becomes injective and stays so under small perturbations, so the mixed kernel has dimension at most one. It contains p and ∂, so $\mathsf{p} = c_d\partial$ in every large degree; inclusions and nonzero free differentiation make the constants equal and give the identity in lower degrees. Rescale the two shell identities by r_k/c_k and $1/c_k$; changing variables $y = r_k z$ and using (16.24) gives

$$\frac{r_k}{c_k}(\mathsf{U}^{\mathrm{sc}}_{k,d+1})^{-1}\mathsf{p}_a\mathsf{U}^{\mathrm{sc}}_{k,d} \to \partial^{\mathrm{Max}}_a, \qquad c_k^{-1}(\mathsf{U}^{\mathrm{sc}}_{k,d})^{-1}\mathsf{Y}_K\mathsf{U}^{\mathrm{sc}}_{k,d} \to \mathsf{L}^{\mathrm{Max}}_d(K), \tag{16.42}$$

since finite separating tests recover operator limits from the shell integrals. In particular $c_\star = c_d \neq 0$ and $c_k \to c_\star$. Put $\mathsf{D}_K = c_\star^{-1}\mathsf{Y}_K$; then (16.41) reads $\partial_b\mathsf{D}_K + \sum_i K_{ib}\partial_i = \mathsf{D}_K\partial_b$.

Step 3: analyticity of one-insertion vectors. By this commutator, every positional derivative of an angular sum is again an angular sum, with differentiated labels; so all jets of the collinear identity vanish on a line. In a coordinate plane write $G_{u,v}(t,s) = S(u(te_a + se_b), v(0))$, (ϱ,θ) for polar coordinates, and $G^{\mathrm{rot}}_{u,v}(t,s) = S((e^{-\theta\mathsf{D}_K}u)(\varrho e_a), (e^{-\theta\mathsf{D}_K}v)(0))$. For positive separation the axial function is a matrix coefficient of $e^{-tH^{\mathrm{OS}}_a}$ in $\mathcal{H}^{\mathrm{OS}}_a$, after fixed positive-time damping of its end vectors, hence analytic by Lemma 16.12 and spectral calculus; so G^{rot} is analytic. The angular equation holds for G^{rot}, and $\mathcal{F} = G - G^{\mathrm{rot}}$ vanishes on the axis. Its angular jets give the recurrence

$$t\partial_s^{m+1}\mathcal{F}_{u,v}(t,0) = -\partial_s^m(\mathcal{F}_{\mathsf{D}_K u,v} + \mathcal{F}_{u,\mathsf{D}_K v})(t,0) + m\partial_t\partial_s^{m-1}\mathcal{F}_{u,v}(t,0),$$

the last term omitted for $m = 0$; induction, for all labels at once, makes every transverse Taylor coefficient of $\mathcal{F}$ zero. In the reflection Hilbert space of axis a, let $v(s)$ be a one-insertion vector translated by se_b; by the derivative law of Proposition 6.4 its difference quotients converge, against a total set of vectors, to the vector of the differentiated label, which is continuous in s by Gram continuity; hence v is smooth. Its Gram kernel depends on $s' - s$, and its Taylor coefficients at zero are those of G^{rot}, which is analytic. Lemma 16.16 makes v real analytic. Pairing with a fixed exterior word proves that an angular sum is analytic in the position of an insertion that is strictly earliest or latest in some coordinate.

Step 4: from a line to arbitrary configurations. If the angular sums vanish for all labels at one end of a segment along which each of them is analytic, they vanish along it, and so do all their jets at the other end. Fix a configuration with distinct first coordinates; such configurations are dense. Treat its points one at a time, always taking the unmoved point with the smallest first coordinate; it is strictly earliest in axis 1, since every moved point has first coordinate larger than all original ones. Move it along e_2 until its second coordinate exceeds all others; then along e_1 until its first coordinate exceeds all others, remaining strictly latest in axis 2; then set its coordinates $2, 3, 4$ to zero in turn, remaining strictly latest in axis 1. Each segment is separated and moves an insertion that is strictly extreme in some axis, so Step 3 applies along it. The final configuration lies on the first axis, where all jets vanish. Reversing the path proves the angular identity at the given configuration, and continuity extends it to Conf_n.

Step 5: the bracket identity. The defect $[\mathsf{D}_K, \mathsf{D}_L] - \mathsf{D}_{[K,L]}$ has vanishing correlation sums by the commutator of two angular identities; it is real, alternating in K, L, covariant under signed permutations of determinant one, kills the unit and preserves the filtration. The finite free tests (16.10) are injective on this defect space, and their injectivity transfers by (16.24); pulling the mixed defect back by the stable frames makes it zero in large degree, and injective inclusions give the lower degrees. Thus $[\mathsf{D}_K, \mathsf{D}_L] = \mathsf{D}_{[K,L]}$.

Step 6: the orthogonal-group representation. Let $C_i = \mathsf{L}^{\mathrm{Max}}_d(K_i)$ and $\mathsf{B}_{k,i} = (\mathsf{U}^{\mathrm{sc}}_{k,d})^{-1}\mathsf{D}_{K_i}\mathsf{U}^{\mathrm{sc}}_{k,d}$ for the six plane generators; $\mathsf{B}_k \to C$ by (16.42). On the endomorphism space define the relative

generators and their quadratic operator

$$\mathcal{A}_{k,i}(X) = \mathsf{B}_{k,i}X - XC_i, \qquad \mathrm{Cas}_k = \sum_i \mathcal{A}_{k,i}^2,$$

and $\mathcal{A}_i$, Cas with C in place of B_k. By Step 5, $K_i \mapsto \mathcal{A}_{k,i}$ is a Lie-algebra representation, so Cas_k commutes with every $\mathcal{A}_{k,i}$: in the commutator, the antisymmetric structure constants cancel in pairs. An invariant inner product makes Cas self-adjoint, with kernel the commutant of the free action, and a real polynomial z with $z(0) = 1$ gives the kernel projection $P = z(\mathrm{Cas})$. We first construct projections $P_k \to P$ commuting with the $\mathcal{A}_{k,i}$, then show that the relative generators vanish on their ranges. Iterate $\varphi(t) = 3t^2 - 2t^3$ on $z(\mathrm{Cas}_k)$. The identities

$$\varphi(t) - t = (1 - 2t)(t^2 - t), \qquad \varphi(t)^2 - \varphi(t) = (4(t^2 - t) - 3)(t^2 - t)^2$$

show that, near P, the errors square and the displacements have a convergent geometric sum; the limits are projections $P_k \to P$, polynomial limits in Cas_k, hence commuting with every $\mathcal{A}_{k,i}$. The restrictions of the $\mathcal{A}_{k,i}$ to $\operatorname{ran} P_k$ satisfy the plane commutator relations and tend to zero, since $\mathcal{A}_i$ vanishes on $\operatorname{ran} P$. Every plane generator is a commutator of two others, so

$$m_k = \max_i \|\mathcal{A}_{k,i}|_{\operatorname{ran} P_k}\| \longrightarrow 0, \qquad m_k \le 2m_k^2,$$

and $m_k = 0$ eventually. Therefore $f_k = P_k(I) \to I$ is invertible and $\mathsf{B}_{k,i}f_k = f_kC_i$. All operations preserve the real signed commutant. Set $\mathcal{V}_d = \mathsf{U}_{k,d}^{\mathrm{sc}} f_k$ for one large k, and

$$R_d(h) = \mathcal{V}_d R_d^{\mathrm{Max}}(h)\mathcal{V}_d^{-1}.$$

This is continuous and real, extends the signed action, and has generator D. The global angular identity makes $S(R(e^{tK})A; e^{tK}x)$ constant in t; plane rotations and coordinate reflections generate $O(4)$, proving covariance. Covariance, unit deletion, inclusion and differentiation give the unit, inclusion and vector laws by injectivity of responses, with $\mathsf{L} = \mathsf{D}$ and the same $c_\star$ in every degree. □

Put $J_0 = c_\star^{-1}J$ and define, with probability Haar measure on $O(4)$,

$$T_{ab} = \int_{O(4)} \sum_{i,j} h_{ai}h_{bj}R_4(h)(J_0)_{ij}\, dh. \tag{16.43}$$

This average imposes exact tensor covariance while retaining the normalized shell actions.

Lemma 16.18 (Normalized tensor)**.** *The tensor T of (16.43) satisfies $R_4(h)T_{ab} = \sum_{i,j} h_{ia}h_{jb}T_{ij}$ for $h \in O(4)$. It is centered, nonzero, real, symmetric and conserved, and has both normalized shell Ward identities of Theorem 6.10.*

Proof. The integrand is a continuous function on a compact group with values in a finite-dimensional space; Haar invariance gives the tensor law. Reality and symmetry pass through the integral, and the vector law for derivatives gives conservation. Changing variables $y = hz$ in each transformed shell and using covariance gives the translation and angular actions ∂_b and $\mathsf{L}_d(K)$; for the latter use $R(h)\mathsf{L}(h^{-1}Kh) = \mathsf{L}(K)R(h)$. Integration preserves both identities, and the positional derivative law of Proposition 6.4 identifies the translation action with $\partial_{x_0^b}S$. The tensor is centered: covariance gives $S_1(R_4(h)J_{ij}) = S_1(J_{ij}) = 0$ in every integrand. If $T = 0$, its translation shell identity would give $S((\partial_b A)(x_0), \boldsymbol{B}(x)) = 0$ for every label and separated spectator list; Definition 2.12 would give $\partial_b = 0$ in degree four, contradicting (16.23) and nonzero free differentiation of a quadratic curvature polynomial. □

Proof of Theorem 6.10. Lemma 16.17 gives the representations and covariance, and Lemma 16.18 gives the tensor with the stated properties. □

Finally the charge normalization removes the factors c_k from stress insertions; no new choice depending on the arity or derivative order is needed.

Lemma 16.19 (Transported infinitesimal actions)**.** *Along the sequence of Lemma 16.10, for every fixed d, a, K,*

$$r_k(\mathsf{U}^{\mathrm{sc}}_{k,d+1})^{-1}\partial_a\mathsf{U}^{\mathrm{sc}}_{k,d} \longrightarrow \partial_a^{\mathrm{Max}}, \qquad (\mathsf{U}^{\mathrm{sc}}_{k,d})^{-1}\mathsf{L}_d(K)\mathsf{U}^{\mathrm{sc}}_{k,d} \longrightarrow \mathsf{L}_d^{\mathrm{Max}}(K)$$

in operator norm, and $(\mathsf{U}^{\mathrm{sc}}_{k,4})^{-1}(r_k^4 T_{ab}) \to T_{ab}^{\mathrm{Max}}$.

Proof. The derivative limit is (16.23); the rotation limit follows from (16.42) and $c_k \to c_\star$. By Lemma 16.18 the tensor T is centered, nonzero, signed-tensorial and conserved; since $J \neq 0$ spans the line of Lemma 16.10, $J = cT$ for a real $c \neq 0$, and comparing translation shells gives $c = c_\star$. Consequently

$$(\mathsf{U}^{\mathrm{sc}}_{k,4})^{-1}(r_k^4 T_{ab}) = \frac{c_k}{c_\star}(\mathsf{U}^{\mathrm{sc}}_{k,4})^{-1}(c_k^{-1} r_k^4 J_{ab}) \longrightarrow T_{ab}^{\mathrm{Max}}.$$ □

Proof of Proposition 6.11. Use the convergent free preimages of Lemma 16.19 in the chosen stress slots and the fixed free labels in the others. Equation (16.24) gives convergence of every positional derivative, uniformly on every separated compact set. The derivative differentiates $x \mapsto S(\boldsymbol{A}_k; r_k x)$, so all chain-rule factors are included. The sequence is the refinement chosen in Lemma 16.10; each fixed arity, degree and derivative order is covered on a tail of that single sequence. □

17 Phases and reconstruction

We prove the phase decomposition of Theorem 6.13 and the reconstruction of Theorem 6.16, starting from the correlations of Theorem 6.1, the quotient fields of Proposition 6.4, the slab operators of Lemma 16.12, the rotations of Lemma 16.17 and the tensor of Theorem 6.10. Three different objects appear, and only the first two are used here.

object	role
separated correlations S_n^ξ of a phase	Euclidean correlations, with reflection positivity on positive-time words (Section 17.1)
ordered-time continuations F_n^ξ and boundary distributions W_n^ξ	construction of the tempered Wightman functions and the relativistic fields (Section 17.2)
contact extensions of Theorem 6.21	a specified Euclidean extension through collisions with controlled local order (Section 19); not used in the reconstruction

The reconstruction thus proceeds from separated Euclidean correlations and slab bounds; it does not require a positive Euclidean extension across collision diagonals.

We keep the words, reflections ϑ, θ, insertion rule and slab operators $\mathsf{B}_h(w)$ of Section 16.4.

17.1 Compression and phases

Put $P_0 = P_4^0$ and $\Omega = \Omega_4^{\mathrm{OS}}$ in this subsection, and write U_y, $y^4 = 0$, for spatial translations on the OS space. Compression does the following. A long empty time interval removes the positive-energy components: $\mathsf{B}_{h+t}(w) = \mathsf{B}_h(w)e^{-tH_4^{\mathrm{OS}}}$ and $e^{-tH_4^{\mathrm{OS}}} \to P_0$. The surviving insertions

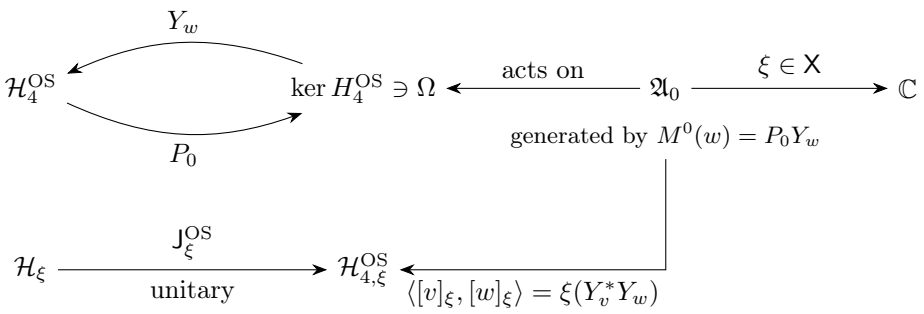


Figure 13: The spaces and maps of this section. A positive word w in a slab gives the insertion Y_w from the full ground space $\ker H_4^{\mathrm{OS}}$ of the mixture, which need not be a line, into its OS space; compression by the ground projection P_0 gives $M^0(w)$, and these operators generate the commutative algebra $\mathfrak{A}_0$. A phase is a character ξ of $\mathfrak{A}_0$. The OS space of the phase is the completion of positive words under the form $\xi(Y_v^* Y_w)$, the value of ξ on the compression of a whole word, which need not be the product of the values on its two halves (17.2). Its ground space is a line for ν-almost every phase by Lemma 17.3 and for every phase by Lemma 17.9. The relativistic space $\mathcal{H}_\xi$ of Section 17.2 is identified with it by the unitary map of Lemma 17.8.

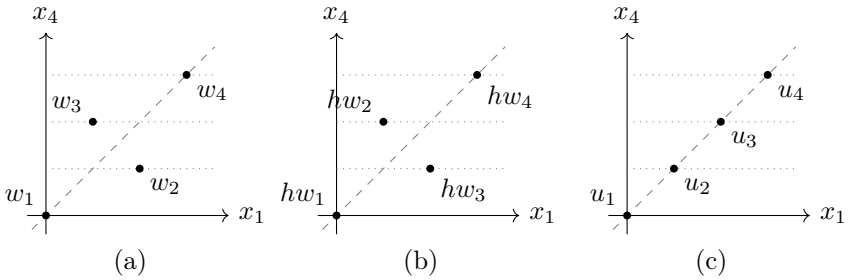


Figure 14: The axis exchange in the proof of Lemma 17.1, in the plane of the first and fourth coordinates, in units of j; time x_4 is vertical, and the dashed line is fixed by the exchange h. (a) Four words, in time order w_1, w_2, w_3, w_4. (b) After h the set of points is the same, but the transformed middle words have traded places, so the time order is hw_1, hw_3, hw_2, hw_4. (c) On the diagonal, h moves no point and changes no time order; this identity removes h from the products.

$M^0(w) = P_0 \mathsf{B}_h(w)|_{\ker H_4^{\mathrm{OS}}}$ commute (Lemma 17.1), and a character of the algebra they generate assigns a compatible scalar to every compressed insertion; integrating over characters against the vacuum state gives the mixture formula. A phase is such a character, not a set of field configurations. Commutativity alone does not show that each character is reflection invariant, or that no field direction is lost in it: a lattice variance inequality gives the first, and the chessboard inequality preserves the rank of finite families of field tests (Lemma 17.3). Figure 13 shows where each of these objects lives.

Rotations make spatial translations trivial at zero energy; compression then removes slab width and placement.

Lemma 17.1 (Compression algebra)**.** *The operators $M^0(w)$ are independent of the slab width and placement. They generate a separable commutative unital C^*-algebra $\mathfrak{A}_0$ on $\ker H_4^{\mathrm{OS}}$, with cyclic and separating vector Ω. Its character space X is compact metric, the vacuum state is given by a*

full-support probability measure ν, and

$$S^{\xi}(w) = \xi(M^0(w)), \qquad S(w) = \int_{\mathsf{X}} \xi(M^0(w))\, d\nu(\xi)\,.$$

Proof. For fixed words and large z^4 the correlation is $\langle v, U_{\vec z} e^{-(z^4-c)H_4^{\mathrm{OS}}} v'\rangle$, and strong heat convergence makes fixed time shifts negligible, uniformly in $\vec z$. Choose an orthogonal reflection K taking $y+e_4$ to $|y+e_4|e_4$. Since $\langle K(y+e_4), Ke_4\rangle = 1$, the ray tKe_4 has positive time component. By the covariance of Lemma 16.17, rotating the displacements te_4+y+e_4 and te_4 makes their difference purely temporal, so their correlations become equal in the limit; removing the extra e_4 gives $U_y P_0 = P_0$ by density.

The identities $\mathsf{B}_{h+t}(w) = \mathsf{B}_h(w) e^{-tH_4^{\mathrm{OS}}}$ and $\mathsf{B}_{h+t}(\tau_{te_4} w) = e^{-tH_4^{\mathrm{OS}}} \mathsf{B}_h(w)$ give width and placement independence of $M^0(w)$, and $M^0(w)\Omega = P_0[w]$, $\langle \Omega, M^0(w)\Omega\rangle = S(w)$. Successive use of (16.25) proves the cluster identity

$$\lim_{j\to\infty} S\big(\tau_{z_1(j)} w_1 \cdots \tau_{z_m(j)} w_m\big) = \langle \Omega, M^0(w_1)\cdots M^0(w_m)\Omega\rangle \tag{17.1}$$

whenever $(z_{i+1}(j) - z_i(j))^4 \to \infty$; in the induction, $\psi_j \to \psi$ and $t_j \to \infty$ imply $U_{y_j} e^{-t_j H_4^{\mathrm{OS}}} \psi_j \to P_0\psi$ for arbitrary y_j.

Commutativity. Let $h \in O(4)$ exchange the first and fourth coordinates, and write hw for the word whose points, internal offsets included, and labels are transformed by h, the labels through the representations of Lemma 16.17. Place four words $w_1, \dots, w_4$ at

$$z_1 = 0, \quad z_2 = j(e_4 + 2e_1), \quad z_3 = j(2e_4 + e_1), \quad z_4 = 3j(e_4 + e_1),$$

with times $0, j, 2j, 3j$ (Figure 14). The points hz_i have times $0, 2j, j, 3j$: the two middle words exchange their time order. Covariance and (17.1), applied before and after h, give

$$\langle \Omega, M^0(w_1)M^0(w_2)M^0(w_3)M^0(w_4)\Omega\rangle = \langle \Omega, M^0(hw_1)M^0(hw_3)M^0(hw_2)M^0(hw_4)\Omega\rangle.$$

The right side still carries h. Words u_k at the points $(k-1)j(e_4+e_1)$, $k = 1, \dots, 4$, which h fixes, keep their time order, so the same two identities give $\langle \Omega, \prod_k M^0(hu_k)\Omega\rangle = \langle \Omega, \prod_k M^0(u_k)\Omega\rangle$. With $(u_k) = (w_1, w_3, w_2, w_4)$ this removes h:

$$\langle \Omega, M^0(w_1)\big[M^0(w_2), M^0(w_3)\big] M^0(w_4)\Omega\rangle = 0.$$

The vectors $M^0(w_4)\Omega = P_0[w_4]$ span a dense subspace of $\ker H_4^{\mathrm{OS}}$. By (16.25), the reflection $w_1 = \theta v$ of a positive word v has $M^0(w_1) = M^0(v)^*$, so the vectors $M^0(w_1)^*\Omega = P_0[v]$ do too. Hence $M^0(w_2)$ and $M^0(w_3)$ commute. Density also gives cyclicity, and the empty word gives the unit. If $a\Omega = 0$, then $ab\Omega = 0$ on this span, so $a = 0$. Basis labels and countable dense position sets generate $\mathfrak{A}_0$ by norm continuity. Gelfand theory and the vacuum state give the compact metric spectrum and the probability measure [19, Theorems 2.4.1 and 2.8.2]; separation gives full support, since $\int |\xi(a)|^2 d\nu = \|a\Omega\|^2$ vanishes only for $a = 0$. □

Compression does not preserve products, and the identities used below have separate jobs. For a positive word v in a slab of width h put $Y_v = \mathsf{B}_h(v)|_{\ker H_4^{\mathrm{OS}}}$, a bounded map from $\ker H_4^{\mathrm{OS}}$ to $\mathcal{H}_4^{\mathrm{OS}}$. Then $M^0(v) = P_0 Y_v$, and (16.25) gives $M^0(\theta v\, w) = Y_v^* Y_w$ for the reflected whole word. For bounded operators A, B and an orthogonal projection P_0 one has $P_0ABP_0 - (P_0AP_0)(P_0BP_0) = P_0A(1-P_0)BP_0$; here it reads

$$M^0(\theta v\, w) - M^0(v)^* M^0(w) = Y_v^*(1 - P_0) Y_w\,. \tag{17.2}$$

A character is multiplicative on $\mathfrak{A}_0$, but a phase evaluates a word at finite separation through the compression of the whole word, $S^\xi(\theta v\, w) = \xi(Y_v^* Y_w)$. In the OS space of the phase, $\xi(M^0(v)^* M^0(w)) = \langle [v]_\xi, \Omega_\xi^{\mathrm{OS}}\rangle\langle \Omega_\xi^{\mathrm{OS}}, [w]_\xi\rangle$, so the character of the right side of (17.2) is the part of $\langle [v]_\xi, [w]_\xi\rangle$ orthogonal to the phase vacuum; it vanishes for all v, w only if that space is the line $\mathbb{C}\Omega_\xi^{\mathrm{OS}}$. A long empty time interval between the two halves replaces $1 - P_0$ by $e^{-tH_4^{\mathrm{OS}}} - P_0$, which tends to zero strongly. This is the mechanism of the cluster identity (17.1), which produces products and hence commutativity; the whole-word identity $M^0(\theta v\, w) = Y_v^* Y_w$ produces the positivity of the phase OS form in Lemma 17.3.

Moment bounds in the vacuum state bound a compressed operator in every phase. This is the form in which lattice chessboard estimates enter here and in Sections 18–19.

Lemma 17.2 (Vacuum moments bound compressed operators)**.** *Let $U \in \mathfrak{A}_0$ and $B \geq 0$ satisfy $\langle \Omega, (U^*U)^n \Omega\rangle \leq B^{2n}$ for every $n \in \mathbb{N}$. Then $\|U\| = \sup_{\xi \in \mathsf{X}} |\xi(U)| \leq B$.*

Proof. By Lemma 17.1, $\langle \Omega, (U^*U)^n \Omega\rangle = \int |\xi(U)|^{2n}\, d\nu(\xi)$. If $|\xi_0(U)| > b > B$, continuity gives an open neighbourhood O of ξ_0 on which $|\xi(U)| > b$, and full support gives $\nu(O) > 0$. Then $\nu(O) b^{2n} \leq B^{2n}$ for every n, which is impossible. Gelfand theory identifies $\|U\|$ with $\sup_\xi |\xi(U)|$. □

The moment hypothesis comes from the lattice as follows. Let a lattice word difference be interior with centered chessboard seminorm at most B_0 in one cell, multiplied by exterior insertions with centered chessboard seminorms at most $B_1, \dots, B_L$ in disjoint finer cells, and let $U \in \mathfrak{A}_0$ be its compressed limit along the ordered limit. Apply (15.4) to $2n$ widely translated copies, alternately conjugated; take the ordered limits and then infinite translation separation, using (17.1); reflection identifies the alternating copies with adjoints. This gives $\langle \Omega, (U^*U)^n \Omega\rangle \leq (B_0 \prod_l B_l)^{2n}$, hence $\|U\| \leq B_0 \prod_l B_l$.

A generic grid word has positions in $a_k \mathbb{Z}^4$ with pairwise distinct coordinates along each axis. For real coefficient representatives (d_i, c_i) write $\langle \cdot \rangle_{N,p}$ for the Wilson expectation of Section 2.1, and put $\mathcal{O}_{N,p}(w) = \prod_i O_{N,p}^{(d_i)}(c_i, x_i/a_N)$. The word differences used below are

$$f_{j,N,p,T} = \mathcal{O}_{N,p}(\tau_{z_j(T)} w) - \mathcal{O}_{N,p}(\tau_{z_j(T)} \vartheta w), \qquad z_j(T) = T j a_k (1,1,1,1). \tag{17.3}$$

For each degree bound and lower grid level m, Lemma 14.30 chooses $k \geq m$ before w; these are the cofinal grids used below. For $\varepsilon > 0$ it supplies real $g_{j,N,p,T}$ and a finite R_ε, independent of N, p, T, j, with $|g_{j,N,p,T}| \leq R_\varepsilon$ eventually in the ordered cutoffs for each fixed T, j, and, for $i < j$ and large T, eventually in the ordered cutoffs, $|\langle g_i g_j \rangle_{N,p} - \mathrm{Re}\langle f_i f_j \rangle_{N,p}| \leq \varepsilon$; the threshold may depend on T, i, j. Bounded g_i control the diagonal variance terms, for which f_i^2 has no uniform bound.

Lemma 17.3 (Phase fields and selection)**.** *For every $\xi \in \mathsf{X}$, its table satisfies E1–E7, retains the inclusions, derivatives, conjugation, rotations and stress identities of the mixture, and*

$$\mathcal{N}_d(S^\xi) = \mathcal{N}_d(S) \qquad (d \in \mathbb{N})\,.$$

Each separated value is continuous in ξ. For ν-almost every ξ, the time-axis OS ground space is one-dimensional.

Proof. Reflection and rotation invariance. We first prove

$$M^0(\vartheta w) = M^0(w), \qquad M^0(hw) = M^0(w) \quad (h \in O(4))\,. \tag{17.4}$$

For a real grid word set $Z = M^0(w) - M^0(\vartheta w)$. Equation (16.25) gives $Z^* = -Z$, so $a = \langle \Omega, Z^2\Omega\rangle = -\|Z\Omega\|^2 \le 0$. Lattice variance gives

$$0 \le \left\langle \left(\sum_{i=0}^{q} g_i \right)^2 \right\rangle \le (q+1)R_\varepsilon^2 + 2\sum_{i<j} (\mathrm{Re}\langle f_i f_j\rangle + \varepsilon) \,.$$

For fixed q, T, intersect the finitely many eventual sets, take the ordered cutoffs, then $T \to \infty$. Each pair expands into four separated moments, whose signed sum tends to a by Lemma 14.33, (15.1) and (17.1). Thus $0 \le (q+1)[R_\varepsilon^2 + q(a+\varepsilon)]$; first $q \to \infty$, then $\varepsilon \downarrow 0$, gives $a \ge 0$, and separation of Ω gives $Z = 0$. Grid density, norm continuity and multilinearity give all words. For the second identity, the cluster identity gives simultaneous orthogonal invariance of two-factor vacuum entries along rays with positive time before and after rotation; such a ray exists unless time is reversed, when composition with ϑ suffices. With $v = M^0(u)\Omega$ and $x = M^0(w)\Omega$, simultaneous invariance and time-reflection invariance give $\langle v, x\rangle = \langle g^{-1}v, \vartheta g^{-1}x\rangle = \langle v, g\vartheta g^{-1}x\rangle$. Density gives every hyperplane reflection, and an orthogonal map is a product of at most four reflections.

Euclidean properties. Testing on positive words gives the permutation, unit and linearity identities and annihilates mixed null labels, so $\mathcal{N}_d(S) \subset \mathcal{N}_d(S^\xi)$. Character evaluation gives reality, covariance and smooth phase values, continuous in ξ. For positive words w_i in a common slab write $Y_i = Y_{w_i} \colon \ker H_4^{\mathrm{OS}} \to \mathcal{H}_4^{\mathrm{OS}}$. Since $M^0(\theta w_i w_j) = Y_i^* Y_j$,

$$\sum_{i,j} \bar{c}_i c_j M^0(\theta w_i w_j) = \left(\sum_i c_i Y_i \right)^* \left(\sum_j c_j Y_j \right) \ge 0 \,.$$

The entries are compressions of whole words, not the products $M^0(w_i)^* M^0(w_j)$ (see (17.2)); a character applied to them gives a positive semidefinite matrix, which need not have rank one. Spectral permanence and character evaluation preserve positivity [19, Theorems 2.5.1(1) and 2.6.4], and coordinate rotations give every axis; integrating the vectors of this positive Gram kernel against compact positive-time joint tests expresses the smeared OS form as a norm square. Inclusions and derivatives pass through $T \mapsto \xi(P_0 T|_{\ker H_4^{\mathrm{OS}}})$. Put a stress shell and its insertions in one buffered slab; its weights vanish near collisions, so the operator integral exists, and testing against exterior positive words gives the identities of Theorem 6.10, which character evaluation transfers to every phase. The bound $|S^\xi(w)| \le \|\mathsf{B}_h(w)\|$, also for derivative labels, gives E7 and separated-support integrability.

Equal null spaces. Use the rows of Lemma 14.22, corrected lifts of Lemma 14.26 and coefficient limits of Lemma 15.5. Each row of trial ℓ consists of a singleton u and optional spectator v, placed at the comparison points $r_\ell y^{\ell,i}$. Write $\mathcal{G}_I^\ell$ for the Maxwell moment of the row labels in slots I at the unscaled points $y^{\ell,i}$, and set $s = \mathcal{G}_{\{u\}}^\ell$, $s' = \mathcal{G}_{\{v\}}^\ell$ and $c = \mathcal{G}_{\{u,v\}}^\ell - ss'$, omitting terms with v in singleton rows. In $\mathfrak{A}_0$ put

$$\Delta_1 = M^0(u) - s1, \quad \Delta_1' = M^0(v) - s'1, \quad \Delta_2 = M^0(uv) - sM^0(v) - s'M^0(u) + (ss' - c)1 \,.$$

The polynomials of (14.39) have centered chessboard seminorms at most $\gamma_\ell + \varepsilon$, preserved under signed images and conjugation by translating the periodic pattern; the lattice moment argument after Lemma 17.2, followed by $\varepsilon \downarrow 0$, gives

$$\int_{\mathsf{X}} |\xi(\Delta)|^{2j} \, d\nu(\xi) \le \gamma_\ell^{2j} \qquad (j \in \mathbb{N}) \,, \tag{17.5}$$

and Lemma 17.2 gives $|\xi(\Delta_1)|, |\xi(\Delta_1')|, |\xi(\Delta_2)| \le \gamma_\ell$ in every phase. Expanding Δ_2 gives $|\xi(M^0(uv)) - ss' - c| \le \gamma_\ell(1 + |s| + |s'|)$. The comparison has three parts of different kinds:

$$\mathcal{N}_d(S) \subset \mathcal{N}_d(S^\xi), \qquad \dim V_d/\mathcal{N}_d(S^\xi) \le \dim F_d = \dim M_d^{\mathrm{Max}} \le \dim V_d/\mathcal{N}_d(S^\xi).$$

The inclusion is exact and was proved above, and with (15.11) it gives the first inequality. The last inequality is perturbative: the phase singleton and pair evaluations of the corrected lifts factor through $V_d/\mathcal{N}_d(S^\xi)$ and converge, by the bounds just proved, to the injective free row map, whose positive minimum on the unit sphere makes them eventually injective. Equal finite codimension and the inclusion give $\mathcal{N}_d(S^\xi) = \mathcal{N}_d(S)$.

Almost every ground space. Reflection positivity and E7 construct the phase heat semigroup as in Lemma 16.12. Write $[b]_\xi$ for a positive-word class and $P^{\mathrm{OS}}_{0,\xi}$ for its ground projection. Strong heat convergence and the mixture identity give, by dominated convergence,

$$\int_{\mathsf{X}} \|P^{\mathrm{OS}}_{0,\xi}[b]_\xi\|^2 \, d\nu(\xi) = \|P_0[b]\|^2 = \int_{\mathsf{X}} |S^\xi(b)|^2 \, d\nu(\xi)\,,$$

the heat entries decreasing from their zero-time bounds. The integrands differ by $\|P^{\mathrm{OS}}_{0,\xi}[b]_\xi - S^\xi(b)\Omega^{\mathrm{OS}}_\xi\|^2$, which therefore vanishes almost everywhere. Rational configurations and basis coefficients give a countable dense family in every phase OS space, by Gram continuity; outside the union of its null sets the ground range is $\mathbb{C}\Omega^{\mathrm{OS}}_\xi$. □

Choose ξ_* from the full-measure set of Lemma 17.3; only its ground-space dimension enters this choice, and it is retained throughout. The slab bound passes to every phase with the same constants, by transferring the repeated insertion estimate through the character and applying the power argument of Lemma 16.12.

Corollary 17.4 (Phase slab operators)**.** *For every $\xi \in \mathsf{X}$ the operators A_w extend to bounded $\mathsf{B}^\xi_h(w)$ on $\mathcal{H}^{\mathrm{OS}}_{4,\xi}$ with $\|\mathsf{B}^\xi_h(w)\| \le \|\mathsf{B}_h(w)\|$, satisfying (16.25) and (16.26).*

Proof. In the phase OS space put $v_j = (A_{w^\dagger}A_w)^j v$. The mixed-space map $Y(v_j) = \mathsf{B}(v_j)|_{\ker H^{\mathrm{OS}}_4}$ equals $(\mathsf{B}_h(w)^*\mathsf{B}_h(w))^j Y(v)$, so $\|[v_j]_\xi\|^2 = \xi(Y(v_j)^*Y(v_j)) \le \|Y(v)\|^2\|\mathsf{B}_h(w)\|^{4j}$. The power argument gives $\mathsf{B}^\xi_h(w)$ of norm at most $\|\mathsf{B}_h(w)\|$, and the dense-vector argument gives its identities and norm derivatives. □

Proof of Theorem 6.13. Lemma 17.1 gives the character space, the full-support probability measure and the mixture formula. Lemma 17.3 gives E1–E7, equal null spaces, the common identities of Proposition 6.4 and Theorem 6.10, covariance and phase continuity. □

17.2 Reconstruction

Use the response labels $\mathcal{L}$ of Definition 2.12. With time reflection acting on labels through R_d, the star is

$$(\iota_d A)^* = \iota_d\big(R_d(\mathsf{s}_4^{\mathrm{refl}})\overline{A}\big)\,. \tag{17.6}$$

The boost generators $-i\mathsf{L}_d(K_{4i})$ and the spatial rotations integrate to $SL(2,\mathbb{C})$, the simply connected cover of $SO^+(1,3)$; a full spatial turn acts trivially, so the action descends to $SO^+(1,3)$. Inclusions intertwine the actions, and the time-reflection sign of K_{4i} and antilinearity make the star (17.6) commute with boosts. Take $\mathcal{P} = SO^+(1,3) \ltimes \mathbb{R}^4$ itself in Definition 2.13, and $\rho_{\mathrm{lab}}(\Lambda, z)$ the continued label action of Λ. Let $\mathfrak{B}$ be the Borchers algebra of finite sequences $F = (F_0, F_1, \dots)$,

$F_n \in \mathcal{L}^{\otimes n} \otimes \mathcal{S}_n$; its product concatenates label tensors and tensors the tests, and its antilinear star reverses order, applies the star (17.6) to labels and complex conjugation to tests. For a phase write U_y^ξ for spatial translation on $\mathcal{H}_{4,\xi}^{\mathrm{OS}}$, and put

$$\mathcal{V}_n = \Big\{ p \in (\mathbb{R}^4)^n : \sum_j p_j = 0, \quad \sum_{k=j}^{n} p_k \in \overline{V}_+ \ (j \geq 2) \Big\}.$$

Fix a phase ξ. The reconstruction follows the scheme of Osterwalder and Schrader [25, 26], with one change: in place of their growth condition, the slab bounds of Lemma 16.12 and Corollary 17.4 give the strip bound (17.8), from which the boundary distributions are obtained directly. Lemma 17.5 constructs the continuation and the boundary distributions, with covariance, spectral support, positivity, continuity in ξ and the barycenter formula. Lemma 17.6 proves locality, Lemma 17.7 realizes the fields, Lemma 17.8 compares the Hilbert spaces, Lemma 17.9 proves the uniform low-energy bound and vacuum uniqueness, Lemma 17.10 proves irreducibility, and Lemma 17.11 continues the stress tensor.

The requirements of Theorem 6.16 come from three sources. Reflection positivity gives the OS spaces, their heat semigroups and the positivity of the boundary distributions. Slab growth gives bounded insertions, hence the strip bound, tempered boundary values and continuity in the phase. The rotations of Lemma 16.17, continued to boosts, give Lorentz covariance and, through imaginary boosts, locality. Vacuum uniqueness first holds only for ν-almost every phase (Lemma 17.3). The low-energy estimate of Lemma 17.9 is uniform in the phase; with the continuity in ξ of the vacuum-subtracted heat pairings and the full support of ν, it removes the exceptional set. That estimate, $\mu_{A\Omega_\xi}\{0 < E \leq u\} \leq M_A u^2$, is polynomial in the energy: no positive lower bound on nonzero energies (spectral gap) is used or proved. Vacuum uniqueness is a statement inside one phase; nothing here shows that ν is a point mass.

Positive-height operators give positive boundary distributions without a Euclidean extension across collisions.

Lemma 17.5 (Continuation and boundary distributions)**.** *For every $\xi \in \mathsf{X}$ and label list a, the functions $\mathsf{F}_n^\xi(a;\cdot)$ of (17.7) satisfy Definition 2.16, and their boundary values in Definition 2.17 define joint Schwartz distributions $W_n^\xi(a;\psi)$, bounded by one Schwartz seminorm uniformly in ξ. These are Poincaré covariant for the label action above, their Fourier transforms are supported in $\mathcal{V}_n$, and $W^\xi(F^*F) \geq 0$ for $F \in \mathfrak{B}$. Each $\xi \mapsto W_n^\xi(a;\psi)$ is continuous, and $\int_{\mathsf{X}} W_n^\xi(a;\psi)\, d\nu(\xi) = W_n^{\mathrm{mix}}(a;\psi)$.*

Here W_n^{mix} is the boundary distribution of the mixed continuation, defined below.

Proof. Positive imaginary times. Write $e^{-tH_{4,\xi}^{\mathrm{OS}}}$ for the phase heat semigroup. For a label a put $B_{\xi,\epsilon}(a) = \mathsf{B}_{2\epsilon}^\xi(a(\epsilon e_4))$; Lemma 16.12 and Corollary 17.4 give $\|B_{\xi,\epsilon}(a)\| \leq C_a \epsilon^{-q_a}$ for $0 < \epsilon < 1$, uniformly in ξ. For $\operatorname{Im} z > 2\epsilon$ set $G_{\xi,\epsilon}(z,y) = e^{-(\operatorname{Im} z - 2\epsilon)H_{4,\xi}^{\mathrm{OS}}} e^{i \operatorname{Re} z H_{4,\xi}^{\mathrm{OS}}} U_y^\xi$, and for $n \geq 1$

$$\mathsf{F}_n^\xi(a;z,\vec{x}) = \langle \Omega_\xi^{\mathrm{OS}}, B_{\xi,\epsilon}(a_1) \prod_{j=2}^{n} \big[G_{\xi,\epsilon}(z_j - z_{j-1}, \vec{x}_j - \vec{x}_{j-1}) B_{\xi,\epsilon}(a_j) \big] \Omega_\xi^{\mathrm{OS}} \rangle, \tag{17.7}$$

with $0 < \epsilon < 1$ and $4\epsilon < \min_j \operatorname{Im}(z_{j+1} - z_j)$; set $\mathsf{F}_0^\xi = 1$. At imaginary times the insertion rule gives S_n^ξ, independently of ϵ, and the identity theorem extends this agreement. The function is

holomorphic in the times and continuous in space; extend it by zero outside ordered times. Taking $\epsilon = \delta/8$ gives, uniformly in x and ξ,

$$|\mathsf{F}_n^\xi(a; x^4 + iy, \vec{x})| \le C\delta^{-m} \quad (0 < \delta \le 1,\ y_{j+1} - y_j \ge \delta)\,. \tag{17.8}$$

Boundary distributions. Write $\phi_\xi(y; f) = \int \mathsf{F}_n^\xi(a; x^4 + iy, \vec{x}) f(x)\, dx$ and $\mathcal{G} = -i \sum_j j\, \partial_{x_j^4}$. By (17.8) and holomorphy, integration by parts gives $\partial_t \phi_\xi(y + t\varrho; f) = \phi_\xi(y + t\varrho; \mathcal{G} f)$ with $\varrho_j = j$, and repeated integration gives

$$\phi_\xi(y; f) = \sum_{k=0}^{m} \frac{(-1)^k}{k!} \phi_\xi(y + \varrho; \mathcal{G}^k f) - \int_0^1 \frac{(-t)^m}{m!} \phi_\xi(y + t\varrho; \mathcal{G}^{m+1} f)\, dt\,. \tag{17.9}$$

The integrand is bounded by $C\|\mathcal{G}^{m+1} f\|_{L^1}/m!$, uniformly in y, ξ. At $y = 0$ this defines W_n^ξ, and dominated convergence gives every ordered boundary limit and a common Schwartz seminorm bound.

Dependence on the phase. The mixed operators of Lemma 16.12 define $\mathsf{F}^{\mathrm{mix}}$ by (17.7) and W^{mix} by (17.9), with the same bound. The mixture formula at imaginary times extends to strict times by the identity theorem and integration of Cauchy's formula. For continuity set $z_j = iy_j + \varsigma x_j^4$; the common strip bound gives a normal family in ς, and if $\xi_\ell \to \xi$, each subsequential limit agrees on the imaginary segment by Lemma 17.3, hence everywhere. Fubini and the common integrable bound in (17.9) give continuity of the distributions and the barycenter formula.

Lorentz covariance. Display the jth label in $\phi_\xi(y; f; \dots, a_j, \dots)$. Differentiating Euclidean $(4, i)$ covariance and using the time Cauchy–Riemann identity gives

$$\sum_j \phi_\xi(y; f; \dots, -i\mathsf{L}(K_{4i}) a_j, \dots) = \phi_\xi\Big(y; \sum_j [x_j^i \partial_{x_j^4} + (x_j^4 + iy_j)\partial_{x_j^i}] f\Big),$$

first at imaginary times, then by holomorphy. At the boundary (17.9) removes y, so along a real boost the derivative of simultaneously transformed labels and tests is zero. Spatial rotations and translations keep their Euclidean identities.

Spectral support. Define $\widetilde{W}_n^\xi(b) = W_n^\xi(\widehat{b}_+(-\cdot))$. For compactly supported smooth frequency tests b, differentiating the height and the exponential multiplier gives cancellation, hence

$$\widetilde{W}_n^\xi(b) = \phi_\xi\big(y; (\widehat{e^{\sum_j y_j p_j^4} b})_+(-\cdot)\big)\,, \tag{17.10}$$

including the zero endpoint. Sending ordered heights to infinity kills negative time-tail momenta, and translation and Lorentz covariance give support in $\mathcal{V}_n$.

Positivity. Apply the propagators and insertions in (17.7) to the vacuum, with an initial heat interval, and smear against a word test b. Reflection and the insertion rule identify mixed-height Gram entries with $\mathsf{F}^\xi(b^* c)$, whose limits as both height lists vanish are $W^\xi(b^* c)$ by (17.9). The vectors are therefore Cauchy, and their limits $v_\xi(b) \in \mathcal{H}_{4,\xi}^{\mathrm{OS}}$ have this Gram form; the common seminorm bound and product-test density extend it to joint tests. Hence $W^\xi(F^* F) = \|\sum_b c_b v_\xi(b)\|^2 \ge 0$. □

Locality is the exchange of two adjacent spacelike points. For $1 \le j < n$ let s be the transposition of j and $j + 1$, write x^s and a^s for configurations and label lists with these entries exchanged, and put $\mathcal{G}_j = \{x \in (\mathbb{R}^4)^n : \mathsf{m}(x_{j+1} - x_j) < 0\}$.

Lemma 17.6 (Exchange of adjacent spacelike points)**.** *For every $\xi \in \mathsf{X}$, label list a, $1 \le j < n$ and $\psi \in \mathcal{S}_n$ supported in $\mathcal{G}_j$,*

$$W_n^\xi(a; \psi) = W_n^\xi(a^s; \psi^s), \qquad \psi^s(x) = \psi(x^s).$$

The two orders give two holomorphic functions on two different tubes. Opposite imaginary boosts move a configuration with tied heights into the two tubes and so define two holomorphic germs at that configuration. At zero real times both germs restrict to the same Euclidean correlation, by permutation symmetry, so they coincide; the equality is carried to arbitrary real times, and partial boundary values give the claim.

Proof. The tube function. Let $\mathcal{T}_n = \{w \in (\mathbb{C}^4)^n : \mathrm{Im}(w_{k+1} - w_k) \in V_+,\ k < n\}$, where V_+ is the interior of $\overline{V}_+$. Let $\chi = \mathbf{1}_{\{\mathrm{dist}(p,\mathcal{V}_n)<2\}} * \varrho_1$ with $\varrho_1 \geq 0$ smooth, of unit integral and supported in the unit ball; then $\chi = 1$ near $\mathcal{V}_n$, has bounded derivatives, and vanishes at distance greater than three. For $w \in \mathcal{T}_n$ define

$$\mathcal{W}_a(w) = (2\pi)^{-4n} \widetilde{W}_n^{\xi}\big(\chi(p) e^{i\sum_k p_k \cdot w_k}\big).$$

With $q_k = \sum_{l \geq k} p_l$, one has $\sum_k p_k \cdot \mathrm{Im}\, w_k = \sum_{k \geq 2} q_k \cdot \mathrm{Im}(w_k - w_{k-1}) \geq c|p|$ on $\mathcal{V}_n$, and up to a bounded subtraction on $\mathrm{supp}\,\chi$; so the test is a holomorphic family of Schwartz functions and $\mathcal{W}_a$ is holomorphic on $\mathcal{T}_n$. By (17.10), $\mathcal{W}_a(w) = \mathsf{F}_n^{\xi}(a; w^4, \vec{w})$ when the imaginary parts are purely temporal and strictly ordered; in particular $\mathcal{W}_a$ equals $S_n^{\xi}(a;\cdot)$ at ordered Euclidean points $w_k = (it_k, \vec{x}_k)$. For nondecreasing heights y, the damping multipliers $e^{-\sum_k y_k p_k^4} = e^{-\sum_{k\geq 2}(y_k - y_{k-1}) q_k^4}$ are bounded with bounded derivatives on $\mathcal{V}_n$, so they act continuously on Schwartz tests. This defines partial boundary distributions $W_{n,y}^{\xi}(a;\cdot)$, continuous in y, equal to $W_n^{\xi}(a;\cdot)$ at $y = 0$, and equal to $\int \mathcal{W}_a(x + iy) f(x)\, dx$ for strictly increasing y. Here $x + iy$ means $(x_k^4 + iy_k, \vec{x}_k)_k$.

Complex boosts. Real Lorentz covariance in Lemma 17.5 gives $\mathcal{W}_a(\Lambda w) = \rho_n^{\vee}(\Lambda)\mathcal{W}_a(w)$ for real Λ and $w \in \mathcal{T}_n$, where $\rho_n^{\vee}$ acts on label tensors. For a spatial unit vector e and rapidity β, the label action of the boost Λ_β along e is $\exp(-i\beta \sum_i e_i \mathsf{L}(K_{4i}))$ on finite-dimensional spaces, an entire function of β. Hence the covariance identity holds for complex β, by the identity theorem in β, whenever $\Lambda_{\beta'} w \in \mathcal{T}_n$ along a path from 0 to β.

Two germs. Fix heights with $y_j = y_{j+1}$ and $y_k < y_{k+1}$ for the other k, and $x \in \mathcal{G}_j$, so $w_0 = x + iy$ lies on the boundary of $\mathcal{T}_n$. Write $x_{j+1} - x_j = (\tau, \vec{v})$ with $|\tau| < |\vec{v}|$ and put $e = \vec{v}/|\vec{v}|$. The imaginary boost Λ_{ib} along e gives this difference the imaginary part $\sin b\,(|\vec{v}|, \tau e)$, in time and space components, which lies in V_+ for $0 < b < \pi$; for small b the other differences stay in V_+. So, for a small $b_0 > 0$ and a small neighbourhood N of w_0, $\Lambda_{ib_0} w \in \mathcal{T}_n$ for $w \in N$, and $\Lambda_{ib} w \in \mathcal{T}_n$ for $0 \leq b \leq b_0$ if moreover $w \in \mathcal{T}_n$; the opposite boost Λ_{-ib} moves the exchanged configuration into $\mathcal{T}_n$. Define, near w_0,

$$G_+(w) = \rho_n^{\vee}(\Lambda_{ib_0})^{-1} \mathcal{W}_a(\Lambda_{ib_0} w), \qquad G_-(w) = \rho_n^{\vee}(\Lambda_{-ib_0})^{-1} \mathcal{W}_{a^s}\big((\Lambda_{-ib_0} w)^s\big).$$

Both are holomorphic in a neighbourhood of w_0. By the complex boost identity along $[0, b_0]$, $G_+ = \mathcal{W}_a$ on the part of that neighbourhood in $\mathcal{T}_n$, and $G_-(w) = \mathcal{W}_{a^s}(w^s)$ where $w^s \in \mathcal{T}_n$.

The Wick slice. Suppose first that all real times of x vanish. Then $w_0 = (iy_k, \vec{x}_k)_k$ is a separated Euclidean configuration, since $\vec{x}_j \neq \vec{x}_{j+1}$. The Wick slice $\{(it_k, \vec{x}_k)\}$ is a real subspace of dimension $4n$ whose complex span is $(\mathbb{C}^4)^n$. Near w_0 on the slice, points with $t_j < t_{j+1}$ lie in $\mathcal{T}_n$, where $G_+ = S_n^{\xi}(a;\cdot)$; points with $t_j > t_{j+1}$ have exchanged configuration in $\mathcal{T}_n$, where $G_- = S_n^{\xi}(a^s; (\cdot)^s) = S_n^{\xi}(a;\cdot)$ by permutation symmetry. Since S_n^{ξ} is smooth at w_0, all jets of G_+ and G_- along the slice at w_0, computed as limits from these open half-spaces, equal those of $S_n^{\xi}(a;\cdot)$. Real jets along the slice are restrictions of complex jets, and the slice spans $(\mathbb{C}^4)^n$; so all complex derivatives of $G_+ - G_-$ vanish at w_0, and $G_+ = G_-$ near w_0.

Propagation. For general $x \in \mathcal{G}_j$ consider $x(t) = (tx_k^4, \vec{x}_k)_k$, $t \in [0,1]$; it stays in $\mathcal{G}_j$, since $|t\tau| \leq |\tau| < |\vec{v}|$, and the strict height margins persist. Along this compact path both germs are defined at every point. Two germs of the same kind at nearby points agree on the overlap of their

domains, since both extend the same function from the open set $\mathcal{T}_n$ (respectively its exchanged image), which accumulates at each point. A finite chain of overlapping neighbourhoods carries $G_+ = G_-$ from $t = 0$ to $t = 1$.

Partial boundary values. For f supported in a small ball around $x \in \mathcal{G}_j$, the representation of $W^{\xi}_{n,y}$ by the tube function and continuity up to the tied heights give $W^{\xi}_{n,y}(a; f) = \int G_+(x' + iy) f(x')\, dx'$, and likewise $W^{\xi}_{n,y^s}(a^s; f^s) = \int G_-(x' + iy) f(x')\, dx'$. These agree, and a partition of unity gives $W^{\xi}_{n,y}(a; \psi) = W^{\xi}_{n,y^s}(a^s; \psi^s)$ for compactly supported ψ with support in $\mathcal{G}_j$; for general $\psi \in \mathcal{S}_n$ with support in $\mathcal{G}_j$, multiply by cutoffs equal to one on growing balls, which converge in $\mathcal{S}_n$ without enlarging the support. Let the heights tend to zero along such tied height lists; continuity of the partial boundary distributions gives the claim. □

Lemma 17.7 (Relativistic fields)**.** *For every $\xi \in \mathsf{X}$, the boundary distributions of Lemma 17.5 define a faithful theory on $\mathcal{L}$ satisfying W1–W3 and the spectral representation and spectral condition $\mathsf{E}(\mathbb{R}^4 \setminus \overline{V}_+) = 0$ of W0, with continuation the functions F^{ξ}_n of that lemma. Its adjoint and Lorentz label maps are the common maps specified above.*

Proof. Complete the form $W^{\xi}(F^*G)$ on $\mathfrak{B}$ modulo null vectors to obtain $\mathcal{H}_{\xi}$, with $\Phi_{\xi}(a, f)[b] = [(a, f)b]$. For null x the inserted vector pairs with c as $\langle x, (a^*, \bar{f})c\rangle = 0$; taking c to be that vector preserves nullity and gives the adjoint law W1. Test continuity gives strongly continuous covariance unitaries, and Lemma 17.5 gives W2. Translations integrate to a nondegenerate representation of $C^*(\mathbb{R}^4) = C_0(\mathbb{R}^4)$ [19, Theorem 3.7.11 and equation (3.89)] with spectral measure from [19, Proposition 4.2.3], and the Fourier support in $\mathcal{V}_n$ puts the spectrum in $\overline{V}_+$. Lemma 17.6, applied to adjacent pairs inside vacuum expectations of words, gives W3 on the word domain. The boundary condition of Definition 2.17 is the definition of W^{ξ}_n in Lemma 17.5. A label acting as zero has zero Euclidean correlations by (17.10), symmetry and continuity; Lemma 17.3 and injectivity of responses give faithfulness. □

Surjectivity of the comparison map transfers ground-space information between the two Hilbert spaces.

Lemma 17.8 (OS comparison)**.** *For every ξ, there is a unique linear isometry $\mathsf{J}^{\mathrm{OS}}_{\xi} : \mathcal{H}_{\xi} \to \mathcal{H}^{\mathrm{OS}}_{4,\xi}$ with the boundary-word values specified before Theorem 6.16. It is surjective, hence unitary, and intertwines spatial translations and the heat semigroups: $\mathsf{J}^{\mathrm{OS}}_{\xi} e^{-sH_{\xi}} = e^{-sH^{\mathrm{OS}}_{4,\xi}} \mathsf{J}^{\mathrm{OS}}_{\xi}$ for $s \geq 0$.*

Proof. The map $J : [b] \mapsto v_{\xi}(b)$ of Lemma 17.5 is an isometry taking vacuum to vacuum and intertwines spatial translations and heat by (17.10). Two linear isometries with these boundary-word values agree on the dense word span, hence everywhere. For compact frequency b let $f_y = (e^{-\widehat{\sum_j y_j p_j^4}} b)_+(-\cdot)$; by (17.10) the norms and mixed pairing of its boundary-word image under J and the corresponding positive-height OS vector agree, so the two vectors coincide. Frequency-test density and closed range include every smeared positive-height vector, approximate identities give point vectors, and permutation and perturbation of equal times give all positive words. So J is surjective. □

The low-energy estimate is the infrared input of the reconstruction. For a fixed compactly supported local word A, locality makes a commutator vanish outside a bounded spatial region for bounded times. An odd time average turns this into a positive spectral weight comparable to the energy E near zero. A momentum cutoff at scale u costs u^3 in three spatial dimensions, and the cone $|\vec{p}| \leq E$ gives $u\, \mu\{u/2 < E \leq u\} \lesssim u^3$; dyadic summation gives u^2.

Lemma 17.9 (Uniform low-energy bounds and vacuum uniqueness)**.** *For every finite complex combination A of field words smeared with compactly supported smooth tests, there are $M_A, M'_A \ge 0$, depending on its words, labels and tests, such that for every $\xi \in \mathsf{X}$,*

$$\begin{aligned} \mu_{A\Omega_\xi}(\{0 < \mathbf{p}^4 \le u\}) &\le M_A u^2 && (0 < u \le 1), \\ 0 \le \langle A\Omega_\xi, (e^{-sH_\xi} - P_{0,\xi})A\Omega_\xi\rangle &\le M'_A s^{-2} && (s \ge 1), \end{aligned}$$

where $P_{0,\xi}$ projects onto $\mathbb{C}\Omega_\xi$. Moreover $\ker H_\xi = \mathbb{C}\Omega_\xi$ *and* $\ker H^{\mathrm{OS}}_{4,\xi} = \mathbb{C}\Omega^{\mathrm{OS}}_\xi$.

Proof. The low-energy measure. Let $Q_\xi = \mathsf{E}_\xi(\{\mathbf{p}^4 = 0\})$ be the full zero-energy projection; until the last step its range is not assumed to be a line. Let A^* be the word adjoint, reversing order and applying the star (17.6) to labels and complex conjugation to tests, and put $\alpha_z A = \mathsf{T}_\xi(z) A \mathsf{T}_\xi(z)^*$. Bound all support times and spatial norms by r and set $R = 4r + 1$, $\mu = \mu_{A\Omega_\xi}$ and $\mu^* = \mu_{A^*\Omega_\xi}$. The common distribution bounds give $\mu(\mathbb{R}^4) + \mu^*(\mathbb{R}^4) \le M$, uniformly in ξ. Covariance and locality give

$$a(y,t) = \widehat{\mu}_+(y,t) - \widehat{\mu^*}_+(-y,-t) = \langle \Omega_\xi, [A^*, \alpha_{(y,t)}A]\Omega_\xi\rangle = 0 \quad (|y| > R,\ |t| \le 1)\,,$$

with $\widehat{\mu}_+(z) = \int e^{ip\cdot z}\, d\mu(p)$. Write $E = p^4$. The time average $\bar{a}(y) = \int_0^1 [a(y,t) - a(y,-t)]/(2i)\, dt$ is bounded by M and supported in $\overline{B}_R$. It is the Fourier transform of two positive measures, the pushforwards of $\varpi(E)\, d\mu$ under $p \mapsto \vec{p}$ and of $\varpi(E)\, d\mu^*$ under $p \mapsto -\vec{p}$, where $\varpi(E) = (1 - \cos E)/E$, $\varpi(0) = 0$; on $0 \le E \le 1$, $\varpi(E) \ge E/\pi$. Choose a smooth momentum bump b_u with $0 \le b_u \le 1$, equal to one on B_u and supported in B_{2u}; $|\widehat{b_{u+}}(-y)| \le 8|B_1|u^3$. Since $|\vec{p}| \le E$ on the spectral cone, $b_u = 1$ on the shell $\{u/2 < E \le u\}$, and Fourier inversion and positivity give

$$(2\pi)^3 \frac{u}{2\pi}\, \mu(\{u/2 < E \le u\}) \le \left| \int \bar{a}(y) \widehat{b_{u+}}(-y)\, dy \right| \le 8|B_1||B_R| M u^3\,.$$

Summing the shells at $u2^{-j}$ gives

$$\mu(\{0 < E \le u\}) \le C R^3 M u^2 \qquad (0 < u \le 1)\,. \tag{17.11}$$

For $u > 1$ use the total mass. Since $e^{-sE} = \int_E^\infty s e^{-st}\, dt$, Tonelli gives

$$0 \le \langle A\Omega_\xi, (e^{-sH_\xi} - Q_\xi)A\Omega_\xi\rangle = \int_{E>0} e^{-sE}\, d\mu \le C_A s \int_0^\infty t^2 e^{-st}\, dt = 2C_A s^{-2}\,. \tag{17.12}$$

From almost every phase to every phase. By Lemma 17.8, $\mathsf{J}^{\mathrm{OS}}_\xi$ is unitary and intertwines heat, so it identifies $\ker H_\xi$ with $\ker H^{\mathrm{OS}}_\xi$; by Lemma 17.3 both are $\mathbb{C}\Omega_\xi$ for ν-almost every ξ, and there $Q_\xi = P_{0,\xi}$. The vacuum-subtracted heat pairing $\langle A\Omega_\xi, e^{-sH_\xi}A\Omega_\xi\rangle - |\langle \Omega_\xi, A\Omega_\xi\rangle|^2$ is continuous in ξ: by (17.9) it is a continuous Schwartz pairing, with error Me^{-sN} outside radius $2N$ on $\overline{V}_+$. It is bounded by $2C_A s^{-2}$ for almost every ξ, hence, by continuity and full support of ν, for every ξ. Letting $s \to \infty$ gives $\|Q_\xi A\Omega_\xi\|^2 = |\langle \Omega_\xi, A\Omega_\xi\rangle|^2$. Since $Q_\xi\Omega_\xi = \Omega_\xi$, density of compact word vectors gives $Q_\xi = P_{0,\xi}$, and (17.12) becomes the second bound. Unitarity and heat intertwining give the OS vacuum line. □

Lemma 17.10 (Irreducibility)**.** *For every ξ, a bounded operator B is scalar if and only if, for every smeared field Φ and $u, v \in \mathcal{D}_\xi$, one has $u \in \operatorname{Dom}\Phi^*$ and $\langle \Phi^* u, Bv\rangle = \langle u, B\Phi v\rangle$.*

Proof. A scalar satisfies the relation by the adjoint law. Conversely, the adjoint law and iteration move words across matrix entries without requiring B to preserve the domain. For a word A put $u = A\Omega_\xi$, $v = A^*\Omega_\xi$ and, in the upper half-plane,

$$p(z) = \langle u, e^{izH_\xi} B\Omega_\xi\rangle, \qquad q(z) = \langle B^*\Omega_\xi, e^{izH_\xi} v\rangle\,.$$

These are bounded and continuous to the boundary, where covariance and the weak relation give $p(t) = q(-t)$. Morera and Liouville make their bounded pasted continuation constant, so $\langle u, e^{-sH_\xi} B\Omega_\xi\rangle = \langle u, B\Omega_\xi\rangle$. Heat convergence and Lemma 17.9 give $B\Omega_\xi = b\Omega_\xi$. Moving words again gives $\langle A'\Omega_\xi, BA\Omega_\xi\rangle = b\langle A'\Omega_\xi, A\Omega_\xi\rangle$, so $B = b1$ by density. □

For the charge law set $\eta = \operatorname{diag}(-1,-1,-1,1)$, $\zeta_i = 1$ for $i \le 3$, $\zeta_4 = -i$, and

$$T^{\mathrm{M}}_{ab} = -\zeta_a\zeta_b T_{ab}, \qquad T^{\mathrm{M}}_{ab}(f) = \Phi_\xi(\iota_4 T^{\mathrm{M}}_{ab}, f)\,.$$

Let $K(x) = k + Vx$ with $V^{\mathsf{T}}\eta + \eta V = 0$, and let Σ_K be its infinitesimal Lorentz label action. For real $f \in C^\infty_c(\mathbb{R})$ of integral one and $c \in C^\infty_c(\mathbb{R}^3)$ put $Q_K(f,c) = \sum_b T^{\mathrm{M}}_{4b}(K^b(c\otimes f))$. Suppose $g \in C^\infty_c(\mathbb{R}^4)$, and $c = 1$ on a ball such that all (y,s) over its complement, $s \in \operatorname{supp} f$, are spacelike to $\operatorname{supp} g$. The charge law is

$$i[Q_K(f,c), \Phi_\xi(\iota_d A, g)] = \Phi_\xi(\Sigma_K \iota_d A, g) - \sum_a \Phi_\xi(\iota_d A, \partial_a(K^a g))\,. \tag{17.13}$$

A common polynomial bound permits the continuation of local stress identities.

Lemma 17.11 (Continued stress)**.** *In every phase, T continues to a symmetric Hermitian conserved Lorentz tensor of operator-valued distributions, with the local charge law* (17.13) *under its support hypothesis.*

Proof. Tensor properties. The factors give $T^{\mathrm{M}}_{44} = T_{44}$, $T^{\mathrm{M}}_{4i} = iT_{4i}$, $T^{\mathrm{M}}_{ij} = -T_{ij}$. Euclidean symmetry and (17.6) give symmetry and Hermiticity, and differentiation and exponentiation of covariance give the Lorentz action. Continuing the derivative identity gives $\Phi_\xi(\iota_{d+1}\partial_a A, f) = -\zeta_a^{-1}\Phi_\xi(\iota_d A, \partial_a f)$, so Euclidean conservation gives $\sum_a \eta_{aa} T^{\mathrm{M}}_{ab}(\partial_a f) = 0$ on the common word domain.

The Euclidean jump. For real c, insert $c(y)h(t-t_0)$ into the shell identities of Theorem 6.10, transferred to the phase by Lemma 17.3, and integrate conservation on each side of t_0. The ∇c term stays away from the distinguished point and remains bounded, so $\sum_b \int c(y) K^b_{\mathrm{E}}(y,t) T_{4b}(y,t)\,dy$ is Cauchy on each side, with jump $\Sigma_K A + \sum_b K^b_{\mathrm{E}}\partial_b A$, where $K^b_{\mathrm{E}}(y) = \zeta_b K^b(\vec{y}, iy^4)$.

Continuation of the jump. Fix strictly ordered heights y_j for all fields other than the current, including the distinguished field with index 0. For their real times t_j put $z_j(\varsigma) = iy_j + \varsigma t_j$, and place the current at $z_0(\varsigma) \pm i\varepsilon$, retaining Euclidean separation $\varepsilon > 0$. Write

$$I^\pm_\varepsilon(\varsigma) = \sum_b \int c(x) K^b_{\mathrm{E}}(x, -iz_0(\varsigma) \pm \varepsilon)\, \mathsf{F}^\xi_\pm(\ldots, T_{4b}(x, z_0(\varsigma) \pm i\varepsilon), \ldots)\,dx,$$

where $\mathsf{F}_\pm$ inserts the current immediately after or before the distinguished field. Choose a complex neighbourhood $\mathcal{U}$ of $[0,1]$ preserving the other strict height differences. On $\operatorname{supp}\nabla c$ the spacelike germs of Lemma 17.6 bound the side integrals, uniformly for small ε and ς in compact $L \subset \mathcal{U}$, and conservation gives $\sup_{\varsigma\in L} |I^\pm_\varepsilon(\varsigma) - I^\pm_{\varepsilon'}(\varsigma)| \le C_L|\varepsilon - \varepsilon'|$. The limits are holomorphic; at small imaginary ς the Euclidean calculation identifies their jump, and the identity theorem extends it to $\varsigma = 1$. Test against g, f and take the continuous partial boundaries; integration by parts differentiates $K^a g$, giving (17.13). Schwartz cutoffs and the joint bounds extend it from compact spectators. For complex c, the charge of its imaginary part commutes with the distinguished field by locality. Constant and rotational K give the required actions. □

Proof of Theorem 6.16. Lemmas 17.5, 17.6 and 17.7 give, in every phase, a faithful theory satisfying W1–W3 and the spectral condition, with continuations satisfying Definitions 2.16 and 2.17. Lemma 17.8 gives the unitary comparison, Lemma 17.9 the vacuum lines and hence W0, and Lemma 17.10 the scalar weak commutant. Lemmas 17.9–17.11 also give the uniform rates and the continued stress, at the retained phase ξ_* as at every other. □

18 The trace law and non-free comparisons

This section proves Theorem 6.18 and the non-free comparisons. The stress tensor T of Section 16 is normalized by its comparison with the free Maxwell stress (Proposition 6.11), and the free stress is traceless. So this comparison cannot determine the first nonzero trace term, which is smaller by the inverse running coupling (Table 9). That term corresponds to the one-loop trace anomaly [9]; we compute its coefficient from a lattice Ward identity, with a rate. The proof of Theorem 6.18 has five steps.

(1) $L = T^{\mathrm{Max}}$, $D = \delta_{ab}\mathcal{O}_{F^2}$ and Ξ span the centered real symmetric H_4-tensorial tensors of degree four; a functional ρ_{tr} reads the trace from the divergence (18.3).

(2) A lattice stress current (Definition 18.4) has divergence equal to the action variation up to a defect, which block transfer (18.4) makes small; block transfer also keeps the trace and antisymmetric part of the current lattice differences.

(3) Lemma 18.5 computes the regulated Ward coefficient $-12\kappa_G$. Lemma 18.7 carries it to continuum fields along the radii of Lemma 18.3, with the rate of Lemma 18.6.

(4) A conserved combination of these fields is a multiple of T; its D-coefficient tends to $d_2 = -\kappa_G/4$ (Proposition 18.8).

(5) Positivity makes G_Θ nonincreasing. This turns comparison along subsequences, which uses frames, into the frame-free limit at every radius (Figure 15).

Section 18.2 proves the non-free comparisons, each with its comparison class: no two-point statement excludes a generalized free field with the same spectral measure. In the selected phase, on no interval $(0, \delta)$ does G_Θ agree with a finite Wick profile (Proposition 6.19).

Throughout, retain the fields, inclusions and units of Definition 2.12 and Proposition 6.4, the tensor of Theorem 6.10, and the selected reconstruction of Theorem 6.16. The symbols $\tau, e_{\mathrm{tr}}, \Theta, G_\Theta$ and A_G have the meanings of Definition 6.17. Write $\Phi_\Theta(f) = \Phi(\iota_4\Theta, f)$, and use the word domain of item W1 of Definition 2.14.

18.1 Trace and spectrum

Centering must hold on the common operator domain, as well as in separated correlations. The proof below identifies the subtraction after continuation.

Lemma 18.1 (Centering as an operator identity)**.** *For every $f \in \mathcal{S}$, on the word domain,*

$$\Phi_\Theta(f) = \Phi(\iota_4\tau, f) - e_{\mathrm{tr}}\left(\int f\right)\mathrm{Id}_{\mathcal{H}}, \qquad \langle \Omega, \Phi_\Theta(f)\Omega\rangle = 0\,.$$

Proof. Unit deletion (Definition 2.9) and the quotient calculus of Lemma 15.5 remove the unit insertion; the continuation of Lemma 17.5 and the field realization of Lemma 17.7 carry this to the operators. Deleting the unit insertion leaves a Euclidean correlation independent of its position.

On each ordered-time region its continuation is therefore constant in that variable. The identity theorem on the connected continuation domain $\mathcal{Z}_n$ of Definition 2.16, applied successively in each time variable, identifies it with the correlation containing the unit. Boundary values and smearing give $\langle u, \Phi(\iota_4 \mathbf{1}_4, f)v\rangle = (\int f)\langle u, v\rangle$ for word vectors u, v. Density and linearity give the identity. The zero Euclidean one-point function continues to zero and has zero boundary distribution, proving the centering assertion. □

We use the Fourier convention of Section 6.4:

$$\widehat{f}_+(\mathbf{p}) = \int e^{i\mathbf{p}\cdot x} f(x)\, dx, \qquad (\mathcal{W}_{\mathrm{rel}}\psi)(\mathbf{p}) = \int e^{i\mathbf{p}\cdot(y-x)}\psi(x,y)\, dx\, dy\,.$$

In particular, $\mathcal{W}_{\mathrm{rel}}(\bar{f}\otimes h) = \overline{\widehat{f}_+}\widehat{h}_+$. Write $W_2^{\xi*}(\Theta,\Theta;\bar{f}\otimes h) = \langle \Phi_\Theta(f)\Omega, \Phi_\Theta(h)\Omega\rangle$ for the joint tempered boundary distribution of Lemma 17.5. The next lemma constructs its positive measure.

Lemma 18.2 (Unique trace spectral measure)**.** *There is a unique positive tempered-growth Borel measure μ_Θ, supported in $\overline{V}_+$, such that*

$$d\mu_{\Phi_\Theta(f)\Omega} = |\widehat{f}_+|^2\, d\mu_\Theta \qquad (f \in \mathcal{S})\,.$$

It is Lorentz invariant in momentum variables, and

$$W_2^{\xi*}(\Theta,\Theta;\psi) = \int \mathcal{W}_{\mathrm{rel}}\psi\, d\mu_\Theta \qquad (\psi \in \mathcal{S}_2),$$
$$G_\Theta(r) = \int e^{-r\mathbf{p}^4}\, d\mu_\Theta \qquad (r > 0)\,.$$

Proof. The measure is built from the positive boundary distributions of Lemma 17.5. For $v_f = \Phi_\Theta(f)\Omega$, translation covariance gives $\widehat{h}_+(P)v_f = \widehat{f}_+(P)v_h$. Here $\widehat{h}_+(P) = \int \widehat{h}_+(\mathbf{p})\, d\mathsf{E}(\mathbf{p})$, with E the joint translation spectral resolution of Theorem 6.16. Both sides smear the field with $f * h$. Choose tests whose transforms do not vanish on successively larger balls. On such a ball define $d\mu_\Theta = |\widehat{f}_+|^{-2} d\mu_{v_f}$. The displayed identity makes these definitions agree on overlaps. Polarization gives $\langle v_f, v_h\rangle = \int \overline{\widehat{f}_+}\widehat{h}_+\, d\mu_\Theta$. Product tests determine the joint distribution.

The spectrum condition gives cone support. Schwartz continuity, applied to inverse transforms of cutoffs equal to one on $\{|\mathbf{p}| \le R\}$, gives $\mu_\Theta(\{|\mathbf{p}| \le R\}) \le C(1+R)^k$. Lemma 17.11 continues T to a symmetric Hermitian Lorentz tensor. Contraction and invariance of the unit (Lemma 16.17) make $\iota_4\Theta$ a Hermitian Lorentz scalar. Covariance of W_2 and uniqueness of the measure give Lorentz invariance.

Tempered growth and $|\mathbf{p}| \le \sqrt{2}\mathbf{p}^4$ on the cone make the Fourier–Laplace integral holomorphic at positive imaginary time, with boundary W_2. To identify it with the Euclidean continuation of Lemma 17.5, take enough primitives to obtain continuous boundary values, using the polynomial boundary bound (17.8). Their zero difference extends by Schwarz reflection [30, Theorem 11.14]; holomorphic uniqueness and differentiation give equality and the Laplace formula. In particular G_Θ is real, nonnegative and nonincreasing. □

Comparison at prescribed radii. Use the scaling frames of Definition 6.6: $\mathsf{U}^{\mathrm{sc}}_{i,d} : M_d^{\mathrm{Max}} \to F_d$. Insert $\mathsf{U}^{\mathrm{sc}}_{i,d}A$ at $r_i x$, and read tensors by $r_i^4(\mathsf{U}^{\mathrm{sc}}_{i,4})^{-1}$. Write $x^{(i)}$ for nearest points to x in $r_i^{-1}a_{n_i}\mathbb{Z}^4$. Jets use positional derivatives of smooth interpolants and their normalized lattice differences, including derivatives of the inserted fields. A shell flux is the integral against $-\partial_a\chi$ in Definition 6.8.

Define $\beta_i = \lim_{(N,p)} \beta_{N,p,k_i}$ along the ordered limits retained in the proof of Lemma 15.5; the radius level n_i precedes the insertion level k_i.

The next lemma extends Lemma 16.8 to prescribed radii. Its proof also transfers the derivative argument of (16.23) to these frames.

Lemma 18.3 (Prescribed radii)**.** *Every positive null sequence has a subsequence (r_i), with frames $\mathsf{U}^{\mathrm{sc}}_{i,d}$ making $(r_i, \mathsf{U}^{\mathrm{sc}}_{i,\cdot})$ a stable scaling sequence (Definition 6.6) and levels $n_i < k_i$ tending to infinity, such that:*

(i) the couplings satisfy

$$\beta_i \to \infty, \qquad \frac{\log(1/r_i)}{\beta_i} \to \kappa_G^{-1}\,; \tag{18.1}$$

(ii) for every arity $n_{\mathrm{ar}} \geq 1$, degrees $d_j \leq D$, labels $A_j^{(i)} \to A_j$ and compact $\mathcal{K} \subset \mathrm{Gen}_{n_{\mathrm{ar}}}$,

$$\sup_{x \in \mathcal{K}} \left| S_{n_{\mathrm{ar}}}(\mathsf{U}^{\mathrm{sc}}_{i,\boldsymbol{d}} \boldsymbol{A}^{(i)}; r_i x^{(i)}) - S^{\mathrm{Max}}_{n_{\mathrm{ar}}}(\boldsymbol{A}; x) \right| \to 0\,,$$

and the same holds for each fixed jet and translation-shell flux;

(iii) for $n_{\mathrm{ar}} = 2$ without derivatives, (ii) holds uniformly over the phases.

Proof. The proof follows the finite-dimensional construction of Lemma 16.8, with radii placed in comparison windows. The source estimates of Lemmas 14.24 and 14.14 control the windows, and Lemma 14.27 moves the insertion points.

Radii and levels. Put $M_n = M_n^{\mathrm{cut}}$. At radius level n allow

$$r(n,u) = a_n M_n^u, \qquad u \in [\alpha_0, \alpha_0 + 1], \qquad k(n) = \min\{m > n : \log(a_n/a_m) \geq (\log \widehat{\beta}_n)^{48}\}\,. \tag{18.2}$$

The separation in $k(n)$ leaves room for all fixed-degree marked transports; its logarithmic size is negligible compared with $\widehat{\beta}_n$. For degree D and an arity bound $n_{\mathrm{ar}} \geq 2$ choose comparison exponents $\alpha_0, \alpha_1, \alpha_2$ and c with

$$c < 0, \quad \alpha_1 + 6 < \alpha_0, \quad \alpha_2 + 5 < \alpha_0 + c, \quad (\alpha_0 - \alpha_1) D n_{\mathrm{ar}} + c < 0, \quad (\alpha_0 - \alpha_2) D n_{\mathrm{ar}} - \alpha_0 - c < 0\,.$$

Choose the differences $\alpha_0 - \alpha_1 > 6$, then $-c > (\alpha_0 - \alpha_1) D n_{\mathrm{ar}}$, then $\alpha_0 - \alpha_2 > 5 - c$, and finally increase all three α_j together until $\alpha_2 \geq D n_{\mathrm{ar}} + 10$.

Comparison windows. We first obtain comparison windows at every sufficiently late radius level, rather than only the intervals selected in Lemma 14.2. For each of the three exponents $b = \alpha_j + u - \alpha_0$, stop backwards from n when the accumulated logarithmic block length first exceeds $(b+2)\log M_n$, then stop again after another $10 \log B_n$. The entry bounds fixed in Section 14.1 give $\ell_m \leq \log \widehat{\beta}_m$ at every level m. Each overshoot is therefore $O(\log\log \widehat{\beta}_n) = o(\log M_n)$, so the resulting cell side is at most $a_n M_n^{b+5}$. All levels in these windows tend to infinity and their couplings have ratio tending to one. The accumulated source loss satisfies

$$\log \prod_{m \text{ in the window}} \mathcal{C}_{J_m} \leq \Big(\sup_{m \text{ in the window}} J_m^{-2} \Big) \sum_{m \text{ in the window}} \log \ell_m = o(\log M_n),$$

by (14.11). Thus these windows satisfy the window hypotheses (Win1)–(Win3) stated before Lemma 14.23: its initial rounding error is M_n^{-b}, its covariance and healthy-chart errors are smaller than any required fixed inverse power of M_n after entry, and its source recurrence (14.45) has total

linear loss $M_n^{o(1)}$. The entry bounds of Section 14.1 include the comparison requirements (14.10) for the finitely many arities admitted at each stage. At mark weight $M_n^{1/(8n_{\mathrm{ar}})}$ the resulting error is at most $M_n^{-3/4}$. The normalized integral is bounded by (14.47); its stage constant is at most $\log M_n/(16n_{\mathrm{ar}})$ after entry. These estimates are uniform in $u \in [\alpha_0, \alpha_0 + 1]$ and do not require a window to remain in one stage.

Chessboard seminorms. At its own normalization a singleton therefore has chessboard seminorm at most $1 + |S_1^{\mathrm{Max}}(A)|$. Indeed coefficient extraction at amplitude e^{2q}, $q = \log M_n/(16n_{\mathrm{ar}})$, bounds its centered seminorm by $e^{q-2q} \le 1$. Expand a degree-D label in weight-homogeneous basis vectors. Changing from exponent $\alpha_1 + u - \alpha_0$ to u multiplies a weight-j coordinate by $M_n^{(\alpha_0-\alpha_1)j}$, hence bounds this seminorm by $CM_n^{(\alpha_0-\alpha_1)D}$. Include all signed and conjugated basis vectors in the finite comparison request. The chessboard inequality (14.19) then bounds their reflected products by the product of these seminorms. The same calculation with α_2 gives seminorms $CM_n^{(\alpha_0-\alpha_2)D}$. Their cell sides are $o(r(n,u)\delta_n)$ for $\delta_n = M_n^c$. Apply Lemma 14.27 only when every coordinate hybrid of the two physical configurations has periodic coordinate separation at least $4r(n,u)\delta_n$. The centered level D_{cen} can be chosen with $a_{D_{\mathrm{cen}}} < r(n,u)\delta_n$; the insertions and their single-axis reflections are interior by (14.41). At small radii these periodic separations equal the ordinary coordinate separations on the fixed compact set. The lemma bounds the positional difference by the product of the seminorms times $4n_{\mathrm{ar}}/\delta_n$. With scaled mesh $h_n^{\mathrm{mesh}} = M_n^{-u}$, the two errors are consequently bounded by

$$CM_n^{(\alpha_0-\alpha_1)Dn_{\mathrm{ar}}}\delta_n, \qquad CM_n^{(\alpha_0-\alpha_2)Dn_{\mathrm{ar}}}h_n^{\mathrm{mesh}}/\delta_n\,.$$

These are the contributions of coordinate-coincidence strips of width $4\delta_n$ and interpolation across one mesh cell; include a strip for each coordinate hybrid, a finite union of volume $O(\delta_n)$ since $h_n^{\mathrm{mesh}} = o(\delta_n)$. Both vanish uniformly in $u \in [\alpha_0, \alpha_0 + 1]$ by the choice of the comparison exponents. For the remaining error, use the corrected Gaussian source of Lemma 14.24 on every subset of the n_{ar} slots, as in (16.16). The weighted mode norm in the proof of Lemma 14.14 is submultiplicative at every arity. Coefficient extraction at the amplitudes just chosen gives $\varepsilon_n \le CM_n^{-1/(16n_{\mathrm{ar}})} \to 0$ for the complete reflected array.

Phase uniformity. By Lemma 17.2 and the lattice paragraph after it, these seminorm bounds on every repeated reflected product bound the norm of the corresponding compressed zero-energy operator, hence its value at every character of the phase algebra of Theorem 6.13. This gives (iii), including moving coefficients by finite-dimensional multilinearity.

Every small radius lies in a window. By the same entry bound $\ell_m \le \log \widehat{\beta}_m$, applied to both adjacent stage ratios, $\ell_n \le M_{n+1}$ eventually. Monotonicity of M_n gives $r(n,\alpha_0) \le r(n+1,\alpha_0+1)$, while $r(n,\alpha_0) \to 0$. First crossing of any prescribed small radius, followed by continuity in u, puts that radius in one of these intervals. Diagonal extraction over degrees, arities, compact sets and jet orders gives $n_i, k_i = k(n_i)$ and frames, using Lemma 14.8 for symmetries and units and the recursive correction (16.22) for inclusions. Fix the exponents before each extraction, and require $(\alpha_0 + 1)\log M_{n_i} \le (\log \widehat{\beta}_{n_i})^{48}$.

Jets and couplings. For jets, the argument of (16.23) applies on these frames with Proposition 6.4: finitely many tested free responses have convergent matrices and bounded left inverses, integration by parts identifies the tested derivative $r_i(\mathsf{U}^{\mathrm{sc}}_{i,d+1})^{-1}\partial_a \mathsf{U}^{\mathrm{sc}}_{i,d} A$ with the response of $\partial_a^{\mathrm{Max}} A$, and iteration and integration on the fixed compact shell give the jet and flux limits. Minimality of $k(n)$ gives $\log(a_n/a_{k(n)}) = O((\log \widehat{\beta}_n)^{48}) = o(\widehat{\beta}_n)$. The running law of Theorem 5.6, in (5.14), gives $\beta_i/\widehat{\beta}_{k_i} \to 1$. Since $\widehat{\beta}_n = \bar{\beta} + \kappa_G \log(1/a_n)$, the imposed bound on $u \log M_n$ proves (18.1). □

Tensor coordinates. Set $L = T^{\mathrm{Max}}$, $D_{ab} = \delta_{ab}\mathcal{O}_{F^2}$, and $\Xi_{aa} = 0$, $\Xi_{ab} = \sum_{c,\alpha} : F^{\alpha}_{ac}F^{\alpha}_{bc} :$ for $a \neq b$. In the notation of (16.2), $L = -Q + \frac{1}{4}\delta Q_{\mathrm{tot}}$, $\Xi_{ab} = (1-\delta_{ab})Q_{ab}$ and $D = \frac{1}{4}\delta Q_{\mathrm{tot}}$; in the basis $Z^{(1)}, Z^{(2)}, Z^{(3)}$ of the proof of Lemma 16.1,

$$L = -\tfrac{1}{2}Z^{(1)} + \tfrac{1}{4}Z^{(2)} - Z^{(3)}, \qquad \Xi = Z^{(3)}, \qquad D = \tfrac{1}{2}Z^{(1)} + \tfrac{1}{4}Z^{(2)}\,.$$

By that proof, these three tensors span the centered real symmetric H_4-tensorial tensors of degree four.

Write $\langle F_{ac}, F_{bc}\rangle = \sum_{\alpha} : F^{\alpha}_{ac}F^{\alpha}_{bc} :$ and similarly with derivatives. Before imposing the Maxwell equation, divergence has three coordinates, represented in component b by

$$\sum_{c}\langle F_{bc}, \partial_b F_{bc}\rangle, \quad \sum_{a,c}\langle F_{ac}, \partial_b F_{ac}\rangle, \quad \sum_{a,c}\langle F_{bc}, \partial_a F_{ac}\rangle\,.$$

For diagonal entries $z_0\sum_c\langle F_{ac}, F_{ac}\rangle + z_1\sum_{b,c}\langle F_{bc}, F_{bc}\rangle$ and off-diagonal entries $z_2\sum_c\langle F_{ac}, F_{bc}\rangle$, the divergence coordinates are $(2(z_0-z_2), 2z_1+z_2/2, z_2)$. Indeed Bianchi gives $\sum_{a,c}\langle F_{ac}, \partial_a F_{bc}\rangle = \frac{1}{2}\sum_{a,c}\langle F_{ac}, \partial_b F_{ac}\rangle$. Thus the functional $\rho_{\mathrm{tr}}(x,y,z) = -2x - 8y$ satisfies

$$\rho_{\mathrm{tr}}(\mathrm{div}_{\mathrm{raw}}\, B)\mathcal{O}_{F^2} = -\operatorname{tr} B, \qquad \rho_{\mathrm{tr}}(\operatorname{div} L) = \rho_{\mathrm{tr}}(\operatorname{div}\Xi) = 0, \qquad \rho_{\mathrm{tr}}(\operatorname{div} D) = -4\,. \tag{18.3}$$

It annihilates the third coordinate, the Maxwell equation direction. Write $R_5 \cong \mathbb{R}^3$ for this raw space and V_5 for the space of real H_4-covariant axis vectors in $(M_5^{\mathrm{Max}})^4$. The free Maxwell quotient map $s : R_5 \to V_5$ is onto, with $\ker s = \mathbb{R}v_{\mathrm{eom}}$, $v_{\mathrm{eom}} = (0,0,1)$ and $\rho_{\mathrm{tr}}(v_{\mathrm{eom}}) = 0$; thus V_5 is two dimensional.

Lattice stress current. We construct the lattice insertions from plaquette curvatures. Let $\mathsf{S} = \mathsf{S}_{N,m}$ be the leading analytic path action without its coupling, defined in Section 12.2, and let $\mathsf{s}_a q(x) = q(x+e_a)$ be the lattice shift.

Definition 18.4 (Lattice stress current)**.** At a site x put $f_{ab}(x) = \mathrm{proj}_{\mathfrak{g}}(U_{\square(x;a,b)} - U^{-1}_{\square(x;a,b)})/2$ and

$$J_{ab} = \frac{\delta_{ab}}{4}\sum_{c,d}\langle f_{cd}, f_{cd}\rangle - \sum_{c}\langle f_{ac}, f_{bc}\rangle, \qquad (Y^x_b)_e = \mathbf{1}_{e^-=x} f_{b,e^{\mathrm{dir}}}(x)\,.$$

With $\nabla_a = \sum_{j=1}^{4} c_j^{\nabla}(\mathsf{s}_a^j - \mathsf{s}_a^{-j})$ and $(c_1^{\nabla}, c_2^{\nabla}, c_3^{\nabla}, c_4^{\nabla}) = (4/5, -1/5, 4/105, -1/280)$, the Ward defect is $\mathcal{E}_b = \sum_a \nabla_a J_{ab} - D\mathsf{S}[Y_b]$.

The current J is the lattice form of T^{Max}, and Y_b is the initial left displacement. Initially $\operatorname{tr} J = 0$ and $J_{ab} - J_{ba} = 0$, so these data satisfy the identities (a)–(b) of Lemma 18.5 below with $q = 0$ and $\mathsf{t} = 0$.

Block transfer. The averaging kernels must respect the particular difference ∇_a. For odd block ratio ℓ, define commuting finite sums

$$\mathsf{M}_a = \ell^{-1}\sum_{j=0}^{\ell-1}\mathsf{s}_a^j, \qquad \mathsf{G}_a = \sum_{j=1}^{4} c_j^{\nabla}\sum_{v=-j}^{j-1}\mathsf{s}_a^v, \qquad \mathsf{N}_a = \mathsf{G}_a^{(\ell)}\mathsf{M}_a, \qquad \mathsf{K}_0 = \prod_a \mathsf{M}_a\mathsf{N}_a\,.$$

Here $\mathsf{G}_a^{(\ell)}$ is G_a with every shift replaced by its ℓth power, and $\mathsf{G}_a(\mathsf{s}_a - 1) = \nabla_a$. Put $\mathsf{K}_a = \mathsf{G}_a\mathsf{M}_a\prod_{c\neq a}\mathsf{M}_c\mathsf{N}_c$ and $\mathsf{K}_a^{\mathrm{edge}} = \mathsf{N}_a\prod_{c\neq a}\mathsf{M}_c\mathsf{N}_c$. Since $\mathsf{M}_a(\mathsf{s}_a - 1) = \ell^{-1}(\mathsf{s}_a^{\ell} - 1)$,

$$\mathsf{K}_0\nabla_a = \ell^{-1}\nabla_a^{\mathrm{coarse}}\mathsf{K}_a, \qquad \mathsf{K}_0(\mathsf{s}_a - 1) = \ell^{-1}(\mathsf{s}_a^{\ell} - 1)\mathsf{K}_a^{\mathrm{edge}}, \qquad \mathsf{K}_a - \mathsf{K}_0 = (\mathsf{s}_a^{\ell} - 1)\mathsf{K}_a^{\mathrm{tr}}\,.$$

The last quotient is a finite Laurent polynomial: $\mathsf{G}_a - \mathsf{G}_a^{(\ell)}\mathsf{M}_a$ vanishes modulo $\mathsf{s}_a^\ell - 1$ after multiplication by M_a, because M_a averages all residue classes and both factors $\mathsf{G}_a, \mathsf{G}_a^{(\ell)}$ have mass one. Polynomial division defines $\mathsf{K}_a^{\mathrm{tr}}$. The support radius of K_0 is at most 56ℓ, its mass is one and its absolute coefficient sum is at most 32^4.

For Lemma 11.6's stationary background $U_{\mathrm{stat}}(V)$, stationary substitution is $\mathcal{I}F(V) = F(U_{\mathrm{stat}}(V))$. The stationary equation is $D_U(\mathsf{S} + \mathsf{S}_{\mathrm{block}}) = 0$ on the gauge-fixed fine variations. Retain the stationary coarse frame h_{fr} from the coordinates preceding Lemma 11.7, and write $Q^{\mathrm{pred}}(U) = (Q_e^{\mathrm{pred}}(U))_e$ and $\widetilde{Q}(U,h) = (Q^{\mathrm{pred}}(U))^h$ for the predictor in that frame. Its fine-coordinate differential is

$$R_f\eta = \frac{d}{ds}\Big|_{s=0} \log\bigl(\widetilde{Q}(e^{s\eta}U, h_{\mathrm{fr}})\widetilde{Q}(U, h_{\mathrm{fr}})^{-1}\bigr)\,.$$

For the root correction, let $\varrho_x(y;\eta)$ be minus the uniform average of the signed sums of η over the frame routes from y to the patch points, as in the proof of Lemma 11.6. Define scalar coefficients $b_{x,e}$ by

$$\sum_e b_{x,e}\eta_e = \varrho_x(0;\eta) - \sum_y w_y^{\mathrm{av}}\varrho_x(y;\eta)\,.$$

Let $g_{x,e} = h_{\mathrm{fr}}(x)U(\gamma_{x,e})$, where $\gamma_{x,e}$ is the fixed coordinate route from the embedded root x to e^-. The root map and its coarse gauge variation are

$$(R_0\eta)(x) = \sum_e b_{x,e}\,\mathrm{Ad}(g_{x,e})\eta_e, \qquad (d_V\phi)_e = \phi(e^-) - \mathrm{Ad}(V_e)\phi(e^+)\,.$$

These formulas transport a fine displacement to the predictor and correct its root frame. Evaluate them at the stationary pair and set $\mathcal{I}_*Y = (R_f + d_V R_0)Y(U_{\mathrm{stat}}(V))$. With $\delta_a = \mathsf{s}_a - 1$, transfer by

$$J'_{ab} = \ell^4\mathsf{K}_a\mathcal{I}J_{ab}, \quad Y'_b = \ell^5\mathsf{K}_0\mathcal{I}_*Y_b, \quad \mathcal{E}'_b = \ell^5\mathsf{K}_0\mathcal{I}\mathcal{E}_b, \quad q'_a = \ell^3\mathsf{K}_a^{\mathrm{edge}}\mathcal{I}q_a + \ell^4\mathsf{K}_a^{\mathrm{tr}}\mathcal{I}J_{aa}\,. \tag{18.4}$$

Starting with $q_a = 0$, this preserves $\operatorname{tr} J = \sum_a \delta_a q_a$. The antisymmetric part, which need not remain zero, is likewise a sum of forward differences: its new correction in direction a is $\ell^4\mathsf{K}_a^{\mathrm{tr}}\mathcal{I}J_{ab}$ and in direction b is $-\ell^4\mathsf{K}_b^{\mathrm{tr}}\mathcal{I}J_{ba}$, in addition to the transferred old differences.

For an analytic local function F, its degree-d Taylor map $\mathrm{Tay}_d F$ (Section 2.3) is, on a radial jet j, $\sum_{v=0}^d [\varepsilon^v]F(U(\mathrm{sc}_\varepsilon\,\mathsf{j}))$, in the basis of Definition 2.7. The weighted Taylor map is $t_m^{\mathsf{N}}\,\mathrm{Tay}_d F$, with the factor-count operator of Definition 2.8. Symmetrize J, average over H_4 and conjugation, apply the degree-four Taylor map and marked transport, and subtract the mixture mean. Call the resulting field B_i^L. The same operations on the local contractions defining Ξ, D give B_i^Ξ, B_i^D.

Extraction of the scalar coefficient. We first calculate the scalar coefficient before marked transport. For a chart function F define its Hessian Fourier kernel by

$$\widehat{F}_{c\alpha,c\alpha}(p,q) = \sum_{x,y} e^{i(p\cdot x + q\cdot y)}D^2F(0)[e_{x,c,\alpha}, e_{y,c,\alpha}]\,.$$

Fix a colour α. For a family F_b write

$$\mathsf{I}(F) = \mathrm{Re}\sum_{a\neq c,b} i[z^2t]\widehat{F_b}_{c\alpha,c\alpha}(ze_a, -ze_a + te_b),$$
$$\mathsf{q}(F) = |\Lambda|^{-1}\,\mathrm{Re}\sum_{a\neq c}[z^2]\widehat{F}_{c\alpha,c\alpha}(ze_a, -ze_a)\,.$$

Coefficients use $[z^r t^s]F = (r!s!)^{-1}\partial_z^r\partial_t^s F(0)$. Put $\mathsf{h}(F) = \mathsf{I}(F) - \frac{1}{2}\operatorname{Re}\sum_{a\neq c}[z^2]\widehat{F}_{cc\alpha,c\alpha}(ze_a, -ze_a)$. The Haar coefficient is $\mathfrak{h}(Y) = \mathsf{h}(\operatorname{div}_{\mathrm{Haar}} Y)$; the subtraction corrects the lattice derivative. Define

$$\Gamma_\mu = \tfrac{1}{2}\log\det(H + dd^* + \mu^2 I) - \log\det(d^*d + \mu^2 I), \quad R_\Lambda(\mu) = \mathsf{q}(\Gamma_\mu), \quad \mathcal{D}_\Lambda(\mu) = -\mu R_\Lambda'(\mu),$$

where H is the left Hessian of S and d is the infinitesimal gauge insertion. For $a > 0$ and $r \in \mathbb{N}$, with $\|x\| = |x|_\infty$ in the kernel's lattice units, put

$$\vartheta_{a,r}(x) = e^{a\|x\|}(1 + \|x\|)^r .$$

For a chart function anchored at o, its weighted order-r seminorm is the sum of absolute link coefficients of $D^r F(0)$ against $\prod_{v=1}^r \vartheta_{a,r}(e_v^- - o)$; for a displacement use its link components. These weights control the spatial moments in the momentum derivatives below. Write $F \equiv_4 G$ if the field derivatives of F and G at zero agree through order four. The next lemma computes the scalar coefficient; its proof follows.

Lemma 18.5 (Regulated Ward coefficient)**.** *Fix a level k, a weight $a > 0$ and $C < \infty$. For $\nu \in \mathbb{N}$ let $N_\nu > k$ and let $(J, Y, \mathcal{E}, q, \mathsf{t})^{(\nu)}$ be translation-invariant local data on the level-k torus of period $n_\nu \to \infty$, with $\mathsf{S} = \mathsf{S}_{N_\nu,k}$, such that*

(a) $D\mathsf{S}[Y_b] \equiv_4 \sum_a \nabla_a J_{ab} - \mathcal{E}_b$;

(b) $\operatorname{tr} J = \sum_a \delta_a q_a$ and $J_{ab} - J_{ba} = \sum_c \delta_c \mathsf{t}_{abc}$;

(c) the a-weighted seminorms of orders 2–4 of J, Y, q, t are at most C;

(d) those of $\mathcal{E}$, and the deviations of the zero-field Hessians of J and Y from their free values on tests of weight at most 4 and 5, are at most $\varepsilon_\nu \to 0$.

Then, for every colour, $\mathfrak{h}(Y^{(\nu)}) \to -12\kappa_G$.

Proof. The inputs are the normalization of κ_G (Definition 2.1), the kinetic and quadratic cancellations and the classical matching of the path action (Lemmas 12.4 and 12.6), and the one-loop coefficient of Lemma 12.9. Write $\mathcal{R}_\mu = (H + dd^* + \mu^2 I)^{-1}$, $\mathcal{G}_\mu = (d^*d + \mu^2 I)^{-1}$. For $\Psi_s(U) = e^{sY(U)}U$ let $\mathcal{Y}_U(z) = \partial_s|_0 \log(\Psi_s(e^z U)\Psi_s(U)^{-1})$ and $\mathcal{V} = D\mathcal{Y}_U(0)$. If $\mathcal{J}_U$ is the mixed derivative of $\log(\mathrm{gauge}_{e^v}(e^z U)U^{-1})$ in v, z at zero, gauge invariance gives $Hd = -\mathcal{N}$, $\mathcal{N} = D\mathsf{S}[\mathcal{J}_U]$, hence

$$\mathcal{R}_\mu d = d\mathcal{G}_\mu + \mathcal{R}_\mu \mathcal{N} \mathcal{G}_\mu .$$

The product rule for $D^2(D\mathsf{S}[Y])$ and cyclicity of trace now give

$$\tfrac{1}{2}\operatorname{tr}(\mathcal{R}_\mu D^2(D\mathsf{S}[Y])) - \operatorname{div}_{\mathrm{Haar}} Y = D\Gamma_\mu[Y] - \mu^2 \operatorname{tr}(\mathcal{R}_\mu \mathcal{V}) + D\mathsf{S}[Z], \tag{18.5}$$

with $Z = \frac{1}{2}\operatorname{contr}(\mathcal{R}_\mu, D^2\mathcal{Y}_U(0)) - \operatorname{contr}(\mathcal{G}_\mu d^*(\mathcal{V} + \mathcal{V}^*)\mathcal{R}_\mu, \mathcal{J}_U)$. Contractions sum the indicated matrix indices.

Cancellation. The leading quadratic kernel of an invariant scalar action at the required momentum order is a colour multiple of $|p|^2 I - p \otimes p$. Write X_{acb} for the coefficient of momentum e_a in the diagonal polarization-c entry of DY_b, with its colour indices contracted against the action kernel. Gauge covariance makes the mixed polarization term $-X_{cac}$ and gives $X_{aaa} = 0$. In the tested variation the three terms, after summing the intermediate polarization, are

$$-2X_{aca} - X_{cac} + \sum_b X_{bcb} \qquad (a \neq c).$$

Summing over $a \neq c$ makes the first and second sums $-3\sum_{b,c} X_{bcb}$, cancelling the third sum. Higher momentum orders do not contribute. This applies to $D\Gamma_\mu[Y]$, $D\mathsf{S}[Z]$ and the further action variation below, uniformly on each fixed $\mu \in [\delta, 1]$. For the first term on the left of (18.5), substitute $D\mathsf{S}[Y_b] \equiv_4 \sum_a \nabla_a J_{ab} - \mathcal{E}_b$. The coefficient linear in the total momentum te_b selects $a = b$; summing b gives $\operatorname{tr} J = \sum_a \delta_a q_a$. Its additional total-momentum factor has zero coefficient linear in t. The defect $\mathcal{E}_b$ tends to zero in the weighted seminorms, so its contracted coefficient does too. The correction in $\mathfrak{h}$ tends to zero by the same identity.

The regulator term. For $\chi(x) = \mathbf{1}_{x=0}$ define $M_b h(x) = \chi(x) h_b(x)$, $(D_b h)_a(x) = \delta_a \chi(x) h_b(x)$ and $(T_b^{(r)} h)(x) = \delta_b \chi(x) h(x)$ on r-forms, $r = 0, 1$. Set

$$W_b^{(1)} = \operatorname{sym}(\mathcal{V}_b - dM_b - D_b) + \tfrac{1}{2} T_b^{(1)}, \qquad W_b^{(0)} = \operatorname{sym}(M_b d) - \tfrac{1}{2} T_b^{(0)},$$

where $\operatorname{sym} A = (A + A^*)/2$. The inverse identity above yields

$$\begin{aligned} \operatorname{tr}(\mathcal{R}_\mu \mathcal{V}_b) = {} & \operatorname{tr}\big(\mathcal{R}_\mu (D_b - \tfrac{1}{2} T_b^{(1)})\big) + \tfrac{1}{2} \operatorname{tr}(\mathcal{G}_\mu T_b^{(0)}) + D\mathsf{S}[Z_b'] + \operatorname{Rem}_b, \\ \operatorname{Rem}_b = {} & \operatorname{tr}(\mathcal{R}_\mu W_b^{(1)}) + \operatorname{tr}(\mathcal{G}_\mu W_b^{(0)}), \qquad Z_b' = \operatorname{contr}(\mathcal{G}_\mu M_b \mathcal{R}_\mu, \mathcal{J}_U). \end{aligned}$$

The tested D_b term tends to $\mathsf{q}(\operatorname{tr} \mathcal{R}_\mu)$; each of the four $T_b^{(r)}$ terms tends to $\mathsf{q}(\operatorname{tr} \mathcal{R}_\mu)$ or $\mathsf{q}(\operatorname{tr} \mathcal{G}_\mu)$ respectively. This follows by summing $\delta_b \chi$ against the two Fourier waves: its symbol is $e^{-it} - 1 = -it + O(t^2)$, paired with the factor i in l. Translation invariance identifies the local anchor sum with the volume-normalized trace in q. Consequently their sum is $-\mathsf{q}(\operatorname{tr} \mathcal{R}_\mu) + 2\mathsf{q}(\operatorname{tr} \mathcal{G}_\mu)$. Differentiation of Γ_μ in μ gives the same expression multiplied by μ^2, namely $\mathcal{D}_\Lambda(\mu)$. We have proved

$$\mathfrak{h}(Y) = \mathcal{D}_\Lambda(\mu) + \mu^2 \mathsf{l}(\operatorname{Rem}) + o(1), \tag{18.6}$$

with error uniform on every fixed $[\delta, 1]$.

For the remainder, the inverse-kernel argument in the proof of (12.3) applies with the scalar regulator $\mu^2 I$: $\|\partial^r R_\mu(q)\| \leq C(|q| + \mu)^{-2-r}$. Differentiating $R_\mu A_\mu = I$ once more gives the same bound for $r = 3$. Hypotheses (c)–(d) give bounded weighted derivatives of $W^{(r)}$. Their zero-background symbols and first momentum derivatives tend to zero, as does the zero-momentum first field insertion: the free curvature displacement cancels $dM_b + D_b - \frac{1}{2} T_b^{(1)}$ in the displayed formula. Put $z = |q| + \mu$. Taylor expansion gives factors $z^2, z, 1$ for the zero-, one- and two-field insertions of these symbols. In the product rule they pair with inverse bounds $Cz^{-4}, Cz^{-3}, Cz^{-2}$, respectively. Thus the scaled derivative bound through order three is $C(z^2 z^{-4} + 2z z^{-3} + z^{-2}) \leq C' z^{-2}$. Unscaling the quadratic coefficient gives $Cz^{-4} \leq C' z^{-5}$. In four dimensions the integral of this bound is at most C/μ: the ball $|q| \leq \mu$ costs C/μ, and the annulus $2^j \mu < |q| \leq 2^{j+1}\mu$ costs $C\mu^{-1} 2^{-j}$. At fixed $\mu > 0$ the finite sums converge; hence $|\mu^2 \mathsf{l}(\operatorname{Rem})| \leq C\mu + o(1)$ with C independent of $\mu \leq 1$.

The proof of (12.3), with $\mu^2 I$ in place of its regulator, gives $|R(\mu) - 12\kappa_G \log \mu| \leq C$ for $0 < \mu \leq 1$. Here $R/12$ is the coefficient α_b of this scalar-regulated kernel: the Hessian is twice the quadratic-action kernel, exactly the factor 2 in that coefficient. At fixed positive μ, Riemann sums give $R_\Lambda(\mu) \to R(\mu)$. Choose $u < 0$, then a length $L_* > 0$. Once the period controls the endpoints, the mean value theorem on $[u - L_*, u]$ selects $\mu_\Lambda \in [e^{u - L_*}, e^u]$ with $|\mathcal{D}_\Lambda(\mu_\Lambda) + 12\kappa_G| \leq (2C + o(1))/L_*$. Apply (18.6) on this fixed interval, then let $u \to -\infty$ and $L_* \to \infty$. This proves the limit without differentiating an asymptotic error. □

A stability lemma. The maps from raw coefficients to continuum vectors only converge to the free quotient map s. The next lemma turns a nearly null vector of these maps into functionals that read ρ_{tr} with a rate.

Lemma 18.6 (Stability of the trace functional)**.** *Let E, H be finite-dimensional real normed spaces, $s : E \to H$ linear with a right inverse t and $\ker s = \mathbb{R}v$, $v \neq 0$, and $\rho : E \to \mathbb{R}$ linear. Let $s_i : E \to H$ and $t_i : H \to E$ be linear with $s_i t_i = I$ and $t_i \to t$, let $v_i \to v$, and let $e_i \geq 1$ satisfy $e_i \|s_i v_i\| \to 0$ and $e_i \rho(v_i) \to 0$. Then $s_i \to s$, the functionals $\lambda_i = \rho t_i$ are uniformly bounded, and $e_i \|\lambda_i s_i - \rho\| \to 0$.*

Proof. Put $k_i = v_i - t_i s_i v_i$. Then $s_i k_i = 0$, and $k_i \to v$ since t_i is bounded and $s_i v_i \to 0$. The maps $\Phi_i(h, c) = t_i h + c k_i$ from $H \times \mathbb{R}$ to E converge to $\Phi(h, c) = th + cv$, which is injective (apply s) and hence invertible, as $\dim E = \dim H + 1$. So $\Phi_i^{-1} \to \Phi^{-1}$; since $s_i \Phi_i(h, c) = h$, also $s_i \to s$. For $y \in E$ write $\Phi_i^{-1} y = (s_i y, c_i(y))$, so that $y - t_i s_i y = c_i(y) k_i$ and $|c_i(y)| \leq C \|y\|$ uniformly in i. Then

$$\lambda_i s_i y - \rho(y) = -c_i(y) \rho(k_i) = -c_i(y) \big(\rho(v_i) - \rho(t_i s_i v_i) \big).$$

Hence $e_i \|\lambda_i s_i - \rho\| \leq C \big(e_i |\rho(v_i)| + \|\rho\| \sup_j \|t_j\| \, e_i \|s_i v_i\| \big) \to 0$. Boundedness of λ_i follows from $t_i \to t$. □

Transfer to the continuum. Put $b_i = \widehat{\beta}_{k_i}$, $E_i = \exp((\log\log b_i)^2/5)$, $R_i = \exp((\log\log b_i)^2/6)$, and $\mathsf{A}_i = \beta_i r_i^4 (\mathsf{U}^{\mathrm{sc}}_{i,4})^{-1}$, $K_i = \beta_i r_i^5 (\mathsf{U}^{\mathrm{sc}}_{i,5})^{-1} \operatorname{div}$. The frames intertwine signed permutations and conjugation, so K_i maps centered real symmetric H_4-tensorial tensors into V_5; on the span of such tensors its kernel is the conserved line of Lemma 16.10. Functionals below act on V_5, with any fixed norm.

Quantity	Behaviour as the radius tends to zero	Proved in
stress insertion $r^4 T_{ab}$	tends to T^{Max}_{ab}, which is traceless	Proposition 6.11
running coupling β_i	$\beta_i / \log(1/r_i) \to \kappa_G$	Lemma 18.3
scalar trace coefficient	$r_i^4 (\mathsf{U}^{\mathrm{sc}}_{i,4})^{-1} \Theta = -\kappa_G \beta_i^{-1} \mathcal{O}_{F^2} + o(\beta_i^{-1})$	Proposition 18.8
trace two-point function	$r^8 \log^2(1/r)\, G_\Theta(r) \to A_G$	Theorem 6.18

Table 9: Orders of the stress comparison: along the sequence of Proposition 6.11, the radii of Lemma 18.3, and all radii. Heuristically the last row is the square of the third times the free correlation $A_G |x|^{-8}$ of $\mathcal{O}_{F^2}$.

Why a rate is needed. The divergence of the amplified stress field $\beta_i B_i^L$ may grow like R_i. Proposition 18.8 cancels it with multiples of B_i^{Ξ}, B_i^D of that size and reads the trace coefficient from that of B_i^D with functionals λ_i. An error of λ_i thus enters multiplied by up to R_i; an error that merely tends to zero would not suffice. It must be $o(E_i^{-1})$, and $R_i / E_i = \exp(-(\log\log b_i)^2/30) \to 0$. The next lemma transfers the scalar coefficient to V_5 at this accuracy; its proof follows. Here Υ ranges over the indicated tensors.

Lemma 18.7 (Fourth-order Ward transfer)**.** *There are uniformly bounded real functionals λ_i on V_5 and $C < \infty$, depending only on the chosen construction, such that the divergences satisfy*

$$\begin{aligned} &\|\beta_i K_i B_i^L\| \leq C R_i, \qquad && \lambda_i(\beta_i K_i B_i^L) \to -\kappa_G, \\ &K_i B_i^{\Upsilon} \to \operatorname{div}^{\mathrm{Max}} \Upsilon, \qquad && E_i \big[\lambda_i(K_i B_i^{\Upsilon}) - \rho_{\mathrm{tr}}(\operatorname{div}_{\mathrm{raw}} \Upsilon) \big] \to 0 \quad (\Upsilon = \Xi, D), \end{aligned} \tag{18.7}$$

and the tensors satisfy

$$\mathsf{A}_i B_i^{\Upsilon} \to \Upsilon \ (\Upsilon = L, \Xi, D), \qquad b_i^p \operatorname{tr}(\mathsf{A}_i B_i^{\Upsilon}) \to 0 \ (\Upsilon = L, \Xi;\ p \in \mathbb{N}), \qquad \beta_i / b_i \to 1 \,. \tag{18.8}$$

Thus λ_i reads ρ_{tr}, that is -4 times the D-coefficient, of a divergence with error $o(E_i^{-1})$, and its value on the stress divergence, of size $O(R_i)$, is the coefficient of Lemma 18.5 divided by 12.

Proof. We verify the hypotheses of Lemma 18.5 for the transferred current, compute the finite coefficient at each level, and read it along the radii of Lemma 18.3. The stationary kernels (Lemmas 11.6–11.10) control the transfer, and Lemmas 14.7–14.9 control marked transport.

The proof has five parts, and each has its own reason for smallness (Table 10). The current estimates are a contraction under repeated block transfer. The finite coefficient is accurate because of the thresholds and the exponential tails. The trace of the current is an exact forward difference; this, with the matching gain of marked transport, gives the equation direction its rates, and Lemma 18.6 turns that rate into the accuracy of λ_i. The tensor traces are small for a different reason: the same forward difference becomes small after averaging over many sites. No single power count covers all five. The choices are made in this order: the block ratios and entry couplings of the stages, once and for all, so that transfer contracts; at a fixed level k, the transfer depth and the period; then the ordered limit at that k; and finally $k = k_i$ along the radii. Since $M_k^{\mathrm{cut}} = e^{h_k}$ with $h_k = (\log\log\widehat{\beta}_k)^2$, the multipliers of the statement are $E_i = (M_{k_i}^{\mathrm{cut}})^{1/5}$ and $R_i = (M_{k_i}^{\mathrm{cut}})^{1/6}$.

Part	Quantity made small	Reason	Bound used
Current estimates	normalized defects of orders 2–4	one transfer acts on them by a triangular map with small entries	$O(2^{-n_{\mathrm{step}}})$
Finite coefficient	the four errors in (18.9)	thresholds, kernel and path tails, forced allowance	each $o(e^{-h_k})$; $\|\Sigma_k\| \le C(M_k^{\mathrm{cut}})^{1/6}$
Equation direction	$\rho_{\mathrm{tr}}(j(k))$ and $s_i(v_i^{\mathrm{eq}})$	trace of the current is a forward difference; matching gain $e^{-H_n/8}$	$Ce^{-h_k/4}$ and $o(b_i^{-p})$
Stability	$\lambda_i s_i - \rho_{\mathrm{tr}}$	Lemma 18.6	$o(E_i^{-1})$
Tensor traces	$\mathrm{tr}(\mathsf{A}_i B_i^L)$	averaging a forward difference over m sites	$O(a_k^{\eta_0/2})$, so $o(b_i^{-p})$

Table 10: The five parts of the proof of Lemma 18.7. The readout error $o(E_i^{-1})$ is multiplied by coefficients of size $O(R_i)$, and $R_i/E_i \to 0$.

Current estimates. The transfer identities give (a)–(b) of Lemma 18.5; we prove (c)–(d) below. Differentiating the stationary equation eliminates its fine variation. Fourth-order matching in Lemma 12.6 preserves $\sum_a \nabla_a J_{ab} - \mathcal{E}_b \equiv_4 D\mathsf{S}[Y_b]$. The kernel identities above preserve the trace and antisymmetric parts as differences. Work after stage $j \ge 7$ and choose $0 < a < c_j$, with c_j the kernel decay exponent of Lemma 11.6. Its kernel bound applies through polynomial degree three. Divide the order-r current defect by $(\log\widehat{\beta}_m)^r$. At orders $r = 3, 4$, the linearized stationary fields contribute ℓ^{-2} each, so transfer costs $C_j\ell^{5-2r}$. At order two the matched low-momentum terms must first be removed. The affine and quadratic parts of a stationary test field have weighted sizes $C_j(1+|x|)/\ell^2$ and $C_j(1+|x|)^2/\ell^3$; the remainder is at most $C_j(1+|x|)^3/\ell^4$, by Lemmas 11.6 and 11.9. Without the subtraction, the affine–affine and affine–quadratic products, with factors ℓ^{-4} and ℓ^{-5}, would cost ℓ and 1 after multiplication by ℓ^5, and transfer would not contract them. Subtracting them leaves quadratic–quadratic and affine–remainder products, each with factor ℓ^{-6}. Multiplication by ℓ^5 gives C_j/ℓ. A bounded matched contribution gives the additional $C_j/(\log\widehat{\beta}_m)^2$.

More precisely the order-two factor is bounded by

$$C_j \left\{ \frac{1}{(\log \widehat{\beta}_m)^2} + \frac{(\log \widehat{\beta}_{m+1})^2 + 1}{\ell_m (\log \widehat{\beta}_m)^2} \right\}.$$

Both terms are small once the ratio and the coupling are large. Thus the three orders gain for different reasons: ℓ^{-3} at order four, ℓ^{-1} at order three, and at order two the two displayed terms, after the subtraction. Nonlinear chain-rule partitions involve fewer than r derivatives and gain inverse logarithmic powers. So one transfer maps the vector of normalized defects of orders $2, 3, 4$ by a lower-triangular matrix: its diagonal entries are the factors just computed, and the entries below the diagonal, which carry the defects of lower orders, contain a quotient $(\log \widehat{\beta}_{m+1})^{r'}/(\log \widehat{\beta}_m)^r$ with $r' < r$. Choose each ratio large, then its entry coupling large; every entry is then small, and the map contracts by $1/2$. After n_{step} transfers the normalized defects are $O(2^{-n_{\mathrm{step}}})$.

The finite coefficient. At level k set $h_k = (\log\log \widehat{\beta}_k)^2$ and use $M_k = M_k^{\mathrm{cut}}$. Define $\mathrm{Tay}_{5,k} = t_k^{-2} t_k^{\mathsf{N}} \mathrm{Tay}_5$, followed by real H_4 averaging. Haar integration by parts in the density gives the degree-five coefficient in the form

$$\Sigma_k = \mathrm{Tay}_{5,k}(\mathrm{div}_{\mathrm{Haar}}\, Y - D\mathcal{L}_k[Y]) + \varepsilon_{\mathrm{stat}} + \varepsilon_{\mathrm{tail}} + \varepsilon_{\mathrm{cl}} + \varepsilon_{\mathrm{mark}}\,. \tag{18.9}$$

Here $\mathcal{L}_k$ is the quadratic loop action of Definition 12.2 at level k. The errors are stationary substitution, omitted loop paths, the classical defect and the marked remainder. Each has norm at most e^{-h_k}: truncate stationary kernels and loop paths at $C_j(\log \widehat{\beta}_k)^2$, using the kernel decay in the proof of Lemma 11.6 and the path bound of Lemma 12.3. These exponential tails bound $\varepsilon_{\mathrm{stat}}$ and $\varepsilon_{\mathrm{tail}}$ after the finite projection losses. The activity caps (5.12) and the threshold (5.2) bound $\varepsilon_{\mathrm{cl}}$ in the seminorm of Definition 5.3:

$$t_k^{-2} \epsilon_k^{\mathrm{reg}} = \widehat{\beta}_k^{-1/10} M_k^{-2}\,.$$

For $\varepsilon_{\mathrm{mark}}$, Lemma 14.9 cancels the radial Taylor coefficients through degree five on nonwrapping supports. The Taylor remainder has an additional small-field factor, at most one after entry. The fixed projection and localization powers of M_k therefore give $C_j \widehat{\beta}_k^{-\eta_j} M_k^{C_j}$ with $\eta_j = 1/10$. For the forced term, Lemma 14.7 leaves $(\epsilon_k^{\mathrm{for}})^{1/2} = e^{-M_k^{1/5}/2}$, and Lemma 14.9 retains the original eligibility event. Thus the bound is $C_j \widehat{\beta}_k^{-\eta_j} M_k^{C_j} + C_j \widehat{\beta}_k^{C_j} e^{-M_k^{1/5}/2}$ after the finite projection losses. A wrapping support reaches distance $n_{p,k}/10$ from its anchor, where $n_{p,k} = L_p D_k$, so its covering number is at least $n_{p,k}/[10(2B_k + 1)] - 1$. Lemma 14.7 at support exponent 98, at which the scale choices of Section 14.1 retain its estimates, bounds its projected contribution by

$$C_j \widehat{\beta}_k^{C_j} M_k^{C_j} \exp\left\{ -\frac{n_{p,k}}{5(2B_k + 1)} + 2 \right\}.$$

The wrapping bound vanishes on the ordered period tails at fixed k. The nonwrapping bounds are $o(e^{-h_k})$ after entry, because $M_k = e^{h_k}$ grows more slowly than every power of $\widehat{\beta}_k$ but its fifth root eventually exceeds every power of $\log \widehat{\beta}_k$. Indeed, with $x = \log\log \widehat{\beta}_k$, so that $h_k = x^2$, the normalization of $\varepsilon_{\mathrm{cl}}$ above is $e^{-2h_k} \widehat{\beta}_k^{-1/10}$, and

$$e^{h_k} C_j \widehat{\beta}_k^{-\eta_j} M_k^{C_j} = C_j e^{(C_j+1)x^2 - e^x/10}, \qquad e^{h_k} C_j \widehat{\beta}_k^{C_j} e^{-M_k^{1/5}/2} \le C_j e^{C_j e^x + x^2 - e^{2x}/2} \quad (x \ge 10),$$

using $M_k^{1/5} = e^{x^2/5} \ge e^{2x} = (\log \widehat{\beta}_k)^2$ for $x \ge 10$. Both tend to zero.

The loop term in (18.9) has zero ρ_{tr} coefficient. If H_{loop} is its quadratic link kernel, differentiation of $D\mathcal{L}_k[Y]$ pairs H_{loop} with DY. The cubic momentum test is a sum of pairings of constant and quadratic polynomial links and of two affine polynomial links. Constant links are pure gauge; H_{loop} also kills affine links by both moment cancellations of Lemma 12.4. Thus every pairing vanishes. The lattice correction in $\mathfrak{h}$ is the corresponding correction to this Taylor coefficient. Choose the transfer depth and period at this fixed k so that $\mathfrak{h}(Y)$ differs from $-12\kappa_G$ by at most $\widehat{\beta}_k^{-100}$, using Lemma 18.5. Choose the momentum-Taylor approximation to its finite local Taylor coefficient within $\widehat{\beta}_k^{-30}$. The exact Taylor-coefficient identity gives the factor $1/12$ for the twelve ordered transverse planes. Delayed entry bounds the retained loops $2M_k^{\theta_{J_k}}$ and stage losses by $M_k^{1/6}$. Therefore

$$\|\Sigma_k\| \le CM_k^{1/6}, \qquad |\rho_{\mathrm{tr}}(\Sigma_k) - \mathfrak{h}(Y)/12| \le 4\|\rho_{\mathrm{tr}}\|e^{-h_k} + \widehat{\beta}_k^{-30}/12\,.$$

At fixed k, take any joint ordered limit point of the bounded coefficient families along the ordered tails (14.28), and denote its real averaged coefficient by Σ_k. All closed bounds survive; S and the maps converging in the proof of Lemma 15.5 retain their fixed limits. Thus $\rho_{\mathrm{tr}}(\Sigma_k) \to -\kappa_G$ and $\|\Sigma_{k_i}\| \le CR_i$.

The equation direction. Let $s_i : R_5 \to V_5$ be degree-five marked transport followed by $\beta_i r_i^5(\mathsf{U}_{i,5}^{\mathrm{sc}})^{-1}$. Let j_i be the raw coefficient of the transferred current's divergence after the same threshold normalization. At fixed k, the Taylor coefficient of the local current has zero ρ_{tr} component by (18.3): simplicity makes the degree-three Taylor coefficient of its trace a unit, whose forward difference vanishes. The stationary and marked remainder estimates compare the transported coefficient with this Taylor coefficient within $Ce^{-h_k/4}$. Include the current components in the joint ordered limit point, denoting their value by $j(k)$. Closedness preserves $|\rho_{\mathrm{tr}}(j(k))| \le Ce^{-h_k/4}$ for every such value. With $j_i = j(k_i)$, Taylor matching consequently gives

$$j_i \to -v_{\mathrm{eom}}, \quad |\rho_{\mathrm{tr}}(j_i)| \le Ce^{-h_{k_i}/4}, \quad v_i^{\mathrm{eq}} = \beta_i^{-1}\Sigma_{k_i} - j_i \to v_{\mathrm{eom}}\,.$$

In particular $E_i\rho_{\mathrm{tr}}(v_i^{\mathrm{eq}}) \to 0$, since $E_i/\beta_i \to 0$ and $\rho_{\mathrm{tr}}(\Sigma_{k_i})$ is bounded. It remains to show $b_i^p s_i(v_i^{\mathrm{eq}}) \to 0$ for every p; this takes four steps.

Step 1: exterior moments vanish. Localize $\Sigma_{k_i} - \beta_i j_i$ before transport, with a fixed degree-four spectator. Its exterior moment is zero by Haar integration by parts, since the varied link is outside the spectator stencil.

Step 2: the spatial tail is small. Let v_X be regular majorants rooted at the insertion anchor and $c_X = \mathrm{cov}_{B_k}(X)$. The support exponent in Lemma 14.7 is $\sigma_j = 100$ (item 1 of Section 14.1). Thus, for every $q > 0$,

$$\sum_{c_X \ge q} e^{98c_X} v_X \le e^{-2q} \sum_X e^{100c_X} v_X.$$

The same inequality holds for the normalized forced majorants. Since every nonzero marked support contains its anchor, a connected term reaching a spectator at distance R in level-k units has $c_X \ge R/(2B_k+1) - 1$. Shorter terms retain the zero exterior moment.

Step 3: the matched residual is $e^{-H_n/100}$. Use (14.34), starting at $k(n)$ and ending at $n \ge m_{\mathrm{ref}}(5)$, and replicate by (14.25). Across logarithmic length $H_n = (\log \widehat{\beta}_n)^{48}$ the matching gain is $e^{-H_n/8}$. The fixed degree-five projections, threshold factors and transport columns cost $e^{o(H_n)}$: the logarithms of the threshold costs are $O(\log \widehat{\beta}_n)$, and the column cost is bounded by $H_n \sup_{m \ge n} J_m^{-2}$. Fix the tiling level $m = m(n) < n$ by the first backward crossing of B_n^2; eventually $m(n) \ge m_{\mathrm{ref}}(5)$. If $\ell = \max_{m \le v < n} \ell_v$, then

$$B_n^2 \le a_m/a_n \le \ell B_n^2, \qquad (a_m/a_n)^4 q_n + 6 \le \ell^4 B_n^8 q_n + 6 = o(H_n)\,.$$

Indeed $\ell \le \log \widehat{\beta}_n$ and $B_n = O((\log \widehat{\beta}_n)^2)$. Also $q_n = O((\log \widehat{\beta}_n)^2)$ uniformly by (14.1), since the stage-cost bound in the order of choice (Section 9) gives $1 + A^{\mathrm{path}}_{J_n} \le \ell_{n-1} \le \log \widehat{\beta}_n$, and $c_{\mathrm{dir},j} \le 1$. Their product is $O((\log \widehat{\beta}_n)^{22}) = o(H_n)$. The allowances $\zeta_{e,k} = \widehat{\beta}_k^{-(q_e^{\mathrm{ht}}+4)}$ fixed in Section 14 before (14.6) give $\log \zeta_{e,n}^{-1} = O(\log \widehat{\beta}_n) = o(H_n)$, so their factor in (14.25) also costs $e^{o(H_n)}$. The support tail of Step 2 is smaller, since R grows exponentially in H_n. At radius level n, multiplication of the physical two-point error by a_n^5 cancels the factor a_n^{-5} in the replicated mark weight. The resulting bound is

$$\exp\{-H_n/8 + o(H_n)\} \le e^{-H_n/100}$$

after entry.

Step 4: rescaling to the radii. To pass to the full radius normalization multiply by $(r_i/a_{n_i})^5$. We may require $u_i \le \log\log \widehat{\beta}_{n_i}$ in the diagonal extraction. The extra scaling factor is then at most $\exp(5(\log\log \widehat{\beta}_{n_i})^3)$, while $b_i/\widehat{\beta}_{n_i} \to 1$. Thus each rescaled evaluation after multiplication by b_i^p is bounded by

$$C b_i^p \exp\{5(\log\log \widehat{\beta}_{n_i})^3 - H_{n_i}/100\} \longrightarrow 0\,.$$

Finite separating free two-point tests give $b_i^p s_i(v_i^{\mathrm{eq}}) \to 0$.

Stability. Construct right inverses $t_i : V_5 \to R_5$ of s_i: lift a free vector by the common frame, undo its threshold normalization and marked transport, and check $s_i t_i = I$ with the frame inverse. These operations commute with signed permutations and conjugation, so lifts of vectors in V_5 lie in R_5. The lifts converge on a finite basis by Lemma 18.3, so $t_i \to t$ with $st = I$. Lemma 18.6, with $v_i = v_i^{\mathrm{eq}}$, $\rho = \rho_{\mathrm{tr}}$ and $e_i = E_i \le b_i$, gives $s_i \to s$, bounded $\lambda_i = \rho_{\mathrm{tr}} t_i$ and $\|\lambda_i s_i - \rho_{\mathrm{tr}}\| = o(E_i^{-1})$. The actual divergence satisfies $\beta_i K_i B_i^L - s_i \Sigma_{k_i} \to 0$ by the same equation estimate; for $\Upsilon = \Xi, D$, $K_i B_i^\Upsilon - s_i \operatorname{div}_{\mathrm{raw}} \Upsilon = o(b_i^{-p})$. The accuracy of the readout is fixed before it meets the growing coefficient: in

$$\lambda_i(\beta_i K_i B_i^L) = \rho_{\mathrm{tr}}(\Sigma_{k_i}) + (\lambda_i s_i - \rho_{\mathrm{tr}})(\Sigma_{k_i}) + \lambda_i\big(\beta_i K_i B_i^L - s_i \Sigma_{k_i}\big),$$

the first term tends to $-\kappa_G$, the second is $o(E_i^{-1}) \cdot O(R_i)$ and the third tends to zero because the λ_i are bounded. Since $\|s_i \Sigma_{k_i}\| \le C R_i$ and $R_i/E_i \to 0$, this and the same decomposition for $\Upsilon = \Xi, D$ prove (18.7).

Tensor traces. Taylor matching gives $\mathsf{A}_i B_i^\Upsilon \to \Upsilon$. Since L is traceless, this alone gives only $\operatorname{tr}(\mathsf{A}_i B_i^L) \to 0$, without a rate. Proposition 18.8 multiplies this trace by β_i, so a rate is required. It does not come from the readout functionals; it comes from the structure of the trace, as follows. The trace of Ξ is zero; the transferred trace of L is a forward difference. Its degree-three projection is a unit, since simplicity leaves no nonconstant invariant below degree four. For the degree-three stencil and its degree-four difference, fix k_0 so late that its backward crossing $m_0 = m(k_0)$ satisfies $m_{\mathrm{ref}}(4) \le m_0 < k_0$ and $B_{k_0}^2 \le a_{m_0}/a_{k_0}$. Match at $m_{\mathrm{ref}}(4)$, stop at k_0 and replicate at the fixed pair (m_0, k_0). Equations (14.34) and (14.25) give

$$C_{k_0} e^{-(s_k - s_{k_0})/8} = O(a_k^{1/8})\,.$$

The constant includes the inverse allowance, thresholds and $\exp\{(a_{m_0}/a_{k_0})^4 q_{k_0} + 6\}$. Coefficient conversions grow as $e^{o(s_k)}$, giving endpoint error $O(a_k^{\eta_0})$ for any fixed $0 < \eta_0 < 1/8$ after entry. Truncating the support sums above at $c_X = \eta_0 s_k$ leaves a tail at most $a_k^{2\eta_0}$, and decreasing η_0 absorbs all finite losses. This estimate is uniform for endpoints in a fixed small neighbourhood of the separating tests. Average the degree-four difference over $m = \lfloor a_k^{-1+\eta_0/2} \rfloor$ consecutive sites. In physical units the average of m consecutive forward differences telescopes to the difference of the

two endpoint values divided by the window length ma_k. The unit parts of the two endpoints are equal and cancel, leaving at most twice the endpoint error; and $ma_k \to 0$, so the window shrinks to a point. Thus

$$\frac{2Ca_k^{\eta_0}}{ma_k} = O(a_k^{\eta_0/2}), \qquad ma_k = O(a_k^{\eta_0/2}) \longrightarrow 0.$$

The averaged testing functionals converge to the original separating functionals by smoothness, so a finite separating matrix still has a bounded left inverse. They therefore bound the trace coefficient by $O(a_k^{\eta_0/2})$. Subexponential inverse transport preserves $o(b_i^{-p})$ for every p, proving the trace assertion in (18.8); Lemma 18.3 gives its coupling assertion. □

Conservation converts these divergence estimates into the scalar coefficient.

Proposition 18.8 (Scalar trace coefficient)**.** *Along the radii of Lemma 18.3,*

$$\beta_i r_i^4 (\mathsf{U}_{i,4}^{\mathrm{sc}})^{-1} \Theta \longrightarrow -\kappa_G \mathcal{O}_{F^2}\,. \tag{18.10}$$

Proof. We cancel the divergence of $\beta_i B_i^L$ with the two other tensors, read the trace coefficient with λ_i, and identify the resulting conserved tensor with T through the conserved line of Lemma 16.10 and the normalized shell Ward identities of Lemma 16.18. The free divergences of Ξ, D are independent: a kernel combination would be a multiple of L, impossible in these two tensor coordinates (Lemma 16.1). Thus $f_i : \mathbb{R}^2 \to V_5$, $f_i(c) = c_0 K_i B_i^{\Xi} + c_1 K_i B_i^D$, are eventually invertible with bounded inverses by (18.7). Set $c_i = -f_i^{-1}(\beta_i K_i B_i^L)$. Then $|c_i| \le C R_i$ and

$$X_i = \beta_i B_i^L + c_{i,0} B_i^{\Xi} + c_{i,1} B_i^D$$

is conserved. Applying λ_i to its zero divergence gives the exact identity

$$4c_{i,1} = \lambda_i(\beta_i K_i B_i^L) + c_{i,0}\lambda_i(K_i B_i^{\Xi}) + c_{i,1}\big[\lambda_i(K_i B_i^D) + 4\big]\,.$$

The last two terms are $O(R_i)\cdot o(E_i^{-1})$ by (18.7) and (18.3); since $R_i/E_i \to 0$ they vanish, and $c_{i,1} \to -\kappa_G/4 = d_2$, where $d_2 = -b_0/2$ is the trace coefficient of Section 6.4.

Equation (18.8) and $R_i/\beta_i \to 0$ give $r_i^4(\mathsf{U}_{i,4}^{\mathrm{sc}})^{-1} X_i \to L$. Its weighted trace $\beta_i \operatorname{tr}(r_i^4(\mathsf{U}_{i,4}^{\mathrm{sc}})^{-1} X_i)$ tends to $4d_2\mathcal{O}_{F^2}$, since the other traces vanish faster than every inverse power. Lemma 16.10 now gives $X_i = \nu_i T$ with $\nu_i \in \mathbb{R}$: the tensors X_i are centered in the mixture by construction, and Lemma 16.18 gives the centered, nonzero tensor T. By Lemma 16.1, choose $A \in M_4^{\mathrm{Max}}$, a separating spectator functional o, and an axis b with $y = o(\partial_b^{\mathrm{Max}} A) \neq 0$. For these fixed tests the flux of $r_i^4(\mathsf{U}_{i,4}^{\mathrm{sc}})^{-1} X_i$ tends to the flux of L, namely y, by Lemma 18.3. The flux of T, multiplied by r_i, tends to y by the normalized Ward identity of Lemma 16.18 and the jet limit of Lemma 18.3. The first flux is ν_i times the second. Dividing by the latter proves $\nu_i \to 1$. This gives the weighted trace limit for T.

Finally $\Theta = \sum_a T_{aa} - e_{\mathrm{tr}}\mathbf{1}_4$ and $(\mathsf{U}_{i,4}^{\mathrm{sc}})^{-1}\mathbf{1}_4 = \mathbf{1}_4$. The extra term has size $|e_{\mathrm{tr}}|\beta_i r_i^4 \to 0$ by (18.1). □

Proof of Theorem 6.18. The phase-table reflection identity E2 of Definition 2.9 and the reality of T in Lemma 16.18 make e_{tr} real. Lemmas 16.17, 16.18 and 17.11 show that Θ is real and $O(4)$ invariant and that its continued label is Hermitian and Lorentz scalar. Lemma 18.1 gives vacuum centering. For the asymptotic, take any prescribed positive null sequence and the subsequence in (18.1). By Proposition 18.8, the phase comparison, and the free normalization in Lemma 16.1,

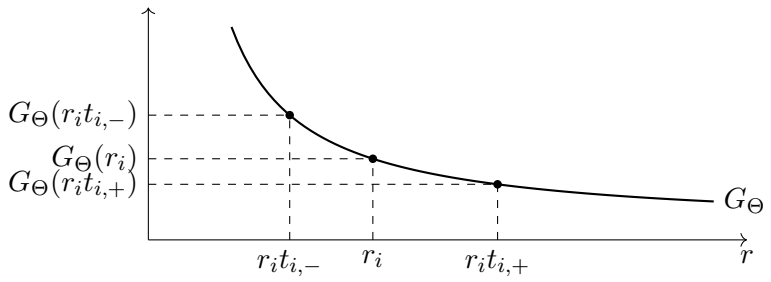


Figure 15: The squeeze. Only G_Θ is monotone (Lemma 18.2). For lattice-rounded separations $t_{i,\pm} \to t_\pm$, $t_- < 1 < t_+$, the outer values, times $r_i^8 \log^2(1/r_i)$, tend to $A_G t_\pm^{-8}$ along the radii of Lemma 18.3; letting $t_\pm \to 1$ gives the limit A_G. Every null sequence has such a subsequence.

every generic pair $x = (x_0, x_1)$ with $|x| < 100$ and $\mathrm{sep}_\infty(x) > 1/100$, with $x^{(i)} \in r_i^{-1} a_{n_i} \mathbb{Z}^4$ nearest to x, satisfies

$$r_i^8 \log^2(1/r_i) G_\Theta(r_i |x_0^{(i)} - x_1^{(i)}|) \longrightarrow \frac{A_G}{|x_0 - x_1|^8} .$$

Translation and $O(4)$ invariance (Lemma 17.3) give $S_2^{\xi *}(\Theta, \Theta; rx_0, rx_1) = G_\Theta(r|x_0 - x_1|)$. Here $\kappa_G^{-2}(4d_2)^2 = 1$, so the amplitude does not depend on κ_G.

Take $x_0 = 0$ and $x_1 = (t/2, t/2, t/2, t/2)$, with $1/2 < t < 2$. This pair is generic and has distance t. Monotonicity from Lemma 18.2 bounds the lower limit at radius r_i by A_G/t^8 for $t > 1$, and the upper limit by that expression for $t < 1$ (Figure 15). Let $t \to 1$. The limit along this subsequence is A_G. The subsequence criterion gives the full-radius limit. Here $A_G = 3d_{\mathfrak{g}}/\pi^4 > 0$ since $d_{\mathfrak{g}} \geq 1$; identifying it uses $d_2 = -\kappa_G/4 \neq 0$. Thus $\Theta \neq 0$. □

The following high-energy law follows from Theorem 6.18 and Lemma 18.2, as proved below.

Lemma 18.9 (High-energy trace limit)**.** *For every $\varphi \in C_c(\mathbb{R}^4; \mathbb{C}) \cup \mathcal{S}$,*

$$\lim_{R_{\mathrm{E}} \to \infty} R_{\mathrm{E}}^{-8} (\log R_{\mathrm{E}})^2 \int \varphi(\mathbf{p}/R_{\mathrm{E}}) \, d\mu_\Theta(\mathbf{p}) = \frac{A_G}{1536\pi} \int_{\overline{V}_+} \mathsf{m}(\mathbf{p})^2 \varphi(\mathbf{p}) \, d^4\mathbf{p} .$$

Proof. The rescaled spectral measures are controlled through their Laplace transforms, which the trace law determines (Definition 6.17, Lemma 18.2). Put $\nu_{R_{\mathrm{E}}} = R_{\mathrm{E}}^{-8} \log^2 R_{\mathrm{E}} \, \mu_\Theta(R_{\mathrm{E}} \cdot)$. For every future timelike Laplace covector u, with the Euclidean pairing $u \cdot \mathbf{p}$ of item W0 of Definition 2.14, which is the Minkowski pairing of $\mathbf{p}$ with the future timelike $(-u^1, -u^2, -u^3, u^4)$, Lorentz invariance reduces its transform to the time-axis transform at $s = \sqrt{\mathsf{m}(u)}$. Hence

$$\int e^{-u \cdot \mathbf{p}} \, d\nu_{R_{\mathrm{E}}} = R_{\mathrm{E}}^{-8} \log^2 R_{\mathrm{E}} \, G_\Theta(s/R_{\mathrm{E}}) \longrightarrow A_G s^{-8} .$$

To identify the constant, write $t = \mathbf{p}^4$ and integrate the spatial radius q:

$$\int_{\overline{V}_+} e^{-st} \mathsf{m}(\mathbf{p})^2 \, d^4\mathbf{p} = 4\pi \int_0^\infty e^{-st} \int_0^t q^2 (t^2 - q^2)^2 \, dq \, dt = 1536\pi s^{-8} .$$

The $s = 1$ transform bounds compact masses, so diagonal extraction and Riesz representation [30, Theorem 2.14] give vague subsequential limits. The $u/2$ transform controls the u-damped tail, so every limit has the displayed transforms. Holomorphic uniqueness in each coordinate for timelike

real part, then Fourier inversion on product Schwartz tests against the finite damped measures [30, Theorems 10.18 and 9.11], identify every limit with the cone density in the statement.

For Schwartz tests positivity gives

$$\nu_{R_{\mathrm{E}}}\{\mathbf{p}^4 \le R\} \le e\, R_{\mathrm{E}}^{-8} \log^2 R_{\mathrm{E}}\, G_\Theta((R_{\mathrm{E}}R)^{-1}) \le CR^8 \qquad (R \ge 1,\ R_{\mathrm{E}} \ge R_{\mathrm{E},0})\,.$$

Use the trace law at $1/(R_{\mathrm{E}}R)$ and $\log R_{\mathrm{E}} \le \log(R_{\mathrm{E}}R)$. A Schwartz bound of order greater than eight sums the dyadic energy tails uniformly, since $|\mathbf{p}| \le \sqrt{2}\mathbf{p}^4$ on the cone. The C_c limit then gives the Schwartz limit. □

For $\psi_{R_{\mathrm{E}}}(x,y) = R_{\mathrm{E}}^8 \psi(R_{\mathrm{E}}x, R_{\mathrm{E}}y)$, $\mathcal{W}_{\mathrm{rel}}\psi_{R_{\mathrm{E}}}(\mathbf{p}) = \mathcal{W}_{\mathrm{rel}}\psi(\mathbf{p}/R_{\mathrm{E}})$. For $\psi \in \mathcal{S}_2$, $\mathcal{W}_{\mathrm{rel}}\psi \in \mathcal{S}$. Substitution in Lemma 18.9 gives the position-space consequence

$$R_{\mathrm{E}}^{-8} \log^2 R_{\mathrm{E}}\, W_2^{\xi*}(\Theta,\Theta;\psi_{R_{\mathrm{E}}}) \longrightarrow \frac{A_G}{1536\pi} \int_{\overline{V}_+} \mathsf{m}(\mathbf{p})^2 \mathcal{W}_{\mathrm{rel}}\psi(\mathbf{p})\, d^4\mathbf{p}\,.$$

18.2 Precise non-free comparisons

A finite Wick profile of Definition 2.18 is, at small distance, a finite sum of terms $r^a \log^k r$ with $a \in \mathbb{Z}$ and $k \ge 0$, plus a bounded remainder; the trace law requires the negative logarithmic power $\log^{-2}(1/r)$. This gives the following exclusion; its proof follows.

Lemma 18.10 (Finite Wick obstruction)**.** *For every finite Wick profile W_{Wick} and every $\delta > 0$, there is $r \in (0,\delta)$ such that*

$$G_\Theta(r) \ne W_{\mathrm{Wick}}(r)\,.$$

Proof. We expand each profile at small r and compare with the positive limit of Theorem 6.18. For one factor, polar coordinates and $\omega_{m_{\mathrm{sh}}} = \sqrt{m_{\mathrm{sh}}^2 + q^2}$ give

$$K_{m_{\mathrm{sh}},k_E,h_{\mathrm{mom}}}(r) = \frac{1}{2}\left(\int_{S^2} h_{\mathrm{mom}}\right) \int_{m_{\mathrm{sh}}}^\infty \omega_{m_{\mathrm{sh}}}^{k_E} (\omega_{m_{\mathrm{sh}}}^2 - m_{\mathrm{sh}}^2)^{(j_h+1)/2} e^{-r\omega_{m_{\mathrm{sh}}}}\, d\omega_{m_{\mathrm{sh}}}\,.$$

For $m_{\mathrm{sh}} = 0$ this is proportional to $r^{-j_h-k_E-2}$. Otherwise split at $2m_{\mathrm{sh}}$; the bounded interval is analytic. On the tail expand $(1 - m_{\mathrm{sh}}^2/\omega_{m_{\mathrm{sh}}}^2)^{(j_h+1)/2}$. Taking enough terms makes the remainder integrable with any prescribed number of energy moments, hence gives a Taylor remainder of the required order. The retained terms are $I_a(r) = \int_{2m_{\mathrm{sh}}}^\infty v^a e^{-rv}\, dv$, $a \in \mathbb{Z}$. Integrate explicitly for $a \ge 0$; $I'_{-1} = -e^{-2m_{\mathrm{sh}}r}/r$ gives the logarithm, and $I'_a = -I_{a+1}$ gives all lower powers. Only nonnegative logarithmic powers occur. For products, choose each remainder order above the other factors' pole orders.

Thus $\mathrm{Re}\, W_{\mathrm{Wick}} = \sum_{a,k} c_{a,k} r^a \log^k r + O(1)$ with finitely many terms. After multiplying by $r^8 \log^2(1/r)$, the terms with $a > -8$ vanish. Among the others, the least a, then largest k, gives a nonzero divergent leading term. A finite limit is therefore zero. Lemma 18.2 makes G_Θ real, so equality with W_{Wick} implies equality with its real part. The positive trace limit contradicts equality on $(0,\delta)$, proving also Proposition 6.19. □

To describe polynomial mass-shell Wick contractions, put $q_m(\mathbf{k}) = (\mathbf{k}, \sqrt{m^2 + |\mathbf{k}|^2})$, $m \ge 0$. Fix a finite set of elementary field indices, with elements x, x'; words u, u' list its elements. For words of length k, π ranges over permutations of $\{1,\dots,k\}$. Comparison data specify finite mass–polynomial lists $\mathcal{M}_{\mathrm{pair}}(\mathsf{x},\mathsf{x}')$, with $P \in \mathbb{C}[\mathbf{p}^1,\dots,\mathbf{p}^4]$, and complex contraction coefficients $c(u,u',\pi)$.

Definition 18.11 (Polynomial mass-shell pairings)**.** Set

$$v_{m,P}(\mathbf{k}) = \frac{P(q_m(\mathbf{k}))}{2\omega_m(\mathbf{k})}, \qquad L_{m,P}(\varphi) = \int v_{m,P}(\mathbf{k})\varphi(q_m(\mathbf{k}))\, d^3\mathbf{k}\,.$$

For words $u = (\mathsf{x}_i)_{i=1}^k$, $u' = (\mathsf{x}'_i)_{i=1}^k$, define the convolutions of the functionals $L_{m,P}$ by

$$\mathsf{P}(u,u';\varphi) = \sum_\pi c(u,u',\pi) \sum_{(m_i,P_i)\in\prod_i \mathcal{M}_{\mathrm{pair}}(\mathsf{x}_i,\mathsf{x}'_{\pi(i)})} \int \prod_i v_{m_i,P_i}(\mathbf{k}_i)\, \varphi\left(\sum_i q_{m_i}(\mathbf{k}_i)\right) \prod_i d^3\mathbf{k}_i\,.$$

For unequal lengths put $\mathsf{P} = 0$. A zero denominator at zero momentum gives value zero.

For finite polynomial fields $Q = \sum_i \alpha_i u_i$, including empty words, the bilinear pairing is $W^{\mathrm{pol}}(Q,Q';\varphi) = \sum_{i,j} \alpha_i\alpha'_j \mathsf{P}(u_i,u'_j;\varphi)$. Empty contractions evaluate φ at zero. The shell measure has no $(2\pi)^{-3}$ factor.

The next lemma separates each comparison by one product test; its proof follows.

Lemma 18.12 (Polynomial countertest)**.** *For every comparison data and every Q, Q', there are $f, h \in \mathcal{S}$ with compactly supported $\mathcal{W}_{\mathrm{rel}}(\bar f \otimes h)$ such that*

$$\langle \Phi_\Theta(f)\Omega, \Phi_\Theta(h)\Omega\rangle \neq W^{\mathrm{pol}}(Q,Q';\mathcal{W}_{\mathrm{rel}}(\bar f \otimes h))\,.$$

Proof. Equality for all product tests would identify the two spectral representations, hence G_Θ with a finite Wick profile, which Lemma 18.10 excludes. In detail, equality for all product tests implies equality for every $\varphi \in C_c^\infty$: choose $\widehat{h}_+ = \varphi$ and real $\widehat{f}_+ = 1$ on its support. Approximate $e^{-r\mathbf{p}^4}$ by compact cutoffs. Dominated convergence applies: Lemma 18.2 controls μ_Θ; exponential damping controls comparison polynomials at infinity, and $d^3\mathbf{k}/|\mathbf{k}|$ is integrable at zero. Expanding the numerators into monomials factors the integrals into the kernels of Definition 2.18, with the empty words giving constants. Hence $G_\Theta(r) = W_{\mathrm{Wick}}(r)$ for all $r > 0$, contrary to Lemma 18.10. □

For the photon comparison use Hilbert tensor products.

Definition 18.13 (Ambient and physical Maxwell Fock spaces)**.** For $n \in \mathbb{N}$ colours put

$$d\sigma(\mathbf{k}) = \frac{d^3\mathbf{k}}{2|\mathbf{k}|}, \qquad \Pi_{ab}(\mathbf{k}) = \delta_{ab} - \frac{\mathbf{k}_a\mathbf{k}_b}{|\mathbf{k}|^2},$$
$$\mathcal{K}_n = L^2(d\sigma;\mathbb{C}^n\otimes\mathbb{C}^3), \qquad \mathcal{F}_n = \bigoplus_{k\geq 0} \mathcal{K}_n^{\otimes k}, \qquad \mathcal{F}_n^{\mathrm{phys}} = \bigoplus_{k\geq 0} \mathrm{Sym}^k(\Pi\mathcal{K}_n)\,.$$

The density vanishes at zero momentum, so the value of Π there is immaterial, and Π acts pointwise. Thus a physical vector is symmetric and transverse in each slot. Translations multiply sector k by $\exp\{iz\cdot\sum_j(\mathbf{k}_j,|\mathbf{k}_j|)\}$. The sector-zero unit is Ω_0.

Definition 18.14 (Coherent comparison vectors)**.** For Schwartz tests $h = (h_{\alpha,a})$, put

$$v_h(\alpha,a;\mathbf{k}) = |\mathbf{k}| \sum_b \Pi_{ab}(\mathbf{k})\widehat{h}_{\alpha,b,+}(\mathbf{k},|\mathbf{k}|), \qquad \mathcal{E}(h) = e^{-\|v_h\|^2/2} \bigoplus_{k\geq 0} \frac{v_h^{\otimes k}}{\sqrt{k!}}\,.$$

A comparison $U : \mathcal{H} \to \mathcal{F}_n$ is a complex linear isometry with image in $\mathcal{F}_n^{\mathrm{phys}}$, $U\Omega = \Omega_0$, intertwining translations. Set $\psi_U(f) = U\Phi_\Theta(f)\Omega$.

Real tests have real components. Causal separation means $\mathsf{m}(x-y)<0$ for every pair of points in their closed supports, where $\mathsf{m}(x)=(x^4)^2-\sum_{a=1}^{3}(x^a)^2$.

Lemma 18.12 compares two-point functions only. The Fock test asks more: an embedding of the interacting states into photon Fock space under which the trace field is local relative to the free Maxwell field. Relative locality would impose the equality excluded below. Formally, let $W(h)$ be Weyl operators of the free Maxwell field with $W(h)\Omega_0=\mathcal{E}(h)$ and $W(h)^*=W(-h)$. If $U\Phi_\Theta(f)U^*$ commuted with $W(h)$ for h causally separated from f, then, since $U^*\Omega_0=\Omega$ and $\iota_4\Theta$ is Hermitian,

$$\begin{aligned}\langle\mathcal{E}(-h),\psi_U(f)\rangle &= \langle\Omega_0,W(h)U\Phi_\Theta(f)U^*\Omega_0\rangle\\ &= \langle\Omega_0,U\Phi_\Theta(f)U^*W(h)\Omega_0\rangle=\langle\psi_U(\bar f),\mathcal{E}(h)\rangle\,.\end{aligned}$$

Weyl operators enter only this motivation. The lemma compares the two matrix elements directly, and its proof uses spectral growth without restricting particle amplitudes.

Lemma 18.15 (Fock countertest)**.** *For every $n\ge 0$ and every comparison U, for every ball about zero there are Schwartz f compactly supported in that ball and real compactly supported Schwartz $h_{\alpha,a}$, each causally separated from f, such that*

$$\langle\mathcal{E}(-h),\psi_U(f)\rangle\neq\langle\psi_U(\bar f),\mathcal{E}(h)\rangle\,.$$

Proof. The proof combines the spectral representation (Lemma 18.2), the high-energy law (Lemma 18.9) and translation covariance (Lemma 17.5) with the dilation behaviour of local photon amplitudes. Fix a ball and suppose equality always holds.

Sector amplitudes. Set $p_j=(\mathbf{k}_j,|\mathbf{k}_j|)$ and $p_{\mathrm{tot}}=\sum_j p_j$. The convolution argument of Lemma 18.2 gives sector amplitudes $Y_{k,s}$, with s listing colour and spatial indices:

$$\psi_U(f)_{k,s}=\widehat{f}_+(p_{\mathrm{tot}})Y_{k,s},\qquad \mu_\Theta=(p_{\mathrm{tot}})_*\sum_{k,s}|Y_{k,s}|^2d\sigma^{\otimes k}\,. \tag{18.11}$$

Box masses are finite. Cauchy–Schwarz and polynomial spectral growth give integrability against sufficiently high inverse polynomials in each momentum.

The common sector unitary i^k preserves the equality, physical vectors and translations. Write $\mathsf{Y}=i^kY_{k,s}$.

Locality and doubling. Expand the coherent equality in a real parameter multiplying h, justified by Cauchy–Schwarz and the exponential series. Polarization and complexification give independent complex slot tests. Transversality removes the projector in contractions with physical waves. Shrink nonnegative unit-integral f to zero. Then $|\widehat{f}_+|\le 1$ and $\widehat{f}_+\to 1$; dominated convergence gives, for g_j supported spacelike to zero,

$$\int\mathsf{Y}(\boldsymbol{k})\prod_j\left[|\mathbf{k}_j|\widehat{g}_{j,-}(p_j)\right]d\sigma^{\otimes k}=\int\overline{\mathsf{Y}(\boldsymbol{k})}\prod_j\left[|\mathbf{k}_j|\widehat{g}_{j,+}(p_j)\right]d\sigma^{\otimes k}\,. \tag{18.12}$$

Here $\widehat{g}_-(p)=\widehat{g}_+(-p)$; the sector factors i^k give exactly these signs.

Fix other slots and use $g(x,t)=a(x)\eta_\epsilon^{(b)}(t)$, with η_ϵ approximating unit mass and $b=0,1$. If $0\notin\operatorname{supp}a$, these are spacelike for small ϵ. Thus both Cauchy data of (18.12) are supported at zero. Weighted integrability bounds their order by J, independently of the other tests.

Each datum depends only on the J-jet: multiplication by a cutoff in $B(0,\epsilon)$ leaves the distribution unchanged. For a zero-J-jet test, Taylor's formula bounds derivatives through order J by $O(\epsilon)$; let $\epsilon\downarrow 0$.

The next paragraph proves a uniqueness implication: a combination of sector amplitudes whose two Cauchy data vanish is zero almost everywhere. It is applied below to a polynomial in the doubling operator applied to Y, not to Y itself. The massless wave equation separates the frequency branches from these two Cauchy data away from zero momentum. After smearing other slots, each branch is integrable, bounded and holomorphic in its timelike tube, with continuous real boundary. For such a combination with zero Cauchy data, vanishing persists under small translations of the other spacelike tests. Schwarz reflection and holomorphic uniqueness [30, Theorems 11.14 and 10.18] make the zero set invariant under adding $\mathbb{R}u$ for every future timelike u. These vectors span $\mathbb{R}^4$, giving vanishing at every real translation. Successive Fourier inversion [30, Theorem 9.11] of the smeared integrable amplitudes gives zero almost everywhere. Countably many shrinking bumps remove the other tests: their transforms are eventually nonzero at each momentum.

Let $D_j\mathsf{Y}(\boldsymbol{k}) = \mathsf{Y}(\mathbf{k}_1, \dots, 2\mathbf{k}_j, \dots, \mathbf{k}_k)$. In the time-order-$b$ Cauchy datum ($b = 0, 1$), this substitution gives 2^{-3-b} times the datum at $a(x) \mapsto 8a(2x)$, from $d\sigma(2\mathbf{k}) = 4d\sigma(\mathbf{k})$ and $|\mathbf{k}|^{1+b}$. Spatial jets of order l scale by 2^{3+l}, giving eigenvalues 2^{l-b}. Choose a polynomial that annihilates this finite-dimensional dilation action on both Cauchy jets. The uniqueness implication then makes the same polynomial in D_j annihilate Y: for some J',

$$\prod_{a=0}^{J'-1} (D_j - 2^{a-1})\mathsf{Y} = 0 \qquad \text{in every slot } j\,. \tag{18.13}$$

Growth. For $R = 2^l$ let $M_{\mathsf{Y}}(R) = \int_{|\mathbf{k}_j| \le R} |\mathsf{Y}|^2 d\sigma^{\otimes k}$. By (18.11) and the trace law,

$$M_{\mathsf{Y}}(2^l) \le \mu_\Theta\{\mathbf{p}^4 \le (k+1)2^l\}, \qquad 2^{-8l} M_{\mathsf{Y}}(2^l) \longrightarrow 0\,. \tag{18.14}$$

For the limit, bound the energy mass by $e\, G_\Theta(((k+1)2^l)^{-1})$.

Lagrange interpolation in the commuting D_j splits Y into joint eigencomponents $D_j\mathsf{Y}_{\boldsymbol{e}} = 2^{e_j - 1}\mathsf{Y}_{\boldsymbol{e}}$. Each is a finite combination of doubled copies, so retains (18.14). Moreover $M_{\mathsf{Y}_{\boldsymbol{e}}}(2^l) = 2^{2l\sum_j e_j} M_{\mathsf{Y}_{\boldsymbol{e}}}(1)$, since $d\sigma(2\mathbf{k}) = 4d\sigma(\mathbf{k})$. If $e_j = 0$, contraction preserves box mass, contradicting local square integrability and absence of an atom at zero unless the component vanishes. Equation (18.14) excludes $\sum_j e_j \ge 4$. Thus $e_j \ge 1$, $\sum_j e_j \le 3$, and $k \le 3$, with box mass $\le CR^6$ for dyadic $R \ge 1$. The vacuum sector vanishes by Lemma 18.1 and $U\Omega = \Omega_0$.

Summing (18.11) gives $\mu_\Theta\{\mathbf{p}^4 \le R\} \le CR^6$ at dyadic $R \ge 1$. Multiplication by $R^{-8}\log^2 R$ gives zero in the limit, contradicting Lemma 18.9 on a nonnegative compact test in $\{0 < \mathbf{p}^4 < 1,\ \mathsf{m}(\mathbf{p}) > 0\}$. □

An averaged form of this countertest, in which the coherent vectors are moved by maps that preserve physical inner products and fix the image of U, is Lemma 8.10 in Section 8.

19 Products and contact subtractions

This section proves Theorems 6.20 and 6.21. The coefficients of the product expansion are read off by probes. A finite family of probe correlations distinguishes the fields of a fixed degree: two fields of F_j with the same correlations against every probe are equal. Applied to a cluster of insertions, the probes return finitely many numbers, and a fixed linear reconstruction map, a decoder, returns a field of F_j; on the probe values of a single field it returns that field. Lower degrees are removed before the next degree is read. The probes are chosen once, before the spectators and the phase, and the coefficients are computed from the mixed table; hence they depend on the cluster alone.

The remainder is bounded in the commutative compression algebra $\mathfrak{A}_0$ of Lemma 17.1 before a character is evaluated, and a norm bound in $\mathfrak{A}_0$ bounds every character and the mixed vacuum. The coefficients do depend on the prescription, that is, on the table and the probes chosen here; Theorem 6.20 concerns this one coefficient system.

For two fields A, B of degrees d_A, d_B at x and 0, with center 0 and $|x| \asymp r$, the expansion tested against spectators reads

$$S^{\zeta}\big(A(x), B(0), \boldsymbol{B}(y)\big) = \sum_{j \le D} \sum_{\lambda \in \mathrm{Ch}_j} c_{j,\lambda}(A, B; (x,0), 0)\, S^{\zeta}\big(O_{j,\lambda}(0), \boldsymbol{B}(y)\big) + \mathcal{R}_D^{\zeta},$$

with $|c_{j,\lambda}| \le r^{j-d_A-d_B-\varepsilon}$ and $|\mathcal{R}_D^{\zeta}| \le r^{D+1-d_A-d_B-\varepsilon}$ for small r; $O_{0,\lambda}$ is a nonzero multiple of the unit. For $D+1 < d_A + d_B$ the remainder bound still permits growth as $r \to 0$; what increases with D is its order, and once $D \ge d_A + d_B$ (with $\varepsilon < 1$) the remainder tends to zero. Nothing is asserted about a series in D. The expansion goes back to Wilson [35]; Fröb and Holland prove its existence for Euclidean Yang–Mills theory to all orders in perturbation theory [13]. The expansion here is nonperturbative, at each finite truncation, with one coefficient system for every phase.

The lattice estimate starts from the signed local and graded quotient bounds of Lemmas 19.1 and 19.2. Each insertion carries a nilpotent external field with a weight. Blocking loses mark weight when insertion points approach one another; grouping nearby markers bounds this loss by a power of the ratio of insertion and fusion scales (Figure 16). Taylor subtraction and transport of the coefficient back to a coarser level produce one field and a small remainder, whose chessboard seminorm reflection positivity converts into a norm bound in $\mathfrak{A}_0$. Derivative induction removes the restriction to grid points; tree induction allows several clusters to shrink at independent rates. The nested and integrated forms (Lemmas 19.13–19.14 and Remark 19.15) are not used in the proofs of Theorems A and B. The contact extension of Section 19.2 is defined from the continuum correlations at distinct points, not as a lattice limit at coincident points; the lattice bounds of Lemma 19.9 make its shell sums converge.

Throughout, the lattice insertions and quotient fields are those of Definitions 2.8 and 2.12; inclusions and derivatives are those of Lemma 15.5. The rotations $R_j(h)$ are those of Lemma 16.17. Write $|\boldsymbol{A}|_{\deg} = \sum_i \deg A_i$ and $\zeta \in \mathsf{X} \sqcup \{\mathrm{mix}\}$, as in Definition 2.19.

Inputs from the marked expansion and the quotient. We first establish two estimates needed when the field to be extracted varies with the cutoff. Throughout this subsection, a local marked expression has the form of Definition 14.5 and the regularity, zero radial jets and forced supports of Lemma 14.9. Write

$$\mu(x) = \sum_{\alpha} |x_\alpha|, \qquad x^{\circ} = x - x_0 \mathbf{1}, \qquad \mathfrak{r}_i = \mathcal{E}_i + (\epsilon_i^{\mathrm{for}})^{-1/2} \mathcal{K}_i\,.$$

The last quantity is the weighted remainder mass of Lemma 14.7. Fix an output degree t. The diagonal rescaling $(\mathsf{d}_i x)_\alpha = \ell_i^{t-\deg\alpha} x_\alpha$ accounts for the engineering degrees inside V_t. Write $C_i = C_{J_i}^{\mathrm{col}}$ for the column bound at level i. A normalized column at level i is the local expression $\mathfrak{c}_i(b)$ of Lemma 14.7 with coefficient vector b; its coefficient map is C_i, and its remainder bound is $C_i \mu((\mathsf{C}_i b)^{\circ})$ by Lemma 14.7.

Let $K_j \ge 1$ be the nondecreasing stage bound fixed before item 1 of Section 14.1. The degree rescaling above is the map S_L in the proof of Lemma 14.6. For an input coefficient v and remainder mass q, its averaged marked estimate gives an output coefficient u and remainder mass r with

$$\begin{aligned} \mu(u - M_i \mathsf{d}_i v) &\le \epsilon_i [\mu(v^{\circ}) + q], \\ r &\le K_{J_{i+1}} \mu((\mathsf{d}_i v)^{\circ}) + \epsilon_i q, \qquad \epsilon_i = \ell_i^{-3/8}. \end{aligned} \tag{19.1}$$

Here M_i is the leading polynomial map, fixes the unit, preserves the nonconstant subspace, and satisfies $\|M_i^{\pm 1}\| \le K_{J_{i+1}}$. These estimates follow from (14.17) and the symmetry averaging in that proof. In the threshold order of Section 9, the construction chooses

$$64K_{J_l+1}^2 \le C_l, \quad 10^5 K_{J_{l+1}+1}^6 (1+C_{l+1})^2 \le \ell_l^{1/8}, \quad 1+2C_l \le \ell_l^{1/40}. \tag{19.2}$$

The column choice imposes the first inequality; the ratio choice imposes the other two. Stage growth gives $J_{l+1} \le J_l + 1$.

The next estimate retains the signed difference produced by subtracting a column. A transported family on $[k_*, m]$ is a sequence of local marked expressions $\mathfrak{c}_l(x_l) + R_l$, $k_* \le l \le m$, in which R_l has coefficient zero and each term arises from the next finer one by one marked integration step of Lemma 14.6; x_l are its coefficients and r_l its remainder bounds. Its free affine parameter is a vector added to the column coefficient at the finest level m. Write q_{fin} for the combined coefficient and remainder bound of the original expression at m, before this affine subtraction. In the next lemma, the interval $[k_*, m]$ is sufficiently late.

Lemma 19.1 (Signed local transport)**.** *(a) Every normalized column plus a zero-coefficient remainder at level m admits a transported family on $[k_*, m]$. Its densities satisfy $H_l = \ell_l^t \mathcal{B}^{(l)}_{\beta_{N,p,l+1}} H_{l+1}$ $(k_* \le l < m)$, each step satisfies (19.1), $r_m \le (1+C_m) q_{\mathrm{fin}} + C_m \mu(x_m)$, and the free affine parameter can be chosen so that $x_{k_*} = 0$.*

(b) If $x_{k_*} = 0$, then, for $k_* \le l \le m$,

$$r_l + \mu(x_l) \le 4(1+C_m) q_{\mathrm{fin}} (a_m/a_l)^{1/8}, \qquad r_m \le 2(1+C_m) q_{\mathrm{fin}}\,.$$

Thus a residual that starts with zero coefficient at the coarse end loses only the factor $4(1+C_m)$ at the fine end, and gains $(a_m/a_l)^{1/8}$ at every coarser level.

Proof. The termwise remainder estimates and the Lipschitz bound in the proof of Lemma 14.6 control one step; the exact column identity of Lemma 14.7 keeps the densities exact; an induction along the interval turns the signed one-step inequalities into the discount.

Let R have coefficient zero and remainder bound q_0. Its regular terms have zero radial jets through t, by the local format of Lemma 14.9; its forced terms retain their eligibility events. In the marked integration of Lemma 14.6, the former gain an inverse block factor before the connected sums, and the latter keep half the forced suppression. The stage constants precede the ratio and its entry threshold, so both resulting bounds $C^{\mathrm{loc}}_{J_{i+1}}/\ell_i + o_{\widehat{\beta}_i}(1)$ are at most $\ell_i^{-2/5}$. Sum the regular masses g_a and forced masses $k_b/(\epsilon^{\mathrm{for}}_{i+1})^{1/2}$ before taking the output norm. Their sum is at most q_0. The termwise bounds in that proof, or its Lipschitz estimate for the difference between a column plus R and the same column, therefore bound the output coefficient and remainder by $\ell_i^{-2/5} q_0$. The rooted clause of Lemma 13.5, in the marked norm defined below, justifies termwise integration and retains the exact density identity. The last inequality of (19.2) gives $(1+2C_i)\ell_i^{-2/5} \le \epsilon_i$. If Z denotes the resulting local expression and z its coefficient, then

$$H(Z) = \ell_i^t \mathcal{B}^{(i)}_{\beta_{N,p,i+1}} H(R), \qquad \mu(z) \le \ell_i^{-2/5} q_0, \qquad \mathfrak{r}_i(Z) \le \ell_i^{-2/5} q_0\,.$$

Write the fine affine family as $\mathfrak{c}_{i+1}(a+b) + R$, with b the free parameter. Set $c = \mathsf{C}_i^{-1} z$ and replace it at the coarse level by

$$\mathfrak{c}_i(a+c+b) + R', \qquad R' = Z - \mathfrak{c}_i(c)\,.$$

The coefficient of R' is zero, and $\mathfrak{r}_i(R') \le (1+C_i)\ell_i^{-2/5}q_0$. Linearity of the column density gives $H(\mathfrak{c}_i(a+c+b)) + H(R') = H(\mathfrak{c}_i(a+b)) + H(Z)$. Together with the exact column identity of Lemma 14.7, this proves the asserted density identity for every b, including all forced terms. The supports remain fixed in b by Lemma 14.7.

For the column alone use (19.1), with fine remainder bound q_{col}. Adding Z changes the coefficient by at most $\ell_i^{-2/5}q_0$. For the remainder, recentering costs at most $C_i\ell_i^{-2/5}q_0$, and shifting the column parameter costs at most another $C_i\ell_i^{-2/5}q_0$. The total extra cost is $(1+2C_i)\ell_i^{-2/5}q_0 \le \epsilon_i q_0$. With $q = q_{\mathrm{col}} + q_0$, these are the asserted bounds. Repeating this construction from the fine end to the chosen coarse end produces the transported family on the whole interval.

For later use put $U = \mu(u)$, $Y = \mu(\mathsf{d}_i v)$, $K = K_{J_{i+1}}$ and retain r, q for the remainder bounds. Applying M_i^{-1} and using $\mu(v^\circ) \le Y$ gives

$$Y \le K[U + \epsilon_i(Y+q)], \qquad r \le KY + \epsilon_i q\,. \tag{19.3}$$

For an interval $[k_*, m]$, choose the free affine parameter to cancel the coefficient at k_*; this uses the invertibility of C_{k_*}. The signed residual then starts with coefficient zero. In step l put $K = K_{J_{l+1}}$ and $A_l = 32(1+C_l)^2$. Induct from k_* toward m on the cone condition $A_l\mu(x_l) \le r_l$, which says that the coefficient is small compared with the remainder bound; it holds at k_* because $x_{k_*} = 0$. Equation (19.2) implies

$$A_l \ge 4K^2, \quad 4K\epsilon_l \le 1, \quad 4KA_{l+1}\epsilon_l \le 1, \quad (4K^2+1)\ell_l^{1/8}\epsilon_l \le 1.$$

In (19.3) at step l, $U = \mu(x_l)$, $r = r_l$ and $q = r_{l+1}$. From $A_lU \le r \le KY + \epsilon_l q$ and $A_l \ge 4K^2$ obtain $4KU \le Y + \epsilon_l q$. Substitution into (19.3) gives $Y \le 4K\epsilon_l q$. Thus $A_{l+1}\mu(x_{l+1}) \le q$, since $\mu(x_{l+1}) \le Y$; this is the cone condition at $l+1$. Also $\ell_l^{1/8}r \le q$. Iteration gives the discount $(a_m/a_{k_*})^{1/8}$. At the finest level the affine subtraction has $r_m \le (1+C_m)q_{\mathrm{fin}} + C_m\mu(x_m)$. The cone condition at m and $A_m \ge 2C_m$ give $2C_m\mu(x_m) \le r_m$, hence $r_m \le 2(1+C_m)q_{\mathrm{fin}}$ and $r_m + \mu(x_m) \le 4(1+C_m)q_{\mathrm{fin}}$. Applying the iterated discount between any l and m proves the interval bound. □

The next estimate recovers the output-degree power after passing to the quotient. Fix any norm on F_t. Put $\|v\|_m = \mu(x_m(v))$, where $x_m(v) = x_{N,p,m;t}(v)$ is the transferred coefficient of Definition 2.8. A refinement below is a nontrivial refinement of the ordered cutoff limits (14.28); it may depend on the cluster geometry.

Lemma 19.2 (Graded quotient bound)**.** *For $0 \le i \le t$ there are positive $C_{t,i}(m)$, independent of the refinement and label family, with $\log C_{t,i}(m)/s_m \to 0$. If labels $v_{N,p} \in V_t$ converge in V_t to v on such a refinement and eventually $\|v_{N,p}\|_m \le \epsilon_{\mathrm{bd}}$, then*

$$\operatorname{dist}_{F_t}([v], \jmath_{i-1,t}F_{i-1}) \le C_{t,i}(m)a_m^{-(t-i)}\epsilon_{\mathrm{bd}}\,. \tag{19.4}$$

Here $[v] \in F_t$ is the class of v, $F_{-1} = 0$, and m is sufficiently large.

A label that is small at level m is therefore, in the quotient and up to subpower factors, within $a_m^{-(t-i)}$ times its size of the fields of degree below i.

Proof. The triangularity of the leading map proved in Lemma 14.6 separates high from low engineering degrees; the marked bounds of Lemma 14.7 control the error; the physical quotient of

Definition 2.12, with the corrected inclusions of Lemma 15.5, identifies the removed low-degree part with a lower field. In the notation of (19.1), the triangularity is

$$(M_l)_{\alpha\beta} = 0 \quad (\alpha \neq \beta,\ \deg\alpha \leq \deg\beta).$$

The diagonal entries are the nonzero fine-to-coarse normalization factors. The full coefficient map differs from this leading map by the marked error. Write P for coordinate projection onto engineering degrees at most $i-1$, first assuming $i > 0$. The leading map M_l and its inverse preserve $\ker P$: triangularity gives preservation by M_l, and injectivity on this finite-dimensional subspace gives preservation by its inverse. Consequently $\mu(Px) \leq K_{J_{l+1}}\mu(PM_l x)$. The degree rescaling expands Px by at least ℓ_l^{t-i+1}, whereas it expands $x - Px$ by at most ℓ_l^{t-i}. The marked error is bounded by $\epsilon_l\mu(x^\circ)$.

The two resulting estimates are

$$\mu(x_{l+1}) \leq C_{l+1}\mu(Px_{l+1}) \implies \begin{cases} \mu(x_l) \leq C_l\mu(Px_l), \\ \ell_l^{t-i+1}\mu(Px_{l+1}) \leq C_{l+1}\mu(Px_l), \end{cases}$$
$$Px_{l+1} = 0 \implies \mu(x_l) \leq C_{l+1}\ell_l^{t-i}\mu(x_{l+1})\,.$$

To obtain the first line, put $Y = \mu(P\mathsf{d}_l x_{l+1})$. The low-cone hypothesis $\mu(x_{l+1}) \leq C_{l+1}\mu(Px_{l+1})$ and $C_{l+1} \leq \ell_l$ give $\mu(\mathsf{d}_l x_{l+1}) \leq 2Y$. Applying the inverse leading map to the low coordinates and using $2K_{J_{l+1}}\epsilon_l(1 + C_{l+1}) \leq 1$ gives $Y \leq 2K_{J_{l+1}}\mu(Px_l)$. The full coefficient is at most $3K_{J_{l+1}}Y$, so $64K_{J_{l+1}}^2 \leq C_l, C_{l+1}$ gives both conclusions. In the second line use $\mu(\mathsf{d}_l x_{l+1}) \leq \ell_l^{t-i}\mu(x_{l+1})$ and $K_{J_{l+1}} + 1 \leq C_{l+1}$. These numerical inequalities follow from (19.2).

At the bare level take the plane PV_t. Its transported image remains in the low cone by the first line. The map from this plane to its low coordinates at any level is injective by its expansion bound, hence onto PV_t. Subtracting a transported vector with the same low coordinates at level m therefore leaves zero low coordinates and increases the full size by at most $1 + C_m$. Repeat the subtraction at every next coarser level, applying the second line between subtractions. This gives a vector $y_{N,p}$ in the transported bare low plane with

$$\|v_{N,p} - y_{N,p}\| \leq G_m(1 + C_m)(a_{m_{\mathrm{ref}}(t)}/a_m)^{t-i}\epsilon_{\mathrm{bd}}, \qquad G_m = \prod_{l<m}(1 + C_l)C_{l+1}\,.$$

Allowing a fixed norm-equivalence factor, put $\mathcal{J}_m = \prod_{l<m} C_{l+1}$. The first-line expansion, iterated from the bare level, shows that bounded vectors $y_{N,p}$ in this plane have bare coefficients bounded by $a_N^{t-i+1}e^{o(s_N)}$. Multiplying the same polynomial by a_N^{i-1} instead of a_N^t therefore gives a degree-$(i-1)$ local observable with subpower healthy derivative and all-configuration supremum bounds. Its stencil is unchanged. To match this observable to the selected lower-degree insertions, apply Lemma 14.17(i): it gives tame degree-$(i-1)$ coefficients $c_{N,p}$ whose insertions differ from these observables by a negligible error (Definition 14.16). Lemma 15.3 removes that error in every separated spectator test.

To pass to a physical lower field, apply Lemma 15.4 in V_{i-1} to the family $c_{N,p}$, with the equation projections $\Pi_{N,p} = \Pi_{i-1,N,p}$ of Lemma 14.18, taken along $N = \mathsf{n}(j)$ and $p = P(k)$, their limit Π_{i-1} from (15.8), and the one-slot spectator functionals, whose kernels intersect in $\ker\Pi_{i-1}$ by (15.11). The normalized equation remainder $(c_{N,p} - \Pi_{N,p}c_{N,p})/(1 + \|\Pi_{N,p}c_{N,p}\|)$ is tame, being a bounded multiple of $c_{N,p}$ minus a family of norm at most one; Lemmas 14.19 and 15.3 make its separated spectator tests tend to zero. If y is a limit of the bounded labels $y_{N,p}$, the lemma gives a label in V_{i-1} with exactly the limiting correlations of y. The corrected inclusions of Lemma 15.5 and

the all-spectator definition of the null space now give $[y] \in \jmath_{i-1,t} F_{i-1}$. Boundedness of $y_{N,p}$ follows from the displayed distance bound and convergence of $v_{N,p}$. If these bounds hold only on the chosen refinement, set $y_{N,p} = 0$ off the set where they hold; this extends the bounded plane family to the original tails without changing its limit. Passing to this limit proves the distance estimate.

For $i = 0$ no plane is removed. Iterating the reverse coefficient estimate of Lemma 14.7 gives $\|v\| \le \mathcal{J}_m (a_{m_{\mathrm{ref}}(t)}/a_m)^t \epsilon_{\mathrm{bd}}$ and hence the same bound in F_t. Finally $\log \mathcal{J}_m = o(s_m)$, since $\log C^{\mathrm{col}}_{J_{m+1}} \le J_{m+1}^{-2} \log \ell_m$ (Section 9) and by Lemma 14.2. Since $C_l \ge 2$, $G_m C_m \le (\mathcal{J}_m)^3 C_0$. Thus $\log G_m = o(s_m)$ as well. Fixed reference-spacing and quotient-norm factors are absorbed in $C_{t,i}(m)$. No estimate depends on the refinement used to take the limits. □

19.1 Products and nested clusters

A probe word is a fixed finite list of fields at distinct nonzero points. Correlating a field at the origin against such a word gives a linear functional on F_j. The intersection of the kernels of all these functionals is zero, by the definition of $\mathcal{N}_j$. Finite dimensionality therefore permits a finite separating collection.

Choose the probe functionals recursively. Given biorthogonal fields u_α and probe functionals ℓ_α below degree j, continue each ℓ_α to F_j by the same correlation formula; inclusions preserve that formula by Lemma 15.5. The retraction

$$\mathtt{t}_j A = \sum_{\deg \alpha < j} \ell_\alpha(A) \jmath_{\deg \alpha, j-1} u_\alpha, \qquad \mathtt{t}_j \jmath_{j-1,j} = 1$$

allows a section s of the new quotient to be replaced by $s - \jmath \mathtt{t}_j s$. Lift a quotient basis with this section and realize its dual by finite combinations of probe functionals, which span F_j^*. The enlarged system is biorthogonal. In particular ℓ_α vanishes on $\jmath F_{\deg \alpha - 1}$. For $j = 0$ take $u_0 = \mathbf{1}$. Let $\mathcal{P}_j$ be the union of the finite probe sets used through degree j, with index γ. Write $\ell_\alpha(A) = \sum_\gamma m_{\alpha\gamma} S(A(0), B_{j,\gamma})$ in this finite collection.

Definition 19.3 (Finite correlation projection)**.** Define the linear reconstruction map $D_j : \mathbb{C}^{\mathcal{P}_j} \to F_j$ by

$$D_j(y) = \sum_{\deg \alpha \le j} \left(\sum_{\gamma \in \mathcal{P}_j} m_{\alpha\gamma} y_\gamma \right) \jmath_{\deg \alpha, j} u_\alpha \,.$$

With $\mathrm{ev}_j(A) = (S(A(0), B_{j,\gamma}))_{\gamma \in \mathcal{P}_j}$, it satisfies $D_j \,\mathrm{ev}_j = \mathrm{id}_{F_j}$: it is a fixed left inverse of field evaluation. For a word w separated from these probes, set

$$P_j^{\mathrm{raw}} w = D_j\big((S(w, B_{j,\gamma}))_{\gamma \in \mathcal{P}_j}\big) \,.$$

This extracts a field from finitely many correlations using the fixed biorthogonal coordinates.

Here and below a word is a finite product of labelled insertions; sums of words are read linearly. Rotations act on every position and label, and conjugation acts on labels and scalar coefficients. Translation of a word by u is denoted by τ_u.

Definition 19.4 (Symmetric projection)**.** When every rotation of w is separated from every probe in $\mathcal{P}_j$, set

$$A_j w = \int_{O(4)} R_j(h) P_j^{\mathrm{raw}}(h^{-1} w)\, dh, \qquad B_j w = \tfrac{1}{2}\big(A_j w + \overline{A_j \bar{w}}\big) \,.$$

The Haar measure has mass one. Outside this domain set both maps to zero. At center o, apply these formulas to $\tau_{-o} w$. These averages impose rotation covariance and reality.

Let $I_q A = A(0)$ for $A \in F_q$. The inclusions $\jmath_{i,j}$ identify the same field in different degrees.

Definition 19.5 (OPE coefficients)**.** Put $L_{i,q} = B_i I_q$ and define

$$Q_0 = 1, \qquad Q_j = 1 - \sum_{i<j} \jmath_{i,j} Q_i L_{i,j}, \qquad E_j = \bigcap_{i<j} \ker(Q_i L_{i,j})\,.$$

For the chosen real basis $(O_{j,\lambda})_{\lambda \in \mathrm{Ch}_j}$ of E_j, let $\omega_{j,\lambda} : E_j \to \mathbb{C}$ be its coordinate functionals and set

$$c_{j,\lambda}(\boldsymbol{A}; x, o) = \omega_{j,\lambda}\left(Q_j B_j \left[\prod_i A_i(x_i - o) \right] \right).$$

The recursion removes lower degrees before reading coordinates and is independent of arity and truncation.

In the first two degrees the recursion reads as follows. In degree zero, $Q_0 = 1$: the degree-zero probes read the multiple of the unit line seen by the cluster. In degree one, $L_{0,1} = B_0 I_1$ reads the degree-zero part of a single degree-one field, and

$$Q_1 B_1 w = B_1 w - \jmath_{0,1} B_0\big[(B_1 w)(0)\big].$$

This is the field read in degree one, minus the multiple of the unit that the degree-zero probes see in it. The degree-one coefficients are the coordinates of this difference in $E_1 = \ker L_{0,1}$.

The following splitting makes the coefficients a single choice for all later estimates.

Lemma 19.6 (One channel prescription)**.** *The real fields $O_{j,\lambda}$ form complex bases of conjugation-invariant complements:*

$$F_j = \jmath_{j-1,j} F_{j-1} \oplus E_j, \qquad E_j = \mathrm{span}_{\mathbb{C}}\{O_{j,\lambda} : \lambda \in \mathrm{Ch}_j\}, \qquad F_{-1} = 0\,.$$

The degree-zero vector spans the unit line and need not equal the unit.

By Definition 19.5, all coefficients are fixed before spectators, states and truncations.

Proof. The biorthogonal construction preceding Definition 19.3 gives a reconstruction formula fixing every field of F_j; the field calculus of Lemma 15.5 makes the old functionals compatible with the inclusions. Continuing an old functional means using the same finite correlations, since $\iota_j \jmath_{q,j} = \iota_q$; no extension of a functional is chosen arbitrarily.

For $i \geq q$, this reconstruction and the rotation and reality laws give $L_{i,q} = \jmath_{q,i}$. Also $L_{i,j}\jmath_{q,j} = L_{i,q}$ for $i < q \leq j$: both sides evaluate the same single-field word before averaging. For all $q \leq j$, induction in j now gives

$$Q_j L_{j,q} = 0 \ (q < j), \qquad \sum_{i \leq j} \jmath_{i,j} Q_i L_{i,q} = \jmath_{q,j}\,. \tag{19.5}$$

For $q = j$ the second identity is the recursion; for $q < j$, compose that recursion with $\jmath_{q,j}$ and use the compatibility just proved and the induction hypothesis. In the same induction assume $Q_k F_k \subset E_k$ and $Q_k = 1$ on E_k for $k < j$. Applying $Q_i L_{i,j}$ to $Q_j A$ then cancels its matching i term; terms with $k < i$ vanish because $Q_i L_{i,j}$ vanishes on $\jmath_{i-1,j} F_{i-1}$, and terms with $i < k < j$ vanish because $Q_k A \in E_k$. Thus $Q_j F_j \subset E_j$. On E_j, the defining sum for Q_j is zero, so $Q_j = 1$. Its kernel is exactly $\jmath F_{j-1}$ by (19.5). Haar invariance and simultaneous input and output conjugation show that every map $Q_i L_{i,j}$ intertwines rotations and conjugation. Thus E_j is invariant under both. Its fixed real subspace spans it, since $v = (v + \bar{v})/2 + i(v - \bar{v})/(2i)$; a real basis is also a complex basis. This proves the stated splitting and the simultaneous choice. □

For $T \to \infty$ write $f \preccurlyeq e^{bT}$ when, for every $\tau > 0$, there is T_0 such that for every $T \ge T_0$ an ordered cutoff tail gives $\|f\| \le e^{(b+\tau)T}$, uniformly in the cutoff and volume on that tail. Products add exponents and finite sums take their maximum. Factors of exponent zero are called subpower. A level k has height b when $s_k/T \to b$; thus $a_k = e^{-bT+o(T)}$. Lemma 14.2 supplies these levels by taking the first with $s_k \ge bT$, or $s_k \ge \sqrt{T}$ if $b = 0$.

Insertions are carried through the integration by nilpotent external fields. This is the device used for observables by Bauerschmidt, Brydges and Slade, who add an external observable field to the interaction and truncate the algebra in it [5, Section 2.3]. A marker is one group of insertions in the reflected cells not yet fused with another; it is live while its weight is positive. To each marker ι attach a commuting variable z_ι with $z_\iota^2 = 0$ and a mark weight $w_\iota > 0$. In this finite scalar algebra put

$$c = \sum_A c_A z_A, \qquad z_A = \prod_{\iota \in A} z_\iota, \qquad \|c\|_{\mathrm{mark}} = \sum_A |c_A| \prod_{\iota \in A} w_\iota, \qquad c_A \in \mathbb{C}\,.$$

Products of monomials with overlapping markers vanish, and the weights of disjoint markers multiply, so $\|cd\|_{\mathrm{mark}} \le \|c\|_{\mathrm{mark}} \|d\|_{\mathrm{mark}}$. The coefficient of z_A in $\rho \prod_\iota (1 + z_\iota O_\iota)$ is $\rho \prod_{\iota \in A} O_\iota$; by linearity, taking this coefficient commutes with integration of integrable terms.

If $F_{X,A}$ is a coefficient with support X and marker set A, its weighted majorant is

$$\sum_{X,A} e^{\sigma \operatorname{cov}(X)} \|F_{X,A}\| \prod_{\iota \in A} w_\iota\,.$$

Here $\|F_{X,A}\|$ is the derivative or forced majorant of Definitions 5.3–5.4; at a support ball use its anchored sum. The factors $e^{\sigma \operatorname{cov}(X)} \|F_{X,A}\|$ enter as v_a in Lemma 13.5. Only the marked norm is used submultiplicatively. A grouped coefficient carries the product of its live mark weights, denoted by ω.

For the marked expansion put

$$\delta_l^{\mathrm{mark}} = \widehat{\beta}_{l+1}^{-2}, \qquad \phi_l = \widehat{\beta}_{l+1}^{-1}, \qquad \Xi_l = \exp \sqrt{\widehat{\beta}_{l+1}}, \qquad y_{t,l} = \widehat{\beta}_l^{-\nu_t}.$$

The fixed integer ν_t bounds the local degree-t polynomial losses, and $y_{t,l} C_l \le 1$ at entry. The four factors pay respectively for support enlargement, regular transport, Taylor extraction and insertion normalization. Their logarithms and $\log C_l$ are $o(s_l)$. For an initial weight $\varpi > 0$ at k_* define its later weights by

$$\varpi_{k_*} = \varpi, \qquad \varpi_{l+1} = \phi_l \varpi_l \quad (l \ge k_*).$$

The marked entry choices require $\phi_l \ell_l^{t+1} y_{t,l} \le y_{t,l+1}$ for $l \ge m_{\mathrm{ref}}(t)$. This follows the zero-jet regular remainder through the integration.

Let B_l be the support radius and q_l the per-site logarithmic bound (14.1) of Section 9. Fix a reflected cell at level k and choose

$$k_* = \min\{l \ge k : B_l^2 \le a_k/a_l\}, \qquad N_k = \exp\{(a_k/a_{k_*})^4 q_{k_*} + 5\}.$$

These quantities ensure that the marked stencils fit in the cell and bound its density normalization.

For a grouped coefficient at level $m+1$ of weight $\omega > 0$, put

$$q_m^{\mathrm{ext}} = 4 \Xi_m a_{m+1}^t / \omega, \qquad \mathfrak{E}_m = 8(1 + C_m) q_m^{\mathrm{ext}}\,.$$

The terminal weight ϖ must satisfy

$$\begin{aligned} \varpi(a_m/a_{k_*})^{t+1} y_{t,m}/y_{t,k_*} &\le \delta_m^{\mathrm{mark}}\omega/(16\Xi_m), \\ 32(1+C_m)\varpi\, a_{k_*}^{-t}\mathfrak{E}_m(a_m/a_{k_*})^{1/8} &\le y_{t,k_*}/2. \end{aligned} \tag{19.6}$$

These conditions pay for extraction at m and for the transported residual, respectively.

Informally, after blocking down to level m, a group of insertions of total weight ω is replaced by one degree-t insertion at the origin, the matched field. The next lemma bounds its coefficient by $2\mathfrak{E}_m$, and the chessboard seminorm of the replacement error in a level-k cell by $2N_k/\varpi$.

Lemma 19.7 (Marked extraction and insertion costs)**.** *A grouped coefficient of weight ω has a degree-t local extraction with coefficient and remainder each at most q_m^{ext}. With k_* as above, sufficiently late, and whenever* (19.6) *holds, the matched field has coefficient size at most $2\mathfrak{E}_m$. Its interior reflected remainder in the level-k cell has chessboard seminorm at most $2N_k/\varpi$.*

Proof. The extraction is a Taylor expansion in the norms of Definitions 5.3–5.4 and 2.8; Lemma 19.1 transports its residual to k_*; the majorants of Lemma 13.12, summed by the rooted clause of Lemma 13.5 in the marked norm, bound the cell integral. Divide a grouped coefficient by its available weight ω. Extract its radial polynomial through t and write the regular complement by the integral Taylor formula. The finite derivative sums, coefficient conversion and remainder constants are at most Ξ_m after the entry levels have been chosen: their logarithms are bounded by stage constants and powers of $\log\widehat{\beta}_{m+1}$, whereas $\log\Xi_m = \sqrt{\widehat{\beta}_{m+1}}$. Multiplication by a_{m+1}^t gives q_m^{ext}. The factor four bounds the polynomial and complement separately. Forced terms keep their original majorants and the forced normalization in (14.16).

The combined cost is at most $2q_m^{\mathrm{ext}}$. Subtract the column matching the transported family at k_*. Lemma 19.1 gives, for $k_* \le l \le m$,

$$\mathfrak{r}_l + \mu(x_l) \le \mathfrak{E}_m(a_m/a_l)^{1/8}, \qquad \mu(x_m^{\mathrm{field}}) \le 2\mathfrak{E}_m.$$

The factor $4(1+C_m)$ multiplying $2q_m^{\mathrm{ext}}$ includes the finest residual and coefficient together. The entry requirement $\phi_l \ell_l^{t+1} y_{t,l} \le y_{t,l+1}$ gives the regular zero-jet gain: iterating $\varpi_{l+1} = \phi_l \varpi_l$ yields $\varpi_m \le \varpi(a_m/a_{k_*})^{t+1} y_{t,m}/y_{t,k_*}$. The first condition of (19.6) permits extraction at m. For the second, the level-l ratio of required to permitted residual weight is bounded by its level-k_* value times $(a_l/a_{k_*})^{7/8} \le 1$: use $\varpi_l \le \varpi(a_l/a_{k_*})^{t+1} y_{t,l}/y_{t,k_*}$, the residual bound $\mathfrak{E}_m(a_m/a_l)^{1/8}$ and divide by $y_{t,l}$. Thus the condition at k_* permits the signed residual and compensating columns throughout the interval. Their density identities are exact by Lemma 19.1.

In the nilpotent variables defined above, a marked polynomial is measured by the sum of the absolute coefficients times the product of their assigned weights. Consequently the coefficient with one residual per cell is at most ϖ^{-1} per cell times that sum. Summing the unmarked majorants and dividing by the two positive density normalizations bounds the integral per cell by N_k. The singleton, forced-profile, connected regular and eligible forced sums contribute at most $2, 1, 1, 1$ per cell, giving the additive constant five. Apply this to the original local difference and the correction representing its column, and use the triangle inequality for the chessboard seminorm, obtained by expanding the reflected product and applying (15.4) to each term. This gives $2N_k/\varpi$.

To verify that k_* has the required height, let $\mathfrak{n}_l = \max(1, q_l)$ bound the normalization cost. The entry choices satisfy

$$B_{l+1}^2 \le 2B_l^2, \qquad (\ell_l B_l^2)^4 \mathfrak{n}_{l+1} = o(s_{l+1}). \tag{19.7}$$

The level k_* exists since $a_l B_l^2$ decreases by at least $2/3$ at each step. If $k_* = k$, the spacing ratio is one. Otherwise $k_* = l+1$, and minimality gives $a_k/a_{k_*} \le \ell_l B_l^2$. Therefore $(a_k/a_{k_*})^4 q_{k_*} = o(s_{k_*})$. Also

$0 \le s_{k_*} - s_k = \log(a_k/a_{k_*}) \le (a_k/a_{k_*})^4 \mathfrak{n}_{k_*} = o(s_{k_*})$. This proves $s_{k_*}/s_k \to 1$ and $\log N_k = o(s_{k_*})$. At height zero it still gives $\log N_k = o(T)$. Finally $\log \Xi_m = o(s_m)$ follows from $\widehat{\beta}_m = \bar{\beta} + \kappa_G s_m$ and $s_{m+1}/s_m \to 1$. The other displayed functions are fixed powers of $\widehat{\beta}$ or the subpower column cost of Lemma 14.2. □

Fix nonempty lattice labels $v_\iota \in V_{d_\iota}$, let n_{cl} be their number and put $W = \sum_\iota d_\iota$. Choose levels k, m, j of heights $0 \le \theta_- < \theta < \theta_+$. The input points lie on the spacing-a_j grid, in the centered level-$(m+1)$ block $[-a_{m+1}/2, a_{m+1}/2)^4$, and their separation is at least $B_j^2 a_j$. The configurations belong to $\mathrm{Gen}_{n_{\mathrm{cl}}}$, and all stencils are interior to the level-k cell. These conditions hold eventually for clusters of extent $e^{-xT+o(T)}$ and reciprocal separation at most $e^{xT+o(T)}$, where $\theta < x < \theta_+$. The chosen k_* has height θ_- by (19.7).

Two contributions make up the exponential cost of bringing the cluster to level m. The initial field normalization contributes $W\theta_+$. Transporting at most n_{cl} groups from height θ_+ to height θ, at four powers of the scale ratio per group, contributes $4n_{\mathrm{cl}}(\theta_+ - \theta)$. We therefore set

$$\mathcal{Z} = W\theta_+ + 4n_{\mathrm{cl}}(\theta_+ - \theta).$$

The support enlargements and the finitely many fusions contribute only subpower factors. Figure 16 shows the three scales and these costs.

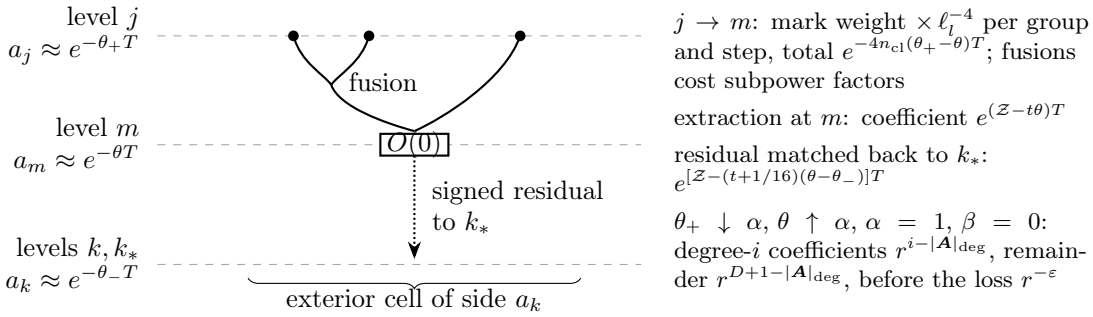


Figure 16: The three scales of Lemma 19.8. Markers start at the insertion resolution a_j, are grouped when they cease to be B_l^2-separated, and reach level m as one group, where one degree-t field is extracted; the signed residual is matched backward to k_*. Right: losses and gains up to subpower factors, at fixed truncation (no series in D).

Lemma 19.8 (Lattice cluster replacement)**.** *For every output degree t, the preceding configurations admit labels $v_{N,p}^{\mathrm{out}} \in V_t$ on an ordered cutoff tail such that*

$$\begin{aligned} \|v_{N,p}^{\mathrm{out}}\|_m &\preccurlyeq e^{(\mathcal{Z}-t\theta)T}, \\ \Big\| \prod_\iota O_{N,p}(v_\iota, x_\iota) - O_{N,p}(v_{N,p}^{\mathrm{out}}, 0) \Big\|_{\mathcal{C}_k^{\mathrm{tile}}(0)} &\preccurlyeq e^{[\mathcal{Z}-(t+1/16)(\theta-\theta_-)]T}. \end{aligned}$$

The difference is interior. The bounds are uniform in cutoff and volume on that tail.

Proof. The insertions are expanded in the marked format of Definition 14.5 and Lemmas 14.7–14.9; Lemmas 19.1 and 19.7 supply the extraction once the markers form one group, and the steps below bound the mark weight lost before that. In the nilpotent variables defined above, the coefficient of

their product in dens $\rho_N \prod_\iota (1 + z_\iota O_{N,p}(v_\iota, x_\iota))$ is the product insertion times the density. Until level j the stencils are disjoint, so their marked integrations are separate.

Moving a marker to its representative through at most h cover balls gives

$$\operatorname{cov}(X \cup \{y_\iota : \iota \in A\}) \le \operatorname{cov}(X) + h|A|. \tag{19.8}$$

If at most $\mathfrak{N}$ representatives occupy a ball, shrinking each affected weight by $\eta e^{-\sigma h}$ gives the original term times $\eta^{|A|}$. In the anchored sum, terms incident on the old ball use its original majorant; any new incidence can be assigned to an old marker on the joining path. There are at most n_{cl} old markers for each new representative and $\mathfrak{N}$ representatives. Taking $\eta \le [(1+\mathfrak{N})n_{\mathrm{cl}}]^{-1}$ pays for these assignments. Unmarked terms keep their supports. Hence the enlarged weighted series is bounded by the original majorant series of Lemma 13.12.

Keep a maximal B_l^2-separated set of representatives at each level, retaining the distinguished one at the origin, and attach each discarded point to a retained neighbor. A fusion decreases the number of groups, so there are at most $n_{\mathrm{cl}} - 1$ fusions. The joining-cover bound can be taken as $h_l = 16[1 + \ell_l(B_l + B_{l+1} + 1)^2]^4$. The entry requirements of Section 9 and Lemma 14.2 give $(\sigma_{J_{l+1}} + 1)h_l = o(s_l)$ and subpower occupancy. Choose one $\alpha_j \le 1$ below all the resulting enlargement allowances for $m \le l < j$. It is subpower: the logarithm of the running minimum is bounded by the maximum logarithmic cost; split this maximum into a fixed initial segment and a tail bounded by τs_j, then send τ to zero. The factor $[(1+\mathfrak{N})n_{\mathrm{cl}}]^{-1}$ above is included in this allowance $\alpha_j/8$.

Between fusions the one-mark integration costs at most $6\ell_l \le \ell_l^4$ in each weight. To derive this bound, let $\mathfrak{m}$ be an incoming marked majorant. Apply the rooted clause of Lemma 13.5 coefficientwise in the submultiplicative marked norm, with the marked factor fixed. Separate the transported regular terms into the zero-jet local integration remainder, of total weighted mass p_l, and the incoming regular contribution carried without Taylor subtraction.

Use the support conversion factor $\mathcal{A}_l$ defined by (14.5). For a fine support X of the unsubtracted incoming contribution, let Y be its coarse support, including the incident windows. The support conversion estimate there gives

$$(\sigma_{J_l} + 2)\operatorname{cov}_{B_l}(Y) \le \log \mathcal{A}_l + \tfrac{1}{2}\sigma_{J_{l+1}} \operatorname{cov}_{B_{l+1}}(X)\,.$$

Exponentiating bounds this incoming mass by $\mathcal{A}_l\mathfrak{m}$; the remaining half of the fine exponent is nonnegative. Item 2 of Section 14.1 imposes $\mathcal{A}_l \le \ell_l$.

The combined regular mass is thus at most $p_l + \mathcal{A}_l\mathfrak{m}$. Splitting a complex marked term into its real and imaginary parts costs at most twice its majorant. For the forced terms, let f_l denote their total contribution after this splitting and the normalization in (14.16); their eligibility tests remain attached. Use the termwise bounds in the proof of Lemma 14.6 for the zero-jet local remainder: its regular part retains the Taylor gain, and its forced part retains half the forced suppression. The unsubtracted incoming contribution is bounded by the support conversion above; the ηr graph bound in that proof is not used for this contribution. Raising the entry coupling gives $2p_l \le \mathfrak{m}$ and $f_l \le \mathfrak{m}$. Adding the two contributions gives

$$2(p_l + \mathcal{A}_l\mathfrak{m}) + f_l \le (2\ell_l + 2)\mathfrak{m} \le 6\ell_l\mathfrak{m}.$$

With g groups remaining, their product weight at level l is consequently at least

$$(a_j/a_l)^{4n_{\mathrm{cl}}}(\alpha_j/8)^{n_{\mathrm{cl}}(n_{\mathrm{cl}}-g)} \prod_\iota w_\iota. \tag{19.9}$$

Indeed a separated step multiplies the weight by $\ell_l^{-4g} \geq \ell_l^{-4n_{\mathrm{cl}}}$. A fusion to $g' < g$ uses at most n_{cl} factors $\alpha_j/8$, whereas $n_{\mathrm{cl}}(g-g') \geq n_{\mathrm{cl}}$.

At $m+1$ all representatives have the same ancestor. The initial normalized-field weights can be chosen with $\prod_\iota w_\iota = e^{-W\theta_+ T+o(T)}$ by Lemma 14.7. Thus (19.9) gives $\omega = e^{-\mathcal{Z}T+o(T)}$ and $\mathfrak{E}_m = e^{(\mathcal{Z}-t\theta)T+o(T)}$. Choose

$$\varpi = \exp\{[(t+1/16)(\theta-\theta_-)-\mathcal{Z}]T\}. \tag{19.10}$$

The second condition of (19.6) has exponent $-(\theta-\theta_-)/16 < 0$; the first has exponent $-\mathcal{Z} - 15(\theta-\theta_-)/16$, strictly below the exponent $-\mathcal{Z}$ of its right side. Both therefore hold, including every subpower factor. Lemma 19.7 gives the coefficient bound $2\mathfrak{E}_m$ and chessboard bound $2N_k/\varpi$ asserted here. All enlargements fit the prescribed cell margin. Reflection and conjugation preserve the majorants, so the same calculation applies to all cells and all cutoffs on the common tail. □

We next state the bounds when the exterior distances also shrink. Write $w = \prod_i A_i(x_i)$, $W = |\boldsymbol{A}|_{\deg}$ and $V = |\boldsymbol{B}|_{\deg}$. Consider families of configurations indexed by $T \to \infty$. Suppose the cluster lies within $e^{-\alpha T+o(T)}$ of o, its pairwise distances are at least $e^{-\alpha T-o(T)}$, and $\alpha > 0$. Exterior points are at distance at least $e^{-\beta T-o(T)}$ from every cluster point and o, and mutually at least $e^{-\upsilon T-o(T)}$ apart, where $0 \leq \beta < \alpha$ and $\upsilon \geq \beta$. The field lists are fixed, or range over fixed compact sets of coefficients. Call these conditions (G). Let $u_{(i,\kappa)}$, κ ranging over the degree-i basis fields, be the biorthogonal fields of Section 19.1, and put $w_o = \tau_{-o}w$. Set

$$\begin{aligned}
\lambda_{i,\kappa}^{\mathrm{raw}} &= \sum_\gamma m_{(i,\kappa)\gamma} S(w_o, B_{i,\gamma}),\\
A_h^{\mathrm{raw}} &= P_h^{\mathrm{raw}} w_o \in F_h,\\
\mathcal{R}_D^{\mathrm{raw}} &= M^0\Big(w\prod_l B_l(y_l)\Big) - M^0\Big(A_D^{\mathrm{raw}}(o)\prod_l B_l(y_l)\Big) \in \mathfrak{A}_0\,.
\end{aligned}$$

Here P_h^{raw} is the projection of Definition 19.3, and M^0 is the compression of Definition 6.12 and Lemma 17.1.

The following estimates retain the exterior distance powers needed for nested clusters and contacts.

Lemma 19.9 (Power-separated estimates)**.** *For every $i, D \geq 0$ and $j \leq h$, families of configurations obeying (G) satisfy*

$$\lambda_{i,\kappa}^{\mathrm{raw}} \preccurlyeq e^{(W-i)\alpha T}, \qquad \mathcal{R}_D^{\mathrm{raw}} \preccurlyeq e^{[W\alpha-(D+1)(\alpha-\beta)+V\upsilon]T}, \tag{19.11}$$

$$Q_j B_j w - Q_j L_{j,h} A_h^{\mathrm{raw}} \preccurlyeq e^{(W-h-1)\alpha T}. \tag{19.12}$$

The final coefficients and tested remainder of Definition 19.5 obey the same respective bounds, in norm before state evaluation.

A degree-i coordinate thus costs the cluster degree less i at the cluster height α, and the remainder gains $(D+1)(\alpha-\beta)$ from the difference of the cluster and exterior heights.

Proof. The proof converts chessboard seminorms into algebra norms by Lemma 17.2, applies Lemma 19.8 on grid configurations, removes the grid restriction by derivative descent with the calculus of Lemma 15.5, and then averages the reconstruction map. Write $n_{\mathrm{cl}} = |I_{\mathrm{cl}}|$.

Step 1: chessboard seminorms bound algebra norms. Suppose a lattice difference has centered chessboard seminorm (Lemma 14.15) at most B_{rt}, and the exterior insertions have chessboard

seminorms at most $B_{\mathrm{rt},l}$. Put the difference in a coarse cell and the exterior insertions in finer disjoint cells; their stencils fit by Lemma 14.11. Let $U \in \mathfrak{A}_0$ be the corresponding compressed difference. The lattice paragraph after Lemma 17.2, with the moment identity of Lemma 17.1, gives for every $n \in \mathbb{N}$

$$\int_{\mathsf{X}} |\xi(U)|^{2n}\, d\nu(\xi) \le (B_{\mathrm{rt}} \prod_l B_{\mathrm{rt},l})^{2n}\,, \tag{19.13}$$

and Lemma 17.2 gives $\|U\| \le B_{\mathrm{rt}} \prod_l B_{\mathrm{rt},l}$. The same argument bounds a word by the chessboard seminorms of its insertions, giving the crude exponent $W\alpha + V\upsilon$ on the stated geometries.

Step 2: grid configurations. Consider the generic grid configurations defined above in a cluster of height $\alpha > 0$. Choose $0 < \theta_- < \theta < \alpha < \theta_+$, with exterior distance exponent $\beta < \theta_-$, and $\mathcal{Z} = W\theta_+ + 4n_{\mathrm{cl}}(\theta_+ - \theta)$. Apply Lemma 19.8 with an auxiliary degree $t \ge D$. For each fixed cluster geometry the output coefficients are bounded at fixed m; the reverse transfer inequality of Lemma 14.7 bounds them also at the reference level. Along a refinement of the ordered cutoff tails they converge to a field $A^{\mathrm{out}} \in F_t$. Step 1 bounds the replacement error by exponent

$$\mathcal{Z} - (t + 1/16)(\theta - \theta_-) + V\upsilon\,.$$

Testing the same difference against the fixed probes bounds the raw coordinate errors. Reference-level coefficient convergence and Lemma 19.2 apply on this geometry-dependent refinement. The degree-i raw coordinate annihilates F_{i-1} by the biorthogonality before Definition 19.3, and is a fixed bounded functional on the quotient. With $\epsilon_{\mathrm{bd}} \preccurlyeq e^{(\mathcal{Z}-t\theta)T}$, (19.4) therefore bounds that coordinate by exponent $\mathcal{Z} - i\theta$.

Choose t large enough that $(t + 1/16)(\theta - \theta_-) \ge (D + 1)\theta$ and also large enough for every coefficient being tested. Exact reconstruction of A^{out} by the prescription separates the degrees $i \le D$ from $D < i \le t$. An omitted degree contributes at most

$$\mathcal{Z} - i\theta + i\beta + V\upsilon \le \mathcal{Z} - (D + 1)(\theta - \beta) + V\upsilon\,.$$

The low-degree coordinate errors satisfy the same bound, by the chosen auxiliary horizon. We have therefore proved grid exponents $\mathcal{Z} - i\theta$ for raw degree-i coordinates and $\mathcal{Z} - (D + 1)(\theta - \beta) + V\upsilon$ for the raw remainder.

Step 3: derivative descent. Run the crude, grid and derivative bounds uniformly over all characters $\xi \in \mathsf{X}$. The algebra $\mathfrak{A}_0$ is commutative, so $\|U\| = \sup_\xi |\xi(U)|$. To remove the grid restriction, consider these character evaluations for every input label list and every positional derivative. A derivative replaces its input label by a field of degree one higher, by Lemma 15.5. For suitable constants c_0, d_0 independent of the input degrees, the crude bound has exponent $\alpha W + c_0$ and the grid bound exponent $\theta_+ W + d_0$. We descend in the total input degree W. The crude bound proves the desired estimate for all W with $(\theta_+ - \alpha)W \ge c_0 - d_0$, because $\theta_+ > \alpha$. For the induction step, move the configuration to a generic level-j grid configuration a distance $O(a_j)$ away; the intervening segment remains separated at exponent α. A positional derivative has total degree $W+1$, so the induction hypothesis and the grid spacing give $e^{[\theta_+(W+1)+d_0]T+o(T)}a_j = e^{(\theta_+ W+d_0)T+o(T)}$: the displacement error has exactly the required degree-W bound, for every slot and axis.

Step 4: limits. Now let $\theta \uparrow \alpha$ and $\theta_+ \downarrow \alpha$, keeping $\beta < \theta_- < \theta$. Translation moves the origin to o. This proves (19.11) for arbitrary configurations. The remainder estimate is in $\mathfrak{A}_0$, and empty clusters give zero remainder.

Step 5: the coefficients of Definition 19.5. Let $A_h^{\mathrm{raw}} \in F_h$ be the raw truncation at an auxiliary degree h. In every rotated probe correlation, the difference between the cluster and $A_h^{\mathrm{raw}}(0)$ has exponent $(W - h - 1)\alpha$. The marked norm bounds and multilinearity make (19.11) uniform on

compact coefficient sets, hence on the rotation orbit. Haar averaging, conjugation and the fixed maps Q_j preserve the exponent. This proves (19.12). With $h = j$, all lower raw modes vanish under $Q_j L_{j,j}$ by (19.5); the remaining modes have exponent $(W - j)\alpha$. With $h = D$, use (19.5) to reconstruct A_D^{raw}, and telescope the raw remainder with the errors (19.12) for $j \le D$. Testing an error in F_j costs $j\beta + V\upsilon$, so their largest exponent is $W\alpha - (D+1)(\alpha - \beta) + V\upsilon$. □

At fixed exterior distances the power-separated estimates become the bounds of Theorem 6.20. The parameters M_{geo} are nonnegative and the separation and error parameters are positive, as in Definition 2.19. For the algebra-valued remainder put

$$\mathcal{R}_D = M^0\Big(\prod_i A_i(x_i) \prod_l B_l(y_l)\Big) - \sum_{j \le D,\, \lambda \in \mathrm{Ch}_j} c_{j,\lambda}\, M^0\Big(O_{j,\lambda}(o) \prod_l B_l(y_l)\Big).$$

Here M^0 is the compression of separated words of Definition 6.12 and Lemma 17.1; its characters and vacuum state give the phase and mixed remainders $\mathcal{R}_D^{\zeta}$ of Definition 2.19.

Lemma 19.10 (Compressed remainder bound)**.** *Fix finite lists $\boldsymbol{A}, \boldsymbol{B}$, $M_{\mathrm{geo}} \ge 0$, $\eta, \delta_{\mathrm{sp}}, \varepsilon > 0$ and $D \in \mathbb{N}$. There is $r_1 > 0$ such that, for $0 < r < r_1$, every $(r, M_{\mathrm{geo}}, \eta)$-cluster (x, o) and every δ_{sp}-clear spectator configuration y,*

$$\|\mathcal{R}_D\|_{\mathfrak{A}_0} \le r^{D+1-|\boldsymbol{A}|_{\deg}-\varepsilon}.$$

Proof. Apply Lemma 19.9 to Definition 2.19, with $T = -\log r$, $\alpha = 1$ and $\beta = \upsilon = 0$. Every admissible configuration has these bounds on extent and reciprocal distances. If the claimed threshold failed, choose violating configurations at radii tending to zero. The estimate just proved along that sequence, with loss $\varepsilon/2$, eventually contradicts the violation with loss ε: constants and subpower factors are absorbed by $r^{\varepsilon/2}$. Taking $M_{\mathrm{geo}} r < \delta_{\mathrm{sp}}/2$ also ensures separation. □

Proof of Theorem 6.20. Lemma 19.6 fixes the fields and coefficients of Definition 2.19 before spectators, state indices or truncation. The coefficient bound follows from the coefficient estimate of Lemma 19.9 by the contradiction argument in the proof of Lemma 19.10; its threshold depends only on the data of the coefficient. Character evaluation and the mixed vacuum are contractions on $\mathfrak{A}_0$, so Lemma 19.10 gives the remainder bound with its own threshold. The argument bounds $\mathcal{R}_D$ in $\mathfrak{A}_0$ before state evaluation. □

A nested configuration retains leaf positions and internal centers, so it also describes partially replaced words. Call an internal vertex other than the root proper, and fix its truncation $D_v \in \mathbb{N}$.

Definition 19.11 (Nested geometry and rates)**.** Give each vertex v of a finite rooted tree a position z_v, each leaf a field A_i, and each internal vertex a scale $\varrho_v > 0$. Its root o_{root} is internal. Require $\varrho_{o_{\mathrm{root}}} = r$, $\varrho_v \le \varrho_{\mathrm{par}(v)}$, $|z_u - z_v| \le M_{\mathrm{geo}} \varrho_v$ for every descendant u, and $|z_u - z_{u'}| \ge \eta \varrho_v$ for positions in different child subtrees. Put

$$W_v = \sum_{i \text{ below } v} \deg A_i, \qquad q_v = \varrho_v / \varrho_{\mathrm{par}(v)},$$

$$\mathcal{Q}_{\varepsilon,q}(\varrho) = r^{q - W_{o_{\mathrm{root}}} - \varepsilon} \prod_{v \ne o_{\mathrm{root}} \text{ internal}} q_v^{-W_v - \varepsilon}, \qquad \mathsf{e}_{\boldsymbol{D}}(\varrho) = \sum_{v \ne o_{\mathrm{root}} \text{ internal}} q_v^{D_v + 1}.$$

Spectators are δ_{sp}-clear of every tree position and each other.
The recursion retains every leaf, including those above a neighboring truncation.

Definition 19.12 (Iterated coefficient)**.** A leaf contributes its field and coefficient one. At a proper internal vertex v, replace its children by

$$\sum_{j\le D_v}\sum_{\lambda\in \mathrm{Ch}_j} c_{j,\lambda}(\text{child fields};\text{child centers}, z_v)O_{j,\lambda},$$

multiplying their coefficients and summing. At the root retain only (q,λ), giving $C^{\mathrm{it},\boldsymbol{D}}_{q,\lambda}$. The direct coefficient is $C^{\mathrm{dir}}_{q,\lambda}=c_{q,\lambda}(\boldsymbol{A};z_{\mathrm{leaves}},z_{o_{\mathrm{root}}})$. This compares successive and direct expansion.

For example, let a cluster v of leaves of total degree W_v first fuse at scale ϱ_v, and let its output then fuse with the remaining leaves at the root scale r. If v is the only proper internal vertex, then

$$\mathcal{Q}_{\varepsilon,q}=r^{q-W_{o_{\mathrm{root}}}-\varepsilon}(\varrho_v/r)^{-W_v-\varepsilon},\qquad \mathsf{e}_{\boldsymbol{D}}=(\varrho_v/r)^{D_v+1}.$$

The first factor is the size allowed for the direct coefficient; the second is the truncation gain at the inner fusion. For a general tree the errors from different vertices are added, as in the definition of $\mathsf{e}_{\boldsymbol{D}}$.

Fix the tree, leaf list, output (q,λ), proper-vertex truncations, $M_{\mathrm{geo}}\ge 0$ and $\eta,\varepsilon>0$. No lower bound on the ratios q_v is imposed.

Lemma 19.13 (Nested coefficient estimate)**.** *There is $r_0>0$ such that all geometries of Definition 19.11 with $0<r<r_0$ satisfy*

$$|C^{\mathrm{dir}}_{q,\lambda}|\le\mathcal{Q}_{\varepsilon,q},\qquad |C^{\mathrm{dir}}_{q,\lambda}-C^{\mathrm{it},\boldsymbol{D}}_{q,\lambda}|\le\mathcal{Q}_{\varepsilon,q}\mathsf{e}_{\boldsymbol{D}}\,.$$

Both expressions are smooth in neighborhoods of these geometries.

Proof. The estimates of Lemma 19.9 are applied vertex by vertex along a tree induction, after reducing to sequences whose scales have convergent logarithmic heights. Along any sequence of admissible radii and scales tending to the root collision, put $T=\sum_{v\ \mathrm{internal}}\log(1/\varrho_v)$. Eventually every summand is positive. After a subsequence, $\log(1/\varrho_v)/T\to a_v\ge 0$, with $\sum_v a_v=1$ and $a_v\ge a_{\mathrm{par}(v)}$. Set

$$\mathcal{B}=W_{o_{\mathrm{root}}}a_{o_{\mathrm{root}}}+\sum_{v\ne o_{\mathrm{root}}\ \mathrm{internal}} W_v(a_v-a_{\mathrm{par}(v)})\,.$$

We prove by induction on the tree that a direct output of degree q has exponent $\mathcal{B}-qa_{o_{\mathrm{root}}}$, and its error against the prescribed finite iteration has that exponent minus $\gamma=\min_{v\ne o_{\mathrm{root}}\ \mathrm{internal}}(D_v+1)(a_v-a_{\mathrm{par}(v)})$. With no proper internal vertices the two coefficients coincide exactly.

First flatten the edges $u\to v$ with $a_u=a_v$: combine their grouping before making any field replacement. The word at v then consists of original leaves in this equal-height component and outputs from strictly finer children; its distinct positions are separated at exponent a_v by the tree geometry. For a child with $a_u>a_v$, expand to an auxiliary horizon H_u. The remainder exponent has the improvement $(H_u+1)(a_u-a_v)$ by (19.11). Choose the children in a fixed order, choosing H_u only after the outputs already used as exterior fields are fixed. Each improvement tends to infinity with H_u, whereas the exterior degree in that step is fixed. It can therefore exceed any prescribed error exponent. The identity

$$\prod_{i=1}^n X_i-\prod_{i=1}^n Y_i=\sum_{i=1}^n\left(\prod_{h<i}Y_h\right)(X_i-Y_i)\left(\prod_{h>i}X_h\right)\tag{19.14}$$

justifies these successive replacements in every correlation and in every probe defining a coefficient. It is an identity of separated expressions, with their scalar coefficient factors retained.

For a fixed collection of child outputs j_u, the child exponents include $-j_u a_u$, and the parent coefficient contributes $j_u a_v$. Their sum contains $-j_u(a_u - a_v)$; the remaining leaf weights telescope to $\mathcal{B} - q a_{o_{\mathrm{root}}}$. Nonnegative output degrees therefore never increase this exponent. If the root height is zero, apply the crude estimate to this root word of child outputs; its finitely many field degrees cost exponent zero, while their coefficients already carry the child contributions to $\mathcal{B}$. This proves the direct bound, by the preceding arbitrary-accuracy replacements and the induction hypothesis. Remove auxiliary outputs above D_u. Each discarded term has at least one $j_u \geq D_u + 1$ and thus gains at least γ. Replacing the remaining direct child coefficients by their iterated coefficients uses (19.14) again and gains γ from one child error. If $\gamma = 0$, the direct and iterated bounds themselves give the error exponent. If $\gamma > 0$, every proper internal scale increase is positive and the auxiliary horizons can be chosen to make the preceding replacement errors smaller than this exponent. This proves the induction for the difference, with one improvement rather than a product of improvements.

The logarithmic exponent of $\mathcal{Q}_{\varepsilon,q}$ is

$$\mathcal{B} - q a_{o_{\mathrm{root}}} + \varepsilon A_*, \qquad A_* = a_{o_{\mathrm{root}}} + \sum_{v \neq o_{\mathrm{root}} \text{ internal}} (a_v - a_{\mathrm{par}(v)}) > 0 \,.$$

Positivity follows because $\sum_v a_v = 1$: if every displayed nonnegative term vanished, every a_v would vanish. The exponent of $\mathsf{e}_{\boldsymbol{D}}$ is $-\gamma$. Thus the estimates with loss $\varepsilon/2$ imply the displayed bounds with loss ε and coefficient one along this sequence. A sequence of violations would have the subsequence just treated, a contradiction. This proves a threshold uniform in the scale vector, even when some scales decrease faster than every power of r. All positions stay separated and, for small r, clear of every finite rotated probe family used in the recursion. The averaging integrands are smooth on an open neighborhood with that clearance; their derivatives are bounded on compact neighborhoods times $O(4)$. Differentiating the finite sums and Haar integrals proves smoothness. □

For tested products, retain the original leaves at the root; only the degree of its output is truncated. Fix additionally the root truncation $m \in \mathbb{N}$, a spectator list $\boldsymbol{B}$ and $\delta_{\mathrm{sp}} > 0$.

Lemma 19.14 (Nested tested estimate)**.** *For some $r_0 > 0$, every smaller radius, admissible nested geometry, admissible spectator configuration and state index ζ satisfy*

$$|\mathcal{R}^{\zeta}_{m}(\boldsymbol{A}, z_{\mathrm{leaves}}; \boldsymbol{B}, y; z_{o_{\mathrm{root}}})| \leq \mathcal{Q}_{\varepsilon,m+1}(\varrho) \,.$$

The same bound holds in the norm of $\mathfrak{A}_0$ before evaluation in a character or the mixed vacuum.

Proof. The tree induction of Lemma 19.13 is run with the spectators kept in every replacement of Definition 2.19. Their reciprocal distances from every tree position and from one another have exponent zero. They therefore add no term to $\mathcal{B}$. After the arbitrary-accuracy child replacements, the root remainder estimate of Lemma 19.9 has exponent $\mathcal{B} - (m+1) a_{o_{\mathrm{root}}}$. When $a_{o_{\mathrm{root}}} = 0$, apply the crude bound to the root word of child outputs after these replacements. The child coefficient powers already account for $\mathcal{B}$; the finitely many root output fields and the spectators cost exponent zero. The positive quantity εA_* used above again absorbs all finite sums and subpower factors. The contradiction argument is unchanged. The norm estimate follows from (19.13); applying a character or the mixed vacuum proves the scalar bounds with the same threshold. □

Remark 19.15 (Integrated forms). The thresholds of Lemmas 19.10, 19.13 and 19.14 precede positions, scale vectors and states, so they survive integration. Fix the data of one of these lemmas, a measurable space Z and $B_0 \ge 0$. Below a threshold preceding the measure, the weight, the scale vector and the maps, the following holds. Let λ_Z be a measure on Z, ϑ_Z an integrable complex weight with $\int |\vartheta_Z|\, d\lambda_Z \le B_0$, and the geometry, spectator and phase maps almost-everywhere measurable and admissible almost everywhere on $\{\vartheta_Z \neq 0\}$; in the nested case the scale vector ϱ is fixed throughout the integral. Then the weighted full, truncated and compared expressions are integrable, and

$$\Big|\int_Z \vartheta_Z \mathcal{R}_D^{\xi(z)}\, d\lambda_Z\Big| \le B_0 r^{D+1-|\boldsymbol{A}|_{\deg}-\varepsilon}, \qquad \Big|\int_Z \vartheta_Z C_{q,\lambda}^{\mathrm{dir}}\, d\lambda_Z\Big| \le B_0 \mathcal{Q}_{\varepsilon,q},$$
$$\Big|\int_Z \vartheta_Z \big(C_{q,\lambda}^{\mathrm{dir}} - C_{q,\lambda}^{\mathrm{it},\boldsymbol{D}}\big)\, d\lambda_Z\Big| \le B_0 \mathcal{Q}_{\varepsilon,q}\mathsf{e}_{\boldsymbol{D}}, \qquad \Big|\int_Z \vartheta_Z \mathcal{R}_m^{\xi(z)}\, d\lambda_Z\Big| \le B_0 \mathcal{Q}_{\varepsilon,m+1}.$$

The first and last bounds also hold for the mixed state. Indeed, for fixed r and ϱ, (19.13) bounds the full correlation uniformly, and the truncated sum is the full expression minus the bounded remainder. Smoothness of the coefficients where their rotated probes are clear, continuity in the phase (Lemma 17.3) and rounding of the geometry to a countable dense set give joint measurability. Integrating the pointwise bounds against $|\vartheta_Z|$ gives the estimates.

19.2 Contact subtractions

Property E7 alone gives some extension through the collision diagonals (see the discussion before Theorem 6.21). This subsection constructs a specific one, by explicit subtraction of normal Taylor jets of the test function read through contact projections, and proves its local order $(2+|\boldsymbol{A}|_{\deg})n$ on compact sets. For each retained field degree we subtract a jet of a different order, and partial collisions require the same operation inside each smaller cluster; absolute convergence follows by summing normal shells.

We work with finite sums of field words, identifying $A(x)$ with $(\jmath_{d,e}A)(x)$ and imposing linearity in A. Their product is concatenation. The correlation of a word is read only at distinct positions; elsewhere give it value zero. This convention is used inside the algebraic formulas below, before any integration. The underlying prescription and its fields are those fixed in Lemma 19.6; the finite probes chosen next are independent of its OPE probes.

Definition 19.16 (Contact projection)**.** Choose contact probes and reconstruction maps as in Definition 19.3. Choose $\rho_q^{\mathrm{ct}} > 0$ with every probe point at distance at least $2\rho_q^{\mathrm{ct}}$ from zero. For a word w, set

$$K_q w = \mathbf{1}_{\{\text{all positions of } w \text{ lie in } \overline{B}(0,\rho_q^{\mathrm{ct}})\}} w\,.$$

Apply the averaging and reality formula of Definition 19.4 to this clipped word with the contact probes; extend linearly to sums, obtaining B_q^{ct}. This extracts contact fields on a fixed region clear of its probes.

For contact expressions $I_q A = A(0)$ still denotes a single field. Unlike the domain restriction in Definition 19.4, clipping here acts on each monomial before summing it.

Definition 19.17 (Coherent contact projections)**.** Put $T_q = P_q I_{q+1}$ and define recursively

$$P_0 = B_0^{\mathrm{ct}}, \qquad P_{q+1} = \jmath_{q,q+1}P_q + (1 - \jmath_{q,q+1}T_q)B_{q+1}^{\mathrm{ct}}, \qquad P_{\le q}^{\mathrm{ct}}(o)w = P_q(\tau_{-o}w)\,.$$

The first term retains the degrees already chosen through q; the second takes the new measurement and removes its component in $\jmath_{q,q+1}F_q$. In particular, $P_{\le q}^{\mathrm{ct}}(o)[A(o)] = A$ for $A \in F_q$.

Definition 19.18 (Contact grades and clearance)**.** The exact-degree components and common clearance are

$$\begin{aligned} P^{\mathrm{ct}}_{=0}(o)w &= B^{\mathrm{ct}}_0(\tau_{-o}w), \\ P^{\mathrm{ct}}_{=q}(o)w &= (1 - \jmath_{q-1,q}T_{q-1})B^{\mathrm{ct}}_q(\tau_{-o}w) \quad (q>0), \\ \delta^{\mathrm{pres}}_q &= \min_{0\le d\le q} \rho^{\mathrm{ct}}_d \,. \end{aligned}$$

These give the graded contact subtractions and their common range of separated evaluation.

The projections are defined before any test function or state. Their strata are the contact degree subspaces $E^{\mathrm{ct}}_0 = F_0$ and $E^{\mathrm{ct}}_q = \ker T_{q-1} \subset F_q$ $(q>0)$. The proof of the next lemma shows that $F_q = \jmath_{q-1,q}F_{q-1} \oplus E^{\mathrm{ct}}_q$; these strata, not the complements E_q of Definition 19.5, carry the real bases of the contact construction.

Lemma 19.19 (Real bases)**.** *The contact projections are covariant under translations, $O(4)$ and conjugation. Each contact stratum E^{ct}_q has a real basis, and the degree-zero basis contains the unit.*

Proof. The inclusions, rotations and conjugation of Part I act as in Lemma 19.6. A ball centered at zero is rotation invariant. Haar invariance proves rotation covariance of B^{ct}_q, and its reality formula proves conjugation covariance. These maps fix $A(0)$: before averaging the reconstruction map returns $R_q(h^{-1})A$, and after multiplication by $R_q(h)$ it returns A. The recursion gives

$$P_qI_q = 1, \qquad T_q\jmath_{q,q+1} = 1, \qquad T_qP_{q+1} = P_q, \qquad P^{\mathrm{ct}}_{\le q}(o) = \sum_{d\le q} \jmath_{d,q}P^{\mathrm{ct}}_{=d}(o)\,. \tag{19.15}$$

For the third identity, apply T_q to the recursion and cancel $T_qB^{\mathrm{ct}}_{q+1}$; the fourth follows by iteration. By the second identity, $1 - \jmath_{q-1,q}T_{q-1}$ projects F_q onto $\ker T_{q-1}$ along $\jmath_{q-1,q}F_{q-1}$, which proves the splitting. Each stratum is invariant under conjugation and rotations. Lemma 19.6 gives real bases; choose the unit in F_0. Covariance passes through the recursion, with translations following from $\tau_{-(o+u)}\tau_u = \tau_{-o}$. □

We give the finite subtraction formula before proving its convergence. Fix $\boldsymbol{A} \in \mathbb{F}^n$, put $d_i = \deg A_i$, $W = \sum_i d_i$, and choose

$$r_* = \min(1, \delta^{\mathrm{pres}}_W), \qquad s_* = r_*/4\,.$$

Lemma 19.21 uses only this single length s_*. Expand each input in the contact stratum bases using (19.15), retaining its original degree tag d_i in every subtraction order. For $h \subset \{1,\dots,n\}$, $|h| \ge 2$, let Δ_h be its collision diagonal, $x^\perp_h$ the orthogonal component normal to Δ_h, and $\bar{x}_h = |h|^{-1}\sum_{i\in h} x_i$. Thus $x^\perp_h$ has entries $x_i - \bar{x}_h$ on h and zero elsewhere. Write

$$d_h = \sum_{i\in h} d_i, \qquad n^\perp_h = 4(|h|-1), \qquad L_h = \max(0, d_h + 1 - n^\perp_h)\,.$$

Here $n^\perp_h$ is the dimension of the normal space of Δ_h. Choose a smooth radial cutoff χ equal to one on the unit ball and zero outside the ball of radius two. The normal Taylor operator is

$$(\mathrm{Tay}^\perp_{h,k}f)(x) = \chi(x^\perp_h/s_*)\sum_{l=0}^{k} \frac{1}{l!} D^l f(x - x^\perp_h)[x^\perp_h, \dots, x^\perp_h]\,. \tag{19.16}$$

It subtracts only normal variables, leaving the centroid and all exterior variables as parameters.

An expression $F \otimes U$ means the field expression $F(x)$ multiplied by $(Uf)(x)$ when tested against f. Products compose the test operators. On a monomial write $F = F_h F_{h^c}$ for its factors in and outside h. Let $\mathsf{R}_{h,q}$ replace the subword of $F(x)$ on h by the field $P^{\mathrm{ct}}_{=q}(\bar{x}_h)F_h(x)$ at $\bar{x}_h$, leaving its test operator unchanged.

For two insertions, $h = \{1,2\}$, the centroid $o = \bar{x}_h$ and the relative coordinate $x_1 - x_2$ parametrize $\mathbb{R}^8$, and $x_h^{\perp} = \frac{1}{2}(x_1 - x_2, x_2 - x_1)$. With $F = A_1(x_1)A_2(x_2)$ the subtracted expression is

$$\Pi_h^{\mathrm{full}} f = Ff - \sum_{q<L_h} \big[P^{\mathrm{ct}}_{=q}(o)F\big](o)\,\mathrm{Tay}^{\perp}_{h,L_h-1-q} f, \qquad L_h = \max(0, d_1 + d_2 - 3).$$

The retained field degree q determines how many normal derivatives of the test are subtracted. On a shell $|x_h^{\perp}| \asymp \rho$, the degree-$q$ component has size ρ^{q-d_h} up to a loss $\rho^{-\varepsilon'}$, the Taylor remainder of f through order $L_h - 1 - q$ is $O(\rho^{L_h-q})$, and the shell has normal volume of order ρ^4. The two arms of the signed expression, the field minus its projected components times f and the projected components times the Taylor remainder, therefore contribute $O(\rho^{4+L_h-d_h-\varepsilon'}) = O(\rho^{1-\varepsilon'})$ per shell, summable over dyadic ρ. The individual terms need not be integrable: for $d_h \geq 4$ the term Ff alone is only $O(\rho^{4-d_h})$ per shell, so each subtraction must stay with the term it subtracts.

In general, define recursively

$$\mathcal{C}^{\mathrm{ct}}_{\{i\}} = A_i(x_i) \otimes 1, \qquad \Pi_h^{\mathrm{punct}} = \sum_{\substack{\pi \text{ partition of } h \\ |\pi| \geq 2}} \prod_{b \in \pi} \mathcal{C}^{\mathrm{ct}}_b, \tag{19.17}$$

$$\mathcal{C}^{\mathrm{ct}}_h = -\sum_{q<L_h} \mathsf{R}_{h,q}\Pi_h^{\mathrm{punct}}\,\mathrm{Tay}^{\perp}_{h,L_h-1-q}, \qquad \Pi_h^{\mathrm{full}} = \Pi_h^{\mathrm{punct}} + \mathcal{C}^{\mathrm{ct}}_h \quad (|h| \geq 2)\,.$$

Singleton and empty full expressions are respectively $A_i(x_i) \otimes 1$ and the unit. In a nested term, the parent Taylor operator acts first on the test and the child operators act afterwards: if $b \subsetneq h$ are a child and its parent, with Taylor orders k_b, k_h, the resulting test factor is

$$\mathrm{Tay}^{\perp}_{b,k_b}\big(\mathrm{Tay}^{\perp}_{h,k_h} f\big),$$

so the parent operator is written on the right. Operators on disjoint slot sets commute; no commutation is asserted for nested subsets. The recursion terminates because every child has fewer slots; expanding it gives forests of disjoint or nested collision subsets. This is a position-space forest formula. Subtracting a Taylor jet of the test function at a diagonal is, in the normal variables, the extension procedure of [7, Section 5.2, Theorem 5.3]; here the subtracted jets are paired with contact projections, and the recursion organizes the partial collisions.

For three insertions the expression before the total-collision subtraction is

$$\Pi^{\mathrm{punct}}_{\{1,2,3\}} = A_1A_2A_3 + \mathcal{C}^{\mathrm{ct}}_{\{1,2\}}A_3 + \mathcal{C}^{\mathrm{ct}}_{\{1,3\}}A_2 + \mathcal{C}^{\mathrm{ct}}_{\{2,3\}}A_1,$$

with the test operators in each $\mathcal{C}^{\mathrm{ct}}$ retained. The total-collision subtraction is applied to this complete expression. Its nested terms have the form $\mathsf{R}_{\{1,2,3\},q}\mathsf{R}_{\{1,2\},q'}(A_1A_2A_3)$ with test factor $\mathrm{Tay}^{\perp}_{\{1,2\},k'}(\mathrm{Tay}^{\perp}_{\{1,2,3\},k} f)$. Where $|x_1 - x_2|$ is much smaller than the distance to x_3, the child arms make the expression integrable in the normal variable of $\Delta_{\{1,2\}}$, and the parent arms then treat the total collision. Figure 17 shows both examples.

Before summing shells we need estimates for complete signed expressions after child subtractions. Normalize $T \to \infty$ so that all separation heights lie in $[0,1]$, and fix a finite set $\mathcal{P}$ of position paths.

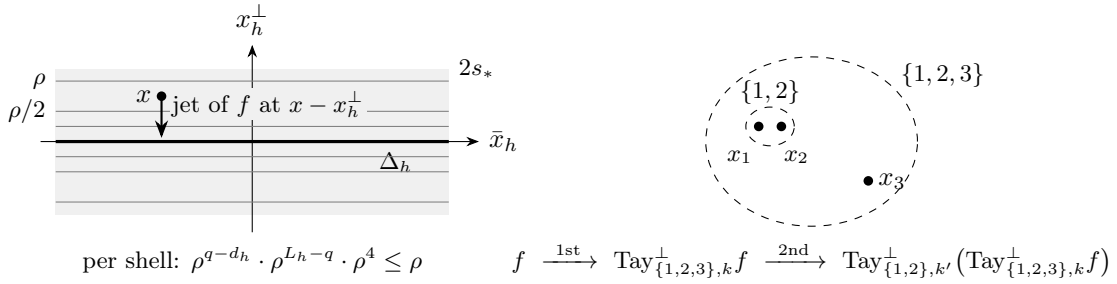


Figure 17: Left: two insertions; the test is Taylor expanded in $x_h^\perp$ at Δ_h where $|x_h^\perp| \le 2s_*$, and each dyadic shell contributes at most ρ, up to $\rho^{-\varepsilon'}$. Right: a nested forest; the parent operator acts on the test first, and only operators on disjoint slot sets commute. Both are subtractions of separated correlations, not lattice limits at coincident points.

A finite separated sum of cost c at height s has terms

$$a(T) \prod_\iota A_\iota(x_\iota(T)), \qquad |a| \preccurlyeq e^{(c - s\sum_\iota \deg A_\iota)T}, \qquad |x_\iota - x_{\iota'}| \ge e^{-sT - o(T)} \quad (\iota \ne \iota'),$$

with $x_\iota \in \mathcal{P}$. This definition permits different terms to use overlapping subsets of $\mathcal{P}$.

The parameter s describes the cluster separation scale e^{-sT}, and c its total exponential growth bound. Membership in the class $\mathbb{A}_s(c)$ defined next means that the signed expression can be replaced, to arbitrarily high accuracy in exterior correlations, by a finite separated sum with this bound. The exterior word may have internal separations as small as e^{-T}, but it stays at distance at least $e^{-sT-o(T)}$ from the cluster. The approximating sum is chosen before the exterior word and its cost.

Write $\mathcal{U} \in \mathbb{A}_s(c)$ if it has a finite height-one representation of some finite cost and satisfies the following approximation property. For every $L > 0$, there is a finite separated sum g of cost c at height s, using $\mathcal{P}$, such that for every $c' \in \mathbb{R}$ and every exterior finite separated sum Y of cost c' at height 1,

$$\big|S^\zeta((\mathcal{U} - g)Y)\big| \preccurlyeq e^{(-L+c')T}. \tag{19.18}$$

Every exterior path must be at distance at least $e^{-sT-o(T)}$ from every path in $\mathcal{P}$. The bound permits any varying character $\zeta(T)$ and the mixed state. Scalar factors of rate b change the cost to $c + b$.

The precision L is relative to the exterior: against an exterior sum of cost c' the error is e^{-LT} times the growth $e^{c'T}$ of that sum. A fixed absolute precision would not do, because exterior sums of every cost occur: in a product $\mathcal{U}\mathcal{V}$, the factor $\mathcal{V}$ belongs to the exterior of $\mathcal{U}$.

Two scales. Consider a child cluster at separation ρ inside a parent region of size R, with $\rho < R \le 1$, and put

$$T = \log(1/\rho), \qquad s = 1, \qquad u = \frac{\log(1/R)}{T},$$

so that $e^{-sT} = \rho$, $e^{-uT} = R$ and $e^{-(s-u)T} = \rho/R$. A product of fields of total degree w at the cluster points, with bounded coefficient, is a separated sum of cost $c = w$ at height s. The next lemma bounds its contact components and residuals; in these scales, and up to a subpower factor

$\rho^{-o(1)}$, its conclusions read

degree-q component at o :	rate $c - qs = w - q$,	size ρ^{q-w},
residual through degree q, seen at height u :	cost $c - (q+1)(s-u)$,	size $\rho^{-w}(\rho/R)^{q+1}$.

The residual cost follows by cancelling lower degrees. Expanding the cluster at o gives, up to an error of any prescribed precision, fields of degree d with coefficient rates $c - ds$. The contact identities (19.15) remove exactly the degrees $d \le q$. A remaining field of degree $d \ge q+1$, and the fields of lower degree that its projection produces, carry the same coefficient rate; seen from height u, they cost at most $c - d(s-u) \le c - (q+1)(s-u)$. In Lemma 19.21 the expression $\mathcal{U}$ is a punctured expression whose child arms carry their test operators. The parent projection acts on the field word and leaves the child Taylor operators in place. The derivatives of these operators are scalar factors of finite rate, which add to the cost.

Suppose the paths in $\mathcal{P}$ lie within $e^{-sT+o(T)}$ and within $2s_*$ of a center $o \in \mathcal{P}$. For contact projections assume $2s_* \le \rho_q^{\mathrm{ct}}$. Write $I_q^o A = A(o)$ for insertion at this center. These fixed-distance requirements ensure that the contact probes are clear; the exponential distances determine the rates.

Lemma 19.20 (Signed layer transfer)**.** *For $0 \le u \le s \le 1$ and $\mathcal{U} \in \mathbb{A}_s(c)$, every clear contact degree q satisfies*

$$P^{\mathrm{ct}}_{=q}(o)\mathcal{U} \preccurlyeq e^{(c-qs)T}, \qquad \mathcal{U} - I_q^o P^{\mathrm{ct}}_{\le q}(o)\mathcal{U} \in \mathbb{A}_u\big(c - (q+1)(s-u)\big).$$

Separated products add their costs; finite signed sums take their maximum. Coarsening from s to u without subtracting retains cost c.

The projected degree-q component thus costs c less the Taylor gain qs, and the residual through degree q gains $(q+1)(s-u)$ when it is viewed from the coarser height u.

Proof. The proof orders the approximation precisions before applying the projections. Lemma 19.9, including its uniformity on compact label sets, expands each approximating term at the center; the telescope (19.14) handles products; the exact contact identities (19.15) give the cancellations.

First check products, which explains the strength of the approximation property. Let $\mathcal{U}, \mathcal{V}$ have costs c_1, c_2 and a separation at height s. Suppose a height-one representation of $\mathcal{V}$ has cost b. Choose an approximant g to $\mathcal{U}$ of precision $L + b$. Its own height-one representation has some finite cost b'. Next choose an approximant g' to $\mathcal{V}$ of precision $L + b'$. The order matters: the precision required of g' depends on the cost b' of g, which is known only after g has been chosen. Then

$$\mathcal{U}\mathcal{V} - gg' = (\mathcal{U} - g)\mathcal{V} + g(\mathcal{V} - g')$$

has precision L against every permitted exterior sum. The product gg' has cost $c_1 + c_2$, since its word positions are separated at height s and its scalar coefficient rates add. Iterate in a fixed order for more factors. Addition and subtraction follow by taking the maximum cost and making every approximation as accurate as required.

For $u < s$, expand a finite separated approximant to $\mathcal{U}$ at o to an auxiliary degree H. Applied to each term, (19.11) gives output fields $A_d(o)$ whose coefficient rates are at most $c - ds$, and an error at exterior height u of rate at most $c - (H+1)(s-u)$. Increasing H gives any specified precision. The original approximation error permits these farther exterior tests. This gives an arbitrarily accurate single-point sum g at o, of cost at most c at height u. If $u = s$, retain the original approximant. This proves coarsening.

For $s > 0$, approximate with exterior height zero. A contact probe of total degree p costs at most p at height one; raise the precision by p. Compactness of $O(4)$ makes this uniform over rotated and conjugated probes, so averaging preserves arbitrary precision. On a degree-d field the degree-q contact row is zero for $d < q$ and is a fixed linear map for $d \geq q$. Its rate is at most $c - ds \leq c - qs$, proving the row bound. At $s = 0$ test directly against the fixed clear probes: their positions have height zero and add no exponent. The same conclusion follows.

For the residual, use $u < s$ and the single-point sum g just constructed. Every field of degree $d \leq q$ cancels exactly under $1 - I_q^o P_{\leq q}^{\mathrm{ct}}$. For $d \geq q+1$ its coefficient rate at the outgoing height is $c - ds$. Viewed at height u, its cost is at most $c - d(s-u) \leq c - (q+1)(s-u)$. The lower fields produced by its contact projection obey the same inequality, since their degrees are at most d. Thus $g - I_q^o P_{\leq q}^{\mathrm{ct}} g$ has the required cost. The exact error is $(\mathcal{U} - g) - I_q^o P_{\leq q}^{\mathrm{ct}}(\mathcal{U} - g)$. The first term has arbitrary precision by choice of H; the finite probes defining the second cost only finitely many additional exponents. Choose the precision to pay for them and for the requested exterior test. This proves the residual's approximation property. Its finite height-one representation follows from the initial one and the finite probe formulas. At $u = s$ the row bound makes the projected finite sum have cost c at height s, so subtraction retains that cost directly. □

The extension and its order are proved together.

Lemma 19.21 (Contact extension and local order)**.** *Fix a state index ζ, an arity n and a list $\boldsymbol{A} \in \mathbb{F}^n$, and write $W = |\boldsymbol{A}|_{\deg}$. The normal subtractions (19.17), with the contact projections, bases and length s_* fixed above, define a complex linear functional $\mathfrak{D}_{\boldsymbol{A}}^{\zeta}$ on $C_c^\infty(\mathbb{R}^{4n})$. For every compact $K \subset \mathbb{R}^{4n}$ there is $C_K < \infty$ such that*

$$|\mathfrak{D}_{\boldsymbol{A}}^{\zeta}(f)| \leq C_K \max_{|\alpha| \leq (W+2)n} \|\partial^\alpha f\|_\infty \qquad (\operatorname{supp} f \subset K).$$

For every $f \in C_c^\infty(\mathrm{Conf}_n)$, $\mathfrak{D}_{\boldsymbol{A}}^{\zeta}(f) = \int f(x) S_n^{\zeta}(\boldsymbol{A}; x)\, dx$.

Proof. We prove absolute integrability of the complete subtracted expression on every compact set, with a finite test seminorm, in four steps. We keep each contact subtraction with the term it subtracts; Lemma 19.20 bounds the resulting signed expressions on one collision sector; normal Taylor vanishing and an induction over partial collisions make the dyadic shell sum converge; a partition of unity covers an arbitrary compact set.

Step 1: keep the signed cancellation. At a group h, reorganize the full expression as

$$\left(1 - \sum_{q < L_h} \mathsf{R}_{h,q}\right) \Pi_h^{\mathrm{punct}} f + \sum_{q < L_h} \mathsf{R}_{h,q} \Pi_h^{\mathrm{punct}} (1 - \mathrm{Tay}_{h, L_h - 1 - q}^{\perp}) f\,. \tag{19.19}$$

These are identities of field expressions with test operators; individual unsubtracted summands need not be integrable. Every nonzero Taylor term lies within $2s_* \leq r_*$ of its diagonal, so all its field projections have their required contact clearance. If $L_h = 0$, no projection is subtracted and $n_h^{\perp} - d_h \geq 1$ already gives integrability. For $L_h > 0$, (19.15) removes every degree below L_h from the first arm. We apply Lemma 19.20 to the punctured expression, including the child subtractions. Here is the induction invariant: after fixing the child arms and their test operators, the field expression belongs to $\mathbb{A}_s(c)$ at its outgoing height; a projected arm of degree q has coefficient rate $c - qs$, and a residual arm through q has cost $c - (q+1)(s-u)$ at a parent height u. Singleton fields start this invariant. Products over separated partition blocks preserve it by the product clause of Lemma 19.20; its projection and residual clauses give the next arms, and its coarsening

clause handles equal heights. Keep every child Taylor operator attached to its arm throughout these operations. Each derivative expansion has finitely many scalar test factors of finite rate. The scalar-factor and product clauses of Lemma 19.20 include these in the approximation cost and retain the invariant for each differentiated arm.

Step 2: bounds on one collision sector. Rescale the patch so all scales are at most one. Fix its finite collision tree, write ρ_b for its scales and $R = \rho_{\mathrm{par}(h)}$, and set

$$T = \log(1/\rho_{\min}), \quad s_b = \log(1/\rho_b)/T, \quad u = \log(1/R)/T, \quad \rho_{\min} = \min_b \rho_b \,.$$

After a subsequence these heights converge in $[0,1]$, with $s_h \geq u$. Retain the child size factor $Z_h = \prod_{b \text{ proper below } h} (\rho_b/\rho_{\mathrm{par}(b)})^{-d_b}$. Before child gains, the incoming cost is $c = d_h s_h + \sum_{b \text{ proper below } h} d_b(s_b - s_{\mathrm{par}(b)})$. Lemma 19.20 gives a degree-q row of rate $c - qs_h$, and a residual through degree $L_h - 1$ of cost $c - L_h(s_h - u)$. Thus, after retaining Z_h and the outer field costs, their physical size bounds are respectively

$$C_{\varepsilon'} Z_h \rho_h^{q-d_h} \rho_{\min}^{-\varepsilon'}, \qquad C_{\varepsilon'} Z_h \rho_h^{-d_h} (\rho_h/R)^{L_h} \rho_{\min}^{-\varepsilon'}. \tag{19.20}$$

With one unresolved scale, $\rho_{\min} = \rho_h = \rho$, giving the row bound $C_{\varepsilon'} \rho^{q-d_h-\varepsilon'}$ and residual gain $(\rho/R)^{L_h}$. With several scales, retain the loss at $\rho_{\min}$ and the child gains; the induction below distributes the loss between ancestors and descendants. If these estimates failed uniformly in a sector, choose a violating sequence and take the convergent height vector just described. The signed invariant, including equal heights, gives the rates with loss $\varepsilon'/2$, contradicting the violation with loss ε'. This proves (19.20) uniformly over the sector.

Taylor's integral remainder gives, on a shell $|x_h^\perp| \asymp \rho$ within the subtraction range,

$$(1 - \mathrm{Tay}^\perp_{h, L_h - 1 - q}) f = O\left(\rho^{L_h - q} \max_{|\alpha| \leq L_h - q} \|\partial^\alpha f\|_\infty \right) .$$

Including normal volume, every term of (19.19) therefore gains $\rho^{n_h^\perp + L_h - d_h - \varepsilon'}$, whose exponent is at least $1 - \varepsilon'$.

Step 3: partial collisions and summable shell bounds. The count $\rho^{q-d_h} \cdot \rho^{L_h - q} \cdot \rho^4$ of the two-insertion example concerns one shell with no further collision. Steps 1 and 2 justify its coefficient size, up to the loss $\rho^{-\varepsilon'}$, for the complete signed expression; the present step supplies the cover of partial collisions and the induction over them. Partial collisions on a normal shell require a further induction. Induct on $|h| - |\pi|$, where π is the currently separated partition. The case of singleton blocks is a separated correlation against a smooth compactly supported test, bounded by Lemma 15.6 and (19.13). For a nonsingleton block, cut its normal coordinate into dyadic shells and rescale to $1/2 \leq |x_h^\perp| \leq 2$. At any point of this compact annulus its equality classes form a proper partition. The centroids of nonempty subgroups contained in different classes are distinct there. There are finitely many such subgroups; continuity gives a neighborhood where their distances have a common positive lower bound. A finite smooth partition of unity on the annulus thus gives sectors with all subsequent subgroup centroids separated. Within a sector apply the induction hypothesis to this strictly finer partition. On overlaps the resulting expressions agree: a group crossing two sectors' separated blocks has a Taylor jet on a diagonal disjoint from the localized test support, and contributes zero. The subdivision therefore preserves the full expression and its ordered test operators.

For the shell estimate, let $a > 0$ be the shell radius divided by its parent scale. The rescaled Taylor remainder through order k obeys

$$\|(1 - \mathrm{Tay}^\perp_{h,k}) f\|_{C^l \text{ on the rescaled shell}} \leq C \|f\|_{C^{l+k+1}} \begin{cases} a^{k+1}, & a \leq 1, \\ a^k, & a > 1. \end{cases} \tag{19.21}$$

For $a \le 1$, differentiate the integral Taylor remainder, including the cutoff; for $a > 1$, use the bounded test and the degree-k polynomial. The cutoff derivatives have fixed bounds at the parent scale. For a projected component q, use Lemma 19.20 to transfer the signed field expression and put $k = L_h - 1 - q$. Its volume and coefficient power $a^{n_h^\perp + q - d_h}$ times the Taylor factor is at most a when $a \le 1$, and at most 1 when $a > 1$. Indeed $n_h^\perp + q - d_h + k = 0$ whenever $q < L_h$. For the unprojected remainder the power is $n_h^\perp + L_h - d_h \ge 1$; its support restricts a above by a fixed geometric constant. Thus, for every $0 < \theta \le 1$, the combined estimate is at most Ca^θ.

Choose $\gamma > 0$ with $(3^{|h|-|\pi|} + 1)\gamma \le 1$. At a finer partition with d' remaining splits, the signed invariant of Step 1 remains valid for every projected and residual arm. Allocate loss $3^{d'}\gamma$ to its coefficient and test estimates. The induction hypothesis is:

$$\begin{aligned}(\mathrm{H}_{d'}) \quad & \text{each projected or residual arm, on a shell} \\ & |x_h^\perp| \asymp \rho \le R \le 1, \text{ contributes at most} \\ & C(\rho/R)^\gamma R^{-3^{d'}\gamma} \|f\|_{C^{(W+2)(d'+1)}} .\end{aligned}$$

Apply the preceding shell estimate with $\varepsilon' = 3^{d'}\gamma$ and $\theta = \gamma + \varepsilon' \le 1$. One split strictly decreases the remaining number of splits, and $3^{d'+1}\gamma \ge 2 \cdot 3^{d'}\gamma + \gamma$ leaves a factor a^γ after the preceding estimate. To see that this is uniform in the parent scale R, combine the child loss $\rho^{-\varepsilon'}$ with its ratio improvement by

$$(\rho/R)^{\gamma+\varepsilon'}\rho^{-\varepsilon'} = (\rho/R)^\gamma R^{-\varepsilon'} .$$

The child keeps the ratio gain $(\rho/R)^\gamma$, and its loss now sits at the parent scale; this factor $R^{-\varepsilon'}$ is charged to the parent's allocated loss. Finite sector sums change only the constant. For shells $\rho \asymp 2^{-j}$, the bound is consequently $C2^{-j\gamma}$ times a fixed test seminorm. Summing this geometric series completes the induction, and gives absolute convergence also for iterated choices of disjoint shells.

Step 4: the distribution on an arbitrary compact set. Cover a given compact set by neighborhoods of the equality classes at each of its points. The same separation argument gives finitely many patches, including patches outside the total-collision subtraction range. On these patches the recursion reduces to the product of the full expressions in the separated classes: any crossing Taylor term has zero jet. A smooth partition of unity and the Leibniz rule combine their bounds. Each descent costs at most $W + 2$ derivatives by (19.21); there are at most n descents. We obtain

$$\int \left| \left\langle \Pi^{\mathrm{full}}_{\{1,\dots,n\}} f \right\rangle_\zeta (x) \right| dx \le C_K \max_{|\alpha| \le (W+2)n} \|\partial^\alpha f\|_\infty \quad (\operatorname{supp} f \subset K) . \tag{19.22}$$

Define $\mathfrak{D}^\zeta_{\boldsymbol{A}}(f)$ by this integral. Absolute convergence preserves complex linearity, and (19.22) with the integral triangle inequality gives the asserted local order.

For $f \in C_c^\infty(\mathrm{Conf}_n)$, every normal jet on a collision diagonal vanishes. Each nontrivial term of (19.17) has a largest Taylor operator acting first; disjoint operators commute. Only $f(x)\prod_i A_i(x_i)$ remains, proving the restriction also for empty and singleton lists. □

Proof of Theorem 6.21. Definitions 19.16–19.18 fix the contact projections on the prescription of Lemma 19.6, and Lemma 19.19 gives their bases. Lemma 19.21 gives the distribution, its local order, and its agreement with the separated correlations. □

The conclusions of Theorem 6.21 assert neither uniqueness, positivity of the extension nor contact Ward identities. Estimates along general collision subspaces are not stated here and are not used in Theorem B.

References

[1] L. F. Abbott, *Introduction to the background field method*, Acta Physica Polonica B **13** (1982), no. 1–2, 33–50.

[2] Tadeusz Bałaban, *The variational problem and background fields in renormalization group method for lattice gauge theories*, Communications in Mathematical Physics **102** (1985), 277–309.

[3] ———, *Renormalization group approach to lattice gauge field theories. I. Generation of effective actions in a small field approximation and a coupling constant renormalization in four dimensions*, Communications in Mathematical Physics **109** (1987), 249–301.

[4] ———, *Large field renormalization. II. Localization, exponentiation, and bounds for the R operation*, Communications in Mathematical Physics **122** (1989), 355–392.

[5] Roland Bauerschmidt, David C. Brydges, and Gordon Slade, *Critical two-point function of the 4-dimensional weakly self-avoiding walk*, Comm. Math. Phys. **338** (2015), 169–193, arXiv:1403.7268.

[6] ———, *Introduction to a renormalisation group method*, Lecture Notes in Mathematics, vol. 2242, Springer, Singapore, 2019.

[7] Romeo Brunetti and Klaus Fredenhagen, *Microlocal analysis and interacting quantum field theories: Renormalization on physical backgrounds*, Communications in Mathematical Physics **208** (2000), 623–661.

[8] David C. Brydges and Horng-Tzer Yau, *Grad φ perturbations of massless Gaussian fields*, Communications in Mathematical Physics **129** (1990), 351–392.

[9] John C. Collins, Anthony Duncan, and Satish D. Joglekar, *Trace and dilatation anomalies in gauge theories*, Physical Review D **16** (1977), 438–449.

[10] Gerald B. Folland, *Real analysis: Modern techniques and their applications*, 2 ed., John Wiley & Sons, New York, 1999.

[11] ———, *A course in abstract harmonic analysis*, 2 ed., Textbooks in Mathematics, CRC Press, Boca Raton, FL, 2015.

[12] Robert L. Foote, *Regularity of the distance function*, Proceedings of the American Mathematical Society **92** (1984), no. 1, 153–155.

[13] Markus B. Fröb and Jan Holland, *All-order existence of and recursion relations for the operator product expansion in Yang–Mills theory*, 2016, arXiv:1603.08012, revised 2021.

[14] Jürg Fröhlich, Robert Israel, E. H. Lieb, and Barry Simon, *Phase transitions and reflection positivity. I. General theory and long range lattice models*, Communications in Mathematical Physics **62** (1978), no. 1, 1–34.

[15] Jürg Fröhlich, Konrad Osterwalder, and Erhard Seiler, *On virtual representations of symmetric spaces and their analytic continuation*, Annals of Mathematics **118** (1983), no. 3, 461–489.

[16] Krzysztof Gawędzki and Antti Kupiainen, *Massless lattice φ^4_4 theory: Rigorous control of a renormalizable asymptotically free model*, Communications in Mathematical Physics **99** (1985), 197–252.

[17] David J. Gross and Frank Wilczek, *Ultraviolet behavior of non-Abelian gauge theories*, Physical Review Letters **30** (1973), 1343–1346.

[18] Arthur Jaffe and Edward Witten, *Quantum Yang–Mills theory*, Clay Mathematics Institute, Millennium problem description, 2006.

[19] N. P. Landsman, *Lecture notes on C*-algebras, Hilbert C*-modules, and quantum mechanics*, 1998.

[20] Jiří Lebl, *Tasty bits of several complex variables: A whirlwind tour of the subject*, 2026, Version 4.4, May 31, 2026.

[21] Jacques Magnen, Vincent Rivasseau, and Roland Sénéor, *Construction of YM4 with an infrared cutoff*, Communications in Mathematical Physics **155** (1993), 325–383.

[22] OpenAI, *The canonical massive continuum limit of the two-dimensional $O(3)$ model*, October 2026, Preprint of October 4, 2026.

[23] ———, *Sharp mass bounds for the two-dimensional $O(4)$ model*, September 2026, Preprint of September 23, 2026.

[24] ———, *Wilson Yang–Mills construction*, Lean 4 formalization, commit 3456e5e, 2026.

[25] Konrad Osterwalder and Robert Schrader, *Axioms for Euclidean Green's functions*, Communications in Mathematical Physics **31** (1973), no. 2, 83–112.

[26] ———, *Axioms for Euclidean Green's functions. II*, Communications in Mathematical Physics **42** (1975), 281–305, With an appendix by Stephen Summers.

[27] Konrad Osterwalder and Erhard Seiler, *Gauge field theories on a lattice*, Annals of Physics **110** (1978), no. 2, 440–471.

[28] H. David Politzer, *Reliable perturbative results for strong interactions?*, Physical Review Letters **30** (1973), 1346–1349.

[29] Walter Rudin, *Principles of mathematical analysis*, 3 ed., McGraw-Hill, New York, 1976.

[30] Walter Rudin, *Real and complex analysis*, 3 ed., McGraw–Hill, New York, 1987.

[31] Erhard Seiler, *Lattice gauge theories*, Gauge Theories as a Problem of Constructive Quantum Field Theory and Statistical Mechanics, Lecture Notes in Physics, vol. 159, Springer, Berlin, Heidelberg, 1982, pp. 3–98.

[32] Gerald Teschl, *Ordinary differential equations and dynamical systems*, Graduate Studies in Mathematics, vol. 140, American Mathematical Society, Providence, RI, 2012.

[33] The Stacks Project Authors, *The Stacks Project*, 2026, Lemma 10.31.1, Tag 00FN; accessed 11 November 2026.

[34] Daniel Ueltschi, *Cluster expansions and correlation functions*, Moscow Mathematical Journal **4** (2004), no. 2, 511–522.

[35] Kenneth G. Wilson, *Non-Lagrangian models of current algebra*, Physical Review **179** (1969), 1499–1512.

[36] ———, *Confinement of quarks*, Physical Review D **10** (1974), 2445–2459.